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United States Patent Application Publication

20250282781

Kind Code

A1

Publication Date

September 11, 2025

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APOL1 INHIBITORS AND METHODS OF USE

Abstract

Provided herein are compounds of formula (II):

##STR00001##

or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein wherein m, n, p, R.sup.1, R.sup.2, R.sup.3, L.sup.1, L.sup.2, L.sup.3, R.sup.4, X.sup.1, X.sup.2, X.sup.3, and X.sup.4 are as defined herein. Also provided are methods of preparing compounds of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing. Also provided are methods of inhibiting APOL1 and methods of treating an APOL1-mediated disease, disorder, or condition in an individual.

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Family ID: 1000008626574

Appl. No.: 19/218105

Filed: May 23, 2025

Related U.S. Application Data

parent US continuation 18617266 20240326 parent-grant-document US 12344610 child US 19218105

parent US continuation 18098070 20230117 parent-grant-document US 11976067 child US 18617266

us-provisional-application US 63300592 20220118

us-provisional-application US 63311668 20220218
us-provisional-application US 63332553 20220419
us-provisional-application US 63400359 20220823
us-provisional-application US 63422341 20221103

Publication Classification

Int. Cl.: **C07D471/10** (20060101); **C07D487/10** (20060101); **C07D519/00** (20060101);
C07F9/53 (20060101)

U.S. Cl.:

CPC **C07D471/10** (20130101); **C07D487/10** (20130101); **C07D519/00** (20130101);
C07F9/5333 (20130101); C07B2200/05 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS [0001] This application is a continuation of U.S. patent application Ser. No. 18/617,266, filed Mar. 26, 2024, which is a continuation application of U.S. patent application Ser. No. 18/098,070, filed Jan. 17, 2023, now U.S. Pat. No. 11,976,067, issued May 7, 2024, which claims priority to U.S. Provisional Application Ser. No. 63/300,592, filed Jan. 18, 2022, U.S. Provisional Application Ser. No. 63/311,668, filed Feb. 18, 2022, U.S. Provisional Application Ser. No. 63/332,553, filed Apr. 19, 2022, U.S. Provisional Application Ser. No. 63/400,359, filed Aug. 23, 2022, and U.S. Provisional Application Ser. No. 63/422,341, filed Nov. 3, 2022, each of which is hereby incorporated herein by reference in its entirety.

BACKGROUND OF THE INVENTION

[0002] Apolipoprotein L1 (APOL1) is a pore forming innate immunity factor, protecting individuals from trypanosome parasites (Vanhamme, L. et al. *Nature* (2003) 422, 83-87). The secreted form of APOL1 circulates in blood as part of distinct high-density lipoprotein (HDL) complexes, known as trypanosome lytic factors (TLFs) (Rifkin, M. R. *Proc. Natl. Acad. Sci. USA*. (1978) 75, 3450-3454; Raper, J. et al. *Infect. Immun.* (1999) 67, 1910-1916). TLFs are internalized by the parasites through endocytosis (Hager, K. M. et al. *J. Cell Biol.* (1994) 126, 155-167). Within trypanosomes, APOL1 forms cation pores, causing ion flux, swelling, and eventual lysis (Rifkin, M. R. *Exp. Parasitol.* (1984) 58, 81-93; Molina-Portela, M. P. et al. *Mol. Biochem. Parasitol.* (2005) 144, 218-226; Pérez-Morga, D. et al. *Science*. (2005) 309, 469-472; Thomson, R. & Finkelstein, A. *Proc. Natl. Acad. Sci. USA*. (2015) 112, 2894-2899).

[0003] Several *Trypanosoma brucei* subspecies (*T.b. rhodesiense* and *T.b. gambiense*) developed resistance mechanisms to APOL1-dependent killing (Pays, E. et al. *Nat. Rev. Microbiol.* (2014) 12, 575-584). Positive selection resulted in APOL1 variants, G1 (S342G, I384M) and G2 (N388A, Y389A), capable of interfering with these resistance mechanisms (Genovese, G. et al. *Science*. (2010) 329, 841-845). However, individuals with any binary combination of these variants (G1/G1, G2/G2, or G1/G2), have a greater risk of developing a variety of chronic kidney diseases, including focal segmental glomerulosclerosis (FSGS), hypertension-attributed kidney disease, human immunodeficiency virus-associated nephropathy (HIVAN) (Genovese, G. et al. *Science*. (2010) 329, 841-845; Tzur, S. et al. *Hum. Genet.* (2010) 128, 345-350; Kopp, J. B. et al. *J. Am. Soc. Nephrol.* (2011) 22, 2129-2137), sickle cell nephropathy (Ashley-Koch, A. E. et al. *Br. J. Haematol.* (2011) 155, 386-394), lupus nephritis (Freedman, B. I. et al. *Arthritis Rheumatol.* (2014)

66, 390-396), and an increased rate of Glomerular Filtration Rate (GFR) decline in diabetic kidney disease (Parsa, A. et al. *N. Engl. J. Med.* (2013) 369, 2183-2196). The APOL1 high-risk genotype has also been associated with COVID-19 associated nephropathy and other viral nephropathies (Shetty, A. et al. *J. Am. Soc. Nephrol.* (2021) 32, 33-40; Chang, J. H. et al. *Am. J. Kidney Dis.* (2019) 73, 134-139). Moreover, decreased renal allograft survival has been observed after deceased-donor kidney transplantations from APOL1 high-risk genotype donors (Freedman, B. I. et al. *Transplantation.* (2016) 100, 194-202). In addition, having two APOL1 risk alleles increases risk for preeclampsia (Reidy, K. J. et al. *Am. J. Hum. Genet.* (2018) 103, 367-376) and sepsis (Chaudhary, N. S. et al. *Clin. J. Am. Soc. Nephrol.* (2019) 14, 1733-1740). There are no approved therapies for APOL1-associated nephropathy, and patients are treated based on the standard of care for their underlying form of chronic kidney disease. This presents a clear unmet need for therapies targeted to people with the APOL1 high-risk genotype.

[0004] Numerous studies have shown that APOL1 risk variants are toxic when overexpressed in human cells (Wan, G. et al. *J. Biol. Chem.* (2008) 283, 21540-21549; Lan, X. et al. *Am. J. Physiol. Renal Physiol.* (2014) 307, F326-F336; Olabisi, O. A. et al. *Proc. Natl. Acad. Sci. USA.* (2016) 113, 830-837; Ma, L. et al. *J. Am. Soc. Nephrol.* (2017) 28, 1093-1105; Lannon, H. et al. *Kidney Int.* (2019) 96, 1303-1307). Recent findings suggest that this toxicity is associated with APOL1 pore function (Giovinazzo, J. A. et al. *eLife.* (2020) 9, e51185). Thus, there is a need to develop compounds suitable for inhibiting APOL1 activity and methods for inhibiting the activity of APOL1 using such compounds.

BRIEF SUMMARY OF THE INVENTION

[0005] This disclosure describes compounds and compositions that may be useful for the treatment of APOL1-mediated diseases, including a variety of chronic kidney diseases such as FSGS, hypertension-attributed kidney disease, HIVAN, sickle cell nephropathy, lupus nephritis, diabetic kidney disease, viral nephropathy, COVID-19 associated nephropathy, and APOL1-associated nephropathy. The compounds and compositions may treat other APOL1-mediated disorders such as preeclampsia and sepsis. Additionally, for individuals with the APOL1 high-risk genotype, the disclosed compounds and may prevent the onset of non-diabetic renal disease and/or delaying the progression of any form of chronic kidney disease. The disclosed chemical matter may also prevent and/or delay progressive renal allograft loss in patients who have received a kidney transplant from a high-risk APOL1 genotype donor.

[0006] In one aspect, provided is a compound of formula (II):

##STR00002## [0007] or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, [0008] wherein: [0009] m is an integer from 0 to 4; [0010] n is an integer from 0 to 2; [0011] p is an integer from 0 to 10; [0012] R^{sup.1}, if present is, independently at each occurrence selected from the group consisting of halo, —CN, C_{sub.1-6}alkoxy, and C_{sub.1-6}alkyl, wherein [0013] the C_{sub.1-6}alkoxy of R^{sup.1} is optionally substituted with one or more halo, and [0014] the C_{sub.1-6}alkyl of R^{sup.1} is optionally substituted with one or more halo; [0015] R^{sup.2} is H, C_{sub.1-6}alkyl, C_{sub.3-10}cycloalkyl, or 3-15 membered heterocyclyl, wherein [0016] the C_{sub.1-6}alkyl of R^{sup.2} is optionally substituted with one or more deuterium, halo, —OH, —NH_{sub.2}, or C_{sub.1-6}alkoxy, and [0017] the C_{sub.3-10}cycloalkyl of R^{sup.2} is optionally substituted with one or more —OH; [0018] R^{sup.3}, if present, is C_{sub.1-6}alkyl; [0019] L^{sup.1} is C_{sub.1-6}alkylene, wherein [0020] the C_{sub.1-6}alkylene of L^{sup.1} is optionally substituted with one or more deuterium or C_{sub.1-6}alkyl, and wherein [0021] the C_{sub.1-6}alkyl is further optionally substituted with one or more —OH or C_{sub.1-6}alkoxy; [0022] L^{sup.2} is O or N(R^{sup.x}), wherein R^{sup.x} is H or C_{sub.1-6}alkyl; and either [0023] (1) L^{sup.3} is absent or is O, C_{sub.3-10}cycloalkyl, 3-10 membered heterocyclyl, or C_{sub.1-6}alkylene, wherein [0024] the C_{sub.3-10}cycloalkyl of L^{sup.3} is optionally substituted with one or more —OH or C_{sub.1-6}alkyl, [0025] the C_{sub.1-6}alkylene of L^{sup.3} is optionally substituted with one or more —OH or C_{sub.1-6}alkyl, wherein the C_{sub.1-6}alkyl is optionally substituted with one or more —OH, and

[0026] the 3-10 membered heterocyclyl of L.sup.3 is optionally substituted with one or more —OH or C.sub.1-6alkyl; [0027] X.sup.1 and X.sup.2 are each independently N or C(R.sup.5); and [0028] R.sup.4 is: [0029] (i) —S(O).sub.2—R.sup.a; [0030] (ii) 5-20 membered heteroaryl, wherein the 5-20 membered heteroaryl of R.sup.4 is optionally substituted with one or more C.sub.1-6alkyl; [0031] (iii) —N(R.sup.d).sub.2, wherein R.sup.d is independently at each occurrence H, C.sub.1-6alkyl, or —S(O).sub.2—R.sup.a, wherein the C.sub.1-6alkyl of R.sup.d is optionally substituted with one or more —OH, [0032] (iv) —NS(O)—(C.sub.1-6alkyl).sub.2, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH, [0033] (v) —C(O)—N(R.sup.e).sub.2 wherein R.sup.e is independently at each occurrence H, C.sub.1-6alkyl, or 3-10 membered heterocycle, wherein [0034] the 3-10 membered heterocycle of R.sup.e is optionally substituted with one or more oxo, or both R.sup.e together with the N to which they are attached are taken together to form a 3-10 membered heterocyclyl, wherein [0035] the 3-10 membered heterocyclyl is optionally substituted with one or more halo, oxo, —OH, NH.sub.2, —NH—S(O).sub.2—R.sup.a, or —S(O).sub.2—R.sup.a, [0036] (vi) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl, —OH, oxo or —S(O).sub.2R.sup.a, [0037] (vii) —S(O)—N(C.sub.1-6alkyl)-(C.sub.1-6alkyl), [0038] (viii) —CN, [0039] (ix) —(CH.sub.2).sub.qOH, wherein q is an integer from 0-6, [0040] (x) —C(O)—C.sub.1-6alkyl, or [0041] (xi) —P(O)(C.sub.1-6alkyl).sub.2; [0042] or [0043] (2) L.sup.3 is absent; and [0044] one of X.sup.1 and X.sup.2 is N or C(R.sup.5); and [0045] the other of X.sup.1 and X.sup.2 is N or C that is taken together with R.sup.4, and the atoms to which they are attached, to form a 5-10 membered heterocyclyl or a 5-20 membered heteroaryl, wherein [0046] the 5-10 membered heterocyclyl is optionally substituted with one or more R.sup.b, wherein [0047] R.sup.b is, independently at each occurrence, selected from the group consisting of —OH, halo, oxo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10cycloalkyl, and 3-10 membered heterocyclyl, wherein [0048] the C.sub.1-6alkyl of R.sup.b is optionally substituted with one or more halo, OH, —S(O).sub.2—C.sub.1-6alkyl, or C.sub.3-10cycloalkyl, and wherein the C.sub.3-10cycloalkyl of the C.sub.1-6alkyl of R.sup.b is further optionally substituted with one or more C.sub.1-6alkyl or —OH and [0049] the C.sub.3-10cycloalkyl of R.sup.b is optionally substituted with one or more —OH, C.sub.3-10cycloalkyl, or C.sub.1-6alkyl, and wherein the C.sub.1-6alkyl of the C.sub.3-10cycloalkyl of R.sup.b is further optionally substituted with one or more —OH, deuterium, or halo, and [0050] the 5-20 membered heteroaryl is optionally substituted with one or more R.sup.e, wherein [0051] R.sup.e is, independently at each occurrence, selected from the group consisting of halo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10cycloalkyl, and 3-10 membered heterocyclyl, wherein [0052] the C.sub.1-6alkyl of R.sup.e is optionally substituted with one or more —S(O).sub.2—C.sub.1-6alkyl, [0053] the C.sub.3-10cycloalkyl of R.sup.e is optionally substituted with one or more —OH or C.sub.1-6alkyl, and [0054] the 3-10 membered heterocyclyl of R.sup.e is optionally substituted with one or more —OH or C.sub.1-6alkyl, and wherein [0055] the C.sub.1-6alkyl of the 3-10 membered heterocyclyl of R.sup.e is further optionally substituted with one or more —OH; [0056] R.sup.a is, independently at each occurrence: [0057] (i) C.sub.1-6alkyl optionally substituted with one or more halo, —OH, —S(O).sub.2—C.sub.1-6alkyl, or —N(C.sub.1-6alkyl)-C(O)—C.sub.1-6alkyl, [0058] (ii) C.sub.3-10cycloalkyl optionally substituted with one or more —OH, —C(O).sub.2—C.sub.1-6alkyl, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, or —C(O)—C.sub.3-10heterocyclyl, or C.sub.1-6alkyl, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH, [0059] (iii) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl, or [0060] (iv) NH(C.sub.1-6alkyl); [0061] R.sup.5 is, independently at each occurrence, H, halo, —CN, 3-10 membered heterocyclyl, C.sub.1-6alkyl, or C.sub.1-6alkoxy, wherein [0062] the C.sub.1-6alkyl of R.sup.5 is optionally substituted with one or more halo or —OH, and [0063] the C.sub.1-6alkoxy of R.sup.5

is optionally substituted with one or more halo; [0064] X.sup.3 is N or C(R.sup.6) [0065] X.sup.4 is N or C(R.sup.7); [0066] and [0067] R.sup.6 and R.sup.7 are each independently H or halo. [0068] In one aspect, provided is a compound of formula (I'):

##STR00003## [0069] or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein: [0070] m is an integer from 0 to 4; [0071] n is an integer from 0 to 2; [0072] p is an integer from 0 to 10; [0073] R.sup.1, if present is, independently at each occurrence selected from the group consisting of halo, —CN, C.sub.1-6alkoxy, or C.sub.1-6alkyl, wherein [0074] the C.sub.1-6alkoxy of R.sup.1 is optionally substituted with one or more halo, and [0075] the C.sub.1-6alkyl of R.sup.1 is optionally substituted with one or more halo; [0076] R.sup.2 is H, C.sub.1-6alkyl, C.sub.3-10cycloalkyl, or 3-15 membered heterocyclyl, wherein [0077] the C.sub.1-6alkyl of R.sup.2 is optionally substituted with one or more halo, —OH, —NH.sub.2, or C.sub.1-6alkoxy, and [0078] the C.sub.3-10cycloalkyl of R.sup.2 is optionally substituted with one or more —OH; [0079] R.sup.3, if present, is C.sub.1-6alkyl; [0080] L.sup.1 is C.sub.1-6alkylene, wherein the C.sub.1-6alkylene of L.sup.1 is optionally substituted with one or more C.sub.1-6alkyl, and wherein the C.sub.1-6alkyl is further optionally substituted with one or more —OH or C.sub.1-6alkoxy; [0081] L.sup.2 is O or N(R.sup.x), wherein R.sup.x is H or C.sub.1-6alkyl; and [0082] either [0083] (1) L.sup.3 is absent or is O, C.sub.3-10cycloalkyl, 3-10 membered heterocyclyl, or C.sub.1-6alkylene, wherein [0084] the C.sub.3-10cycloalkyl is optionally substituted with one or more —OH or C.sub.1-6alkyl, [0085] the C.sub.1-6alkylene of L.sup.3 is optionally substituted with one or more —OH or C.sub.1-6alkyl, and [0086] the 3-10 membered heterocyclyl is optionally substituted with one or more —OH; [0087] X.sup.1 and X.sup.2 are each independently N or C(R.sup.5); and [0088] R.sup.4 is: [0089] (i) —S(O).sub.2—R.sup.a; [0090] (ii) 5-20 membered heteroaryl, wherein the 5-20 membered heteroaryl of R.sup.4 is optionally substituted with one or more C.sub.1-6alkyl; [0091] (iii) —N(R.sup.d).sub.2, wherein R.sup.d is independently at each occurrence H, C.sub.1-6 alkyl, or —S(O).sub.2—R.sup.a, wherein the C.sub.1-6alkyl of R.sup.d is optionally substituted with one or more —OH, [0092] (iv) —NS(O)—(C.sub.1-6alkyl).sub.2, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH, [0093] (v) —C(O)—N(R.sup.e).sub.2 wherein R.sup.e is independently at each occurrence H, C.sub.1-6 alkyl, or 3-10 membered heterocycle, wherein the 3-10 membered heterocycle is optionally substituted with one or more oxo, or both R.sup.e together with the N to which they are attached are taken together to form a 3-10 membered heterocyclyl, wherein the 3-10 membered heterocyclyl is optionally substituted with one or more halo, oxo, —OH, NH.sub.2, —NH—S(O).sub.2—R.sup.a, or —S(O).sub.2—R.sup.a, [0094] (vi) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl, —OH, oxo, or —S(O).sub.2R.sup.a, [0095] (vii) —S(O)—N(C.sub.1-6alkyl)-(C.sub.1-6alkyl), [0096] (viii) —CN, [0097] (ix) —(CH.sub.2).sub.qOH, wherein q is an integer from 0-6, [0098] (x) —C(O)—C.sub.1-6alkyl, or [0099] (xi) —P(O)(C.sub.1-6alkyl).sub.2; [0100] or [0101] (2) L.sup.3 is absent; and [0102] one of X.sup.1 and X.sup.2 is N or C(R.sup.5); and [0103] the other of X.sup.1 and X.sup.2 is N or C that is taken together with R.sup.4 and the atoms to which they are attached to form a 5-10 membered heterocyclyl or a 5-20 membered heteroaryl, wherein [0104] the 5-10 membered heterocyclyl is optionally substituted with one or more R.sup.b, wherein R.sup.b is, independently at each occurrence, selected from the group consisting of halo, oxo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10cycloalkyl, and 3-10 membered heterocyclyl, wherein [0105] the C.sub.1-6alkyl of R.sup.b is optionally substituted with one or more halo, OH, —S(O).sub.2—C.sub.1-6alkyl, or C.sub.3-10cycloalkyl, and wherein the C.sub.3-10cycloalkyl of the C.sub.1-6alkyl of R.sup.b is further optionally substituted with one or more C.sub.1-6alkyl or —OH and [0106] the C.sub.3-10cycloalkyl of R.sup.b is optionally substituted with one or more —OH, C.sub.3-10 cycloalkyl, or C.sub.1-6alkyl, and wherein the C.sub.1-6alkyl of the C.sub.3-10cycloalkyl of R.sup.b is further optionally substituted with one or more —OH, and [0107] the 5-

20 membered heteroaryl is optionally substituted with one or more R.sup.e, wherein R.sup.e is, independently at each occurrence, selected from the group consisting of halo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10 cycloalkyl, and 3-10 membered heterocyclyl, wherein [0108] the C.sub.1-6alkyl of R.sup.e is optionally substituted with one or more —S(O).sub.2—C.sub.1-6alkyl, [0109] the C.sub.3-10cycloalkyl of R.sup.e is optionally substituted with one or more —OH or C.sub.1-6alkyl, [0110] and the 3-10 membered heterocyclyl of R.sup.e is optionally substituted with one or more —OH or C.sub.1-6alkyl, and wherein the C.sub.1-6alkyl of the 3-10 membered heterocyclyl of R.sup.e is further optionally substituted with one or more —OH; [0111] R.sup.a is, independently at each occurrence: [0112] (i) C.sub.1-6alkyl optionally substituted with one or more halo, —OH, —S(O).sub.2—C.sub.1-6alkyl, or —N(C.sub.1-6alkyl)-C(O)—C.sub.1-6alkyl, [0113] (ii) C.sub.3-10cycloalkyl optionally substituted with one or more —OH, —C(O).sub.2—C.sub.1-6alkyl, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, or —C(O)—C.sub.3-10heterocyclyl, or C.sub.1-6alkyl, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH, or [0114] (iii) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl; R.sup.5 is, independently at each occurrence, H, halo, —CN, 3-10 membered heterocyclyl, C.sub.1-6alkyl, or C.sub.1-6alkoxy, wherein the C.sub.1-6alkyl of R.sup.5 is optionally substituted with one or more halo, or —OH and wherein the C.sub.1-6alkoxy of R.sup.5 is optionally substituted with one or more halo; and R.sup.6 and R.sup.7 are each independently H or halo.

[0115] In one aspect, provided is a compound of formula (I):

##STR00004## [0116] or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, [0117] wherein: [0118] m is an integer from 0 to 4; [0119] n is an integer from 0 to 2; [0120] p is an integer from 0 to 10; [0121] R.sup.1 if present is, independently at each occurrence, selected from the group consisting of halo, —CN, C.sub.1-6alkoxy or —C.sub.1-6alkyl, wherein [0122] the C.sub.1-6alkoxy of R.sup.1 is optionally substituted with one or more halo, and [0123] the C.sub.1-6alkyl of R.sup.1 is optionally substituted with one or more halo; [0124] R.sup.2 is H, C.sub.1-6alkyl, C.sub.3-10cycloalkyl, or 3-15 membered heterocyclyl, wherein [0125] the C.sub.1-6alkyl of R.sup.2 is optionally substituted with one or more halo, —OH, —NH.sub.2, or C.sub.1-6alkoxy, and [0126] the C.sub.3-10cycloalkyl of R.sup.2 is optionally substituted with one or more —OH; [0127] R.sup.3, if present, is C.sub.1-6alkyl; [0128] L.sup.1 is C.sub.1-6alkylene, wherein the C.sub.1-6alkylene of L.sup.1 is optionally substituted with one or more C.sub.1-6alkyl, and wherein the C.sub.1-6alkyl is further optionally substituted with one or more —OH or C.sub.1-6alkoxy; L.sup.2 is O or N(R.sup.x), wherein R.sup.x is H or C.sub.1-6alkyl; and [0129] either [0130] (1) L.sup.3 is absent or is O, C.sub.3-10cycloalkyl, 3-10 membered heterocyclyl, or C.sub.1-6alkylene, wherein [0131] the C.sub.1-6alkylene of L.sup.3 is optionally substituted with one or more C.sub.1-6alkyl, and [0132] the 3-10 membered heterocyclyl is optionally substituted with one or more —OH; [0133] X.sup.1 and X.sup.2 are each independently N or C(R.sup.5); and [0134] R.sup.4 is: [0135] (i) —S(O).sub.2—R.sup.a, [0136] (ii) 5-20 membered heteroaryl, wherein the 5-20 membered heteroaryl of R.sup.4 is optionally substituted with one or more C.sub.1-6alkyl, [0137] (iii) —N(R.sup.d).sub.2, wherein R.sup.d is independently at each occurrence H, C.sub.1-6 alkyl, or —S(O).sub.2—R.sup.a, wherein the C.sub.1-6alkyl of R.sup.d is optionally substituted with one or more —OH, [0138] (iv) —NS(O)—(C.sub.1-6alkyl).sub.2, wherein the C.sub.1-6alkyl of R.sup.d is optionally substituted with one or more —OH, [0139] (v) —C(O)—N(R.sup.e).sub.2 wherein R.sup.e is independently at each occurrence H, C.sub.1-6 alkyl, or 3-10 membered heterocycle, wherein the 3-10 membered heterocycle is optionally substituted with one or more oxo, or both R.sup.e together with the N to which they are attached are taken together to form a 3-10 membered heterocyclyl, wherein the 3-10 membered heterocyclyl is optionally substituted with one or more halo, oxo, —OH, NH.sub.2, NH—S(O).sub.2—R.sup.a, or —S(O).sub.2—R.sup.a, [0140] (vi) 3-10 membered heterocyclyl

optionally substituted with one or more C.sub.1-6alkyl or oxo, [0141] (vii) —S(O)—N(C.sub.1-6alkyl)-(C.sub.1-6alkyl), or [0142] (viii) —CN; [0143] or [0144] (2) L.sup.3 is absent; and [0145] one of X.sup.1 and X.sup.2 is N or C(R.sup.5); and [0146] the other of X.sup.1 and X.sup.2 is N or C that is taken together with R.sup.4, and the atoms to which they are attached, to form a 5-10 membered heterocyclyl or a 5-20 membered heteroaryl, wherein [0147] the 5-10 membered heterocyclyl is optionally substituted with one or more R.sup.b, wherein R.sup.b is, independently at each occurrence, selected from the group consisting of halo, oxo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10cycloalkyl, and 3-10 membered heterocyclyl, wherein [0148] the C.sub.1-6alkyl of R.sup.b is optionally substituted with one or more halo, OH, or —S(O).sub.2—C.sub.1-6alkyl, and [0149] the C.sub.3-10cycloalkyl of R.sup.b is optionally substituted with one or more —OH, and the 5-20 membered heteroaryl is optionally substituted with one or more R.sup.e, wherein [0150] R.sup.e is, independently at each occurrence, selected from the group consisting of halo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10 cycloalkyl, and 3-10 membered heterocyclyl, wherein [0151] the C.sub.1-6alkyl of R.sup.e is optionally substituted with one or more —S(O).sub.2—C.sub.1-6alkyl, and [0152] the C.sub.3-10cycloalkyl of R.sup.e is optionally substituted with one or more —OH; R.sup.a is, independently at each occurrence: [0153] (i) C.sub.1-6alkyl optionally substituted with one or more halo, —OH, —S(O).sub.2—C.sub.1-6alkyl, or —N(C.sub.1-6alkyl)-C(O)—C.sub.1-6alkyl, or [0154] (ii) C.sub.3-10cycloalkyl optionally substituted with one or more —OH, —C(O).sub.2—C.sub.1-6alkyl, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —C(O)—C.sub.3-10heterocyclyl, or C.sub.1-6alkyl, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH, or [0155] (iii) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl; [0156] R.sup.5 is, independently at each occurrence, H, halo, —CN, 3-10 membered heterocyclyl, C.sub.1-6alkyl, or C.sub.1-6alkoxy, wherein the C.sub.1-6alkyl of R.sup.5 is optionally substituted with one or more halo or —OH, and wherein the C.sub.1-6alkoxy is optionally substituted with one or more halo; and [0157] R.sup.6 and R.sup.7 are each independently H or halo.

[0158] Any embodiments provided herein of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof, are also embodiments of a compound of formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0159] Any embodiments provided herein of a compound of formula (I) or (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof, are also embodiments of a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0160] In one aspect, provided herein is a compound of formula (I-A):

##STR00005##

or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein: n is 1 or 2; and wherein m, p, R.sup.1, R.sup.2, R.sup.3, L.sup.1, L.sup.3, R.sup.4, X.sup.1, X.sup.2, R.sup.6, and R.sup.7 are as defined elsewhere herein. In another variation, m, p, R.sup.1, R.sup.2, R.sup.3, L.sup.1, L.sup.3, R.sup.4, X.sup.1, X.sup.2, R.sup.6, and R.sup.7 of formula (I-A) are as defined for a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0161] In one aspect, provided herein is a compound of formula (I-B):

##STR00006##

or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein: m, p, R.sup.1, R.sup.2, R.sup.3, L.sup.1, L.sup.3, R.sup.4, R.sup.5, R.sup.6, and R.sup.7 are as defined elsewhere herein. In another variation, m, p, R.sup.1, R.sup.2, R.sup.3, L.sup.1, L.sup.3, R.sup.4, R.sup.5, R.sup.6, and R.sup.7 of formula (I-B) are as defined for a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0162] In one aspect, provided herein is a compound of formula (I-C):

##STR00007##

or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein m, p, R.sup.1, R.sup.2, R.sup.3, L.sup.1, L.sup.3, R.sup.4, and R.sup.5 are as defined elsewhere herein. In another variation, m, p, R.sup.1, R.sup.2, R.sup.3, L.sup.1, L.sup.3, R.sup.4, and R.sup.5 of formula (I-C) are as defined for a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0163] In one aspect, provided herein is a compound of formula (I-D):

##STR00008##

or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein m, p, R.sup.1, R.sup.2, R.sup.3, L.sup.1, L.sup.3, and R.sup.4 are as defined elsewhere herein. In another variation, m, p, R.sup.1, R.sup.2, R.sup.3, L.sup.1, L.sup.3, and R.sup.4 of formula (I-D) are as defined for a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0164] In one aspect, provided herein is a compound of formula (I-E):

##STR00009##

or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein m, p, R.sup.1, R.sup.2, R.sup.3, L.sup.1, X.sup.2, R.sup.6, R.sup.7, and ring A are as defined elsewhere herein. In another variation, m, p, R.sup.1, R.sup.2, R.sup.3, L.sup.1, X.sup.2, R.sup.6, R.sup.7, and ring A of formula (I-E) are as defined for a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0165] In one aspect, provided herein is a compound of formula (I-F):

##STR00010##

or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein m, p, R.sup.1, R.sup.2, R.sup.3, L.sup.1, R.sup.5, and ring A are as defined elsewhere herein. In another variation, m, p, R.sup.1, R.sup.2, R.sup.3, L.sup.1, R.sup.5, and ring A of formula (I-F) are as defined for a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0166] In one aspect, provided herein is a compound of formula (I-G):

##STR00011##

or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein m, p, R.sup.1, R.sup.2, R.sup.3, L.sup.1, and ring A are as defined elsewhere herein. In another variation, m, p, R.sup.1, R.sup.2, R.sup.3, L.sup.1, and ring A of formula (I-G) are as defined for a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0167] In one aspect, provided herein is a compound of formula (II-A):

##STR00012##

or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein m, p, R.sup.1, R.sup.2, R.sup.3, L.sup.1, X.sup.2, X.sup.3, X.sup.4, and ring A are as defined elsewhere herein.

acceptable salt of any of the foregoing, or a pharmaceutical composition comprising (i) a compound of formula (I), or any embodiment or variation thereof, such as a compound of formula (I), (I-A) (I-B), (I-C), (I-D), (I-E), (I-F), or (I-G), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, and (ii) one or more pharmaceutically acceptable excipients. In another variation, provided herein is a method of inhibiting APOL1 in a cell, comprising exposing the cell to a composition comprising an effective amount of a compound of formula (II), or any embodiment or variation thereof, such as a compound of formula (I), (I'), (I-A) (I-B), (I-C), (I-D), (I-E), (I-F), (I-G), or (II-A), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or a pharmaceutical composition comprising (i) a compound of formula (II), or any embodiment or variation thereof, such as a compound of formula (I), (I'), (I-A) (I-B), (I-C), (I-D), (I-E), (I-F), (I-G), or (II-A), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, and (ii) one or more pharmaceutically acceptable excipients.

[0171] In one aspect, provided herein is a method of treating an APOL1-mediated disease, disorder, or condition in an individual in need thereof, comprising administering to the individual an effective amount of a compound of formula (I), or any embodiment or variation thereof, such as a compound of formula (I), (I-A) (I-B), (I-C), (I-D), (I-E), (I-F), or (I-G), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or a pharmaceutical composition comprising (i) a compound of formula (I), or any embodiment or variation thereof, such as a compound of formula (I), (I-A) (I-B), (I-C), (I-D), (I-E), (I-F), or (I-G), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, and (ii) one or more pharmaceutically acceptable excipients. In another variation, provided herein is a method of treating an APOL1-mediated disease, disorder, or condition in an individual in need thereof, comprising administering to the individual an effective amount of a compound of formula (I'), or any embodiment or variation thereof, such as a compound of formula (I), (I-A) (I-B), (I-C), (I-D), (I-E), (I-F), or (I-G), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or a pharmaceutical composition comprising (i) a compound of formula (I'), or any embodiment or variation thereof, such as a compound of formula (I), (I-A) (I-B), (I-C), (I-D), (I-E), (I-F), or (I-G), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, and (ii) one or more pharmaceutically acceptable excipients. In another variation, provided herein is a method of treating an APOL1-mediated disease, disorder, or condition in an individual in need thereof, comprising administering to the individual an effective amount of a compound of formula (II), or any embodiment or variation thereof, such as a compound of formula (I), (I'), (I-A) (I-B), (I-C), (I-D), (I-E), (I-F), (I-G), or (II-A), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or a pharmaceutical composition comprising (i) a compound of formula (II), or any embodiment or variation thereof, such as a compound of formula (I), (I'), (I-A) (I-B), (I-C), (I-D), (I-E), (I-F), (I-G), or (II-A), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, and (ii) one or more pharmaceutically acceptable excipients.

[0172] In one aspect, provided herein is a kit, comprising (i) a compound of formula (I), or any embodiment or variation thereof, such as a compound of formula (I), (I-A) (I-B), (I-C), (I-D), (I-E), (I-F), or (I-G), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, and (ii) instructions for use in treating an APOL1-mediated disease, disorder, or condition in an individual in need thereof. In another variation, provided herein is a kit, comprising (i) a compound of formula (I'), or any embodiment or variation thereof, such as a compound of formula (I), (I-A) (I-B), (I-C), (I-D), (I-E), (I-F), or (I-G), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, and (ii) instructions for use in treating an APOL1-mediated disease, disorder, or condition in an individual in need thereof. In another variation, provided herein is a kit, comprising (i) a compound of formula (II), or any embodiment or variation thereof, such as a compound of (I), (I'), (I-A) (I-B), (I-C), (I-D), (I-E), (I-F), (I-G), or

(II-A), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, and (ii) instructions for use in treating an APOL1-mediated disease, disorder, or condition in an individual in need thereof.

[0173] In some aspect, provided herein are methods of preparing a compound of formula (I), or any embodiment or variation thereof, such as a compound of formula (I), (I-A) (I-B), (I-C), (I-D), (I-E), (I-F), or (I-G), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing. In another variation, provided herein are methods of preparing a compound of formula (I'), or any embodiment or variation thereof, such as a compound of formula (I), (I-A) (I-B), (I-C), (I-D), (I-E), (I-F), or (I-G), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing. In another variation, provided herein are methods of preparing a compound of formula (I'), or any embodiment or variation thereof, such as a compound of formula of (I), (I'), (I-A) (I-B), (I-C), (I-D), (I-E), (I-F), (I-G), or (II-A), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing.

Description

DETAILED DESCRIPTION OF THE INVENTION

[0174] Unless clearly indicated otherwise, the terms “a,” “an,” and the like, refer to one or more.

[0175] As used herein, “about” a parameter or value includes and describes that parameter or value per se. For example, “about X” includes and describes X per se.

[0176] “Individual” refers to mammals and includes humans and non-human mammals. Examples of individuals include, but are not limited to, some primates and humans. In some embodiments, individual refers to a human.

[0177] As used herein, an “at risk” individual is an individual who is at risk of developing a disease or condition. An individual “at risk” may or may not have a detectable disease or condition, and may or may not have displayed detectable disease prior to the treatment methods described herein. “At risk” denotes that an individual has one or more so-called risk factors, which are measurable parameters that correlate with development of a disease or condition and are known in the art. An individual having one or more of these risk factors has a higher probability of developing the disease or condition than an individual without these risk factor(s).

[0178] “Treatment” or “treating” is an approach for obtaining beneficial or desired results including clinical results. Beneficial or desired results may include one or more of the following: decreasing one or more symptom resulting from the disease or condition; diminishing the extent of the disease or condition; slowing or arresting the development of one or more symptom associated with the disease or condition (e.g., stabilizing the disease or condition, preventing or delaying the worsening or progression of the disease or condition); and relieving the disease, such as by causing the regression of clinical symptoms (e.g., ameliorating the disease state, enhancing the effect of another medication, delaying the progression of the disease, increasing the quality of life, and/or prolonging survival).

[0179] As used herein, “delaying” development of a disease or condition means to defer, hinder, slow, retard, stabilize and/or postpone development of the disease or condition. This delay can be of varying lengths of time, depending on the history of the disease and/or individual being treated. As is evident to one skilled in the art, a sufficient or significant delay can, in effect, encompass prevention, in that the individual does not develop the disease or condition.

[0180] As used herein, the term “therapeutically effective amount” or “effective amount” intends such amount of a compound of the disclosure or a pharmaceutically salt thereof sufficient to effect treatment when administered to an individual. As is understood in the art, an effective amount may be in one or more doses, e.g., a single dose or multiple doses may be required to achieve the desired treatment endpoint. An effective amount may be considered in the context of administering

one or more therapeutic agents, and a single agent may be considered to be given in an effective amount if, in conjunction with one or more other agents, a desirable or beneficial result may be or is achieved.

[0181] As used herein, “unit dosage form” refers to physically discrete units, suitable as unit dosages, each unit containing a predetermined quantity of active ingredient, or compound, which may be in a pharmaceutically acceptable carrier.

[0182] As used herein, by “pharmaceutically acceptable” is meant a material that is not biologically or otherwise undesirable, e.g., the material may be incorporated into a pharmaceutical composition administered to an individual without causing significant undesirable biological effects.

[0183] The term “alkyl”, as used herein, refers to an unbranched or branched saturated hydrocarbon chain. As used herein, alkyl has 1-20 carbons (i.e., C.sub.1-20alkyl), 1-16 carbons (i.e., C.sub.1-16alkyl), 1-12 carbons (i.e., C.sub.1-12alkyl), 1-10 carbons (i.e., C.sub.1-10alkyl), 1-8 carbons (i.e., C.sub.1-8alkyl), 1-6 carbons (i.e., C.sub.1-6alkyl), 1-4 carbons (i.e., C.sub.1-4alkyl), or 1-3 carbons (i.e., C.sub.1-3alkyl). Examples of alkyl groups include, but are not limited to, methyl, ethyl, propyl, iso-propyl, n-butyl, sec-butyl, iso-butyl, tert-butyl, pentyl, 2-pentyl, iso-pentyl, neo-pentyl, hexyl, 2-hexyl, 3-hexyl, and 3-methylpentyl. When an alkyl residue having a specific number of carbons is named by chemical name or molecular formula, all positional isomers having that number of carbon atoms may be encompassed—for example, “butyl” includes n-butyl, sec-butyl, iso-butyl, and tert-butyl; and “propyl” includes n-propyl and iso-propyl. Certain commonly used alternative names may be used and will be understood by those of ordinary skill in the art. For instance, a divalent group, such as a divalent “alkyl” group, may be referred to as an “alkylene”.

[0184] The term “alkoxy”, as used herein, refers to an —O-alkyl moiety. Examples of alkoxy groups include, but are not limited to, methoxy, ethoxy, n-propoxy, iso-propoxy, n-butoxy, tert-butoxy, sec-butoxy, n-pentoxy, n-hexoxy, and 1,2-dimethylbutoxy.

[0185] The term “aryl”, as used herein, refers to a fully unsaturated carbocyclic ring moiety. The term “aryl” encompasses monocyclic and polycyclic fused-ring moieties. As used herein, aryl encompasses ring moieties comprising, for example, 6 to 20 annular carbon atoms (i.e., C.sub.6-20aryl), 6 to 16 annular carbon atoms (i.e., C.sub.6-16aryl), 6 to 12 annular carbon atoms (i.e., C.sub.6-12aryl), or 6 to 10 annular carbon atoms (i.e., C.sub.6-10aryl). Examples of aryl moieties include, but are not limited to, phenyl, naphthyl, fluorenyl, and anthryl.

[0186] The term “cycloalkyl”, as used herein, refers to a saturated or partially unsaturated carbocyclic ring moiety. The term “cycloalkyl” encompasses monocyclic and polycyclic ring moieties, wherein the polycyclic moieties may be fused, branched, or spiro. Cycloalkyl includes cycloalkenyl groups, wherein the ring moiety comprises at least one annular double bond. Cycloalkyl includes any polycyclic carbocyclic ring moiety comprising at least one non-aromatic ring, regardless of the point of attachment to the remainder of the molecule. As used herein, cycloalkyl includes rings comprising, for example, 3 to 20 annular carbon atoms (i.e., a C.sub.3-20cycloalkyl), 3 to 16 annular carbon atoms (i.e., a C.sub.3-16cycloalkyl), 3 to 12 annular carbon atoms (i.e., a C.sub.3-12cycloalkyl), 3 to 10 annular carbon atoms (i.e., a C.sub.3-10cycloalkyl), 3 to 8 annular carbon atoms (i.e., a C.sub.3-8cycloalkyl), 3 to 6 annular carbon atoms (i.e., a C.sub.3-6cycloalkyl), or 3 to 5 annular carbon atoms (i.e., a C.sub.3-5cycloalkyl). Monocyclic cycloalkyl ring moieties include, for example, cyclopropyl, cyclobutyl, cyclopentyl, cyclohexyl, cycloheptyl, and cyclooctyl. Polycyclic groups include, for example, bicyclo[2.2.1]heptanyl, bicyclo[2.2.2]octanyl, adamantyl, norbomyl, decalanyl, 7,7-dimethyl-bicyclo [2.2.1]heptanyl, and the like. Still further, cycloalkyl also includes spiro cycloalkyl ring moieties, for example, spiro[2.5]octanyl, spiro[4.5]decanyl, or spiro [5.5]undecanyl.

[0187] The term “halo”, as used herein, refers to atoms occupying group VIIA of The Periodic Table and includes fluorine (fluoro), chlorine (chloro), bromine (bromo), and iodine (iodo).

[0188] The term “heteroaryl”, as used herein, refers to an aromatic (fully unsaturated) ring moiety that comprises one or more annular heteroatoms independently selected from the group consisting

of nitrogen, oxygen, and sulfur. The term “heteroaryl” includes both monocyclic and polycyclic fused-ring moieties. As used herein, a heteroaryl comprises, for example, 5 to 20 annular atoms (i.e., a 5-20 membered heteroaryl), 5 to 16 annular atoms (i.e., a 5-16 membered heteroaryl), 5 to 12 annular atoms (i.e., a 5-12 membered heteroaryl), 5 to 10 annular atoms (i.e., a 5-10 membered heteroaryl), 5 to 8 annular atoms (i.e., a 5-8 membered heteroaryl), or 5 to 6 annular atoms (i.e., a 5-6 membered heteroaryl). Any monocyclic or polycyclic aromatic ring moiety comprising one or more annular heteroatoms is considered a heteroaryl, regardless of the point of attachment to the remainder of the molecule (i.e., the heteroaryl moiety may be attached to the remainder of the molecule through any annular carbon or any annular heteroatom of the heteroaryl moiety). Examples of heteroaryl groups include, but are not limited to, acridinyl, benzimidazolyl, benzindolyl, benzofuranyl, benzonaphthofuranyl, benzoxazolyl, benzotriazolyl, benzo[4,6]imidazo[1,2-a]pyridyl, carbazolyl, dibenzofuranyl, dibenzothiophenyl, furanyl, imidazolyl, indazolyl, indolyl, indazolyl, isoindolyl, isoquinolyl, isoxazolyl, naphthyridinyl, oxadiazolyl, oxazolyl, 1-oxidopyridinyl, 1-oxidopyrimidinyl, 1-oxidopyrazinyl, 1-oxidopyridazinyl, phenazinyl, phthalazinyl, pteridinyl, purinyl, pyrrolyl, pyrazolyl, pyridinyl, pyrazinyl, pyrimidinyl, pyridazinyl, quinazolinyl, quinoxalinyl, quinolinyl, quinuclidinyl, isoquinolinyl, triazolyl, tetrazolyl, and triazinyl. Examples of the fused-heteroaryl rings include, but are not limited to, quinolinyl, isoquinolinyl, benzo[b]thiophenyl, indazolyl, benzo[d]imidazolyl, pyrazolo[1,5-a]pyridinyl, and imidazo[1,5-a]pyridinyl, wherein the heteroaryl can be bound via either ring of the fused system.

[0189] The term “heterocyclyl”, as used herein, refers to a saturated or partially unsaturated cyclic moiety that encompasses one or more annular heteroatoms independently selected from the group consisting of nitrogen, oxygen, and sulfur. The term “heterocyclyl” includes both monocyclic and polycyclic ring moieties, wherein the polycyclic ring moieties may be fused, bridged, or spiro. Any non-aromatic monocyclic or polycyclic ring moiety comprising at least one annular heteroatom is considered a heterocyclyl, regardless of the point of attachment to the remainder of the molecule (i.e., the heterocyclyl moiety may be attached to the remainder of the molecule through any annular carbon or any annular heteroatom of the heterocyclyl moiety). Further, the term heterocyclyl is intended to encompass any polycyclic ring moiety comprising at least one annular heteroatom wherein the polycyclic ring moiety comprises at least one non-aromatic ring, regardless of the point of attachment to the remainder of the molecule. As used herein, a heterocyclyl comprises, for example, 3 to 20 annular atoms (i.e., a 3-20 membered heterocyclyl), 3 to 16 annular atoms (i.e., a 3-16 membered heterocyclyl), 3 to 12 annular atoms (i.e., a 3-12 membered heterocyclyl), 3 to 10 annular atoms (i.e., a 3-10 membered heterocyclyl), 3 to 8 annular atoms (i.e., a 3-8 membered heterocyclyl), 3 to 6 annular atoms (i.e., a 3-6 membered heterocyclyl), 3 to 5 annular atoms (i.e., a 3-5 membered heterocyclyl), 5 to 8 annular atoms (i.e., a 5-8 membered heterocyclyl), or 5 to 6 annular atoms (i.e., a 5-6 membered heterocyclyl). Examples of heterocyclyl groups include, e.g., azetidiny, azepiny, benzodioxolyl, benzo[b][1,4]dioxepiny, 1,4-benzodioxany, benzopyrany, benzodioxiny, benzopyranony, benzofuranony, dioxolany, dihydropyrany, hydropyrany, thienyl[1,3]dithianyl, decahydroisoquinolyl, furanony, imidazoliny, imidazolidiny, indoliny, indoliziny, isoindoliny, isothiazolidiny, isoxazolidiny, morpholiny, octahydroindolyl, octahydroisoindolyl, 2-oxopiperazinyl, 2-oxopiperidinyl, 2-oxopyrrolidinyl, oxazolidiny, oxirany, oxetany, phenothiaziny, phenoxazinyl, piperidinyl, piperazinyl, 4-piperidonyl, pyrrolidinyl, pyrazolidiny, quinuclidiny, thiazolidiny, tetrahydrofuryl, tetrahydropyrany, trithianyl, tetrahydroquinoliny, thiophenyl (i.e., thienyl), thiomorpholiny, thiamorpholiny, 1-oxo-thiomorpholiny, and 1,1-dioxo-thiomorpholiny. Examples of spiro heterocyclyl rings include, but are not limited to, bicyclic and tricyclic ring systems, such as oxabicyclo[2.2.2]octany, 2-oxa-7-azaspiro[3.5]nonany, 2-oxa-6-azaspiro[3.4]octany, and 6-oxa-1-azaspiro[3.3]heptany. Examples of fused heterocyclyl rings include, but are not limited to, 1,2,3,4-tetrahydroisoquinoliny, 4,5,6,7-tetrahydrothieno[2,3-c]pyridiny, indoliny, and isoindoliny, where the heterocyclyl can be bound

via either ring of the fused system.

[0190] The terms “optional” and “optionally”, as used herein, mean that the subsequently described event or circumstance may or may not occur and that the description includes instances where the event or circumstance occurs and instances where it does not. Accordingly, the term “optionally substituted” infers that any one or more (e.g., 1, 2, 1 to 5, 1 to 3, 1 to 2, etc.) hydrogen atoms on the designated atom or moiety or group may be replaced or not replaced by an atom or moiety or group other than hydrogen. By way of illustration and not limitation, the phrase “methyl optionally substituted with one or more chloro” encompasses —CH₃, —CH₂Cl, —CHCl₂, and —CCl₃ moieties.

[0191] It is understood that aspects and embodiments described herein as “comprising” include “consisting of” and “consisting essentially of” embodiments.

[0192] The term “pharmaceutically acceptable salt”, as used herein, of a given compound refers to salts that retain the biological effectiveness and properties of the given compound and which are not biologically or otherwise undesirable. “Pharmaceutically acceptable salts” include, for example, salts with inorganic acids, and salts with an organic acid. In addition, if the compounds described herein are obtained as an acid addition salt, the free base can be obtained by basifying a solution of the acid salt. Conversely, if the product is a free base, an addition salt, particularly a pharmaceutically acceptable addition salt, may be produced by dissolving the free base in a suitable organic solvent and treating the solution with an acid, in accordance with conventional procedures for preparing acid addition salts from base compounds. Such compositions are well known in the pharmaceutical art. See, e.g., *Handbook of Pharmaceutical Salts Properties, Selection, and Use*, International Union of Pure and Applied Chemistry, John Wiley & Sons (2008), which is incorporated herein by reference. Those skilled in the art will recognize various synthetic methodologies that may be used to prepare nontoxic pharmaceutically acceptable addition salts. Pharmaceutically acceptable acid addition salts may be prepared from inorganic or organic acids. Salts derived from inorganic acids include, e.g., hydrochloric acid, hydrobromic acid, sulfuric acid, nitric acid, phosphoric acid, and the like. Salts derived from organic acids include, e.g., acetic acid, propionic acid, gluconic acid, glycolic acid, pyruvic acid, oxalic acid, malic acid, malonic acid, succinic acid, maleic acid, fumaric acid, tartaric acid, citric acid, benzoic acid, cinnamic acid, mandelic acid, methanesulfonic acid, ethanesulfonic acid, p-toluene-sulfonic acid, salicylic acid, trifluoroacetic acid, and the like. Likewise, pharmaceutically acceptable base addition salts can be prepared from inorganic or organic bases. Salts derived from inorganic bases include, by way of example only, sodium, potassium, lithium, aluminum, ammonium, calcium, and magnesium salts. Salts derived from organic bases include, but are not limited to, salts of primary, secondary, and tertiary amines. Specific examples of suitable amines include, by way of example only, isopropylamine, trimethyl amine, diethyl amine, tri(iso-propyl) amine, tri(n-propyl) amine, ethanolamine, 2-dimethylaminoethanol, piperazine, piperidine, morpholine, N-ethylpiperidine, and the like.

[0193] Isotopically labeled forms of the compounds depicted herein may be prepared. Isotopically labeled compounds have structures depicted herein, except that one or more atoms are replaced by an atom having a selected atomic mass or mass number. Examples of isotopes that can be incorporated into the disclosed compounds include isotopes of hydrogen, carbon, nitrogen, oxygen, phosphorous, fluorine, chlorine, and iodine, such as ²H, ³H, ¹¹C, ¹³C, ¹⁴C, ¹³N, ¹⁵N, ¹⁵O, ¹⁷O, ¹⁸O, ³¹P, ³²P, ³⁵S, ¹⁸F, ³⁶Cl, ¹²³I, and ¹²⁵I, respectively. In some embodiments, a compound of formula (I), or formula (I') is provided wherein one or more hydrogen is replaced by deuterium or tritium.

[0194] Some of the compounds provided herein may exist as tautomers. Tautomers are in equilibrium with one another. By way of illustration, amide containing compounds may exist in equilibrium with imidic acid tautomers. Regardless of which tautomer is shown and regardless of the nature of the equilibrium among tautomers, the compounds of this disclosure are understood by

one of ordinary skill in the art to comprise both amide and imidic acid tautomers. Thus, for example, amide-containing compounds are understood to include their imidic acid tautomers. Likewise, imidic-acid containing compounds are understood to include their amide tautomers. [0195] Also provided herein are prodrugs of the compounds depicted herein, or a pharmaceutically acceptable salt thereof. Prodrugs are compounds that may be administered to an individual and release, in vivo, a compound depicted herein as the parent drug compound. It is understood that prodrugs may be prepared by modifying a functional group on a parent drug compound in such a way that the modification is cleaved in vivo to release the parent drug compound. The development of prodrug compounds is well known in the pharmaceutical art. See, e.g., Rautio, J., Kumpulainen, H., Heimbach, T. et al. Prodrugs: design and clinical applications. *Nat. Rev. Drug. Discov.* 7, 255-270 (2008), which is incorporated herein by reference.

[0196] The compounds of the present disclosure, or their pharmaceutically acceptable salts, may include an asymmetric center and may thus give rise to enantiomers, diastereomers, and other stereoisomeric forms that may be defined, in terms of absolute stereochemistry, as (R)- or (S)- (or as (D)- or (L)- for amino acids). The present disclosure is meant to include all such possible isomers, as well as their racemic and optically pure forms and mixtures thereof in any ratio. Optically active (+) and (−), (R)- and (S)-, or (D)- and (L)-isomers may be prepared using chiral synthons or chiral reagents, or may be resolved using conventional techniques, for example, chromatography and/or fractional crystallization. Conventional techniques for the preparation/isolation of individual enantiomers include chiral synthesis from a suitable optically pure precursor or the resolution of the racemate (or the racemate of a salt or derivative) using, for example, chiral high pressure liquid chromatography (HPLC) or chiral supercritical fluid chromatography (SFC). When the compounds described herein contain olefinic double bonds or other centers of geometric asymmetry, unless specified otherwise, it is intended that the present disclosure includes both E and Z geometric isomers. Likewise, cis- and trans- are used in their conventional sense to describe relative spatial relationships.

[0197] A “stereoisomer” refers to a compound made up of the same atoms bonded by the same bonds, but having different three-dimensional structures, which are not interchangeable. The present disclosure contemplates various stereoisomers, or mixtures thereof, and includes “enantiomers,” which refers to two stereoisomers whose structures are non-superimposable mirror images of one another. “Diastereomers” are stereoisomers that have at least two asymmetric atoms, but which are not mirror images of each other.

[0198] Where enantiomeric and/or diastereomeric forms exist of a given structure, flat bonds indicate that all stereoisomeric forms of the depicted structure may be present, e.g.,

##STR00013##

[0199] Where enantiomeric forms exist of a given structure, flat bonds and the presence of a “*” symbol indicate that the composition is made up of at least 90%, by weight, of a single isomer with unknown absolute stereochemistry, e.g.,

##STR00014##

[0200] Where enantiomeric and/or diastereomeric forms exist of a given structure with two or more stereocenters, flat bonds and the presence of two or more “*” symbols indicate the composition is made up of at least 90%, by weight, of a single enantiomer or diastereomer with unknown absolute stereochemistry, e.g.,

##STR00015##

[0201] Where enantiomeric and/or diastereomeric forms exist of a given structure, the composition is made up of at least 90%, by weight, dashes or wedges indicate a single enantiomer or diastereomer with known relative or absolute stereochemistry, e.g

##STR00016##

[0202] Abbreviations used are those conventional in the art and are in accordance with the Periodic Table of the Elements, CAS version, *Handbook of Chemistry and Physics*, 75th Ed, hereby

incorporated herein by reference in its entirety. The following examples are intended to be illustrative only and not limiting in any way.

TABLE-US-00001 ° C. degrees Celsius μ L microliter μ W microwave [M + XX].sup.+ observed mass AC.sub.50 half-maximal activity concentration Ac.sub.2O acetic anhydride AIBN azobisisobutyronitrile app apparent (NMR) br broad (NMR) BH.sub.3•THF borane-tetrahydrofuran complex BBr.sub.3 boron tribromide BnCl benzyl chloride B(Pin).sub.2 bis(pinacolato)diboron Calc'd calculated Cbz-Cl benzyl chloroformate CCl.sub.4 carbontetrachloride CDI carbonyldiimidazole CHCl.sub.3 chloroform CO.sub.2 carbon dioxide CS.sub.2 CO.sub.3 cesium carbonate CuI copper iodide d deuterated (NMR solvents) d doublet (NMR) dd doublet of doublets (NMR) DAST diethylaminosulfur trifluoride DMA N,N-dimethylacetamide DCE 1,2-dichloroethane DCM dichloromethane DIAD diisopropyl azodicarboxylate DIEA N,N-diisopropylethylamine DME dimethoxyethane DMEDA N,N-dimethylethylenediamine DMA dimethylacetamide DMF N,N-dimethylformamide DPPF 1,1'-bis(diphenylphosphino) ferrocene EC.sub.50 half-maximal effective concentration EDCI 1-ethyl-3-(3-dimethylamino-propyl)carbodiimide Et.sub.3N triethyl amine EtOAc ethyl acetate EtOH ethanol g grams h hours H hydrogen H.sub.2 hydrogen gas H.sub.2O water H.sub.2O.sub.2 hydrogen peroxide HATU N-[(Dimethylamino)-1H-1,2,3- triazolo-[4,5-b]pyridin-1- ylmethylene]-N- methylmethanaminium hexafluorophosphate N-oxide HCl hydrochloric acid HOBt hydroxybenzotriazole HPLC high-performance liquid chromatography In vacuo in a vacuum (Ir[dF(CF.sub.3)ppy].sub.2(dtbpy))PF.sub.6 [4,4'-Bis(1,1-dimethylethyl)- 2,2'-bipyridine- N1,N1']bis[3,5- difluoro-2-[5-(trifluoromethyl)- 2-pyridinyl-N]phenyl-C] Iridium(III) hexafluorophosphate IUPAC International Union of Pure and Applied Chemistry J J-coupling value (NMR) K.sub.2CO.sub.3 potassium carbonate KI potassium iodide KOAc potassium acetate KOH potassium hydroxide LiAlH.sub.4 lithium aluminum hydride LiHMDS lithium bis(trimethylsilyl)amide LiOH lithium hydroxide MeOH methanol MeCN acetonitrile MHz megahertz m multiplet (NMR) mg milligrams min minutes mL milliliter mmol millimole mM millimolar M molarity or molar m-CPBA meta-chloroperoxybenzoic acid MeCN acetonitrile MeSO.sub.2Na sodium methanesulfinate MS mass spectrometry MsCl methanesulfonyl chloride Ms.sub.2O methanesulfonic anhydride MTBE methyl tert-butyl ether n/a not applicable NaBH(OAc).sub.3 sodium triacetoxyborohydride NaH sodium hydride NBS N-bromosuccinimide NH.sub.4 ammonium NH.sub.4OH ammonium hydroxide NH.sub.4HCO.sub.3 ammonium bicarbonate Na.sub.2SO.sub.4 sodium sulfate NaBH.sub.3CN sodium cyanoborohydride NMP 1-methyl-2-pyrrolidinone NMR nuclear magnetic resonance NaIO.sub.4 sodium periodate NaOH sodium hydroxide OsO.sub.4 osmium tetroxide Pd/C palladium on carbon PCy.sub.3 tricyclohexylphosphine Pd.sub.2(dba).sub.3 tris(dibenzylideneacetone) dipalladium(0) Pd(PPh.sub.3).sub.2 bis(triphenylphosphine) palladium(II) chloride Pd(dba)₃ tris(dibenzylideneacetone) dipalladium(0) pH potential of hydrogen PPh.sub.3 triphenyl phosphine PSI pounds per square inch s singlet (NMR) SEMCl (2-chloromethoxyethyl) trimethylsilane SFC super fluid chromatography SOCl.sub.2 thionyl chloride t triplet (NMR) T3P propanephosphonic acid anhydride TBAB tetrabutylammonium bromide TBAI tetrabutylammonium iodide TEA triethylamine TFA trifluoroacetic acid THF tetrahydrofuran TMSCl trimethylsilyl chloride Xantphos 4,5-bis(diphenylphosphino)-9,9- dimethylxanthene Zn(CN).sub.2 zinc cyanide Compounds

[0203] Provided herein is a compound of formula (II):

##STR00017## [0204] or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, [0205] wherein: [0206] m is an integer from 0 to 4; [0207] n is an integer from 0 to 2; [0208] p is an integer from 0 to 10; [0209] R.sup.1, if present is, independently at each occurrence selected from the group consisting of halo, —CN, C.sub.1-6alkoxy, and C.sub.1-6alkyl, wherein [0210] the C.sub.1-6alkoxy of R.sup.1 is optionally substituted with one or more halo, and [0211] the C.sub.1-6alkyl of R.sup.1 is optionally substituted with one or more halo; [0212]

R.sup.2 is H, C.sub.1-6alkyl, C.sub.3-10cycloalkyl, or 3-15 membered heterocyclyl, wherein [0213] the C.sub.1-6alkyl of R.sup.2 is optionally substituted with one or more deuterium, halo, —OH, —NH.sub.2, or C.sub.1-6alkoxy, and [0214] the C.sub.3-10cycloalkyl of R.sup.2 is optionally substituted with one or more —OH; [0215] R.sup.3, if present, is C.sub.1-6alkyl; [0216] L.sup.1 is C.sub.1-6alkylene, wherein [0217] the C.sub.1-6alkylene of L.sup.1 is optionally substituted with one or more deuterium or C.sub.1-6alkyl, and wherein [0218] the C.sub.1-6alkyl is further optionally substituted with one or more —OH or C.sub.1-6alkoxy; [0219] L.sup.2 is O or N(R.sup.x), wherein R.sup.x is H or C.sub.1-6alkyl; and [0220] either [0221] (1) L.sup.3 is absent or is O, C.sub.3-10cycloalkyl, 3-10 membered heterocyclyl, or C.sub.1-6alkylene, wherein [0222] the C.sub.3-10cycloalkyl of L.sup.3 is optionally substituted with one or more —OH or C.sub.1-6alkyl, [0223] the C.sub.1-6alkylene of L.sup.3 is optionally substituted with one or more —OH or C.sub.1-6alkyl, wherein [0224] the C.sub.1-6alkyl is optionally substituted with one or more —OH, and [0225] the 3-10 membered heterocyclyl of L.sup.3 is optionally substituted with one or more —OH or C.sub.1-6alkyl; [0226] X.sup.1 and X.sup.2 are each independently N or C(R.sup.5); and [0227] R.sup.4 is: [0228] (i) —S(O).sub.2—R.sup.a; [0229] (ii) 5-20 membered heteroaryl, wherein the 5-20 membered heteroaryl of R.sup.4 is optionally substituted with one or more C.sub.1-6alkyl; [0230] (iii) —N(R.sup.d).sub.2, wherein R.sup.d is independently at each occurrence H, C.sub.1-6 alkyl, or —S(O).sub.2—R.sup.a, wherein the C.sub.1-6alkyl of R.sup.d is optionally substituted with one or more —OH, [0231] (iv) —NS(O)—(C.sub.1-6alkyl).sub.2, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH, [0232] (v) —C(O)—N(R.sup.e).sub.2 wherein R.sup.e is independently at each occurrence H, C.sub.1-6 alkyl, or 3-10 membered heterocycle, wherein [0233] the 3-10 membered heterocycle of R.sup.e is optionally substituted with one or more oxo, or both R.sup.e together with the N to which they are attached are taken together to form a 3-10 membered heterocyclyl, wherein [0234] the 3-10 membered heterocyclyl is optionally substituted with one or more halo, oxo, —OH, NH.sub.2, —NH—S(O).sub.2—R.sup.a, or —S(O).sub.2—R.sup.a, [0235] (vi) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl, —OH, oxo or —S(O).sub.2R.sup.a, [0236] (vii) —S(O)—N(C.sub.1-6alkyl)-(C.sub.1-6alkyl), [0237] (viii) —CN, [0238] (ix) —(CH.sub.2).sub.qOH, wherein q is an integer from 0-6, [0239] (x) —C(O)—C.sub.1-6alkyl, or [0240] (xi) —P(O)(C.sub.1-6alkyl).sub.2; [0241] or [0242] (2) L.sup.3 is absent; and [0243] one of X.sup.1 and X.sup.2 is N or C(R.sup.5); and [0244] the other of X.sup.1 and X.sup.2 is N or C that is taken together with R.sup.4, and the atoms to which they are attached, to form a 5-10 membered heterocyclyl or a 5-20 membered heteroaryl, wherein [0245] the 5-10 membered heterocyclyl is optionally substituted with one or more R.sup.b, wherein [0246] R.sup.b is, independently at each occurrence, selected from the group consisting of —OH, halo, oxo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10cycloalkyl, and 3-10 membered heterocyclyl, wherein [0247] the C.sub.1-6alkyl of R.sup.b is optionally substituted with one or more halo, OH, —S(O).sub.2—C.sub.1-6alkyl, or C.sub.3-10cycloalkyl, and wherein the C.sub.3-10cycloalkyl of the C.sub.1-6alkyl of R.sup.b is further optionally substituted with one or more C.sub.1-6alkyl or —OH and [0248] the C.sub.3-10cycloalkyl of R.sup.b is optionally substituted with one or more —OH, C.sub.3-10cycloalkyl, or C.sub.1-6alkyl, and wherein the C.sub.1-6alkyl of the C.sub.3-10cycloalkyl of R.sup.b is further optionally substituted with one or more —OH, deuterium, or halo, and [0249] the 5-20 membered heteroaryl is optionally substituted with one or more R.sup.e, wherein [0250] R.sup.e is, independently at each occurrence, selected from the group consisting of halo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10cycloalkyl, and 3-10 membered heterocyclyl, wherein [0251] the C.sub.1-6alkyl of R.sup.e is optionally substituted with one or more —S(O).sub.2—C.sub.1-6alkyl, [0252] the C.sub.3-10cycloalkyl of R.sup.e is optionally substituted with one or more —OH or C.sub.1-6alkyl, and [0253] the 3-10 membered

heterocyclyl of R.sup.e is optionally substituted with one or more —OH or C.sub.1-6alkyl, and wherein the C.sub.1-6alkyl of the 3-10 membered heterocyclyl of R.sup.e is further optionally substituted with one or more —OH; [0254] R.sup.a is, independently at each occurrence: [0255] (i) C.sub.1-6alkyl optionally substituted with one or more halo, —OH, —S(O).sub.2—C.sub.1-6alkyl, or —N(C.sub.1-6alkyl)-C(O)—C.sub.1-6alkyl, [0256] (ii) C.sub.3-10cycloalkyl optionally substituted with one or more —OH, —C(O).sub.2—C.sub.1-6alkyl, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, or —C(O)—C.sub.3-10heterocyclyl, or C.sub.1-6alkyl, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH, [0257] (iii) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl, or [0258] (iv) NH(C.sub.1-6alkyl); [0259] R.sup.5 is, independently at each occurrence, H, halo, —CN, 3-10 membered heterocyclyl, C.sub.1-6alkyl, or C.sub.1-6alkoxy, wherein [0260] the C.sub.1-6alkyl of R.sup.5 is optionally substituted with one or more halo or —OH, and [0261] the C.sub.1-6alkoxy of R.sup.5 is optionally substituted with one or more halo; [0262] X.sup.3 is N or C(R.sup.6) [0263] X.sup.4 is N or C(R.sup.7); [0264] and [0265] R.sup.6 and R.sup.7 are each independently H or halo. [0266] Provided herein is a compound of formula (I'):

##STR00018## [0267] or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, [0268] wherein: [0269] m is an integer from 0 to 4; [0270] n is an integer from 0 to 2; [0271] p is an integer from 0 to 10; [0272] R.sup.1, if present is, independently at each occurrence selected from the group consisting of halo, —CN, C.sub.1-6alkoxy, or C.sub.1-6alkyl, wherein [0273] the C.sub.1-6alkoxy of R.sup.1 is optionally substituted with one or more halo, and [0274] the C.sub.1-6alkyl of R.sup.1 is optionally substituted with one or more halo; [0275] R.sup.2 is H, C.sub.1-6alkyl, C.sub.3-10cycloalkyl, or 3-15 membered heterocyclyl, wherein [0276] the C.sub.1-6alkyl of R.sup.2 is optionally substituted with one or more halo, —OH, —NH.sub.2, or C.sub.1-6alkoxy, and [0277] the C.sub.3-10cycloalkyl of R.sup.2 is optionally substituted with one or more —OH; [0278] R.sup.3, if present, is C.sub.1-6alkyl; [0279] L.sup.1 is C.sub.1-6alkylene, wherein the C.sub.1-6alkylene of L.sup.1 is optionally substituted with one or more C.sub.1-6alkyl, and wherein the C.sub.1-6alkyl is further optionally substituted with one or more —OH or C.sub.1-6alkoxy; [0280] L.sup.2 is O or N(R.sup.x), wherein R.sup.x is H or C.sub.1-6alkyl; and [0281] either [0282] (1) L.sup.3 is absent or is O, C.sub.3-10cycloalkyl, 3-10 membered heterocyclyl, or C.sub.1-6alkylene, wherein [0283] the C.sub.3-10cycloalkyl is optionally substituted with one or more —OH or C.sub.1-6alkyl, [0284] the C.sub.1-6alkylene of L.sup.3 is optionally substituted with one or more —OH or C.sub.1-6alkyl, and [0285] the 3-10 membered heterocyclyl is optionally substituted with one or more —OH; [0286] X.sup.1 and X.sup.2 are each independently N or C(R.sup.5); and [0287] R.sup.4 is: [0288] (i) —S(O).sub.2—R.sup.a; [0289] (ii) 5-20 membered heteroaryl, wherein the 5-20 membered heteroaryl of R.sup.4 is optionally substituted with one or more C.sub.1-6alkyl; [0290] (iii) —N(R.sup.d).sub.2, wherein R.sup.d is independently at each occurrence H, C.sub.1-6 alkyl, or —S(O).sub.2—R.sup.a, wherein the C.sub.1-6alkyl of R.sup.d is optionally substituted with one or more —OH, [0291] (iv) —NS(O)—(C.sub.1-6alkyl).sub.2, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH, [0292] (v) —C(O)—N(R.sup.e).sub.2 wherein R.sup.e is independently at each occurrence H, C.sub.1-6 alkyl, or 3-10 membered heterocycle, wherein the 3-10 membered heterocycle is optionally substituted with one or more oxo, or both R.sup.e together with the N to which they are attached are taken together to form a 3-10 membered heterocyclyl, wherein the 3-10 membered heterocyclyl is optionally substituted with one or more halo, oxo, —OH, NH.sub.2, —NH—S(O).sub.2—R.sup.a, or —S(O).sub.2—R.sup.a, [0293] (vi) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl, —OH, oxo or —S(O).sub.2R.sup.a, [0294] (vii) —S(O)—N(C.sub.1-6alkyl)-(C.sub.1-6alkyl), [0295] (viii) —CN, [0296] (ix) —(CH.sub.2).sub.qOH, wherein q is an integer from 0-6, [0297] (x) —C(O)—C.sub.1-6alkyl, or [0298] (xi) —P(O)(C.sub.1-6alkyl).sub.2; [0299] or [0300] (2) L.sup.3 is absent; and [0301] one of X.sup.1 and X.sup.2 is N or C(R.sup.5); and [0302] the other of X.sup.1 and X.sup.2 is N or C that

is taken together with R.sup.4 and the atoms to which they are attached to form a 5-10 membered heterocyclyl or a 5-20 membered heteroaryl, wherein [0303] the 5-10 membered heterocyclyl is optionally substituted with one or more R.sup.b, wherein R.sup.b is, independently at each occurrence, selected from the group consisting of halo, oxo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10cycloalkyl, and 3-10 membered heterocyclyl, wherein [0304] the C.sub.1-6alkyl of R.sup.b is optionally substituted with one or more halo, OH, —S(O).sub.2—C.sub.1-6alkyl, or C.sub.3-10cycloalkyl, and wherein the C.sub.3-10cycloalkyl of the C.sub.1-6alkyl of R.sup.b is further optionally substituted with one or more C.sub.1-6alkyl or —OH and [0305] the C.sub.3-10cycloalkyl of R.sup.b is optionally substituted with one or more —OH, C.sub.3-10 cycloalkyl, or C.sub.1-6alkyl, and wherein the C.sub.1-6alkyl of the C.sub.3-10cycloalkyl of R.sup.b is further optionally substituted with one or more —OH, and [0306] the 5-20 membered heteroaryl is optionally substituted with one or more R.sup.e, wherein [0307] R.sup.e is, independently at each occurrence, selected from the group consisting of halo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10 cycloalkyl, and 3-10 membered heterocyclyl, wherein [0308] the C.sub.1-6alkyl of R.sup.e is optionally substituted with one or more —S(O).sub.2—C.sub.1-6alkyl, [0309] the C.sub.3-10cycloalkyl of R.sup.e is optionally substituted with one or more —OH or C.sub.1-6alkyl, [0310] and the 3-10 membered heterocyclyl of R.sup.e is optionally substituted with one or more —OH or C.sub.1-6alkyl, and [0311] wherein the C.sub.1-6alkyl of the 3-10 membered heterocyclyl of R.sup.e is further optionally substituted with one or more —OH; [0312] R.sup.a is, independently at each occurrence: [0313] (i) C.sub.1-6alkyl optionally substituted with one or more halo, —OH, —S(O).sub.2—C.sub.1-6alkyl, or —N(C.sub.1-6alkyl)-C(O)—C.sub.1-6alkyl, [0314] (ii) C.sub.3-10cycloalkyl optionally substituted with one or more —OH, —C(O).sub.2—C.sub.1-6alkyl, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, or —C(O)—C.sub.3-10heterocyclyl, or C.sub.1-6alkyl, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH, or [0315] (iii) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl; [0316] R.sup.5 is, independently at each occurrence, H, halo, —CN, 3-10 membered heterocyclyl, C.sub.1-6alkyl, or C.sub.1-6alkoxy, wherein the C.sub.1-6alkyl of R.sup.5 is optionally substituted with one or more halo, or —OH and wherein the C.sub.1-6alkoxy of R.sup.5 is optionally substituted with one or more halo; and [0317] R.sup.6 and R.sup.7 are each independently H or halo.

[0318] Provided herein is a compound of formula (I):

##STR00019## [0319] or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, [0320] wherein: [0321] m is an integer from 0 to 4; [0322] n is an integer from 0 to 2; [0323] p is an integer from 0 to 10; [0324] R.sup.1, if present is, independently at each occurrence selected from the group consisting of halo, —CN, C.sub.1-6alkoxy or —C.sub.1-6alkyl, wherein [0325] the C.sub.1-6alkoxy of R.sup.1 is optionally substituted with one or more halo, and [0326] the C.sub.1-6alkyl of R.sup.1 is optionally substituted with one or more halo; [0327] R.sup.2 is H, C.sub.1-6alkyl, C.sub.3-10cycloalkyl, or 3-15 membered heterocyclyl, wherein [0328] the C.sub.1-6alkyl of R.sup.2 is optionally substituted with one or more halo, —OH, —NH.sub.2, or C.sub.1-6alkoxy, and [0329] the C.sub.3-10cycloalkyl of R.sup.2 is optionally substituted with one or more —OH; [0330] R.sup.3, if present, is C.sub.1-6alkyl; [0331] L.sup.1 is C.sub.1-6alkylene, wherein the C.sub.1-6alkylene of L.sup.1 is optionally substituted with one or more C.sub.1-6alkyl, and wherein the C.sub.1-6alkyl is further optionally substituted with one or more —OH or C.sub.1-6alkoxy; [0332] L.sup.2 is O or N(R.sup.x), wherein R.sup.x is H or C.sub.1-6alkyl; and [0333] either [0334] (1) L.sup.3 is absent or is O, C.sub.3-10cycloalkyl, 3-10 membered heterocyclyl, or C.sub.1-6alkylene, wherein [0335] the C.sub.1-6alkylene of L.sup.3 is optionally substituted with one or more C.sub.1-6alkyl, and [0336] the 3-10 membered heterocyclyl is optionally substituted with one or more —OH; [0337] X.sup.1 and X.sup.2 are each

independently N or C(R.sup.5); and [0338] R.sup.4 is: [0339] (i) —S(O).sub.2—R.sup.a, [0340] (ii) 5-20 membered heteroaryl, wherein the 5-20 membered heteroaryl of R.sup.4 is optionally substituted with one or more C.sub.1-6alkyl, [0341] (iii) —N(R.sup.d).sub.2, wherein R.sup.d is independently at each occurrence H, C.sub.1-6 alkyl, or —S(O).sub.2—R.sup.a, wherein the C.sub.1-6alkyl of R.sup.d is optionally substituted with one or more —OH, [0342] (iv) —NS(O)—(C.sub.1-6alkyl).sub.2, wherein the C.sub.1-6alkyl of R.sup.d is optionally substituted with one or more —OH, [0343] (v) —C(O)—N(R.sup.e).sub.2 wherein R.sup.e is independently at each occurrence H, C.sub.1-6 alkyl, or 3-10 membered heterocycle, wherein the 3-10 membered heterocycle is optionally substituted with one or more oxo, or both R.sup.e together with the N to which they are attached are taken together to form a 3-10 membered heterocyclyl, wherein the 3-10 membered heterocyclyl is optionally substituted with one or more halo, oxo, —OH, NH.sub.2, NH—S(O).sub.2—R.sup.a, or —S(O).sub.2—R.sup.a, [0344] (vi) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl or oxo, [0345] (vii) —S(O)—N(C.sub.1-6alkyl)-(C.sub.1-6alkyl), or [0346] (viii) —CN; [0347] or [0348] (2) L.sup.3 is absent; and [0349] one of X.sup.1 and X.sup.2 is N or C(R.sup.5); and [0350] the other of X.sup.1 and X.sup.2 is N or C that is taken together with R.sup.4 and the atoms to which they are attached to form a 5-10 membered heterocyclyl or a 5-20 membered heteroaryl, wherein [0351] the 5-10 membered heterocyclyl is optionally substituted with one or more R.sup.b, wherein R.sup.b is, independently at each occurrence, selected from the group consisting of halo, oxo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10cycloalkyl, and 3-10 membered heterocyclyl, wherein [0352] the C.sub.1-6alkyl of R.sup.b is optionally substituted with one or more halo, OH, or —S(O).sub.2—C.sub.1-6alkyl, and [0353] the C.sub.3-10cycloalkyl of R.sup.b is optionally substituted with one or more —OH, and [0354] the 5-20 membered heteroaryl is optionally substituted with one or more R.sup.e, wherein R.sup.e is, independently at each occurrence, selected from the group consisting of halo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10 cycloalkyl, and 3-10 membered heterocyclyl, wherein [0355] the C.sub.1-6alkyl of R.sup.e is optionally substituted with one or more —S(O).sub.2—C.sub.1-6alkyl, and [0356] the C.sub.3-10cycloalkyl of R.sup.e is optionally substituted with one or more —OH; [0357] R.sup.a is, independently at each occurrence: [0358] (i) C.sub.1-6alkyl optionally substituted with one or more halo, —OH, —S(O).sub.2—C.sub.1-6alkyl, or —N(C.sub.1-6alkyl)—C(O)—C.sub.1-6alkyl, or [0359] (ii) C.sub.3-10cycloalkyl optionally substituted with one or more —OH, —C(O).sub.2—C.sub.1-6alkyl, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, or —C(O)—C.sub.3-10heterocyclyl, or C.sub.1-6alkyl, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH, or [0360] (iii) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl; [0361] R.sup.5 is, independently at each occurrence, H, halo, —CN, 3-10 membered heterocyclyl, C.sub.1-6alkyl, or C.sub.1-6alkoxy, wherein the C.sub.1-6alkyl of R.sup.5 is optionally substituted with one or more halo, or —OH and wherein the C.sub.1-6alkoxy is optionally substituted with one or more halo; and [0362] R.sup.6 and R.sup.7 are each independently H or halo.

[0363] Any embodiments provided herein of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof, are also embodiments of a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0364] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, n is an integer from 0 to 2. In some embodiments, n is 0. In some embodiments, n is 1. In some embodiments, n is 2. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a

stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof. In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.2} is H, C_{sub.1-6}alkyl, C_{sub.3-10}cycloalkyl, or 3-15 membered heterocyclyl, wherein the C_{sub.1-6}alkyl of R^{sup.2} is optionally substituted with one or more halo, —OH, —NH_{sub.2}, or C_{sub.1-6}alkoxy, and the C_{sub.3-10}cycloalkyl of R^{sup.2} is optionally substituted with one or more —OH. In some embodiments, R^{sup.2} is H, C_{sub.1-3}alkyl, C_{sub.3-6}cycloalkyl, or 3-6 membered heterocyclyl, wherein the C_{sub.1-3}alkyl of R^{sup.2} is optionally substituted with one or more halo, —OH, —NH_{sub.2}, or C_{sub.1-3}alkoxy, and the C_{sub.3-6}cycloalkyl of R^{sup.2} is optionally substituted with one or more —OH. In some embodiments, R^{sup.2} is selected from the group consisting of H,

##STR00020##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0365] In some embodiments of a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.2} is H, C_{sub.1-6}alkyl, C_{sub.3-10} cycloalkyl, or 3-15 membered heterocyclyl, wherein the C_{sub.1-6}alkyl of R^{sup.2} is optionally substituted with one or more deuterium, halo, —OH, —NH_{sub.2}, or C_{sub.1-6}alkoxy, and the C_{sub.3-10}cycloalkyl of R^{sup.2} is optionally substituted with one or more —OH. In some embodiments, R^{sup.2} is H, C_{sub.1-3}alkyl, C_{sub.3-6}cycloalkyl, or 3-6 membered heterocyclyl, wherein the C_{sub.1-3}alkyl of R^{sup.2} is optionally substituted with one or more deuterium, halo, —OH, —NH_{sub.2}, or C_{sub.1-3}alkoxy, and the C_{sub.3-6}cycloalkyl of R^{sup.2} is optionally substituted with one or more —OH. In some embodiments, R^{sup.2} is selected from the group consisting of H,

##STR00021##

[0366] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.2} is H. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0367] In some embodiments of a compound of formula (I), formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.2} is C_{sub.1-6}alkyl. In some embodiments, R^{sup.2} is C_{sub.1-3}alkyl. In some embodiments R^{sup.2} is methyl or ethyl. In some embodiments, R^{sup.2} is methyl. In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0368] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.2} is C_{sub.1-6}alkyl optionally substituted with one or more halo, —OH, —NH_{sub.2}, or C_{sub.1-6}alkoxy. In some embodiments, R^{sup.2} is C_{sub.1-3}alkyl optionally substituted with one or more halo, —OH, —NH_{sub.2}, or C_{sub.1-3}alkoxy. In some embodiments, R^{sup.2} is methyl. In some embodiments, R^{sup.2} is ethyl optionally substituted with one or more halo —OH, —NH_{sub.2} or C_{sub.1-6}alkoxy. In some embodiments, R^{sup.2} is selected from the group consisting of

##STR00022##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0369] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.2} is C_{sub.1-6}alkyl optionally

substituted with one or more deuterium, halo, —OH, —NH.sub.2, or C.sub.1-6alkoxy. In some embodiments, R.sup.2 is C.sub.1-3alkyl optionally substituted with one or more deuterium, halo, —OH, —NH.sub.2, or C.sub.1-3alkoxy. In some embodiments, R.sup.2 is methyl. In some embodiments, R.sup.2 is ethyl optionally substituted with one or more halo, —OH, —NH.sub.2, or C.sub.1-6alkoxy. In some embodiments, R.sup.2 is selected from the group consisting of
##STR00023##

[0370] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.2 is C.sub.3-10cycloalkyl, wherein the C.sub.3-10cycloalkyl of R.sup.2 is optionally substituted with one or more —OH. In some embodiments, R.sup.2 is C.sub.3-6cycloalkyl, wherein the C.sub.3-6cycloalkyl of R.sup.2 is optionally substituted with one or more —OH. In some embodiments, R.sup.2 is
##STR00024##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0371] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.2 is 3-15 membered heterocyclyl. In some embodiments, R.sup.2 is 3-6 membered heterocyclyl. In some embodiments, R.sup.2 is
##STR00025##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0372] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, m is an integer from 0 to 4. In some embodiments, m is an integer from 0 to 2. In some embodiments, m is 0. In some embodiments m is 1. In some embodiments, m is 2. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0373] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.1 is halo, —CN, C.sub.1-6alkoxy or —C.sub.1-6alkyl, wherein the C.sub.1-6alkoxy or R.sup.1 is optionally substituted with one or more halo, and wherein the C.sub.1-6alkyl of R.sup.1 is optionally substituted with one or more halo. In some embodiments, R.sup.1 is halo, —CN, C.sub.1-3alkoxy, or —C.sub.1-3alkyl, wherein the C.sub.1-3alkoxy of R.sup.1 is optionally substituted with one or more halo, and wherein the C.sub.1-3alkyl of R.sup.1 is optionally substituted with one or more halo. In some embodiments, R.sup.1 is selected from the group consisting of Cl, Br, —CN, methyl,
##STR00026##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0374] In some embodiments of a compound of formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.1 is halo, —CN, C.sub.1-6alkoxy or —C.sub.1-6alkyl, wherein the C.sub.1-6alkoxy or R.sup.1 is optionally substituted with one or more halo, and wherein the C.sub.1-6alkyl of R.sup.1 is optionally substituted with one or more halo. In some embodiments, R.sup.1 is halo, —CN, C.sub.1-3alkoxy, or —C.sub.1-3alkyl, wherein the C.sub.1-3alkoxy of R.sup.1 is optionally substituted with one or more halo, and wherein the C.sub.1-3alkyl of R.sup.1 is optionally substituted with one or more halo. In some embodiments, R.sup.1 is selected from the group consisting of Cl, Br, F, I, —CN, methyl,
##STR00027##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0375] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.1} is halo. In some embodiments R^{sup.1} is Cl. In some embodiments, R^{sup.1} is Br. In some embodiments, R^{sup.1} is F. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0376] In some embodiments of a compound of formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.1} is halo. In some embodiments R^{sup.1} is Cl. In some embodiments, R^{sup.1} is Br. In some embodiments, R^{sup.1} is F. In some embodiments R^{sup.1} is I. In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0377] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.1} is C_{sub.1-6}alkyl, wherein the C_{sub.1-6}alkyl of R^{sup.1} is optionally substituted with one or more halo. In some embodiments, R^{sup.1} is C_{sub.1-3}alkyl, wherein the C_{sub.1-3}alkyl of R^{sup.1} is optionally substituted with one or more halo. In some embodiments, R^{sup.1} is C_{sub.1-3}alkyl, wherein the C_{sub.1-3}alkyl of R^{sup.1} is optionally substituted with one or more F. In some embodiments, R^{sup.1} is methyl, wherein the methyl of R^{sup.1} is optionally substituted with one or more F. In some embodiments, R^{sup.1} is methyl. In some embodiments, R^{sup.1} is

##STR00028##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0378] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.1} is —CN. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0379] In some embodiments of a compound of formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.1} is C_{sub.1-6}alkoxy, wherein the C_{sub.1-6}alkoxy of R^{sup.1} is optionally substituted with one or more halo. In some embodiments, R^{sup.1} is C_{sub.1-3}alkoxy, wherein the C_{sub.1-3}alkoxy of R^{sup.1} is optionally substituted with one or more halo. In some embodiments, R^{sup.1} is C_{sub.1-3}alkoxy, wherein the C_{sub.1-3}alkoxy of R^{sup.1} is optionally substituted with one or more F. In some embodiments, R^{sup.1} is methoxy, wherein the methoxy of R^{sup.1} is optionally substituted with one or more F. In some embodiments, R^{sup.1} is

##STR00029##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0380] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, p is an integer from 0 to 10. In some embodiments, p is 0 or 1. In some embodiments, p is 0. In some embodiments, p is 1. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0381] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.3 is C.sub.1-6alkyl. In some embodiments, R.sup.3 is C.sub.1-3alkyl. In some embodiments, R.sup.3 is methyl. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0382] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, the ring bearing (R.sup.1).sub.m and R.sup.2 is a 2-indolinone optionally substituted at one or more of positions 1, 5, and 7. In some embodiments, the 2-indolinone ring is unsubstituted. In some embodiments, the 2-indolinone ring is substituted at position 1. In some embodiments, the 2-indolinone ring is substituted at position 5. In some embodiments, the 2-indolinone ring is substituted at position 7. In some embodiments, the 2-indolinone ring is substituted at positions 1, and 5. In some embodiments, the 2-indolinone ring is substituted at positions 5 and 7. In some embodiments, the 2-indolinone ring is substituted at positions 1, 5, and 7. In some embodiments positions 1, 5 and 7 are defined as indicated in the structure,

##STR00030##

wherein position 1 is a N atom, each of positions 5 and 7 is a C atom, and ## represent the point of attachment to the remainder of the molecule. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0383] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, the ring bearing (R.sup.1).sub.m and R.sup.2 is selected from the group consisting of

##STR00031## ##STR00032##

wherein ## represents the point of attachment to the remainder of the molecule. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0384] In some embodiments of a compound of formula (I), formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, the ring bearing (R.sup.1).sub.m and R.sup.2 is selected from the group consisting of

##STR00033## ##STR00034##

wherein ## represents the point of attachment to the remainder of the molecule. In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0385] In some embodiments of a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, the ring bearing (R.sup.1).sub.m and R.sup.2 is selected from the group consisting of

##STR00035## ##STR00036##

wherein ## represents the point of attachment to the remainder of the molecule.

[0386] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, the ring bearing (R.sup.3).sub.p is selected from the group consisting of

##STR00037##

wherein ## represents the point of attachment to the remainder of the molecule. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0387] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, the ring bearing (R.sup.1).sub.m, R.sup.2, and (R.sup.3).sub.p is selected from the group consisting of
##STR00038## ##STR00039## ##STR00040##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0388] In some embodiments of a compound of formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, the ring bearing (R.sup.1).sub.m, R.sup.2, and (R.sup.3).sub.p is selected from the group consisting of
##STR00041## ##STR00042## ##STR00043##

In some embodiments, the ring bearing (R.sup.1).sub.m, R.sup.2, and (R.sup.3).sub.p is selected from the group consisting of
##STR00044##

In some embodiments, the ring bearing (R.sup.1).sub.m, R.sup.2, and (R.sup.3).sub.p is selected from the group consisting of
##STR00045##

In some embodiments, the ring bearing (R.sup.1).sub.m, R.sup.2, and (R.sup.3).sub.p is
##STR00046##

In some embodiments, the ring bearing (R.sup.1).sub.m, R.sup.2, and (R.sup.3).sub.p is
##STR00047##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0389] In some embodiments of a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, the ring bearing (R.sup.1).sub.m, R.sup.2, and (R.sup.3).sub.p is selected from the group consisting of
##STR00048## ##STR00049## ##STR00050## ##STR00051##

In some embodiments, the ring bearing (R.sup.1).sub.m, R.sup.2, and (R.sup.3).sub.p is selected from the group consisting of
##STR00052##

In some embodiments, the ring bearing (R.sup.1).sub.m, R.sup.2, and (R.sup.3).sub.p is selected from the group consisting of
##STR00053##

In some embodiments, the ring bearing (R.sup.1).sub.m, R.sup.2, and (R.sup.3).sub.p is
##STR00054##

In some embodiments, the ring bearing (R.sup.1).sub.m, R.sup.2, and (R.sup.3).sub.p is
##STR00055##

In some embodiments, the ring bearing (R.sup.1).sub.m, R.sup.2, and (R.sup.3).sub.p is
##STR00056##

[0390] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.1 is C.sub.1-6alkylene optionally substituted with one or more C.sub.1-6alkyl, wherein the C.sub.1-6alkyl is optionally substituted with —OH or C.sub.1-6alkoxy. In some embodiments, L.sup.1 is C.sub.1-3alkylene optionally substituted with one or more C.sub.1-3alkyl, wherein the C.sub.1-3alkyl is optionally substituted with —OH or C.sub.1-3alkoxy. In some embodiments, L.sup.1 is methylene. In some embodiments, L.sup.1 is ethylene. In some embodiments, L.sup.1 is ethylene optionally substituted with one or more C.sub.1-3alkyl, wherein the C.sub.1-3alkyl is optionally substituted with one or more —OH or C.sub.1-3alkoxy. In some embodiments, L.sup.1 is selected from the group consisting of

##STR00057##

wherein, for each L.sup.1, # denotes the point of attachment to L.sup.2 and ## denotes the point of attachment to the remainder of the molecule. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof. [0391] In some embodiments of a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.1 is C.sub.1-6alkylene optionally substituted with one or more deuterium, or C.sub.1-6alkyl, wherein the C.sub.1-6alkyl is optionally substituted with —OH or C.sub.1-6alkoxy. In some embodiments, L.sup.1 is C.sub.1-3alkylene optionally substituted with one or more deuterium, or C.sub.1-3alkyl, wherein the C.sub.1-3alkyl is optionally substituted with —OH or C.sub.1-3alkoxy. In some embodiments, L.sup.1 is methylene. In some embodiments, L.sup.1 is ethylene. In some embodiments, L.sup.1 is ethylene optionally substituted with one or more deuterium, or C.sub.1-3alkyl, wherein the C.sub.1-3alkyl is optionally substituted with one or more —OH or C.sub.1-3alkoxy. In some embodiments, L.sup.1 is selected from the group consisting of

##STR00058##

wherein, for each L.sup.1, # denotes the point of attachment to L.sup.2 and ## denotes the point of attachment to the remainder of the molecule.

[0392] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.2 is O or N(R.sup.x), wherein R.sup.x is H or C.sub.1-6alkyl. In some embodiments, L.sup.2 is O or N(R.sup.x), wherein R.sup.x is H or C.sub.1-3alkyl. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0393] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.2 is O. In some variations, the embodiments provided herein also apply to a compound of formula (I') or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0394] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.3 is absent or is O, C.sub.3-10 cycloalkyl, 3-10 membered heterocyclyl, or C.sub.1-6alkylene, wherein the C.sub.1-6alkylene of L.sup.3 is optionally substituted with one or more C.sub.1-6alkyl, the C.sub.3-10cycloalkyl of L.sup.3 is optionally substituted with one or more —OH, and the 3-10 membered heterocyclyl of L.sup.3 is optionally substituted with one or more —OH. In some embodiments, L.sup.3 absent or is selected from the group consisting of O,

##STR00059##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0395] In some embodiments of a compound of formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.3 is absent or is O, C.sub.3-10 cycloalkyl, 3-10 membered heterocyclyl, or C.sub.1-6alkylene, wherein the C.sub.3-10cycloalkyl is optionally substituted with one or more —OH, or C.sub.1-6alkyl, the C.sub.1-6alkylene of L.sup.3 is optionally substituted with one or more —OH, or C.sub.1-6alkyl, and the 3-10 membered heterocyclyl is optionally substituted with one or more —OH. In some embodiments, L.sup.3 absent or is selected from the group consisting of O,

##STR00060##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or

any variation or embodiment thereof.

[0396] In some embodiments of a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.3} is absent or is O, C_{sub.3-10}cycloalkyl, 3-10 membered heterocyclyl, or C_{sub.1-6}alkylene, wherein the C_{sub.3-10}cycloalkyl of L^{sup.3} is optionally substituted with one or more —OH, or C_{sub.1-6}alkyl; the C_{sub.1-6}alkylene of L^{sup.3} is optionally substituted with one or more —OH, or C_{sub.1-6}alkyl wherein the C_{sub.1-6}alkyl is optionally substituted with one or more —OH; and the 3-10 membered heterocyclyl is optionally substituted with one or more —OH or C_{sub.1-6}alkyl. In some embodiments, L^{sup.3} is absent or is selected from the group consisting of O,

##STR00061##

[0397] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.3} is absent. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0398] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.3} is O. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0399] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.3} is C_{sub.1-6}alkylene. In some embodiments, L^{sup.3} is C_{sub.1-3}alkylene. In some embodiments, L^{sup.3} is selected from the group consisting of

##STR00062##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0400] In some embodiments of a compound of formula (I), formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.3} is C_{sub.1-6}alkylene, wherein the C_{sub.1-6}alkylene of L^{sup.3} is optionally substituted with one or more —OH, or C_{sub.1-6}alkyl. In some embodiments, L^{sup.3} is C_{sub.1-3}alkylene, wherein the C_{sub.1-6}alkylene of L^{sup.3} is optionally substituted with one or more —OH, or C_{sub.1-6}alkyl. In some embodiments, L^{sup.3} is selected from the group consisting of

##STR00063##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0401] In some embodiments of a compound of formula (II) or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.3} is C_{sub.1-6}alkylene, wherein the C_{sub.1-6}alkylene of L^{sup.3} is optionally substituted with one or more —OH, or C_{sub.1-6}alkyl wherein the C_{sub.1-6}alkyl is optionally substituted with one or more —OH. In some embodiments, L^{sup.3} is C_{sub.1-3}alkylene, wherein the C_{sub.1-6}alkylene of L^{sup.3} is optionally substituted with one or more —OH, or C_{sub.1-6}alkyl. In some embodiments, L^{sup.3} is selected from the group consisting of

##STR00064##

[0402] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.3} is C_{sub.3-10}cycloalkyl, wherein the C_{sub.3-10}cycloalkyl of L^{sup.3} is optionally substituted with one or more —OH. In some embodiments, L^{sup.3} is C_{sub.3-8}cycloalkyl, wherein

the C.sub.3-8cycloalkyl of L.sup.3 is optionally substituted with one or more —OH. In some embodiments, L.sup.3 is selected from the group consisting of

##STR00065##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0403] In some embodiments of a compound of formula (I'), (I), (I-A), (I-A2), (I-B), or (I-B1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.3 is C.sub.3-10cycloalkyl, wherein the C.sub.3-10cycloalkyl of L.sup.3 is optionally substituted with one or more —OH. In some embodiments, L.sup.3 is C.sub.3-8cycloalkyl, wherein the C.sub.3-8cycloalkyl of L.sup.3 is optionally substituted with one or more —OH. In some embodiments, L.sup.3 is selected from the group consisting of

##STR00066##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0404] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.3 is 3-10 membered heterocyclyl. In some embodiments, L.sup.3 is 3-6 membered heterocyclyl. In some embodiments, L.sup.3 is selected from the group consisting of

##STR00067##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0405] In some embodiments of a compound of formula (I), formula (I') or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.3 is 3-10 membered heterocyclyl. In some embodiments, L.sup.3 is 3-6 membered heterocyclyl. In some embodiments, L3 is selected from the group consisting of

##STR00068##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0406] In some embodiments of a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L3 is 3-10 membered heterocyclyl. In some embodiments, L.sup.3 is 3-6 membered heterocyclyl. In some embodiments, L3 is selected from the group consisting of

##STR00069##

[0407] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing R.sup.4 is —S(O).sub.2—R.sup.a, 5-20 membered heteroaryl, —N(R.sup.d).sub.2, —NS(O)—(C.sub.1-6alkyl).sub.2, —C(O)—N(R.sup.e).sub.2, 3-10 membered heterocyclyl, —S(O)(N—C.sub.1-6alkyl)-(C.sub.1-6alkyl), or —CN. In some embodiments, R.sup.4 is S(O).sub.2—R.sup.a, 5-10 membered heteroaryl, —N(R.sup.d).sub.2, —NS(O)—(C.sub.1-3alkyl).sub.2, —C(O)—N(R.sup.e).sub.2, 3-6 membered heterocyclyl, —S(O)(N—C.sub.1-3alkyl)-(C.sub.1-3alkyl), or —CN. In some embodiments, R.sup.4 is selected from the group consisting of —NH.sub.2, —CN, —C(O)—N(CH.sub.3).sub.2,

##STR00070## ##STR00071##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0408] In some embodiments of a compound of formula (II), or a stereoisomer or tautomer thereof,

or a pharmaceutically acceptable salt of any of the foregoing R.sup.4 is —S(O).sub.2—R.sup.a, 5-20 membered heteroaryl, —N(R.sup.d).sub.2, —NS(O)—(C.sub.1-6alkyl).sub.2, —C(O)—N(R.sup.e).sub.2, 3-10 membered heterocyclyl, —S(O)—N(C.sub.1-6alkyl)-(C.sub.1-6alkyl), —CN, —(CH.sub.2).sub.qOH, —C(O)—C.sub.1-6alkyl, or —P(O)(C.sub.1-6alkyl).sub.2. In some embodiments, R.sup.4 is S(O).sub.2—R.sup.a, 5-10 membered heteroaryl, —N(R.sup.d).sub.2, —NS(O)—(C.sub.1-3alkyl).sub.2, —C(O)—N(R.sup.e).sub.2, 3-6 membered heterocyclyl, —S(O)(N—C.sub.1-3alkyl)-(C.sub.1-3alkyl), —CN, —OH, —CH.sub.2OH, —C(O)—C.sub.1-3alkyl, or —P(O)(C.sub.1-3alkyl).sub.2. In some embodiments, R.sup.4 is selected from the group consisting of —NH.sub.2, —CN, —OH, —CH.sub.2OH, —C(O)—N(CH.sub.3).sub.2,

##STR00072## ##STR00073##

[0409] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.4 is —S(O).sub.2—R.sup.a, wherein R.sup.a is C.sub.1-6alkyl. In some embodiments, R.sup.4 is —S(O).sub.2—R.sup.a, wherein R.sup.a is C.sub.1-6alkyl optionally substituted with one or more halo, —OH, —S(O).sub.2—C.sub.1-6alkyl, or —N(C.sub.1-6alkyl)-C(O)—C.sub.1-6alkyl. In some embodiments, R.sup.4 is —S(O).sub.2—R.sup.a, wherein R.sup.a is C.sub.1-3alkyl optionally substituted with one or more halo, —OH, —S(O).sub.2—C.sub.1-3alkyl, or —N(C.sub.1-3alkyl)-C(O)—C.sub.1-3alkyl. In some embodiments, R.sup.4 is selected from the group consisting of

##STR00074##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0410] In some embodiments of a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.4 is —S(O).sub.2—R.sup.a, wherein R.sup.a is C.sub.1-6alkyl. In some embodiments, R.sup.4 is —S(O).sub.2—R.sup.a, wherein R.sup.a is C.sub.1-6alkyl optionally substituted with one or more halo, —OH, —S(O).sub.2—C.sub.1-6alkyl, or —N(C.sub.1-6alkyl)-C(O)—C.sub.1-6alkyl. In some embodiments, R.sup.4 is —S(O).sub.2—R.sup.a, wherein R.sup.a is C.sub.1-3alkyl optionally substituted with one or more halo, —OH, —S(O).sub.2—C.sub.1-3alkyl, or —N(C.sub.1-3alkyl)-C(O)—C.sub.1-3alkyl. In some embodiments, R.sup.4 is selected from the group consisting of

##STR00075##

[0411] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.4 is —S(O).sub.2—R.sup.a, wherein R.sup.a is C.sub.3-10cycloalkyl. In some embodiments, R.sup.4 is —S(O).sub.2—R.sup.a, wherein R.sup.a is C.sub.3-10cycloalkyl optionally substituted with one or more —OH, C(O).sub.2—C.sub.1-6alkyl, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —C(O)—C.sub.3-10heterocyclyl or C.sub.1-6alkyl, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH. In some embodiments, R.sup.4 is —S(O).sub.2—R.sup.a, wherein R.sup.a is C.sub.3-6cycloalkyl optionally substituted with one or more —OH, C(O).sub.2—C.sub.1-6alkyl, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —C(O)—C.sub.3-10heterocyclyl, or C.sub.1-6alkyl, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH. In some embodiments, R.sup.4 is —S(O).sub.2—R.sup.a, wherein R.sup.a is C.sub.3-6cycloalkyl optionally substituted with one or more —OH, C(O).sub.2—C.sub.1-3alkyl, —C(O)—NH(C.sub.1-3alkyl), —C(O)—N(C.sub.1-3alkyl).sub.2, —C(O)—C.sub.3-6heterocyclyl, or C.sub.1-3alkyl, wherein the C.sub.1-3alkyl is optionally substituted with one or more —OH. In some embodiments, R.sup.4 is selected from the group consisting of

##STR00076##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0412] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.4} is —S(O)_{sub.2}—R^{sup.a}, wherein R^{sup.a} is 3-10 membered heterocyclyl. In some embodiments R^{sup.4} is —S(O)_{sub.2}—R^{sup.a}, wherein R^{sup.a} is 3-10 membered heterocyclyl optionally substituted with one or more C_{sub.1-6}alkyl. In some embodiments, R^{sup.4} is —S(O)_{sub.2}—R^{sup.a}, wherein R^{sup.a} is 3-6 membered heterocyclyl optionally substituted with one or more C_{sub.1-6}alkyl. In some embodiments, wherein R^{sup.4} is —S(O)_{sub.2}—R^{sup.a}, wherein R^{sup.a} is 3-6 membered heterocyclyl optionally substituted with one or more C_{sub.1-3}alkyl. In some embodiments, R^{sup.4} is

##STR00077##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0413] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.4} is 5-20 membered heteroaryl. In some embodiments, R^{sup.4} is 5-20 membered heteroaryl optionally substituted with one or more C_{sub.1-6}alkyl. In some embodiments, R^{sup.4} is 5-6 membered heteroaryl optionally substituted with one or more C_{sub.1-6}alkyl. In some embodiments, R^{sup.4} is 5-6 membered heteroaryl optionally substituted with one or more C_{sub.1-3}alkyl. In some embodiments, R^{sup.4} is 5-6 membered heteroaryl optionally substituted with one or more methyl. In some embodiments, R^{sup.4} is selected from the group consisting of

##STR00078##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0414] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.4} is —N(R^{sup.d})_{sub.2}, wherein each of R^{sup.d} is independently H, C_{sub.1-6} alkyl, or —S(O)_{sub.2}—R^{sup.a}, wherein the C_{sub.1-6}alkyl of R^{sup.d} is optionally substituted with one or more —OH, and wherein R^{sup.a} is C_{sub.1-6}alkyl. In some embodiments, R^{sup.4} is —N(R^{sup.d})_{sub.2}, wherein each of R^{sup.d} is independently H, C_{sub.1-3} alkyl, or —S(O)_{sub.2}—R^{sup.a}, wherein the C_{sub.1-3}alkyl of R^{sup.d} is optionally substituted with one or more —OH, and wherein R^{sup.a} is C_{sub.1-3}alkyl. In some embodiments, R^{sup.4} is —N(R^{sup.d})_{sub.2}, wherein each of R^{sup.d} is independently H, C_{sub.1-6} alkyl, or —S(O)_{sub.2}—R^{sup.a}, wherein the C_{sub.1-3}alkyl of R^{sup.d} is optionally substituted with one or more —OH, and wherein R^{sup.a} is methyl. In some embodiments, R^{sup.4} is selected from the group consisting of —NH_{sub.2},

##STR00079##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0415] In some embodiments of a compound of formula (I), formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.4} is —N(R^{sup.d})_{sub.2}, wherein each of R^{sup.d} is independently H, C_{sub.1-6} alkyl, or —S(O)_{sub.2}—R^{sup.a}, wherein the C_{sub.1-6}alkyl of R^{sup.d} is optionally substituted with one or more —OH, and wherein R^{sup.a} is C_{sub.1-6}alkyl. In some embodiments, R^{sup.4} is —N(R^{sup.d})_{sub.2}, wherein each of R^{sup.d} is independently H, C_{sub.1-3} alkyl, or —S(O)_{sub.2}—R^{sup.a}, wherein the C_{sub.1-3}alkyl of R^{sup.d} is optionally substituted with one or more —OH, and wherein R^{sup.a} is C_{sub.1-3}alkyl. In some embodiments, R^{sup.4} is —N(R^{sup.d})_{sub.2}, wherein each of R^{sup.d} is independently H, C_{sub.1-6} alkyl, or —S(O)_{sub.2}—R^{sup.a}, wherein the C_{sub.1-3}alkyl of R^{sup.d} is optionally substituted with one or more —OH, and wherein R^{sup.a} is methyl. In

some embodiments, R.sup.4 is selected from the group consisting of —NH.sub.2,

##STR00080##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0416] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.4 is —C(O)—

N(R.sup.e).sub.2, wherein each of R.sup.e is independently H or C.sub.1-6 alkyl. In some embodiments, R.sup.4 is —C(O)—N(R.sup.e).sub.2, wherein each of R.sup.e is independently H or C.sub.1-3 alkyl. In some embodiments, R.sup.4 is —C(O)—N(R.sup.e).sub.2, wherein each of R.sup.e is independently H or methyl. In some embodiments, R.sup.4 is —C(O)—NH.sub.2. In some embodiments, R.sup.4 is —C(O)—NH(CH.sub.3). In some embodiments, R.sup.4 is —C(O)—N(CH.sub.3).sub.2. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0417] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.4 is —C(O)—

N(R.sup.e).sub.2, wherein each of R.sup.e is independently H or 3-10 membered heterocycle, wherein the 3-10 membered heterocycle is optionally substituted with one or more oxo. In some embodiments, R.sup.4 is —C(O)—N(R.sup.e).sub.2, wherein each of R.sup.e is independently H or 3-6 membered heterocycle, wherein the 3-6 membered heterocycle is optionally substituted with one or more oxo. In some embodiments, R.sup.4 is selected from the group consisting of

##STR00081##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0418] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.4 is —C(O)—

N(R.sup.e).sub.2, wherein both R.sup.e together with the N to which they are attached are taken together to form a 3-10 membered heterocyclyl. In some embodiments R.sup.4 is —C(O)—N(R.sup.e).sub.2, wherein both R.sup.e together with the N to which they are attached are taken together to form a 3-10 membered heterocyclyl, wherein the 3-10 membered heterocyclyl is optionally substituted with one or more halo, oxo, —OH, —NH.sub.2, —NH—S(O).sub.2—R.sup.a, or —S(O).sub.2—R.sup.a, wherein R.sup.a is C.sub.1-6alkyl. In some embodiments R.sup.4 is —C(O)—N(R.sup.e).sub.2, wherein both R.sup.e together with the N to which they are attached are taken together to form a 3-7 membered heterocyclyl, wherein the 3-7 membered heterocyclyl is optionally substituted with one or more halo, oxo, —OH, NH.sub.2, NH—S(O).sub.2—R.sup.a, or —S(O).sub.2—R.sup.a, wherein R.sup.a is C.sub.1-3alkyl. In some embodiments R.sup.4 is —C(O)—N(R.sup.e).sub.2, wherein both R.sup.e together with the N to which they are attached are taken together to form a 3-7 membered heterocyclyl, wherein the 3-7 membered heterocyclyl is optionally substituted with one or more halo, oxo, —OH, —NH.sub.2, —NH—S(O).sub.2—R.sup.a, or —S(O).sub.2—R.sup.a, wherein R.sup.a is methyl. In some embodiments, R.sup.4 is selected from the group consisting of

##STR00082##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0419] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.4 is 3-10 membered heterocyclyl optionally substituted with one or more oxo or C.sub.1-6alkyl. In some embodiments,

R.sup.4 is 3-7 membered heterocyclyl optionally substituted with one or more oxo or C.sub.1-6alkyl. In some embodiments, R.sup.4 is 3-7 membered heterocyclyl optionally substituted with one or more oxo or C.sub.1-3 alkyl. In some embodiments, R.sup.4 is selected from the group consisting of

##STR00083##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0420] In some embodiments of a compound of formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.4 is 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl, —OH, oxo or —S(O).sub.2R.sup.a. In some embodiments, R.sup.4 is 3-7 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl, —OH, oxo or —S(O).sub.2R.sup.a. In some embodiments, R.sup.4 is 3-7 membered heterocyclyl optionally substituted with one or more C.sub.1-3alkyl, —OH, oxo or —S(O).sub.2R.sup.a. In some embodiments, R.sup.4 is selected from the group consisting of

##STR00084##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0421] In some embodiments of a compound of formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.4 is 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl, —OH, oxo or —S(O).sub.2R.sup.a. In some embodiments, R.sup.4 is 3-7 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl, —OH, oxo or —S(O).sub.2R.sup.a. In some embodiments, R.sup.4 is 3-7 membered heterocyclyl optionally substituted with one or more C.sub.1-3alkyl, —OH, oxo or —S(O).sub.2R.sup.a. In some embodiments, R.sup.4 is selected from the group consisting of

##STR00085##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0422] In some embodiments of a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.4 is 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl, —OH, oxo or —S(O).sub.2R.sup.a. In some embodiments, R.sup.4 is 3-7 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl, —OH, oxo or —S(O).sub.2R.sup.a. In some embodiments, R.sup.4 is 3-7 membered heterocyclyl optionally substituted with one or more C.sub.1-3alkyl, —OH, oxo or —S(O).sub.2R.sup.a. In some embodiments, R.sup.4 is selected from the group consisting of

##STR00086##

[0423] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.4 is —NS(O)—(C.sub.1-6alkyl).sub.2. In some embodiments, R.sup.4 is —NS(O)—(C.sub.1-3alkyl).sub.2. In some embodiments, R.sup.4 is

##STR00087##

[0424] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.4 is —S(O)(N—C.sub.1-6alkyl)-(C.sub.1-6 alkyl). In some embodiments, R.sup.4 is —S(O)(N—C.sub.1-3alkyl)-(C.sub.1-3alkyl). In some embodiments, R.sup.4 is

##STR00088##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0425] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.4} is —CN. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0426] In some embodiments of a compound of formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.4} is —(CH_{sub.2})_{sub.q}OH, wherein q is an integer from 0-6. R^{sup.4} is —(CH_{sub.2})_{sub.q}OH, wherein q is an integer from 0-2. In some embodiments, R^{sup.4} is —OH. In some embodiments, R^{sup.4} is selected from the group consisting of

##STR00089##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0427] In some embodiments of a compound of formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.4} is —C(O)—C_{sub.1-6}alkyl. In some embodiments, R^{sup.4} is —C(O)—C_{sub.1-3}alkyl. In some embodiments, R^{sup.4} is —C(O)CH_{sub.3}. In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0428] In some embodiments of a compound of formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.4} is —P(O)(C_{sub.1-6}alkyl)_{sub.2}. In some embodiments, R^{sup.4} is —P(O)(C_{sub.1-3}alkyl)_{sub.2}. In some embodiments, R^{sup.4} is —P(O)(CH_{sub.3})_{sub.2}. In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0429] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, each of X^{sup.1} and X^{sup.2} is independently N or C(R^{sup.5}). In some embodiments, each of X^{sup.1} and X^{sup.2} is N. In some embodiments, each of X^{sup.1} and X^{sup.2} is C(R^{sup.5}). In some embodiments, one of X^{sup.1} and X^{sup.2} is CR^{sup.5}, and the other of X^{sup.1} and X^{sup.2} is N. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0430] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, one of X^{sup.1} and X^{sup.2} is CR^{sup.5}, and the other of X^{sup.1} and X^{sup.2} is C that is taken together with R^{sup.4}, and the atoms to which they are attached, to form a 5-10 membered heterocyclyl optionally substituted with one or more of R^{sup.b}. In some embodiments, one of X^{sup.1} and X^{sup.2} is CR^{sup.5}, and the other of X^{sup.1} and X^{sup.2} is C that is taken together with R^{sup.4}, and the atoms to which they are attached, to form a 5-8 membered heterocyclyl optionally substituted with one or more of R^{sup.b}. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0431] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.b} is halo, oxo, C_{sub.1-6}alkyl,

—C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10cycloalkyl, and 3-10 membered heterocyclyl, wherein the C.sub.1-6alkyl of R.sup.b is optionally substituted with one or more halo, OH, or —S(O).sub.2—C.sub.1-6alkyl, and the C.sub.3-10cycloalkyl of R.sup.b is optionally substituted with one or more —OH. In some embodiments, R.sup.b is halo, oxo, C.sub.1-3alkyl, —C(O)—C.sub.1-3alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-3alkyl), —C(O)—N(C.sub.1-3alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-6cycloalkyl, and 3-6 membered heterocyclyl, wherein the C.sub.1-3alkyl of R.sup.b is optionally substituted with one or more halo, OH, or —S(O).sub.2—C.sub.1-3alkyl, and the C.sub.3-6cycloalkyl of R.sup.b is optionally substituted with one or more —OH. In some embodiments, R.sup.b is selected from the group consisting of oxo, —S(O).sub.2CH.sub.3, ##STR00090##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0432] In some embodiments of a compound of formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.b is, independently at each occurrence, selected from the group consisting of halo, oxo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10cycloalkyl, and 3-10 membered heterocyclyl, wherein the C.sub.1-6alkyl of R.sup.b is optionally substituted with one or more halo, OH, —S(O).sub.2—C.sub.1-6alkyl, or C.sub.3-10cycloalkyl, wherein the C.sub.3-10cycloalkyl of the C.sub.1-6alkyl of R.sup.b is further optionally substituted with one or more C.sub.1-6alkyl or —OH, and the C.sub.3-10cycloalkyl of R.sup.b is optionally substituted with one or more —OH, C.sub.3-10cycloalkyl, or C.sub.1-6alkyl, wherein the C.sub.1-6alkyl of the C.sub.3-10 cycloalkyl of R.sup.b is further optionally substituted with one or more —OH. In some embodiments, R.sup.b is, independently at each occurrence, selected from the group consisting of halo, oxo, C.sub.1-3alkyl, —C(O)—C.sub.1-3alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-3alkyl), —C(O)—N(C.sub.1-3alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-6cycloalkyl, and 3-6 membered heterocyclyl, wherein the C.sub.1-3alkyl of R.sup.b is optionally substituted with one or more halo, OH, —S(O).sub.2—C.sub.1-3alkyl, or C.sub.3-6cycloalkyl, wherein the C.sub.3-6cycloalkyl of the C.sub.1-3alkyl of R.sup.b is further optionally substituted with one or more C.sub.1-3alkyl or —OH, and the C.sub.3-6cycloalkyl of R.sup.b is optionally substituted with one or more —OH, C.sub.3-6cycloalkyl, or C.sub.1-3alkyl, wherein the C.sub.1-3alkyl of the C.sub.3-6cycloalkyl of R.sup.b is further optionally substituted with one or more —OH. In some embodiments, R.sup.b is selected from the group consisting of oxo, —S(O).sub.2CH.sub.3, ##STR00091##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0433] In some embodiments of a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.b is, independently at each occurrence, selected from the group consisting of —OH, halo, oxo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10cycloalkyl, and 3-10 membered heterocyclyl, wherein the C.sub.1-6alkyl of R.sup.b is optionally substituted with one or more halo, OH, —S(O).sub.2—C.sub.1-6alkyl, or C.sub.3-10cycloalkyl, wherein the C.sub.3-10cycloalkyl of the C.sub.1-6alkyl of R.sup.b is further optionally substituted with one or more C.sub.1-6alkyl or —OH, and the C.sub.3-10cycloalkyl of R.sup.b is optionally substituted with one or more —OH, C.sub.3-10cycloalkyl, or C.sub.1-6alkyl, wherein the C.sub.1-6alkyl of the C.sub.3-10-cycloalkyl of R.sup.b is further optionally substituted with one or more —OH, deuterium, or halo. In some

embodiments, R.sup.b is, independently at each occurrence, selected from the group consisting of —OH, halo, oxo, C.sub.1-3alkyl, —C(O)—C.sub.1-3alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-3alkyl), —C(O)—N(C.sub.1-3alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-6cycloalkyl, and 3-6 membered heterocyclyl, wherein the C.sub.1-3alkyl of R.sup.b is optionally substituted with one or more halo, OH, —S(O).sub.2—C.sub.1-3alkyl, or C.sub.3-6cycloalkyl, wherein the C.sub.3-6cycloalkyl of the C.sub.1-3alkyl of R.sup.b is further optionally substituted with one or more C.sub.1-3alkyl or —OH, and the C.sub.3-6cycloalkyl of R.sup.b is optionally substituted with one or more —OH, C.sub.3-6cycloalkyl, or C.sub.1-3alkyl, wherein the C.sub.1-3alkyl of the C.sub.3-6cycloalkyl of R.sup.b is further optionally substituted with one or more —OH, deuterium or halo. In some embodiments, R.sup.b is selected from the group consisting of —OH, oxo, —S(O).sub.2CH.sub.3,

##STR00092##

[0434] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.b is oxo. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0435] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.b is C.sub.1-6alkyl. In some embodiments, R.sup.b is C.sub.1-6alkyl optionally substituted with one or more halo, OH, or —S(O).sub.2—C.sub.1-6alkyl. In some embodiments, R.sup.b is C.sub.1-3alkyl optionally substituted with one or more halo, OH, or —S(O).sub.2—C.sub.1-3alkyl. In some embodiments, R.sup.b is selected from the group consisting of

##STR00093##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0436] In some embodiments of a compound of formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.b is C.sub.1-6alkyl. In some embodiments, the C.sub.1-6alkyl of R.sup.b is optionally substituted with one or more halo, OH, —S(O).sub.2—C.sub.1-6alkyl, or C.sub.3-10cycloalkyl, wherein the C.sub.3-10cycloalkyl of the C.sub.1-6alkyl of R.sup.b is further optionally substituted with one or more C.sub.1-6alkyl or —OH. In some embodiments, R.sup.b is C.sub.1-3alkyl optionally substituted with one or more halo, OH, —S(O).sub.2—C.sub.1-3alkyl, or C.sub.3-6cycloalkyl, wherein the C.sub.3-6cycloalkyl of the C.sub.1-3alkyl of R.sup.b is further optionally substituted with one or more C.sub.1-3alkyl or —OH. In some embodiments, R.sup.b is selected from the group consisting of

##STR00094##

In some variations the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0437] In some embodiments of a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.b is C.sub.1-6alkyl. In some embodiments, the C.sub.1-6alkyl of R.sup.b is optionally substituted with one or more halo, OH, —S(O).sub.2—C.sub.1-6alkyl, or C.sub.3-10cycloalkyl, wherein the C.sub.3-10cycloalkyl of the C.sub.1-6alkyl of R.sup.b is further optionally substituted with one or more C.sub.1-6alkyl or —OH. In some embodiments, R.sup.b is C.sub.1-3alkyl optionally substituted with one or more halo, OH, —S(O).sub.2—C.sub.1-3alkyl, or C.sub.3-6cycloalkyl, wherein the C.sub.3-6cycloalkyl of the C.sub.1-3alkyl of R.sup.b is further optionally substituted with one or more C.sub.1-3alkyl or —OH. In some embodiments, R.sup.b is selected from the group consisting of

##STR00095##

[0438] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.b} is —C(O)—NH(C_{sub.1-6}alkyl). In some embodiments, R^{sup.b} is —C(O)—NH(C_{sub.1-3}alkyl). In some embodiments, R^{sup.b} is

##STR00096##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0439] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.b} is —C(O)—C_{sub.1-6}alkyl. In some embodiments, R^{sup.b} is —C(O)—C_{sub.1-3}alkyl. In some embodiments, R^{sup.b} is

##STR00097##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0440] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing R^{sup.b} is —S(O)_{sub.2}—R^{sup.a}, wherein R^{sup.a} is C_{sub.1-6}alkyl. In some embodiments, R^{sup.b} is —S(O)_{sub.2}—R^{sup.a}, wherein R^{sup.a} is C_{sub.1-3}alkyl. In some embodiments, R^{sup.b} is —S(O)_{sub.2}—R^{sup.a}, R^{sup.a} is methyl. In some embodiments, R^{sup.b} is —S(O)_{sub.2}CH_{sub.3}. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0441] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.b} is C_{sub.3-10}cycloalkyl optionally substituted with one or more —OH. In some embodiments, R^{sup.b} is C_{sub.3-6}cycloalkyl optionally substituted with one or more —OH. In some embodiments, R^{sup.b} is

##STR00098##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0442] In some embodiments of a compound of formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.b} is C_{sub.3-10}cycloalkyl optionally substituted with one or more —OH, C_{sub.3-10}cycloalkyl, or C_{sub.1-6}alkyl, wherein the C_{sub.1-6}alkyl is further optionally substituted with one or more —OH. In some embodiments, R^{sup.b} is C_{sub.3-6}cycloalkyl optionally substituted with one or more —OH, C_{sub.3-6}cycloalkyl, or C_{sub.1-3}alkyl, wherein the C_{sub.1-3}alkyl is further optionally substituted with one or more —OH. In some embodiments, R^{sup.b} is selected from the group consisting of

##STR00099##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0443] In some embodiments of a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.b} is C_{sub.3-10}cycloalkyl optionally substituted with one or more —OH, C_{sub.3-10}cycloalkyl, or C_{sub.1-6}alkyl, wherein the C_{sub.1-6}alkyl is further optionally substituted with one or more —OH, deuterium, or halo. In some embodiments, R^{sup.b} is C_{sub.3-6}cycloalkyl optionally substituted with one or more —OH, C_{sub.3-6}cycloalkyl, or C_{sub.1-3}alkyl, wherein the C_{sub.1-3}alkyl is further optionally substituted with one or more —OH, deuterium, or halo. In some embodiments, R^{sup.b} is selected from the group consisting of OH

##STR00100##

[0444] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.b} is 3-10 membered heterocyclyl. In some embodiments, R^{sup.b} is 3-6 membered heterocyclyl. In some embodiments, R^{sup.b} is

##STR00101##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0445] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, one of X^{sup.1} and X^{sup.2} is CR^{sup.5}, and the other of X^{sup.1} and X^{sup.2} is C that is taken together with R^{sup.4}, and the atoms to which they are attached, to form a 5-10 membered heteroaryl. In some embodiments, one of X^{sup.1} and X^{sup.2} is CR^{sup.5}, and the other of X^{sup.1} and X^{sup.2} is C that is taken together with R^{sup.4}, and the atoms to which they are attached, to form a 5-6 membered heteroaryl. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0446] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, one of X^{sup.1} and X^{sup.2} is CR^{sup.5}, and the other of X^{sup.1} and X^{sup.2} is C that is taken together with R^{sup.4}, and the atoms to which they are attached, to form a 5-10 membered heteroaryl optionally substituted with one or more R^{sup.e}. In some embodiments one of X^{sup.1} and X^{sup.2} is CR^{sup.5}, and the other of X^{sup.1} and X^{sup.2} is C that is taken together with R^{sup.4}, and the atoms to which they are attached, to form a 5-6 membered heteroaryl optionally substituted with one or more R^{sup.e}. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0447] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.e} is halo, C_{sub.1-6}alkyl, —C(O)—C_{sub.1-6}alkyl, —C(O)—NH_{sub.2}, —C(O)—NH(C_{sub.1-6}alkyl), —C(O)—N(C_{sub.1-6}alkyl)_{sub.2}, —S(O)_{sub.2}—R^{sup.a}, C_{sub.3-10}cycloalkyl, and 3-10 membered heterocyclyl, wherein the C_{sub.1-6}alkyl of R^{sup.e} is optionally substituted with one or more —S(O)_{sub.2}—C_{sub.1-6}alkyl, and the C_{sub.3-10}cycloalkyl of R^{sup.e} is optionally substituted with one or more —OH. In some embodiments, R^{sup.e} is halo, C_{sub.1-3}alkyl, —C(O)—C_{sub.1-3}alkyl, —C(O)—NH_{sub.2}, —C(O)—NH(C_{sub.1-3}alkyl), —C(O)—N(C_{sub.1-3}alkyl)_{sub.2}, —S(O)_{sub.2}—R^{sup.a}, C_{sub.3-6}cycloalkyl, and 3-6 membered heterocyclyl, wherein the C_{sub.1-6}alkyl of R^{sup.e} is optionally substituted with one or more —S(O)_{sub.2}—C_{sub.1-3}alkyl, and the C_{sub.3-6}cycloalkyl of R^{sup.e} is optionally substituted with one or more —OH. In some embodiments, R^{sup.e} is selected from the group consisting of methyl, isopropyl, —S(O)_{sub.2}CH_{sub.3}, and

##STR00102##

OH. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0448] In some embodiments of a compound of formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.e} is independently at each occurrence, selected from the group consisting of halo, C_{sub.1-6}alkyl, —C(O)—C_{sub.1-6}alkyl, —C(O)—NH_{sub.2}, —C(O)—NH(C_{sub.1-6}alkyl), —C(O)—N(C_{sub.1-6}alkyl)_{sub.2}, —S(O)_{sub.2}—R^{sup.a}, C_{sub.3-10}cycloalkyl, and 3-10 membered heterocyclyl, wherein the C_{sub.1-6}alkyl of R^{sup.e} is optionally substituted with one or more —S(O)_{sub.2}—C_{sub.1-6}alkyl, the

C.sub.3-10cycloalkyl of R.sup.e is optionally substituted with one or more —OH, or C.sub.1-6alkyl, and the 3-10 membered heterocyclyl of R.sup.e is optionally substituted with one or more —OH, or C.sub.1-6alkyl, wherein the C.sub.1-6alkyl is further optionally substituted with one or more —OH. In some embodiments, R.sup.e is independently at each occurrence, selected from the group consisting of halo, C.sub.1-3alkyl, —C(O)—C.sub.1-3 alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-3alkyl), —C(O)—N(C.sub.1-3alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-6cycloalkyl, and 3-6 membered heterocyclyl, wherein the C.sub.1-3alkyl of R.sup.e is optionally substituted with one or more —S(O).sub.2—C.sub.1-3alkyl, the C.sub.3-6cycloalkyl of R.sup.e is optionally substituted with one or more —OH, or C.sub.1-3alkyl, and the 3-6 membered heterocyclyl of R.sup.e is optionally substituted with one or more —OH, or C.sub.1-3alkyl, wherein the C.sub.1-3alkyl is further optionally substituted with one or more —OH. In some embodiments, R.sup.e is selected from the group consisting of methyl, isopropyl, —S(O).sub.2CH.sub.3, and

##STR00103##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0449] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.e is C.sub.1-6alkyl, —S(O).sub.2—R.sup.a. In some embodiments, R.sup.e is C.sub.1-3alkyl. In some embodiments, R.sup.e is methyl. In some embodiments, R.sup.e is isopropyl. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0450] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.e is C.sub.1-6alkyl optionally substituted with one or more —S(O).sub.2—R.sup.a, wherein R.sup.a is C.sub.1-6alkyl. In some embodiments, R.sup.e is C.sub.1-3alkyl optionally substituted with one or more —S(O).sub.2—R.sup.a, wherein R.sup.a is C.sub.1-3alkyl. In some embodiments, R.sup.e is C.sub.1-3alkyl optionally substituted with one or more —S(O).sub.2—R.sup.a, wherein R.sup.a is methyl. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0451] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.e is C.sub.3-10cycloalkyl optionally substituted with one or more —OH. In some embodiments, R.sup.e is C.sub.3-6cycloalkyl optionally substituted with one or more —OH. In some embodiments, R.sup.e is

##STR00104##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0452] In some embodiments of a compound of formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.e is C.sub.3-10cycloalkyl optionally substituted with one or more —OH, or C.sub.1-6alkyl. In some embodiments, R.sup.e is C.sub.3-6cycloalkyl optionally substituted with one or more —OH, or C.sub.1-3alkyl. In some embodiments, R.sup.e is selected from the group consisting of

##STR00105##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0453] In some embodiments of a compound of formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R_{sup.e} is 3-10 membered heterocyclyl optionally substituted with one or more —OH, or C_{sub.1-6}alkyl, wherein the C_{sub.1-6}alkyl is further optionally substituted with one or more —OH. In some embodiments, R_{sup.e} is 3-6 membered heterocyclyl optionally substituted with one or more —OH, or C_{sub.1-3}alkyl, wherein the C_{sub.1-3}alkyl is further optionally substituted with one or more —OH. In some embodiments, R_{sup.e} is selected from the group consisting of

##STR00106##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0454] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R_{sup.5} is independently at each occurrence, H, halo, —CN, 3-10 membered heterocyclyl, C_{sub.1-6}alkyl, or C_{sub.1-6}alkoxy, wherein the C_{sub.1-6}alkyl of R_{sup.5} is optionally substituted with one or more halo, or —OH and wherein the C_{sub.1-6}alkoxy is optionally substituted with one or more halo. In some embodiments, R_{sup.5} is independently at each occurrence, H, halo, —CN, 3-6 membered heterocyclyl, C_{sub.1-3}alkyl, or C_{sub.1-3}alkoxy, wherein the C_{sub.1-3}alkyl of R_{sup.5} is optionally substituted with one or more halo, or —OH and wherein the C_{sub.1-3}alkoxy is optionally substituted with one or more halo. In some embodiments, R_{sup.5} is selected from the group consisting of H, Cl, F, —CN,

##STR00107##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0455] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R_{sup.5} is H. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0456] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R_{sup.5} is halo. In some embodiments, R_{sup.5} is Cl, or F. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0457] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R_{sup.5} is —CN. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0458] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R_{sup.5} is C_{sub.1-6}alkyl. In some embodiments, R_{sup.5} is C_{sub.1-3}alkyl. In some embodiments, R_{sup.5} is methyl. In some embodiments, R_{sup.5} is C_{sub.1-6}alkyl, wherein the C_{sub.1-6}alkyl of R_{sup.5} is optionally substituted with one or more halo, or —OH. In some embodiments, R_{sup.5} is C_{sub.1-3}alkyl, wherein the C_{sub.1-3}alkyl of R_{sup.5} is optionally substituted with one or more halo, or —OH. In some embodiments, R_{sup.5} is C_{sub.1-3}alkyl, wherein the C_{sub.1-3}alkyl of R_{sup.5} is optionally substituted with one or more fluoro, or —OH. In some embodiments, R_{sup.5} is independently selected from the group consisting of methyl,

##STR00108##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0459] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.5} is 3-10 membered heterocyclyl. In some embodiments, R^{sup.5} is 3-6 membered heterocyclyl. In some embodiments, R^{sup.5} is

##STR00109##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0460] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.5} is C_{sub.1-6}alkoxy optionally substituted with one or more halo. In some embodiments, R^{sup.5} is C_{sub.1-3}alkoxy optionally substituted with one or more halo. In some embodiments, R^{sup.5} is C_{sub.1-3}alkoxy optionally substituted with one or more fluoro. In some embodiments, R^{sup.5} is

##STR00110##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0461] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.6} and R^{sup.7} are each independently H or halo. In some embodiments, R^{sup.6} and R^{sup.7} are each independently H or fluoro. In some embodiments, each of R^{sup.6} and R^{sup.7} is H. In some embodiments, each of R^{sup.6} and R^{sup.7} is fluoro. In some embodiments, one of R^{sup.6} and R^{sup.7} is H and the other of R^{sup.6} and R^{sup.7} is fluoro. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0462] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.1}, L^{sup.2}, and the ring bearing L^{sup.3}, R^{sup.4}, R^{sup.6}, and R^{sup.7} together form a bi-substituted phenyl with one group bound at the para position relative to the phenyl's attachment to L^{sup.2}. In some embodiments, the bi-substituted phenyl is selected from the group consisting of

##STR00111## ##STR00112## ##STR00113## ##STR00114## ##STR00115##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0463] In some embodiments of a compound of formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.1}, L^{sup.2}, and the ring bearing L^{sup.3}, R^{sup.4}, R^{sup.6}, and R^{sup.7} together form a bi-substituted phenyl with one group bound at the para position relative to the phenyl's attachment to L^{sup.2}. In some embodiments, the bi-substituted phenyl is selected from the group consisting of

##STR00116## ##STR00117## ##STR00118## ##STR00119## ##STR00120##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0464] In some embodiments of a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.1}, L^{sup.2}, and the ring bearing L^{sup.3}, R^{sup.4}, R^{sup.6}, and R^{sup.7} together form a bi-substituted phenyl with one group bound at the para position relative to the phenyl's attachment to L^{sup.2}. In some embodiments, the bi-

substituted phenyl is selected from the group consisting of

##STR00121## ##STR00122## ##STR00123## ##STR00124## ##STR00125##

[0465] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form a tri-substituted phenyl with two groups bound at a meta, and the para positions relative to the phenyl's attachment to L.sup.2. In some embodiments, the tri-substituted phenyl is selected from the group consisting of

##STR00126## ##STR00127## ##STR00128## ##STR00129##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0466] In some embodiments of a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form a tri-substituted phenyl with two groups bound at a meta, and the para positions relative to the phenyl's attachment to L.sup.2. In some embodiments, the tri-substituted phenyl is selected from the group consisting of

##STR00130## ##STR00131## ##STR00132## ##STR00133## ##STR00134##

[0467] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form a tetra-substituted phenyl with three groups bound at the meta, and para positions relative to the phenyl's attachment to L.sup.2. In some embodiments, the tetra-substituted phenyl is selected from the group consisting of

##STR00135##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0468] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form a bi-substituted pyridine with one group bound at the para position relative to the pyridine's attachment to L.sup.2. In some embodiments, the bi-substituted pyridine is selected from the group consisting of

##STR00136##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0469] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form a tri-substituted pyridine with two groups bound at the meta, and para positions relative to the pyridine's attachment to L.sup.2. In some embodiments, the tri-substituted pyridine is

##STR00137##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0470] In some embodiments of a compound of formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form a tri-substituted pyridine with two groups bound at the meta, and para positions relative to the pyridine's attachment to L.sup.2. In some embodiments, the tri-substituted pyridine is selected from the group consisting of

##STR00138##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0471] In some embodiments of a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.1}, L^{sup.2}, and the ring bearing L^{sup.3}, R^{sup.4}, R^{sup.6}, and R^{sup.7} together form a tri-substituted pyridine with two groups bound at the meta, and para positions relative to the pyridine's attachment to L^{sup.2}. In some embodiments, the tri-substituted pyridine is selected from the group consisting of

##STR00139## ##STR00140##

[0472] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.1}, L^{sup.2}, and the ring bearing L^{sup.3}, R^{sup.4}, R^{sup.6}, and R^{sup.7} together form a bi-substituted pyrimidine with two groups bound at the para position relative to the pyrimidine's attachment to L^{sup.2}. In some embodiments, the bi-substituted pyrimidine is selected from the group consisting of

##STR00141##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0473] In some embodiments of a compound of formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.1}, L^{sup.2}, and the ring bearing L^{sup.3}, R^{sup.4}, R^{sup.6}, and R^{sup.7} together form a bi-substituted pyrimidine with two groups bound at the para position relative to the pyrimidine's attachment to L^{sup.2}. In some embodiments, the bi-substituted pyrimidine is selected from the group consisting of

##STR00142## ##STR00143## ##STR00144##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0474] In some embodiments of a compound of formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.1}, L^{sup.2}, and the ring bearing L^{sup.3}, R^{sup.4}, R^{sup.6}, and R^{sup.7} together form a bi-substituted pyrimidine with two groups bound at the para position relative to the pyrimidine's attachment to L^{sup.2}. In some embodiments, the bi-substituted pyrimidine is selected from the group consisting of

##STR00145## ##STR00146## ##STR00147##

In some embodiments, the bi-substituted pyrimidine is selected from the group consisting of

##STR00148## ##STR00149##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0475] In some embodiments of a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.1}, L^{sup.2}, and the ring bearing L^{sup.3}, R^{sup.4}, R^{sup.6}, and R^{sup.7} together form a bi-substituted pyrimidine with two groups bound at the para position relative to the pyrimidine's attachment to L^{sup.2}. In some embodiments, the bi-substituted pyrimidine is selected from the group consisting of

##STR00150## ##STR00151## ##STR00152## ##STR00153##

[0476] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.1}, L^{sup.2}, and the ring bearing L^{sup.3}, R^{sup.4}, R^{sup.6}, and R^{sup.7} together form a heterocycle selected from the group consisting of

##STR00154## ##STR00155## ##STR00156## ##STR00157## ##STR00158##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or

(II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0477] In some embodiments of a compound of formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form a heterocycle selected from the group consisting of

##STR00159## ##STR00160## ##STR00161## ##STR00162## ##STR00163## ##STR00164##

In some embodiments L.sup.1, L.sup.2 and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form a heterocycle selected from the group consisting of

##STR00165##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0478] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form a heteroaryl selected from the group consisting of

##STR00166##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0479] In some embodiments of a compound of formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form a heteroaryl selected from the group consisting of

##STR00167##

In some embodiments, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form a heteroaryl selected from the group consisting of

##STR00168##

In some embodiments, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form a heteroaryl selected from the group consisting of

##STR00169##

In some embodiments, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form a heteroaryl selected from the group consisting of

##STR00170##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0480] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form a heterocycle selected from the group consisting of

##STR00171##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0481] In some embodiments of a compound of formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form a heterocycle selected from the group consisting of

##STR00172##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0482] In some embodiments of a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form a heterocycle selected from the group consisting of

##STR00173## ##STR00174##

In some embodiments, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form a heterocycle selected from the group consisting of

##STR00175##

In some embodiments, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form a heterocycle selected from the group consisting of

##STR00176##

[0483] In some embodiments of a compound of formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form

##STR00177##

In some embodiments, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form

##STR00178##

In some embodiments, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form

##STR00179##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0484] In some embodiments of a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form

##STR00180## ##STR00181##

In some embodiments, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form

##STR00182## ##STR00183##

In some embodiments, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form

##STR00184## ##STR00185##

[0485] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form a heterocycle selected from the group consisting of

##STR00186##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0486] In some embodiments of a compound of formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form a heterocycle selected from the group consisting of

##STR00187## ##STR00188## ##STR00189##

In some embodiments, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form a heterocycle selected from the group consisting of

##STR00190## ##STR00191## ##STR00192##

In some embodiments, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form a heterocycle selected from the group consisting of

##STR00193## ##STR00194## ##STR00195##

In some embodiments, L.sup.1, L.sup.2, and the ring bearing L, R.sup.4, R.sup.6 and R.sup.7 together form a heterocycle selected from the group consisting of

##STR00196## ##STR00197## ##STR00198##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0487] In some embodiments of a compound of formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form a heterocycle selected from the group consisting of

##STR00199## ##STR00200##

In some embodiments, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form a heterocycle selected from the group consisting of

##STR00201## ##STR00202##

In some embodiments, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form a heterocycle selected from the group consisting of

##STR00203## ##STR00204##

In some embodiments, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form

##STR00205##

In some variations, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form a heterocycle selected from the group consisting of

##STR00206##

In certain variations, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form

##STR00207##

In certain variations, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form

##STR00208##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0488] In some embodiments of a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.1, L.sup.2, and the ring bearing L.sup.3, and R.sup.4 together form a heterocycle selected from the group consisting of

##STR00209##

In certain variations, L.sup.1, L.sup.2, and the ring bearing L.sup.3, and R.sup.4 together form

##STR00210##

In certain variations, L.sup.1, L.sup.2, and the ring bearing L.sup.3, and R.sup.4 together form

##STR00211##

[0489] In some embodiments of a compound of formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.1, L.sup.2, and the ring bearing L.sup.3, R.sup.4, R.sup.6, and R.sup.7 together form a heteroaryl selected from the group

consisting of

##STR00212## ##STR00213##

In some embodiments, L^{sup.1}, L^{sup.2}, and the ring bearing L^{sup.3}, R^{sup.4}, R^{sup.6}, and R^{sup.7} together form a heteroaryl selected from the group consisting of

##STR00214## ##STR00215##

In some embodiments, L^{sup.1}, L^{sup.2}, and the ring bearing L^{sup.3}, R^{sup.4}, R^{sup.6}, and R^{sup.7} together form a heteroaryl selected from the group consisting of

##STR00216## ##STR00217##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0490] In some embodiments of a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.1}, L^{sup.2}, and the ring bearing R^{sup.6}, and R^{sup.7} together form a heteroaryl selected from the group consisting of

##STR00218## ##STR00219## ##STR00220##

In some embodiments, L^{sup.1}, L^{sup.2} and the ring bearing R^{sup.6}, and R^{sup.7} together form

##STR00221##
In some embodiments, L^{sup.1}, L^{sup.2}, and the ring bearing R^{sup.6}, and R^{sup.7} together form

##STR00222##
[0491] In some embodiments of a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.1}, L^{sup.2}, and the ring bearing R^{sup.6}, and R^{sup.7} together form a heteroaryl selected from the group consisting of

##STR00223##

In some embodiments, L^{sup.1}, L^{sup.2}, and the ring bearing R^{sup.6}, and R^{sup.7} together form

##STR00224##
In some embodiments, L^{sup.1}, L^{sup.2}, and the ring bearing R^{sup.6}, and R^{sup.7} together

##STR00225##
[0492] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, n is 1 or 2, and L^{sup.2} is —O—. In some embodiments, n is 2, and L^{sup.2} is —O—. In some embodiments, n is 1, and L^{sup.2} is 0. In some embodiments, n is 1 or 2, L^{sup.2} is —O— and R^{sup.4} is —S(O)_{sub.2}—R^{sup.a}. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0493] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, n is 1, p is 0, L^{sup.2} is 0 and each of X^{sup.1} and X^{sup.2} is C(R^{sup.5}), wherein each R^{sup.5} is independently H, halo, —CN, 3-10 membered heterocyclyl, C_{sub.1-6}alkyl, or C_{sub.1-6}alkoxy, wherein the C_{sub.1-6}alkyl of R^{sup.5} is optionally substituted with one or more halo, or —OH, and wherein the C_{sub.1-6}alkoxy of R^{sup.5} is optionally substituted with one or more halo. In some embodiments, n is 1, p is 0, L^{sup.2} is 0 and each of X^{sup.1} and X^{sup.2} is C(R^{sup.5}), wherein each R^{sup.5} is independently H. In some embodiments, n is 1, p is 0, L^{sup.2} is 0 and each of X^{sup.1} and X^{sup.2} is C(R^{sup.5}), wherein one of R^{sup.5} is H and the other of R^{sup.5} is halo, —CN, 3-10 membered heterocyclyl, C_{sub.1-6}alkyl, or C_{sub.1-6}alkoxy, wherein the C_{sub.1-6}alkyl of R^{sup.5} is optionally substituted with one or more halo, or —OH, and wherein the C_{sub.1-6}alkoxy is optionally substituted with one or more halo. In some embodiments, n is 1, p is 0, L^{sup.2} is 0 and each of X^{sup.1} and X^{sup.2} is C(R^{sup.5}), wherein each R^{sup.5} is independently halo, —CN, 3-10 membered heterocyclyl, C_{sub.1-6}alkyl, or C_{sub.1-6}alkoxy, wherein the C_{sub.1-6}alkyl of R^{sup.5} is optionally substituted with one or more halo, or —OH, and wherein the C_{sub.1-6}alkoxy is optionally substituted with one or more halo. In some variations, the embodiments provided herein also apply to a compound

of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0494] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, n is 1, p is 0, L^{sup.2} is O, one of X^{sup.1} and X^{sup.2} is N, and the other of X^{sup.1} and X^{sup.2} is C(R^{sup.5}), wherein R^{sup.5} is H, halo, —CN, 3-10 membered heterocyclyl, C_{sub.1-6}alkyl, or C_{sub.1-6}alkoxy, wherein the C_{sub.1-6}alkyl of R^{sup.5} is optionally substituted with one or more halo, or —OH, and wherein the C_{sub.1-6}alkoxy of R^{sup.5} is optionally substituted with one or more halo. In some embodiments, n is 1, p is 0, L^{sup.2} is O, one of X^{sup.1} and X^{sup.2} is N, and the other of X^{sup.1} and X^{sup.2} is C(R^{sup.5}), wherein R^{sup.5} is H. In some embodiments, n is 1, p is 0, L^{sup.2} is O, one of X^{sup.1} and X^{sup.2} is N, and the other of X^{sup.1} and X^{sup.2} is C(R^{sup.5}), wherein R^{sup.5} is halo, —CN, 3-10 membered heterocyclyl, C_{sub.1-6}alkyl, or C_{sub.1-6}alkoxy, wherein the C_{sub.1-6}alkyl of R^{sup.5} is optionally substituted with one or more halo, or —OH, and wherein the C_{sub.1-6}alkoxy of R^{sup.5} is optionally substituted with one or more halo. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0495] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, n is 1, L^{sup.2} is O, and each of X^{sup.1} and X^{sup.2} is N. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0496] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, n is 1, p is 0, L^{sup.2} is O, one of X^{sup.1} and X^{sup.2} is C(R^{sup.5}), wherein R^{sup.5} is H, the other of X^{sup.1} and X^{sup.2} is C that is taken together with R^{sup.4}, and the atoms to which they are attached, to form (i) a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more of R^{sup.b}, or (ii) a 5-10 membered heteroaryl, wherein the 5-10 membered heteroaryl is optionally substituted with one or more R^{sup.e}, and each of R^{sup.6} and R^{sup.7} is H. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0497] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, n is 1, L^{sup.2} is O, one of X^{sup.1} and X^{sup.2} is C(R^{sup.5}), wherein R^{sup.5} is H, halo, —CN, 3-10 membered heterocyclyl, C_{sub.1-6}alkyl, or C_{sub.1-6}alkoxy, wherein the C_{sub.1-6}alkyl of R^{sup.5} is optionally substituted with one or more halo, or —OH, and wherein the C_{sub.1-6}alkoxy of R^{sup.5} is optionally substituted with one or more halo, the other of X^{sup.1} and X^{sup.2} is C that is taken together with R^{sup.4}, and the atoms to which they are attached, to form (i) a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more of R^{sup.b}, or (ii) a 5-10 membered heteroaryl, wherein the 5-10 membered heteroaryl is optionally substituted with one or more R^{sup.e}, and each of R^{sup.6} and R^{sup.7} is H. In some embodiments, n is 1, L^{sup.2} is O, one of X^{sup.1} and X^{sup.2} is C(R^{sup.5}), wherein R^{sup.5} is H, the other of X^{sup.1} and X^{sup.2} is C that is taken together with R^{sup.4}, and the atoms to which they are attached, to form (i) a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more of R^{sup.b}, or (ii) a 5-10 membered heteroaryl, wherein the 5-10 membered heteroaryl is optionally substituted with one or more R^{sup.e}, and each of R^{sup.6} and R^{sup.7} is H. In some embodiments, n is 1, L^{sup.2} is O, one of X^{sup.1} and X^{sup.2} is C(R^{sup.5}), wherein R^{sup.5} is halo, —CN, 3-10 membered heterocyclyl, C_{sub.1-6}alkyl, or C_{sub.1-6}alkoxy, wherein the C_{sub.1-6}alkyl of R^{sup.5} is optionally substituted with one or more halo, or —OH,

and wherein the C.sub.1-3alkoxy of R.sup.5 is optionally substituted with one or more halo, the other of X.sup.1 and X.sup.2 is C that is taken together with R.sup.4, and the atoms to which they are attached, to form (i) a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more of R.sup.b, or (ii) a 5-10 membered heteroaryl, wherein the 5-10 membered heteroaryl is optionally substituted with one or more R.sup.e, and each of R.sup.6 and R.sup.7 is H. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0498] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, n is 1, p is 0, L.sup.2 is O, one of X.sup.1 and X.sup.2 is N, the other of X.sup.1 and X.sup.2 is C that is taken together with R.sup.4, and the atoms to which they are attached, to form (i) a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more of R.sup.b, or (ii) a 5-10 membered heteroaryl, wherein the 5-10 membered heteroaryl is optionally substituted with one or more R.sup.e, and each of R.sup.6 and R.sup.7 is H. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0499] In some embodiments of a compound of formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing: [0500] m is an integer from 0 to 2; [0501] n is an integer from 1 to 2; [0502] p is an integer from 0 to 2; [0503] R.sup.1, if present is, independently at each occurrence selected from the group consisting of halo, —CN, C.sub.1-3alkoxy, or C.sub.1-3alkyl, wherein [0504] the C.sub.1-3alkoxy of R.sup.1 is optionally substituted with one or more halo, and [0505] the C.sub.1-3alkyl of R.sup.1 is optionally substituted with one or more halo; [0506] R.sup.2 is H, C.sub.1-3alkyl, C.sub.3-6cycloalkyl, or 3-10 membered heterocyclyl, wherein [0507] the C.sub.1-3alkyl of R.sup.2 is optionally substituted with one or more halo, —OH, —NH.sub.2, or C.sub.1-3alkoxy, and [0508] the C.sub.3-6cycloalkyl of R.sup.2 is optionally substituted with one or more —OH; [0509] R.sup.3, if present, is C.sub.1-3alkyl; [0510] L.sup.1 is C.sub.1-3alkylene, wherein the C.sub.1-3alkylene of L.sup.1 is optionally substituted with one or more C.sub.1-3alkyl, and wherein the C.sub.1-3alkyl is further optionally substituted with one or more —OH or C.sub.1-3alkoxy; [0511] L.sup.2 is O or N(R.sup.x), wherein R.sup.x is H or C.sub.1-3alkyl; and [0512] either [0513] (1) L.sup.3 is absent or is O, C.sub.3-6cycloalkyl, 3-10 membered heterocyclyl, or C.sub.1-6alkylene, wherein [0514] the C.sub.3-6cycloalkyl is optionally substituted with one or more —OH, or C.sub.1-3alkyl, [0515] the C.sub.1-3alkylene of L.sup.3 is optionally substituted with one or more —OH or C.sub.1-3alkyl, and [0516] the 3-10 membered heterocyclyl is optionally substituted with one or more —OH; [0517] X.sup.1 and X.sup.2 are each independently N or —C(R.sup.5); and [0518] R.sup.4 is: [0519] (i) —S(O).sub.2—R.sup.a; [0520] (ii) 5-10 membered heteroaryl, wherein the 5-10 membered heteroaryl of R.sup.4 is optionally substituted with one or more C.sub.1-3alkyl; [0521] (iii) —N(R.sup.d).sub.2, wherein R.sup.d is independently at each occurrence H, C.sub.1-3alkyl, or —S(O).sub.2—R.sup.a, wherein the C.sub.1-3alkyl of R.sup.d is optionally substituted with one or more —OH, [0522] (iv) —NS(O)—(C.sub.1-3alkyl).sub.2, wherein the C.sub.1-3alkyl is optionally substituted with one or more —OH, [0523] (v) —C(O)—N(R.sup.e).sub.2 wherein R.sup.e is independently at each occurrence H, C.sub.1-3alkyl, or 3-10 membered heterocycle, wherein the 3-10 membered heterocycle is optionally substituted with one or more oxo, or both R.sup.e together with the N to which they are attached are taken together to form a 3-10 membered heterocyclyl, wherein the 3-10 membered heterocyclyl is optionally substituted with one or more halo, oxo, —OH, NH.sub.2, —NH—S(O).sub.2—R.sup.a, or —S(O).sub.2—R.sup.a, [0524] (vi) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-3alkyl, —OH, oxo or —S(O).sub.2R.sup.a, [0525] (vii) —S(O)—N(C.sub.1-3alkyl)-(C.sub.1-3alkyl), [0526] (viii) —CN, [0527] (ix) —(CH.sub.2).sub.qOH, wherein q is an integer from 0-4, [0528] (x) —C(O)—

C.sub.1-3alkyl, or [0529] (xi) —P(O)(C.sub.1-3alkyl).sub.2; [0530] or [0531] (2) L.sup.3 is absent; and [0532] one of X.sup.1 and X.sup.2 is N or C(R.sup.5); and [0533] the other of X.sup.1 and X.sup.2 is N or C that is taken together with R.sup.4 and the atoms to which they are attached to form a 5-10 membered heterocyclyl or a 5-10 membered heteroaryl, wherein [0534] the 5-10 membered heterocyclyl is optionally substituted with one or more R.sup.b, [0535] wherein R.sup.b is, independently at each occurrence, selected from the group consisting of halo, oxo, C.sub.1-3alkyl, —C(O)—C.sub.1-3alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-3alkyl), —C(O)—N(C.sub.1-3alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-6cycloalkyl, and 3-6 membered heterocyclyl, wherein [0536] the C.sub.1-3alkyl of R.sup.b is optionally substituted with one or more halo, OH, —S(O).sub.2—C.sub.1-3alkyl, or C.sub.3-6cycloalkyl, and [0537] wherein the C.sub.3-6cycloalkyl of the C.sub.1-3alkyl of R.sup.b is further optionally substituted with one or more C.sub.1-3alkyl or —OH and [0538] the C.sub.3-6cycloalkyl of R.sup.b is optionally substituted with one or more —OH, C.sub.3-6cycloalkyl, or C.sub.1-3alkyl, and [0539] wherein the C.sub.1-3alkyl of the C.sub.3-6cycloalkyl of R.sup.b is further optionally substituted with one or more —OH, and [0540] the 5-10 membered heteroaryl is optionally substituted with one or more R.sup.e, wherein R.sup.e is, independently at each occurrence, selected from the group consisting of halo, C.sub.1-3alkyl, —C(O)—C.sub.1-3alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-3alkyl), —C(O)—N(C.sub.1-3alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-6cycloalkyl, and 3-6 membered heterocyclyl, wherein [0541] the C.sub.1-3alkyl of R.sup.e is optionally substituted with one or more —S(O).sub.2—C.sub.1-6alkyl, [0542] the C.sub.3-6cycloalkyl of R.sup.e is optionally substituted with one or more —OH or C.sub.1-6alkyl, [0543] and the 3-6 membered heterocyclyl of R.sup.e is optionally substituted with one or more —OH or C.sub.1-3alkyl, and [0544] wherein the C.sub.1-3alkyl of the 3-6 membered heterocyclyl of R.sup.e is further optionally substituted with one or more —OH; [0545] R.sup.a is, independently at each occurrence: [0546] (i) C.sub.1-3alkyl optionally substituted with one or more halo, —OH, —S(O).sub.2—C.sub.1-3alkyl, or —N(C.sub.1-3alkyl)-C(O)—C.sub.1-3alkyl, [0547] (ii) C.sub.3-6cycloalkyl optionally substituted with one or more —OH, —C(O).sub.2—C.sub.1-3alkyl, —C(O)—NH(C.sub.1-3alkyl), —C(O)—N(C.sub.1-3alkyl).sub.2, or —C(O)—C.sub.3-6heterocyclyl, or C.sub.1-3alkyl, wherein the C.sub.1-3alkyl is optionally substituted with one or more —OH, or [0548] (iii) 3-6 membered heterocyclyl optionally substituted with one or more C.sub.1-3alkyl; [0549] R.sup.5 is, independently at each occurrence, H, halo, —CN, 3-6 membered heterocyclyl, C.sub.1-3alkyl, or C.sub.1-3alkoxy, wherein the C.sub.1-3alkyl of R.sup.5 is optionally substituted with one or more halo or —OH, and wherein the C.sub.1-3alkoxy of R.sup.5 is optionally substituted with one or more halo; and [0550] R.sup.6 and R.sup.7 are each independently H or halo. In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0551] In some embodiments of a compound of formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing: [0552] m is an integer from 0 to 2; [0553] n is an integer from 1 to 2; [0554] p is an integer from 0 to 1; [0555] R.sup.1, if present is, independently at each occurrence selected from the group consisting of Cl, Br, F, I, —CN, C.sub.1-3alkoxy, or C.sub.1-3alkyl, wherein [0556] the C.sub.1-3alkoxy of R.sup.1 is optionally substituted with one or more F, and [0557] the C.sub.1-3alkyl of R.sup.1 is optionally substituted with one or more F; [0558] R.sup.2 is H, C.sub.1-3alkyl, C.sub.3-4cycloalkyl, or 3-4 membered heterocyclyl, wherein [0559] the C.sub.1-3alkyl of R.sup.2 is optionally substituted with one or more F, —OH, —NH.sub.2, or —OCH.sub.3, and [0560] the C.sub.3-4cycloalkyl of R.sup.2 is optionally substituted with one or more —OH; [0561] R.sup.3, if present, is C.sub.1-3alkyl; [0562] L.sup.1 is C.sub.1-3alkylene, wherein the C.sub.1-3alkylene of L.sup.1 is optionally substituted with one or more methyl, and wherein the methyl is further optionally substituted with one or more —OH or —OCH.sub.3; [0563] L.sup.2 is O; and [0564] either [0565] (1) L.sup.3 is absent or is O, C.sub.3-

4cycloalkyl, 3-7 membered heterocyclyl, or C.sub.1-4alkylene, wherein [0566] the C.sub.3-4cycloalkyl is optionally substituted with one or more —OH, or —CH.sub.3, [0567] the C.sub.1-6alkylene of L.sup.3 is optionally substituted with one or more —OH, —CH.sub.3, and [0568] X.sup.1 and X.sup.2 are each independently N or C(R.sup.5); and [0569] R.sup.4 is: [0570] (i) —S(O).sub.2—R.sup.a; [0571] (ii) 5-6 membered heteroaryl, wherein the 5-6 membered heteroaryl of R.sup.4 is optionally substituted with one or more —CH.sub.3; [0572] (iii) —N(R.sup.d).sub.2, wherein R.sup.d is independently at each occurrence H, C.sub.1-3alkyl, or —S(O).sub.2—R.sup.a, wherein the C.sub.1-3alkyl of R.sup.d is optionally substituted with one or more —OH, [0573] (iv) —NS(O)—(CH.sub.3).sub.2 [0574] (v) —C(O)—N(R.sup.e).sub.2 wherein R.sup.e is independently at each occurrence H, CH.sub.3, or 3-6 membered heterocycle, wherein the 3-6 membered heterocycle is optionally substituted with one or more oxo, or both R.sup.e together with the N to which they are attached are taken together to form a 3-7 membered heterocyclyl, wherein the 3-7 membered heterocyclyl is optionally substituted with one or more F, oxo, —OH, NH.sub.2, —NH—S(O).sub.2—R.sup.a, or —S(O).sub.2—R.sup.a, [0575] (vi) 3-7 membered heterocyclyl optionally substituted with one or more C.sub.1-3alkyl, —OH, oxo or —S(O).sub.2R.sup.a, [0576] (vii) —S(O)—N(CH.sub.3)—(CH.sub.3), [0577] (viii) —CN, [0578] (ix) —(CH.sub.2).sub.qOH, wherein q is an integer from 0-2, or [0579] (x) —C(O)—CH.sub.3, or [0580] (xi) —P(O)(CH.sub.3).sub.2; [0581] or [0582] (2) L.sup.3 is absent; and [0583] one of X.sup.1 and X.sup.2 is N or C(R.sup.5); and [0584] the other of X.sup.1 and X.sup.2 is N or C that is taken together with R.sup.4 and the atoms to which they are attached to form a 5-8 membered heterocyclyl or a 5-6 membered heteroaryl, wherein [0585] the 5-8 membered heterocyclyl is optionally substituted with one or more R.sup.b, wherein R.sup.b is, independently at each occurrence, selected from the group consisting of oxo, C.sub.1-3alkyl, —C(O)—CH.sub.3, —C(O)—NH.sub.2, —C(O)—NH(CH.sub.3), —C(O)—N(CH.sub.3).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-4cycloalkyl, and 3-4 membered heterocyclyl, wherein [0586] the C.sub.1-3alkyl of R.sup.b is optionally substituted with one or more F, OH, —S(O).sub.2—CH.sub.3, or C.sub.3-4cycloalkyl, and wherein the C.sub.3-4cycloalkyl of the C.sub.1-3alkyl of R.sup.b is further optionally substituted with one or more CH.sub.3 or —OH and [0587] the C.sub.3-4cycloalkyl of R.sup.b is optionally substituted with one or more —OH, C.sub.3-4cycloalkyl, or C.sub.1-3alkyl, and wherein the C.sub.1-3alkyl of the C.sub.3-4cycloalkyl of R.sup.b is further optionally substituted with one or more —OH, and [0588] the 5-6 membered heteroaryl is optionally substituted with one or more R.sup.e, wherein R.sup.e is, independently at each occurrence, selected from the group consisting of C.sub.1-3alkyl, —S(O).sub.2—R.sup.a, C.sub.3-4cycloalkyl, and 3-4 membered heterocyclyl, wherein [0589] the C.sub.1-6alkyl of R.sup.e is optionally substituted with one or more —S(O).sub.2—CH.sub.3 [0590] the C.sub.3-4cycloalkyl of R.sup.e is optionally substituted with one or more —OH or C.sub.1-3alkyl, [0591] and the 3-4 membered heterocyclyl of R.sup.e is optionally substituted with one or more —OH or methyl, and wherein the methyl of the 3-4 membered heterocyclyl of R.sup.e is further optionally substituted with one or more —OH; [0592] R.sup.a is, independently at each occurrence: [0593] (i) C.sub.1-3alkyl optionally substituted with one or more F, —OH, —S(O).sub.2—CH.sub.3, or —N(CH.sub.3)—C(O)—CH.sub.3, [0594] (ii) C.sub.3-4cycloalkyl optionally substituted with one or more —OH, —C(O).sub.2—CH.sub.3, —C(O)—NH(CH.sub.3), —C(O)—N(CH.sub.3).sub.2, or —C(O)—C.sub.3-4heterocyclyl, or methyl, wherein the methyl is optionally substituted with one or more —OH, or [0595] (iii) 3-4 membered heterocyclyl optionally substituted with one or more methyl; [0596] R.sup.5 is, independently at each occurrence, H, Cl, F, —CN, 3-4 membered heterocyclyl, C.sub.1-3alkyl, or C.sub.1-3alkoxy, wherein the C.sub.1-3alkyl of R.sup.5 is optionally substituted with one or more F or —OH and wherein the C.sub.1-3alkoxy of R.sup.5 is optionally substituted with one or more F; and [0597] R.sup.6 and R.sup.7 are each independently H or F. In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0598] In some embodiments of a compound of formula (II), or a stereoisomer or a tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, [0599] m is an integer from 0 to 2; [0600] n is an integer from 1 to 2; [0601] p is an integer from 0 to 1; [0602] R^{sup.1}, if present is, independently at each occurrence selected from the group consisting of halo, —CN, C_{sub.1}-3alkoxy, and C_{sub.1}-3alkyl, wherein [0603] the C_{sub.1}-3alkoxy of R^{sup.1} is optionally substituted with one or more halo, and [0604] the C_{sub.1}-3alkyl of R^{sup.1} is optionally substituted with one or more halo; [0605] R^{sup.2} is H, C_{sub.1}-3alkyl, C_{sub.3}-6cycloalkyl, or 3-6 membered heterocyclyl, wherein [0606] the C_{sub.1}-6alkyl of R^{sup.2} is optionally substituted with one or more deuterium, halo, —OH, —NH_{sub.2}, or C_{sub.1}-3alkoxy, and [0607] the C_{sub.3}-6cycloalkyl of R^{sup.2} is optionally substituted with one or more —OH; [0608] R^{sup.3}, if present, is C_{sub.1}-3alkyl; [0609] L^{sup.1} is C_{sub.1}-3alkylene, wherein [0610] the C_{sub.1}-6alkylene of L^{sup.1} is optionally substituted with one or more deuterium or C_{sub.1}-3alkyl, and wherein [0611] the C_{sub.1}-3alkyl is further optionally substituted with one or more —OH or C_{sub.1}-3alkoxy; [0612] L^{sup.2} is O; and [0613] either [0614] (1) L^{sup.3} is absent or is O, C_{sub.3}-6cycloalkyl, 3-6 membered heterocyclyl, or C_{sub.1}-3alkylene, wherein [0615] the C_{sub.3}-6cycloalkyl of L^{sup.3} is optionally substituted with one or more —OH or C_{sub.1}-3alkyl, [0616] the C_{sub.1}-3alkylene of L^{sup.3} is optionally substituted with one or more —OH or C_{sub.1}-3alkyl, wherein [0617] the C_{sub.1}-3alkyl is optionally substituted with one or more —OH, and [0618] the 3-6 membered heterocyclyl of L^{sup.3} is optionally substituted with one or more —OH or C_{sub.1}-3alkyl; [0619] X^{sup.1} and X^{sup.2} are each independently N or C(R^{sup.5}); and [0620] R^{sup.4} is: [0621] (i) —S(O)_{sub.2}—R^{sup.a}; [0622] (ii) 5-10 membered heteroaryl, wherein the 5-10 membered heteroaryl of R^{sup.4} is optionally substituted with one or more C_{sub.1}-3alkyl; [0623] (iii) —N(R^{sup.d})_{sub.2}, wherein R^{sup.d} is independently at each occurrence H, C_{sub.1}-3alkyl, or —S(O)_{sub.2}—R^{sup.a}, wherein the C_{sub.1}-3alkyl of R^{sup.d} is optionally substituted with one or more —OH, [0624] (iv) —NS(O)—(C_{sub.1}-3alkyl)_{sub.2}, wherein the C_{sub.1}-3alkyl is optionally substituted with one or more —OH, [0625] (v) —C(O)—N(R^{sup.e})_{sub.2} wherein R^{sup.e} is independently at each occurrence H, C_{sub.1}-3alkyl, or 3-6 membered heterocycle, wherein [0626] the 3-6 membered heterocycle of R^{sup.e} is optionally substituted with one or more oxo, or both R^{sup.e} together with the N to which they are attached are taken together to form a 3-6 membered heterocyclyl, wherein [0627] the 3-6 membered heterocyclyl is optionally substituted with one or more halo, oxo, —OH, NH_{sub.2}, —NH—S(O)_{sub.2}—R^{sup.a}, or —S(O)_{sub.2}—R^{sup.a}, [0628] (vi) 3-6 membered heterocyclyl optionally substituted with one or more C_{sub.1}-3alkyl, —OH, oxo or —S(O)_{sub.2}—R^{sup.a}, [0629] (vii) —S(O)—N(C_{sub.1}-3alkyl)-(C_{sub.1}-3alkyl), [0630] (viii) —CN, [0631] (ix) —(CH_{sub.2})_{sub.q}OH, wherein q is an integer from 0-3, [0632] (x) —C(O)—C_{sub.1}-3alkyl, or [0633] (xi) —P(O)(C_{sub.1}-3alkyl)_{sub.2}; [0634] or [0635] (2) L^{sup.3} is absent; and [0636] one of X^{sup.1} and X^{sup.2} is N or C(R^{sup.5}); and [0637] the other of X^{sup.1} and X^{sup.2} is N or C that is taken together with R^{sup.4}, and the atoms to which they are attached, to form a 5-8 membered heterocyclyl or a 5-10 membered heteroaryl, wherein [0638] the 5-8 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein [0639] R^{sup.b} is, independently at each occurrence, selected from the group consisting of —OH, halo, oxo, C_{sub.1}-3alkyl, —C(O)—C_{sub.1}-3alkyl, —C(O)—NH_{sub.2}, —C(O)—NH(C_{sub.1}-3alkyl), —C(O)—N(C_{sub.1}-3alkyl)_{sub.2}, —S(O)_{sub.2}—R^{sup.a}, C_{sub.3}-6cycloalkyl, and 3-6 membered heterocyclyl, wherein [0640] the C_{sub.1}-3alkyl of R^{sup.b} is optionally substituted with one or more halo, OH, —S(O)_{sub.2}—C_{sub.1}-3alkyl, or C_{sub.3}-6cycloalkyl, and wherein the C_{sub.3}-6cycloalkyl of the C_{sub.1}-3alkyl of R^{sup.b} is further optionally substituted with one or more C_{sub.1}-3alkyl or —OH and [0641] the C_{sub.3}-6cycloalkyl of R^{sup.b} is optionally substituted with one or more —OH, C_{sub.3}-6cycloalkyl, or C_{sub.1}-3alkyl, and wherein the C_{sub.1}-3alkyl of the C_{sub.3}-6cycloalkyl of R^{sup.b} is further optionally substituted with one or more —OH, deuterium, or halo, and [0642] the 5-10 membered heteroaryl is optionally substituted with one or more R^{sup.e}, wherein [0643] R^{sup.e} is, independently at each occurrence, selected

from the group consisting of halo, C.sub.1-3alkyl, —C(O)—C.sub.1-3alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-3alkyl), —C(O)—N(C.sub.1-3alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-6cycloalkyl, and 3-6 membered heterocyclyl, wherein [0644] the C.sub.1-3alkyl of R.sup.e is optionally substituted with one or more —S(O).sub.2—C.sub.1-3alkyl, [0645] the C.sub.3-6cycloalkyl of R.sup.e is optionally substituted with one or more —OH or C.sub.1-3alkyl, and [0646] the 3-6 membered heterocyclyl of R.sup.e is optionally substituted with one or more —OH or C.sub.1-3alkyl, and wherein the C.sub.1-3alkyl of the 3-6 membered heterocyclyl of R.sup.e is further optionally substituted with one or more —OH; [0647] R.sup.a is, independently at each occurrence: [0648] (i) C.sub.1-3alkyl optionally substituted with one or more halo, —OH, —S(O).sub.2—C.sub.1-3alkyl, or —N(C.sub.1-3alkyl)-C(O)—C.sub.1-3alkyl, [0649] (ii) C.sub.3-6cycloalkyl optionally substituted with one or more —OH, —C(O).sub.2—C.sub.1-3alkyl, —C(O)—NH(C.sub.1-3alkyl), —C(O)—N(C.sub.1-3alkyl).sub.2, or —C(O)—C.sub.3-6heterocyclyl, or C.sub.1-3alkyl, wherein the C.sub.1-3alkyl is optionally substituted with one or more —OH, [0650] (iii) 3-6 membered heterocyclyl optionally substituted with one or more C.sub.1-3alkyl, or [0651] (iv) NH(C.sub.1-3alkyl); R.sup.5 is, independently at each occurrence, H, halo, —CN, 3-6 membered heterocyclyl, C.sub.1-3alkyl, or C.sub.1-3alkoxy, wherein [0652] the C.sub.1-3alkyl of R.sup.5 is optionally substituted with one or more halo or —OH, and [0653] the C.sub.1-3alkoxy of R.sup.5 is optionally substituted with one or more halo; [0654] X.sup.3 is N or C(R.sup.6) [0655] X.sup.4 is N or C(R.sup.7); [0656] and [0657] R.sup.6 and R.sup.7 are each independently H or halo.

[0658] In some embodiments of a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, [0659] m is an integer from 0 to 2; [0660] n is 1; [0661] p is an integer from 0; [0662] R.sup.1, if present is, independently at each occurrence selected from the group consisting of halo, —CN, C.sub.1-3alkoxy, and C.sub.1-3alkyl, wherein [0663] the C.sub.1-3alkoxy of R.sup.1 is optionally substituted with one or more halo, and [0664] the C.sub.1-3alkyl of R.sup.1 is optionally substituted with one or more halo; [0665] R.sup.2 is H, C.sub.1-3alkyl, C.sub.3-6cycloalkyl, or 3-6 membered heterocyclyl, wherein [0666] the C.sub.1-6alkyl of R.sup.2 is optionally substituted with one or more deuterium, halo, —OH, —NH.sub.2, or C.sub.1-3alkoxy, and [0667] the C.sub.3-6cycloalkyl of R.sup.2 is optionally substituted with one or more —OH; [0668] L.sup.1 is C.sub.1-3alkylene, wherein [0669] the C.sub.1-6alkylene of L.sup.1 is optionally substituted with one or more deuterium; [0670] L.sup.2 is O; and [0671] either [0672] (1) L.sup.3 is absent or is O, C.sub.3-6cycloalkyl, 3-6 membered heterocyclyl, or C.sub.1-3alkylene, wherein [0673] the C.sub.3-6cycloalkyl of L.sup.3 is optionally substituted with one or more —OH or C.sub.1-3alkyl, [0674] the C.sub.1-3alkylene of L.sup.3 is optionally substituted with one or more —OH or C.sub.1-3alkyl, wherein [0675] the C.sub.1-3alkyl is optionally substituted with one or more —OH, and [0676] the 3-6 membered heterocyclyl of L.sup.3 is optionally substituted with one or more —OH or C.sub.1-3alkyl; [0677] X.sup.1 and X.sup.2 are each independently N or C(R.sup.5); and [0678] R.sup.4 is: [0679] (i) —S(O).sub.2—R.sup.a; [0680] (ii) 5-10 membered heteroaryl, wherein the 5-10 membered heteroaryl of R.sup.4 is optionally substituted with one or more C.sub.1-3alkyl; [0681] (iii) —N(R.sup.d).sub.2, wherein R.sup.d is independently at each occurrence H, C.sub.1-3alkyl, or —S(O).sub.2—R.sup.a, wherein the C.sub.1-3alkyl of R.sup.d is optionally substituted with one or more —OH, [0682] (iv) —NS(O)—(C.sub.1-3alkyl).sub.2, wherein the C.sub.1-3alkyl is optionally substituted with one or more —OH, [0683] (v) —C(O)—N(R.sup.e).sub.2 wherein R.sup.e is independently at each occurrence H, C.sub.1-3alkyl, or 3-6 membered heterocycle, wherein [0684] the 3-6 membered heterocycle of R.sup.e is optionally substituted with one or more oxo, or both R.sup.e together with the N to which they are attached are taken together to form a 3-6 membered heterocyclyl, wherein [0685] the 3-6 membered heterocyclyl is optionally substituted with one or more halo, oxo, —OH, NH.sub.2, —NH—S(O).sub.2—R.sup.a, or —S(O).sub.2—R.sup.a, [0686] (vi) 3-6 membered heterocyclyl optionally substituted with one or more C.sub.1-3alkyl, —OH, oxo or —

S(O).sub.2R.sup.a, [0687] (ix) —(CH.sub.2).sub.qOH, wherein q is an integer from 0-3; [0688] or [0689] (2) L.sup.3 is absent; and [0690] one of X.sup.1 and X.sup.2 is N or C(R.sup.5); and [0691] the other of X.sup.1 and X.sup.2 is N or C that is taken together with R.sup.4, and the atoms to which they are attached, to form a 5-8 membered heterocyclyl or a 5-10 membered heteroaryl, wherein [0692] the 5-10 membered heterocyclyl is optionally substituted with one or more R.sup.b, wherein [0693] R.sup.b is, independently at each occurrence, selected from the group consisting of —OH, halo, oxo, C.sub.1-3alkyl, —C(O)—C.sub.1-3alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-3alkyl), —C(O)—N(C.sub.1-3alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-6cycloalkyl, and 3-6 membered heterocyclyl, wherein [0694] the C.sub.1-3alkyl of R.sup.b is optionally substituted with one or more halo, OH, —S(O).sub.2—C.sub.1-3alkyl, or C.sub.3-6cycloalkyl, and wherein the C.sub.3-6cycloalkyl of the C.sub.1-3alkyl of R.sup.b is further optionally substituted with one or more C.sub.1-3alkyl or —OH and [0695] the C.sub.3-6cycloalkyl of R.sup.b is optionally substituted with one or more —OH, C.sub.3-6cycloalkyl, or C.sub.1-3alkyl, and wherein the C.sub.1-3alkyl of the C.sub.3-6cycloalkyl of R.sup.b is further optionally substituted with one or more —OH, deuterium, or halo, and [0696] the 5-10 membered heteroaryl is optionally substituted with one or more R.sup.e, wherein [0697] R.sup.e is, independently at each occurrence, selected from the group consisting of halo, C.sub.1-3alkyl, —C(O)—C.sub.1-3alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-3alkyl), —C(O)—N(C.sub.1-3alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-6cycloalkyl, and 3-6 membered heterocyclyl, wherein [0698] the C.sub.1-3alkyl of R.sup.e is optionally substituted with one or more —S(O).sub.2—C.sub.1-3alkyl, [0699] the C.sub.3-6cycloalkyl of R.sup.e is optionally substituted with one or more —OH or C.sub.1-3alkyl, and [0700] the 3-6 membered heterocyclyl of R.sup.e is optionally substituted with one or more —OH or C.sub.1-3alkyl, and wherein the C.sub.1-3alkyl of the 3-6 membered heterocyclyl of R.sup.e is further optionally substituted with one or more —OH; [0701] R.sup.a is, independently at each occurrence: [0702] (i) C.sub.1-3alkyl optionally substituted with one or more halo, —OH, —S(O).sub.2—C.sub.1-3alkyl, or —N(C.sub.1-3alkyl)—C(O)—C.sub.1-3alkyl, [0703] (ii) C.sub.3-6cycloalkyl optionally substituted with one or more —OH, —C(O).sub.2—C.sub.1-3alkyl, —C(O)—NH(C.sub.1-3alkyl), —C(O)—N(C.sub.1-3alkyl).sub.2, or —C(O)—C.sub.3-6heterocyclyl, or C.sub.1-3alkyl, wherein the C.sub.1-3alkyl is optionally substituted with one or more —OH, [0704] (iii) 3-6 membered heterocyclyl optionally substituted with one or more C.sub.1-3alkyl, or [0705] (iv) NH(C.sub.1-3alkyl); R.sup.5 is, independently at each occurrence, H, halo, —CN, 3-6 membered heterocyclyl, C.sub.1-3alkyl, or C.sub.1-3alkoxy, wherein [0706] the C.sub.1-3alkyl of R.sup.5 is optionally substituted with one or more halo or —OH, and [0707] the C.sub.1-3alkoxy of R.sup.5 is optionally substituted with one or more halo; [0708] X.sup.3 is N or C(R.sup.6) [0709] X.sup.4 is N or C(R.sup.7); [0710] and [0711] R.sup.6 and R.sup.7 are each independently H or halo.

[0712] In some embodiments of a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, [0713] m is an integer from 0 to 2; [0714] n is 1; [0715] p is an integer from 0; [0716] R.sup.1, if present is, independently at each occurrence selected from the group consisting of halo, —CN, C.sub.1-3alkoxy, and C.sub.1-3alkyl, wherein [0717] the C.sub.1-3alkoxy of R.sup.1 is optionally substituted with one or more halo, and [0718] the C.sub.1-3alkyl of R.sup.1 is optionally substituted with one or more halo; [0719] R.sup.2 is H; [0720] L.sup.1 is C.sub.1-3alkylene, wherein [0721] the C.sub.1-6alkylene of L.sup.1 is optionally substituted with one or more deuterium; [0722] L.sup.2 is O; and [0723] either [0724] (1) L.sup.3 is absent or is C.sub.1-3alkylene, wherein [0725] the C.sub.1-3alkylene of L.sup.3 is optionally substituted with one or more —OH or C.sub.1-3alkyl; [0726] X.sup.1 and X.sup.2 are each independently N or C(R.sup.5); and [0727] R.sup.4 is: [0728] (i) —S(O).sub.2—R.sup.a; [0729] (ii) —(CH.sub.2).sub.qOH, wherein q is an integer from 0-3; [0730] or [0731] (2) L.sup.3 is absent; and [0732] one of X.sup.1 and X.sup.2 is N or C(R.sup.5); and [0733] the other of X.sup.1 and X.sup.2 is N or C that is taken together with R.sup.4, and the atoms to which they are attached,

to form a 5-8 membered heterocyclyl or a 5-20 membered heteroaryl, wherein [0734] the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein R^{sup.b} is, independently at each occurrence oxo, or C_{sub}.3-6cycloalkyl, wherein the C_{sub}.3-6cycloalkyl is optionally substituted with one or more —OH, C_{sub}.3-6cycloalkyl, or C_{sub}.1-3alkyl, and [0735] the 5-10 membered heteroaryl is optionally substituted with one or more R^{sup.e}, wherein R^{sup.e} is, independently at each occurrence, selected from the group consisting of C_{sub}.1-3alkyl, C_{sub}.3-6cycloalkyl, wherein the C_{sub}.3-6cycloalkyl of R^{sup.e} is optionally substituted with one or more —OH or C_{sub}.1-3alkyl; [0736] R^{sup.a} is, independently at each occurrence: [0737] (i) C_{sub}.1-3alkyl optionally substituted with one or more halo, —OH, —S(O)_{sub}.2—C_{sub}.1-3alkyl, or —N(C_{sub}.1-3alkyl)-C(O)—C_{sub}.1-3alkyl; [0738] R^{sup.5} is, independently at each occurrence, H, halo, C_{sub}.1-3alkyl, or C_{sub}.1-3alkoxy, wherein [0739] the C_{sub}.1-3alkyl of R^{sup.5} is optionally substituted with one or more halo or —OH, and [0740] the C_{sub}.1-3alkoxy of R^{sup.5} is optionally substituted with one or more halo; [0741] X^{sup.3} is N or C(R^{sup.6}) [0742] X^{sup.4} is N or C(R^{sup.7}); [0743] and [0744] R^{sup.6} and R^{sup.7} are each independently H or halo. [0745] In some embodiments, provided herein is a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein the compound is a compound of formula (I-A):

##STR00226## [0746] or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein: n is 1 or 2; and m, p, R^{sup.1}, R^{sup.2}, R^{sup.3}, L^{sup.1}, L^{sup.3}, R^{sup.4}, X^{sup.1}, X^{sup.2}, R^{sup.6}, and R^{sup.7} are as defined for formula (I). In some variations m, p, R^{sup.1}, R^{sup.2}, R^{sup.3}, L^{sup.1}, L^{sup.3}, R^{sup.4}, X^{sup.1}, X^{sup.2}, R^{sup.6}, and R^{sup.7} of formula (I-A) are as defined for a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0747] In some embodiments, provided herein is a compound of formula (I) or formula (I-A), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein the compound is a compound of formula (I-A1):

##STR00227## [0748] or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein: m, p, R^{sup.1}, R^{sup.2}, R^{sup.3}, L^{sup.1}, L^{sup.3}, R^{sup.4}, X^{sup.1}, X^{sup.2}, R^{sup.6}, and R^{sup.7} are as defined for formula (I). In some variations m, p, R^{sup.1}, R^{sup.2}, R^{sup.3}, L^{sup.1}, L^{sup.3}, R^{sup.4}, X^{sup.1}, X^{sup.2}, R^{sup.6}, and R^{sup.7} of formula (I-A1) are as defined for a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0749] In some embodiments of a compound of formula (I), (I-A) or (I-A1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.2} is H. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0750] In some embodiments of a compound of formula (I), (I-A) or (I-A1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, m is an integer from 0 to 4. In some embodiments, m is an integer from 0 to 2. In some embodiments, m is 0. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0751] In some embodiments of a compound of formula (I), (I-A) or (I-A1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, p is an integer from 0 to 10. In some embodiments, p is 0 or 1. In some embodiments, p is 0. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or

embodiment thereof.

[0752] In some embodiments of a compound of formula (I), (I-A) or (I-A1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.1} is C_{sub.1-6}alkylene. In some embodiments, L^{sup.1} is C_{sub.1-3}alkylene. In some embodiments, L^{sup.1} is ethylene. In some embodiments, L^{sup.1} is

##STR00228##

wherein, # denotes the point of attachment to —O— and ## denotes the point of attachment to the remainder of the molecule. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0753] In some embodiments of a compound of formula (I), (I-A) or (I-A1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.3} is absent. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0754] In some embodiments of a compound of formula (I), (I-A) or (I-A1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.4} is —S(O)_{sub.2}—R^{sup.a}, wherein R^{sup.a} is C_{sub.1-6}alkyl. In some embodiments, R^{sup.4} is —S(O)_{sub.2}—R^{sup.a}, wherein R^{sup.a} is C_{sub.1-3}alkyl. In some embodiments, R^{sup.4} is —S(O)_{sub.2}—R^{sup.a}, wherein R^{sup.a} is methyl. In some embodiments, R^{sup.4} is

##STR00229##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0755] In some embodiments of a compound of formula (I), (I-A) or (I-A1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, each of X^{sup.1} and X^{sup.2} is C(R^{sup.5}). In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0756] In some embodiments of a compound of formula (I), (I-A) or (I-A1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.5} is H. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0757] In some embodiments of a compound of formula (I), (I-A) or (I-A1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, each of R^{sup.6} and R^{sup.7} is H. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0758] In some embodiments, provided herein is a compound of formula (I) or formula (I-A), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein the compound is a compound of formula (I-A2):

##STR00230##

or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein m, p, R^{sup.1}, R^{sup.2}, R^{sup.3}, L^{sup.1}, L^{sup.3}, R^{sup.4}, X^{sup.1}, X^{sup.2}, R^{sup.6}, and R^{sup.7} are as defined for formula (I). In some variations, m, p, R^{sup.1}, R^{sup.2}, R^{sup.3}, L^{sup.1}, L^{sup.3}, R^{sup.4}, X^{sup.1}, X^{sup.2}, R^{sup.6}, and R^{sup.7} of formula (I-A2) are as defined for a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0759] In some embodiments, provided herein is a compound of formula (I) or formula (I-A), or a

stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein the compound is a compound of formula (I-A3):

##STR00231##

or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein: n is 1 or 2; and wherein m, n, p, R^{sup.1}, R^{sup.2}, R^{sup.3}, L^{sup.1}, X^{sup.1}, X^{sup.2}, R^{sup.6}, R^{sup.7}, and R^{sup.a} are as defined for formula (I). In some variations, m, n, p, R^{sup.1}, R^{sup.2}, R^{sup.3}, L^{sup.1}, X^{sup.1}, X^{sup.2}, R^{sup.6}, R^{sup.7}, and R^{sup.a} of formula (I-A3) are as defined for a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof. [0760] In some embodiments, provided herein is a compound of formula (I), (I-A), or (I-A2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt thereof, wherein the compound is a compound of formula (I-B):

##STR00232##

or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein: m, p, R^{sup.1}, R^{sup.2}, R^{sup.3}, L^{sup.1}, L^{sup.3}, R^{sup.4}, R^{sup.5}, R^{sup.6}, and R^{sup.7} are as defined for formula (I). In some variations, m, p, R^{sup.1}, R^{sup.2}, R^{sup.3}, L^{sup.1}, L^{sup.3}, R^{sup.4}, R^{sup.5}, R^{sup.6}, and R^{sup.7} of formula (I-B) are as defined for a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0761] In some embodiments, provided herein is a compound of formula (I), (I-A), (I-A2), or (I-B), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt thereof, wherein the compound is a compound of formula (I-B1):

##STR00233##

or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein: m, R^{sup.1}, R^{sup.2}, L^{sup.1}, L^{sup.3}, R^{sup.4}, R^{sup.6}, and R^{sup.7} are as defined for formula (I). In some variations, m, R^{sup.1}, R^{sup.2}, L^{sup.1}, L^{sup.3}, R^{sup.4}, R^{sup.6}, and R^{sup.7} of formula (I-B1) are as defined for a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0762] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.2} is H. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0763] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.2} is C_{sub.1-6}alkyl. In some embodiments, R^{sup.2} is C_{sub.1-3}alkyl. In some embodiments, R^{sup.2} is methyl. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0764] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.2} is C_{sub.1-6}alkyl optionally substituted with one or more halo, —OH, or C_{sub.1-6}alkoxy. In some embodiments, R^{sup.2} is C_{sub.1-3}alkyl optionally substituted with one or more halo, —OH, or C_{sub.1-3}alkoxy. In some embodiments, R^{sup.2} is ethyl optionally substituted with one or more halo, —OH, NH_{sub.2}, or C_{sub.1-3}alkoxy. In some embodiments, R^{sup.2} is selected from the group consisting of

##STR00234##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the

foregoing, or any variation or embodiment thereof.

[0765] In some embodiments of a compound of formula (II), such as a compound of formula (I-A), (I-A2), (I-B), or (I-B1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.2} is C_{sub.1-6}alkyl optionally substituted with one or more deuterium, halo, —OH, or C_{sub.1-6}alkoxy. In some embodiments, R^{sup.2} is C_{sub.1-3}alkyl optionally substituted with one or more deuterium, halo, —OH, or C_{sub.1-3}alkoxy. In some embodiments, R^{sup.2} is methyl optionally substituted with one or more deuterium halo, —OH, NH_{sub.2}, or C_{sub.1-3}alkoxy. In some embodiments, R^{sup.2} is ethyl optionally substituted with one or more deuterium halo, —OH, NH_{sub.2}, or C_{sub.1-3}alkoxy. In some embodiments, R^{sup.2} is selected from the group consisting of

##STR00235##

[0766] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.2} is C_{sub.3-10}cycloalkyl optionally substituted with one or more —OH. In some embodiments, R^{sup.2} is C_{sub.3-6} cycloalkyl optionally substituted with one or more —OH. In some embodiments, R^{sup.2} is

##STR00236##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0767] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.2} is 3-15 membered heterocyclyl. In some embodiments, R^{sup.2} is 3-6 membered heterocyclyl. In some embodiments, R^{sup.2} is

##STR00237##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0768] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, m is an integer from 0 to 4. In some embodiments, m is an integer from 0 to 2. In some embodiments, m is 0. In some embodiments m is 1. In some embodiments, m is 2. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0769] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.1} is halo. In some embodiments R^{sup.1} is Cl. In some embodiments, R^{sup.1} is Br. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0770] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.1} is C_{sub.1-6}alkyl. In some embodiments, R^{sup.1} is methyl. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0771] In some embodiments of a compound of formula (I'), (I-A), (I-A2), (I-B), or (I-B1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.1} is C_{sub.1-6}alkyl, wherein the C_{sub.1-6}alkyl of R^{sup.1} is optionally substituted with one

or more halo. In some embodiments, R.sup.1 is C.sub.1-3alkyl, wherein the C.sub.1-3alkyl of R.sup.1 is optionally substituted with one or more halo. In some embodiments, R.sup.1 is methyl, wherein the methyl of R.sup.1 is optionally substituted with one or more F. In some embodiments, R.sup.1 is

##STR00238##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0772] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.1 is —CN. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0773] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, p is an integer from 0 to 10. In some embodiments, p is 0. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0774] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.1 is C.sub.1-6alkylene, wherein the C.sub.1-6alkylene of L.sup.1 is optionally substituted with one or more C.sub.1-6alkyl. In some embodiments, L.sup.1 is C.sub.1-3alkylene optionally substituted with one or more C.sub.1-3alkyl. In some embodiments, L.sup.1 is ethylene optionally substituted with one or more C.sub.1-6alkyl. In some embodiments, L.sup.1 is selected from the group consisting of

##STR00239##

wherein, for each L.sup.1, # denotes the point of attachment to —O— and ## denotes the point of attachment to the remainder of the molecule. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0775] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.3 is absent. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0776] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.3 is C.sub.1-6alkylene. In some embodiments, L.sup.3 is C.sub.1-3alkylene. In some embodiments, L.sup.3 is selected from the group consisting of

##STR00240##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0777] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.3 is C.sub.3-10cycloalkyl, wherein the C.sub.3-10cycloalkyl of L.sup.3 is optionally substituted with one or more —OH. In some embodiments, L.sup.3 is C.sub.3-8cycloalkyl, wherein the C.sub.3-8cycloalkyl of L.sup.3 is optionally substituted with one or more —OH. In some embodiments, L.sup.3 is selected from the group consisting of

##STR00241##

In some variations, the embodiments provided herein also apply to a compound of formula (I') or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0778] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.3 is 3-10 membered heterocyclyl. In some embodiments, L.sup.3 is 3-6 membered heterocyclyl. In some embodiments, L.sup.3 is selected from the group consisting of
##STR00242##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0779] In some embodiments of a compound of formula (II), such as a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.3 is 3-10 membered heterocyclyl. In some embodiments, L.sup.3 is 3-6 membered heterocyclyl. In some embodiments, L.sup.3 is selected from the group consisting of

##STR00243##

[0780] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.4 is —S(O).sub.2—R.sup.a, wherein R.sup.a is C.sub.1-6alkyl. In some embodiments, R.sup.4 is —S(O).sub.2—R.sup.a, wherein R.sup.a is C.sub.1-6alkyl, wherein the C.sub.1-6alkyl of R.sup.a is optionally substituted with one or more halo, —OH, —S(O).sub.2—C.sub.1-6alkyl, or —N(C.sub.1-6alkyl)-C(O)—C.sub.1-6alkyl. In some embodiments, R.sup.4 is —S(O).sub.2—R.sup.a, wherein R.sup.a is C.sub.1-3alkyl, wherein the C.sub.1-3alkyl of R.sup.a is optionally substituted with one or more halo, —OH, —S(O).sub.2—C.sub.1-3alkyl, or —N(C.sub.1-3alkyl)-C(O)—C.sub.1-3alkyl. In some embodiments, R.sup.4 is selected from the group consisting of
##STR00244##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0781] In some embodiments of a compound of formula (II), such as a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.4 is selected from the group consisting of
##STR00245##

[0782] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.4 is —S(O).sub.2—R.sup.a, wherein R.sup.a is C.sub.3-10cycloalkyl. R.sup.4 is —S(O).sub.2—R.sup.a, wherein R.sup.a is C.sub.3-10cycloalkyl, wherein the C.sub.3-10cycloalkyl of R.sup.a is optionally substituted with one or more —OH, C(O).sub.2—C.sub.1-6alkyl, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —C(O)—C.sub.3-10heterocyclyl or C.sub.1-6alkyl, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH. In some embodiments, R.sup.4 is —S(O).sub.2—R.sup.a, wherein R.sup.a is C.sub.3-6cycloalkyl, wherein the C.sub.3-6cycloalkyl of R.sup.a is optionally substituted with one or more —OH, C(O).sub.2—C.sub.1-6alkyl, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —C(O)—C.sub.3-10heterocyclyl, or C.sub.1-6alkyl, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH. In some embodiments, R.sup.4 is —S(O).sub.2—R.sup.a, wherein R.sup.a is C.sub.3-6cycloalkyl, wherein the C.sub.3-6cycloalkyl of R.sup.a is optionally substituted with one or more —OH, C(O).sub.2—C.sub.1-3alkyl, —C(O)—NH(C.sub.1-3alkyl), —C(O)—N(C.sub.1-3alkyl).sub.2, —C(O)—C.sub.3-6heterocyclyl, or C.sub.1-3alkyl, wherein the C.sub.1-3alkyl is optionally substituted with one or more —OH. In some embodiments, R.sup.4 is

selected from the group consisting of

##STR00246##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0783] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.4} is —S(O)_{sub.2}—R^{sup.a}, wherein R^{sup.a} is 3-10 membered heterocyclyl. In some embodiments R^{sup.4} is —S(O)_{sub.2}—R^{sup.a}, wherein R^{sup.a} is 3-10 membered heterocyclyl, wherein the 3-10 membered heterocyclyl of R^{sup.a} is optionally substituted with one or more C_{sub.1-6}alkyl. In some embodiments, R^{sup.4} is —S(O)_{sub.2}—R^{sup.a}, wherein R^{sup.a} is 3-6 membered heterocyclyl, wherein the 3-6 membered heterocyclyl of R^{sup.a} is optionally substituted with one or more C_{sub.1-6}alkyl. In some embodiments, wherein R^{sup.4} is —S(O)_{sub.2}—R^{sup.a}, wherein R^{sup.a} is 3-6 membered heterocyclyl, wherein the 3-6 membered heterocyclyl of R^{sup.a} is optionally substituted with one or more C_{sub.1-3}alkyl. In some embodiments, R^{sup.4} is

##STR00247##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0784] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B1) or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.4} is 5-20 membered heteroaryl. In some embodiments, R^{sup.4} is 5-20 membered heteroaryl optionally substituted with one or more C_{sub.1-6}alkyl. In some embodiments, R^{sup.4} is 5-6 membered heteroaryl optionally substituted with one or more C_{sub.1-6}alkyl. In some embodiments, R^{sup.4} is 5-6 membered heteroaryl optionally substituted with one or more C_{sub.1-3}alkyl. In some embodiments, R^{sup.4} is 5-6 membered heteroaryl optionally substituted with one or more methyl. In some embodiments, R^{sup.4} is selected from the group consisting of

##STR00248##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0785] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.4} is —N(R^{sup.d})_{sub.2}, wherein each of R^{sup.d} is independently H, or C_{sub.1-6}alkyl, wherein the C_{sub.1-6}alkyl of R^{sup.d} is optionally substituted with one or more —OH or —S(O)_{sub.2}—R^{sup.a}, wherein R^{sup.a} is C_{sub.1-6}alkyl. In some embodiments, R^{sup.4} is —N(R^{sup.d})_{sub.2}, wherein each of R^{sup.d} is independently H, or C_{sub.1-3}alkyl, wherein the C_{sub.1-3}alkyl of R^{sup.d} is optionally substituted with one or more —OH, or —S(O)_{sub.2}—R^{sup.a}, wherein R^{sup.a} is C_{sub.1-3}alkyl. In some embodiments, R^{sup.4} is —N(R^{sup.d})_{sub.2}, wherein each of R^{sup.d} is independently H, or C_{sub.1-3}alkyl, wherein the C_{sub.1-3}alkyl of R^{sup.d} is optionally substituted with one or more —OH or —S(O)_{sub.2}—R^{sup.a}, wherein R^{sup.a} is methyl. In some embodiments, R^{sup.4} is selected from the group consisting of —NH_{sub.2},

##STR00249##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0786] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing,

R.sup.4 is —C(O)—N(R.sup.e).sub.2, wherein each of R.sup.e is independently H or C.sub.1-6 alkyl. In some embodiments, R.sup.4 is —C(O)—N(R.sup.e).sub.2, wherein each of R.sup.e is independently H or C.sub.1-3 alkyl. In some embodiments, R.sup.4 is —C(O)—N(R.sup.e).sub.2, wherein each of R.sup.e is independently H or methyl. In some embodiments, R.sup.4 is —C(O)—NH.sub.2. In some embodiments, R.sup.4 is —C(O)—NH(CH.sub.3). In some embodiments, R.sup.4 is —C(O)—N(CH.sub.3).sub.2. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0787] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.4 is —C(O)—N(R.sup.e).sub.2, wherein both R.sup.e together with the N to which they are attached are taken together to form a 3-10 membered heterocyclyl. In some embodiments R.sup.4 is —C(O)—N(R.sup.e).sub.2, wherein both R.sup.e together with the N to which they are attached are taken together to form a 3-10 membered heterocyclyl, wherein the 3-10 membered heterocyclyl is optionally substituted with one or more halo, oxo, —OH, NH.sub.2, NH—S(O).sub.2—R.sup.a, or —S(O).sub.2—R.sup.a, wherein R.sup.a is C.sub.1-6alkyl. In some embodiments R.sup.4 is —C(O)—N(R.sup.e).sub.2, wherein both R.sup.e together with the N to which they are attached are taken together to form a 3-7 membered heterocyclyl, wherein the 3-7 membered heterocyclyl is optionally substituted with one or more halo, oxo, —OH, NH.sub.2, NH—S(O).sub.2—R.sup.a, or —S(O).sub.2—R.sup.a, wherein R.sup.a is C.sub.1-3alkyl. In some embodiments R.sup.4 is —C(O)—N(R.sup.e).sub.2, wherein both R.sup.e together with the N to which they are attached are taken together to form a 3-7 membered heterocyclyl, wherein the 3-7 membered heterocyclyl is optionally substituted with one or more halo, oxo, —OH, NH.sub.2, NH—S(O).sub.2—R.sup.a, or —S(O).sub.2—R.sup.a, wherein R.sup.a is methyl. In some embodiments, R.sup.4 is selected from the group consisting of

##STR00250##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0788] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.4 is 3-10 membered heterocyclyl optionally substituted with one or more oxo. In some embodiments, R.sup.4 is 3-7 membered heterocyclyl optionally substituted with one or more oxo. In some embodiments, R.sup.4 is selected from the group consisting of

##STR00251##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0789] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.4 is —NS(O)—(C.sub.1-6alkyl).sub.2. In some embodiments, R.sup.4 is —NS(O)—(C.sub.1-3alkyl).sub.2. In some embodiments, wherein R.sup.4 is

##STR00252##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0790] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.4 is —S(O)—N(C.sub.1-6alkyl)-(C.sub.1-6alkyl). In some embodiments, R.sup.4 is —S(O)—N(C.sub.1-3alkyl)-(C.sub.1-3alkyl). In some embodiments, R.sup.4 is

##STR00253##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0791] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.4} is —CN. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0792] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.4} is —P(O)(C_{sub.1-6}alkyl)_{sub.2}. In some embodiments, R^{sup.4} is —P(O)(C_{sub.1-3}alkyl)_{sub.2}. In some embodiments, R^{sup.4} is —P(O)(CH_{sub.3})_{sub.2}. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0793] In some embodiments, provided herein is a compound of formula (I), (I-A), (I-A2), or (I-B), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt thereof, wherein the compound is a compound of formula (I-B2):

##STR00254## [0794] or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein: R^{sup.5} is halo, —CN, 3-10 membered heterocyclyl, C_{sub.1-6}alkyl, or C_{sub.1-6}alkoxy, wherein the C_{sub.1-6}alkyl of R^{sup.5} is optionally substituted with one or more halo, or —OH and wherein the C_{sub.1-6}alkoxy is optionally substituted with one or more halo; and m, R^{sup.1}, R^{sup.2}, L^{sup.1}, L^{sup.3}, R^{sup.4}, R^{sup.6}, and R^{sup.7} are as defined for formula (I). In some variations, m, R^{sup.1}, R^{sup.2}, L^{sup.1}, L^{sup.3}, R^{sup.4}, R^{sup.6}, and R^{sup.7} of formula (I-B2) are as defined for a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0795] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.2} is H. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0796] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.2} is C_{sub.1-6}alkyl, wherein the C_{sub.1-6}alkyl of R^{sup.2} is optionally substituted with one or more halo, —OH. In some embodiments, R^{sup.2} is C_{sub.1-3}alkyl, wherein the C_{sub.1-3}alkyl of R^{sup.2} is optionally substituted with one or more halo, —OH. In some embodiments, R^{sup.2} is methyl. In some embodiments, R^{sup.2} is ethyl, wherein the ethyl of R^{sup.2} is optionally substituted with one or more halo, —OH. In some embodiments, R^{sup.2} is

##STR00255##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0797] In some embodiments of a compound of formula (II), such as a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.2} is C_{sub.1-6}alkyl, wherein the C_{sub.1-6}alkyl of R^{sup.2} is optionally substituted with one or more deuterium. In some embodiments, R^{sup.2} is C_{sub.1-3}alkyl, wherein the C_{sub.1-3}alkyl of R^{sup.2} is optionally substituted with one or more deuterium. In some embodiments, R^{sup.2} is methyl, wherein the methyl of R^{sup.2} is optionally substituted with one or more deuterium. In some embodiments, R^{sup.2} is

##STR00256##

[0798] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, m is 1. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0799] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.1 is halo. In some embodiments R.sup.1 is Cl. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0800] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.1 is —CN. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0801] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.1 is C.sub.1-6alkylene optionally substituted with one or more C.sub.1-6alkyl, wherein the C.sub.1-6alkyl is optionally substituted with —OH or C.sub.1-6alkoxy. In some embodiments, L.sup.1 is methylene. In some embodiments, L.sup.1 is ethylene optionally substituted with one or more C.sub.1-6alkyl, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH or C.sub.1-6alkoxy. In some embodiments, L.sup.1 is selected from the group consisting of

##STR00257##

and # wherein, for each L.sup.1, # denotes the point of attachment to —O— and ## denotes the point of attachment to the remainder of the molecule. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0802] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.3 is absent. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0803] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.3 is 3-10 membered heterocyclyl. In some embodiments, L.sup.3 is 3-6 membered heterocyclyl. In some embodiments, L.sup.3 is

##STR00258##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0804] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.4 is —S(O).sub.2—R.sup.a, wherein R.sup.a is C.sub.1-6alkyl. In some embodiments, R.sup.4 is —S(O).sub.2—R.sup.a, wherein R.sup.a is C.sub.1-6alkyl. In some embodiments, R.sup.4 is —S(O).sub.2—R.sup.a, wherein R.sup.a is C.sub.1-3alkyl. In some embodiments, R.sup.4 is

##STR00259##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or

(II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0805] In some embodiments of a compound of formula (II) such as a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.4} is —NS(O)—(C_{sub.1-6}alkyl)_{sub.2}. In some embodiments, R^{sup.4} is —NS(O)—(C_{sub.1-6}alkyl)_{sub.2}. In some embodiments, R^{sup.4} is —NS(O)—(C_{sub.1-3}alkyl)_{sub.2}. In some embodiments, R^{sup.4} is

##STR00260##

[0806] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.4} is —N(R^{sup.d})_{sub.2}, wherein each of R^{sup.d} is independently H, C_{sub.1-6} alkyl, or —S(O)_{sub.2}—R^{sup.a}, wherein R^{sup.a} is C_{sub.1-6}alkyl. In some embodiments, R^{sup.4} is —N(R^{sup.d})_{sub.2}, wherein each of R^{sup.d} is independently H, C_{sub.1-3} alkyl, or —S(O)_{sub.2}—R^{sup.a}, wherein R^{sup.a} is C_{sub.1-3}alkyl. In some embodiments, R^{sup.4} is —N(R^{sup.d})_{sub.2}, wherein each of R^{sup.d} is independently H, C_{sub.1-6} alkyl, or —S(O)_{sub.2}—R^{sup.a}, wherein R^{sup.a} is methyl. In some embodiments, R^{sup.4} is selected from the group consisting of —NH_{sub.2}, and

##STR00261##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0807] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.4} is —C(O)—N(R^{sup.e})_{sub.2}, wherein each of R^{sup.e} is independently H or C_{sub.1-6} alkyl. In some embodiments, R^{sup.4} is —C(O)—N(R^{sup.e})_{sub.2}, wherein each of R^{sup.e} is independently H or C_{sub.1-3} alkyl. In some embodiments, R^{sup.4} is —C(O)—N(R^{sup.e})_{sub.2}, wherein each of R^{sup.e} is independently H or methyl. In some embodiments, R^{sup.4} is —C(O)—NH_{sub.2}. In some embodiments, R^{sup.4} is —C(O)—NH(CH_{sub.3}). In some embodiments, R^{sup.4} is —C(O)—N(CH_{sub.3})_{sub.2}. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0808] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.4} is —C(O)—N(R^{sup.e})_{sub.2}, wherein each of R^{sup.e} is independently H or 3-10 membered heterocycle, wherein the 3-10 membered heterocycle is optionally substituted with one or more oxo. In some embodiments, R^{sup.4} is —C(O)—N(R^{sup.e})_{sub.2}, wherein each of R^{sup.e} is independently H or 3-6 membered heterocycle, wherein the 3-6 membered heterocycle is optionally substituted with one or more oxo. In some embodiments, R^{sup.4} is selected from the group consisting of

##STR00262##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0809] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.4} is —C(O)—N(R^{sup.e})_{sub.2}, wherein both R^{sup.e} together with the N to which they are attached are taken together to form a 3-10 membered heterocyclyl. In some embodiments R^{sup.4} is —C(O)—N(R^{sup.e})_{sub.2}, wherein both R^{sup.e} together with the N to which they are attached are taken together to form a 3-10 membered heterocyclyl, wherein the 3-10 membered heterocyclyl is optionally substituted with one or more oxo, —OH, NH_{sub.2}, or —S(O)_{sub.2}—R^{sup.a},

wherein R.sup.a is C.sub.1-6alkyl. In some embodiments R.sup.4 is —C(O)—N(R.sup.e).sub.2, wherein both R.sup.e together with the N to which they are attached are taken together to form a 3-7 membered heterocyclyl, wherein the 3-7 membered heterocyclyl is optionally substituted with one or more halo, oxo, —OH, NH.sub.2, NH—S(O).sub.2—R.sup.a, or —S(O).sub.2—R.sup.a, wherein R.sup.a is C.sub.1-3alkyl. In some embodiments R.sup.4 is —C(O)—N(R.sup.e).sub.2, wherein both R.sup.e together with the N to which they are attached are taken together to form a 3-7 membered heterocyclyl, wherein the 3-7 membered heterocyclyl is optionally substituted with one or more halo, oxo, —OH, NH.sub.2, NH—S(O).sub.2—R.sup.a, or —S(O).sub.2—R.sup.a, wherein R.sup.a is methyl. In some embodiments, R.sup.4 is selected from the group consisting of

##STR00263##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0810] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.5 is halo. In some embodiments, R.sup.5 is Cl, or F. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0811] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.5 is —CN. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0812] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.5 is C.sub.1-6alkyl, wherein the C.sub.1-6alkyl of R.sup.5 is optionally substituted with one or more halo, or —OH. In some embodiments, R.sup.5 is C.sub.1-3alkyl, wherein the C.sub.1-3alkyl of R.sup.5 is optionally substituted with one or more halo, or —OH. In some embodiments, R.sup.5 is C.sub.1-3alkyl, wherein the C.sub.1-3alkyl of R.sup.5 is optionally substituted with one or more fluoro, or —OH. In some embodiments, R.sup.5 is selected from the group consisting of methyl,

##STR00264##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0813] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.5 is 3-10 membered heterocyclyl. In some embodiments, R.sup.5 is 3-6 membered heterocyclyl. In some embodiments, R.sup.5 is

##STR00265##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0814] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.5 is C.sub.1-6alkoxy, wherein the C.sub.1-6alkoxy is optionally substituted with one or more halo. In some embodiments R.sup.5 is C.sub.1-3alkoxy, wherein the C.sub.1-3alkoxy is optionally substituted with one or more halo. In some embodiments, R.sup.5 is C.sub.1-3alkoxy, wherein the C.sub.1-3alkoxy is optionally substituted with one or more fluoro. In some embodiments, R.sup.5

is

##STR00266##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0815] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.6} and R^{sup.7} are each independently H or halo. In some embodiments, R^{sup.6} and R^{sup.7} are each independently H or fluoro. In some embodiments, each of R^{sup.6} and R^{sup.7} is H. In some embodiments, one of R^{sup.6} and R^{sup.7} is H and the other of R^{sup.6} and R^{sup.7} is fluoro. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0816] In some embodiments, provided herein is a compound of formula (I), (I-A), (I-A2), or (I-B), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt thereof, wherein the compound is a compound of formula (I-B3):

##STR00267## [0817] or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein: each R^{sup.5} is, independently at each occurrence, halo, —CN, 3-10 membered heterocyclyl, C_{sub.1-6}alkyl, or C_{sub.1-6}alkoxy, wherein the C_{sub.1-6}alkyl of R^{sup.5} is optionally substituted with one or more halo, or —OH and wherein the C_{sub.1-6}alkoxy is optionally substituted with one or more halo; and m, R^{sup.1}, R^{sup.2}, L^{sup.1}, L^{sup.3}, R^{sup.4}, R^{sup.6}, and R^{sup.7} are as defined for formula (I). In some variations, m, R^{sup.1}, R^{sup.2}, L^{sup.1}, L^{sup.3}, R^{sup.4}, R^{sup.6}, and R^{sup.7} of formula (I-B3) are as defined for a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0818] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B3), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.2} is H. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0819] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B3), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.2} is C_{sub.1-6}alkyl. In some embodiments, R^{sup.2} is methyl. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0820] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B3), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, m is an integer from 0 to 4. In some embodiments, m is 0 or 1. In some embodiments, m is 0. In some embodiments m is 1. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0821] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B3), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.1} is halo. In some embodiments R^{sup.1} is Cl. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0822] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B3), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.1} is —CN. In some variations, the embodiments provided herein also apply to a compound

of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0823] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B3), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.1} is C_{sub.1-6}alkylene optionally substituted with one or more C_{sub.1-6}alkyl, wherein the C_{sub.1-6}alkyl is optionally substituted with —OH or C_{sub.1-6}alkoxy. In some embodiments, L^{sup.1} is methylene. In some embodiments, L^{sup.1} is ethylene optionally substituted with one or more C_{sub.1-6}alkyl, wherein the C_{sub.1-6}alkyl is optionally substituted with one or more —OH or C_{sub.1-6}alkoxy. In some embodiments, L^{sup.1} is selected from the group consisting of
##STR00268##

wherein, for each L^{sup.1}, # denotes the point of attachment to —O— and ## denotes the point of attachment to the remainder of the molecule. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0824] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B3), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.3} is absent. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0825] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B3), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.3} is C_{sub.1-6}alkylene. In some embodiments, L^{sup.3} is C_{sub.1-3}alkylene. In some embodiments, L^{sup.3} is

##STR00269##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0826] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B3), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.3} is 3-10 membered heterocyclyl. In some embodiments, L^{sup.3} is 3-6 membered heterocyclyl. In some embodiments, L^{sup.3} is

##STR00270##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0827] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B3), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.4} is —S(O)_{sub.2}—R^{sup.a}, wherein R^{sup.a} is C_{sub.1-6}alkyl. In some embodiments, R^{sup.4} is —S(O)_{sub.2}—R^{sup.a}, wherein R^{sup.a} is C_{sub.1-6}alkyl. In some embodiments, R^{sup.4} is —S(O)_{sub.2}—R^{sup.a}, wherein R^{sup.a} is C_{sub.1-3}alkyl. In some embodiments, R^{sup.4} is

##STR00271##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0828] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B3), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.4} is —N(R^{sup.d})_{sub.2}, wherein each of R^{sup.d} is independently C_{sub.1-6} alkyl, or —S(O)_{sub.2}—R^{sup.a}, and wherein R^{sup.a} is C_{sub.1-6}alkyl. In some embodiments, R^{sup.4} is —N(R^{sup.d})_{sub.2}, wherein each of R^{sup.d} is independently C_{sub.1-3} alkyl, or —S(O)_{sub.2}—

R.sup.a, wherein R.sup.a is C.sub.1-3alkyl. In some embodiments, R.sup.4 is —N(R.sup.d).sub.2, wherein each of R.sup.d is methyl, or —S(O).sub.2—R.sup.a, wherein R.sup.a is methyl. In some embodiments, R.sup.4 is

##STR00272##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0829] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B3), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.4 is —C(O)—N(R.sup.e).sub.2, wherein both R.sup.e together with the N to which they are attached are taken together to form a 3-10 membered heterocyclyl, wherein the 3-10 membered heterocyclyl is optionally substituted with one or more oxo. In some embodiments R.sup.4 is —C(O)—N(R.sup.e).sub.2, wherein both R.sup.e together with the N to which they are attached are taken together to form a 3-7 membered heterocyclyl, wherein the 3-7 membered heterocyclyl is optionally substituted with one or more oxo. In some embodiments, R.sup.4 is

##STR00273##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0830] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-B), or (I-B3), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.5 is, independently at each occurrence selected from the group consisting of, halo and C.sub.1-6alkyl. In some embodiments, R.sup.5 is, independently at each occurrence selected from the group consisting of halo and C.sub.1-3alkyl. In some embodiments, R.sup.5 is, independently at each occurrence, selected from the group consisting of F and methyl. In some embodiments, each R.sup.5 is F. In some embodiments, each R.sup.5 is methyl. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0831] In some embodiments, provided herein is a compound of formula (I), (I-A), or (I-A2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt thereof, wherein the compound is a compound of formula (I-C):

##STR00274## [0832] or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein: m, p, R.sup.1, R.sup.2, R.sup.3, L.sup.1, L.sup.3, R.sup.4, and R.sup.5 are as defined for formula (I). In some variations m, p, R.sup.1, R.sup.2, R.sup.3, L.sup.1, L.sup.3, R.sup.4, and R.sup.5 of formula (I-C) are as defined for a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0833] In some embodiments, provided herein is a compound of formula (I), (I-A), (I-A2), or (I-C), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt thereof, wherein the compound is a compound of formula (I-C1):

##STR00275## [0834] or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein: m, R.sup.1, R.sup.2, R.sup.3, L.sup.1, L.sup.3, and R.sup.4, are as defined for formula (I). In some variations m, R.sup.1, R.sup.2, R.sup.3, L.sup.1, L.sup.3, and R.sup.4 of formula (I-C) are as defined for a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0835] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-C), or (I-C1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.2 is H. In some variations, the embodiments provided herein also apply to a compound of

formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0836] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-C), or (I-C1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, m is an integer from 0 to 4. In some embodiments, m is an integer from 0 to 2. In some embodiments m is 1. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0837] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-C), or (I-C1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.1} is halo. In some embodiments R^{sup.1} is Cl. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0838] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-C), or (I-C1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.1} is —CN. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0839] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-C), or (I-C1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.1} is C_{sub.1-6}alkylene. In some embodiments, L^{sup.1} is ethylene. In some embodiments, L^{sup.1} is

##STR00276##

wherein, #denotes the point of attachment to —O— and ## denotes the point of attachment to the remainder of the molecule. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0840] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-C), or (I-C1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.3} is absent. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0841] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-C), or (I-C1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing L^{sup.3} is C_{sub.1-6}alkylene. In some embodiments, L^{sup.3} is C_{sub.1-3}alkylene. In some embodiments, L^{sup.3} is

##STR00277##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0842] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-C), or (I-C1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.4} is —S(O)_{sub.2}—R^{sup.a}, wherein R^{sup.a} is C_{sub.1-6}alkyl. In some embodiments, R^{sup.4} is —S(O)_{sub.2}—R^{sup.a}, wherein R^{sup.a} is C_{sub.1-3}alkyl. In some embodiments, R^{sup.4} is

##STR00278##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0843] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-C), or (I-C1), or a

stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.4 is —S(O).sub.2—R.sup.a, wherein R.sup.a is C.sub.3-10cycloalkyl. R.sup.4 is —S(O).sub.2—R.sup.a, wherein R.sup.a is C.sub.3-10cycloalkyl, wherein the C.sub.3-10cycloalkyl of R.sup.a is optionally substituted with one or more C.sub.1-6alkyl, and wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH. In some embodiments, R.sup.4 is —S(O).sub.2—R.sup.a, wherein R.sup.a is C.sub.3-6cycloalkyl, wherein the C.sub.3-6cycloalkyl of R.sup.a is optionally substituted with one or more C.sub.1-6 alkyl, and wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH. In some embodiments, R.sup.4 is —S(O).sub.2—R.sup.a, wherein R.sup.a is C.sub.3-6cycloalkyl, wherein the C.sub.3-6cycloalkyl of R.sup.a is optionally substituted with one or more C.sub.1-3alkyl, wherein the C.sub.1-3alkyl is optionally substituted with one or more —OH. In some embodiments, R.sup.4 is

##STR00279##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0844] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-C), or (I-C1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.4 is —S(O).sub.2—R.sup.a, wherein R.sup.a is 3-10 membered heterocyclyl. In some embodiments R.sup.4 is —S(O).sub.2—R.sup.a, wherein R.sup.a is 3-10 membered heterocyclyl, wherein the 3-10 membered heterocyclyl of R.sup.a is optionally substituted with one or more C.sub.1-6alkyl. In some embodiments, R.sup.4 is —S(O).sub.2—R.sup.a, wherein R.sup.a is 3-6 membered heterocyclyl, wherein the 3-6 membered heterocyclyl of R.sup.a is optionally substituted with one or more C.sub.1-6alkyl. In some embodiments, wherein R.sup.4 is —S(O).sub.2—R.sup.a, wherein R.sup.a is 3-6 membered heterocyclyl, wherein the 3-6 membered heterocyclyl of R.sup.a is optionally substituted with one or more C.sub.1-3alkyl. In some embodiments, R.sup.4 is

##STR00280##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0845] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-C), or (I-C1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.4 is —C(O)—N(R.sup.e).sub.2, wherein each of R.sup.e is independently H or C.sub.1-6 alkyl. In some embodiments, R.sup.4 is —C(O)—N(R.sup.e).sub.2, wherein each of R.sup.e is independently H or C.sub.1-3 alkyl. In some embodiments, R.sup.4 is —C(O)—N(R.sup.e).sub.2, wherein each of R.sup.e is independently H or methyl. In some embodiments, R.sup.4 is —C(O)—NH.sub.2. In some embodiments, R.sup.4 is —C(O)—N(CH.sub.3).sub.2. In some embodiments, R.sup.4 is —C(O)—NH.sub.2. In some embodiments, R.sup.4 is —C(O)—NH(CH.sub.3). In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0846] In some embodiments, provided herein is a compound of formula (I), (I-A), or (I-A2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt thereof, wherein the compound is a compound of formula (I-C2):

##STR00281##

or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein: R.sup.5 is, halo, —CN, 3-10 membered heterocyclyl, C.sub.1-6alkyl, or C.sub.1-6alkoxy, wherein the C.sub.1-6alkyl of R.sup.5 is optionally substituted with one or more halo, or —OH and wherein the C.sub.1-6alkoxy is optionally substituted with one or more halo; m, p, R.sup.1, R.sup.2, R.sup.3, L.sup.1, L.sup.3, and R.sup.4, are as defined for formula (I). In some variations,

R.sup.5 is halo, —CN, 3-10 membered heterocyclyl, C.sub.1-6alkyl, or C.sub.1-6alkoxy, wherein the C.sub.1-6alkyl of R.sup.5 is optionally substituted with one or more halo, or —OH and wherein the C.sub.1-6alkoxy of R.sup.5 is optionally substituted with one or more halo; m, p, R.sup.1, R.sup.2, R.sup.3, L.sup.1, L.sup.3, and R.sup.4 of formula (I-C2) are as defined for a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0847] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-C), or (I-C2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.2 is H. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0848] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-C), or (I-C2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, m is an integer from 0 to 4. In some embodiments, m is an integer from 0 to 2. In some embodiments m is 1. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0849] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-C), or (I-C2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.1 is halo. In some embodiments R.sup.1 is Cl. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0850] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-C), or (I-C2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.1 is —CN. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0851] In some embodiments of a compound of formula (I'), (I-A), (I-A2), (I-C), or (I-C2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.1 is C.sub.1-6alkyl, wherein the C.sub.1-6alkyl of R.sup.1 is optionally substituted with one or more halo. In some embodiments, R.sup.1 is C.sub.1-3alkyl, wherein the C.sub.1-3alkyl of R.sup.1 is optionally substituted with one or more halo. In some embodiments, R.sup.1 is methyl, wherein the methyl of R.sup.1 is optionally substituted with one or more F. In some embodiments, R.sup.1 is

##STR00282##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0852] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-C), or (I-C2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.1 is C.sub.1-6alkylene. In some embodiments, L.sup.1 is ethylene. In some embodiments, L.sup.1 is

##STR00283##

wherein, # denotes the point of attachment to —O— and ## denotes the point of attachment to the remainder of the molecule. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0853] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-C), or (I-C2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.3 is absent. In some variations, the embodiments provided herein also apply to a compound

of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0854] In some embodiments of a compound formula (I'), (I-A), (I-A2), (I-C), or (I-C2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.3} is C_{sub.1-6}alkylene, wherein the C_{sub.1-6}alkylene of L^{sup.3} is optionally substituted with one or more —OH, or C_{sub.1-6}alkyl. In some embodiments, L^{sup.3} is C_{sub.1-3}alkylene, wherein the C_{sub.1-6}alkylene of L^{sup.3} is optionally substituted with one or more —OH, or C_{sub.1-6}alkyl. In some embodiments, L^{sup.3} is selected from the group consisting of

##STR00284##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0855] In some embodiments of a compound formula (II), such as a compound of formula (I'), (I-A), (I-A2), (I-C), or (I-C2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.3} is C_{sub.1-6}alkylene, wherein the C_{sub.1-6}alkylene of L^{sup.3} is optionally substituted with one or more —OH, or C_{sub.1-6}alkyl, wherein the C_{sub.1-6}alkyl is optionally substituted with one or more —OH. In some embodiments, L^{sup.3} is C_{sub.1-3}alkylene, wherein the C_{sub.1-6}alkylene of L^{sup.3} is optionally substituted with one or more —OH, or C_{sub.1-3}alkyl, wherein the C_{sub.1-3}alkyl is optionally substituted with one or more —OH. In some embodiments, L^{sup.3} is selected from the group consisting of

##STR00285##

[0856] In some embodiments of a compound of formula (I'), (I), (I-A), (I-A2), (I-B), or (I-B1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.3} is C_{sub.3-10}cycloalkyl, wherein the C_{sub.3-10}cycloalkyl of L^{sup.3} is optionally substituted with one or more —OH. In some embodiments, L^{sup.3} is C_{sub.3-8}cycloalkyl, wherein the C_{sub.3-8}cycloalkyl of L^{sup.3} is optionally substituted with one or more —OH. In some embodiments, L^{sup.3} is

##STR00286##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0857] In some embodiments of a compound formula (II), such as a compound of formula (I'), (I-A), (I-A2), (I-C), or (I-C2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.3} is 3-10 membered heterocyclyl, optionally substituted with one or more —OH or C_{sub.1-6}alkyl. In some embodiments, L^{sup.3} is 3-6 membered heterocyclyl, optionally substituted with one or more —OH or C_{sub.1-3}alkyl. In some embodiments, L^{sup.3} is

##STR00287##

[0858] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-C), or (I-C2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.4} is —S(O)_{sub.2}—R^{sup.a}, wherein R^{sup.a} is C_{sub.1-6}alkyl. In some embodiments, R^{sup.4} is —S(O)_{sub.2}—R^{sup.a}, wherein R^{sup.a} is C_{sub.1-3}alkyl. In some embodiments, R^{sup.4} is

##STR00288##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0859] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-C), or (I-C2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.4} is —(CH_{sub.2})_{sub.q}OH, wherein q is an integer from 0-6. R^{sup.4} is —(CH_{sub.2})_{sub.q}OH, wherein q is an integer from 0-2. In some embodiments, R^{sup.4} is —OH. In

some embodiments, R^{sup.4} is selected from the group consisting of

##STR00289##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0860] In some embodiments of a compound of formula (I'), (I-A), (I-A2), (I-C), or (I-C2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.4} is —C(O)—C_{sub.1-6}alkyl. In some embodiments, R^{sup.4} is —C(O)—C_{sub.1-3}alkyl. In some embodiments, R^{sup.4} is —C(O)CH_{sub.3}. In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0861] In some embodiments of a compound of formula (I), (I-A), (I-A2), (I-C), or (I-C2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.5} is C_{sub.1-6}alkyl. In some embodiments, R^{sup.5} is C_{sub.1-3}alkyl. In some embodiments, R^{sup.5} is methyl. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0862] In some embodiments of a compound of formula (I'), (I-A), (I-A2), (I-C), or (I-C2) or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.5} is C_{sub.1-6}alkyl, wherein the C_{sub.1-6}alkyl of R^{sup.5} is optionally substituted with one or more halo. In some embodiments, R^{sup.5} is C_{sub.1-3}alkyl, wherein the C_{sub.1-3}alkyl of R^{sup.5} is optionally substituted with one or more halo. In some embodiments, R^{sup.5} is C_{sub.1-3}alkyl, wherein the C_{sub.1-3}alkyl of R^{sup.5} is optionally substituted with one or more fluoro. In some embodiments, R^{sup.5} is

##STR00290##

In some embodiments, R^{sup.5} is methyl. In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0863] In some embodiments of a compound of formula (I'), (I-A), (I-A2), (I-C), or (I-C2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.5} is halo. In some embodiments, R^{sup.5} is C_{sub.1}, or F. In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0864] In some embodiments, provided herein is a compound of formula (I), (I-A), or (I-A2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt thereof, wherein the compound is a compound of formula (I-D):

##STR00291##

or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein: m, p, R^{sup.1}, R^{sup.2}, R^{sup.3}, L^{sup.1}, L^{sup.3}, and R^{sup.4}, are as defined for formula (I). In some variations, m, p, R^{sup.1}, R^{sup.2}, R^{sup.3}, L^{sup.1}, L^{sup.3}, and R^{sup.4} of formula (I-D) are as defined for a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0865] In some embodiments of a compound of formula (I), (I-A), (I-A2), or (I-D), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.2} is H. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0866] In some embodiments of a compound of formula (II), such as a compound of formula (I), (I-A), (I-A2), or (I-D), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of

any of the foregoing, R.sup.2 is C.sub.1-6alkyl optionally substituted with one or more deuterium. In some embodiments, R.sup.2 is C.sub.1-3alkyl optionally substituted with one or more deuterium. In some embodiments, R.sup.2 is methyl optionally substituted with one or more deuterium. In some embodiments, R.sup.2 is

##STR00292##

[0867] In some embodiments of a compound of formula (I), (I-A), (I-A2), or (I-D), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, m is an integer from 0 to 4. In some embodiments, m is an integer from 0 to 2. In some embodiments m is 1. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0868] In some embodiments of a compound of formula (I), (I-A), (I-A2), or (I-D), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.1 is halo. In some embodiments R.sup.1 is Cl. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0869] In some embodiments of a compound of formula (II), such as a compound of formula (I), (I-A), (I-A2), or (I-D), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.1 is Cl, or I.

[0870] In some embodiments of a compound of formula (I), (I-A), (I-A2), or (I-D), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.1 is —CN. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0871] In some embodiments of a compound of formula (I), (I-A), (I-A2), or (I-D), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.1 is C.sub.1-6alkyl, wherein the C.sub.1-6alkyl of R.sup.1 is optionally substituted with one or more halo. In some embodiments, R.sup.1 is C.sub.1-3alkyl, wherein the C.sub.1-3alkyl of R.sup.1 is optionally substituted with one or more halo. In some embodiments, R.sup.1 is C.sub.1-3alkyl, wherein the C.sub.1-3alkyl of R.sup.1 is optionally substituted with one or more F. In some embodiments, R.sup.1 is methyl, wherein the methyl of R.sup.1 is optionally substituted with one or more F. In some embodiments, R.sup.1 is

##STR00293##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0872] In some embodiments of a compound of formula (I), (I-A), (I-A2), or (I-D), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.1 is C.sub.1-6alkylene optionally substituted with one or more C.sub.1-6alkyl. In some embodiments, L.sup.1 is ethylene optionally substituted with one or more C.sub.1-6alkyl. In some embodiments, L.sup.1 is selected from the group consisting of

##STR00294##

wherein, for each L.sup.1, # denotes the point of attachment to —O— and ## denotes the point of attachment to the remainder of the molecule. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0873] In some embodiments of a compound of formula (I), (I-A), (I-A2), or (I-D), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.3 is absent. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt

of any of the foregoing, or any variation or embodiment thereof.

[0874] In some embodiments of a compound of formula (I), (I-A), (I-A2), or (I-D), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.3 is —O—. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0875] In some embodiments of a compound of formula (I), (I-A), (I-A2), or (I-D), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.3 is C.sub.1-6alkylene. In some embodiments, L.sup.3 is C.sub.1-3alkylene. In some embodiments, L.sup.3 is

##STR00295##

In some variations, the embodiments provided herein also apply to a compound of formula (I') or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0876] In some embodiments of a compound of formula (I'), (I-A), (I-A2), or (I-D), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.3 is C.sub.1-6alkylene, wherein the C.sub.1-6alkylene of L.sup.3 is optionally substituted with one or more —OH, or C.sub.1-6alkyl. In some embodiments, L.sup.3 is C.sub.1-3alkylene, wherein the C.sub.1-6alkylene of L.sup.3 is optionally substituted with one or more —OH, or C.sub.1-6alkyl. In some embodiments, L.sup.3 is selected from the group consisting of

##STR00296##

[0877] In some embodiments of a compound of formula (I), (I-A), (I-A2), or (I-D), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.3 is C.sub.3-10cycloalkyl. In some embodiments, L.sup.3 is C.sub.3-8cycloalkyl. In some embodiments, L.sup.3 is

##STR00297##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0878] In some embodiments of a compound of formula (I'), (I-A), (I-A2), or (I-D), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.3 is C.sub.3-10cycloalkyl, wherein the C.sub.3-10cycloalkyl of L.sup.3 is optionally substituted with one or more —OH. In some embodiments, L.sup.3 is C.sub.3-8cycloalkyl, wherein the C.sub.3-8cycloalkyl of L.sup.3 is optionally substituted with one or more —OH. In some embodiments, L.sup.3 is selected from the group consisting of

##STR00298##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0879] In some embodiments of a compound of formula (II), such as a compound of formula (I'), (I-A), (I-A2), or (I-D), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.3 is C.sub.3-10cycloalkyl, wherein the C.sub.3-10cycloalkyl of L.sup.3 is optionally substituted with one or more —OH. In some embodiments, L.sup.3 is C.sub.3-8cycloalkyl, wherein the C.sub.3-8-cycloalkyl of L.sup.3 is optionally substituted with one or more —OH. In some embodiments, L.sup.3 is selected from the group consisting of

##STR00299##

[0880] In some embodiments of a compound of formula (I'), (I-A), (I-A2), or (I-D) or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.3 is 3-10 membered heterocyclyl. In some embodiments, L.sup.3 is 3-6 membered heterocyclyl. In some embodiments, L.sup.3 is selected from the group consisting of

##STR00300##

some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0881] In some embodiments of a compound of formula (II), such as a compound of (I'), (I-A), (I-A2), or (I-D) or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.3} is 3-10 membered heterocyclyl. In some embodiments, L^{sup.3} is 3-6 membered heterocyclyl. In some embodiments, L^{sup.3} is selected from the group consisting of

##STR00301##

[0882] In some embodiments of a compound of formula (I), (I-A), (I-A2), or (I-D), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.4} is —S(O)₂—R^{sup.a}, wherein R^{sup.a} is C₁₋₆alkyl. In some embodiments, R^{sup.4} is —S(O)₂—R^{sup.a}, wherein R^{sup.a} is C₁₋₃alkyl. In some embodiments, R^{sup.4} is

##STR00302##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0883] In some embodiments of a compound of formula (I), (I-A), (I-A2), or (I-D), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.4} is —C(O)—N(R^{sup.e})₂, wherein each of R^{sup.e} is H. In some embodiments, R^{sup.4} is —C(O)—NH₂. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0884] In some embodiments of a compound of formula (I), (I-A), (I-A2), or (I-D), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.4} is 3-10 membered heterocyclyl optionally substituted with one or more C₁₋₆ alkyl. In some embodiments, R^{sup.4} is 3-6 membered heterocyclyl optionally substituted with one or more C₁₋₆ alkyl. In some embodiments, R^{sup.4} is 3-6 membered heterocyclyl optionally substituted with one or more C₁₋₃ alkyl. In some embodiments, R^{sup.4} is 3-6 membered heterocyclyl optionally substituted with one or more methyl. In some embodiments, R^{sup.4} is

##STR00303##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0885] In some embodiments of a compound of formula (II), such as a compound of formula (I), (I-A), (I-A2), or (I-D), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.4} is

##STR00304##

[0886] In some embodiments of a compound of formula (I'), (I-A), (I-A2), or (I-D), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.4} is 3-10 membered heterocyclyl optionally substituted with one or more C₁₋₆ alkyl. In some embodiments, R^{sup.4} is 3-6 membered heterocyclyl optionally substituted with one or more C₁₋₆ alkyl. In some embodiments, R^{sup.4} is 3-6 membered heterocyclyl optionally substituted with one or more C₁₋₃ alkyl. In some embodiments, R^{sup.4} is 3-6 membered heterocyclyl optionally substituted with one or more methyl. In some embodiments, R^{sup.4} is selected from the group consisting of

##STR00305##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or

any variation or embodiment thereof.

[0887] In some embodiments of a compound of formula (I'), (I-A), (I-A2), or (I-D) or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.4} is —(CH_{sub.2})_{sub.q}OH, wherein q is an integer from 0-6. R^{sup.4} is —(CH_{sub.2})_{sub.q}OH, wherein q is an integer from 0-2. In some embodiments, R^{sup.4} is —OH. In some embodiments, R^{sup.4} is selected from the group consisting of

##STR00306##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0888] In some embodiments of a compound of formula (I'), (I-A), (I-A2), or (I-D) or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.4} is —N(R^{sup.d})_{sub.2}, wherein each of R^{sup.d} is independently H. In some embodiments, R^{sup.4} is —NH_{sub.2}. In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0889] In some embodiments of a compound of formula (I'), (I-A), (I-A2), or (I-D) or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.4} is —P(O)(C_{sub.1-6}alkyl)_{sub.2}. In some embodiments, R^{sup.4} is —P(O)(C_{sub.1-3}alkyl)_{sub.2}. In some embodiments, R^{sup.4} is —P(O)(CH_{sub.3})_{sub.2}. In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0890] In some embodiments, provided herein is a compound of formula (I), (I-A) or (I-A2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt thereof, wherein the compound is a compound of formula (I-E):

##STR00307##

or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein: ring A is 5-10 membered heterocyclyl or a 5-10 membered heteroaryl, wherein the 5-10 membered heterocyclyl of ring A is optionally substituted with one or more R^{sup.b}, and wherein the 5-10 membered heteroaryl of ring A is optionally substituted with one or more R^{sup.e}; and m, p, R^{sup.1}, R^{sup.2}, R^{sup.3}, L^{sup.1}, X^{sup.2}, R^{sup.b}, R^{sup.c}, R^{sup.6}, and R^{sup.7} are as defined for formula (I). In some variations, ring A is 5-10 membered heterocyclyl or a 5-10 membered heteroaryl, wherein the 5-10 membered heterocyclyl of ring A is optionally substituted with one or more R^{sup.b}, and wherein the 5-10 membered heteroaryl of ring A is optionally substituted with one or more R^{sup.c}; and m, p, R^{sup.1}, R^{sup.2}, R^{sup.3}, L_{sub.1}, X^{sup.2}, R^{sup.b}, R^{sup.c}, R^{sup.6}, and R^{sup.7} of formula (I-E) are as defined for a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0891] In some embodiments, provided herein is a compound of formula (I), (I-A) or (I-A2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt thereof, wherein the compound is a compound of formula (I-E1):

##STR00308##

or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein: ring A is 5-10 membered heterocyclyl or a 5-10 membered heteroaryl, wherein the 5-10 membered heterocyclyl of ring A is optionally substituted with one or more R^{sup.b}, and wherein the 5-10 membered heteroaryl of ring A is optionally substituted with one or more R^{sup.c}; and m, p, R^{sup.1}, R^{sup.2}, R^{sup.3}, L^{sup.1}, X^{sup.1}, R^{sup.b}, R^{sup.c}, R^{sup.6}, and R^{sup.7} are as defined for formula (I). In some variations, ring A is 5-10 membered heterocyclyl or a 5-10 membered heteroaryl, wherein the 5-10 membered heterocyclyl of ring A is optionally substituted

with one or more R.sup.b, and wherein the 5-10 membered heteroaryl of ring A is optionally substituted with one or more R.sup.c; and m, p, R.sup.1, R.sup.2, R.sup.3, L.sup.1, X.sup.1, R.sup.b, R.sup.c, R.sup.6, and R.sup.7 of formula (I-E1) are as defined for a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0892] In some embodiments, provided herein is a compound of formula (I'), (I-A), (I-A2), or (I-E) or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt thereof, wherein the compound is a compound of formula (I-E2):

##STR00309##

or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein: r is an integer from 0-1; Y.sup.1 and Y.sup.2 are each independently C or N, optionally substituted by one or more H or R.sup.b; and m, R.sup.1, R.sup.2, L.sup.1, X.sup.2, and R.sup.b are as defined for formula (I'), or (II); and wherein the dashed line represents a single or double bond.

[0893] In some embodiments, provided herein is a compound of formula (I'), (I-A), (I-A2), or (I-E) or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt thereof, wherein the compound is a compound of formula (I-E3):

##STR00310##

or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein: r is an integer from 0-1; Y.sup.1 and Y.sup.2 are each independently C or N, optionally substituted by one or more H or R.sup.b; R.sup.b1 is OH; R.sup.b2 is H, C.sub.1-6alkyl, or C.sub.3-10cycloalkyl, wherein the C.sub.1-6alkyl of R.sup.b2 is optionally substituted with one or more OH; and m, R.sup.1, R.sup.2, L.sup.1, X.sup.2, and R.sup.b are as defined for formula (I'), or (II); and wherein the dashed line represents a single or double bond.

[0894] In some embodiments of a compound of formula (I'), (I-A) (I-A2), or (I-E3) or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, the moiety represented by

##STR00311##

has a stereochemical configuration represented by

##STR00312##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0895] In some embodiments of a compound of formula (I'), (I-A) (I-A2), or (I-E3) or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, the moiety represented by

##STR00313##

has a stereochemical configuration represented by

##STR00314##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0896] In some embodiments, provided herein is a compound of formula (I'), (I-A), (I-A2), or (I-E) or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt thereof, wherein the compound is a compound of formula (I-E4):

##STR00315##

or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein: r is an integer from 0-1; Y.sup.3, Y.sup.4, and Y.sup.5 are each independently C or N, optionally substituted by one or more H or R.sup.e; and m, R.sup.1, R.sup.2, L.sup.1, X.sup.2, and R.sup.e are as defined for formula (I'), or (II); and wherein the dashed line represents a single or

double bond.

[0897] In some embodiments, provided herein is a compound of formula (I'), (I-A), (I-A2), or (I-E) or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt thereof, wherein the compound is a compound of formula (I-E5):

##STR00316##

or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein: r is an integer from 0-1; Y^{sup.3}, Y^{sup.4} and Y^{sup.5} are each independently C or N, optionally substituted by one or more H or R^{sup.c}; R^{sup.c1} is OH; R^{sup.2} is H, C_{sub.1-6}alkyl, or C_{sub.3-10}cycloalkyl, wherein the C_{sub.1-6}alkyl of R^{sup.c2} is optionally substituted with one or more OH; and m, R^{sup.1}, R^{sup.2}, L^{sup.1}, X^{sup.2}, and R^{sup.e} are as defined for formula (I'), or (II); and wherein the dashed line represents a single or double bond.

[0898] In some embodiments of a compound of formula (I'), (I-A) (I-A2), or (I-E5) or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, the moiety represented by

##STR00317##

has a stereochemical configuration represented by

##STR00318##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0899] In some embodiments of a compound of formula (I'), (I-A) (I-A2), or (I-E) or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, the moiety represented by

##STR00319##

has a stereochemical configuration represented by

##STR00320##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0900] In some embodiments, provided herein is a compound of formula (I), (I-A) or (I-A2), (I-E), or (I-E1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt thereof, wherein the compound is a compound of formula (I-F):

##STR00321##

or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein: ring A is 5-10 membered heterocyclyl or a 5-10 membered heteroaryl, wherein the 5-10 membered heterocyclyl of ring A is optionally substituted with one or more R^{sup.b}, and wherein the 5-10 membered heteroaryl of ring A is optionally substituted with one or more R^{sup.c}; and m, p, R^{sup.1}, R^{sup.2}, R^{sup.3}, L^{sup.1}, R^{sup.5}, R^{sup.b}, and R^{sup.c}, are as defined for formula (I). In some variations, ring A is 5-10 membered heterocyclyl or a 5-10 membered heteroaryl, wherein the 5-10 membered heterocyclyl of ring A is optionally substituted with one or more R^{sup.b}, and wherein the 5-10 membered heteroaryl of ring A is optionally substituted with one or more R^{sup.c}; and m, p, R^{sup.1}, R^{sup.2}, R^{sup.3}, L^{sup.1}, R^{sup.5}, R^{sup.b}, and R^{sup.c} of formula (I-F) are as defined for a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0901] In some embodiments, provided herein is a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), or (I-F), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt thereof, wherein the compound is a compound of formula (I-F1):

##STR00322##

or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein: ring A is 5-10 membered heterocyclyl or a 5-10 membered heteroaryl, wherein the 5-10

membered heterocyclyl of ring A is optionally substituted with one or more R.sup.b, and wherein the 5-10 membered heteroaryl of ring A is optionally substituted with one or more R.sup.c; and m, R.sup.1, R.sup.2, L.sup.1, R.sup.b, and R.sup.c, are as defined for formula (I). In some variations, ring A is 5-10 membered heterocyclyl or a 5-10 membered heteroaryl, wherein the 5-10 membered heterocyclyl of ring A is optionally substituted with one or more R.sup.b, and wherein the 5-10 membered heteroaryl of ring A is optionally substituted with one or more R.sup.c; and m, R.sup.1, R.sup.2, L.sup.1, R.sup.b, and R.sup.c of formula (I-F1) are as defined for a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0902] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.2 is H. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof

[0903] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.2 is C.sub.1-6alkyl, wherein the C.sub.1-6alkyl of R.sup.2 is optionally substituted with one or more —OH, —NH.sub.2, or C.sub.1-6alkoxy. In some embodiments, R.sup.2 is methyl. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0904] In some embodiments of a compound of formula (II), such as a compound of formula (I'), (I), (I-A) or (I-A2), (I-E), (I-E1), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.2 is C.sub.1-6alkyl, wherein the C.sub.1-6alkyl of R.sup.2 is optionally substituted with one or more deuterium, —OH, —NH.sub.2, or C.sub.1-6alkoxy. In some embodiments, R.sup.2 is methyl optionally substituted with one or more deuterium. In some embodiments, R.sup.2 is

##STR00323##

[0905] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.2 is C.sub.3-10cycloalkyl, wherein the C.sub.3-10cycloalkyl of R.sup.2 is optionally substituted with one or more —OH. In some embodiments, R.sup.2 is C.sub.3-6cycloalkyl, wherein the C.sub.3-6cycloalkyl of R.sup.2 is optionally substituted with one or more —OH. In some embodiments, R.sup.2 is

##STR00324##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0906] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, m is an integer from 0 to 4. In some embodiments, m is an integer from 0 to 2. In some embodiments, m is 0 or 1. In some embodiments, m is 0. In some embodiments m is 1. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0907] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.1 is halo. In some embodiments R.sup.1 is Cl. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or

embodiment thereof.

[0908] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.1} is —CN. In some variations, the embodiments provided herein also apply to a compound of formula (I') or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof. In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.1} is C_{sub.1-6}alkylene, wherein the C_{sub.1-6}alkylene of L^{sup.1} is optionally substituted with one or more C_{sub.1-6}alkyl. In some embodiments, L^{sup.1} is ethylene, wherein the ethylene of L^{sup.1} is optionally substituted with one or more C_{sub.1-6}alkyl. In some embodiments, L^{sup.1} is selected from the group consisting of

##STR00325##

wherein, for each L^{sup.1}, # denotes the point of attachment to —O— and ## denotes the point of attachment to the remainder of the molecule. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0909] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more of R^{sup.b}, wherein R^{sup.b} is oxo. In some embodiments, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more of R^{sup.b}, wherein R^{sup.b} is oxo. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0910] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is selected from the group consisting of

##STR00326##

wherein ## represent a point of attachment to the remainder of the molecule and the dashed line represents a single or double bond. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0911] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E1), (I-E2), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein each R^{sup.b} is independently oxo, or C_{sub.1-6}alkyl. In some embodiments, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein each R^{sup.b} is independently oxo, or C_{sub.1-6}alkyl, wherein the C_{sub.1-6}alkyl of R^{sup.b} is optionally substituted with one or more more halo, OH, or —S(O)_{sub.2}—C_{sub.1-6}alkyl. In some embodiments, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein wherein each R^{sup.b} is independently oxo, or C_{sub.1-3}alkyl, wherein the C_{sub.1-3}alkyl of R^{sup.b} is optionally substituted with one or more more halo, —OH, or —S(O)_{sub.2}—C_{sub.1-3}alkyl. In some embodiments, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein each R^{sup.b} is independently selected from the group consisting of oxo,

##STR00327##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or

(II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0912] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E1), (I-E2), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is selected from the group consisting of

##STR00328##

wherein ## represent a point of attachment to the remainder of the molecule and the dashed line represents a single or double bond. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0913] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E1), (I-E2), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein one or more of R^{sup.b} is —C(O)—NH(C_{sub.1-6}alkyl). In some embodiments, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein one or more of R^{sup.b} is —C(O)—NH(C_{sub.1-3}alkyl). In some embodiments, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein one or more of R^{sup.b} is

##STR00329##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0914] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E1), (I-E2), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is selected from the group consisting of

##STR00330##

wherein ## represent a point of attachment to the remainder of the molecule and the dashed line represents a single or double bond. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0915] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E1), (I-E2), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein one or more of R^{sup.b} is —C(O)—C_{sub.1-6}alkyl. In some embodiments, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein one or more of R^{sup.b} is —C(O)—C_{sub.1-3}alkyl. In some embodiments, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein one or more of R^{sup.b} is

##STR00331##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0916] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E1), (I-E2), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is selected from the group consisting of

##STR00332##

wherein ## represent a point of attachment to the remainder of the molecule and the dashed line represents a single or double bond. In some variations, the embodiments provided herein also apply

to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0917] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E1), (I-E2), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein one or more of R^{sup.b} is —S(O)_{sub.2}—R^{sup.a}, wherein R^{sup.a} is C_{sub.1-6}alkyl. In some embodiments, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein one or more of R^{sup.b} is —S(O)_{sub.2}—R^{sup.a}, wherein R^{sup.a} is C_{sub.1-3}alkyl. In some embodiments, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein one or more of R^{sup.b} is —S(O)_{sub.2}—R^{sup.a}, wherein R^{sup.a} is methyl. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0918] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E1), (I-E2), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is selected from the group consisting of
##STR00333##

wherein ## represent a point of attachment to the remainder of the molecule and the dashed line represents a single or double bond. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0919] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E1), (I-E2), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein R^{sup.b} is C_{sub.3-10}cycloalkyl, wherein the C_{sub.3-10}-cycloalkyl of R^{sup.b} is optionally substituted with one or more —OH. In some embodiments, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein R^{sup.b} is C_{sub.3-6}cycloalkyl, wherein the C_{sub.3-6}cycloalkyl of R^{sup.b} is optionally substituted with one or more —OH. In some embodiments, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein one or more of R^{sup.b} is
##STR00334##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0920] In some embodiments of a compound of formula (I'), (I-A) or (I-A2), (I-E1), (I-E2), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein R^{sup.b} is C_{sub.3-10}cycloalkyl, wherein the C_{sub.3-10}-cycloalkyl of R^{sup.b} is optionally substituted with one or more —OH or C_{sub.1-6}alkyl. In some embodiments, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein R^{sup.b} is C_{sub.3-6}cycloalkyl, wherein the C_{sub.3-6}cycloalkyl of R^{sup.b} is optionally substituted with one or more —OH or C_{sub.1-3}alkyl. In some embodiments, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein one or more of R^{sup.b} is selected from the group consisting of
##STR00335##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or

any variation or embodiment thereof

[0921] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E1), (I-E2), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is

##STR00336##

wherein ## represent a point of attachment to the remainder of the molecule and the dashed line represents a single or double bond. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0922] In some embodiments of a compound of formula (I'), (I-A) or (I-A2), (I-E1), (I-E2), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is selected from the group consisting of

##STR00337##

wherein ## represent a point of attachment to the remainder of the molecule and the dashed line represents a single or double bond. In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0923] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E1), (I-E2), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein R^{sup.b} is 3-10 membered heterocyclyl. In some embodiments, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein R^{sup.b} is 3-6 membered heterocyclyl. In some embodiments, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein one or more of R^{sup.b} is

##STR00338##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0924] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E1), (I-E2), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is

##STR00339##

wherein ## represent a point of attachment to the remainder of the molecule and the dashed line represents a single or double bond. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0925] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E1), (I-E2), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is a 5-10 membered heteroaryl. In some embodiments, ring A is a 5-6 membered heteroaryl. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0926] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E1), (I-E2), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is selected from the group consisting of

##STR00340##

wherein ## represent a point of attachment to the remainder of the molecule and the dashed line represents a single or double bond. In some variations, the embodiments provided herein also apply

to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0927] In some embodiments of a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is a 5-10 membered heteroaryl, wherein the 5-10 membered heteroaryl is optionally substituted with one or more R^{sup.c}, wherein one or more R^{sup.e} is C_{sub.1-6}alkyl. In some embodiments ring A is a 5-6 membered heteroaryl, wherein the 5-6 membered heteroaryl is optionally substituted with one or more R^{sup.c}, wherein one or more R^{sup.e} is C_{sub.1-3}alkyl. In some embodiments, ring A is a 5-6 membered heteroaryl, wherein the 5-6 membered heteroaryl is optionally substituted with one or more R^{sup.e}, wherein one or more R^{sup.e} is methyl. In some embodiments, ring A is a 5-6 membered heteroaryl, wherein the 5-6 membered heteroaryl is optionally substituted with one or more R^{sup.e}, wherein one or more R^{sup.e} is isopropyl. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0928] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E1), (I-E2), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is selected from the group consisting of
##STR00341##

wherein ## represent a point of attachment to the remainder of the molecule and the dashed line represents a single or double bond. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0929] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E1), (I-E2), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is a 5-10 membered heteroaryl, wherein the 5-10 membered heteroaryl is optionally substituted with one or more R^{sup.c}, wherein one or more R^{sup.e} is C_{sub.1-6}alkyl, wherein the C_{sub.1-6} alkyl of R^{sup.e} is optionally substituted with one or more —S(O)_{sub.2}—R^{sup.a}, wherein R^{sup.a} is C_{sub.1-6}alkyl. In some embodiments, ring A is a 5-6 membered heteroaryl, wherein the 5-6 membered heteroaryl is optionally substituted with one or more R^{sup.c}, wherein one or more R^{sup.e} is C_{sub.1-3}alkyl, wherein the C_{sub.1-3}alkyl of R^{sup.e} is optionally substituted with one or more —S(O)_{sub.2}—R^{sup.a}, wherein R^{sup.a} is C_{sub.1-3}alkyl. In some embodiments, ring A is a 5-6 membered heteroaryl, wherein the 5-6 membered heteroaryl is optionally substituted with one or more R^{sup.c}, wherein one or more R^{sup.e} is C_{sub.1-3}alkyl, wherein the C_{sub.1-3}alkyl of R^{sup.e} is optionally substituted with one or more —S(O)_{sub.2}—R^{sup.a}, wherein R^{sup.a} is methyl. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0930] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E1), (I-E2), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is
##STR00342##

wherein ## represent a point of attachment to the remainder of the molecule and the dashed line represents a single or double bond. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0931] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E1), (I-E2), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is a 5-10 membered heteroaryl, wherein the 5-10 membered heteroaryl is optionally substituted with one or more R^{sup.c}, wherein one or more R^{sup.e} is C_{sub.3-10}cycloalkyl, wherein the C_{sub.3-10}cycloalkyl of R^{sup.e} is optionally substituted with one or

more —OH. In some embodiments, ring A is a 5-10 membered heteroaryl, wherein the 5-6 membered heteroaryl is optionally substituted with one or more R^{sup.c}, wherein one or more R^{sup.e} is C_{sub}.3-6cycloalkyl, wherein the C_{sub}.3-6cycloalkyl of R^{sup.e} is optionally substituted with one or more —OH. In some embodiments, ring A is a 5-10 membered heteroaryl, wherein the 5-6 membered heteroaryl is optionally substituted with one or more R^{sup.c}, wherein one or more R^{sup.e} is

##STR00343##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0932] In some embodiments of a compound of formula (I'), (I-A) or (I-A2), (I-E1), (I-E2), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is a 5-10 membered heteroaryl, wherein the 5-10 membered heteroaryl is optionally substituted with one or more R^{sup.c}, wherein one or more R^{sup.e} is C_{sub}.3-10cycloalkyl, wherein the C_{sub}.3-10cycloalkyl of R^{sup.e} is optionally substituted with one or more —OH, or C_{sub}.1-6alkyl. In some embodiments, ring A is a 5-10 membered heteroaryl, wherein the 5-6 membered heteroaryl is optionally substituted with one or more R^{sup.c}, wherein one or more R^{sup.e} is C_{sub}.3-6cycloalkyl, wherein the C_{sub}.3-6cycloalkyl of R^{sup.e} is optionally substituted with one or more —OH, or C_{sub}.1-3alkyl. In some embodiments, ring A is a 5-10 membered heteroaryl, wherein the 5-6 membered heteroaryl is optionally substituted with one or more R^{sup.c}, wherein one or more R^{sup.e} is selected from the group consisting of

##STR00344##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0933] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E1), (I-E2), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is

##STR00345##

wherein ## represent a point of attachment to the remainder of the molecule and the dashed line represents a single or double bond. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0934] In some embodiments of a compound of formula (I'), (I-A) or (I-A2), (I-E1), (I-E2), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is selected from the group consisting of

##STR00346##

wherein ## represent a point of attachment to the remainder of the molecule and the dashed line represents a single or double bond. In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0935] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E1), (I-E2), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is a 5-10 membered heteroaryl, wherein the 5-10 membered heteroaryl is optionally substituted with one or more R^{sup.b}, wherein R^{sup.b} is 3-10 membered heterocyclyl. In some embodiments, ring A is a 5-10 membered heteroaryl, wherein the 5-10 membered heteroaryl is optionally substituted with one or more R^{sup.b}, wherein R^{sup.b} is 3-6 membered heterocyclyl. In some embodiments, ring A is a 5-10 membered heteroaryl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein one or more of R^{sup.b} is

##STR00347##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0936] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E1), (I-E2), (I-F), or (I-F1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is

##STR00348##

wherein ## represent a point of attachment to the remainder of the molecule and the dashed line represents a single or double bond. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0937] In some embodiments, provided herein is a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), or (I-F), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt thereof, wherein the compound is a compound of formula (I-F2):

##STR00349##

or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein: R^{sup.5} is halo, —CN, 3-10 membered heterocyclyl, C_{sub.1-6}alkyl, or C_{sub.1-6}alkoxy, wherein the C_{sub.1-6}alkyl of R^{sup.5} is optionally substituted with one or more halo, or —OH and wherein the C_{sub.1-6}alkoxy is optionally substituted with one or more halo; ring A is 5-10 membered heterocyclyl or a 5-10 membered heteroaryl, wherein the 5-10 membered heterocyclyl of ring A is optionally substituted with one or more R^{sup.b}, and wherein the 5-10 membered heteroaryl of ring A is optionally substituted with one or more R^{sup.c}; and m, R^{sup.1}, R^{sup.2}, L^{sup.1}, R^{sup.b}, and R^{sup.c}, are as defined for formula (I). In some variations, R^{sup.5} is halo, —CN, 3-10 membered heterocyclyl, C_{sub.1-6}alkyl, or C_{sub.1-6}alkoxy, wherein the C_{sub.1-6}alkyl of R^{sup.5} is optionally substituted with one or more halo, or —OH and wherein the C_{sub.1-6}alkoxy of R^{sup.5} is optionally substituted with one or more halo; ring A is 5-10 membered heterocyclyl or a 5-10 membered heteroaryl, wherein the 5-10 membered heterocyclyl of ring A is optionally substituted with one or more R^{sup.b}, and wherein the 5-10 membered heteroaryl of ring A is optionally substituted with one or more R^{sup.c}; and m, R^{sup.1}, R^{sup.2}, L^{sup.1}, R^{sup.b}, and R^{sup.c}, of formula (I-F2) are as defined for a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0938] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), (I-F), or (I-F2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.2} is H. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0939] In some embodiments of a compound of formula (II), such as a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), (I-F), or (I-F2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.2} is C_{sub.1-6}alkyl, wherein the C_{sub.1-6}alkyl of R^{sup.2} is optionally substituted with one or more deuterium, —OH, —NH_{sub.2}, or C_{sub.1-6}alkoxy. In some embodiments, R^{sup.2} is methyl optionally substituted with one or more deuterium. In some embodiments, R^{sup.2} is

##STR00350##

[0940] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), (I-F), or (I-F2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, m is an integer from 0 to 4. In some embodiments, m is an integer from 0 to 2. In some embodiments m is 1. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically

acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0941] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), (I-F), or (I-F2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.1} is halo. In some embodiments R^{sup.1} is Cl. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0942] In some embodiments of a compound of formula (II), such as a compound of formula (I'), (I), (I-A) or (I-A2), (I-E), (I-E1), (I-F), or (I-F2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.1} is halo. In some embodiments R^{sup.1} is C_{sub.1}, Br, or I.

[0943] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), (I-F), or (I-F2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.1} is —CN. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0944] In some embodiments of a compound of formula (II), such as a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), (I-F), or (I-F2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.1} is C_{sub.1}-6alkyl, wherein the C_{sub.1}-6alkyl of R^{sup.1} is optionally substituted with one or more halo. In some embodiments, R^{sup.1} is C_{sub.1}-3alkyl, wherein the C_{sub.1}-3alkyl of R^{sup.1} is optionally substituted with one or more halo. In some embodiments, R^{sup.1} is C_{sub.1}-3 alkyl, wherein the C_{sub.1}-3alkyl of R^{sup.1} is optionally substituted with one or more F. In some embodiments, R^{sup.1} is methyl, wherein the methyl of R^{sup.1} is optionally substituted with one or more F. In some embodiments, R^{sup.1} is

##STR00351##

[0945] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), (I-F), or (I-F2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.1} is C_{sub.1}-6alkylene. In some embodiments, L^{sup.1} is ethylene. In some variations, the embodiments provided herein also apply to a compound of formula (I') or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof. In some embodiments, L^{sup.1} is

##STR00352##

wherein, for each L^{sup.1}, # denotes the point of attachment to L^{sup.2} and ## denotes the point of attachment to the remainder of the molecule. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0946] In some embodiments of a compound of formula (II), such as a compound of formula (I'), (I), (I-A) or (I-A2), (I-E), (I-E1), (I-F), or (I-F2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.1} is C_{sub.1}-6alkylene, optionally substituted with one or more deuterium. In some embodiments, L^{sup.1} is ethylene optionally substituted with one or more deuterium. In some embodiments, L^{sup.1} is

##STR00353##

wherein, for each L^{sup.1}, # denotes the point of attachment to L^{sup.2} and ## denotes the point of attachment to the remainder of the molecule.

[0947] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), (I-F), or (I-F2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more of R^{sup.b}, wherein one or more R^{sup.b} is oxo. In some embodiments, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is

optionally substituted with one or more of R.sup.b, wherein one or more R.sup.b is oxo. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0948] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), (I-F), or (I-F2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is

##STR00354##

wherein ## represent a point of attachment to the remainder of the molecule and the dashed line represents a single or double bond. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0949] In some embodiments of a compound of formula (II), such as a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), (I-F), or (I-F2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is

##STR00355##

wherein ## represent a point of attachment to the remainder of the molecule and the dashed line represents a single or double bond.

[0950] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), (I-F), or (I-F2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R.sup.b, wherein each R.sup.b is independently oxo, or C.sub.1-6alkyl. In some embodiments, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R.sup.b, wherein each R.sup.b is independently oxo, or C.sub.1-6alkyl, wherein the C.sub.1-6alkyl of R.sup.b is optionally substituted with one or more halo, OH, or —S(O).sub.2—C.sub.1-6alkyl. In some embodiments, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R.sup.b, wherein each R.sup.b is independently oxo, or C.sub.1-3alkyl. In some embodiments, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R.sup.b, wherein each R.sup.b is independently selected from the group consisting of oxo, and

##STR00356##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0951] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), (I-F), or (I-F2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is

##STR00357##

wherein ## represent a point of attachment to the remainder of the molecule and the dashed line represents a single or double bond. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0952] In some embodiments of a compound of formula (I'), (I-A) or (I-A2), (I-E), (I-E1), (I-F), or (I-F2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R.sup.b, wherein R.sup.b is C.sub.3-10cycloalkyl, wherein the C.sub.3-10-cycloalkyl of R.sup.b is optionally substituted with one or more —OH or C.sub.1-6alkyl. In some embodiments, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R.sup.b, wherein R.sup.b is C.sub.3-

6cycloalkyl, wherein the C.sub.3-6cycloalkyl of R.sup.b is optionally substituted with one or more —OH or C.sub.1-3alkyl. In some embodiments, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R.sup.b, wherein one or more of R.sup.b is selected from the group consisting of

##STR00358##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0953] In some embodiments of a compound of formula (I'), (I-A) or (I-A2), (I-E), (I-E1), (I-F), or (I-F2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is selected from the group consisting of

##STR00359##

wherein ## represent a point of attachment to the remainder of the molecule and the dashed line represents a single or double bond. In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0954] In some embodiments of a compound of formula (II), such as a compound of formula (I'), (I-A) or (I-A2), (I-E), (I-E1), (I-F), or (I-F2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R.sup.b, wherein R.sup.b is C.sub.1-6alkyl and wherein, the C.sub.1-6alkyl of R.sup.b is optionally substituted with one or more —OH. In some embodiments, ring A is 5-6 membered heterocyclyl, wherein the 5-6 membered heterocyclyl is optionally substituted with one or more R.sup.b, wherein R.sup.b is C.sub.1-3alkyl and wherein, the C.sub.1-3alkyl of R.sup.b is optionally substituted with one or more —OH. In some embodiments, ring A is a 5-10 membered heterocyclyl, wherein the 5-6 membered heterocyclyl is optionally substituted with one or more R.sup.b, wherein one or more of R.sup.b is —CH.sub.2OH.

[0955] In some embodiments of a compound of formula (II), such as a compound of formula (I'), (I-A) or (I-A2), (I-E), (I-E1), (I-F), or (I-F2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is selected from the group consisting of

##STR00360##

wherein ## represent a point of attachment to the remainder of the molecule and the dashed line represents a single or double bond.

[0956] In some embodiments of a compound of formula (I'), (I-A) or (I-A2), (I-E), (I-E1), (I-F), or (I-F2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is a 5-10 membered heteroaryl, wherein the 5-10 membered heteroaryl is optionally substituted with one or more R.sup.c, wherein one or more R.sup.e is C.sub.3-10cycloalkyl, wherein the C.sub.3-10cycloalkyl of R.sup.e is optionally substituted with one or more —OH, or C.sub.1-6alkyl. In some embodiments, ring A is a 5-10 membered heteroaryl, wherein the 5-6 membered heteroaryl is optionally substituted with one or more R.sup.c, wherein one or more R.sup.e is C.sub.3-6cycloalkyl, wherein the C.sub.3-6cycloalkyl of R.sup.e is optionally substituted with one or more —OH, or C.sub.1-3alkyl. In some embodiments, ring A is a 5-10 membered heteroaryl, wherein the 5-6 membered heteroaryl is optionally substituted with one or more R.sup.c, wherein one or more R.sup.e is selected from the group consisting of

##STR00361##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0957] In some embodiments of a compound of formula (I'), (I-A) or (I-A2), (I-E), (I-E1), (I-F), or

(I-F2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is selected from the group consisting of

##STR00362##

wherein ## represent a point of attachment to the remainder of the molecule and the dashed line represents a single or double bond. In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0958] In some embodiments of a compound of formula (II), such as a compound of formula (I'), (I-A) or (I-A2), (I-E), (I-E1), (I-F), or (I-F2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is selected from the group consisting of

##STR00363##

wherein ## represent a point of attachment to the remainder of the molecule and the dashed line represents a single or double bond.

[0959] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), (I-F), or (I-F2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.5} is halo. In some embodiments, R^{sup.5} is fluoro. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0960] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), (I-F), or (I-F2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.5} is C_{sub.1-6}alkyl. In some embodiments, R^{sup.5} is C_{sub.1-3}alkyl. In some embodiments, R^{sup.5} is methyl. In some embodiments, R^{sup.5} is C_{sub.1-6}alkyl, wherein the C_{sub.1-6}alkyl of R^{sup.5} is optionally substituted with one or more halo. In some embodiments, R^{sup.5} is C_{sub.1-3}alkyl, wherein the C_{sub.1-3}alkyl of R^{sup.5} is optionally substituted with one or more halo. In some embodiments, R^{sup.5} is C_{sub.1-3}alkyl, wherein the C_{sub.1-3}alkyl of R^{sup.5} is optionally substituted with one or more fluoro. In some embodiments, R^{sup.5} is

##STR00364##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0961] In some embodiments of a compound of formula (II), such as a compound of formula (I'), (I-A) or (I-A2), (I-E), (I-E1), (I-F), or (I-F2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.5} is halo, —CN, or C_{sub.1-6}alkoxy, wherein the C_{sub.1-6}alkoxy of R^{sup.5} is optionally substituted with one or more halo. In some embodiments, R^{sup.5} is F, —CN, or C_{sub.1-3}alkoxy, wherein the C_{sub.1-3}alkoxy of R^{sup.5} is optionally substituted with one or more F. In some embodiments, R^{sup.5} is —OCHF_{sub.2}.

[0962] In some embodiments, provided herein is a compound of formula (I), (I-A) or (I-A2), (I-E1), or (I-E2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt thereof, wherein the compound is a compound of formula (I-G):

##STR00365##

or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein: ring A is 5-10 membered heterocyclyl or a 5-10 membered heteroaryl, wherein the 5-10 membered heterocyclyl of ring A is optionally substituted with one or more R^{sup.b}, and wherein the 5-10 membered heteroaryl of ring A is optionally substituted with one or more R^{sup.c}; and m, R^{sup.1}, R^{sup.2}, L^{sup.1}, R^{sup.b}, and R^{sup.c}, are as defined for formula (I). In some variations, ring A is 5-10 membered heterocyclyl or a 5-10 membered heteroaryl, wherein the 5-10 membered heterocyclyl of ring A is optionally substituted with one or more R^{sup.b}, and wherein the 5-10 membered heteroaryl of ring A is optionally substituted with one or more R^{sup.c}; and m, R^{sup.1},

R.sup.2, L.sup.1, R.sup.b, and R.sup.c of formula (I-G) are as defined for a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0963] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), or (I-G), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.2 is H. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0964] In some embodiments of a compound of formula (II), such as a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), or (I-G), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.2 is C.sub.1-6alkyl, wherein the C.sub.1-6alkyl of R.sup.2 is optionally substituted with one or more deuterium. In some embodiments, R.sup.2 is methyl optionally substituted with one or more deuterium. In some embodiments, R.sup.2 is
##STR00366##

[0965] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), or (I-G), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, m is an integer from 0 to 4. In some embodiments, m is an integer from 0 to 2. In some embodiments, m is 0 or 1. In some embodiments m is 1. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0966] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), or (I-G), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.1 is halo. In some embodiments R.sup.1 is Cl. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0967] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), or (I-G), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.1 is —CN. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0968] In some embodiments of a compound of formula (I'), (I-A) or (I-A2), (I-E), (I-E1), or (I-G), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R.sup.1 is C.sub.1-6alkyl, wherein the C.sub.1-6alkyl of R.sup.1 is optionally substituted with one or more halo. In some embodiments, R.sup.1 is C.sub.1-3alkyl, wherein the C.sub.1-3alkyl of R.sup.1 is optionally substituted with one or more halo. In some embodiments, R.sup.1 is C.sub.1-3alkyl, wherein the C.sub.1-3alkyl of R.sup.1 is optionally substituted with one or more F. In some embodiments, R.sup.1 is methyl, wherein the methyl of R.sup.1 is optionally substituted with one or more F. In some embodiments, R.sup.1 is methyl. In some embodiments, R.sup.1 is
##STR00367##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0969] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), or (I-G), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L.sup.1 is C.sub.1-6alkylene. In some embodiments, L.sup.1 is C.sub.1-3alkylene. In some embodiments, L.sup.1 is ethylene. In some embodiments, L.sup.1 is
##STR00368##

wherein, for each L.sup.1, # denotes the point of attachment to —O— and ## denotes the point of attachment to the remainder of the molecule. In some variations, the embodiments provided herein

also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof. [0970] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), or (I-G), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein each R^{sup.b} is independently oxo, or C_{sub.1-6}alkyl. In some embodiments, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein each R^{sup.b} is independently oxo, or C_{sub.1-6}alkyl, wherein the C_{sub.1-6}alkyl of R^{sup.b} is optionally substituted with one or more —OH. In some embodiments, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein each R^{sup.b} is independently oxo, or C_{sub.1-3}alkyl, wherein the C_{sub.1-3}alkyl of R^{sup.b} is optionally substituted with one or more —OH. In some embodiments, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein each R^{sup.b} is independently selected from the group consisting of oxo,

##STR00369##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0971] In some embodiments of a compound of formula (I'), (I-A) or (I-A2), (I-E), (I-E1), or (I-G), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein each R^{sup.b} is independently oxo, or C_{sub.1-6}alkyl. In some embodiments, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein each R^{sup.b} is independently oxo, or C_{sub.1-6}alkyl. In some embodiments, the C_{sub.1-6}alkyl of R^{sup.b} is optionally substituted with one or more halo, OH, —S(O)_{sub.2}—C_{sub.1-6}alkyl, or C_{sub.3-10}cycloalkyl, wherein the C_{sub.3-10}cycloalkyl of the C_{sub.1-6}alkyl of R^{sup.b} is further optionally substituted with one or more C_{sub.1-6}alkyl or —OH. In some embodiments, each R^{sup.b} is independently oxo, or C_{sub.1-3}alkyl. In some embodiments the C_{sub.1-3}alkyl of R^{sup.b} is optionally substituted with one or more halo, OH, —S(O)_{sub.2}—C_{sub.1-3}alkyl, or C_{sub.3-6}cycloalkyl, wherein the C_{sub.3-6}cycloalkyl of the C_{sub.1-3}alkyl of R^{sup.b} is further optionally substituted with one or more C_{sub.1-3}alkyl or —OH. In some embodiments, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein each R^{sup.b} is independently selected from the group consisting of oxo,

##STR00370##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0972] In some embodiments of a compound of formula (II), such as a compound of formula (I'), (I-A) or (I-A2), (I-E), (I-E1), or (I-G), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b} wherein each R^{sup.b} is independently selected from the group consisting of oxo,

##STR00371##

[0973] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), or (I-G), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is selected from the group consisting of

##STR00372##

wherein ## represent a point of attachment to the remainder of the molecule and the dashed line represents a single or double bond. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0974] In some embodiments of a compound of formula (I'), (I-A) or (I-A2), (I-E), (I-E1), or (I-G), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is selected from the group consisting of

##STR00373##

wherein ## represent a point of attachment to the remainder of the molecule and the dashed line represents a single or double bond. In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0975] In some embodiments of a compound of formula (II), such as a compound of formula (I'), (I-A) or (I-A2), (I-E), (I-E1), or (I-G), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is selected from the group consisting of

##STR00374##

wherein ## represent a point of attachment to the remainder of the molecule and the dashed line represents a single or double bond.

[0976] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), or (I-G), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein one or more of R^{sup.b} is —C(O)—C_{sub.1-6}alkyl. In some embodiments, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein one or more of R^{sup.b} is —C(O)—C_{sub.1-3}alkyl. In some embodiments, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein one or more of R^{sup.b} is

##STR00375##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0977] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), or (I-G), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is

##STR00376##

wherein ## represent a point of attachment to the remainder of the molecule and the dashed line represents a single or double bond. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0978] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), or (I-G), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein R^{sup.b} is C_{sub.3-10}cycloalkyl, wherein the C_{sub.3-10}-cycloalkyl of R^{sup.b} is optionally substituted with one or more —OH. In some embodiments, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein R^{sup.b} is C_{sub.3-6}cycloalkyl, wherein the C_{sub.3-6}cycloalkyl of R^{sup.b} is optionally substituted with one or more —OH. In some embodiments, ring A is a 5-10 membered heterocyclyl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein one or more of R^{sup.b} is

##STR00377##

In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0979] In some embodiments of a compound of formula (I'), (I-A) or (I-A2), (I-E), (I-E1), or (I-G), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.b} is C_{sub}.3-10cycloalkyl optionally substituted with one or more —OH, C_{sub}.3-10cycloalkyl, or C_{sub}.1-6alkyl, wherein the C_{sub}.1-6alkyl is further optionally substituted with one or more —OH. In some embodiments, R^{sup.b} is C_{sub}.3-6cycloalkyl optionally substituted with one or more —OH, C_{sub}.3-6cycloalkyl, or C_{sub}.1-3alkyl, wherein the C_{sub}.1-3alkyl is further optionally substituted with one or more —OH. In some embodiments, R^{sup.b} is selected from the group consisting of

##STR00378##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0980] In some embodiments of a compound of formula (II), such as a compound of formula (I'), (I-A) or (I-A2), (I-E), (I-E1), or (I-G), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.b} is C_{sub}.3-10cycloalkyl optionally substituted with one or more —OH, C_{sub}.3-10cycloalkyl, or C_{sub}.1-6alkyl, wherein the C_{sub}.1-6alkyl is further optionally substituted with one or more —OH or deuterium. In some embodiments, R^{sup.b} is C_{sub}.3-6cycloalkyl optionally substituted with one or more —OH, C_{sub}.3-6cycloalkyl, or C_{sub}.1-3alkyl, wherein the C_{sub}.1-3alkyl is further optionally substituted with one or more —OH or deuterium. In some embodiments, R^{sup.b} is selected from the group consisting of

##STR00379##

[0981] In some embodiments of a compound of formula (I), (I-A) or (I-A2), (I-E), (I-E1), or (I-G), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is

##STR00380##

wherein ## represent a point of attachment to the remainder of the molecule and the dashed line represents a single or double bond. In some variations, the embodiments provided herein also apply to a compound of formula (I'), or (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0982] In some embodiments of a compound of formula (I'), (I-A) or (I-A2), (I-E), (I-E1), or (I-G), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is selected from the group consisting of

##STR00381##

wherein ## represent a point of attachment to the remainder of the molecule and the dashed line represents a single or double bond. In some embodiments, ring A is selected from the group consisting of

##STR00382##

wherein ## represent a point of attachment to the remainder of the molecule and the dashed line represents a single or double bond. In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0983] In some embodiments of a compound of formula (II), such as a compound of formula (I'), (I-A) or (I-A2), (I-E), (I-E1), or (I-G), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is selected from the group consisting

##STR00383##

wherein ## represent a point of attachment to the remainder of the molecule and the dashed line

represents a single or double bond.

[0984] In some embodiments of a compound of formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.e} is independently at each occurrence, selected from the group consisting of halo, C_{sub.1-6}alkyl, —C(O)—C_{sub.1-6}alkyl, —C(O)—NH_{sub.2}, —C(O)—NH(C_{sub.1-6}alkyl), —C(O)—N(C_{sub.1-6}alkyl)_{sub.2}, —S(O)_{sub.2}—R^{sup.a}, C_{sub.3-10}cycloalkyl, and 3-10 membered heterocyclyl, wherein the C_{sub.1-6}alkyl of R^{sup.e} is optionally substituted with one or more —S(O)_{sub.2}—C_{sub.1-6}alkyl, the C_{sub.3-10}cycloalkyl of R^{sup.e} is optionally substituted with one or more —OH, or C_{sub.1-6}alkyl, and the 3-10 membered heterocyclyl of R^{sup.e} is optionally substituted with one or more —OH, or C_{sub.1-6}alkyl, wherein the C_{sub.1-6}alkyl is further optionally substituted with one or more —OH. In some embodiments, R^{sup.e} is independently at each occurrence, selected from the group consisting of halo, C_{sub.1-3}alkyl, —C(O)—C_{sub.1-3}alkyl, —C(O)—NH_{sub.2}, —C(O)—NH(C_{sub.1-3}alkyl), —C(O)—N(C_{sub.1-3}alkyl)_{sub.2}, —S(O)_{sub.2}—R^{sup.a}, C_{sub.3-6}cycloalkyl, and 3-6 membered heterocyclyl, wherein the C_{sub.1-3}alkyl of R^{sup.e} is optionally substituted with one or more —S(O)_{sub.2}—C_{sub.1-3}alkyl, the C_{sub.3-6}cycloalkyl of R^{sup.e} is optionally substituted with one or more —OH, or C_{sub.1-3}alkyl, and the 3-6 membered heterocyclyl of R^{sup.e} is optionally substituted with one or more —OH, or C_{sub.1-3}alkyl, wherein the C_{sub.1-3}alkyl is further optionally substituted with one or more —OH. In some embodiments, R^{sup.e} is selected from the group consisting of methyl, isopropyl, —S(O)_{sub.2}CH_{sub.3}, and

##STR00384##

In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0985] In some embodiments of a compound of formula (I'), (I-A) or (I-A2), (I-E), (I-E1), or (I-G), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is selected from the group consisting of

##STR00385##

wherein ## represent a point of attachment to the remainder of the molecule and the dashed line represents a single or double bond. In some variations, the embodiments provided herein also apply to a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or any variation or embodiment thereof.

[0986] In some embodiments of a compound of formula (II), such as a compound of formula (I'), (I-A) or (I-A2), (I-E), (I-E1), or (I-G), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is selected from the group consisting of

##STR00386##

N wherein ## represent a point of attachment to the remainder of the molecule and the dashed line represents a single or double bond.

[0987] In some embodiments, provided herein is a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt thereof, wherein the compound is a compound of formula (II-A):

##STR00387##

or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein: ring A is 5-10 membered heterocyclyl or a 5-10 membered heteroaryl, wherein the 5-10 membered heterocyclyl of ring A is optionally substituted with one or more R^{sup.b}, and wherein the 5-10 membered heteroaryl of ring A is optionally substituted with one or more R^{sup.c}; and m, p, R^{sup.1}, R^{sup.2}, R^{sup.3}, L^{sup.1}, R^{sup.b}, R^{sup.c}, X^{sup.2}, X^{sup.3}, and X^{sup.4} are as defined for formula (II).

[0988] In some embodiments, provided herein is a compound of formula (II), or (II-A), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt thereof, wherein the

compound is a compound of formula (II-A1):

##STR00388##

or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein: ring A is 5-10 membered heterocyclyl or a 5-10 membered heteroaryl, wherein the 5-10 membered heterocyclyl of ring A is optionally substituted with one or more R^{sup.b}, and wherein the 5-10 membered heteroaryl of ring A is optionally substituted with one or more R^{sup.c}; and m, R^{sup.1}, R^{sup.2}, R^{sup.6}, L^{sup.1}, R^{sup.b}, and R^{sup.e} are as defined for formula (II).

[0989] In some embodiments, provided herein is a compound of formula (II), or (II-A), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt thereof, wherein the compound is a compound of formula (II-A2):

##STR00389##

or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein: ring A is 5-10 membered heterocyclyl or a 5-10 membered heteroaryl, wherein the 5-10 membered heterocyclyl of ring A is optionally substituted with one or more R^{sup.b}, and wherein the 5-10 membered heteroaryl of ring A is optionally substituted with one or more R^{sup.c}; and m, R^{sup.1}, R^{sup.2}, R^{sup.7}, L^{sup.1}, R^{sup.b}, and R^{sup.e} are as defined for formula (II).

[0990] In some embodiments of a compound of formula (II), (II-A), (II-A1), or (II-A2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.2} is H.

[0991] In some embodiments of a compound of formula (II), (II-A), (II-A1), or (II-A2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, m is an integer from 0 to 4. In some embodiments, m is an integer from 0 to 2. In some embodiments, m is 0 or 1. In some embodiments m is 1.

[0992] In some embodiments of a compound of formula (II), (II-A), (II-A1), or (II-A2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.1} is halo. In some embodiments R^{sup.1} is Cl.

[0993] In some embodiments of a compound of formula (II), (II-A), (II-A1), or (II-A2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, L^{sup.1} is C_{sub.1-6}alkylene. In some embodiments, L^{sup.1} is C_{sub.1-3}alkylene. In some embodiments, L^{sup.1} is ethylene. In some embodiments, L^{sup.1} is

##STR00390##

wherein, for each L^{sup.1}, # denotes the point of attachment to —O— and ## denotes the point of attachment to the remainder of the molecule.

[0994] In some embodiments of a compound of formula (II), (II-A), (II-A1), or (II-A2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.e} is independently at each occurrence C_{sub.3-10}cycloalkyl optionally substituted with one or more —OH, or C_{sub.1-6}alkyl. In some embodiments, R^{sup.e} is independently at each occurrence, C_{sub.3-6}cycloalkyl optionally substituted with one or more —OH, or C_{sub.1-3}alkyl. In some embodiments, R^{sup.c} is OH

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[0995] In some embodiments of a compound of formula (II), (II-A), (II-A1), or (II-A2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, ring A is

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wherein ## represent a point of attachment to the remainder of the molecule and the dashed line represents a single or double bond.

[0996] In some embodiments of a compound of formula (II), (II-A), (II-A1), or (II-A2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.5} is C_{sub.1-6}alkyl optionally substituted with one or more halo. In some embodiments R^{sup.5} is CF_{sub.3}.

[0997] In some embodiments of a compound of formula (II), (II-A), or (II-A1), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.6} is H.

[0998] In some embodiments of a compound of formula (II), (II-A), or (II-A2), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, R^{sup.7} is H.

[0999] It is to be understood that any variation or embodiment of m, n, p, R^{sup.1}, R^{sup.2}, R^{sup.3}, L^{sup.1}, L^{sup.2}, L^{sup.3}, X^{sup.1}, X^{sup.2}, X^{sup.3}, X^{sup.4}, R^{sup.4}, R^{sup.5}, R^{sup.6}, R^{sup.7}, R^{sup.a}, R^{sup.b}, R^{sup.c}, Y^{sup.1}, Y^{sup.2}, and Y^{sup.3} provided herein can be combined with every other variation or embodiment of m, n, p, R^{sup.1}, R^{sup.2}, R^{sup.3}, L^{sup.1}, L^{sup.2}, L^{sup.3}, X^{sup.1}, X^{sup.2}, X^{sup.3}, X^{sup.4}, R^{sup.4}, R^{sup.5}, R^{sup.6}, R^{sup.7}, R^{sup.a}, R^{sup.b}, R^{sup.c}, Y^{sup.1}, Y^{sup.2}, and Y^{sup.3}, the same as if each and every combination had been individually and specifically described. For example, embodiments where m is 1, n is 1, p is 0, R^{sup.1} is Cl, R^{sup.2} is H, L^{sup.1} is ethylene, L^{sup.2} is O, L^{sup.3} is C_{sub.1-6}alkyl, R^{sup.4} is SO_{sub.2}R^{sup.a} wherein R^{sup.a} is C_{sub.1-6}alkyl, X^{sup.1} is N, X^{sup.2} is N, R^{sup.6} is H, and R^{sup.7} is H, can be combined to give

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[1000] In some embodiments, provided herein is a compound of formula (I), or any variation of embodiment thereof, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein the compound is a compound of Table 1, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing.

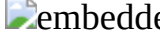
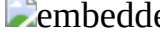
TABLE-US-00002 TABLE 1 Compound Number Structure Name 1 [00394]  embedded image 5-chloro-1'-(2-([2-(1-methanesulfonyl)ethyl]pyrimidin-5-yl)oxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 2 [00395]  embedded image 5-chloro-1'-(2-([2-((1R) or (1S)-1-methanesulfonyl)ethyl]pyrimidin-5-yl)oxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 3 [00396]  embedded image 5-chloro-1'-(2-([2-((1R) or (1S)-1-methanesulfonyl)ethyl]pyrimidin-5-yl)oxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 4 [00397]  embedded image 5-chloro-1'-(2-([2-(1-methanesulfonyl)cyclopropyl]pyrimidin-5-yl)oxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 5 [00398]  embedded image 5-(difluoromethyl)-1'-(2-([2-(1-methanesulfonyl)cyclopropyl]pyrimidin-5-yl)oxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 6 [00399]  embedded image 5-(difluoromethoxy)-1'-(2-([2-(1-methanesulfonyl)cyclopropyl]pyrimidin-5-yl)oxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 7 [00400]  embedded image 5,7-dichloro-1'-(2-(4-methanesulfonylphenoxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 8 [00401]  embedded image 7-chloro-1'-(2-(4-methanesulfonylphenoxy)ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 9 [00402]  embedded image 5-chloro-1'-(2-(4-methanesulfonylphenoxy)ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-7-carbonitrile 10 [00403]  embedded image 5-chloro-1'-((2S)-1-(4-methanesulfonylphenoxy)propan-2-yl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 11 [00404]  embedded image 5-chloro-1'-((2S)-2-(4-methanesulfonylphenoxy)propyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 12 [00405]  embedded image 1'-((2S)-1-(4-methanesulfonylphenoxy)propan-2-yl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 13 [00406]  embedded image 1'-((2S)-2-(4-methanesulfonylphenoxy)propyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 14 [00407]  embedded image 5-chloro-1'-(2-([1-(3-methanesulfonyl)propyl]-1H-indazol-5-yl)oxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 15 [00408]  embedded image 5-chloro-1'-(2-([1-(oxetan-3-yl)-1H-indazol-5-yl]oxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 16 [00409]  embedded image 5-chloro-1'-(2-([1-(propan-2-yl)-1H-indazol-5-yl]oxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 17 [00410]  embedded image 5-chloro-1-(2-hydroxyethyl)-1'-(2-(4-methanesulfonylphenoxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 18 [00411]  embedded image 1'-2-(4-methanesulfonylphenoxy)ethyl)-1-[(cis)-3-hydroxycyclobutyl]-1,2-

dihydrospiro[indole-3,4'- piperidin]-2-one 20 [00413]  embedded image 5-chloro-1'-[2-(4-methanesulfonylphenoxy)ethyl]-1- (2-methoxyethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 21 [00414]  embedded image 1'-[2-(4- methanesulfonylphenoxy)ethyl]-1- (2-methoxyethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 22 [00415]  embedded image 5-chloro-1'-[2-(4- methanesulfonylphenoxy)ethyl]-1- (oxetan-3-yl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 23 [00416]  embedded image 5-chloro-1'-{2-[4-(3-methanesulfonylpropanesulfonyl)phenoxy]ethyl}- 1-methyl-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 24 [00417]  embedded image 5-bromo-1-(2-hydroxyethyl)-1'-[2- (4-methanesulfonylphenoxy)ethyl]- 1,2-dihydrospiro[indole-3,4'- piperidin]-2-one 25 [00418]  embedded image 1'-[2-(4- methanesulfonylphenoxy)ethyl]-1- (2-methoxyethyl)-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 26 [00419]  embedded image 1-[2-(4-methanesulfonylphenoxy)ethyl]- 1',2'-dihydrospiro[azepane-4,3'- indol]-2'-one 27 [00420]  embedded image 5-chloro-1'-{2-[(1-methyl-2-oxo- 1,2,3,4-tetrahydroquinolin-6-yl)oxy]ethyl}-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 28 [00421]  embedded image 1'-[1-(4- methanesulfonylphenoxy)propan- 2-yl]-1,2-dihydrospiro[indole-3,4'- piperidin]-2-one 29 [00422]  embedded image 1-methyl-1'-{2-[(3-methyl-4-oxo- 3,4-dihydroquinazolin-6-yl)oxy]ethyl}-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 30 [00423]  embedded image 1-methyl-1'-{2-[(2-methyl-1-oxo- 2,3-dihydro-1H-indol-5- yl)oxy]ethyl}-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 31 [00424]  embedded image 1-methyl-1'-{2-[4-(1-methyl-1H- pyrazol-4-yl)phenoxy]ethyl}-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 32 [00425]  embedded image 5-(2-{1-methyl-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'-yl}ethoxy)-2,3- dihydro-1λ^{sup}.6-benzothiophene-1,1- dione 33 [00426]  embedded image 1-methyl-1'-{2-[(1-methyl-2-oxo- 2,3-dihydro-1H-indol-5- yl)oxy]ethyl}-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 34 [00427]  embedded image 1-methyl-1'-[2-(quinoxalin-6-yloxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 35 [00428]  embedded image 1-methyl-1'-{2-[4-(5-methyl- 1,2,4-oxadiazol-3- yl)phenoxy]ethyl}-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 36 [00429]  embedded image 1'-{2-[(1-acetyl-1,2,3,4- tetrahydroquinolin-6-yl)oxy]ethyl}-1-methyl-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 37 [00430]  embedded image 1-methyl-1'-{2-[4-(2- oxopyrrolidin-1- yl)phenoxy]ethyl}-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 38 [00431]  embedded image 5-chloro-1'-{2-[4-(1-methanesulfonyl)ethyl]phe- noxy}ethyl}-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 39 [00432]  embedded image 5-chloro-1'-(2-{4-[(1S or 1R)-1- methanesulfonyl)ethyl]phe- noxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 40 [00433]  embedded image 5-chloro-1'-(2-{4-[(1R or 1S)-1- methanesulfonyl)ethyl]phe- noxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 41 [00434]  embedded image 5-chloro-1'-{2-[4-(3- methanesulfonylpropanesulfonyl)phenoxy]ethyl}-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 42 [00435]  embedded image 1'-[2-(3,5-difluoro-4- methanesulfonylphenoxy)ethyl]- 1,2-dihydrospiro[indole-3,4'- piperidin]-2-one 43 [00436]  embedded image 5-chloro-1'-{2-[4- (cyclopropanesulfonyl)phe- noxy]ethyl}-1,2- dihydrospiro[indole- 3,4'-piperidin]-2-one 44 [00437]  embedded image 1'-[2-(4-difluoromethanesulfonyl- phenoxy)ethyl]-1,2- dihydrospiro[indole- 3,4'-piperidin]-2-one 45 [00438]  embedded image 4-(2-{5-chloro-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'-yl}ethoxy)benzamide 46 [00439]  embedded image 4-(2-{5-chloro-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'-yl}ethoxy)-2- fluorobenzamide 47 [00440] embedded image 5-(2-{5-chloro-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'-yl}ethoxy)pyridine- 2-carboxamide 48 [00441] embedded image 5-chloro-1'-[2-(2-fluoro-4-methanesulfonylphenoxy)ethyl]- 1,2-dihydrospiro[indole-3,4'- piperidin]-2-one 49 [00442] embedded image 4-(2-{5-chloro-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'-yl}ethoxy)-2- methylbenzamide 50 [00443] embedded image 5-(2-{5-chloro-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'-yl}ethoxy)-2- methanesulfonylbenzonitrile 51 [00444] embedded image 5-(2-{5-chloro-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'-yl}ethoxy)-N,N- dimethylpyridine-2-

carboxamide 52 [00445]  embedded image 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-methylbenzamide 53 [00446]  embedded image 5-chloro-1'-{2-[(1-methanesulfonyl-1,2,3,4-tetrahydroquinolin-6-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 54 [00447]  embedded image 5-chloro-1'-{2-[(2-methanesulfonyl-2,3-dihydro-1H-indol-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 55 [00448]  embedded image 1'-{2-[(2-methanesulfonyl-1,2,3,4-tetrahydroisoquinolin-6-yl)oxy]ethyl}-1-methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 56 [00449]  embedded image 1'-{2-[(1-acetyl-1,2,3,4-tetrahydroquinolin-6-yl)oxy]ethyl}-5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 57 [00450]  embedded image 1'-{2-[(2-acetyl-2,3-dihydro-1H-indol-5-yl)oxy]ethyl}-5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 58 [00451]  embedded image 1'-(2-[(2-acetyl-1,2,3,4-tetrahydroisoquinolin-6-yl)oxy]ethyl)-1-methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 59 [00452]  embedded image 1'-{2-[(1-acetyl-1,2,3,4-tetrahydroquinolin-6-yl)oxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 60 [00453]  embedded image 6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-methyl-1,2,3,4-tetrahydroquinoline-1-carboxamide 61 [00454]  embedded image 5-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-methyl-2,3-dihydro-1H-indole-2-carboxamide 62 [00455]  embedded image 5-chloro-1'-[2-(quinoxalin-6-yloxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 63 [00456]  embedded image 5-chloro-1'-[2-(3-fluoro-4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 64 [00457]  embedded image 5-chloro-1'-{2-[(6-methanesulfonylpyridin-3-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 65 [00458]  embedded image 5-chloro-1'-{2-[(1-methyl-2-oxo-2,3-dihydro-1H-indol-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 66 [00459]  embedded image 1'-[2-(4-methanesulfonylphenoxy)ethyl]-5-methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 67 [00460]  embedded image 1-2-(4-methanesulfonylphenoxy)ethyl]-1-methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 68 [00461]  embedded image 5-chloro-1'-{2-[(1-methyl-1H-indazol-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 69 [00462]  embedded image 5-chloro-1'-[2-(4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 70 [00463]  embedded image 1'-{2-[(1-methyl-1H-indazol-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 71 [00464]  embedded image 1'-[2-(4-methanesulfonylphenoxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 72 [00465]  embedded image 5-bromo-1'-[2-(4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-({grave over ()})one 73 [00466]  embedded image 1'-[2-(4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 74 [00467]  embedded image 1'-[2-(3-fluoro-4-methanesulfonylphenoxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 75 [00468]  embedded image N-[4-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]-N-methylmethanesulfonamide 76 [00469]  embedded image N-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]-N-methylmethanesulfonamide 77 [00470]  embedded image 1'-[2-(4-{[dimethyl(oxo)-λ^{sup}.6-sulfanylidene]amino}phenoxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 78 [00471]  embedded image 5-chloro-1'-{2-[(8-methyl-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 79 [00472]  embedded image 1'-{2-[(8-methyl-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl)oxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 80 [00473]  embedded image 1'-{2-[(8-acetyl-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl)oxy]ethyl}-5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 81 [00474]  embedded image 1'-(2-[(8-acetyl-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl)oxy]ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 82 [00475]  embedded image 5-chloro-1'-[2-[(2-oxo-1,2,3,4-tetrahydroquinolin-6-yl)oxy]ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 83 [00476]  embedded image 2-oxo-1'-{2-[(1,3,3-trimethyl-2-oxo-2,3-dihydro-1H-indol-5-

yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 84 [00477]  embedded image 1'-{2-(3,5-difluoro-4-(3-methanesulfonyloxetan-3-yl)phenoxy)ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 85 [00478]  embedded image 5-chloro-1'-[2-({1-[(cis)-3-hydroxycyclobutyl]-1H-indazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 86 [00479]  embedded image 5-chloro-1'-[2-({1-[(trans)-3-hydroxycyclobutyl]-1H-indazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 87 [00480]  embedded image 5-chloro-1'-[2-(1H-indazol-5-yloxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 88 [00481]  embedded image 5-chloro-1'-{2-[(1-methyl-2-oxo-2,3-dihydro-1H-1,3-benzodiazol-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 89 [00482]  embedded image 5-chloro-1'-[2-(3,5-difluoro-4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 90 [00483]  embedded image 1'-[2-(3,5-difluoro-4-methanesulfonylphenoxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 91 [00484]  embedded image methyl 1-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)benzenesulfonyl]cyclopropane-1-carboxylate 92 [00485]  embedded image 5-chloro-1'-(2-{4-[(3-methyloxetan-3-yl)sulfonyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 93 [00486]  embedded image 1'-(2-{4-[(3-methyloxetan-3-yl)sulfonyl]phenoxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 94 [00487]  embedded image 5-chloro-1'-{2-[(2-methanesulfonyl-1,2,3,4-tetrahydroisoquinolin-6-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 95 [00488]  embedded image 1'-{2-[(1-methyl-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl)oxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 96 [00489]  embedded image 2-oxo-1'-(2-{[2-oxo-1-(propan-2-yl)-1,2,3,4-tetrahydroquinolin-6-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 97 [00490]  embedded image 5-chloro-1'-(2-{[2-oxo-1-(propan-2-yl)-1,2,3,4-tetrahydroquinolin-6-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 98 [00491]  embedded image N-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]-N-(2-hydroxyethyl)methanesulfonamide 99 [00492]  embedded image 5-chloro-1'-(2-{[1-(oxetan-3-yl)-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 100 [00493]  embedded image 1'-{2-[3-(difluoromethyl)-4-methanesulfonylphenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 101 [00494]  embedded image 1'-{2-[4-(1-methanesulfonylcyclopropyl)phenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 102 [00495]  embedded image 5-chloro-1'-{2-[4-(1-methanesulfonylcyclopropyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 103 [00496]  embedded image 6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-1-methyl-3,4-dihydro-1H-2λ^{sup}.6,1-benzothiazine-2,2-dione 104 [00497]  embedded image 1'-{2-[(1-methyl-2,2-dioxo-3,4-dihydro-1H-2λ^{sup}.6,1-benzothiazin-6-yl)oxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 105 [00498]  embedded image 1'-(2-{[1-(2,2-difluoroethyl)-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 106 [00499]  embedded image 5-chloro-1'-(2-{[1-(2,2-difluoroethyl)-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 107 [00500]  embedded image 1'-{2-[4-methanesulfonyl-3-(trifluoromethyl)phenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 108 [00501]  embedded image 5-chloro-1'-{2-[4-methanesulfonyl-3-(trifluoromethyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 109 [00502]  embedded image 1'-{2-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 110 [00503]  embedded image 5-chloro-1'-[2-(4-methanesulfonyl-3-methylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 111 [00504]  embedded image 1'-[2-(4-methanesulfonyl-3-methylphenoxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 112 [00505]  embedded image 5-chloro-1'-[2-(4-difluoromethanesulfonylphe-noxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 113

[00506]  embedded image 1'-[2-(4-difluoromethanesulfonylphenoxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 114 [00507]  embedded image 5-chloro-1'-[2-({2-oxo-1-[(cis)-3-hydroxycyclobutyl]-1,2,3,4-tetrahydroquinolin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 115 [00508]  embedded image 5-chloro-1'-[2-({3-oxo-1H,2H,3H,5H,6H,10bH-imidazo[4,3-a]isoquinolin-8-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 116 [00509]  embedded image (S) or (R)-5-chloro-1'-[2-({3-oxo-1H,2H,3H,5H,6H,10bH-imidazo[4,3-a]isoquinolin-8-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 117 [00510]  embedded image (R) or (S)-5-chloro-1'-[2-({3-oxo-1H,2H,3H,5H,6H,10bH-imidazo[4,3-a]isoquinolin-8-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 118 [00511]  embedded image 2-oxo-1'-[2-({3-oxo-1H,2H,3H,5H,6H,10bH-imidazo[4,3-a]isoquinolin-8-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 119 [00512]  embedded image (S) or (R)-2-oxo-1'-[2-({3-oxo-1H,2H,3H,5H,6H,10bH-imidazo[4,3-a]isoquinolin-8-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 120 [00513]  embedded image (R) or (S)-2-oxo-1'-[2-({3-oxo-1H,2H,3H,5H,6H,10bH-imidazo[4,3-a]isoquinolin-8-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 121 [00514]  embedded image 5-chloro-1'-[2-({3-oxo-1H,2H,3H,5H,6H,10bH-pyrrolo[2,1-a]isoquinolin-8-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 122 [00515]  embedded image (S) or (R)-5-chloro-1'-[2-({3-oxo-1H,2H,3H,5H,6H,10bH-pyrrolo[2,1-a]isoquinolin-8-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 123 [00516]  embedded image (R) or (S)-5-chloro-1'-[2-({3-oxo-1H,2H,3H,5H,6H,10bH-pyrrolo[2,1-a]isoquinolin-8-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 124 [00517]  embedded image 2-oxo-1'-(2-({3-oxo-1H,2H,3H,5H,6H,10bH-pyrrolo[2,1-a]isoquinolin-8-yl}oxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 125 [00518]  embedded image (S) or (R)-2-oxo-1'-(2-({3-oxo-1,2,3,5,6,10b-hexahydropyrrolo[2,1-a]isoquinolin-8-yl}oxy)ethyl)spiro[indoline-3,4'-piperidine]-5-carbonitrile 126 [00519]  embedded image (R) or (S)-2-oxo-1'-(2-({3-oxo-1,2,3,5,6,10b-hexahydropyrrolo[2,1-a]isoquinolin-8-yl}oxy)ethyl)spiro[indoline-3,4'-piperidine]-5-carbonitrile 127 [00520]  embedded image N-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2,6-difluorophenyl]-N-methylmethanesulfonamide 128 [00521]  embedded image N-[4-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2,6-difluorophenyl]-N-methylmethanesulfonamide 129 [00522]  embedded image 5-chloro-1'-[2-(4-methanesulfonyl-3,5-dimethylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 130 [00523]  embedded image 1'-[2-(4-methanesulfonyl-3,5-dimethylphenoxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 131 [00524]  embedded image 5-chloro-1'-[2-[(3-methyl-2-oxo-2,3-dihydro-1,3-benzoxazol-6-yl)oxy]ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 132 [00525]  embedded image 1'-[2-[(3-methyl-2-oxo-2,3-dihydro-1,3-benzoxazol-6-yl)oxy]ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 133 [00526]  embedded image 1'-(2-[4-(1-methanesulfonyl)ethyl]phenoxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 134 [00527]  embedded image (S) or (R)-1'-[2-[4-(1-methanesulfonyl)ethyl]phenoxy]ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 135 [00528]  embedded image (R) or (S)-1'-[2-[4-(1-methanesulfonyl)ethyl]phenoxy]ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 136 [00529]  embedded image 5-chloro-1'-[2-[3,5-difluoro-4-(3-methanesulfonyloxetan-3-yl)phenoxy]ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 137 [00530]  embedded image 1'-(2-[3,5-difluoro-4-(1-methanesulfonyl)ethyl]phenoxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 138 [00531]  embedded image (S) or (R)-1'-[2-[3,5-difluoro-4-(1-methanesulfonyl)ethyl]phenoxy]ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 139 [00532]  embedded image (R) or (S)-1'-[2-[3,5-difluoro-4-(1-methanesulfonyl)ethyl]phenoxy]ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 140 [00533]  embedded image 1'-[2-[(6-methanesulfonyl-5-

methyipyridin-3-yl)oxy}ethyl}-2- oxo-1',2-dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 141 [00534]  5-chloro-1'-{2-[3,5-difluoro-4-(1- methanesulfonyl)ethyl]phenoxy}ethyl}-1,2- dihydrospiro[indole- 3,4'-piperidin]-2-one 142 [00535]  (S) or (R)-5-chloro-1'-{2-[3,5- difluoro-4-(1- methanesulfonyl)ethyl]phenoxy}ethyl}-1,2- dihydrospiro[indole- 3,4'-piperidin]-2-one 143 [00536]  (R) or (S)-5-chloro-1'-{2-[3,5- difluoro-4-(1- methanesulfonyl)ethyl]phenoxy}ethyl}-1,2- dihydrospiro[indole- 3,4'- piperidin]-2-one 144 [00537]  2-oxo-1'-(2-{[2-oxo-1-(propan-2- yl)-2,3-dihydro-1H-indol-5- yl]oxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 145 [00538]  5-chloro-1'-(2-{[2-oxo-1-(propan- 2-yl)-2,3-dihydro-1H-indol-5- yl]oxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 146 [00539]  5-chloro-1'-{2-[(2-methyl-1-oxo- 2,3-dihydro-1H-isoindol-5- yl)oxy]ethyl}-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 147 [00540]  5-chloro-1'-(2-{4- [methyl(methylimino)oxo-λ.sup.6- sulfanyl]phenoxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 148 [00541]  (S) or (R)-5-chloro-1'-(2-{4- [methyl(methylimino)oxo-λ.sup.6- sulfanyl]phenoxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 149 [00542]  (R) or (S)-5-chloro-1'-(2-{4- [methyl(methylimino)oxo-λ.sup.6- sulfanyl]phenoxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 150 [00543]  1'-(2-{4- [methyl(methylimino)oxo-λ.sup.6- sulfanyl]phenoxy}ethyl)-2-oxo- 1,2-dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 151 [00544]  (S) or (R)-1'-(2-{4- [methyl(methylimino)oxo-λ.sup.6- sulfanyl]phenoxy}ethyl)-2-oxo- 1,2- dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 152 [00545]  (R) or (S)-1'-(2-{4- [methyl(methylimino)oxo-λ.sup.6- sulfanyl]phenoxy}ethyl)-2-oxo- 1,2-dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 153 [00546]  6-(2-{5-chloro-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'-yl}ethoxy)-2-methyl- 3,4-dihydro-2H-1λ.sup.6,2- benzothiazine-1,1-dione 154 [00547]  1'-(2-{(2-methyl-1,1-dioxo-3,4- dihydro-2H-1λ.sup.6,2-benzothiazin-6- yl)oxy}ethyl)-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 155 [00548]  5-chloro-1'-[2-(3-chloro-4- methanesulfonylphenoxy)ethyl]- 1,2-dihydrospiro[indole-3,4'- piperidin]-2-one 156 [00549]  1'-(2-(3-chloro-4- methanesulfonylphenoxy)ethyl)-2- oxo-1,2- dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 157 [00550]  1'-(2-[4-(1- cyano-1- methylethyl)phenoxy]ethyl)-2- oxo-1,2-dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 158 [00551]  5-chloro-1'-[2-({6-[(3- methyloxetan-3- yl)sulfonyl]pyridin-3- yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 159 [00552]  1-(2-hydroxyethyl)-1'-{2-[4- methanesulfonyl-3- (trifluoromethyl)phenoxy]ethyl}- 2-oxo-1,2-dihydrospiro[indole- 3,4'-piperidine]-5-carbonitrile 160 [00553]  5-chloro-1-(2-hydroxyethyl)-1'- {2-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]ethyl}-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 161 [00554]  5-chloro-1'-{2-[(6- methanesulfonyl-5-methylpyridin- 3-yl)oxy]ethyl}-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 162 [00555]  5-chloro-1'-(2-{[1-(2- hydroxyethyl)-2-oxo-1,2,3,4- tetrahydroquinolin-6- yl]oxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 163 [00556]  5-chloro-1'-{2-[3- (difluoromethoxy)-4- methanesulfonylphenoxy]ethyl}- 1,2-dihydrospiro[indole-3,4'- piperidin]-2-one 164 [00557]  1'-(2-[3-(difluoromethoxy)-4- methanesulfonylphenoxy]ethyl)- 2-oxo-1,2- dihydrospiro[indole- 3,4'-piperidine]-5-carbonitrile 165 [00558]  5-chloro-1'-{2-[4-(3- methanesulfonyloxetan-3-yl)-3- (trifluoromethyl)phenoxy]ethyl}- 1,2-dihydrospiro[indole-3,4'- piperidin]-2-one 166 [00559]  1'-(2-[4-(3- methanesulfonyloxetan-3-yl)-3- (trifluoromethyl)phenoxy]ethyl)- 2-oxo-1,2-dihydrospiro[indole- 3,4'-piperidine]-5-carbonitrile 167 [00560]  6-(2-{5-chloro-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'-yl}ethoxy)-3,4- dihydro-2H-1λ.sup.6,2-benzothiazine- 1,1-dione 168 [00561]  1'-(2-{(1,1-dioxo-3,4-dihydro-2H- 1λ.sup.6,2-benzothiazin-6- yl)oxy}ethyl)-2-oxo-1,2-

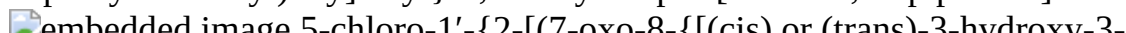
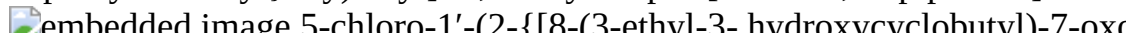
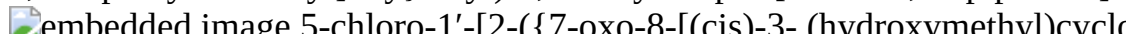
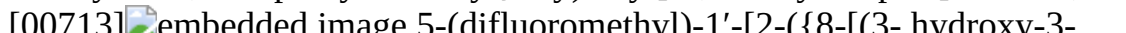
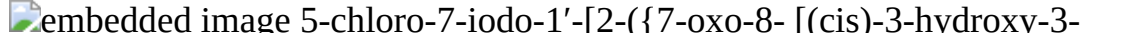
dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 169 [00562]  embedded image 1'-{2-[(7-fluoro-1-methyl-2-oxo- 2,3-dihydro-1H-indol-5- yl)oxy]ethyl}-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 170 [00563]  embedded image 2-oxo-1'-{2-[(1-oxo-2,3-dihydro-1H-isoindol-5-yl)oxy]ethyl}-1,2- dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 171 [00564]  embedded image 5-chloro-1'-{2-[(1-oxo-2,3- dihydro-1H-isoindol-5- yl)oxy]ethyl}-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 172 [00565]  embedded image 5-chloro-1'-[2-({2-[(3-methyloxetan-3-yl)oxy]pyrimidin- 5-yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 173 [00566]  embedded image 1'-[2-({2-[(3-methyloxetan-3- yl)oxy]pyrimidin-5-yl}oxy)ethyl]-2-oxo-1,2-dihydrospiro[indole- 3,4'-piperidine]-5-carbonitrile 174 [00567]  embedded image 1'- (2-{4-[3- (ethanesulfonyl)oxetan-3- yl]phenoxy}ethyl)-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 175 [00568]  embedded image 2-[4-(2-{5-chloro-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'-yl}ethoxy)phenyl]- 1λ.sup.6-thiolane-1,1-dione 176 [00569]  embedded image (S) or (R)-2-[4-(2-{5-chloro-2- oxo-1,2-dihydrospiro[indole-3,4'- piperidin]-1'-yl}ethoxy)phenyl]- 1λ.sup.6-thiolane-1,1-dione 177 [00570]  embedded image (R) or (S)-2-[4-(2-{5-chloro-2- oxo-1,2-dihydrospiro[indole-3,4'- piperidin]-1'-yl}ethoxy)phenyl]- 1λ.sup.6-thiolane-1,1-dione 178 [00571]  embedded image 1'-{2-[4-(1,1-dioxo-1λ.sup.6-thiolan-2- yl)phenoxy]ethyl}-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 179 [00572]  embedded image 5-chloro-1'-(2-{3-oxo-3H-spiro[2- benzofuran-1,3'-oxetan]-5- yloxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 180 [00573]  embedded image 5-chloro-1'-{2-[4-(1- methanesulfonylcyclo- butyl)phenoxy]ethyl}-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 181 [00574]  embedded image 1'-{2-[4-(1- methanesulfonylcyclo- butyl)phenoxy]ethyl}- 2-oxo-1,2- dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 182 [00575]  embedded image 5-chloro-1'-[2-(4-{6- methanesulfonyl-2- oxaspiro[3.3]heptan-6- yl}phenoxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 183 [00576]  embedded image 1'-[2-(4-{6-methanesulfonyl-2- oxaspiro[3.3]heptan-6- yl}phenoxy)ethyl]-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 184 [00577]  embedded image 1'-{2-[4-(3- methanesulfonyloxetan-3- yl)phenoxy]ethyl}-2-oxo-1-(2,2,2- trifluoroethyl)-1,2- dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 185 [00578]  embedded image 5-chloro-1'-{2-[4-(3- methanesulfonyloxetan-3- yl)phenoxy]ethyl}-1-(2,2,2- trifluoroethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 186 [00579]  embedded image N-(4-{2-[5-chloro-1-(2- hydroxyethyl)-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'-yl]ethoxy}-2- (trifluoromethyl)phe- nyl)methanesulfonamide 187 [00580]  embedded image 1'-{2-[4-(4- methanesulfonyloxan- 4-yl)phenoxy]ethyl}-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidine]-5- carbonitrile 188 [00581]  embedded image 5-chloro-1'-{2-[4-(4- methanesulfonyloxan-4- yl)phenoxy]ethyl}-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 189 [00582]  embedded image 1'-{2-[3,5-difluoro-4-(3- methanesulfonyloxetan-3- yl)phenoxy]ethyl}-1-methyl-2- oxo-1,2- dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 190 [00583]  embedded image 5-chloro-1'-[2- ({7-oxo-8-[(cis)-3- hydroxycyclobutyl]-5,6,7,8- tetrahydro-1,8-naphthyridin-3- yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 191 [00584]  embedded image 1'-(2-{[(6-(2- methanesulfonylpropan-2- yl)pyridin-3-yl)oxy}ethyl]-2-oxo- 1,2-dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 192 [00585]  embedded image 5-chloro-1'-(2-{[8-(2- hydroxyethyl)-7- oxo-5,6,7,8- tetrahydro-1,8-naphthyridin-3- yl]oxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 193 [00586]  embedded image N-{2-[4-(2-{5-chloro-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'- yl}ethoxy)benzenesulfonyl]ethyl}- N-methylacetamide 194 [00587]  embedded image 1'-{2-[(2-acetyl-1,2,3,4- tetrahydroisoquinolin-6- yl)oxy]ethyl}-5- chloro-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 195 [00588]  embedded image 5-chloro-1'- {2-[4-(3- methanesulfonyloxetan-3- yl)phenoxy]ethyl}-1,2- dihydrospiro[indole-3,4'- piperidin]-2- one 196 [00589]  embedded image 5-(2-{5-chloro-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'-yl}ethoxy)-2,3- dihydro-1λ.sup.6-benzothiophene-1,1- dione 197 [00590]  embedded image 5-chloro-1'-{2-[3- (difluoromethyl)-4- methanesulfonylphenoxy]ethyl}- 1,2-

dihydrospiro[indole-3,4'-piperidin]-2-one 198 [00591]  embedded image 5-chloro-1'-{2-[4-(2-hydroxyethanesulfonyl)phe- noxy]ethyl}-1,2- dihydrospiro[indole- 3,4'-piperidin]-2-one 199 [00592]  embedded image 5-chloro-1'-{1-[(1-methyl-2-oxo- 1,2,3,4-tetrahydroquinolin-6-yl)oxy]propan-2-yl}-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 200 [00593]  embedded image chloro-1'-[(2S) or (2R)-1-[(1- methyl-2-oxo-1,2,3,4- tetrahydroquinolin-6-yl)oxy]propan-2-yl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 201 [00594]  embedded image chloro-1'-[(2R) or (2S)-1-[(1- methyl-2-oxo-1,2,3,4- tetrahydroquinolin-6- yl)oxy]propan-2-yl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 202 [00595]  embedded image 1'-{1-[(1- methyl-2-oxo-1,2,3,4- tetrahydroquinolin-6- yl)oxy]propan-2-yl}-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 203 [00596]  embedded image (S) or (R)-1'-{1-[(1-methyl-2-oxo- 1,2,3,4-tetrahydroquinolin-6- yl)oxy]propan-2-yl}-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 204 [00597]  embedded image (R) or (S)-1'-{1-[(1-methyl-2-oxo- 1,2,3,4-tetrahydroquinolin-6- yl)oxy]propan-2-yl}-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 205 [00598]  embedded image 5-chloro-1'-[1-(4- methanesulfonylphenoxy)propan- 2-yl]-1-[3-hydroxycyclobutyl]- 1,2-dihydrospiro[indole-3,4'- piperidin]-2-one 206 [00599]  embedded image 5-chloro-1'-[(2S) or (2R)-1-(4- methanesulfonylphenoxy)propan- 2-yl]-1-[(cis) or (trans)-3- hydroxycyclobutyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 207 [00600]  embedded image 5-chloro-1'-[(2S) or (2R)-1-(4- methanesulfonylphenoxy)propan- 2-yl]-1-[(trans) or (cis)-3- hydroxycyclobutyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 208 [00601]  embedded image 5-chloro-1'-[(2R) or (2S)-1-(4- methanesulfonylphenoxy)propan- 2-yl]-1-[(cis) or (trans)-3- hydroxycyclobutyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 209 [00602]  embedded image 5-chloro-1'-[(2R) or (2S)-1-(4- methanesulfonylphenoxy)propan- 2-yl]-1-[(trans) or (cis)-3- hydroxycyclobutyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 210 [00603]  embedded image 5-chloro-1'-[(2S)-1-[3-(difluoromethyl)-4- methanesulfonylphenoxy]propan- 2-yl]-1,2-dihydrospiro[indole-3,4'- piperidin]-2-one 211 [00604]  embedded image 5-chloro-1'-[(2S)-1-[(1-methyl- 1H-indazol-5-yl)oxy]propan-2-yl]- 1,2-dihydrospiro[indole-3,4'- piperidin]-2-one 212 [00605]  embedded image 1'-[(2S)-1-[3-(difluoromethyl)-4- methanesulfonylphenoxy]propan- 2-yl]-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 213 [00606]  embedded image 1'-[(2S)-1-(3,5- difluoro-4- methanesulfonylphenoxy)propan- 2-yl]-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 214 [00607]  embedded image N-[4-(2-{5-chloro-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'- yl}ethoxy)phenyl]methane- sulfonamide 215 [00608]  embedded image 5-chloro-1'-[2-(4-{[(trans)-3- hydroxycyclobutyl]sulfonyl}phe- noxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 216 [00609]  embedded image 5- chloro-1'-(2-[3-(1,2- dihydroxyethyl)-4- methanesulfonylphenoxy]ethyl)- 1,2-dihydrospiro[indole- 3,4'- piperidin]-2-one 217 [00610]  embedded image 5-chloro-1'-{2-[3-(1,2- difluoroethyl)-4- methanesulfonylphenoxy]ethyl}- 1,2-dihydrospiro[indole-3,4'- piperidin]-2-one 218 [00611]  embedded image 5-chloro-1'-(2-{3-[(1S or 1R)-1,2- difluoroethyl]-4- methanesulfonylphenoxy}ethyl)- 1,2-dihydrospiro[indole-3,4'- piperidin]-2-one 219 [00612]  embedded image 5-chloro-1'-(2-{3-[(1R or 1S)-1,2- difluoroethyl]-4- methanesulfonylphenoxy}ethyl)- 1,2-dihydrospiro[indole-3,4'- piperidin]-2-one 220 [00613]  embedded image 1'-{2-[3-(1,2-difluoroethyl)-4- methanesulfonylphenoxy]ethyl}- 2-oxo-1,2- dihydrospiro[indole- 3,4'-piperidine]-5-carbonitrile 221 [00614]  embedded image 1'-(2-(3-[(1S or 1R)-1,2- difluoroethyl]-4- methanesulfonylphenoxy}ethyl)- 2-oxo-1,2-dihydrospiro[indole- 3,4'- piperidine]-5-carbonitrile 222 [00615]  embedded image 1'-(2-(3-[(1R or 1S)-1,2- difluoroethyl]-4- methanesulfonylphenoxy}ethyl)- 2-oxo-1,2-dihydrospiro[indole- 3,4'- piperidine]-5-carbonitrile 223 [00616]  embedded image 5-chloro-1'-{2-[4- methanesulfonyl-3-(oxetan-3- yl)phenoxy]ethyl}-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 224 [00617]  embedded image 5-chloro-1'-[2-(2-oxo-1-[(trans)- 3-hydroxycyclobutyl]-1,2,3,4- tetrahydroquinolin-6- yl)oxy]ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 225 [00618]

 embedded image 5-chloro-1'-[2-(4-{[1-(hydroxymethyl)cyclo-propyl]sulfonyl}phenoxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 226 [00619]  embedded image 1-[4-(2-{5-chloro-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'- yl}ethoxy)benzenesulfonyl]-N- methylcyclopropane-1- carboxamide 227 [00620]  embedded image 1-[4-(2-{5-chloro-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'- yl}ethoxy)benzenesulfonyl]-N,N- dimethylcyclopropane-1- carboxamide 228 [00621]  embedded image 1'-[2-(4-{[1-(azetidine-1- carbonyl)cyclopropyl]sulfo- nyl}phenoxy)ethyl]- 5-chloro-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 229 [00622]  embedded image 4-(2-{5-chloro-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'-yl}ethoxy)-2-fluoro- N-methylbenzamide 230 [00623]  embedded image 4-(2-{5-chloro-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'-yl}ethoxy)-2-fluoro- N,N- dimethylbenzamide 231 [00624]  embedded image 5-chloro-1'-{2-[3-fluoro-4- (morpholine-4- carbonyl)phenoxy]ethyl}-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 232 [00625]  embedded image 4-[4-(2-{5-chloro-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'- yl}ethoxy)-2- fluorobenzoyl]-1λ.sup.6- thiomorpholine-1,1-dione 233 [00626]  embedded image 5-chloro-1'-(2-[3-fluoro-4-(3- methanesulfonylazetidine-1- carbonyl)phenoxy]ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 234 [00627]  embedded image 1'-{2-[4-(azetidine-1- carbonyl)-3- fluorophenoxy]ethyl}-5-chloro- 1,2-dihydrospiro[indole-3,4'- piperidin]-2-one 235 [00628]  embedded image 5-chloro-1'-{2-[3-fluoro-4-(3- hydroxyazetidine-1- carbonyl)phenoxy]ethyl}-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 236 [00629]  embedded image 1'-{2-[4-(3-aminoazetidine-1- carbonyl)-3-fluorophenoxy]ethyl}- 5-chloro-1,2- dihydrospiro[indole- 3,4'-piperidin]-2-one 237 [00630]  embedded image N-{1-[4-(2-{5-chloro-2- oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'-yl}ethoxy)-2- fluorobenzoyl]azetidin-3- yl}methanesulfonamide 238 [00631]  embedded image 4-(2-{5-cyano-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'-yl}ethoxy)-2- fluorobenzamide 239 [00632]  embedded image 6-[4-(2-{5-chloro-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'- yl}ethoxy)-2- fluorobenzoyl]-2λ.sup.6-thia-6- azaspiro[3.3]heptane-2,2-dione 240 [00633]  embedded image 4-(2-{5-chloro-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'-yl}ethoxy)- N-(1,1- dioxo-1λ.sup.6-thian-4-yl)-2- fluorobenzamide 241 [00634]  embedded image 4-(2-{5- chloro-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'-yl}ethoxy)-N-(1- dioxo-1λ.sup.6-thiolan- 3-yl)-2- fluorobenzamide 242 [00635]  embedded image 4-(2-{5-chloro-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'-yl}ethoxy)-N-[(3S) or (3R)-1,1-dioxo-1λ.sup.6-thiolan-3- yl]-2-fluorobenzamide 243 [00636]  embedded image 4-(2-{5-chloro-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'-yl}ethoxy)-N-[(3R) or (3S)-1,1-dioxo-1λ.sup.6-thiolan-3- yl]-2-fluorobenzamide 244 [00637]  embedded image 5-chloro-1'-{2-[4-(3-methyl-1H- pyrazol-4- yl)phenoxy]ethyl}-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 245 [00638]  embedded image 6-(2-{5-chloro-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'-yl}ethoxy)-N- methyl-1,2,3,4- tetrahydroisoquinoline-2- carboxamide 246 [00639]  embedded image 6-(2-{5-cyano-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'-yl}ethoxy)-N- methyl-1,2,3,4- tetrahydroisoquinoline-2- carboxamide 247 [00640]  embedded image 5-chloro-1'-[2-({3-methyl-2-oxo- 1H,2H,3H- imidazo[4,5-b]pyridin- 6-yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 248 [00641]  embedded image 5-chloro-1'-(2-[(6-{[1-(hydroxymethyl)cyclo- propyl]sulfonyl}pyridin- 3- yl)oxy]ethyl)- 1,2-dihydrospiro[indole-3,4'- piperidin]-2-one 249 [00642]  embedded image 4-(2-{5-chloro-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'-yl}ethoxy)-2- (difluoromethyl)benzamide 250 [00643]  embedded image 4-(2-{5-cyano-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'-yl}ethoxy)-2- (difluoromethyl)benzamide 251 [00644]  embedded image 1'-{2-[4-(3- methanesulfonyloxetan-3- yl)phenoxy]ethyl}-1-methyl-2- oxo-1,2- dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 252 [00645]  embedded image 1'-(2-{4-[3-(3- fluoroazetidine-1- carbonyl)oxetan-3- yl]phenoxy}ethyl)-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 253 [00646]  embedded image 1'-(2-{4-[3-(azetidine-1- carbonyl)oxetan-3- yl]phenoxy}ethyl)-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidine]-5-

carbonitrile 254 [00647]  embedded image 3-[4-(2-{5-cyano-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'-yl}ethoxy)phenyl]- N,N-dimethyloxetane-3- carboxamide 255 [00648]  embedded image 3-[4-(2-{5-cyano-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'-yl}ethoxy)phenyl]- N-methyloxetane-3-carboxamide 256 [00649]  embedded image 5-chloro-1'-{2-[(7-fluoro-2-oxo- 2,3-dihydro-1H-indol-5- yl)oxy]ethyl}-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 257 [00650]  embedded image 2-[4-(2-{5-cyano-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'-yl}ethoxy)phenyl]-2- methylpropanamide 258 [00651]  embedded image 5-(2-{5-chloro-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'- yl}ethoxy)pyrimidine-2- carboxamide 259 [00652]  embedded image 4-[4-(2-{5-chloro-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'-yl}ethoxy)-2,6- difluorobenzoyl]-1λ.sup.6- thiomorpholine-1,1-dione 260 [00653]  embedded image 1'-{2-[4-(1,1-dioxo-1λ.sup.6- thiomorpholine-4-carbonyl)-3,5- difluorophenoxy]ethyl}-1-methyl- 2-oxo-1,2-dihydrospiro[indole- 3,4'-piperidine]-5-carbonitrile 261 [00654]  embedded image 1'-{2-[4-(1,1-dioxo-1λ.sup.6- thiomorpholine-4-carbonyl)-3- (trifluoromethyl)phenoxy]ethyl}- 1-methyl-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidine]-5- carbonitrile 262 [00655]  embedded image 1'-{1-[4-(3- methanesulfonyloxetan-3- yl)phenoxy]propan-2-yl}-2-oxo- 1,2-dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 263 [00656]  embedded image (S) or (R)-1'-{1-[4-(3- methanesulfonyloxetan-3- yl)phenoxy]propan-2-yl}-2-oxo- 1,2-dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 264 [00657]  embedded image (R) or (S)-1'-(1-[4-(3- methanesulfonyloxetan-3- yl)phenoxy]propan-2-yl)-2-oxo- 1,2- dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 265 [00658]  embedded image 5-chloro-1'-(2-[3- (difluoromethyl)-4- methanesulfonylphenoxy]ethyl)- 1-(2-hydroxyethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 266 [00659]  embedded image 1'-{2-[3- (difluoromethyl)-4- methanesulfonylphenoxy]ethyl}- 1-(2-hydroxyethyl)-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 267 [00660]  embedded image 5-chloro-1'-[2- ({7-oxo-8-[(trans)- 3-hydroxycyclobutyl]-5,6,7,8- tetrahydro-1,8-naphthyridin-3- yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 268 [00661]  embedded image 5-chloro-1'-{2-[4-(3-hydroxy-1- methanesulfonylcyclo- butyl)phenoxy]ethyl}- 1,2- dihydrospiro[indole- 3,4'-piperidin]-2-one 269 [00662]  embedded image 5-chloro-1'-(2-{4-[(cis) or (trans)- 3-hydroxy-1- methanesulfonylcyclo- butyl]phenoxy}ethyl)- 1,2-dihydrospiro[indole- 3,4'-piperidin]-2-one 270 [00663]  embedded image 5-chloro-1'-(2-{4-[(trans) or (cis)- 3-hydroxy- 1- methanesulfonylcyclo- butyl]phenoxy}ethyl)- 1,2-dihydrospiro[indole- 3,4'-piperidin]-2-one 271 [00664]  embedded image 4-(4-{2-[5-chloro-1-(2- hydroxyethyl)-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'-yl]ethoxy}-2- (trifluoromethyl)benzoyl)-1λ.sup.6- thiomorpholine-1,1-dione 272 [00665]  embedded image 4-[4-(2-{5-chloro-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'-yl}ethoxy)-2- (trifluoromethyl)benzoyl]-1λ.sup.6- thiomorpholine-1,1-dione 273 [00666]  embedded image 5-chloro-1'-[2-(3-fluoro-4-{6- methanesulfonyl-2,6- diazaspiro[3.3]heptane-2- carbonyl}phenoxy)ethyl]-1,2- dihydrospiro[indole- 3,4'- piperidin]-2-one 274 [00667]  embedded image 5-(difluoromethyl)-1'-[2-({7-oxo- 8-[(cis)-3- hydroxy-3- methylcyclobutyl]-5,6,7,8- tetrahydro-1,8-naphthyridin-3- yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 275 [00668]  embedded image 5-chloro-1'-[2-({7-oxo- 8-[(cis)-3- hydroxy-3-methylcyclobutyl]- 5,6,7,8-tetrahydro-1,8- naphthyridin-3-yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 276 [00669]  embedded image 5-chloro-1'-(2-{[2-(3- methanesulfonyloxetan-3- yl)pyrimidin-5-yl]oxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 277 [00670]  embedded image 5-(difluoromethyl)-1'-{2-[4-(3- hydroxy-1- methanesulfonylcyclo- butyl)phenoxy]ethyl}- 1-methyl-1,2- dihydrospiro[indole-3,4'- piperidin]-2- one 278 [00671]  embedded image 5-(difluoromethyl)-1-methyl-1'-(2- {4-[(cis) or (trans)-3- hydroxy-1- methanesulfonylcyclo- butyl]phenoxy}ethyl)- 1,2-dihydrospiro[indole- 3,4'- piperidin]-2-one 279 [00672]  embedded image 5-(difluoromethyl)-1-methyl-1'-(2- {4-[(trans) or (cis)-3-hydroxy-1- methanesulfonylcyclo- butyl]phenoxy}ethyl)- 1,2-dihydrospiro[indole- 3,4'- piperidin]-2-one 280 [00673]  embedded image 5-(difluoromethyl)-1'-{2-[4-(3- hydroxy-1-

methanesulfonylcyclo- butyl]phenoxy}ethyl)- 1,2-dihydrospiro[indole- 3,4'-piperidin]-2-one 281 [00674]  embedded image 5-(difluoromethyl)-1'-(2-{4- [(cis) or (trans)-3-hydroxy-1-methanesulfonylcyclo- butyl]phenoxy}ethyl)- 1,2-dihydrospiro[indole- 3,4'-piperidin]-2-one 282 [00675]  embedded image 5-(difluoromethyl)-1'-(2-{4- [(trans) or (cis)-3-hydroxy-1-methanesulfonylcyclo- butyl]phenoxy}ethyl)- 1,2-dihydrospiro[indole- 3,4'-piperidin]-2-one 283 [00676]  embedded image 1'-(2-{[5-fluoro-6-(1- methanesulfonylcyclo- propyl)pyridin-3-yl]oxy}ethyl)-2-oxo- 1,2-dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 284 [00677]  embedded image 5-chloro-1'-(2-{[2-(1- hydroxyethyl)pyrimidin-5- yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'- piperidin]-2-one 285 [00678]  embedded image 5-chloro-1'-[2-({2-[(1S) or (1R)-1- hydroxyethyl]pyrimidin-5- yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 286 [00679]  embedded image 5-chloro-1'-[2-((2-[(1R) or (1S)-1- hydroxyethyl]pyrimidin-5- yl}oxy)ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 287 [00680]  embedded image 5-(difluoromethyl)-1'-[2-({6- [(1S) or (1R)-1-hydroxyethyl]-5- (trifluoromethyl)pyridin-3-yl]oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 288 [00681]  embedded image 5-(difluoromethyl)-1'-[2-({6- [(1R) or (1S)-1-hydroxyethyl]-5- (trifluoromethyl)pyridin-3-yl]oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 289 [00682]  embedded image 5-chloro-1'-[2-({2-oxo-1-[(cis)-3- hydroxy-3-methylcyclobutyl]- 1H,2H,4H-pyrido[2,3- d][1,3]oxazin-6-yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 290 [00683]  embedded image 5-chloro-1'-[2-({7-oxo-8-[(trans)- 3-ethyl-3-hydroxycyclobutyl]- 5,6,7,8-tetrahydro-1,8- naphthyridin-3-yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 291 [00684]  embedded image 5-chloro-1'-[2-({6-[(1S) or (1R)-1- hydroxyethyl]-5- (trifluoromethyl)pyridin-3- yl]oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 292 [00685]  embedded image 5-chloro-1'-[2-({6-[(1R) or (1S)-1- hydroxyethyl]-5- (trifluoromethyl)pyridin-3- yl]oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 293 [00686]  embedded image 5-chloro-1'-[2-[(7-oxo-5,6,7,8- tetrahydro-1,8-naphthyridin-3-yl)oxy]ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 294 [00687]  embedded image 1'-(2-{[6-acetyl-5- (trifluoromethyl)pyridin-3- yl]oxy}ethyl)-5-chloro-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 295 [00688]  embedded image 1'-(2-{[6-acetyl-5- (trifluoromethyl)pyridin-3- yl]oxy}ethyl)-5-(difluoromethyl)- 1,2-dihydrospiro[indole-3,4'- piperidin]-2-one 296 [00689]  embedded image 5-(difluoromethyl)-1'-(2-{[6-(1- hydroxyethyl)-5- (trifluoromethyl)pyridin-3- yl]oxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 297 [00690]  embedded image 5-chloro-1'-[2-({7-oxo-8-[(trans)- 3-cyclopropyl-3- hydroxycyclobutyl]-5,6,7,8- tetrahydro-1,8- naphthyridin-3- yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 298 [00691]  embedded image 1'-(2-{[6-acetyl-5- (trifluoromethyl)pyridin-3- yl]oxy}ethyl)-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 299 [00692]  embedded image 5-chloro-1'-(2-{[8-(3-cyclopropyl- 3-hydroxycyclobutyl)-7-oxo- 5,6,7,8-tetrahydro-1,8- naphthyridin-3- yl]oxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 300 [00693]  embedded image 5-chloro-1'-(2-{[2-(3-hydroxy-3- methylazetidin-1-yl)pyrimidin-5- yl]oxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 301 [00694]  embedded image 5-chloro-1'-(2-{[2-(3- hydroxyazetidin-1-yl)pyrimidin-5- yl]oxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 302 [00695]  embedded image 5-chloro-1'-(2-{[2-(3- methanesulfonylazetidin-1- yl)pyrimidin-5- yl]oxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 303 [00696]  embedded image 5-chloro-1'-(2-{[2- (dimethylamino)pyrimidin-5- yl]oxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 304 [00697]  embedded image 6-[5-(2-{5-chloro-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'-yl}ethoxy)pyrimidin- 2-yl]-2λ_{sup}.6-thia-6- azaspiro[3.3]heptane-2,2-dione 305 [00698]  embedded image 5-chloro-1'-(2-{[2-(2- hydroxypropan-2-yl)pyrimidin-5- yl]oxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 306 [00699]  embedded image 5-chloro-1'-(2-{[8-(3-hydroxy-3- methylazetidin-1-yl)-1,7- naphthyridin-3-yl]oxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 307 [00700]  embedded image 5-chloro-1'-[2-({7-oxo-8-[(trans)- 3-hydroxy-3-methylcyclobutyl]- 5,6,7,8-tetrahydro-1,8- naphthyridin-3-

yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 308 [00701] embedded image 5-chloro-1'-[2-([2-(1- hydroxycyclopropyl)pyrimidin-5- yl]oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 309 [00702] embedded image 5-(difluoromethyl)-1'-[2-({7-oxo- 8-[(cis) or (trans)-3- (hydroxymethyl)cyclobutyl]- 5,6,7,8-tetrahydro-1,8- naphthyridin-3-yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 310 [00703] embedded image 5-(difluoromethyl)-1'-[2-({7-oxo- 8-[(trans) or (cis)-3- (hydroxymethyl)cyclobutyl]- 5,6,7,8-tetrahydro-1,8- naphthyridin-3-yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 311 [00704] embedded image 5-(difluoromethyl)-1'-{2-[(7-oxo- 8-[(cis) or (trans)-3-hydroxy-3-methylcyclobutyl]methyl}-5,6,7,8- tetrahydro-1,8-naphthyridin-3- yl)oxy]ethyl}-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 312 [00705] embedded image 5-(difluoromethyl)-1'-{2-[(7-oxo- 8-[(trans) or (cis)-3-hydroxy-3- methylcyclobutyl]methyl}-5,6,7,8- tetrahydro-1,8-naphthyridin-3- yl)oxy]ethyl}-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 313 [00706] embedded image 5-chloro-1'-{2-[(7-oxo-8-[(cis) or (trans)-3-hydroxy-3-methylcyclobutyl]methyl}-5,6,7,8- tetrahydro-1,8-naphthyridin-3- yl)oxy]ethyl}-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 314 [00707] embedded image 5-chloro-1'-[2-([8-(3-hydroxyazetidin-1-yl)-1,7- naphthyridin-3-yl]oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 315 [00708] embedded image 5-chloro-1'-[2-([8-(3-hydroxy-3-methylcyclobutyl)-7-oxo-5,6,7,8- tetrahydro-1,8-naphthyridin-3- yl]oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 316 [00709] embedded image 5-(difluoromethyl)-1'-[2-([8-[3- (hydroxymethyl)cyclobutyl]-7- oxo-5,6,7,8-tetrahydro-1,8- naphthyridin-3-yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 317 [00710] embedded image 5-chloro-1'-[2-([7-oxo-8-[(cis)-3- hydroxy-3-(propan-2- yl)cyclobutyl]-5,6,7,8-tetrahydro- 1,8-naphthyridin-3-yl}oxy)ethyl]- 1,2-dihydrospiro[indole-3,4'- piperidin]-2-one 318 [00711] embedded image 5-chloro-1'-[2-([8-(3-ethyl-3- hydroxycyclobutyl)-7-oxo-5,6,7,8- tetrahydro-1,8-naphthyridin-3- yl]oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 319 [00712] embedded image 5-chloro-1'-[2-([7-oxo-8-[(cis)-3- (hydroxymethyl)cyclobutyl]- 5,6,7,8-tetrahydro-1,8- naphthyridin-3-yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 320 [00713] embedded image 5-(difluoromethyl)-1'-[2-([8-[(3- hydroxy-3-methylcyclobutyl]methyl]-7-oxo- 5,6,7,8-tetrahydro-1,8- naphthyridin-3-yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 321 [00714] embedded image 5-chloro-1'-[2-([7-oxo-8-[(cis)-3- ethyl-3-hydroxycyclobutyl]- 5,6,7,8-tetrahydro-1,8- naphthyridin-3-yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 322 [00715] embedded image 5-chloro-1'-{2-[(7-oxo-8- [(trans) or (cis)-3-hydroxy-3- methylcyclobutyl]methyl}-5,6,7,8- tetrahydro-1,8-naphthyridin-3- yl)oxy]ethyl}-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 323 [00716] embedded image 5-chloro-1'-[2-([7-oxo-8-[(cis)-3- hydroxy-3-methylcyclobutyl]-7,8- dihydro-1,8-naphthyridin-3-yl]oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 324 [00717] embedded image 5-(difluoromethyl)-1'-[2-([7-oxo- 8-[(cis)-3-hydroxy-3- methylcyclobutyl]-7,8-dihydro- 1,8-naphthyridin-3-yl}oxy)ethyl]- 1,2-dihydrospiro[indole-3,4'- piperidin]-2-one 325 [00718] embedded image 5-(difluoromethoxy)-1'-[2-([7- oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-5,6,7,8- tetrahydro-1,8-naphthyridin-3- yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 326 [00719] embedded image 5-chloro-1'-[2-([8-[3- (hydroxymethyl)cyclobutyl]-7- oxo-5,6,7,8-tetrahydro-1,8- naphthyridin-3-yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 327 [00720] embedded image 5-chloro-1'-[2-([8-[(3- hydroxy-3- methylcyclobutyl]methyl]-7-oxo- 5,6,7,8-tetrahydro-1,8- naphthyridin-3-yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 328 [00721] embedded image 5-(difluoromethoxy)-1'-[2-([7- oxo-8-[(cis)-3-hydroxy-3- methylcyclobutyl]-7,8-dihydro- 1,8-naphthyridin-3-yl}oxy)ethyl]- 1,2-dihydrospiro[indole-3,4'- piperidin]-2-one 329 [00722] embedded image 5-chloro-7-iodo-1'-[2-([7-oxo-8- [(cis)-3-hydroxy-3-methylcyclobutyl]-5,6,7,8- tetrahydro-1,8-naphthyridin-3- yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 330 [00723] embedded image 5-chloro-1'-[2-([2-(1,2-

dihydroxyethyl)pyrimidin-5-yl}oxy)ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 331 [00724]  embedded image 5-chloro-1'-(2-([2-(2- hydroxyethyl)pyrimidin-5- yl}oxy)ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 332 [00725]  embedded image 1'-(2-([2-(2- hydroxyethyl)pyrimidin-5- yl}oxy)ethyl)-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidine]-5- carbonitrile 333 [00726]  embedded image 5-chloro-1'-[2-([1- (hydroxymethyl)cyclo- propyl]pyrimidin-5- yl}oxy)ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 334 [00727]  embedded image 5-chloro-1'-(2-([2-(1-hydroxy-2- methylpropan-2-yl)pyrimidin-5- yl}oxy)ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 335 [00728]  embedded image 5- chloro-1'-[2-([7-oxo-8-[(trans)- 3-hydroxy-3-(propan-2- yl)cyclobutyl]-5,6,7,8-tetrahydro- 1,8- naphthyridin-3-yl}oxy)ethyl]- 1,2-dihydrospiro[indole-3,4'- piperidin]-2-one 336 [00729]  embedded image 5-chloro-1'-[2-([7-oxo-8-[(trans)- 3-(hydroxymethyl)cyclobutyl]- 5,6,7,8- tetrahydro-1,8- naphthyridin-3-yl}oxy)ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 338 [00730]  embedded image 1'-(2-([6-(1-hydroxyethyl)-5- (trifluoromethyl)pyridin-3- yl}oxy)ethyl)-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 339 [00731]  embedded image 5-chloro-1'-(2-([5- (difluoromethyl)-7-oxo-8-[(cis)-3- hydroxy-3- methylcyclobutyl]-7,8- dihydro-1,8-naphthyridin-3- yl}oxy)ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 340 [00732]  embedded image 5-(difluoromethyl)-1'-(2-([5- (difluoromethyl)-7- oxo-8-[(cis)-3- hydroxy-3-methylcyclobutyl]-7,8- dihydro-1,8-naphthyridin-3- yl}oxy)ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 341 [00733]  embedded image 5-chloro-1'-[2-([7-oxo- 8-[(cis)-3- hydroxy-3-methylcyclobutyl]-5- (trifluoromethyl)-7,8-dihydro-1,8- naphthyridin-3- yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 342 [00734]  embedded image 5- (difluoromethyl)-1'-[2-([7-oxo- 8-[(cis)-3-hydroxy-3- methylcyclobutyl]-5- (trifluoromethyl)-7,8- dihydro-1,8- naphthyridin-3-yl}oxy)ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 343 [00735]  embedded image 5-chloro-1'-(2-([4-[1- (dimethylphosphoryl)cyclo- propyl]phenoxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 344 [00736]  embedded image 5-chloro-1'-[2-([2-[1- (dimethylphosphoryl)cyclo- propyl]pyrimidin-5- yl}oxy)ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 345 [00737]  embedded image 5- chloro-1'-[2-([2-oxo-1-[(cis)-3- hydroxy-3-methylcyclobutyl]- 1H,2H,3H-pyrrolo[2,3-b]pyridin- 5- yl}oxy)ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 346 [00738]  embedded image 5- (difluoromethyl)-1'-[2-([2-oxo- 1-[(cis)-3-hydroxy-3- methylcyclobutyl]-1H,2H,3H- pyrrolo[2,3- b]pyridin-5- yl}oxy)ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 347 [00739]  embedded image 5-chloro-1'-[2-([2-oxo-1-[(trans)- 3-hydroxy-3-methylcyclobutyl]- 1H,2H,3H- pyrrolo[2,3-b]pyridin- 5-yl}oxy)ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 348 [00740]  embedded image 5-(difluoromethyl)-1'-[2-([2-oxo- 1-[(trans)-3-hydroxy-3- methylcyclobutyl]-1H,2H,3H- pyrrolo[2,3-b]pyridin-5- yl}oxy)ethyl)-1,2- dihydrospiro[indole- 3,4'- piperidin]-2-one 349 [00741]  embedded image 5-chloro-1'-[2-([1-[(cis)-3- hydroxy-3- methylcyclobutyl]-1H- pyrrolo[2,3-b]pyridin-5- yl}oxy)ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 350 [00742]  embedded image 5-(difluoromethyl)-1'-[2-([1- [(cis)-3-hydroxy-3- methylcyclobutyl]-1H- pyrrolo[2,3-b]pyridin-5- yl}oxy)ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 351 [00743]  embedded image 5-chloro-1'-[2-([1-[(cis)-3- hydroxy-3- methylcyclobutyl]-1H- 1,3-benzodiazol-5-yl}oxy)ethyl]- 1,2-dihydrospiro[indole-3,4'- piperidin]-2-one 352 [00744]  embedded image 5-chloro-1'-[2-([1-[(cis)-3- hydroxy-3- methylcyclobutyl]-7- (trifluoromethyl)-1H-1,3- benzodiazol-5-yl}oxy)ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 353 [00745]  embedded image 5-chloro-1'-[2-([2-oxo- 1-[(cis)-3- hydroxy-3-methylcyclobutyl]-8- (trifluoromethyl)-1,2,3,4- tetrahydroquinolin-6- yl}oxy)ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 354 [00746]  embedded image 2- oxo-1'-[2-([2-oxo-1-[(cis)-3- hydroxy-3-methylcyclobutyl]-8- (trifluoromethyl)-1,2,3,4- tetrahydroquinolin-6- yl}oxy)ethyl)-1,2- dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 355 [00747]  embedded image 5-chloro-1'-[2-([2-oxo-1-[(cis)-3- hydroxy-3-methylcyclobutyl]- 1H,2H-pyrido[2,3-d]pyrimidin-6- yl}oxy)ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one

356 [00748]  embedded image 5-chloro-1'-[2-({2-oxo-1-[(cis)-3- hydroxy-3-methylcyclobutyl]-1H,2H,3H,4H-pyrido[2,3- d]pyrimidin-6-yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 357 [00749]  embedded image 5-chloro-1'-[2-({1-[(cis)-3- hydroxy-3-methylcyclobutyl]-1H-pyrazolo[3,4-b]pyridin-5- yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 358 [00750]  embedded image 5-chloro-1'-[2-({3-oxo-4-[(cis)-3- hydroxy-3-methylcyclobutyl]-3H,4H-pyrido[2,3-b]pyrazin-7- yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 359 [00751]  embedded image 5-chloro-1'-[2-({2-oxo-1-[(cis)-3- hydroxy-3-methylcyclobutyl]-1,2-dihydroquinoxalin-6- yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 360 [00752]  embedded image 5-chloro-1'-[2-({2-oxo-1-[(cis)-3- hydroxy-3-methylcyclobutyl]-8-(trifluoromethyl)-1,2- dihydroquinoxalin-6- yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 361 [00753]  embedded image 5-chloro-1'-(2-{[4-(3-hydroxy-3-methylcyclobutyl)pyrido[3,2- d]pyrimidin-7-yl]oxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 362 [00754]  embedded image 5-chloro-1'-(2-{[8-(3-hydroxy-3-methylcyclobutyl)-1,7- naphthyridin-3-yl]oxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 363 [00755]  embedded image 5-chloro-1'-(2-{[8-(3-hydroxy-3- methylazetidin-1-yl)pyrido[2,3- d]pyridazin-3-yl]oxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 364 [00756]  embedded image 5-chloro-1'-(2-{[4-(3-hydroxy-3- methylazetidin-1-yl)pyrido[3,2- d]pyrimidin-7-yl]oxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 365 [00757]  embedded image 5-chloro-1'-(2-{[8-(3- hydroxyazetidin-1-yl)pyrido[2,3- d]pyridazin-3-yl]oxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 366 [00758]  embedded image 5-chloro-1'-(2-{[4-(3- hydroxyazetidin-1-yl)pyrido[3,2- d]pyrimidin-7-yl]oxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 367 [00759]  embedded image 5-chloro-1'-[2-((3-[(cis)-3- hydroxy-3-methylcyclobutyl]-3H- imidazo[4,5-b]pyridin-6- yl}oxy)ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 368 [00760]  embedded image 5-chloro-1'-(2-{[2-(1,2-dihydroxypropan-2-yl)pyrimidin- 5-yl]oxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 369 [00761]  embedded image 5-chloro-1'-(2-((6-(1,2- dihydroxyethyl)-5-(trifluoromethyl)pyridin-3- yl)oxy)ethyl)spiro[indoline-3,4'- piperidin]-2-one 370 [00762]  embedded image 5-chloro-1'-(2-{[6-(1,2- dihydroxypropan-2-yl)-5- (trifluoromethyl)pyridin-3- yl]oxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 371 [00763]  embedded image 5-chloro-1'-(2-{[6-(2- hydroxypropan-2-yl)-5- (trifluoromethyl)pyridin-3- yl]oxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 372 [00764]  embedded image 5-chloro-1'-(2-{[6-(1- hydroxycyclopropyl)-5- (trifluoromethyl)pyridin-3- yl]oxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 373 [00765]  embedded image 5-chloro-1'-(2-{[2-(3-hydroxy-3-methylcyclobutyl)pyrimidin-5- yl]oxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 374 [00766]  embedded image 5-chloro-1'-[2-({2-oxo-3-[(cis)-3- hydroxy-3-methylcyclobutyl]-1H,2H,3H-imidazo[4,5-b]pyridin- 6-yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 375 [00767]  embedded image 5-chloro-1'-[2-((8-[3- (hydroxymethyl)azetidin-1-yl]- 1,7-naphthyridin-3-yl]oxy)ethyl)- 1,2-dihydrospiro[indole-3,4'- piperidin]-2-one 376 [00768]  embedded image 1'-[2-({6-[(1S) or (1R)-1- hydroxyethyl]-5- (trifluoromethyl)pyridin-3- yl}oxy)ethyl]-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 377 [00769]  embedded image 1'-[2-({6-[(1R) or (1S)-1- hydroxyethyl]-5- (trifluoromethyl)pyridin-3- yl}oxy)ethyl]-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 378 [00770]  embedded image 5-chloro-1'-[2-({2-[(1S) or (1R)- 1,2-dihydroxyethyl]pyrimidin-5- yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 379 [00771]  embedded image 5-chloro-1'-[2-((2-[(1R) or (1S)- 1,2-dihydroxyethyl]pyrimidin-5- yl}oxy)ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 380 [00772]  embedded image 5-chloro-1'-[2-({2-[(2S) or (2R)- 1,2-dihydroxypropan-2- yl]pyrimidin-5-yl}oxy)ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 381 [00773]  embedded image 5-chloro-1'-[2-({2-[(2R) or (2S)- 1,2-dihydroxypropan-2- yl]pyrimidin-5-yl}oxy)ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 382 [00774]  embedded image 5-chloro-1'-[2-({6-[(1S) or (1R)- 1,2-dihydroxyethyl]-5-

(trifluoromethyl)pyridin-3-yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 383 [00775]  embedded image 5-chloro-1'-[2-({6-[(1R) or (1S)- 1,2-dihydroxyethyl]-5- (trifluoromethyl)pyridin-3- yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 384 [00776]  embedded image 5-chloro-1'-[2-({6-[(2S) or (2R)- 1,2-dihydroxypropan-2-yl]-5- (trifluoromethyl)pyridin-3- yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 385 [00777]  embedded image 5-chloro-1'-[2-({6-[(2R) or (2S)- 1,2-dihydroxypropan-2-yl]-5- (trifluoromethyl)pyridin-3- yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 386 [00778]  embedded image 5-chloro-1'-[2-({8-[(cis) or (trans)-3-hydroxy-3- methylcyclobutyl]-1,7- naphthyridin-3-yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 387 [00779]  embedded image 5-chloro-1'-[2-({8-[(trans) or (cis)-3-hydroxy-3- methylcyclobutyl]-1,7- naphthyridin-3-yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 388 [00780]  embedded image 5-chloro-1'-[2-({4-[(cis) or (trans)-3-hydroxy-3- methylcyclobutyl]pyrido[3,2- d]pyrimidin-7-yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 389 [00781]  embedded image 5-chloro-1'-[2-({4-[(trans) or (cis)-3-hydroxy-3- methylcyclobutyl]pyrido[3,2- d]pyrimidin-7-yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 390 [00782]  embedded image 5-chloro-1'-[2-({8-[(cis) or (trans)-3-hydroxy-3- methylcyclobutyl]pyrido[2,3- d]pyridazin-3-yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 391 [00783]  embedded image 5-chloro-1'-[2-({8-[(trans) or (cis)-3-hydroxy-3- methylcyclobutyl]pyrido[2,3- d]pyridazin-3-yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 392 [00784]  embedded image 5-chloro-1'-[2-({2-[(cis) or (trans)-3-hydroxy-3- methylcyclobutyl]pyrimidin-5- yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 393 [00785]  embedded image 5-chloro-1'-[2-({2-[(trans) or (cis)-3-hydroxy-3- methylcyclobutyl]pyrimidin-5- yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 394 [00786]  embedded image 5-chloro-1'-[2-({2-oxo-3-[(cis)-3- hydroxy-3-methylcyclobutyl]- 2H,3H-[1,3]oxazolo[4,5- b]pyridin-6-yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 395 [00787]  embedded image 5-(difluoromethyl)-1'- (2-{[2-(3- hydroxy-3-methylazetidin-1- yl)pyrimidin-5-yl]oxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 396 [00788]  embedded image 5-chloro-1'-{2-[(2-{6-hydroxy-2- azaspiro[3.3]heptan-2- yl}pyrimidin-5-yl)oxy]ethyl}-1,2- dihydrospiro[indole-3,4'- piperidin]-2- one 397 [00789]  embedded image 5-chloro-1'-(2-{[2-(3-ethyl-3- hydroxyazetidin-1-yl)pyrimidin-5- yl]oxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 398 [00790]  embedded image 5-chloro-1'-[2-({7-oxo-8-[(cis)-3- cyclopropyl-3- hydroxycyclobutyl]-5,6,7,8- tetrahydro-1,8- naphthyridin-3- yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 399 [00791]  embedded image 5-chloro-1'-(2-{[6-(1- hydroxyethyl)-5- (trifluoromethyl)pyridin-3- yl]oxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 400 [00792]  embedded image 5,7- dichloro-1'-(2-{[2-(1- methanesulfonylcyclo- propyl)pyrimidin-5- yl]oxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 401 [00793]  embedded image 5-chloro-1'-(2-{[6-(1- hydroxy-2- methanesulfonylpropan-2-yl)-5- (trifluoromethyl)pyridin-3- yl]oxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 402 [00794]  embedded image 5-chloro-1'-[2-({6-[(2S or 2R)-1- hydroxy-2- methanesulfonylpropan-2-yl]-5- (trifluoromethyl)pyridin-3- yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 403 [00795]  embedded image 5-chloro-1'-[2-({6-[(2R or 2S)-1- hydroxy-2- methanesulfonylpropan-2-yl]-5- (trifluoromethyl)pyridin-3- yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 404 [00796]  embedded image 2-oxo-1'-[2-({7-oxo-8-[(cis)-3- hydroxy-3-methylcyclobutyl]- 5,6,7,8- tetrahydro-1,8- naphthyridin-3-yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidine]-5- carbonitrile 405 [00797]  embedded image 5-chloro-1'-(2-{[6-(3-hydroxy-3- methylazetidin-1- yl)-5- (trifluoromethyl)pyridin-3- yl]oxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 406 [00798]  embedded image 2-oxo-1'-[2-({7-oxo-8-[(cis)-3- hydroxy-3-methylcyclobutyl]-7,8- dihydro-1,8-naphthyridin-3- yl]oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidine]-5-carbonitrile 407 [00799]  embedded image 5-chloro-1'-(2-{[1-(3-hydroxy-3- methylcyclobutyl)-1H- pyrazolo[3,4-b]pyridin-5- yl]oxy}ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one 408

[00800]  embedded image 5-chloro-1'-[2-({1-[(trans)-3-hydroxy-3-methylcyclobutyl]-1H-pyrazolo[3,4-b]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 409

[00801]  embedded image 5,7-dichloro-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 410

[00802]  embedded image 2-oxo-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-pyrrolo[2,3-b]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 411

[00803]  embedded image 1'-(2-{[6-(2-hydroxypropan-2-yl)-5-(trifluoromethyl)pyridin-3-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 412

[00804]  embedded image 5-chloro-1'-{2-[4-(1-methanesulfonylpiperidin-4-yl)phenoxy]ethyl}-1-(.sup.2H.sub.3)methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 413

[00805]  embedded image 5-chloro-1-(.sup.2H.sub.3)methyl-1'-(2-{4-[methyl(methylimino)oxo-λ.sup.6-sulfanyl]-3-(trifluoromethyl)phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 414

[00806]  embedded image 5-chloro-1-(.sup.2H.sub.3)methyl-1'-(2-{4-[3-(propane-2-sulfonyl)oxetan-3-yl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 415

[00807]  embedded image 5-chloro-1'-{2-[4-(1-methanesulfonylcyclobutyl)phenoxy]ethyl}-1-(.sup.2H.sub.3)methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 416

[00808]  embedded image 5-chloro-1-(.sup.2H.sub.3)methyl-1'-(2-{(2-methyl-1-oxo-2,3-dihydro-1H-isindol-5-yl)oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 417

[00809]  embedded image 5-chloro-1'-(2-{[8-(2-hydroxy-2-methylpropyl)-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy}ethyl)-1-(.sup.2H.sub.3)methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 418

[00810]  embedded image 5-chloro-1'-[2-(4-methanesulfonylphenoxy)ethyl]-1-(.sup.2H.sub.3)methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 419

[00811]  embedded image 5-chloro-1-(.sup.2H.sub.3)methyl-1'-(2-({7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 420

[00812]  embedded image 5-chloro-1'-[2-({2-[1-(hydroxymethyl)cyclopropyl]pyrimidin-5-yl}oxy)ethyl]-1-(.sup.2H.sub.3)methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 421

[00813]  embedded image 6-{2-[5-chloro-1-(.sup.2H.sub.3)methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl]ethoxy}-8-fluoro-3,4-dihydro-2H-1λ.sup.6,2-benzothiazine-1,1-dione 422

[00814]  embedded image 5-chloro-1'-(2-{[2-(3-methanesulfonyloxetan-3-yl)pyrimidin-5-yl]oxy}ethyl)-1-(.sup.2H.sub.3)methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 423

[00815]  embedded image 5-chloro-1-(.sup.2H.sub.3)methyl-1'-[2-({7-oxo-8-[(cis)-3-hydroxycyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 424

[00816]  embedded image 5-chloro-1'-(2-{[2-(1-methanesulfonylcyclopropyl)pyrimidin-5-yl]oxy}ethyl)-1-(.sup.2H.sub.3)methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 425

[00817]  embedded image 5-chloro-1'-(2-{[2-(1-methanesulfonylethyl)pyrimidin-5-yl]oxy}ethyl)-1-(.sup.2H.sub.3)methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 426

[00818]  embedded image 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)(1,1,2,2-.sup.2H.sub.4)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 427

[00819]  embedded image 5-chloro-1'-[2-({2-(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-2H-indazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 428

[00820]  embedded image 2-oxo-1'-[2-({2-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-2H-indazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 429

[00821]  embedded image 5-chloro-1'-(2-{[2-(3-hydroxyoxetan-3-yl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 430

[00822]  embedded image 5-chloro-1'-(2-{[2-(1-hydroxycyclobutyl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 431

[00823]  embedded image 1'-(2-{[6-(3-hydroxy-3-methylazetidin-1-yl)-5-(trifluoromethyl)pyridin-3-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 432

[00824]  embedded image 1'-(2-{[2-(1-hydroxy-2-methylpropan-2-yl)pyrimidin-5-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-

carbonitrile 433 [00825]  embedded image 2-oxo-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 434 [00826]  embedded image 5-chloro-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,2,3-benzotriazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 435 [00827]  embedded image 2-oxo-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,2,3-benzotriazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 436 [00828]  embedded image 1'-(2-({1-(3-hydroxy-3-methylcyclobutyl)-1H-pyrazolo[3,4-b]pyridin-5-yl}oxy)ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 437 [00829]  embedded image 2-oxo-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-pyrazolo[3,4-b]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 438 [00830]  embedded image 1'-(2-({1-(3-hydroxy-3-methylcyclobutyl)-1H-pyrazolo[3,4-b]pyridin-5-yl}oxy)ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 439 [00831]  embedded image 5-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-1,3-benzodiazole-7-carbonitrile 440 [00832]  embedded image 5-chloro-1'-[2-({3-[(cis)-3-hydroxy-3-methylcyclobutyl]-3H-[1,2,3]triazolo[4,5-b]pyridin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 441 [00833]  embedded image 5-chloro-1'-(2-({2-(2,2-dimethyl-1,3-dioxolan-4-yl)pyrimidin-5-yl}oxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 442 [00834]  embedded image 5-chloro-1'-[2-({2-methyl-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 443 [00835]  embedded image 1'-[2-({2-methyl-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 444 [00836]  embedded image 5-chloro-1'-{2-[4-(1-methanesulfonylpiperidin-4-yl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 445 [00837]  embedded image 5-chloro-1'-(2-({7-(difluoromethyl)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-1,3-benzodiazol-5-yl}oxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 446 [00838]  embedded image 1'-(2-({7-(difluoromethyl)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-1,3-benzodiazol-5-yl}oxy)ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 447 [00839]  embedded image 5-chloro-1'-(2-({7-(difluoromethoxy)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-1,3-benzodiazol-5-yl}oxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 448 [00840]  embedded image 1'-(2-({7-(difluoromethoxy)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-1,3-benzodiazol-5-yl}oxy)ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 449 [00841]  embedded image 5-chloro-1'-(2-({2-(2-hydroxypropan-2-yl)pyrimidin-5-yl}oxy)ethyl)-7-iodo-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 450 [00842]  embedded image 5-(difluoromethyl)-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 451 [00843]  embedded image 5-chloro-1'-[2-({2-methyl-3-[(cis)-3-hydroxy-3-methylcyclobutyl]-3H-imidazo[4,5-b]pyridin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 452 [00844]  embedded image 5-chloro-1'-(2-({8-(2-hydroxy-2-methylpropyl)-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 453 [00845]  embedded image 5-chloro-7-iodo-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 454 [00846]  embedded image 5-chloro-1'-[2-({7-fluoro-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 455 [00847]  embedded image 1'-[2-({7-fluoro-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 456 [00848]  embedded image 5-chloro-1'-[2-({2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-8-(trifluoromethyl)-1,2,3,4-tetrahydroquinazolin-6-

yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-2-one 457 [00849]  embedded image 2-oxo-1'-[2-({2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-8-(trifluoromethyl)-1,2,3,4-tetrahydroquinazolin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 458 [00850]  embedded image 5-chloro-1'-[2-({7,7-dimethyl-5-oxo-5H,7H-furo[3,4-b]pyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 459 [00851]  embedded image 5-chloro-1'-{2-[(2-{6-hydroxy-6-methyl-2-azaspiro[3.3]heptan-2-yl}pyrimidin-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 460 [00852]  embedded image 5-chloro-1'-(2-{2'-oxo-1'-[(cis)-3-hydroxy-3-methylcyclobutyl]-1',2'-dihydrospiro[cyclopropane-1,3'-pyrrolo[2,3-b]pyridin]-5'-yloxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 461 [00853]  embedded image 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-(.sup.2H.sub.3)methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 462 [00854]  embedded image 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-(.sup.2H.sub.3)methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)(1,1,2,2-.sup.2H.sub.4)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 463 [00855]  embedded image 5-chloro-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl}oxy)(1,1,2,2-.sup.2H.sub.4)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 464 [00856]  embedded image 5-chloro-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-indazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 465 [00857]  embedded image 2-oxo-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-indazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 466 [00858]  embedded image 5-chloro-1'-[2-({2-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 467 [00859]  embedded image 5-chloro-1'-(2-{[3-(hydroxymethyl)-2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-2,3-dihydro-1H-1,3-benzodiazol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 468 [00860]  embedded image 5-chloro-1'-[2-({2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-2,3-dihydro-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 469 [00861]  embedded image 1'-(2-{[3-(hydroxymethyl)-2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-2,3-dihydro-1H-1,3-benzodiazol-5-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 470 [00862]  embedded image 2-oxo-1'-[2-({2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-2,3-dihydro-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile 471 [00863]  embedded image 5-chloro-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-imidazo[4,5-b]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 472 [00864]  embedded image 5-chloro-1'-(2-{3-[(cis)-3-hydroxy-3-methylcyclobutyl]-4-(trifluoromethyl)-3H-1,3,5-triazainden-6-yloxy}ethyl)spiro[indoline-3,4'-piperidin]-2-one

[1001] In some embodiments, a compound of formula (I) is selected from the group consisting of:

[1002] 1'-[2-(4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one;

[1003] 1'-[2-(4-methanesulfonylphenoxy)ethyl]-1-(2-methoxyethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1004] 5-bromo-1'-[2-(4-methanesulfonylphenoxy)ethyl]-1,2-

dihydrospiro[indole-3,4'-piperidin]-2-one; [1005] 1'-[2-(4-methanesulfonylphenoxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1006] 1'-{2-[(1-methyl-1H-indazol-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1007] 5-bromo-1-(2-

hydroxyethyl)-1'-[2-(4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1008] 5-chloro-1'-[2-(4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-

piperidin]-2-one; [1009] 5-chloro-1'-[2-(4-methanesulfonylphenoxy)ethyl]-1-(2-methoxyethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1010] 1'-[2-(4-

methanesulfonylphenoxy)ethyl]-1-(2-methoxyethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1011] 1'-{2-[(2-methanesulfonyl-1,2,3,4-tetrahydroisoquinolin-6-

yl)oxy]ethyl}-1-methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1012] 1'-{2-[(2-acetyl-1,2,3,4-tetrahydroisoquinolin-6-yl)oxy]ethyl}-1-methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1013] 1'-[1-(4-methanesulfonylphenoxy)propan-2-yl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1014] 1-[2-(4-methanesulfonylphenoxy)ethyl]-1',2'-dihydrospiro[azepane-4,3'-indol]-2'-one; [1015] 1'-[1-(4-methanesulfonylphenoxy)propan-2-yl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1016] 1'-[2-(4-methanesulfonylphenoxy)propyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1017] 5-chloro-1'-{2-[(1-methyl-1H-indazol-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1018] 1'-[2-(4-difluoromethanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1019] 1'-[2-(3,5-difluoro-4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1020] 1-(3-hydroxycyclobutyl)-1'-[2-(4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1021] 5-chloro-1'-{2-[4-(3-methanesulfonylpropanesulfonyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1022] 5-chloro-1'-{2-[4-(cyclopropanesulfonyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1023] 5,7-dichloro-1'-[2-(4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1024] 1-methyl-1'-{2-[4-(2-oxopyrrolidin-1-yl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1025] 1'-{2-[(1-acetyl-1,2,3,4-tetrahydroquinolin-6-yl)oxy]ethyl}-1-methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1026] 1-methyl-1'-{2-[4-(5-methyl-1,2,4-oxadiazol-3-yl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1027] 1-methyl-1'-[2-(quinoxalin-6-yloxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1028] 1-methyl-1'-{2-[(1-methyl-2-oxo-2,3-dihydro-1H-indol-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1029] 5-(2-{1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2,3-dihydro-1 λ 6-benzothiophene-1,1-dione; [1030] 5-chloro-1-(2-hydroxyethyl)-1'-[2-(4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1031] 5-chloro-1'-{2-[4-(1-methanesulfonyl)ethyl]phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1032] 5-chloro-1'-[2-(4-methanesulfonylphenoxy)ethyl]-1-(oxetan-3-yl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1033] 7-chloro-1'-[2-(4-methanesulfonylphenoxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1034] 5-chloro-1'-{2-[4-(1-methanesulfonyl)ethyl]phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1035] 5-chloro-1'-{2-[4-(1-methanesulfonyl)ethyl]phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1036] 5-chloro-1'-(2-{[1-(propan-2-yl)-1H-indazol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1037] 5-chloro-1'-(2-{[1-(3-methanesulfonylpropyl)-1H-indazol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1038] 5-chloro-1'-{2-[(6-methanesulfonylpyridin-3-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1039] 5-chloro-1'-{2-[(2-methanesulfonyl-2,3-dihydro-1H-isoindol-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1040] 5-chloro-1'-[1-(4-methanesulfonylphenoxy)propan-2-yl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1041] 5-chloro-1'-[2-(4-methanesulfonylphenoxy)propyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1042] 5-chloro-1'-{2-[4-(3-methanesulfonylpropanesulfonyl)phenoxy]ethyl}-1-methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1043] 5-chloro-1'-{2-[(1-methyl-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1044] 1-methyl-1'-{2-[4-(1-methyl-1H-pyrazol-4-yl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1045] 1-methyl-1'-{2-[(2-methyl-1-oxo-2,3-dihydro-1H-isoindol-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1046] 1-methyl-1'-{2-[(3-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1047] 5-chloro-1'-[2-(3-fluoro-4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1048] 1'-[2-(4-methanesulfonylphenoxy)ethyl]-1-methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1049] 5-chloro-1'-[2-(4-methanesulfonylphenoxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-7-carbonitrile;

[1050] 5-chloro-1'-(2-{4-[(3-hydroxycyclobutyl)sulfonyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1051] 5-chloro-1'-(2-{[1-(oxetan-3-yl)-1H-indazol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1052] 5-chloro-1'-(2-{4-(2-hydroxyethanesulfonyl)phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1053] 5-chloro-1'-(1-{[1-(1-methyl-1H-indazol-5-yl)oxy]propan-2-yl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1054] 1'-(2-(4-methanesulfonylphenoxy)ethyl)-5-methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1055] 5-chloro-1'-(2-(quinoxalin-6-yloxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1056] 5-chloro-1'-(2-{(1-methyl-2-oxo-2,3-dihydro-1H-indol-5-yl)oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1057] 1'-(2-(3,5-difluoro-4-methanesulfonylphenoxy)ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1058] 5-chloro-1'-(2-{4-(3-methyl-1H-pyrazol-4-yl)phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1059] 5-chloro-1'-(2-(3,5-difluoro-4-methanesulfonylphenoxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1060] 5-chloro-1'-(2-{(1-methyl-2-oxo-2,3-dihydro-1H-1,3-benzodiazol-5-yl)oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1061] 5-chloro-1'-(2-{3-(difluoromethyl)-4-methanesulfonylphenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1062] 5-chloro-1'-(2-{(1-methanesulfonyl-1,2,3,4-tetrahydroquinolin-6-yl)oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1063] 1'-(2-{(1-acetyl-1,2,3,4-tetrahydroquinolin-6-yl)oxy}ethyl)-5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1064] 5-chloro-1'-(2-(1H-indazol-5-yloxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1065] 6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-methyl-1,2,3,4-tetrahydroquinoline-1-carboxamide; [1066] 1'-(2-{(2-acetyl-2,3-dihydro-1H-isoindol-5-yl)oxy}ethyl)-5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1067] 5-chloro-1'-(2-(4-methanesulfonyl-3-methylphenoxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1068] 5-chloro-1'-(2-(4-difluoromethanesulfonylphenoxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1069] 1'-(2-(4-difluoromethanesulfonylphenoxy)ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1070] N-{2-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)benzenesulfonyl]ethyl}-N-methylacetamide; [1071] 5-chloro-1'-(2-{4-(3-methanesulfonyloxetan-3-yl)phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1072] 1'-(2-(4-methanesulfonyl-3-methylphenoxy)ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1073] 1'-(2-{(1-methyl-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl)oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1074] 5-chloro-1'-(2-{(2-methanesulfonyl-1,2,3,4-tetrahydroisoquinolin-6-yl)oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1075] 5-chloro-1'-(2-{4-[(3-methyloxetan-3-yl)sulfonyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1076] 6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-methyl-1,2,3,4-tetrahydroisoquinoline-2-carboxamide; [1077] 5-chloro-1'-(2-{[1-(3-hydroxycyclobutyl)-1H-indazol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1078] 5-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2,3-dihydro-1 λ 6-benzothiophene-1,1-dione; [1079] 1'-(2-{3-(difluoromethyl)-4-methanesulfonylphenoxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1080] 1'-(2-{(2-acetyl-1,2,3,4-tetrahydroisoquinolin-6-yl)oxy}ethyl)-5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1081] 5-chloro-1'-(2-{4-[(3-hydroxycyclobutyl)sulfonyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1082] 5-chloro-1'-(2-{[1-(3-hydroxycyclobutyl)-1H-indazol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1083] 5-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-methyl-2,3-dihydro-1H-isoindole-2-carboxamide; [1084] 1'-(2-{4-[(3-methyloxetan-3-yl)sulfonyl]phenoxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1085] 1'-(1-{[1-(1-methyl-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl)oxy]propan-2-yl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1086] 5-chloro-1'-(2-(2-fluoro-4-methanesulfonylphenoxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1087] 2-oxo-1'-(2-{[2-oxo-1-(propan-2-yl)-1,2,3,4-tetrahydroquinolin-6-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-

piperidine]-5-carbonitrile; [1088] 1'-{2-[4-(1-methanesulfonylcyclopropyl)phenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1089] 5-chloro-1'-(2-{[1-(2-hydroxyethyl)-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1090] 5-chloro-1'-(2-{[2-oxo-1-(propan-2-yl)-1,2,3,4-tetrahydroquinolin-6-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1091] 1'-{1-[(1-methyl-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl)oxy]propan-2-yl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1092] 1'-{1-[(1-methyl-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl)oxy]propan-2-yl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1093] 5-chloro-1'-{1-[3-(difluoromethyl)-4-methanesulfonylphenoxy]propan-2-yl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1094] 5-chloro-1'-{1-[(1-methyl-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl)oxy]propan-2-yl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1095] 5-chloro-1'-{2-[4-(1-methanesulfonylcyclopropyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1096] 1'-{1-[3-(difluoromethyl)-4-methanesulfonylphenoxy]propan-2-yl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1097] 5-chloro-1-(3-hydroxycyclobutyl)-1'-[1-(4-methanesulfonylphenoxy)propan-2-yl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1098] 5-chloro-1'-{1-[(1-methyl-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl)oxy]propan-2-yl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1099] 5-chloro-1'-{1-[(1-methyl-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl)oxy]propan-2-yl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1100] N-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]-N-methylmethanesulfonamide; [1101] N-[4-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]-N-methylmethanesulfonamide; [1102] 5-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-methanesulfonylbenzonitrile; [1103] 5-chloro-1'-(2-{[1-(2,2-difluoroethyl)-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1104] 1'-{2-[4-methanesulfonyl-3-(trifluoromethyl)phenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1105] 5-chloro-1'-{2-[4-methanesulfonyl-3-(trifluoromethyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1106] methyl 1-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)benzenesulfonyl]cyclopropane-1-carboxylate; [1107] 1'-{2-[3-(difluoromethyl)-4-methanesulfonylphenoxy]ethyl}-1-(2-hydroxyethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1108] 5-chloro-1'-{2-[3-(difluoromethyl)-4-methanesulfonylphenoxy]ethyl}-1-(2-hydroxyethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1109] 1'-(2-{[1-(2,2-difluoroethyl)-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1110] 5-chloro-1-(3-hydroxycyclobutyl)-1'-[1-(4-methanesulfonylphenoxy)propan-2-yl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1111] 2-oxo-1'-{2-[(1,3,3-trimethyl-2-oxo-2,3-dihydro-1H-indol-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1112] 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)benzamide; [1113] 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-fluorobenzamide; [1114] 5-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)pyridine-2-carboxamide; [1115] 6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-1-methyl-3,4-dihydro-1H-2lambda6,1-benzothiazine-2,2-dione; [1116] N-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]methanesulfonamide; [1117] 1'-{2-[(1-methyl-2,2-dioxo-3,4-dihydro-1H-2lambda6,1-benzothiazin-6-yl)oxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1118] 5-chloro-1'-(2-{[1-(oxetan-3-yl)-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1119] 5-chloro-1-(3-hydroxycyclobutyl)-1'-[1-(4-methanesulfonylphenoxy)propan-2-yl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1120] 1'-{2-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1121] 5-chloro-1-(3-hydroxycyclobutyl)-1'-[1-(4-methanesulfonylphenoxy)propan-2-yl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1122] 5-

chloro-1-(3-hydroxycyclobutyl)-1'-[1-(4-methanesulfonylphenoxy)propan-2-yl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1123] 5-chloro-1'-[2-(4-{[1-(hydroxymethyl)cyclopropyl)sulfonyl}phenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1124] 1'-[2-(3-fluoro-4-methanesulfonylphenoxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1125] 1'-{2-[(1-acetyl-1,2,3,4-tetrahydroquinolin-6-yl)oxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1126] 5-chloro-1'-{2-[(8-methyl-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1127] 1'-{2-[(8-methyl-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl)oxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1128] 1'-[2-(4-methanesulfonyl-3,5-dimethylphenoxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1129] 5-chloro-1'-[2-(3-oxo-1H,2H,3H,5H,6H,10bH-pyrrolo[2,1-a]isoquinolin-8-yl)oxy]ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1130] 2-oxo-1'-[2-(3-oxo-1H,2H,3H,5H,6H,10bH-pyrrolo[2,1-a]isoquinolin-8-yl)oxy]ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1131] 6-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-methyl-1,2,3,4-tetrahydroisoquinoline-2-carboxamide; [1132] 5-chloro-1'-(2-{[1-(3-hydroxycyclobutyl)-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1133] 5-chloro-1'-[2-(4-methanesulfonyl-3,5-dimethylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1134] 5-chloro-1'-{2-[(3-methyl-2-oxo-2,3-dihydro-1,3-benzoxazol-6-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1135] 2-oxo-1'-[2-(3-oxo-1H,2H,3H,5H,6H,10bH-pyrrolo[2,1-a]isoquinolin-8-yl)oxy]ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1136] 2-oxo-1'-[2-(3-oxo-1H,2H,3H,5H,6H,10bH-pyrrolo[2,1-a]isoquinolin-8-yl)oxy]ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1137] 1'-{2-[4-(1-methanesulfonyl)ethyl]phenoxy}ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1138] 5-chloro-1'-[2-(3-oxo-1H,2H,3H,5H,6H,10bH-pyrrolo[2,1-a]isoquinolin-8-yl)oxy]ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1139] 1'-{2-[(3-methyl-2-oxo-2,3-dihydro-1,3-benzoxazol-6-yl)oxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1140] 5-chloro-1'-[2-(3-oxo-1H,2H,3H,5H,6H,10bH-pyrrolo[2,1-a]isoquinolin-8-yl)oxy]ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1141] 1'-{2-[4-(1-cyano-1-methylethyl)phenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1142] 2-oxo-1'-(2-{[2-oxo-1-(propan-2-yl)-2,3-dihydro-1H-indol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1143] 5-chloro-1'-{2-[(6-methanesulfonyl-5-methylpyridin-3-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1144] 1'-{2-[4-(1-methanesulfonyl)ethyl]phenoxy}ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1145] 1'-{2-[4-(1-methanesulfonyl)ethyl]phenoxy}ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1146] 5-chloro-1'-[2-(3-methyl-2-oxo-1H,2H,3H-imidazo[4,5-b]pyridin-6-yl)oxy]ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1147] 1'-{2-[(6-methanesulfonyl-5-methylpyridin-3-yl)oxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1148] 5-chloro-1'-(2-{4-[methyl(methylimino)oxo-lambda6-sulfanyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1149] 1'-(2-{4-[methyl(methylimino)oxo-lambda6-sulfanyl]phenoxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1150] 1'-{2-[(8-acetyl-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl)oxy]ethyl}-5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1151] 1'-{2-[(8-acetyl-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl)oxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1152] 5-chloro-1'-{2-[4-methanesulfonyl-3-(oxetan-3-yl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1153] 5-chloro-1'-[2-(3-oxo-1H,2H,3H,5H,6H,10bH-imidazo[4,3-a]isoquinolin-8-yl)oxy]ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1154] 1'-{2-[3,5-difluoro-4-(1-methanesulfonyl)ethyl]phenoxy}ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1155] 5-chloro-1'-[2-(3-chloro-4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1156] 1'-[2-(3-chloro-4-methanesulfonylphenoxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-

carbonitrile; [1157] 5-chloro-1'-[2-([2-oxo-1-(propan-2-yl)-2,3-dihydro-1H-indol-5-yl]oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1158] 5-chloro-1'-[2-[(2-methyl-1-oxo-2,3-dihydro-1H-isoindol-5-yl)oxy]ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1159] 2-oxo-1'-[2-([3-oxo-1H,2H,3H,5H,6H,10bH-imidazo[4,3-a]isoquinolin-8-yl]oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1160] 1'-[1-(3,5-difluoro-4-methanesulfonylphenoxy)propan-2-yl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1161] 2-oxo-1'-[2-([3-oxo-1H,2H,3H,5H,6H,10bH-imidazo[4,3-a]isoquinolin-8-yl]oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1162] 5-chloro-1'-[2-([3-oxo-1H,2H,3H,5H,6H,10bH-imidazo[4,3-a]isoquinolin-8-yl]oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1163] 2-oxo-1'-[2-([3-oxo-1H,2H,3H,5H,6H,10bH-imidazo[4,3-a]isoquinolin-8-yl]oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1164] 5-chloro-1'-[2-([3-oxo-1H,2H,3H,5H,6H,10bH-imidazo[4,3-a]isoquinolin-8-yl]oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1165] 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-methylbenzamide; [1166] 5-chloro-1'-[2-[(2-oxo-1,2,3,4-tetrahydroquinolin-6-yl)oxy]ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1167] 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-fluoro-N,N-dimethylbenzamide; [1168] 5-chloro-1'-[2-[3-fluoro-4-(morpholine-4-carbonyl)phenoxy]ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1169] 4-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-fluorobenzoyl]-1lambda6-thiomorpholine-1,1-dione; [1170] 1'-[2-(4-{[1-(azetidine-1-carbonyl)cyclopropyl]sulfonyl}phenoxy)ethyl]-5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1171] 5-chloro-1'-[2-[3-fluoro-4-(3-methanesulfonylazetidine-1-carbonyl)phenoxy]ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1172] 6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-methyl-3,4-dihydro-2H-1lambda6,2-benzothiazine-1,1-dione; [1173] 1'-[2-[(2-methyl-1,1-dioxo-3,4-dihydro-2H-1lambda6,2-benzothiazin-6-yl)oxy]ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1174] 1'-[2-[4-(azetidine-1-carbonyl)-3-fluorophenoxy]ethyl]-5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1175] 5-chloro-1'-[2-[3,5-difluoro-4-(3-methanesulfonyloxetan-3-yl)phenoxy]ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1176] 5-chloro-1'-[2-[3-fluoro-4-(3-hydroxyazetidine-1-carbonyl)phenoxy]ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1177] 1'-[2-[3,5-difluoro-4-(1-methanesulfonylethyl)phenoxy]ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1178] 1'-[2-[3,5-difluoro-4-(1-methanesulfonylethyl)phenoxy]ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1179] N-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2,6-difluorophenyl]-N-methylmethanesulfonamide; [1180] 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-fluoro-N-methylbenzamide; [1181] 1-(2-hydroxyethyl)-1'-[2-[4-methanesulfonyl-3-(trifluoromethyl)phenoxy]ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1182] N-[4-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2,6-difluorophenyl]-N-methylmethanesulfonamide; [1183] 1-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)benzenesulfonyl]-N-methylcyclopropane-1-carboxamide; [1184] 5-chloro-1'-[2-[3-(1,2-difluoroethyl)-4-methanesulfonylphenoxy]ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1185] 1-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)benzenesulfonyl]-N,N-dimethylcyclopropane-1-carboxamide; [1186] 1'-[2-[3,5-difluoro-4-(3-methanesulfonyloxetan-3-yl)phenoxy]ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1187] 5-chloro-1'-[2-[3-(1,2-dihydroxyethyl)-4-methanesulfonylphenoxy]ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1188] 5-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N,N-dimethylpyridine-2-carboxamide; [1189] 4-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-fluorobenzamide; [1190] 1'-[2-[4-(3-aminoazetidine-1-carbonyl)-3-fluorophenoxy]ethyl]-5-chloro-

1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1191] 5-chloro-1'-[2-({6-[(3-methyloxetan-3-yl)sulfonyl]pyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1192] 5-chloro-1'-{2-[3,5-difluoro-4-(1-methanesulfonyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1193] 5-chloro-1'-{2-[3,5-difluoro-4-(1-methanesulfonyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1194] 5-chloro-1'-{2-[3,5-difluoro-4-(1-methanesulfonyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1195] 5-chloro-1'-{2-[3-(1,2-difluoroethyl)-4-methanesulfonyl]phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1196] 5-chloro-1'-(2-{[1-(3-hydroxycyclobutyl)-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1197] 5-chloro-1'-{2-[3-(1,2-difluoroethyl)-4-methanesulfonyl]phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1198] N-{1-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-fluorobenzoyl]azetidin-3-yl}methanesulfonamide; [1199] 1'-{2-[3-(1,2-difluoroethyl)-4-methanesulfonyl]phenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1200] 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-methylbenzamide; [1201] 1'-(2-{4-[methyl(methylimino)oxo-lambda6-sulfanyl]phenoxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1202] 1'-{1-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]propan-2-yl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1203] 5-chloro-1'-(2-{4-[methyl(methylimino)oxo-lambda6-sulfanyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1204] 5-chloro-1-(2-hydroxyethyl)-1'-{2-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1205] 5-chloro-1'-(2-{4-[methyl(methylimino)oxo-lambda6-sulfanyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1206] 1'-(2-{4-[methyl(methylimino)oxo-lambda6-sulfanyl]phenoxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1207] 1'-[2-(4-{[dimethyl(oxo)-lambda6-sulfanylidene]amino}phenoxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1208] 1'-{2-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]ethyl}-2-oxo-1-(2,2,2-trifluoroethyl)-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1209] 5-chloro-1'-{2-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]ethyl}-1-(2,2,2-trifluoroethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1210] 1'-{2-[4-(1-methanesulfonylcyclobutyl)phenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1211] 1'-{2-[3-(difluoromethoxy)-4-methanesulfonyl]phenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1212] 5-chloro-1'-{2-[3-(difluoromethoxy)-4-methanesulfonyl]phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1213] 5-chloro-1'-{2-[(6-{[1-(hydroxymethyl)cyclopropyl]sulfonyl}pyridin-3-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1214] 1'-{1-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]propan-2-yl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1215] 1'-{2-[3-(1,2-difluoroethyl)-4-methanesulfonyl]phenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1216] 1'-{1-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]propan-2-yl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1217] 5-chloro-1'-{2-[4-(1-methanesulfonylcyclobutyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1218] 1'-{2-[(7-fluoro-1-methyl-2-oxo-2,3-dihydro-1H-indol-5-yl)oxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1219] 1'-{2-[3-(1,2-difluoroethyl)-4-methanesulfonyl]phenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1220] 1'-{2-[4-(4-methanesulfonyloxan-4-yl)phenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1221] 5-chloro-1'-{2-[4-(4-methanesulfonyloxan-4-yl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1222] 1'-{2-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]ethyl}-1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1223] 5-chloro-1'-(2-{3-oxo-3H-spiro[2-benzofuran-1,3'-oxetan]-5-yloxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1224] 5-chloro-1'-{2-[4-(3-methanesulfonyloxetan-3-yl)-3-(trifluoromethyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-

piperidin]-2-one; [1225] N-(4-{2-[5-chloro-1-(2-hydroxyethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl]ethoxy}-2-(trifluoromethyl)phenyl)methanesulfonamide; [1226] 1'-{2-[4-(3-methanesulfonyloxetan-3-yl)-3-(trifluoromethyl)phenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1227] 6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-3,4-dihydro-2H-1lambda6,2-benzothiazine-1,1-dione; [1228] 2-oxo-1'-{2-[(1-oxo-2,3-dihydro-1H-isoindol-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1229] 5-chloro-1'-{2-[(1-oxo-2,3-dihydro-1H-isoindol-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1230] 5-chloro-1'-{2-[(7-fluoro-2-oxo-2,3-dihydro-1H-indol-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1231] 1'-{2-[(1,1-dioxo-3,4-dihydro-2H-1lambda6,2-benzothiazin-6-yl)oxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1232] 5-chloro-1'-[2-({2-[(3-methyloxetan-3-yl)oxy]pyrimidin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1233] 3-[4-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]-N-methyloxetane-3-carboxamide; [1234] 3-[4-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]-N,N-dimethyloxetane-3-carboxamide; [1235] 2-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]-1lambda6-thiolane-1,1-dione; [1236] 2-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]-1lambda6-thiolane-1,1-dione; [1237] 1'-[2-(4-{6-methanesulfonyl-2-oxaspiro[3.3]heptan-6-yl}phenoxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1238] 1'-(2-{4-[3-(azetidine-1-carbonyl)oxetan-3-yl]phenoxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1239] 5-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)pyrimidine-2-carboxamide; [1240] 2-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]-1lambda6-thiolane-1,1-dione; [1241] 1'-(2-{4-[3-(ethanesulfonyl)oxetan-3-yl]phenoxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1242] 2-[4-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]-2-methylpropanamide; [1243] 1'-{2-[4-(1,1-dioxo-1lambda6-thiolan-2-yl)phenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1244] 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-(difluoromethyl)benzamide; [1245] 1'-{2-[3,5-difluoro-4-(3-methanesulfonyloxetan-3-yl)phenoxy]ethyl}-1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1246] 1'-[2-({2-[(3-methyloxetan-3-yl)oxy]pyrimidin-5-yl}oxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1247] 1'-(2-{4-[3-(3-fluoroazetidine-1-carbonyl)oxetan-3-yl]phenoxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1248] 4-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-(difluoromethyl)benzamide; [1249] 5-chloro-1'-[2-(4-{6-methanesulfonyl-2-oxaspiro[3.3]heptan-6-yl}phenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1250] 5-chloro-1'-(2-{[2-(1-methanesulfonylethyl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1251] N-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]-N-(2-hydroxyethyl)methanesulfonamide; [1252] 6-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-fluorobenzoyl]-2lambda6-thia-6-azaspiro[3.3]heptane-2,2-dione; [1253] 5-chloro-1'-[2-(3-fluoro-4-{6-methanesulfonyl-2,6-diazaspiro[3.3]heptane-2-carbonyl}phenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1254] 4-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2,6-difluorobenzoyl]-1lambda6-thiomorpholine-1,1-dione; [1255] 5-chloro-1'-(2-{[8-(2-hydroxyethyl)-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1256] 5-chloro-1'-(2-{[2-(1-methanesulfonylcyclopropyl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1257] 1'-{2-[4-(1,1-dioxo-1lambda6-thiomorpholine-4-carbonyl)-3,5-difluorophenoxy]ethyl}-1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1258] 4-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-(trifluoromethyl)benzoyl]-1lambda6-thiomorpholine-1,1-dione; [1259] 1'-{2-[4-(1,1-dioxo-

1-lambda6-thiomorpholine-4-carbonyl)-3-(trifluoromethyl)phenoxy]ethyl}-1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1260] 1'-(2-([6-(2-methanesulfonylpropan-2-yl)pyridin-3-yl]oxy)ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1261] 5-chloro-1'-{2-[4-(3-hydroxy-1-methanesulfonylcyclobutyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1262] 5-(difluoromethyl)-1'-(2-([2-(1-methanesulfonylcyclopropyl)pyrimidin-5-yl]oxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1263] 5-chloro-1'-(2-([8-(3-hydroxycyclobutyl)-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1264] 4-(4-{2-[5-chloro-1-(2-hydroxyethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl]ethoxy}-2-(trifluoromethyl)benzoyl)-1-lambda6-thiomorpholine-1,1-dione; [1265] 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-(1,1-dioxo-1-lambda6-thiolan-3-yl)-2-fluorobenzamide; [1266] 5-chloro-1'-{2-[4-(3-hydroxy-1-methanesulfonylcyclobutyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1267] 5-chloro-1'-{2-[4-(3-hydroxy-1-methanesulfonylcyclobutyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1268] 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-(1,1-dioxo-1-lambda6-thian-4-yl)-2-fluorobenzamide; [1269] 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-(1,1-dioxo-1-lambda6-thiolan-3-yl)-2-fluorobenzamide; [1270] 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-(1,1-dioxo-1-lambda6-thiolan-3-yl)-2-fluorobenzamide; [1271] 5-chloro-1'-(2-([8-(3-hydroxycyclobutyl)-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1272] 5-(difluoromethoxy)-1'-(2-([2-(1-methanesulfonylcyclopropyl)pyrimidin-5-yl]oxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1273] 5-chloro-1'-(2-([2-(1-methanesulfonylethyl)pyrimidin-5-yl]oxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1274] 5-chloro-1'-(2-([2-(1-methanesulfonylethyl)pyrimidin-5-yl]oxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1275] 5-(difluoromethyl)-1'-[2-([7-oxo-8-[3-hydroxy-3-methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1276] 5-chloro-1'-[2-([7-oxo-8-[3-hydroxy-3-methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1277] 5-chloro-1'-[2-([2-oxo-1-[3-hydroxy-3-methylcyclobutyl]-1H,2H,4H-pyrido[2,3-d][1,3]oxazin-6-yl]oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1278] 5-chloro-1'-[2-([7-oxo-8-[3-ethyl-3-hydroxycyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1279] 5-chloro-1'-{2-[(7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1280] 5-chloro-1'-[2-([7-oxo-8-[3-cyclopropyl-3-hydroxycyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1281] 5-chloro-1'-(2-([8-(3-cyclopropyl-3-hydroxycyclobutyl)-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1282] 5-chloro-1'-(2-([8-(3-hydroxy-3-methylazetidin-1-yl)-1,7-naphthyridin-3-yl]oxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1283] 5-chloro-1'-[2-([7-oxo-8-[3-hydroxy-3-methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1284] 5-(difluoromethyl)-1'-[2-([7-oxo-8-[3-(hydroxymethyl)cyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1285] 5-(difluoromethyl)-1'-{2-[(7-oxo-8-{[3-hydroxy-3-methylcyclobutyl]methyl}-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1286] 5-chloro-1'-{2-[(7-oxo-8-{[3-hydroxy-3-methylcyclobutyl]methyl}-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1287] 5-chloro-1'-(2-([8-(3-hydroxyazetidin-1-yl)-1,7-naphthyridin-3-yl]oxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1288] 5-chloro-1'-(2-([8-(3-hydroxy-3-methylcyclobutyl)-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1289] 5-(difluoromethyl)-1'-[2-([8-

[3-(hydroxymethyl)cyclobutyl]-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1290] 5-chloro-1'-[2-({7-oxo-8-[3-hydroxy-3-(propan-2-yl)cyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1291] 5-chloro-1'-(2-({8-(3-ethyl-3-hydroxycyclobutyl)-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1292] 5-chloro-1'-[2-({7-oxo-8-[3-(hydroxymethyl)cyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1293] 5-(difluoromethyl)-1'-[2-({8-[(3-hydroxy-3-methylcyclobutyl)methyl]-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1294] 5-chloro-1'-[2-({7-oxo-8-[3-ethyl-3-hydroxycyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1295] 5-chloro-1'-[2-({7-oxo-8-[3-hydroxy-3-methylcyclobutyl]-7,8-dihydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1296] 5-(difluoromethyl)-1'-[2-({7-oxo-8-[3-hydroxy-3-methylcyclobutyl]-7,8-dihydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1297] 5-(difluoromethoxy)-1'-[2-({7-oxo-8-[3-hydroxy-3-methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1298] 5-chloro-1'-[2-({8-[3-(hydroxymethyl)cyclobutyl]-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1299] 5-chloro-1'-[2-({8-[(3-hydroxy-3-methylcyclobutyl)methyl]-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1300] 5-(difluoromethoxy)-1'-[2-({7-oxo-8-[3-hydroxy-3-methylcyclobutyl]-7,8-dihydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1301] 5-chloro-7-iodo-1'-[2-({7-oxo-8-[3-hydroxy-3-methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1302] 5-chloro-1'-[2-({7-oxo-8-[3-hydroxy-3-(propan-2-yl)cyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1303] 5-chloro-1'-[2-({7-oxo-8-[3-(hydroxymethyl)cyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1304] 5-(difluoromethyl)-1'-(2-({6-(1-hydroxyethyl)-5-(trifluoromethyl)pyridin-3-yl}oxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1305] 1'-(2-({6-(1-hydroxyethyl)-5-(trifluoromethyl)pyridin-3-yl}oxy)ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1306] 5-chloro-1'-(2-({5-(difluoromethyl)-7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-7,8-dihydro-1,8-naphthyridin-3-yl}oxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1307] 5-(difluoromethyl)-1'-(2-({5-(difluoromethyl)-7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-7,8-dihydro-1,8-naphthyridin-3-yl}oxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1308] 5-chloro-1'-[2-({7-oxo-8-[3-hydroxy-3-methylcyclobutyl]-5-(trifluoromethyl)-7,8-dihydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1309] 5-(difluoromethyl)-1'-[2-({7-oxo-8-[3-hydroxy-3-methylcyclobutyl]-5-(trifluoromethyl)-7,8-dihydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1310] 5-chloro-1'-[2-({2-oxo-1-[3-hydroxy-3-methylcyclobutyl]-1H,2H,3H-pyrrolo[2,3-b]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1311] 5-(difluoromethyl)-1'-[2-({2-oxo-1-[3-hydroxy-3-methylcyclobutyl]-1H,2H,3H-pyrrolo[2,3-b]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1312] 5-chloro-1'-[2-({2-oxo-1-[3-hydroxy-3-methylcyclobutyl]-1H,2H,3H-pyrrolo[2,3-b]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1313] 5-(difluoromethyl)-1'-[2-({2-oxo-1-[3-hydroxy-3-methylcyclobutyl]-1H,2H,3H-pyrrolo[2,3-b]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1314] 5-chloro-1'-[2-({1-[3-hydroxy-3-methylcyclobutyl]-1H-pyrrolo[2,3-b]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1315] 5-(difluoromethyl)-1'-[2-({1-[3-hydroxy-3-methylcyclobutyl]-1H-pyrrolo[2,3-b]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1316] 5-chloro-1'-[2-({2-oxo-1-[3-hydroxy-3-methylcyclobutyl]-1H,2H-pyrido[2,3-d]pyrimidin-6-yl}oxy)ethyl]-1,2-

dihydrospiro[indole-3,4'-piperidin]-2-one; [1317] 5-chloro-1'-[2-({2-oxo-1-[3-hydroxy-3-methylcyclobutyl]-1H,2H,3H,4H-pyrido[2,3-d]pyrimidin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1318] 5-chloro-1'-[2-({1-[3-hydroxy-3-methylcyclobutyl]-1H-pyrazolo[3,4-b]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1319] 5-chloro-1'-[2-({3-oxo-4-[3-hydroxy-3-methylcyclobutyl]-3H,4H-pyrido[2,3-b]pyrazin-7-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1320] 5-chloro-1'-(2-{{4-(3-hydroxy-3-methylcyclobutyl)pyrido[3,2-d]pyrimidin-7-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1321] 5-chloro-1'-(2-{{8-(3-hydroxy-3-methylcyclobutyl)-1,7-naphthyridin-3-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1322] 5-chloro-1'-(2-{{8-(3-hydroxy-3-methylazetidin-1-yl)pyrido[2,3-d]pyridazin-3-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1323] 5-chloro-1'-(2-{{4-(3-hydroxy-3-methylazetidin-1-yl)pyrido[3,2-d]pyrimidin-7-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1324] 5-chloro-1'-(2-{{8-(3-hydroxyazetidin-1-yl)pyrido[2,3-d]pyridazin-3-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1325] 5-chloro-1'-(2-{{4-(3-hydroxyazetidin-1-yl)pyrido[3,2-d]pyrimidin-7-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1326] 5-chloro-1'-[2-({3-[3-hydroxy-3-methylcyclobutyl]-3H-imidazo[4,5-b]pyridin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1327] 5-chloro-1'-[2-({2-oxo-3-[3-hydroxy-3-methylcyclobutyl]-1H,2H,3H-imidazo[4,5-b]pyridin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1328] 5-chloro-1'-[2-({8-[3-(hydroxymethyl)azetidin-1-yl]-1,7-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1329] 5-chloro-1'-[2-({2-oxo-3-[(cis)-3-hydroxy-3-methylcyclobutyl]-2H,3H-[1,3]oxazolo[4,5-b]pyridin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1330] 5-(difluoromethyl)-1'-(2-{{2-(3-hydroxy-3-methylazetidin-1-yl)pyrimidin-5-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1331] 5-chloro-1'-[2-((2-{{6-hydroxy-2-azaspiro[3.3]heptan-2-yl}pyrimidin-5-yl}oxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1332] 5-chloro-1'-(2-{{2-(3-ethyl-3-hydroxyazetidin-1-yl)pyrimidin-5-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1333] 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-cyclopropyl-3-hydroxycyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1334] 5-chloro-1'-(2-{{6-(1-hydroxyethyl)-5-(trifluoromethyl)pyridin-3-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1335] 5,7-dichloro-1'-(2-{{2-(1-methanesulfonylcyclopropyl)pyrimidin-5-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1336] 5-chloro-1'-(2-{{6-(1-hydroxy-2-methanesulfonylpropan-2-yl)-5-(trifluoromethyl)pyridin-3-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1337] 5-chloro-1'-(2-{{6-(1-hydroxy-2-methanesulfonylpropan-2-yl)-5-(trifluoromethyl)pyridin-3-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1338] 5-chloro-1'-(2-{{6-(1-hydroxy-2-methanesulfonylpropan-2-yl)-5-(trifluoromethyl)pyridin-3-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1339] 1'-(2-{{8-(3-hydroxy-3-methylcyclobutyl)-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1340] 5-chloro-1'-(2-{{6-(3-hydroxy-3-methylazetidin-1-yl)-5-(trifluoromethyl)pyridin-3-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1341] 1'-(2-{{8-(3-hydroxy-3-methylcyclobutyl)-7-oxo-7,8-dihydro-1,8-naphthyridin-3-yl}oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1342] 5-chloro-1'-(2-{{1-(3-hydroxy-3-methylcyclobutyl)-1H-pyrazolo[3,4-b]pyridin-5-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1343] 5-chloro-1'-(2-{{1-(3-hydroxy-3-methylcyclobutyl)-1H-pyrazolo[3,4-b]pyridin-5-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1344] 5,7-dichloro-1'-(2-{{8-(3-hydroxy-3-methylcyclobutyl)-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1345] 1'-(2-{{1-(3-hydroxy-3-methylcyclobutyl)-1H-pyrrolo[2,3-b]pyridin-5-yl}oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1346] 1'-(2-{{6-(2-hydroxypropan-2-yl)-5-(trifluoromethyl)pyridin-3-yl}oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1347] 5-chloro-1'-[2-

[4-(1-methanesulfonylpiperidin-4-yl)phenoxy]ethyl}-1-(2H3)methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1348] 5-chloro-1-(2H3)methyl-1'-(2-{4-[methyl(methylimino)oxo-lambda6-sulfanyl]-3-(trifluoromethyl)phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1349] 5-chloro-1-(2H3)methyl-1'-(2-{4-[3-(propane-2-sulfonyl)oxetan-3-yl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1350] 5-chloro-1'-(2-{4-(1-methanesulfonylcyclobutyl)phenoxy}ethyl)-1-(2H3)methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1351] 5-chloro-1-(2H3)methyl-1'-(2-{(2-methyl-1-oxo-2,3-dihydro-1H-isoindol-5-yl)oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1352] 5-chloro-1'-(2-{[8-(2-hydroxy-2-methylpropyl)-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy}ethyl)-1-(2H3)methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1353] 5-chloro-1'-(2-{4-(methanesulfonylphenoxy)ethyl}-1-(2H3)methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1354] 5-chloro-1'-(2-{[8-(3-hydroxy-3-methylcyclobutyl)-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy}ethyl)-1-(2H3)methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1355] 5-chloro-1'-(2-{(2-{1-(hydroxymethyl)cyclopropyl}pyrimidin-5-yl}oxy)ethyl)-1-(2H3)methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1356] 6-{2-[5-chloro-1-(2H3)methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl]ethoxy}-8-fluoro-3,4-dihydro-2H-1lambda6,2-benzothiazine-1,1-dione; [1357] 5-chloro-1'-(2-{[2-(3-methanesulfonyloxetan-3-yl)pyrimidin-5-yl]oxy}ethyl)-1-(2H3)methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1358] 5-chloro-1'-(2-{[8-(3-hydroxycyclobutyl)-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy}ethyl)-1-(2H3)methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1359] 5-chloro-1'-(2-{[2-(1-methanesulfonylcyclopropyl)pyrimidin-5-yl]oxy}ethyl)-1-(2H3)methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1360] 5-chloro-1'-(2-{[2-(1-methanesulfonylethyl)pyrimidin-5-yl]oxy}ethyl)-1-(2H3)methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1361] 5-chloro-1'-(2-{[8-(3-hydroxy-3-methylcyclobutyl)-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy}(1,1,2,2-H4)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1362] 5-chloro-1'-(2-{[2-(3-hydroxy-3-methylcyclobutyl)-7-(trifluoromethyl)-2H-indazol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1363] 1'-(2-{[2-(3-hydroxy-3-methylcyclobutyl)-7-(trifluoromethyl)-2H-indazol-5-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1364] 5-chloro-1'-(2-{[2-(3-hydroxyoxetan-3-yl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1365] 5-chloro-1'-(2-{[2-(1-hydroxycyclobutyl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1366] 1'-(2-{[6-(3-hydroxy-3-methylazetidin-1-yl)-5-(trifluoromethyl)pyridin-3-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1367] 1'-(2-{[2-(1-hydroxy-2-methylpropan-2-yl)pyrimidin-5-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1368] 1'-(2-{[1-(3-hydroxy-3-methylcyclobutyl)-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1369] 5-chloro-1'-(2-{[1-(3-hydroxy-3-methylcyclobutyl)-7-(trifluoromethyl)-1H-1,2,3-benzotriazol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1370] 1'-(2-{[1-(3-hydroxy-3-methylcyclobutyl)-7-(trifluoromethyl)-1H-1,2,3-benzotriazol-5-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1371] 1'-(2-{[1-(3-hydroxy-3-methylcyclobutyl)-1H-pyrazolo[3,4-b]pyridin-5-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1372] 1'-(2-{[1-(3-hydroxy-3-methylcyclobutyl)-1H-pyrazolo[3,4-b]pyridin-5-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1373] 1'-(2-{[1-(3-hydroxy-3-methylcyclobutyl)-1H-pyrazolo[3,4-b]pyridin-5-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1374] 5-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-1-(3-hydroxy-3-methylcyclobutyl)-1H-1,3-benzodiazole-7-carbonitrile; [1375] 5-chloro-1'-(2-{[3-(3-hydroxy-3-methylcyclobutyl)-3H-[1,2,3]triazolo[4,5-b]pyridin-6-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1376] 5-chloro-1'-(2-{[2-(2,2-dimethyl-1,3-dioxolan-4-yl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1377] 5-chloro-1'-(2-{[1-(3-hydroxy-3-methylcyclobutyl)-2-methyl-7-(trifluoromethyl)-1H-

1,3-benzodiazol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1378] 1'-(2-{[1-(3-hydroxy-3-methylcyclobutyl)-2-methyl-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1379] 5-chloro-1'-(2-[4-(1-methanesulfonylpiperidin-4-yl)phenoxy]ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1380] 5-chloro-1'-(2-{[7-(difluoromethyl)-1-(3-hydroxy-3-methylcyclobutyl)-1H-1,3-benzodiazol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1381] 1'-(2-{[7-(difluoromethyl)-1-(3-hydroxy-3-methylcyclobutyl)-1H-1,3-benzodiazol-5-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1382] 5-chloro-1'-(2-{[7-(difluoromethoxy)-1-(3-hydroxy-3-methylcyclobutyl)-1H-1,3-benzodiazol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1383] 1'-(2-{[7-(difluoromethoxy)-1-(3-hydroxy-3-methylcyclobutyl)-1H-1,3-benzodiazol-5-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1384] 5-chloro-1'-(2-{[2-(2-hydroxypropan-2-yl)pyrimidin-5-yl]oxy}ethyl)-7-iodo-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1385] 5-(difluoromethyl)-1'-(2-{[1-(3-hydroxy-3-methylcyclobutyl)-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1386] 5-chloro-1'-(2-{[3-(3-hydroxy-3-methylcyclobutyl)-2-methyl-3H-imidazo[4,5-b]pyridin-6-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1387] 5-chloro-1'-(2-{[8-(2-hydroxy-2-methylpropyl)-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1388] 5-chloro-1'-(2-{[1-(3-hydroxy-3-methylcyclobutyl)-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl]oxy}ethyl)-7-iodo-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1389] 5-chloro-1'-(2-{[7-fluoro-1-(3-hydroxy-3-methylcyclobutyl)-1H-1,3-benzodiazol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1390] 1'-(2-{[7-fluoro-1-(3-hydroxy-3-methylcyclobutyl)-1H-1,3-benzodiazol-5-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1391] 5-chloro-1'-(2-{[1-(3-hydroxy-3-methylcyclobutyl)-2-oxo-8-(trifluoromethyl)-1,2,3,4-tetrahydroquinazolin-6-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1392] 1'-(2-{[1-(3-hydroxy-3-methylcyclobutyl)-2-oxo-8-(trifluoromethyl)-1,2,3,4-tetrahydroquinazolin-6-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1393] 5-chloro-1'-(2-{[7,7-dimethyl-5-oxo-5H,7H-furo[3,4-b]pyridin-3-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1394] 5-chloro-1'-(2-{[2-(2-{6-hydroxy-6-methyl-2-azaspiro[3.3]heptan-2-yl]pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1395] 5-chloro-1'-(2-{[1'-(3-hydroxy-3-methylcyclobutyl)-2'-oxo-1',2'-dihydrospiro[cyclopropane-1,3'-pyrrolo[2,3-b]pyridin]-5'-yloxy]ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1396] 5-chloro-1'-(2-{[8-[3-hydroxy-3-(2H3)methylcyclobutyl]-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1397] 5-chloro-1'-(2-{[8-[3-hydroxy-3-(2H3)methylcyclobutyl]-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy}(1,1,2,2-2H4)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1398] 5-chloro-1'-(2-{[1-(3-hydroxy-3-methylcyclobutyl)-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl]oxy}(1,1,2,2-2H4)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1399] 5-chloro-1'-(2-{[1-(3-hydroxy-3-methylcyclobutyl)-7-(trifluoromethyl)-1H-indazol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1400] 1'-(2-{[1-(3-hydroxy-3-methylcyclobutyl)-7-(trifluoromethyl)-1H-indazol-5-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1401] 5-chloro-1'-(2-{[2-(3-hydroxy-3-methylcyclobutyl)-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1402] 5-chloro-1'-(2-{[1-(3-hydroxy-3-methylcyclobutyl)-3-(hydroxymethyl)-2-oxo-7-(trifluoromethyl)-2,3-dihydro-1H-1,3-benzodiazol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1403] 5-chloro-1'-(2-{[1-(3-hydroxy-3-methylcyclobutyl)-2-oxo-7-(trifluoromethyl)-2,3-dihydro-1H-1,3-benzodiazol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1404] 1'-(2-{[1-(3-hydroxy-3-methylcyclobutyl)-3-(hydroxymethyl)-2-oxo-7-(trifluoromethyl)-2,3-dihydro-1H-1,3-benzodiazol-5-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1405] 1'-(2-{[1-(3-hydroxy-3-methylcyclobutyl)-2-oxo-7-(trifluoromethyl)-2,3-dihydro-1H-1,3-

5-diazol-5-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1406] 5-chloro-1'-(2-{[1-(3-hydroxy-3-methylcyclobutyl)-7-(trifluoromethyl)-1H-imidazo[4,5-b]pyridin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; and [1407] 5-chloro-1'-(2-{[3-(3-hydroxy-3-methylcyclobutyl)-4-(trifluoromethyl)-3H-1,3,5-triazainden-6-yl]oxy}ethyl)spiro[indoline-3,4'-piperidin]-2-one.

[1408] In some embodiments, a compound of formula (I) is selected from the group consisting of:

[1409] 5-chloro-1'-(2-{[2-(1-methanesulfonyl)ethyl]pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1410] 5-chloro-1'-[2-({2-[(1S)-1-methanesulfonyl]ethyl]pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1411] 5-chloro-1'-[2-({2-[(1R)-1-methanesulfonyl]ethyl]pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1412] 5-chloro-1'-(2-{[2-(1-methanesulfonylcyclopropyl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1413] 5-(difluoromethyl)-1'-(2-{[2-(1-methanesulfonylcyclopropyl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1414] 5-(difluoromethoxy)-1'-(2-{[2-(1-methanesulfonylcyclopropyl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1415] 5,7-dichloro-1'-[2-(4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1416] 7-chloro-1'-[2-(4-methanesulfonylphenoxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1417] 5-chloro-1'-[2-(4-methanesulfonylphenoxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-7-carbonitrile; [1418] 5-chloro-1'-[(2S)-1-(4-methanesulfonylphenoxy)propan-2-yl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1419] 5-chloro-1'-[(2S)-2-(4-methanesulfonylphenoxy)propyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1420] 1'-[(2S)-1-(4-methanesulfonylphenoxy)propan-2-yl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1421] 1'-[(2S)-2-(4-methanesulfonylphenoxy)propyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1422] 5-chloro-1'-(2-{[1-(3-methanesulfonylpropyl)-1H-indazol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1423] 5-chloro-1'-(2-{[1-(oxetan-3-yl)-1H-indazol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1424] 5-chloro-1'-(2-{[1-(propan-2-yl)-1H-indazol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1425] 5-chloro-1'-[2-(4-{[(cis)-3-hydroxycyclobutyl]sulfonyl}phenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1426] 5-chloro-1-(2-hydroxyethyl)-1'-[2-(4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1427] 1'-[2-(4-methanesulfonylphenoxy)ethyl]-1-[(cis)-3-hydroxycyclobutyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1428] 5-chloro-1'-[2-(4-methanesulfonylphenoxy)ethyl]-1-(2-methoxyethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1429] 1'-[2-(4-methanesulfonylphenoxy)ethyl]-1-(2-methoxyethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1430] 5-chloro-1'-[2-(4-methanesulfonylphenoxy)ethyl]-1-(oxetan-3-yl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1431] 5-chloro-1'-[2-[4-(3-methanesulfonylpropanesulfonyl)phenoxy]ethyl]-1-methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1432] 5-bromo-1-(2-hydroxyethyl)-1'-[2-(4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1433] 1'-[2-(4-methanesulfonylphenoxy)ethyl]-1-(2-methoxyethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1434] 1-[2-(4-methanesulfonylphenoxy)ethyl]-1',2'-dihydrospiro[azepane-4,3'-indol]-2'-one; [1435] 5-chloro-1'-[2-[(1-methyl-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl)oxy]ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1436] 1'-[1-(4-methanesulfonylphenoxy)propan-2-yl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1437] 1-methyl-1'-[2-[(3-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)oxy]ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1438] 1-methyl-1'-[2-[(2-methyl-1-oxo-2,3-dihydro-1H-isoindol-5-yl)oxy]ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1439] 1-methyl-1'-[2-[4-(1-methyl-1H-pyrazol-4-yl)phenoxy]ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1440] 5-(2-{1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2,3-dihydro-1).sub.6-benzothiophene-1,1-dione; [1441] 1-methyl-1'-[2-[(1-methyl-2-oxo-2,3-dihydro-1H-indol-5-yl)oxy]ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1442] 1-

methyl-1'-[2-(quinoxalin-6-yloxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1443] 1-methyl-1'-{2-[4-(5-methyl-1,2,4-oxadiazol-3-yl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1444] 1'-{2-[(1-acetyl-1,2,3,4-tetrahydroquinolin-6-yl)oxy]ethyl}-1-methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1445] 1-methyl-1'-{2-[4-(2-oxopyrrolidin-1-yl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1446] 5-chloro-1'-{2-[4-(1-methanesulfonyl)ethyl]phenoxy}ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1447] 5-chloro-1'-(2-{4-[(1S)-1-methanesulfonyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1448] 5-chloro-1'-(2-{4-[(1R)-1-methanesulfonyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1449] 5-chloro-1'-{2-[4-(3-methanesulfonylpropanesulfonyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1450] 1'-[2-(3,5-difluoro-4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1451] 5-chloro-1'-{2-[4-(cyclopropanesulfonyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1452] 1'-[2-(4-difluoromethanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1453] 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)benzamide; [1454] 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-fluorobenzamide; 5-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)pyridine-2-carboxamide; 5-chloro-1'-[2-(2-fluoro-4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1455] 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-methylbenzamide; 5-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-methanesulfonylbenzamide; [1456] 5-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N,N-dimethylpyridine-2-carboxamide; [1457] 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-methylbenzamide; 5-chloro-1'-{2-[(1-methanesulfonyl-1,2,3,4-tetrahydroquinolin-6-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1458] 5-chloro-1'-{2-[(2-methanesulfonyl-2,3-dihydro-1H-indol-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1459] 1'-{2-[(2-methanesulfonyl-1,2,3,4-tetrahydroisoquinolin-6-yl)oxy]ethyl}-1-methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1460] 1'-{2-[(1-acetyl-1,2,3,4-tetrahydroquinolin-6-yl)oxy]ethyl}-5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1461] 1'-{2-[(2-acetyl-2,3-dihydro-1H-indol-5-yl)oxy]ethyl}-5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1462] 1'-{2-[(2-acetyl-1,2,3,4-tetrahydroisoquinolin-6-yl)oxy]ethyl}-1-methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1463] 1'-{2-[(1-acetyl-1,2,3,4-tetrahydroquinolin-6-yl)oxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1464] 6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-methyl-1,2,3,4-tetrahydroquinoline-1-carboxamide; [1465] 5-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-methyl-2,3-dihydro-1H-indole-2-carboxamide; [1466] 5-chloro-1'-[2-(quinoxalin-6-yloxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1467] 5-chloro-1'-[2-(3-fluoro-4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1468] 5-chloro-1'-{2-[(6-methanesulfonylpyridin-3-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1469] 5-chloro-1'-{2-[(1-methyl-2-oxo-2,3-dihydro-1H-indol-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1470] 1'-[2-(4-methanesulfonylphenoxy)ethyl]-5-methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1471] 1'-[2-(4-methanesulfonylphenoxy)ethyl]-1-methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1472] 5-chloro-1'-{2-[(1-methyl-1H-indazol-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1473] 5-chloro-1'-[2-(4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1474] 1'-{2-[(1-methyl-1H-indazol-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1475] 1'-[2-(4-methanesulfonylphenoxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1476] 5-bromo-1'-[2-(4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1477] 1'-[2-(4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1478] 1'-[2-(3-fluoro-4-

methanesulfonylphenoxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1479] N-[4-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]-N-methylmethanesulfonamide; [1480] N-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]-N-methylmethanesulfonamide; [1481] 1'-[2-(4-{[dimethyl(oxo)- λ .sup.6-sulfanylidene]amino}phenoxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1482] 5-chloro-1'-{2-[(8-methyl-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1483] 1'-{2-[(8-methyl-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl)oxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1484] 1'-{2-[(8-acetyl-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl)oxy]ethyl}-5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1485] 1'-{2-[(8-acetyl-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl)oxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1486] 5-chloro-1'-{2-[(2-oxo-1,2,3,4-tetrahydroquinolin-6-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1487] 2-oxo-1'-{2-[(1,3,3-trimethyl-2-oxo-2,3-dihydro-1H-indol-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1488] 1'-{2-[3,5-difluoro-4-(3-methanesulfonyloxetan-3-yl)phenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1489] 5-chloro-1'-[2-({1-[(cis)-3-hydroxycyclobutyl]-1H-indazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1490] 5-chloro-1'-[2-({1-[(trans)-3-hydroxycyclobutyl]-1H-indazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1491] 5-chloro-1'-[2-(1H-indazol-5-yloxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1492] 5-chloro-1'-{2-[(1-methyl-2-oxo-2,3-dihydro-1H-1,3-benzodiazol-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1493] 5-chloro-1'-[2-(3,5-difluoro-4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1494] 1'-[2-(3,5-difluoro-4-methanesulfonylphenoxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1495] methyl 1-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)benzenesulfonyl]cyclopropane-1-carboxylate; [1496] 5-chloro-1'-(2-{4-[(3-methyloxetan-3-yl)sulfonyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1497] 1'-(2-{4-[(3-methyloxetan-3-yl)sulfonyl]phenoxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1498] 5-chloro-1'-{2-[(2-methanesulfonyl-1,2,3,4-tetrahydroisoquinolin-6-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1499] 1'-{2-[(1-methyl-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl)oxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1500] 2-oxo-1'-(2-{[2-oxo-1-(propan-2-yl)-1,2,3,4-tetrahydroquinolin-6-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1501] 5-chloro-1'-(2-{[2-oxo-1-(propan-2-yl)-1,2,3,4-tetrahydroquinolin-6-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1502] N-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]-N-(2-hydroxyethyl)methanesulfonamide; [1503] 5-chloro-1'-(2-{[1-(oxetan-3-yl)-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1504] 1'-{2-[3-(difluoromethyl)-4-methanesulfonylphenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1505] 1'-{2-[4-(1-methanesulfonylcyclopropyl)phenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1506] 5-chloro-1'-{2-[4-(1-methanesulfonylcyclopropyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1507] 6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-1-methyl-3,4-dihydro-1H-2 λ .sup.6,1-benzothiazine-2,2-dione; [1508] 1'-{2-[(1-methyl-2,2-dioxo-3,4-dihydro-1H-2 λ .sup.6,1-benzothiazin-6-yl)oxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1509] 1'-(2-{[1-(2,2-difluoroethyl)-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1510] 5-chloro-1'-(2-{[1-(2,2-difluoroethyl)-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1511] 1'-{2-[4-methanesulfonyl-3-(trifluoromethyl)phenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1512] 5-chloro-1'-{2-[4-methanesulfonyl-3-(trifluoromethyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1513] 1'-{2-[4-

(3-methanesulfonyloxetan-3-yl)phenoxyethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1514] 5-chloro-1'-[2-(4-methanesulfonyl-3-methylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1515] 1'-[2-(4-methanesulfonyl-3-methylphenoxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1516] 5-chloro-1'-[2-(4-difluoro-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1517] 1'-[2-(4-difluoromethanesulfonylphenoxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1518] 5-chloro-1'-[2-({2-oxo-1-[(cis)-3-hydroxycyclobutyl]-1,2,3,4-tetrahydroquinolin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1519] 5-chloro-1'-[2-({3-oxo-1H,2H,3H,5H,6H,10bH-imidazo[4,3-a]isoquinolin-8-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1520] (S)-5-chloro-1'-[2-({3-oxo-1H,2H,3H,5H,6H,10bH-imidazo[4,3-a]isoquinolin-8-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1521] (R)-5-chloro-1'-[2-({3-oxo-1H,2H,3H,5H,6H,10bH-imidazo[4,3-a]isoquinolin-8-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1522] 2-oxo-1'-[2-({3-oxo-1H,2H,3H,5H,6H,10bH-imidazo[4,3-a]isoquinolin-8-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1523] (S)-2-oxo-1'-[2-({3-oxo-1H,2H,3H,5H,6H,10bH-imidazo[4,3-a]isoquinolin-8-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1524] (R)-2-oxo-1'-[2-({3-oxo-1H,2H,3H,5H,6H,10bH-imidazo[4,3-a]isoquinolin-8-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1525] 5-chloro-1'-[2-({3-oxo-1H,2H,3H,5H,6H,10bH-pyrrolo[2,1-a]isoquinolin-8-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1526] (S)-5-chloro-1'-[2-({3-oxo-1H,2H,3H,5H,6H,10bH-pyrrolo[2,1-a]isoquinolin-8-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1527] (R)-5-chloro-1'-[2-({3-oxo-1H,2H,3H,5H,6H,10bH-pyrrolo[2,1-a]isoquinolin-8-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1528] 2-oxo-1'-[2-({3-oxo-1H,2H,3H,5H,6H,10bH-pyrrolo[2,1-a]isoquinolin-8-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1529] (S)-2-oxo-1'-(2-((3-oxo-1,2,3,5,6,10b-hexahydropyrrolo[2,1-a]isoquinolin-8-yl)oxy)ethyl)spiro[indoline-3,4'-piperidine]-5-carbonitrile; [1530] (R)-2-oxo-1'-(2-((3-oxo-1,2,3,5,6,10b-hexahydropyrrolo[2,1-a]isoquinolin-8-yl)oxy)ethyl)spiro[indoline-3,4'-piperidine]-5-carbonitrile; [1531] N-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2,6-difluorophenyl]-N-methylmethanesulfonamide; [1532] N-[4-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2,6-difluorophenyl]-N-methylmethanesulfonamide; [1533] 5-chloro-1'-[2-(4-methanesulfonyl-3,5-dimethylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1534] 1'-[2-(4-methanesulfonyl-3,5-dimethylphenoxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1535] 5-chloro-1'-{2-[(3-methyl-2-oxo-2,3-dihydro-1,3-benzoxazol-6-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1536] 1'-{2-[(3-methyl-2-oxo-2,3-dihydro-1,3-benzoxazol-6-yl)oxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1537] 1'-{2-[4-(1-methanesulfonylethyl)phenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1538] (S)-1'-{2-[4-(1-methanesulfonylethyl)phenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1539] (R)-1'-{2-[4-(1-methanesulfonylethyl)phenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1540] 5-chloro-1'-{2-[3,5-difluoro-4-(3-methanesulfonyloxetan-3-yl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1541] 1'-{2-[3,5-difluoro-4-(1-methanesulfonylethyl)phenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1542] (S)-1'-{2-[3,5-difluoro-4-(1-methanesulfonylethyl)phenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1543] (R)-1'-{2-[3,5-difluoro-4-(1-methanesulfonylethyl)phenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1544] 1'-{2-[(6-methanesulfonyl-5-methylpyridin-3-yl)oxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1545] 5-chloro-1'-{2-[3,5-difluoro-4-(1-methanesulfonylethyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1546] (S)-5-chloro-1'-{2-[3,5-difluoro-4-(1-

methanesulfonylphenoxylethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1547] (R)-5-chloro-1'-{2-[3,5-difluoro-4-(1-methanesulfonylphenoxylethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1548] 2-oxo-1'-(2-{[2-oxo-1-(propan-2-yl)-2,3-dihydro-1H-indol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1549] 5-chloro-1'-(2-{[2-oxo-1-(propan-2-yl)-2,3-dihydro-1H-indol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1550] 5-chloro-1'-(2-{(2-methyl-1-oxo-2,3-dihydro-1H-isoindol-5-yl)oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1551] 5-chloro-1'-(2-{4-[methyl(methylimino)oxo- λ .sup.6-sulfanyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1552] (S)-5-chloro-1'-(2-{4-[methyl(methylimino)oxo- λ .sup.6-sulfanyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1553] (R)-5-chloro-1'-(2-{4-[methyl(methylimino)oxo- λ .sup.6-sulfanyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1554] 1'-(2-{4-[methyl(methylimino)oxo- λ .sup.6-sulfanyl]phenoxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1555] (S)-1'-(2-{4-[methyl(methylimino)oxo- λ .sup.6-sulfanyl]phenoxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1556] (R)-1'-(2-{4-[methyl(methylimino)oxo- λ .sup.6-sulfanyl]phenoxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1557] 6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-methyl-3,4-dihydro-2H-1 λ .sup.6,2-benzothiazine-1,1-dione; [1558] 1'-(2-{(2-methyl-1,1-dioxo-3,4-dihydro-2H-1 λ .sup.6,2-benzothiazin-6-yl)oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1559] 5-chloro-1'-[2-(3-chloro-4-methanesulfonylphenoxylethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1560] 1'-[2-(3-chloro-4-methanesulfonylphenoxylethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1561] 1'-(2-[4-(1-cyano-1-methylethyl)phenoxy]ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1562] 5-chloro-1'-[2-({6-[(3-methyloxetan-3-yl)sulfonyl]pyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1563] 1-(2-hydroxyethyl)-1'-{2-[4-methanesulfonyl-3-(trifluoromethyl)phenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1564] 5-chloro-1-(2-hydroxyethyl)-1'-{2-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1565] 5-chloro-1'-{2-[(6-methanesulfonyl-5-methylpyridin-3-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1566] 5-chloro-1'-(2-{[1-(2-hydroxyethyl)-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1567] 5-chloro-1'-{2-[3-(difluoromethoxy)-4-methanesulfonylphenoxylethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1568] 1'-{2-[3-(difluoromethoxy)-4-methanesulfonylphenoxylethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1569] 5-chloro-1'-{2-[4-(3-methanesulfonyloxetan-3-yl)-3-(trifluoromethyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1570] 1'-{2-[4-(3-methanesulfonyloxetan-3-yl)-3-(trifluoromethyl)phenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1571] 6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-3,4-dihydro-2H-1 λ .sup.6,2-benzothiazine-1,1-dione; [1572] 1'-{2-[(1,1-dioxo-3,4-dihydro-2H-1 λ .sup.6,2-benzothiazin-6-yl)oxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1573] 1'-{2-[(7-fluoro-1-methyl-2-oxo-2,3-dihydro-1H-indol-5-yl)oxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1574] 2-oxo-1'-{2-[(1-oxo-2,3-dihydro-1H-isoindol-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1575] 5-chloro-1'-{2-[(1-oxo-2,3-dihydro-1H-isoindol-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1576] 5-chloro-1'-[2-({2-[(3-methyloxetan-3-yl)oxy]pyrimidin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1577] 1'-[2-({2-[(3-methyloxetan-3-yl)oxy]pyrimidin-5-yl}oxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1578] 1'-(2-{4-[3-(ethanesulfonyl)oxetan-3-yl]phenoxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1579] 2-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]-1 λ .sup.6-thiolane-1,1-dione; [1580] (S)-2-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-

yl}ethoxy)phenyl]-1λ.sup.6-thiolane-1,1-dione; [1581] (R)-2-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]-1λ.sup.6-thiolane-1,1-dione; [1582] 1'-{2-[4-(1,1-dioxo-1λ.sup.6-thiolan-2-yl)phenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1583] 5-chloro-1'-(2-{3-oxo-3H-spiro[2-benzofuran-1,3'-oxetan]-5-yloxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1584] 5-chloro-1'-{2-[4-(1-methanesulfonylcyclobutyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1585] 1'-{2-[4-(1-methanesulfonylcyclobutyl)phenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1586] 5-chloro-1'-[2-(4-{6-methanesulfonyl-2-oxaspiro[3.3]heptan-6-yl}phenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1587] 1'-[2-(4-{6-methanesulfonyl-2-oxaspiro[3.3]heptan-6-yl}phenoxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1588] 1'-{2-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]ethyl}-2-oxo-1-(2,2,2-trifluoroethyl)-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1589] 5-chloro-1'-{2-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]ethyl}-1-(2,2,2-trifluoroethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1590] N-(4-{2-[5-chloro-1-(2-hydroxyethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl]ethoxy}-2-(trifluoromethyl)phenyl)methanesulfonamide; [1591] 1'-{2-[4-(4-methanesulfonyloxan-4-yl)phenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1592] 5-chloro-1'-{2-[4-(4-methanesulfonyloxan-4-yl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1593] 1'-{2-[3,5-difluoro-4-(3-methanesulfonyloxetan-3-yl)phenoxy]ethyl}-1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1594] 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-hydroxycyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1595] 1'-(2-{[6-(2-methanesulfonylpropan-2-yl)pyridin-3-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1596] 5-chloro-1'-(2-{[8-(2-hydroxyethyl)-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1597] N-{2-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)benzenesulfonyl]ethyl}-N-methylacetamide; [1598] 1'-{2-[(2-acetyl-1,2,3,4-tetrahydroisoquinolin-6-yl)oxy]ethyl}-5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1599] 5-chloro-1'-{2-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1600] 5-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2,3-dihydro-1λ.sup.6-benzothiophene-1,1-dione; [1601] 5-chloro-1'-{2-[3-(difluoromethyl)-4-methanesulfonylphenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1602] 5-chloro-1'-{2-[4-(2-hydroxyethanesulfonyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1603] 5-chloro-1'-{1-[(1-methyl-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl)oxy]propan-2-yl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1604] chloro-1'-[(2S)-1-[(1-methyl-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl)oxy]propan-2-yl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1605] chloro-1'-[(2R)-1-[(1-methyl-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl)oxy]propan-2-yl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1606] 1'-{1-[(1-methyl-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl)oxy]propan-2-yl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1607] (S)-1'-{1-[(1-methyl-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl)oxy]propan-2-yl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1608] (R)-1'-{1-[(1-methyl-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl)oxy]propan-2-yl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1609] 5-chloro-1'-[1-(4-methanesulfonylphenoxy)propan-2-yl]-1-[3-hydroxycyclobutyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1610] 5-chloro-1'-[(2S)-1-(4-methanesulfonylphenoxy)propan-2-yl]-1-[(cis)-3-hydroxycyclobutyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1611] 5-chloro-1'-[(2S)-1-(4-methanesulfonylphenoxy)propan-2-yl]-1-[(trans)-3-hydroxycyclobutyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1612] 5-chloro-1'-[(2R)-1-(4-methanesulfonylphenoxy)propan-2-yl]-1-[(cis)-3-hydroxycyclobutyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1613] 5-chloro-1'-[(2R)-1-(4-methanesulfonylphenoxy)propan-2-yl]-1-[(trans)-3-hydroxycyclobutyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1614] 5-chloro-1'-[(2S)-1-[3-(difluoromethyl)-4-

methanesulfonylphenoxy]propan-2-yl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1615] 5-chloro-1'-[(2S)-1-[(1-methyl-1H-indazol-5-yl)oxy]propan-2-yl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1616] 1'-[(2S)-1-[3-(difluoromethyl)-4-methanesulfonylphenoxy]propan-2-yl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1617] 1'-[(2S)-1-(3,5-difluoro-4-methanesulfonylphenoxy)propan-2-yl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1618] N-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]methanesulfonamide; [1619] 5-chloro-1'-[2-(4-{[(trans)-3-hydroxycyclobutyl]sulfonyl}phenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1620] 5-chloro-1'-{2-[3-(1,2-dihydroxyethyl)-4-methanesulfonylphenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1621] 5-chloro-1'-{2-[3-(1,2-difluoroethyl)-4-methanesulfonylphenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1622] 5-chloro-1'-(2-{3-[(1S)-1,2-difluoroethyl]-4-methanesulfonylphenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1623] 5-chloro-1'-(2-{3-[(1R)-1,2-difluoroethyl]-4-methanesulfonylphenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1624] 1'-{2-[3-(1,2-difluoroethyl)-4-methanesulfonylphenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1625] 1'-(2-{3-[(1S)-1,2-difluoroethyl]-4-methanesulfonylphenoxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1626] 1'-(2-{3-[(1R)-1,2-difluoroethyl]-4-methanesulfonylphenoxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1627] 5-chloro-1'-{2-[4-methanesulfonyl-3-(oxetan-3-yl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1628] 5-chloro-1'-[2-({2-oxo-1-[(trans)-3-hydroxycyclobutyl]-1,2,3,4-tetrahydroquinolin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1629] 5-chloro-1'-[2-(4-{[1-(hydroxymethyl)cyclopropyl]sulfonyl}phenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1630] 1-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)benzenesulfonyl]-N-methylcyclopropane-1-carboxamide; [1631] 1-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)benzenesulfonyl]-N,N-dimethylcyclopropane-1-carboxamide; [1632] 1'-[2-(4-{[1-(azetidine-1-carbonyl)cyclopropyl]sulfonyl}phenoxy)ethyl]-5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1633] 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-fluoro-N-methylbenzamide; [1634] 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-fluoro-N,N-dimethylbenzamide; [1635] 5-chloro-1'-{2-[3-fluoro-4-(morpholine-4-carbonyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1636] 4-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-fluorobenzoyl]-1 λ .sup.6-thiomorpholine-1,1-dione; [1637] 5-chloro-1'-{2-[3-fluoro-4-(3-methanesulfonylazetidine-1-carbonyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1638] 1'-{2-[4-(azetidine-1-carbonyl)-3-fluorophenoxy]ethyl}-5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1639] 5-chloro-1'-{2-[3-fluoro-4-(3-hydroxyazetidine-1-carbonyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1640] 1'-{2-[4-(3-aminoazetidine-1-carbonyl)-3-fluorophenoxy]ethyl}-5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1641] N-{1-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-fluorobenzoyl]azetidin-3-yl}methanesulfonamide; [1642] 4-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-fluorobenzamide; [1643] 6-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-fluorobenzoyl]-2 λ .sup.6-thia-6-azaspiro[3.3]heptane-2,2-dione; [1644] 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-(1,1-dioxo-1 λ .sup.6-thian-4-yl)-2-fluorobenzamide; [1645] 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-(1,1-dioxo-1 λ .sup.6-thiolan-3-yl)-2-fluorobenzamide; [1646] 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-[(3S)-1,1-dioxo-1 λ .sup.6-thiolan-3-yl]-2-fluorobenzamide; [1647] 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-[(3R)-1,1-dioxo-1 λ .sup.6-thiolan-3-yl]-2-fluorobenzamide; [1648] 5-chloro-1'-{2-[4-(3-methyl-1H-pyrazol-4-yl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-

piperidin]-2-one; [1649] 6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-methyl-1,2,3,4-tetrahydroisoquinoline-2-carboxamide; [1650] 6-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-methyl-1,2,3,4-tetrahydroisoquinoline-2-carboxamide; [1651] 5-chloro-1'-[2-({3-methyl-2-oxo-1H,2H,3H-imidazo[4,5-b]pyridin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1652] 5-chloro-1'-{2-[(6-{[1-(hydroxymethyl)cyclopropyl]sulfonyl}pyridin-3-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1653] 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-(difluoromethyl)benzamide; [1654] 4-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-(difluoromethyl)benzamide; [1655] 1'-{2-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]ethyl}-1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1656] 1'-(2-{4-[3-(3-fluoroazetidine-1-carbonyl)oxetan-3-yl]phenoxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1657] 1'-(2-{4-[3-(azetidine-1-carbonyl)oxetan-3-yl]phenoxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1658] 3-[4-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]-N,N-dimethyloxetane-3-carboxamide; [1659] 3-[4-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]-N-methyloxetane-3-carboxamide; [1660] 5-chloro-1'-{2-[(7-fluoro-2-oxo-2,3-dihydro-1H-indol-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1661] 2-[4-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]-2-methylpropanamide; [1662] 5-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)pyrimidine-2-carboxamide; [1663] 4-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2,6-difluorobenzoyl]-1 λ .sup.6-thiomorpholine-1,1-dione; [1664] 1'-{2-[4-(1,1-dioxo-1 λ .sup.6-thiomorpholine-4-carbonyl)-3,5-difluorophenoxy]ethyl}-1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1665] 1'-{2-[4-(1,1-dioxo-1 λ .sup.6-thiomorpholine-4-carbonyl)-3-(trifluoromethyl)phenoxy]ethyl}-1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1666] 1'-{1-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]propan-2-yl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1667] (S)-1'-{1-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]propan-2-yl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1668] (R)-1'-{1-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]propan-2-yl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1669] 5-chloro-1'-{2-[3-(difluoromethyl)-4-methanesulfonylphenoxy]ethyl}-1-(2-hydroxyethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1670] 1'-{2-[3-(difluoromethyl)-4-methanesulfonylphenoxy]ethyl}-1-(2-hydroxyethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1671] 5-chloro-1'-[2-({7-oxo-8-[(trans)-3-hydroxycyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1672] 5-chloro-1'-{2-[4-(3-hydroxy-1-methanesulfonylcyclobutyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1673] 5-chloro-1'-(2-{4-[(cis)-3-hydroxy-1-methanesulfonylcyclobutyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1674] 5-chloro-1'-(2-{4-[(trans)-3-hydroxy-1-methanesulfonylcyclobutyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1675] 4-(4-{2-[5-chloro-1-(2-hydroxyethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl]ethoxy}-2-(trifluoromethyl)benzoyl)-1 λ .sup.6-thiomorpholine-1,1-dione; [1676] 4-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-(trifluoromethyl)benzoyl]-1 λ .sup.6-thiomorpholine-1,1-dione; [1677] 5-chloro-1'-[2-(3-fluoro-4-{6-methanesulfonyl-2,6-diazaspiro[3.3]heptane-2-carbonyl}phenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1678] 5-(difluoromethyl)-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1679] 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1680] 5-chloro-1'-[2-({2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H,2H,4H-pyrido[2,3-d][1,3]oxazin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-

3,4'-piperidin]-2-one; [1681] 5-chloro-1'-[2-({7-oxo-8-[(trans)-3-ethyl-3-hydroxycyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1682] 5-chloro-1'-{2-[(7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1683] 5-chloro-1'-[2-({7-oxo-8-[(trans)-3-cyclopropyl-3-hydroxycyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1684] 5-chloro-1'-(2-{[8-(3-cyclopropyl-3-hydroxycyclobutyl)-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1685] 5-chloro-1'-(2-{[8-(3-hydroxy-3-methylazetidin-1-yl)-1,7-naphthyridin-3-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1686] 5-chloro-1'-[2-({7-oxo-8-[(trans)-3-hydroxy-3-methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1687] 5-(difluoromethyl)-1'-[2-({7-oxo-8-[(cis)-3-(hydroxymethyl)cyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1688] 5-(difluoromethyl)-1'-[2-({7-oxo-8-[(trans)-3-(hydroxymethyl)cyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1689] 5-(difluoromethyl)-1'-{2-[(7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]methyl)-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy}ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1690] 5-(difluoromethyl)-1'-{2-[(7-oxo-8-[(trans)-3-hydroxy-3-methylcyclobutyl]methyl)-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy}ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1691] 5-chloro-1'-{2-[(7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]methyl)-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy}ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1692] 5-chloro-1'-(2-{[8-(3-hydroxyazetidin-1-yl)-1,7-naphthyridin-3-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1693] 5-chloro-1'-(2-{[8-(3-hydroxy-3-methylcyclobutyl)-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1694] 5-(difluoromethyl)-1'-[2-({8-[3-(hydroxymethyl)cyclobutyl]-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1695] 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-(propan-2-yl)cyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1696] 5-chloro-1'-(2-{[8-(3-ethyl-3-hydroxycyclobutyl)-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1697] 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-(hydroxymethyl)cyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1698] 5-(difluoromethyl)-1'-[2-({8-[3-(hydroxy-3-methylcyclobutyl)methyl]-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1699] 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-ethyl-3-hydroxycyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1700] 5-chloro-1'-{2-[(7-oxo-8-[(trans)-3-hydroxy-3-methylcyclobutyl]methyl)-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy}ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1701] 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-7,8-dihydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1702] 5-(difluoromethyl)-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-7,8-dihydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1703] 5-(difluoromethoxy)-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1704] 5-chloro-1'-[2-({8-[3-(hydroxymethyl)cyclobutyl]-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1705] 5-chloro-1'-[2-({8-[3-(hydroxy-3-methylcyclobutyl)methyl]-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1706] 5-(difluoromethoxy)-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-7,8-dihydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1707] 5-chloro-7-iodo-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-

yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1708] 5-chloro-1'-[2-({7-oxo-8-[(trans)-3-hydroxy-3-(propan-2-yl)cyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1709] 5-chloro-1'-[2-({7-oxo-8-[(trans)-3-(hydroxymethyl)cyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1710] 5-(difluoromethyl)-1'-(2-{{6-(1-hydroxyethyl)-5-(trifluoromethyl)pyridin-3-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1711] 1'-(2-{{6-(1-hydroxyethyl)-5-(trifluoromethyl)pyridin-3-yl}oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1712] 5-chloro-1'-(2-{{5-(difluoromethyl)-7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-7,8-dihydro-1,8-naphthyridin-3-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1713] 5-(difluoromethyl)-1'-(2-{{5-(difluoromethyl)-7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-7,8-dihydro-1,8-naphthyridin-3-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1714] 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-5-(trifluoromethyl)-7,8-dihydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1715] 5-(difluoromethyl)-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-5-(trifluoromethyl)-7,8-dihydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1716] 5-chloro-1'-[2-({2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H,2H,3H-pyrrolo[2,3-b]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1717] 5-(difluoromethyl)-1'-[2-({2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H,2H,3H-pyrrolo[2,3-b]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1718] 5-chloro-1'-[2-({2-oxo-1-[(trans)-3-hydroxy-3-methylcyclobutyl]-1H,2H,3H-pyrrolo[2,3-b]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1719] 5-(difluoromethyl)-1'-[2-({2-oxo-1-[(trans)-3-hydroxy-3-methylcyclobutyl]-1H,2H,3H-pyrrolo[2,3-b]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1720] 5-chloro-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-pyrrolo[2,3-b]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1721] 5-(difluoromethyl)-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-pyrrolo[2,3-b]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1722] 5-chloro-1'-[2-({2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H,2H-pyrido[2,3-d]pyrimidin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1723] 5-chloro-1'-[2-({2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H,2H,3H,4H-pyrido[2,3-d]pyrimidin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1724] 5-chloro-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-pyrazolo[3,4-b]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1725] 5-chloro-1'-[2-({3-oxo-4-[(cis)-3-hydroxy-3-methylcyclobutyl]-3H,4H-pyrido[2,3-b]pyrazin-7-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1726] 5-chloro-1'-(2-{{4-(3-hydroxy-3-methylcyclobutyl)pyrido[3,2-d]pyrimidin-7-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1727] 5-chloro-1'-(2-{{8-(3-hydroxy-3-methylcyclobutyl)-1,7-naphthyridin-3-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1728] 5-chloro-1'-(2-{{8-(3-hydroxy-3-methylazetidin-1-yl)pyrido[2,3-d]pyridazin-3-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1729] 5-chloro-1'-(2-{{4-(3-hydroxy-3-methylazetidin-1-yl)pyrido[3,2-d]pyrimidin-7-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1730] 5-chloro-1'-(2-{{8-(3-hydroxyazetidin-1-yl)pyrido[2,3-d]pyridazin-3-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1731] 5-chloro-1'-(2-{{4-(3-hydroxyazetidin-1-yl)pyrido[3,2-d]pyrimidin-7-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1732] 5-chloro-1'-[2-({3-[(cis)-3-hydroxy-3-methylcyclobutyl]-3H-imidazo[4,5-b]pyridin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1733] 5-chloro-1'-[2-({2-oxo-3-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H,2H,3H-imidazo[4,5-b]pyridin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1734] 5-chloro-1'-[2-({8-[3-(hydroxymethyl)azetidin-1-yl]-1,7-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1735] 1'-[2-({6-[(1S)-1-hydroxyethyl]-5-(trifluoromethyl)pyridin-3-yl}oxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1736] 1'-[2-({6-

[(1R)-1-hydroxyethyl]-5-(trifluoromethyl)pyridin-3-yl}oxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1737] 5-chloro-1'-[2-({2-[(1S)-1,2-dihydroxyethyl]pyrimidin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1738] 5-chloro-1'-[2-({2-[(1R)-1,2-dihydroxyethyl]pyrimidin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1739] 5-chloro-1'-[2-({2-[(2S)-1,2-dihydroxypropan-2-yl]pyrimidin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1740] 5-chloro-1'-[2-({2-[(2R)-1,2-dihydroxypropan-2-yl]pyrimidin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1741] 5-chloro-1'-[2-({6-[(1S)-1,2-dihydroxyethyl]-5-(trifluoromethyl)pyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1742] 5-chloro-1'-[2-({6-[(1R)-1,2-dihydroxyethyl]-5-(trifluoromethyl)pyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1743] 5-chloro-1'-[2-({6-[(2S)-1,2-dihydroxypropan-2-yl]-5-(trifluoromethyl)pyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1744] 5-chloro-1'-[2-({6-[(2R)-1,2-dihydroxypropan-2-yl]-5-(trifluoromethyl)pyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1745] 5-chloro-1'-[2-({8-[(cis)-3-hydroxy-3-methylcyclobutyl]-1,7-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1746] 5-chloro-1'-[2-({8-[(trans)-3-hydroxy-3-methylcyclobutyl]-1,7-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1747] 5-chloro-1'-[2-({4-[(cis)-3-hydroxy-3-methylcyclobutyl]pyrido[3,2-d]pyrimidin-7-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1748] 5-chloro-1'-[2-({4-[(trans)-3-hydroxy-3-methylcyclobutyl]pyrido[3,2-d]pyrimidin-7-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1749] 5-chloro-1'-[2-({8-[(cis)-3-hydroxy-3-methylcyclobutyl]pyrido[2,3-d]pyridazin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1750] 5-chloro-1'-[2-({8-[(trans)-3-hydroxy-3-methylcyclobutyl]pyrido[2,3-d]pyridazin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1751] 5-chloro-1'-[2-({2-[(cis)-3-hydroxy-3-methylcyclobutyl]pyrimidin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1752] 5-chloro-1'-[2-({2-[(trans)-3-hydroxy-3-methylcyclobutyl]pyrimidin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1753] 5-chloro-1'-[2-({2-oxo-3-[(cis)-3-hydroxy-3-methylcyclobutyl]-2H,3H-[1,3]oxazolo[4,5-b]pyridin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1754] 5-(difluoromethyl)-1'-(2-{[2-(3-hydroxy-3-methylazetidin-1-yl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1755] 5-chloro-1'-{2-[(2-{6-hydroxy-2-azaspiro[3.3]heptan-2-yl}pyrimidin-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1756] 5-chloro-1'-(2-{[2-(3-ethyl-3-hydroxyazetidin-1-yl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1757] 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-cyclopropyl-3-hydroxycyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1758] 5-chloro-1'-(2-{[6-(1-hydroxyethyl)-5-(trifluoromethyl)pyridin-3-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1759] 5,7-dichloro-1'-(2-{[2-(1-methanesulfonylcyclopropyl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1760] 5-chloro-1'-(2-{[2-(1,2-dihydroxyethyl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1761] 5-chloro-1'-[2-({2-[1-(hydroxymethyl)cyclopropyl]pyrimidin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1762] 5-chloro-1'-(2-{[2-(1-hydroxy-2-methylpropan-2-yl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1763] 5-chloro-1'-(2-{[5-(difluoromethyl)-7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-7,8-dihydro-1,8-naphthyridin-3-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1764] 5-(difluoromethyl)-1'-(2-{[5-(difluoromethyl)-7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-7,8-dihydro-1,8-naphthyridin-3-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1765] 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-5-(trifluoromethyl)-7,8-dihydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1766] 5-(difluoromethyl)-1'-(2-{[7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-5-(trifluoromethyl)-7,8-dihydro-1,8-naphthyridin-3-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1767] 5-chloro-1'-(2-{4-[1-(dimethylphosphoryl)cyclopropyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-

piperidin]-2-one; [1768] 5-chloro-1'-[2-({2-[1-(dimethylphosphoryl)cyclopropyl]pyrimidin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1769] 5-chloro-1'-[2-({2-oxo-1-[(1*r*,3*s*)-3-hydroxy-3-methylcyclobutyl]-1*H*,2*H*,3*H*-pyrrolo[2,3-*b*]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1770] 5-(difluoromethyl)-1'-[2-({2-oxo-1-[(*cis*)-3-hydroxy-3-methylcyclobutyl]-1*H*,2*H*,3*H*-pyrrolo[2,3-*b*]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1771] 5-chloro-1'-[2-({1-[(*cis*)-3-hydroxy-3-methylcyclobutyl]-1*H*-pyrrolo[2,3-*b*]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1772] 5-chloro-1'-[2-({1-[(*cis*)-3-hydroxy-3-methylcyclobutyl]-1*H*-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1773] 5-chloro-1'-[2-({2-oxo-1-[(*cis*)-3-hydroxy-3-methylcyclobutyl]-8-(trifluoromethyl)-1,2,3,4-tetrahydroquinolin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1774] 2-oxo-1'-[2-({2-oxo-1-[(*cis*)-3-hydroxy-3-methylcyclobutyl]-8-(trifluoromethyl)-1,2,3,4-tetrahydroquinolin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1775] 5-chloro-1'-[2-({2-oxo-1-[(1*r*,3*s*)-3-hydroxy-3-methylcyclobutyl]-1*H*,2*H*,3*H*,4*H*-pyrido[2,3-*d*]pyrimidin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1776] 5-chloro-1'-[2-({1-[(*cis*)-3-hydroxy-3-methylcyclobutyl]-1*H*-pyrazolo[3,4-*b*]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1777] 5-chloro-1'-[2-({3-oxo-4-[(*cis*)-3-hydroxy-3-methylcyclobutyl]-3*H*,4*H*-pyrido[2,3-*b*]pyrazin-7-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1778] 5-chloro-1'-[2-({3-[(*cis*)-3-hydroxy-3-methylcyclobutyl]-3*H*-imidazo[4,5-*b*]pyridin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1779] 5-chloro-1'-(2-{[6-(2-hydroxypropan-2-yl)-5-(trifluoromethyl)pyridin-3-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1780] 5-chloro-1'-(2-{[2-(3-hydroxy-3-methylcyclobutyl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1781] 5-chloro-1'-[2-({2-oxo-3-[(*cis*)-3-hydroxy-3-methylcyclobutyl]-1*H*,2*H*,3*H*-imidazo[4,5-*b*]pyridin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1782] 5-chloro-1'-[2-({2-[(1*R* or 1*S*)-1,2-dihydroxyethyl]pyrimidin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1783] 5-chloro-1'-[2-({2-[(1*S* or 1*R*)-1,2-dihydroxyethyl]pyrimidin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1784] 5-chloro-1'-[2-({2-[(*cis*)-3-hydroxy-3-methylcyclobutyl]pyrimidin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1785] 5-chloro-1'-[2-({2-[(*cis*) or (*trans*)-3-hydroxy-3-methylcyclobutyl]pyrimidin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1786] 5-chloro-1'-(2-{[6-(1-hydroxy-2-methanesulfonylpropan-2-yl)-5-(trifluoromethyl)pyridin-3-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1787] 5-chloro-1'-[2-({6-[(2*S* or 2*R*)-1-hydroxy-2-methanesulfonylpropan-2-yl]-5-(trifluoromethyl)pyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1788] 5-chloro-1'-[2-({6-[(2*R* or 2*S*)-1-hydroxy-2-methanesulfonylpropan-2-yl]-5-(trifluoromethyl)pyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1789] 2-oxo-1'-[2-({7-oxo-8-[(*cis*)-3-hydroxy-3-methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1790] 5-chloro-1'-(2-{[6-(3-hydroxy-3-methylazetidin-1-yl)-5-(trifluoromethyl)pyridin-3-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1791] 2-oxo-1'-[2-({7-oxo-8-[(*cis*)-3-hydroxy-3-methylcyclobutyl]-7,8-dihydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1792] 5-chloro-1'-(2-{[1-(3-hydroxy-3-methylcyclobutyl)-1*H*-pyrazolo[3,4-*b*]pyridin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1793] 5-chloro-1'-[2-({1-[(*trans*)-3-hydroxy-3-methylcyclobutyl]-1*H*-pyrazolo[3,4-*b*]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1794] 5,7-dichloro-1'-[2-({7-oxo-8-[(*cis*)-3-hydroxy-3-methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1795] 2-oxo-1'-[2-({1-[(*cis*)-3-hydroxy-3-methylcyclobutyl]-1*H*-pyrrolo[2,3-*b*]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1796] 1'-[2-({[6-(2-hydroxypropan-2-yl)-5-(trifluoromethyl)pyridin-3-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1797] 5-chloro-1'-[2-

[4-(1-methanesulfonylpiperidin-4-yl)phenoxy]ethyl}-1-(2H3)methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1798] 5-chloro-1-(2H3)methyl-1'-(2-{4-[methyl(methylimino)oxo- λ .sup.6-sulfanyl]-3-(trifluoromethyl)phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1799] 5-chloro-1-(2H3)methyl-1'-(2-{4-[3-(propane-2-sulfonyl)oxetan-3-yl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1800] 5-chloro-1'-{2-[4-(1-methanesulfonylcyclobutyl)phenoxy]ethyl}-1-(2H3)methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1801] 5-chloro-1-(2H3)methyl-1'-{2-[(2-methyl-1-oxo-2,3-dihydro-1H-isoindol-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1802] 5-chloro-1'-(2-{[8-(2-hydroxy-2-methylpropyl)-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy}ethyl)-1-(2H3)methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1803] 5-chloro-1'-[2-(4-methanesulfonylphenoxy)ethyl]-1-(2H3)methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1804] 5-chloro-1-(2H3)methyl-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1805] 5-chloro-1'-[2-({2-[1-(hydroxymethyl)cyclopropyl]pyrimidin-5-yl}oxy)ethyl]-1-(2H3)methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1806] 6-{2-[5-chloro-1-(2H3)methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl]ethoxy}-8-fluoro-3,4-dihydro-2H-1 λ .sup.6,2-benzothiazine-1,1-dione; [1807] 5-chloro-1'-(2-{[2-(3-methanesulfonyloxetan-3-yl)pyrimidin-5-yl]oxy}ethyl)-1-(2H3)methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1808] 5-chloro-1-(2H3)methyl-1'-[2-({7-oxo-8-[(cis)-3-hydroxycyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1809] 5-chloro-1'-(2-{[2-(1-methanesulfonylcyclopropyl)pyrimidin-5-yl]oxy}ethyl)-1-(2H3)methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1810] 5-chloro-1'-(2-{[2-(1-methanesulfonylethyl)pyrimidin-5-yl]oxy}ethyl)-1-(2H3)methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1811] 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)(1,1,2,2-2H4)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1812] 5-chloro-1'-[2-({2-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-2H-indazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1813] 2-oxo-1'-[2-({2-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-2H-indazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1814] 5-chloro-1'-(2-{[2-(3-hydroxyoxetan-3-yl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1815] 5-chloro-1'-(2-{[2-(1-hydroxycyclobutyl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1816] 1'-(2-{[6-(3-hydroxy-3-methylazetidin-1-yl)-5-(trifluoromethyl)pyridin-3-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1817] 1'-(2-{[2-(1-hydroxy-2-methylpropan-2-yl)pyrimidin-5-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1818] 2-oxo-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1819] 5-chloro-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,2,3-benzotriazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1820] 2-oxo-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,2,3-benzotriazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1821] 1'-(2-{[1-(3-hydroxy-3-methylcyclobutyl)-1H-pyrazolo[3,4-b]pyridin-5-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1822] 2-oxo-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-pyrazolo[3,4-b]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1823] 1'-(2-{[1-(3-hydroxy-3-methylcyclobutyl)-1H-pyrazolo[3,4-b]pyridin-5-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1824] 5-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-1,3-benzodiazole-7-carbonitrile; [1825] 5-chloro-1'-[2-({3-[(cis)-3-hydroxy-3-methylcyclobutyl]-3H-[1,2,3]triazolo[4,5-b]pyridin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1826] 5-chloro-1'-(2-{[2-(2,2-dimethyl-1,3-dioxolan-4-yl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1827] 5-chloro-1'-[2-({2-methyl-1-[(cis)-3-hydroxy-3-

methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1828] 1'-[2-({2-methyl-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1829] 5-chloro-1'-[2-[4-(1-methanesulfonylpiperidin-4-yl)phenoxy]ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1830] 5-chloro-1'-(2-{[7-(difluoromethyl)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-1,3-benzodiazol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1831] 1'-(2-{[7-(difluoromethyl)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-1,3-benzodiazol-5-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1832] 5-chloro-1'-(2-{[7-(difluoromethoxy)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-1,3-benzodiazol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1833] 1'-(2-{[7-(difluoromethoxy)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-1,3-benzodiazol-5-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1834] 5-chloro-1'-(2-{[2-(2-hydroxypropan-2-yl)pyrimidin-5-yl]oxy}ethyl)-7-iodo-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1835] 5-(difluoromethyl)-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1836] 5-chloro-1'-[2-({2-methyl-3-[(cis)-3-hydroxy-3-methylcyclobutyl]-3H-imidazo[4,5-b]pyridin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1837] 5-chloro-1'-(2-{[8-(2-hydroxy-2-methylpropyl)-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1838] 5-chloro-7-iodo-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1839] 5-chloro-1'-[2-({7-fluoro-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1840] 1'-[2-({7-fluoro-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1841] 5-chloro-1'-[2-({2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-8-(trifluoromethyl)-1,2,3,4-tetrahydroquinazolin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1842] 2-oxo-1'-[2-({2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-8-(trifluoromethyl)-1,2,3,4-tetrahydroquinazolin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1843] 5-chloro-1'-[2-({7,7-dimethyl-5-oxo-5H,7H-furo[3,4-b]pyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1844] 5-chloro-1'-[2-({2-[6-hydroxy-6-methyl-2-azaspiro[3.3]heptan-2-yl]pyrimidin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1845] 5-chloro-1'-(2-{2'-oxo-1'-[(cis)-3-hydroxy-3-methylcyclobutyl]-1',2'-dihydrospiro[cyclopropane-1,3'-pyrrolo[2,3-b]pyridin]-5'-yloxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1846] 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-(2H3)methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1847] 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-(2H3)methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)(1,1,2,2-2H4)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1848] 5-chloro-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl}oxy)(1,1,2,2-2H4)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1849] 5-chloro-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-indazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1850] 2-oxo-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-indazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1851] 5-chloro-1'-[2-({2-[(1r,3s)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1852] 5-chloro-1'-(2-{[3-(hydroxymethyl)-2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-2,3-dihydro-1H-1,3-benzodiazol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1853] 5-chloro-1'-[2-({2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-2,3-dihydro-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1854] 1'-(2-{[3-(hydroxymethyl)-2-oxo-1-[(cis)-3-hydroxy-3-

methylcyclobutyl]-7-(trifluoromethyl)-2,3-dihydro-1H-1,3-benzodiazol-5-yl}oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1855] 2-oxo-1'-[2-({2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-2,3-dihydro-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile; [1856] 5-chloro-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-imidazo[4,5-b]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one; [1857] 5-chloro-1'-(2-{3-[(cis)-3-hydroxy-3-methylcyclobutyl]-4-(trifluoromethyl)-3H-1,3,5-triazainden-6-yloxy}ethyl)spiro[indoline-3,4'-piperidin]-2-one,

or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing. [1858] Isotopically labeled forms of any of the foregoing are also embraced, such as deuterated or tritiated forms (wherein at least one hydrogen is replaced by at least one deuterium or tritium) of any of the specific compounds detailed herein. Mixtures of any of the foregoing are also embraced and described. Prodrugs of any of the foregoing are also embraced herein.

[1859] As a non-limiting example, compounds of formula (II), formula (I'), or formula (I), or any embodiment or variation thereof, are provided, wherein any one or more H atoms are replaced with deuterium. For example, compounds of formula (II), formula (I'), or formula (I), or any embodiment or variation thereof, are provided wherein L^{sup.1} is C_{sub.1-6}alkylene, wherein one or more H atoms of the C_{sub.1-6}alkylene are replaced with deuterium. For example, compounds of formula (II), formula (I'), or formula (I), or any embodiment or variation thereof, are provided wherein L^{sup.1} is —(CD_{sub.2})_{sub.1-6}—. In some embodiments of formula (II), formula (I'), or formula (I), or any embodiment or variation thereof, L^{sup.1} is —(CD_{sub.2})—(CD_{sub.2})—.

[1860] In some embodiments, compounds of formula (II), formula (I'), or formula (I) contain one or more hydrogen atoms that are replaced with deuterium, wherein deuterium is present in an amount that is greater than its natural abundance. Thus, as used herein, designation of an atom as deuterium at a position indicates that the abundance of deuterium is significantly greater than the natural abundance of deuterium. Unless otherwise stated, when a position is designated specifically as “H” or “hydrogen,” the position is understood to have hydrogen at its naturally abundant isotopic composition. Also, unless otherwise stated, when a position is designated specifically as “D” or “deuterium,” the position is understood to have deuterium at an abundance that is significantly greater than the natural abundance of deuterium, e.g., at least 3000 times greater than the natural abundance of deuterium, which is about 0.015% (i.e., the term “D” or “deuterium” indicates at least about 45% incorporation of deuterium).

[1861] Compound Names included in Table 1 and in the list in the paragraph above were generated ChemDraw® software version 18.1.0.458, ChemDraw® software version 18.0.0.231, or Collaborative Drug Discovery Inc. (CDD) CDD Vault update #3.

Compositions

[1862] Provided herein are pharmaceutical compositions comprising one or more compounds of formula (II), formula (I), formula (I'), or any variation or embodiment thereof, as described elsewhere herein, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing. In some embodiments, provided herein is a pharmaceutical composition comprising (i) a compound of formula (II), formula (I), formula (I') or any variation or embodiment thereof, as described elsewhere herein, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, and (ii) one or more pharmaceutically acceptable excipients.

[1863] Suitable pharmaceutically acceptable excipients may include, for example, fillers, diluents, sterile aqueous solutions and various organic solvents, permeation enhancers, solubilizers, and adjuvants. Various substances may be embraced by the term excipient, including without limitation any substance used as a binder, disintegrant, coating, compression/encapsulation aid, cream or lotion, lubricant, solutions for parenteral administration, materials for chewable tablets, sweetener or flavoring, suspending/gelling agent, or wet granulation agent. Examples of suitable excipients

are well-known to those skilled in the art. Such compositions are prepared in a manner well known in the pharmaceutical art. See, e.g., *Remington's Pharmaceutical Sciences*, Academic Press, 23rd ed. (2020), which is incorporated herein by reference.

[1864] The pharmaceutical compositions may be administered in either single or multiple doses. The pharmaceutical composition may be administered by various methods including, for example, oral, rectal, buccal, intranasal, and transdermal routes. In certain embodiments, the pharmaceutical composition may be administered by intra-arterial injection, intravenously, intraperitoneally, parenterally, intramuscularly, subcutaneously, orally, topically, or as an inhalant.

[1865] Compounds as described herein may be administered to individuals in a form of generally accepted oral compositions, such as tablets, coated tablets, gel capsules in a hard or in soft shell, emulsions or suspensions. Examples of carriers, which may be used for the preparation of such compositions, are lactose, corn starch or its derivatives, talc, stearate or its salts, etc. Acceptable carriers for gel capsules with soft shell are, for instance, plant oils, wax, fats, semisolid and liquid polyols, and so on. In addition, pharmaceutical formulations may contain preservatives, solubilizers, stabilizers, re-wetting agents, emulgators, sweeteners, dyes, adjusters, salts for the adjustment of osmotic pressure, buffers, coating agents or antioxidants.

[1866] The specific dose level of a compound as described herein will depend upon a variety of factors such as the age, body weight and sex of the individual as well as the route of administration and other factors. In some embodiments, a dosage is expressed as a number of milligrams of a compound described herein per kilogram of the individual's body weight (mg/kg). Dosages of between about 0.1 mg/kg and 100-150 mg/kg may be appropriate.

[1867] The compound may be administered to an individual in accordance with an effective dosing regimen for a desired period of time or duration, such as at least about one month, at least about 2 months, at least about 3 months, at least about 6 months, or at least about 12 months or longer, which in some variations may be for the duration of the individual's life.

Methods of Treatment

[1868] Provided herein is a method of modulating APOL1 in a cell, comprising exposing the cell to an effective amount of a compound of formula (I), formula (I'), formula (II), or any variation or embodiment thereof, as described elsewhere herein, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing. Also provided herein is a method of modulating APOL1 in a cell, comprising exposing the cell to a composition comprising an effective amount of a compound of formula (I), formula (I'), formula (II), or any variation or embodiment thereof, as described elsewhere herein, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, and one or more pharmaceutically acceptable excipients. Isotopically labeled forms of any of the foregoing are also embraced, including, but not limited to, deuterated or tritiated forms (wherein at least one hydrogen is replaced by at least one deuterium, or tritium) of any of the specific compounds detailed herein.

[1869] Provided herein is a method of inhibiting APOL1 in a cell, comprising exposing the cell to an effective amount of a compound of formula (I), formula (I'), formula (II), or any variation or embodiment thereof, as described elsewhere herein, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing. Also provided herein is a method of inhibiting APOL1 in a cell, comprising exposing the cell to a pharmaceutical composition comprising an effective amount of a compound of formula (I), formula (I'), formula (II), or any variation or embodiment thereof, as described elsewhere herein, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, and one or more pharmaceutically acceptable excipients.

[1870] Provided herein is a method of inhibiting APOL1 in an individual, comprising administering to the individual an effective amount of a compound of formula (I), formula (I'), or any variation or embodiment thereof, as described elsewhere herein, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing. Also provided herein is a

method of inhibiting APOL1 in an individual, comprising administering to the individual a pharmaceutical composition comprising an effective amount of a compound of formula (I), formula (I'), formula (II), or any variation or embodiment thereof, as described elsewhere herein, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, and one or more pharmaceutically acceptable excipients.

[1871] In some embodiments, the compounds provided herein inhibit APOL1 at a concentration of less than 10 μM , less than 1 μM , less than 0.5 μM , or less than 0.1 μM . In some embodiments, the compounds provided herein inhibit APOL1 at a concentration of 1 to 10 μM , 0.01 to 1 μM , or 0.01 to 10 μM .

[1872] In some embodiments, the compounds provided herein reduce cell death caused by overexpression of APOL1. In some embodiments, the compounds provided herein reduce cell death caused by overexpression APOL1 at a concentration of less than 10 μM , less than 1 μM , less than 0.5 μM , or less than 0.1 μM . In some embodiments, the compounds provided herein reduce cell death caused by APOL1 overexpression at a concentration of 1 to 10 μM , 0.01 to 1 μM , or 0.01 to 10 μM .

[1873] In some embodiments, compounds provided herein have an EC₅₀ of less than 1 μM , less than 0.5 μM , or less than 0.1 μM . In some embodiments, the compounds provided herein have an EC₅₀ of 1 to 10 μM , 0.01 to 1 μM , or 0.01 to 10 μM .

[1874] In some embodiments, compounds provided herein have an AC₅₀ of less than 1 μM , less than 0.5 μM , or less than 0.1 μM . In some embodiments, the compounds provided herein have an AC₅₀ of 1 to 10 μM , 0.01 to 1 μM , or 0.01 to 10 μM . In some embodiments, the AC₅₀ value reflects the compound's ability to prevent calcium influx by inhibiting APOL1.

[1875] In some embodiments, the compounds provided herein inhibit a cation channel. In some embodiments, the compounds of the present disclosure inhibit a calcium channel. In some embodiments, the compounds of the present disclosure reduce calcium transport.

[1876] Provided herein is a method of treating an APOL1-mediated disease, disorder, or condition in an individual in need thereof, comprising administering to the individual a therapeutically effective amount of a compound of formula (I), formula (I'), formula (II), or any variation or embodiment thereof, as described elsewhere herein, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing. Also provided herein is a method of treating an APOL1-mediated disease, disorder, or condition in an individual in need thereof, comprising administering to the individual a pharmaceutical composition comprising a therapeutically effective amount of a compound of formula (I), formula (I'), formula (II), or any variation or embodiment thereof, as described elsewhere herein, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, and one or more pharmaceutically acceptable excipients.

[1877] Provided herein is a method of treating a kidney disease, disorder, or condition in an individual in need thereof, comprising administering to the individual a therapeutically effective amount of a compound of formula (I), formula (I'), formula (II), or any variation or embodiment thereof, as described elsewhere herein, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing. Also provided herein is a method of treating a kidney disease, disorder, or condition in an individual in need thereof, comprising administering to the individual a pharmaceutical composition comprising a therapeutically effective amount of a compound of formula (I), formula (I'), formula (II), or any variation or embodiment thereof, as described elsewhere herein, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, and one or more pharmaceutically acceptable excipients.

[1878] In some embodiments, the individual has a chronic kidney disease. In some embodiments, the individual has hypertension-attributed kidney disease. In some embodiments, the kidney disease, disorder, or condition is an APOL1-mediated kidney disease, disorder, or condition. In some embodiments, the kidney disease, disorder, or condition is selected from the group consisting

of focal segmental glomerulosclerosis (FSGS), hypertension-attributed kidney disease, viral nephropathy, COVID-19 associated nephropathy, human immunodeficiency virus-associated nephropathy (HIVAN), sickle-cell nephropathy, lupus nephritis, and diabetic kidney disease. [1879] Also provided herein is a method of treating an APOL1-mediated disorder, such as preeclampsia and sepsis, comprising administering to an individual in need thereof a therapeutically effective amount of a compound of formula (I), formula (I'), formula (II), or any variation or embodiment thereof, as described elsewhere herein, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing. In some embodiments, the individual is genetically predisposed to developing the APOL1-mediated disorder.

[1880] Also provided herein is a method of delaying development of progressive renal allograft loss in a kidney transplant recipient comprising administering to the kidney transplant recipient a therapeutically effective amount of a compound of formula (I), formula (I'), or any variation or embodiment thereof, as described elsewhere herein, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing. In some embodiments, the kidney transplant recipient receives a kidney from a high-risk APOL1 genotype donor. In some embodiments, the kidney transplant recipient is administered a therapeutically effective amount of the compound for a period of time before receiving the kidney transplant. In some embodiments, the kidney transplant recipient is administered a therapeutically effective amount of the compound subsequent to receiving the kidney transplant.

[1881] Provided herein is a method of treating a kidney disease, disorder, or condition in an individual in need thereof, comprising administering to the individual a therapeutically effective amount of a compound of formula (I), formula (I'), formula (II), or any variation or embodiment thereof, as described elsewhere herein, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein the individual has an APOL1 mutation. Also provided herein is a method of treating a kidney disease, disorder, or condition in an individual in need thereof, comprising administering to the individual a pharmaceutical composition comprising a therapeutically effective amount of a compound of formula (I), formula (I'), formula (II), or any variation or embodiment thereof, as described elsewhere herein, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, and one or more pharmaceutically acceptable excipients, wherein the individual has an APOL1 mutation.

[1882] The compounds provided herein may also be used in a method of delaying the development of an APOL1-mediated disease, disorder, or condition, comprising administering a compound of formula (I), formula (I'), formula (II), or any variation or embodiment thereof, as described elsewhere herein, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, to an individual who is at risk of developing an APOL1-mediated disease, disorder, or condition. In some embodiments, the APOL1-mediated disease, disorder, or condition is preeclampsia or sepsis and the individual has two APOL1 risk alleles. In some embodiments, the APOL1-mediated disease, disorder, or condition is a chronic kidney disease and the individual has any binary combination of G1 and G2 APOL1 risk alleles. In some embodiments, the chronic kidney disease is focal segmental glomerulosclerosis (FSGS), hypertension-attributed kidney disease, human immunodeficiency virus-associated nephropathy (HIVAN), hypertension-attributed kidney disease, sickle cell nephropathy, viral nephropathy, COVID-19 associated nephropathy, lupus nephritis, diabetic kidney disease, or APOL1-associated nephropathy. The compounds as provided herein may also be used in a method of delaying the development of progressive renal allograft loss in an individual who has received a kidney transplantation from a high-risk APOL1 genotype donor.

[1883] In some embodiments, the individual has a gain-of-function mutation in APOL1. In some embodiments, the individual has an APOL1 risk allele. In some embodiments, the APOL1 risk allele is a missense variant. In some embodiments, the APOL1 risk allele is a G1 variant. In some embodiments, the G1 variant is G1.sup.G (p.S342 G) or G1M (p.I384 M). In some embodiments,

the APOL1 risk allele is the G2 variant. In some embodiments, the G2 variant is NYK388-389K. In some embodiments, the APOL1 risk variant is a mutation in the serum resistance-associated (SRA) binding domain of the APOL1 protein. In some embodiments, the individual has two APOL1 risk alleles.

[1884] Also provided herein is a method of inhibiting APOL1 in an individual comprising administering to the individual a therapeutically effective amount of a compound of formula (I), formula (I'), formula (II), or any variation or embodiment thereof, as described elsewhere herein, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing.

[1885] Also provided herein is method of preventing kidney failure in an individual comprising administering a therapeutically effective amount of a compound of formula (I), formula (I'), formula (II), or any variation or embodiment thereof, as described elsewhere herein, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing to the individual. In some embodiments, the compound prevents tissue necrosis. In some embodiments, the compound prevents apoptosis. In some embodiments, the compound reduces inflammation.

[1886] In some embodiments, the compounds provided herein reduce or eliminate one or more symptoms of a kidney disease. In some embodiments, the compounds reduce nausea, vomiting, loss of appetite, fatigue and weakness, sleep problems, urinary frequency issues, muscle twinges and cramps, swelling, itching, chest pain, shortness of breath, and/or high blood pressure.

[1887] In some embodiments, the compounds provided herein reduce the rate of kidney damage and/or progression of kidney damage. In some embodiments, the compounds provided herein reduce the rate of kidney failure. In some embodiments, the compounds provided herein reverse kidney damage. In some embodiments, the compounds reduce the need for dialysis. In some embodiments, the compounds provided herein delay the need for dialysis at least one month, at least two months, at least three months, or at least one year.

[1888] In some embodiments, the compounds reduce the rate of or delay the need for a kidney transplant. For example, in some embodiments, the compounds provided herein delay the need for a kidney transplant at least one month, at least two months, at least three months, at least six months, or at least one year. In some embodiments, the compounds provided herein eliminate the need for a kidney transplant.

[1889] In some embodiments, the individual has stage 1, stage 2, stage 3A, stage 3B, stage 4, or stage 5 chronic kidney disease. In some embodiments, kidney function is evaluated using an estimated glomerular filtration rate (eGFR) kidney function test.

[1890] The compounds and compositions comprising the compounds provided herein may also be used in a method of delaying or preventing proteinuria, the method comprising administering the compound, or any variation or embodiment thereof, as described elsewhere herein, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, to an individual. In one aspect, the methods herein comprise preventing or reducing protein in the urine, e.g., proteinuria. In some embodiments, the methods provided herein prevent proteinuria. In some embodiments, the methods reduce proteinuria. In some embodiments, the methods provided herein prevent albuminuria. In some embodiments, the methods reduce albuminuria. In some embodiments, the methods reduce urine albumin. In some embodiments, urine albumin is reduced by at least about 50%, about 60%, about 70%, about 80%, or about 90%, or greater. In some embodiments, urine albumin is reduced by at least about 50%. In some embodiments, urine albumin is reduced by at least about 60%. In some embodiments, urine albumin is reduced by at least about 70%. In some embodiments, urine albumin is reduced by at least about 80%. In some embodiments, urine albumin is reduced by at least about 90%. In some embodiments, reduction of urine albumin is dose-dependent. In some embodiments, the methods provided herein reduce urine albumin/creatinine ratio. In some embodiments, urine albumin/creatinine ratio is reduced by at least about 50%, about 60%, about 70%, about 80%, or about 90%, or greater. In some embodiments,

urine albumin/creatinine ratio is reduced by at least about 50%. In some embodiments, urine albumin/creatinine ratio is reduced by at least about 60%. In some embodiments, urine albumin/creatinine ratio is reduced by at least about 70%. In some embodiments, urine albumin/creatinine ratio is reduced by at least about 80%. In some embodiments, urine albumin/creatinine ratio is reduced by at least about 90%. In some embodiments, reduction of urine albumin/creatinine ratio is dose-dependent. In some embodiments, the reduction and/or ratios are measured according to assays detailed herein. In any of the aforementioned methods, the individual is an individual in need thereof, such as an individual having an APOL1-mediated disease, disorder, or condition. In some embodiments, the APOL1-mediated disease, disorder, or condition is a kidney disease. In some embodiments, the APOL1-mediated disease, disorder, or condition is a chronic kidney disease. In some embodiments, the individual has hypertension-attributed kidney disease. In some embodiments, the kidney disease, disorder, or condition is selected from the group consisting of focal segmental glomerulosclerosis (FSGS), hypertension-attributed kidney disease, viral nephropathy, COVID-19 associated nephropathy, human immunodeficiency virus-associated nephropathy (HIVAN), sickle-cell nephropathy, lupus nephritis, and diabetic kidney disease. [1891] In some embodiments, the administration is oral administration.

Kits

[1892] The present disclosure further provides kits for carrying out the methods of the invention. The kits may comprise a compound or pharmaceutically acceptable salt thereof as described herein and suitable packaging. The kits may comprise one or more containers comprising any compound described herein. In one aspect, a kit includes a compound of the disclosure or a pharmaceutically acceptable salt thereof, and a label and/or instructions for use of the compound in the treatment of a disease or disorder described herein. The kits may comprise a unit dosage form of the compound.

[1893] Provided herein are kits, comprising (i) a compound of formula (I), formula (I'), formula (II), or any variation or embodiment thereof, as described elsewhere herein, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, and (ii) instructions for use in treating an APOL1-mediated disease, disorder, or condition in an individual in need thereof. Also provided herein are kits, comprising (i) a pharmaceutical composition comprising a compound of formula (I), or any variation or embodiment thereof, as described elsewhere herein, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, and one or more pharmaceutically acceptable excipients; and (ii) instructions for use in treating an APOL1-mediated disease, disorder, or condition in an individual in need thereof.

[1894] Articles of manufacture are also provided, wherein the article of manufacture comprises a compound of formula (I), formula (I'), formula (II), or any variation or embodiment thereof, as described elsewhere herein, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, in a suitable container. Also provided herein are articles of manufacture, comprising a pharmaceutical composition comprising a compound of formula (I), formula (I'), formula (II), or any variation or embodiment thereof, as described elsewhere herein, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, in a suitable container. The container may be a vial, jar, ampoule, preloaded syringe, or intravenous bag.

ENUMERATED EMBODIMENTS

[1895] Embodiment 1A. A compound of formula (I):

##STR00865## [1896] or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein: [1897] m is an integer from 0 to 4; [1898] n is an integer from 0 to 2; [1899] p is an integer from 0 to 10; [1900] R^{sup.1}, if present is, independently at each occurrence selected from the group consisting of halo, —CN, C_{sub.1-6}alkoxy, or —C_{sub.1-6}alkyl, wherein [1901] the C_{sub.1-6}alkoxy of R^{sup.1} is optionally substituted with one or more halo, and [1902] the C_{sub.1-6}alkyl of R^{sup.1} is optionally substituted with one or more halo; [1903] R^{sup.2} is H, C_{sub.1-6}alkyl, C_{sub.3-10}cycloalkyl, or 3-15 membered heterocyclyl,

wherein [1904] the C.sub.1-6alkyl of R.sup.2 is optionally substituted with one or more halo, —OH, —NH.sub.2, or C.sub.1-6alkoxy, and [1905] the C.sub.3-10cycloalkyl of R.sup.2 is optionally substituted with one or more —OH; [1906] R.sup.3, if present, is C.sub.1-6alkyl; [1907] L.sup.1 is C.sub.1-6alkylene, wherein the C.sub.1-6alkylene of L.sup.1 is optionally substituted with one or more C.sub.1-6alkyl, and wherein the C.sub.1-6alkyl is further optionally substituted with one or more —OH or C.sub.1-6alkoxy; [1908] L.sup.2 is O or N(R.sup.x), wherein R.sup.x is H or C.sub.1-6alkyl; and [1909] either [1910] (1) L.sup.3 is absent or is O, C.sub.3-10cycloalkyl, 3-10 membered heterocyclyl, or C.sub.1-6alkylene, wherein [1911] the C.sub.1-6alkylene of L.sup.3 is optionally substituted with one or more C.sub.1-6alkyl, and [1912] the 3-10 membered heterocyclyl is optionally substituted with one or more —OH; [1913] X.sup.1 and X.sup.2 are each independently N or C(R.sup.5); and [1914] R.sup.4 is: [1915] (i) —S(O).sub.2—R.sup.a; [1916] (ii) 5-20 membered heteroaryl, wherein the 5-20 membered heteroaryl of R.sup.4 is optionally substituted with one or more C.sub.1-6alkyl; [1917] (iii) —N(R.sup.d).sub.2, wherein R.sup.d is independently at each occurrence H, C.sub.1-6 alkyl, or —S(O).sub.2—R.sup.a, wherein the C.sub.1-6alkyl of R.sup.d is optionally substituted with one or more —OH, [1918] (iv) —NS(O)—(C.sub.1-6alkyl).sub.2, wherein the C.sub.1-6alkyl of R.sup.d is optionally substituted with one or more —OH, [1919] (v) —C(O)—N(R.sup.e).sub.2 wherein R.sup.e is independently at each occurrence H, C.sub.1-6 alkyl, or 3-10 membered heterocycle, wherein the 3-10 membered heterocycle is optionally substituted with one or more oxo, or both R.sup.e together with the N to which they are attached are taken together to form a 3-10 membered heterocyclyl, wherein the 3-10 membered heterocyclyl is optionally substituted with one or more halo, oxo, —OH, NH.sub.2, NH—S(O).sub.2—R.sup.a, or —S(O).sub.2—R.sup.a, [1920] (vi) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl or oxo, [1921] (vii) —S(O)—N(C.sub.1-6alkyl)-(C.sub.1-6alkyl), [1922] (viii) —CN; [1923] or [1924] (2) L.sup.3 is absent; and [1925] one of X.sup.1 and X.sup.2 is N or C(R.sup.5); and [1926] the other of X.sup.1 and X.sup.2 is N or C that is taken together with R.sup.4, and the atoms to which they are attached, to form a 5-10 membered heterocyclyl or a 5-20 membered heteroaryl, wherein [1927] the 5-10 membered heterocyclyl is optionally substituted with one or more R.sup.b, [1928] wherein R.sup.b is, independently at each occurrence, selected from the group consisting of halo, oxo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10cycloalkyl, and 3-10 membered heterocyclyl, wherein [1929] the C.sub.1-6alkyl of R.sup.b is optionally substituted with one or more halo, OH, or —S(O).sub.2—C.sub.1-6alkyl, and [1930] the C.sub.3-10cycloalkyl of R.sup.b is optionally substituted with one or more —OH, and [1931] the 5-20 membered heteroaryl is optionally substituted with one or more R.sup.e, wherein R.sup.e is, independently at each occurrence, selected from the group consisting of halo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10 cycloalkyl, and 3-10 membered heterocyclyl, wherein [1932] the C.sub.1-6alkyl of R.sup.e is optionally substituted with one or more —S(O).sub.2—C.sub.1-6alkyl, and [1933] the C.sub.3-10cycloalkyl of R.sup.e is optionally substituted with one or more —OH; [1934] R.sup.a is, independently at each occurrence: [1935] (i) C.sub.1-6alkyl optionally substituted with one or more halo, —OH, —S(O).sub.2—C.sub.1-6alkyl, or —N(C.sub.1-6alkyl)-C(O)—C.sub.1-6alkyl, or [1936] (ii) C.sub.3-10cycloalkyl optionally substituted with one or more —OH, —C(O).sub.2—C.sub.1-6alkyl, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, or —C(O)—C.sub.3-10heterocyclyl, or C.sub.1-6alkyl, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH, or [1937] (iii) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl; [1938] R.sup.5 is, independently at each occurrence, H, halo, —CN, 3-10 membered heterocyclyl, C.sub.1-6alkyl, or C.sub.1-6alkoxy, wherein the C.sub.1-6alkyl of R.sup.5 is optionally substituted with one or more halo, or —OH and wherein the C.sub.1-6alkoxy is optionally substituted with one or more halo; and [1939] R.sup.6

and R.sup.7 are each independently H or halo.

[1940] Embodiment 2A. The compound of embodiment 1A, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein L.sup.2 is O, such that the compound is of formula (I-A):

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or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing.

[1941] Embodiment 3A. The compound of embodiment 1A or embodiment 2A, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein n is 1, or 2.

[1942] Embodiment 4A. The compound of any one of embodiments 1A-3A, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R.sup.2 is H, C.sub.1-6alkyl, C.sub.3-10cycloalkyl, or 3-15 membered heterocyclyl, wherein the C.sub.1-6alkyl of R.sup.2 is optionally substituted with one or more halo, —OH, —NH.sub.2, or C.sub.1-6alkoxy, and the C.sub.3-10cycloalkyl of R.sup.2 is optionally substituted with one or more —OH.

[1943] Embodiment 5A. The compound of any one of embodiments 1A-4A, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein m is 0, 1, or 2.

[1944] Embodiment 6A. The compound of any one of embodiments 1A-5A, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R.sup.1 is halo, —CN, or —C.sub.1-6alkyl, wherein the C.sub.1-6alkyl of R.sup.1 is optionally substituted with one or more halo.

[1945] Embodiment 7A. The compound of any one of embodiments 1A-6A, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein p is 0 or 1.

[1946] Embodiment 8A. The compound of any one of embodiments 1A-7A, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R.sup.3 is C.sub.1-6alkyl.

[1947] Embodiment 9A. The compound of any one of embodiments 1A-8A, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein L.sup.1 is C.sub.1-6alkylene, wherein the C.sub.1-6alkylene of L.sup.1 is optionally substituted with one or more C.sub.1-6alkyl, wherein the C.sub.1-6alkyl is optionally substituted with —OH or C.sub.1-6alkoxy.

[1948] Embodiment 10A. The compound of any one of embodiments 1A-9A, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein L.sup.1 is selected from the group consisting of

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wherein, for each L.sup.1, * denotes the point of attachment to L.sup.2 and ** denotes the point of attachment to the remainder of the molecule.

[1949] Embodiment 11A. The compound of any one of embodiments 1A-10A, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein L.sup.2 is O.

[1950] Embodiment 12A. The compound of any one of embodiments 1A-11A, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein L.sup.3 is absent.

[1951] Embodiment 13A. The compound of any one of embodiments 1A-12A, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein L.sup.3 is —O—, C.sub.3-10cycloalkyl, 3-10 membered heterocyclyl, or C.sub.1-6alkylene, wherein [1952] the C.sub.1-6alkylene of L.sup.3 is optionally substituted with one or more C.sub.1-6alkyl, [1953] the C.sub.3-10cycloalkyl membered heterocyclyl of L.sup.3 is optionally substituted with one or more —OH, and [1954] the 3-10 membered heterocyclyl of L.sup.3 is optionally substituted with one or more —OH.

[1955] Embodiment 14A. The compound of any one of embodiments 1A-11A, and 13A or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein L.sup.3 is selected from the group consisting of —O—,

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[1956] Embodiment 15A. The compound of any one of embodiments 1A-14A or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R.sup.4 is S(O).sub.2—R.sup.a, 5-10 membered heteroaryl, —N(R.sup.d).sub.2, —NS(O)—(C.sub.1-3alkyl).sub.2, —C(O)—N(R.sup.e).sub.2, 3-6 membered heterocyclyl, —S(O)(N—C.sub.1-3alkyl)-(C.sub.1-3alkyl), or —CN, wherein [1957] the 5-20 membered heteroaryl of R.sup.4 is optionally substituted with one or more C.sub.1-6alkyl, and [1958] the 3-6 membered heterocyclyl optionally substituted with one or more oxo or C.sub.1-6alkyl.

[1959] Embodiment 16A. The compound of embodiment 15A or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R.sup.a is C.sub.1-6alkyl, is C.sub.3-10cycloalkyl, or is 3-10 membered heterocyclyl, wherein [1960] the C.sub.1-6alkyl of R.sup.a is optionally substituted with one or more halo, —OH, —S(O).sub.2—C.sub.1-6alkyl, or —N(C.sub.1-6alkyl)-C(O)—C.sub.1-6alkyl, [1961] the C.sub.3-10cycloalkyl of R.sup.a is optionally substituted with one or more —OH, C(O).sub.2—C.sub.1-6alkyl, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —C(O)—C.sub.3-10heterocyclyl or C.sub.1-6alkyl, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH, and [1962] the 3-10 membered heterocyclyl of R.sup.a is optionally substituted with one or more C.sub.1-6alkyl.

[1963] Embodiment 17A. The compound of embodiment 15A, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein each of R.sup.d is independently H, C.sub.1-6 alkyl, or —S(O).sub.2—R.sup.a, wherein the C.sub.1-6alkyl of R.sup.d is optionally substituted with one or more —OH, and wherein R.sup.a is C.sub.1-6alkyl.

[1964] Embodiment 18A. The compound of embodiment 15A, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein each of R.sup.e is independently H, C.sub.1-6alkyl, or 3-10 membered heterocycle, wherein the 3-6 membered heterocycle is optionally substituted with one or more oxo.

[1965] Embodiment 19A. The compound of embodiment 15A or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein both R.sup.e together with the N to which they are attached are taken together to form a 3-10 membered heterocyclyl, wherein the 3-10 membered heterocyclyl is optionally substituted with one or more halo, oxo, —OH, —NH.sub.2, —NH—S(O).sub.2—R.sup.a, or —S(O).sub.2—R.sup.a, wherein R.sup.a is C.sub.1-6alkyl.

[1966] Embodiment 20A. The compound of any one of embodiments 1A-19A, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein each of X.sup.1 and X.sup.2 is C(R.sup.5).

[1967] Embodiment 21A. The compound of any one of embodiments 1A-19A, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein one of X.sup.1 and X.sup.2 is N and the other is C(R.sup.5).

[1968] Embodiment 22A. The compound of any one of embodiments 1A-19A, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein one of X.sup.1 and X.sup.2 is CR.sup.5, and the other of X.sup.1 and X.sup.2 is C that is taken together with R.sup.4, and the atoms to which they are attached, to form a 5-10 membered heterocyclyl optionally substituted with one or more of R.sup.b.

[1969] Embodiment 23A. The compound of embodiment 22A, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R.sup.b is halo, oxo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10cycloalkyl, and 3-10

membered heterocyclyl, wherein the C.sub.1-6alkyl of R.sup.b is optionally substituted with one or more halo, OH, or —S(O).sub.2—C.sub.1-6alkyl, and the C.sub.3-10cycloalkyl of R.sup.b is optionally substituted with one or more —OH.

[1970] Embodiment 24A. The compound of any one of embodiments 1A-19A, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein one of X.sup.1 and X.sup.2 is CR.sup.5, and the other of X.sup.1 and X.sup.2 is C that is taken together with R.sup.4, and the atoms to which they are attached, to form a 5-10 membered heteroaryl optionally substituted with one or more R.sup.e.

[1971] Embodiment 25A. The compound of embodiment 24A, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein, R.sup.e is halo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10cycloalkyl, and 3-10 membered heterocyclyl, wherein the C.sub.1-6alkyl of R.sup.e is optionally substituted with one or more —S(O).sub.2—C.sub.1-6alkyl, and the C.sub.3-10cycloalkyl of R.sup.e is optionally substituted with one or more —OH.

[1972] Embodiment 26A. The compound of any one of embodiments 1A-25A, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R.sup.5 is independently at each occurrence, H, halo, —CN, 3-10 membered heterocyclyl, C.sub.1-6alkyl, or C.sub.1-6alkoxy, wherein the C.sub.1-6alkyl of R.sup.5 is optionally substituted with one or more halo, or —OH and wherein the C.sub.1-6alkoxy is optionally substituted with one or more halo.

[1973] Embodiment 27A. The compound of embodiment 1A, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein the compound, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, is selected from Compounds 1-273 of Table 1.

[1974] Embodiment 29A. A method for preparing a compound of embodiment 1A, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, comprising a step of reacting a compound of formula I'-A:

##STR00869## [1975] wherein: [1976] m is an integer from 0 to 4; [1977] n is an integer from 0 to 2; [1978] p is an integer from 0 to 10; [1979] R.sup.1, if present is, independently at each occurrence selected from the group consisting of halo, —CN, C.sub.1-6alkoxy or —C.sub.1-6alkyl, wherein [1980] the C.sub.1-6alkoxy of R.sup.1 is optionally substituted with one or more halo, and [1981] the C.sub.1-6alkyl of R.sup.1 is optionally substituted with one or more halo; R.sup.2 is H, C.sub.1-6alkyl, C.sub.3-10cycloalkyl, or 3-15 membered heterocyclyl, wherein [1982] the C.sub.1-6alkyl of R.sup.2 is optionally substituted with one or more halo, —OH, —NH.sub.2, or C.sub.1-6alkoxy, and [1983] the C.sub.3-10cycloalkyl of R.sup.2 is optionally substituted with one or more —OH; and [1984] R.sup.3, if present, is C.sub.1-6alkyl; [1985] with: [1986] a compound of formula (I'-B):

##STR00870## [1987] wherein: [1988] the dashed line represents a single or double bond; [1989] Y.sup.1 is halo, oxo, or a sulfonate ester [1990] L.sup.1 is C.sub.1-6alkylene, wherein the C.sub.1-6alkylene of L.sup.1 is optionally substituted with one or more C.sub.1-6alkyl, and wherein the C.sub.1-6alkyl is further optionally substituted with one or more —OH or C.sub.1-6alkoxy; [1991] L.sup.2 is O or N(R.sup.x), wherein R.sup.x is H or C.sub.1-6alkyl; and [1992] either [1993] (i) L.sup.3 is absent or is O, C.sub.3-10cycloalkyl, 3-10 membered heterocyclyl, or C.sub.1-6alkylene, wherein [1994] the C.sub.1-6alkylene of L.sup.3 is optionally substituted with one or more C.sub.1-6alkyl, and [1995] the 3-10 membered heterocyclyl is optionally substituted with one or more —OH; X.sup.1 and X.sup.2 are each independently N or C(R.sup.5); and [1996] R.sup.4 is: [1997] (i) —S(O).sub.2—R.sup.a; [1998] (ii) 5-20 membered heteroaryl, wherein the 5-20 membered heteroaryl of R.sup.4 is optionally substituted with one or more C.sub.1-6alkyl; [1999] (iii) —N(R.sup.d).sub.2, wherein R.sup.d is independently at each occurrence H, C.sub.1-6alkyl, or —S(O).sub.2—R.sup.a, wherein the C.sub.1-6alkyl of R.sup.d is optionally substituted with one

or more —OH, [2000] (iv) —NS(O)—(C.sub.1-6alkyl).sub.2, wherein the C.sub.1-6alkyl of R.sup.d is optionally substituted with one or more —OH, [2001] (v) —C(O)—N(R.sup.e).sub.2 wherein R.sup.e is independently at each occurrence H, C.sub.1-6 alkyl, or 3-10 membered heterocycle, wherein the 3-10 membered heterocycle is optionally substituted with one or more oxo, or both R.sup.e together with the N to which they are attached are taken together to form a 3-10 membered heterocyclyl, wherein the 3-10 membered heterocyclyl is optionally substituted with one or more halo, oxo, —OH, NH.sub.2, NH—S(O).sub.2—R.sup.a, or —S(O).sub.2—R.sup.a, [2002] (vi) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl or oxo, [2003] (vii) —S(O)—N(C.sub.1-6alkyl)-(C.sub.1-6alkyl), [2004] (viii) —CN [2005] or [2006] (2) L.sup.3 is absent; and [2007] one of X.sup.1 and X.sup.2 is N or C(R.sup.5); and [2008] the other of X.sup.1 and X.sup.2 is N or C that is taken together with R.sup.4, and the atoms to which they are attached, to form a 5-10 membered heterocyclyl or a 5-20 membered heteroaryl, wherein [2009] the 5-10 membered heterocyclyl is optionally substituted with one or more R.sup.b, wherein R.sup.b is, independently at each occurrence, selected from the group consisting of halo, oxo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10cycloalkyl, and 3-10 membered heterocyclyl, wherein [2010] the C.sub.1-6alkyl of R.sup.b is optionally substituted with one or more halo, OH, or —S(O).sub.2—C.sub.1-6alkyl, and [2011] the C.sub.3-10cycloalkyl of R.sup.b is optionally substituted with one or more —OH, and [2012] the 5-20 membered heteroaryl is optionally substituted with one or more R.sup.e, wherein R.sup.c is, independently at each occurrence, selected from the group consisting of halo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10 cycloalkyl, and 3-10 membered heterocyclyl, wherein [2013] the C.sub.1-6alkyl of R.sup.e is optionally substituted with one or more —S(O).sub.2—C.sub.1-6alkyl, and [2014] the C.sub.3-10cycloalkyl of R.sup.e is optionally substituted with one or more —OH; [2015] R.sup.a is, independently at each occurrence: [2016] (i) C.sub.1-6alkyl optionally substituted with one or more halo, —OH, —S(O).sub.2—C.sub.1-6alkyl, or —N(C.sub.1-6alkyl)-C(O)—C.sub.1-6alkyl, or [2017] (ii) C.sub.3-10cycloalkyl optionally substituted with one or more —OH, —C(O).sub.2—C.sub.1-6alkyl, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, or —C(O)—C.sub.3-10heterocyclyl, or C.sub.1-6alkyl, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH, or [2018] (iii) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl; [2019] R.sup.5 is, independently at each occurrence, H, halo, —CN, 3-10 membered heterocyclyl, C.sub.1-6alkyl, or C.sub.1-6alkoxy, wherein the C.sub.1-6alkyl of R.sup.5 is optionally substituted with one or more halo, or —OH and wherein the C.sub.1-6alkoxy is optionally substituted with one or more halo; [2020] R.sup.6 and R.sup.7 are each independently H or halo; [2021] to give a compound of embodiment 1, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing. [2022] Embodiment 30A. The method of embodiment 29A, wherein the compound of embodiment 1 is prepared by a step comprising: [2023] a) alkylation of an amine of formula (I'-A) with an alkyl halide, or sulfonate ester compound of formula (I'-B) in the presence of an inorganic base; or [2024] b) reductive amination of a ketone of formula (I'-B) with an amine of formula (I'-A). [2025] Embodiment 31A. The method of embodiment 30A, wherein the inorganic base is selected from the group consisting of potassium carbonate, and sodium bicarbonate. [2026] Embodiment 32A. The method of embodiment 30A, wherein the reductive amination proceeds under the action of sodium triacetoxyborohydride, titanium tetraisopropoxide and acetic acid. [2027] Embodiment 33A. A method for preparing a compound of embodiment 1A, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, comprising a step of reacting a compound of formula I'—C: ##STR00871## [2028] wherein: [2029] m is an integer from 0 to 4; [2030] n is an integer from 0 to

2; [2031] p is an integer from 0 to 10; [2032] R.sup.1, if present is, independently at each occurrence selected from the group consisting of halo, —CN, C.sub.1-6alkoxy or —C.sub.1-6alkyl, wherein [2033] the C.sub.1-6alkoxy of R.sup.1 is optionally substituted with one or more halo, and [2034] the C.sub.1-6alkyl of R.sup.1 is optionally substituted with one or more halo; [2035] R.sup.2 is H, C.sub.1-6alkyl, C.sub.3-10cycloalkyl, or 3-15 membered heterocyclyl, wherein [2036] the C.sub.1-6alkyl of R.sup.2 is optionally substituted with one or more halo, —OH, —NH.sub.2, or C.sub.1-6alkoxy, and [2037] the C.sub.3-10cycloalkyl of R.sup.2 is optionally substituted with one or more —OH; [2038] R.sup.3, if present, is C.sub.1-6alkyl; [2039] L.sup.1 is C.sub.1-6alkylene, wherein the C.sub.1-6alkylene of L.sup.1 is optionally substituted with one or more C.sub.1-6alkyl, and wherein the C.sub.1-6alkyl is further optionally substituted with one or more —OH or C.sub.1-6alkoxy; and [2040] Y.sup.2 is halo, —OH or NH.sub.2; [2041] with: [2042] a compound of formula (I'-D):

##STR00872## [2043] wherein: [2044] Y.sup.3 is —OH or NH(R.sup.x), wherein R.sup.x is H or C.sub.1-6alkyl; and [2045] either [2046] (1) L.sup.3 is absent or is O, C.sub.3-10cycloalkyl, 3-10 membered heterocyclyl, or C.sub.1-6alkylene, wherein [2047] the C.sub.1-6alkylene of L.sup.3 is optionally substituted with one or more C.sub.1-6alkyl, and [2048] the 3-10 membered heterocyclyl is optionally substituted with one or more —OH; [2049] X.sup.1 and X.sup.2 are each independently N or C(R.sup.5); and [2050] R.sup.4 is: [2051] (i) —S(O).sub.2—R.sup.a; [2052] (ii) 5-20 membered heteroaryl, wherein the 5-20 membered heteroaryl of R.sup.4 is optionally substituted with one or more C.sub.1-6alkyl; [2053] (iii) —N(R.sup.d).sub.2, wherein R.sup.d is independently at each occurrence H, C.sub.1-6 alkyl, or —S(O).sub.2—R.sup.a, wherein the C.sub.1-6alkyl of R.sup.d is optionally substituted with one or more —OH, [2054] (iv) —NS(O)—(C.sub.1-6alkyl).sub.2, wherein the C.sub.1-6alkyl of R.sup.d is optionally substituted with one or more —OH, [2055] (v) —C(O)—N(R.sup.e).sub.2 wherein R.sup.e is independently at each occurrence H, C.sub.1-6 alkyl, or 3-10 membered heterocycle, wherein the 3-10 membered heterocycle is optionally substituted with one or more oxo, or both R.sup.e together with the N to which they are attached are taken together to form a 3-10 membered heterocyclyl, wherein the 3-10 membered heterocyclyl is optionally substituted with one or more halo, oxo, —OH, NH.sub.2, NH —S(O).sub.2—R.sup.a, or —S(O).sub.2—R.sup.a, [2056] (vi) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl or oxo, [2057] (vii) —S(O)—N(C.sub.1-6alkyl)-(C.sub.1-6alkyl), [2058] (viii) —CN; [2059] or [2060] (2) L.sup.3 is absent; and [2061] one of X.sup.1 and X.sup.2 is N or C(R.sup.5); and [2062] the other of X.sup.1 and X.sup.2 is N or C that is taken together with R.sup.4, and the atoms to which they are attached, to form a 5-10 membered heterocyclyl or a 5-20 membered heteroaryl, wherein [2063] the 5-10 membered heterocyclyl is optionally substituted with one or more R.sup.b, wherein R.sup.b is, independently at each occurrence, selected from the group consisting of halo, oxo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10cycloalkyl, and 3-10 membered heterocyclyl, wherein [2064] the C.sub.1-6alkyl of R.sup.b is optionally substituted with one or more halo, OH, or —S(O).sub.2—C.sub.1-6alkyl, and [2065] the C.sub.3-10cycloalkyl of R.sup.b is optionally substituted with one or more —OH, and [2066] the 5-20 membered heteroaryl is optionally substituted with one or more R.sup.e, wherein R.sup.e is, independently at each occurrence, selected from the group consisting of halo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10 cycloalkyl, and 3-10 membered heterocyclyl, wherein [2067] the C.sub.1-6alkyl of R.sup.e is optionally substituted with one or more —S(O).sub.2—C.sub.1-6alkyl, and [2068] the C.sub.3-10cycloalkyl of R.sup.e is optionally substituted with one or more —OH; [2069] R.sup.a is, independently at each occurrence: [2070] (i) C.sub.1-6alkyl optionally substituted with one or more halo, —OH, —S(O).sub.2—C.sub.1-6alkyl, or —N(C.sub.1-6alkyl)-C(O)—C.sub.1-6alkyl, or [2071] (ii) C.sub.3-10cycloalkyl optionally substituted with one or more

—OH, —C(O).sub.2—C.sub.1-6alkyl, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, or —C(O)—C.sub.3-10heterocyclyl, or C.sub.1-6alkyl, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH, or [2072] (iii) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl; [2073] R.sup.5 is, independently at each occurrence, H, halo, —CN, 3-10 membered heterocyclyl, C.sub.1-6alkyl, or C.sub.1-6alkoxy, wherein the C.sub.1-6alkyl of R.sup.5 is optionally substituted with one or more halo, or —OH and wherein the C.sub.1-6alkoxy is optionally substituted with one or more halo; and [2074] R.sup.6 and R.sup.7 are each independently H or halo; [2075] to give a compound of embodiment 1, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing.

[2076] Embodiment 34A. The method of embodiment 33A, wherein the compound of embodiment 1 is prepared by a step comprising: [2077] a) coupling of an alcohol compound of formula (I'—C) with a phenol compound of formula (I'-D), or a heterocyclic variant, under Mitsunobu-type reaction conditions; or [2078] b) reacting an alkyl halide compound of formula (I'—C) with a phenol or amine compound of formula (I'-D), in the presence of a catalyst.

[2079] Embodiment 35A. The method of embodiment 34A, wherein the Mitsunobi-type reaction conditions comprises coupling of an alcohol compound of formula (I'—C) with a phenol compound of formula (I'-D), or a heterocyclic variant, in the presence of triphenylphosphine and diisopropyl azodicarboxylate.

[2080] Embodiment 36A. The method of embodiment 34A, wherein the catalyst is silver oxide or potassium carbonate.

[2081] Embodiment 37A. A pharmaceutical composition, comprising (i) a compound of any one of embodiments 1A-28A, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, and (ii) one or more pharmaceutically acceptable excipients.

[2082] Embodiment 38A. A method of modulating APOL1 in a cell, comprising exposing the cell to a composition comprising an effective amount of a compound of any one or embodiments 1A-28A, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or a pharmaceutical composition of embodiment 37A.

[2083] Embodiment 39A. A method of inhibiting APOL1 in a cell, comprising exposing the cell to a composition comprising an effective amount of a compound of any one or embodiments 1A-28A or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or a pharmaceutical composition of embodiment 37A.

[2084] Embodiment 40A. A method of treating an APOL1-mediated disease, disorder, or condition in an individual in need thereof, comprising administering to the individual a compound of any one of embodiments 1A-28A, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or a pharmaceutical composition of embodiment 37A.

[2085] Embodiment 41A. The method of embodiment 40A, wherein the disease, disorder, or condition is selected from the group consisting of chronic kidney disease, focal segmental glomerulosclerosis (FSGS), hypertension-attributed kidney disease, human immunodeficiency virus-associated nephropathy (HIVAN), sickle-cell nephropathy, lupus nephritis, diabetic kidney disease, APOL1-associated nephropathy, viral nephropathy, COVID-19 associated nephropathy, preeclampsia, and sepsis.

[2086] Embodiment 42A. The method of embodiment 40A or embodiment 41A, wherein the disease, disorder, or condition is a kidney disease.

[2087] Embodiment 43A. The method of any one of embodiments 40A-42A, wherein the disease, disorder, or condition is a chronic kidney disease (CKD).

[2088] Embodiment 44A. A method of delaying the development of an APOL1-mediated disease, disorder, or condition, comprising administering a compound of any one of embodiments 1A-28A, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or a pharmaceutical composition of embodiment 37A, to an individual who is at risk of developing an APOL1-mediated disease, disorder, or condition.

[2089] Embodiment 45A. The method of embodiment 44A, wherein the APOL1-mediated disease, disorder, or condition is a kidney disease.

[2090] Embodiment 46A. The method of embodiment 44A or embodiment 45A, wherein the APOL1-mediated disease, disorder, or condition is a chronic kidney disease.

[2091] Embodiment 47A. The method of embodiment 44A, wherein the APOL1-mediated disease, disorder, or condition is selected from the group consisting of chronic kidney disease, focal segmental glomerulosclerosis (FSGS), hypertension-attributed kidney disease, human immunodeficiency virus-associated nephropathy (HIVAN), sickle-cell nephropathy, lupus nephritis, diabetic kidney disease, APOL1-associated nephropathy, viral nephropathy, COVID-19 associated nephropathy, preeclampsia, and sepsis.

[2092] Embodiment 48A. The method of any one of embodiments 40A-47A, wherein the individual has an APOL1 mutation.

[2093] Embodiment 49A. The method of embodiment 48A, wherein the APOL1 mutation is a gain-of-function mutation.

[2094] Embodiment 50A. The method of any one of embodiments 40A-49A, wherein a therapeutically effective amount of a compound of any one of embodiments 1A-28A, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or a pharmaceutical composition of embodiment 37A, is administered.

[2095] Embodiment 51A. A kit, comprising (i) a compound of any one of embodiments 1A-28A, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or a pharmaceutical composition of embodiment 37A, and (ii) instructions for use in treating an APOL1-mediated disease, disorder, or condition in an individual in need thereof.

[2096] Embodiment 52A. The kit of embodiment 51A, wherein the disease, disorder, or condition is a kidney disease.

[2097] Embodiment 53A. The kit of embodiment 51A or embodiment 52A, wherein the disease, disorder, or condition is a chronic kidney disease (CKD).

[2098] Embodiment 54A. The kit of any one of embodiments 51A-53A, wherein the disease, disorder, or condition is selected from the group consisting of chronic kidney disease, focal segmental glomerulosclerosis (FSGS), hypertension-attributed kidney disease, human immunodeficiency virus-associated nephropathy (HIVAN), sickle-cell nephropathy, lupus nephritis, diabetic kidney disease, APOL1-associated nephropathy, viral nephropathy, COVID-19 associated nephropathy, preeclampsia, and sepsis.

[2099] Embodiment 55A. The kit of any one of embodiments 51A-54A, wherein the individual has an APOL1 mutation.

[2100] Embodiment 56A. The kit of embodiment 55A, wherein the APOL1 mutation is a gain-of-function mutation.

[2101] Embodiment 1B. A compound of formula (I):

##STR00873## [2102] or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, [2103] wherein: [2104] m is an integer from 0 to 4; [2105] n is an integer from 0 to 2; [2106] p is an integer from 0 to 10; [2107] R_{sup.1}, if present is, independently at each occurrence selected from the group consisting of halo, —CN, C_{sub.1-6}alkoxy, or C_{sub.1-6}alkyl, wherein [2108] the C_{sub.1-6}alkoxy of R_{sup.1} is optionally substituted with one or more halo, and [2109] the C_{sub.1-6}alkyl of R_{sup.1} is optionally substituted with one or more halo; [2110] R_{sup.2} is H, C_{sub.1-6}alkyl, C_{sub.3-10}cycloalkyl, or 3-15 membered heterocyclyl, wherein [2111] the C_{sub.1-6}alkyl of R_{sup.2} is optionally substituted with one or more halo, —OH, —NH_{sub.2}, or C_{sub.1-6}alkoxy, and [2112] the C_{sub.3-10}cycloalkyl of R_{sup.2} is optionally substituted with one or more —OH; [2113] R_{sup.3}, if present, is C_{sub.1-6}alkyl; [2114] L_{sup.1} is C_{sub.1-6}alkylene, wherein the C_{sub.1-6}alkylene of L_{sup.1} is optionally substituted with one or more C_{sub.1-6}alkyl, and wherein the C_{sub.1-6}alkyl is further optionally substituted with one or more —OH or C_{sub.1-6}alkoxy; [2115] L_{sup.2} is O or N(R_{sup.x}), wherein R_{sup.x} is H or

C.sub.1-6alkyl; and [2116] either [2117] (1) L.sup.3 is absent or is O, C.sub.3-10cycloalkyl, 3-10 membered heterocyclyl, or C.sub.1-6alkylene, wherein [2118] the C.sub.3-10cycloalkyl is optionally substituted with one or more —OH, or C.sub.1-6alkyl, [2119] the C.sub.1-6alkylene of L.sup.3 is optionally substituted with one or more —OH, or C.sub.1-6alkyl, and [2120] the 3-10 membered heterocyclyl is optionally substituted with one or more —OH; [2121] X.sup.1 and X.sup.2 are each independently N or C(R.sup.5); and [2122] R.sup.4 is: [2123] (i) —S(O).sub.2—R.sup.a; [2124] (ii) 5-20 membered heteroaryl, wherein the 5-20 membered heteroaryl of R.sup.4 is optionally substituted with one or more C.sub.1-6alkyl; [2125] (iii) —N(R.sup.d).sub.2, wherein R.sup.d is independently at each occurrence H, C.sub.1-6 alkyl, or —S(O).sub.2—R.sup.a, wherein the C.sub.1-6alkyl of R.sup.d is optionally substituted with one or more —OH, [2126] (iv) —NS(O)—(C.sub.1-6alkyl).sub.2, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH, [2127] (v) —C(O)—N(R.sup.e).sub.2 wherein R.sup.e is independently at each occurrence H, C.sub.1-6 alkyl, or 3-10 membered heterocycle, wherein the 3-10 membered heterocycle is optionally substituted with one or more oxo, or both R.sup.e together with the N to which they are attached are taken together to form a 3-10 membered heterocyclyl, wherein the 3-10 membered heterocyclyl is optionally substituted with one or more halo, oxo, —OH, NH.sub.2, —NH—S(O).sub.2—R.sup.a, or —S(O).sub.2—R.sup.a, [2128] (vi) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl, —OH, oxo or —S(O).sub.2R.sup.a, [2129] (vii) —S(O)—N(C.sub.1-6alkyl)-(C.sub.1-6alkyl), [2130] (viii) —CN, [2131] (ix) —(CH.sub.2).sub.qOH, wherein q is an integer from 0-6, [2132] (x) —C(O)—C.sub.1-6alkyl, or [2133] (xi) —P(O)(C.sub.1-6alkyl).sub.2; [2134] or [2135] (2) L.sup.3 is absent; and [2136] one of X.sup.1 and X.sup.2 is N or C(R.sup.5); and [2137] the other of X.sup.1 and X.sup.2 is N or C that is taken together with R.sup.4, and the atoms to which they are attached, to form a 5-10 membered heterocyclyl or a 5-20 membered heteroaryl, wherein [2138] the 5-10 membered heterocyclyl is optionally substituted with one or more R.sup.b, wherein R.sup.b is, independently at each occurrence, selected from the group consisting of halo, oxo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10cycloalkyl, and 3-10 membered heterocyclyl, wherein [2139] the C.sub.1-6alkyl of R.sup.b is optionally substituted with one or more halo, OH, —S(O).sub.2—C.sub.1-6alkyl, or C.sub.3-10cycloalkyl, and wherein the C.sub.3-10cycloalkyl of the C.sub.1-6alkyl of R.sup.b is further optionally substituted with one or more C.sub.1-6alkyl or —OH and [2140] the C.sub.3-10cycloalkyl of R.sup.b is optionally substituted with one or more —OH, C.sub.3-10 cycloalkyl, or C.sub.1-6alkyl, and wherein the C.sub.1-6alkyl of the C.sub.3-10cycloalkyl of R.sup.b is further optionally substituted with one or more —OH, and [2141] the 5-20 membered heteroaryl is optionally substituted with one or more R.sup.e, wherein R.sup.e is, independently at each occurrence, selected from the group consisting of halo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10 cycloalkyl, and 3-10 membered heterocyclyl, wherein [2142] the C.sub.1-6alkyl of R.sup.e is optionally substituted with one or more —S(O).sub.2—C.sub.1-6alkyl, [2143] the C.sub.3-10cycloalkyl of R.sup.e is optionally substituted with one or more —OH, or C.sub.1-6alkyl, [2144] and the 3-10 membered heterocyclyl of R.sup.e is optionally substituted with one or more —OH, or C.sub.1-6alkyl, and wherein the C.sub.1-6alkyl of 3-10 membered heterocyclyl of R.sup.e is further optionally substituted with one or more —OH; [2145] R.sup.a is, independently at each occurrence: [2146] (i) C.sub.1-6alkyl optionally substituted with one or more halo, —OH, —S(O).sub.2—C.sub.1-6alkyl, or —N(C.sub.1-6alkyl)—C(O)—C.sub.1-6alkyl, [2147] (ii) C.sub.3-10cycloalkyl optionally substituted with one or more —OH, —C(O).sub.2—C.sub.1-6alkyl, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, or —C(O)—C.sub.3-10heterocyclyl, or C.sub.1-6alkyl, wherein [2148] the C.sub.1-6alkyl is optionally substituted with one or more —OH, or [2149] (iii) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl; [2150] R.sup.5 is, independently at each

occurrence, H, halo, —CN, 3-10 membered heterocyclyl, C.sub.1-6alkyl, or C.sub.1-6alkoxy, wherein the C.sub.1-6alkyl of R.sup.5 is optionally substituted with one or more halo, or —OH and wherein the C.sub.1-6alkoxy of R.sup.5 is optionally substituted with one or more halo; and [2151] R.sup.6 and R.sup.7 are each independently H or halo.

[2152] Embodiment 15B. The compound of embodiments 1B or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R.sup.4 is S(O).sub.2—R.sup.a, 5-10 membered heteroaryl, —N(R.sup.d).sub.2, —NS(O)—(C.sub.1-3alkyl).sub.2, —C(O)—N(R.sup.e).sub.2, 3-6 membered heterocyclyl, —S(O)(N—C.sub.1-3alkyl)-(C.sub.1-3alkyl), —CN, or —OH, wherein the 5-20 membered heteroaryl of R.sup.4 is optionally substituted with one or more C.sub.1-6alkyl, and the 3-6 membered heterocyclyl optionally substituted with one or more oxo or C.sub.1-6alkyl.

[2153] Embodiment 23B. The compound of embodiment 1B, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein one of X.sup.1 and X.sup.2 is CR.sup.5, and the other of X.sup.1 and X.sup.2 is C that is taken together with R.sup.4, and the atoms to which they are attached, to form a 5-10 membered heterocyclyl optionally substituted with one or more of R.sup.b.

[2154] Embodiments 24B. The compound of embodiment 23B, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R.sup.b is halo, oxo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10cycloalkyl, and 3-10 membered heterocyclyl, wherein the C.sub.1-6alkyl of R.sup.b is optionally substituted with one or more halo, OH, or —S(O).sub.2—C.sub.1-6alkyl, and the C.sub.3-10cycloalkyl of R.sup.b is optionally substituted with one or more —OH, C.sub.3-10cycloalkyl, or C.sub.1-6alkyl.

[2155] Embodiments 25B. The compound of embodiment 24B, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein, R.sup.e is halo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10 cycloalkyl, and 3-10 membered heterocyclyl, wherein the C.sub.1-6alkyl of R.sup.e is optionally substituted with one or more —S(O).sub.2—C.sub.1-6alkyl, and the C.sub.3-10cycloalkyl of R.sup.e is optionally substituted with one or more —OH, or C.sub.1-6alkyl.

Methods of Preparing

[2156] The present disclosure further provides methods for preparing the compounds of present invention. In some aspect, provided herein are methods of preparing a compound of formula (II), or any embodiment or variation thereof, such as a compound of formula (I'), (I), (I-A) (I-B), (I-C), (I-D), (I-E), (I-F), (I-G), or (II-A), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing.

[2157] In some embodiments, a method for preparing a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, comprises a step of reacting a compound of formula I'-A:

##STR00874## [2158] wherein: [2159] m is an integer from 0 to 4; [2160] n is an integer from 0 to 2; [2161] p is an integer from 0 to 10; [2162] R.sup.1, if present is, independently at each occurrence selected from the group consisting of halo, —CN, C.sub.1-6alkoxy, or —C.sub.1-6alkyl, wherein [2163] the C.sub.1-6alkoxy of R.sup.1 is optionally substituted with one or more halo, and [2164] the C.sub.1-6alkyl of R.sup.1 is optionally substituted with one or more halo; [2165] R.sup.2 is H, C.sub.1-6alkyl, C.sub.3-10cycloalkyl, or 3-15 membered heterocyclyl, wherein [2166] the C.sub.1-6alkyl of R.sup.2 is optionally substituted with one or more halo, —OH, —NH.sub.2, or C.sub.1-6alkoxy, and [2167] the C.sub.3-10cycloalkyl of R.sup.2 is optionally substituted with one or more —OH; and [2168] R.sup.3, if present, is C.sub.1-6alkyl; [2169] with: [2170] a compound of formula (I'-B):

##STR00875## [2171] wherein: [2172] the dashed line represents a single or double bond; [2173]

Y.sup.1 is halo, oxo, or a sulfonate ester [2174] L.sup.1 is C.sub.1-6alkylene, wherein the C.sub.1-6alkylene of L.sup.1 is optionally substituted with one or more C.sub.1-6alkyl, and wherein the C.sub.1-6alkyl is further optionally substituted with one or more —OH or C.sub.1-6alkoxy; [2175] L.sup.2 is O or N(R.sup.x), wherein R.sup.x is H or C.sub.1-6alkyl; and [2176] either [2177] (1) L.sup.3 is absent or is O, C.sub.3-10cycloalkyl, 3-10 membered heterocyclyl, or C.sub.1-6alkylene, wherein [2178] the C.sub.1-6alkylene of L.sup.3 is optionally substituted with one or more C.sub.1-6alkyl, and [2179] the 3-10 membered heterocyclyl is optionally substituted with one or more —OH; [2180] X.sup.1 and X.sup.2 are each independently N or C(R.sup.5); and [2181] R.sup.4 is: [2182] (i) —S(O).sub.2—R.sup.a, [2183] (ii) 5-20 membered heteroaryl, wherein the 5-20 membered heteroaryl of R.sup.4 is optionally substituted with one or more C.sub.1-6alkyl, [2184] (iii) —N(R.sup.d).sub.2, wherein R.sup.d is independently at each occurrence H, C.sub.1-6alkyl, or —S(O).sub.2—R.sup.a, wherein the C.sub.1-6alkyl of R.sup.d is optionally substituted with one or more —OH, [2185] (iv) —NS(O)—(C.sub.1-6alkyl).sub.2, wherein the C.sub.1-6alkyl of R.sup.d is optionally substituted with one or more —OH, [2186] (v) —C(O)—N(R.sup.e).sub.2 wherein R.sup.e is independently at each occurrence H, C.sub.1-6alkyl, or 3-10 membered heterocycle, wherein the 3-10 membered heterocycle is optionally substituted with one or more oxo, or both R.sup.e together with the N to which they are attached are taken together to form a 3-10 membered heterocyclyl, wherein the 3-10 membered heterocyclyl is optionally substituted with one or more halo, oxo, —OH, NH.sub.2, NH—S(O).sub.2—R.sup.a, or —S(O).sub.2—R.sup.a, [2187] (vi) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl or oxo, [2188] (vii) —S(O)—N(C.sub.1-6alkyl)-(C.sub.1-6alkyl), or [2189] (viii) —CN; [2190] or [2191] (2) L.sup.3 is absent; and [2192] one of X.sup.1 and X.sup.2 is N or C(R.sup.5); and [2193] the other of X.sup.1 and X.sup.2 is N or C that is taken together with R.sup.4, and the atoms to which they are attached, to form a 5-10 membered heterocyclyl or a 5-20 membered heteroaryl, wherein [2194] the 5-10 membered heterocyclyl is optionally substituted with one or more R.sup.b, wherein R.sup.b is, independently at each occurrence, selected from the group consisting of halo, oxo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10cycloalkyl, and 3-10 membered heterocyclyl, wherein [2195] the C.sub.1-6alkyl of R.sup.b is optionally substituted with one or more halo, OH, or —S(O).sub.2—C.sub.1-6alkyl, and [2196] the C.sub.3-10cycloalkyl of R.sup.b is optionally substituted with one or more —OH, and [2197] the 5-20 membered heteroaryl is optionally substituted with one or more R.sup.e, wherein R.sup.e is, independently at each occurrence, selected from the group consisting of halo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10 cycloalkyl, and 3-10 membered heterocyclyl, wherein [2198] the C.sub.1-6alkyl of R.sup.e is optionally substituted with one or more —S(O).sub.2—C.sub.1-6alkyl, and [2199] the C.sub.3-10cycloalkyl of R.sup.e is optionally substituted with one or more —OH; [2200] R.sup.a is, independently at each occurrence: [2201] (i) C.sub.1-6alkyl optionally substituted with one or more halo, —OH, —S(O).sub.2—C.sub.1-6alkyl, or —N(C.sub.1-6alkyl)—C(O)—C.sub.1-6alkyl, or [2202] (ii) C.sub.3-10cycloalkyl optionally substituted with one or more —OH, —C(O).sub.2—C.sub.1-6alkyl, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, or —C(O)—C.sub.3-10heterocyclyl, or C.sub.1-6alkyl, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH, or [2203] (iii) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl; [2204] R.sup.5 is, independently at each occurrence, H, halo, —CN, 3-10 membered heterocyclyl, C.sub.1-6alkyl, or C.sub.1-6alkoxy, wherein the C.sub.1-6alkyl of R.sup.5 is optionally substituted with one or more halo, or —OH and wherein the C.sub.1-6alkoxy is optionally substituted with one or more halo; [2205] R.sup.6 and R.sup.7 are each independently H or halo; [2206] to give a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing. [2207] In some embodiments, a method for preparing a compound of formula (I'), or a

stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, comprises a step of reacting a compound of formula I'-A':

##STR00876## [2208] wherein: [2209] m is an integer from 0 to 4; [2210] n is an integer from 0 to 2; [2211] p is an integer from 0 to 10; [2212] R.sup.1, if present is, independently at each occurrence selected from the group consisting of halo, —CN, C.sub.1-6alkoxy, or —C.sub.1-6alkyl, wherein [2213] the C.sub.1-6alkoxy of R.sup.1 is optionally substituted with one or more halo, and [2214] the C.sub.1-6alkyl of R.sup.1 is optionally substituted with one or more halo; [2215] R.sup.2 is H, C.sub.1-6alkyl, C.sub.3-10cycloalkyl, or 3-15 membered heterocyclyl, wherein [2216] the C.sub.1-6alkyl of R.sup.2 is optionally substituted with one or more halo, —OH, —NH.sub.2, or C.sub.1-6alkoxy, and [2217] the C.sub.3-10cycloalkyl of R.sup.2 is optionally substituted with one or more —OH; and [2218] R.sup.3, if present, is C.sub.1-6alkyl; [2219] with: [2220] a compound of formula (I'-B'):

##STR00877## [2221] wherein: [2222] the dashed line represents a single or double bond; [2223] Y.sup.1 is halo, oxo, or a sulfonate ester [2224] L.sup.1 is C.sub.1-6alkylene, wherein the C.sub.1-6alkylene of L.sup.1 is optionally substituted with one or more C.sub.1-6alkyl, and wherein the C.sub.1-6alkyl is further optionally substituted with one or more —OH or C.sub.1-6alkoxy; [2225] L.sup.2 is O or N(R.sup.x), wherein R.sup.x is H or C.sub.1-6alkyl; and [2226] either [2227] (1) L.sup.3 is absent or is O, C.sub.3-10cycloalkyl, 3-10 membered heterocyclyl, or C.sub.1-6alkylene, wherein [2228] the C.sub.3-10cycloalkyl is optionally substituted with one or more —OH, or C.sub.1-6alkyl, [2229] the C.sub.1-6alkylene of L.sup.3 is optionally substituted with one or more —OH, or C.sub.1-6alkyl, and [2230] the 3-10 membered heterocyclyl is optionally substituted with one or more —OH; [2231] X.sup.1 and X.sup.2 are each independently N or C(R.sup.5); and [2232] R.sup.4 is: [2233] (i) —S(O).sub.2—R.sup.a; [2234] (ii) 5-20 membered heteroaryl, wherein the 5-20 membered heteroaryl of R.sup.4 is optionally substituted with one or more C.sub.1-6alkyl; [2235] (iii) —N(R.sup.d).sub.2, wherein R.sup.d is independently at each occurrence H, C.sub.1-6 alkyl, or —S(O).sub.2—R.sup.a, wherein the C.sub.1-6alkyl of R.sup.d is optionally substituted with one or more —OH, [2236] (iv) —NS(O)—(C.sub.1-6alkyl).sub.2, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH, [2237] (v) —C(O)—N(R.sup.e).sub.2 wherein R.sup.e is independently at each occurrence H, C.sub.1-6 alkyl, or 3-10 membered heterocycle, wherein the 3-10 membered heterocycle is optionally substituted with one or more oxo, or both R.sup.e together with the N to which they are attached are taken together to form a 3-10 membered heterocyclyl, wherein the 3-10 membered heterocyclyl is optionally substituted with one or more halo, oxo, —OH, NH.sub.2, —NH—S(O).sub.2—R.sup.a, or —S(O).sub.2—R.sup.a, [2238] (vi) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl, —OH, oxo or —S(O).sub.2R.sup.a, [2239] (vii) —S(O)—N(C.sub.1-6alkyl)-(C.sub.1-6alkyl), [2240] (viii) —CN, [2241] (ix) —(CH.sub.2).sub.qOH, wherein q is an integer from 0-6, [2242] (x) —C(O)—C.sub.1-6alkyl, or [2243] (xi) —P(O)(C.sub.1-6alkyl).sub.2; [2244] or [2245] (2) L.sup.3 is absent; and [2246] one of X.sup.1 and X.sup.2 is N or C(R.sup.5); and [2247] the other of X.sup.1 and X.sup.2 is N or C that is taken together with R.sup.4, and the atoms to which they are attached, to form a 5-10 membered heterocyclyl or a 5-20 membered heteroaryl, wherein [2248] the 5-10 membered heterocyclyl is optionally substituted with one or more R.sup.b, wherein R.sup.b is, independently at each occurrence, selected from the group consisting of halo, oxo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10cycloalkyl, and 3-10 membered heterocyclyl, wherein [2249] the C.sub.1-6alkyl of R.sup.b is optionally substituted with one or more halo, OH, —S(O).sub.2—C.sub.1-6alkyl, or C.sub.3-10cycloalkyl, and wherein the C.sub.3-10cycloalkyl of the C.sub.1-6alkyl of R.sup.b is further optionally substituted with one or more C.sub.1-6alkyl or —OH and [2250] the C.sub.3-10cycloalkyl of R.sup.b is optionally substituted with one or more —OH, C.sub.3-10 cycloalkyl, or C.sub.1-6alkyl, and wherein the C.sub.1-6alkyl of the C.sub.3-10cycloalkyl of R.sup.b is further optionally substituted with one or

more —OH, and [2251] the 5-20 membered heteroaryl is optionally substituted with one or more R.sup.e, wherein R.sup.e is, independently at each occurrence, selected from the group consisting of halo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10 cycloalkyl, and 3-10 membered heterocyclyl, wherein [2252] the C.sub.1-6alkyl of R.sup.e is optionally substituted with one or more —S(O).sub.2—C.sub.1-6alkyl, [2253] the C.sub.3-10cycloalkyl of R.sup.e is optionally substituted with one or more —OH, or C.sub.1-6alkyl, [2254] and the 3-10 membered heterocyclyl of R.sup.e is optionally substituted with one or more —OH, or C.sub.1-6alkyl, and wherein the C.sub.1-6alkyl of the 3-10 membered heterocyclyl of R.sup.e is further optionally substituted with one or more —OH; [2255] R.sup.a is, independently at each occurrence: [2256] (i) C.sub.1-6alkyl optionally substituted with one or more halo, —OH, —S(O).sub.2—C.sub.1-6alkyl, or —N(C.sub.1-6alkyl)-C(O)—C.sub.1-6alkyl, [2257] (ii) C.sub.3-10cycloalkyl optionally substituted with one or more —OH, —C(O).sub.2—C.sub.1-6alkyl, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, or —C(O)—C.sub.3-10heterocyclyl, or C.sub.1-6alkyl, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH, or [2258] (iii) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl; [2259] R.sup.5 is, independently at each occurrence, H, halo, —CN, 3-10 membered heterocyclyl, C.sub.1-6alkyl, or C.sub.1-6alkoxy, wherein the C.sub.1-6alkyl of R.sup.5 is optionally substituted with one or more halo, or —OH and wherein the C.sub.1-6alkoxy of R.sup.5 is optionally substituted with one or more halo; and [2260] R.sup.6 and R.sup.7 are each independently H or halo; [2261] to give a compound of formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing.

[2262] In some embodiments, the compound of formula (I) is prepared by a step comprising:

[2263] a) alkylation of an amine of formula (I'-A) with an alkyl halide, or sulfonate ester compound of formula (I'-B) in the presence of an inorganic base; or [2264] b) reductive amination of a ketone of formula (I'-B) with an amine of formula (I'-A).

[2265] In some embodiments, the compound of formula (I') is prepared by a step comprising:

[2266] a) alkylation of an amine of formula (I'-A') with an alkyl halide, or sulfonate ester compound of formula (I'-B') in the presence of an inorganic base; or [2267] b) reductive amination of a ketone of formula (I'-B') with an amine of formula (I'-A').

[2268] In some embodiments, the compound of formula (I) is prepared by a step comprising alkylation of an amine of formula (I'-A) with an alkyl halide, or sulfonate ester compound of formula (I'-B) in the presence of an inorganic base. In some embodiments, the inorganic base is selected from the group consisting of potassium carbonate, and sodium bicarbonate.

[2269] In some embodiments, the compound of formula (I') is prepared by a step comprising alkylation of an amine of formula (I'-A') with an alkyl halide, or sulfonate ester compound of formula (I'-B') in the presence of an inorganic base. In some embodiments, the inorganic base is selected from the group consisting of potassium carbonate, and sodium bicarbonate.

[2270] In some embodiments, the sulfonate ester compound of formula (I'-B) is a mesylate or a tosylate. In some embodiments, the sulfonate ester compound of formula (I'-B) is a mesylate, or CH.sub.3SO.sub.3—. In some embodiments, the sulfonate ester compound of formula (I'-B) is a tosylate or CH.sub.3C.sub.6H.sub.4SO.sub.3—.

[2271] In some embodiments, the sulfonate ester compound of formula (I'-B') is a mesylate or a tosylate. In some embodiments, the sulfonate ester compound of formula (I'-B') is a mesylate, or CH.sub.3SO.sub.3—. In some embodiments, the sulfonate ester compound of formula (I'-B') is a tosylate or CH.sub.3C.sub.6H.sub.4SO.sub.3—.

[2272] In some embodiments, the compound of formula (I) is prepared by a step comprising reductive amination of a ketone of formula (I'-B) with an amine of formula (I'-A). In some embodiments, the reductive amination proceeds under the action of sodium triacetoxyborohydride, titanium tetraisopropoxide and acetic acid.

[2273] In some embodiments, the compound of formula (I') is prepared by a step comprising reductive amination of a ketone of formula (I'-B') with an amine of formula (I'-A'). In some embodiments, the reductive amination proceeds under the action of sodium triacetoxyborohydride, titanium tetraisopropoxide and acetic acid.

[2274] It is understood that the present disclosure also provides for methods of preparing compounds of formula (II). For example, compounds of formula (II) may be prepared by reacting a compound of formula (II'-A):

##STR00878##

wherein and m, n, p, R^{sup.1}, and R^{sup.2} are as defined for a compound of formula (II), with a compound of formula (II'-B):

##STR00879##

wherein Y^{sup.1} is halo, oxo, or a sulfonate ester; and L^{sup.1}, L^{sup.2}, L^{sup.3}, R^{sup.4}, X^{sup.1}, X^{sup.2}, X^{sup.3}, and X^{sup.4} are as defined for a compound of formula (II), to give a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing.

[2275] In some embodiments, a method for preparing a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, comprises a step of reacting a compound of formula (I'-C):

##STR00880## [2276] wherein: [2277] m is an integer from 0 to 4; [2278] n is an integer from 0 to 2; [2279] p is an integer from 0 to 10; [2280] R^{sup.1}, if present is, independently at each occurrence selected from the group consisting of halo, —CN, C_{sub.1-6}alkoxy, or —C_{sub.1-6}alkyl, wherein [2281] the C_{sub.1-6}alkoxy of R^{sup.1} is optionally substituted with one or more halo, and [2282] the C_{sub.1-6}alkyl of R^{sup.1} is optionally substituted with one or more halo; [2283] R^{sup.2} is H, C_{sub.1-6}alkyl, C_{sub.3-10}cycloalkyl, or 3-15 membered heterocyclyl, wherein [2284] the C_{sub.1-6}alkyl of R^{sup.2} is optionally substituted with one or more halo, —OH, —NH_{sub.2}, or C_{sub.1-6}alkoxy, and [2285] the C_{sub.3-10}cycloalkyl of R^{sup.2} is optionally substituted with one or more —OH; [2286] R^{sup.3}, if present, is C_{sub.1-6}alkyl; [2287] L^{sup.1} is C_{sub.1-6}alkylene, wherein the C_{sub.1-6}alkylene of L^{sup.1} is optionally substituted with one or more C_{sub.1-6}alkyl, and wherein the C_{sub.1-6}alkyl is further optionally substituted with one or more —OH or C_{sub.1-6}alkoxy; and [2288] Y^{sup.2} is halo, —OH or —NH_{sub.2}; [2289] with: [2290] a compound of formula (I'-D):

##STR00881## [2291] wherein: [2292] Y^{sup.3} is —OH or —NH(R^{sup.x}), wherein each R^{sup.x} is independently H or C_{sub.1-6}alkyl; and either [2293] (1) L^{sup.3} is absent or is O, C_{sub.3-10}cycloalkyl, 3-10 membered heterocyclyl, or C_{sub.1-6}alkylene, wherein [2294] the C_{sub.1-6}alkylene of L^{sup.3} is optionally substituted with one or more C_{sub.1-6}alkyl, and [2295] the 3-10 membered heterocyclyl is optionally substituted with one or more —OH; X^{sup.1} and X^{sup.2} are each independently N or C(R^{sup.5}); and [2296] R^{sup.4} is: [2297] (i) —S(O)_{sub.2}—R^{sup.a}; [2298] (ii) 5-20 membered heteroaryl, wherein the 5-20 membered heteroaryl of R^{sup.4} is optionally substituted with one or more C_{sub.1-6}alkyl; [2299] (iii) N(R^{sup.d})_{sub.2}, wherein R^{sup.d} is independently at each occurrence H, C_{sub.1-6}alkyl, or —S(O)_{sub.2}—R^{sup.a}, wherein the C_{sub.1-6}alkyl of R^{sup.d} is optionally substituted with one or more —OH, [2300] (iv) —NS(O)—(C_{sub.1-6}alkyl)_{sub.2}, wherein the C_{sub.1-6}alkyl of R^{sup.d} is optionally substituted with one or more —OH, [2301] (v) —C(O)—N(R^{sup.e})_{sub.2} wherein R^{sup.e} is independently at each occurrence H, C_{sub.1-6}alkyl, or 3-10 membered heterocycle, wherein the 3-10 membered heterocycle is optionally substituted with one or more oxo, or both R^{sup.e} together with the N to which they are attached are taken together to form a 3-10 membered heterocyclyl, wherein the 3-10 membered heterocyclyl is optionally substituted with one or more halo, oxo, —OH, NH_{sub.2}, NH—S(O)_{sub.2}—R^{sup.a}, or —S(O)_{sub.2}—R^{sup.a}, [2302] (vi) 3-10 membered heterocyclyl optionally substituted with one or more C_{sub.1-6}alkyl or oxo, [2303] (vii) —S(O)—N(C_{sub.1-6}alkyl)-(C_{sub.1-6}alkyl), [2304] (viii) —CN; [2305] or [2306] (2) L^{sup.3} is absent; and [2307]

one of X.sup.1 and X.sup.2 is N or C(R.sup.5); and [2308] the other of X.sup.1 and X.sup.2 is N or C that is taken together with R.sup.4, and the atoms to which they are attached, to form a 5-10 membered heterocyclyl or a 5-20 membered heteroaryl, wherein [2309] the 5-10 membered heterocyclyl is optionally substituted with one or more R.sup.b, wherein R.sup.b is, independently at each occurrence, selected from the group consisting of halo, oxo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10cycloalkyl, and 3-10 membered heterocyclyl, wherein [2310] the C.sub.1-6alkyl of R.sup.b is optionally substituted with one or more halo, OH, or —S(O).sub.2—C.sub.1-6alkyl, and [2311] the C.sub.3-10cycloalkyl of R.sup.b is optionally substituted with one or more —OH, and [2312] the 5-20 membered heteroaryl is optionally substituted with one or more R.sup.e, wherein R.sup.e is, independently at each occurrence, selected from the group consisting of halo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10 cycloalkyl, and 3-10 membered heterocyclyl, wherein [2313] the C.sub.1-6alkyl of R.sup.e is optionally substituted with one or more —S(O).sub.2—C.sub.1-6alkyl, and [2314] the C.sub.3-10cycloalkyl of R.sup.e is optionally substituted with one or more —OH; [2315] R.sup.a is, independently at each occurrence: [2316] (i) C.sub.1-6alkyl optionally substituted with one or more halo, —OH, —S(O).sub.2—C.sub.1-6alkyl, or —N(C.sub.1-6alkyl)—C(O)—C.sub.1-6alkyl, or [2317] (ii) C.sub.3-10cycloalkyl optionally substituted with one or more —OH, —C(O).sub.2—C.sub.1-6alkyl, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, or —C(O)—C.sub.3-10heterocyclyl, or C.sub.1-6alkyl, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH, or [2318] (iii) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl; [2319] R.sup.5 is, independently at each occurrence, H, halo, —CN, 3-10 membered heterocyclyl, C.sub.1-6alkyl, or C.sub.1-6alkoxy, wherein the C.sub.1-6alkyl of R.sup.5 is optionally substituted with one or more halo, or —OH and wherein the C.sub.1-6alkoxy is optionally substituted with one or more halo; and [2320] R.sup.6 and R.sup.7 are each independently H or halo; [2321] to give a compound of formula (I), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing. [2322] In some embodiments, a method for preparing a compound of formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, comprises a step of reacting a compound of formula (I'—C'):

##STR00882## [2323] wherein: [2324] m is an integer from 0 to 4; [2325] n is an integer from 0 to 2; [2326] p is an integer from 0 to 10; [2327] R.sup.1, if present is, independently at each occurrence selected from the group consisting of halo, —CN, C.sub.1-6alkoxy, or —C.sub.1-6alkyl, wherein [2328] the C.sub.1-6alkoxy of R.sup.1 is optionally substituted with one or more halo, and [2329] the C.sub.1-6alkyl of R.sup.1 is optionally substituted with one or more halo; [2330] R.sup.2 is H, C.sub.1-6alkyl, C.sub.3-10cycloalkyl, or 3-15 membered heterocyclyl, wherein [2331] the C.sub.1-6alkyl of R.sup.2 is optionally substituted with one or more halo, —OH, —NH.sub.2, or C.sub.1-6alkoxy, and [2332] the C.sub.3-10cycloalkyl of R.sup.2 is optionally substituted with one or more —OH; [2333] R.sup.3, if present, is C.sub.1-6alkyl; [2334] L.sup.1 is C.sub.1-6alkylene, wherein the C.sub.1-6alkylene of L.sup.1 is optionally substituted with one or more C.sub.1-6alkyl, and wherein the C.sub.1-6alkyl is further optionally substituted with one or more —OH or C.sub.1-6alkoxy; and [2335] Y.sup.2 is halo, —OH or —NH.sub.2; [2336] with: [2337] a compound of formula (I'—D'):

##STR00883## [2338] wherein: [2339] Y.sup.3 is —OH or NH(R.sup.x), wherein each R.sup.x is independently H or C.sub.1-6alkyl; and either [2340] (1) L.sup.3 is absent or is O, C.sub.3-10cycloalkyl, 3-10 membered heterocyclyl, or C.sub.1-6alkylene, wherein [2341] the C.sub.3-10cycloalkyl is optionally substituted with one or more —OH, or C.sub.1-6alkyl, [2342] the C.sub.1-6alkylene of L.sup.3 is optionally substituted with one or more —OH, or C.sub.1-6alkyl, and [2343] the 3-10 membered heterocyclyl is optionally substituted with one or more —OH;

X.sup.1 and X.sup.2 are each independently N or C(R.sup.5); and [2344] R.sup.4 is: [2345] (i) —S(O).sub.2—R.sup.a; [2346] (ii) 5-20 membered heteroaryl, wherein the 5-20 membered heteroaryl of R.sup.4 is optionally substituted with one or more C.sub.1-6alkyl; [2347] (iii) —N(R.sup.d).sub.2, wherein R.sup.d is independently at each occurrence H, C.sub.1-6 alkyl, or —S(O).sub.2—R.sup.a, wherein the C.sub.1-6alkyl of R.sup.d is optionally substituted with one or more —OH, [2348] (iv) —NS(O)—(C.sub.1-6alkyl).sub.2, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH, [2349] (v) —C(O)—N(R.sup.e).sub.2 wherein R.sup.e is independently at each occurrence H, C.sub.1-6 alkyl, or 3-10 membered heterocycle, wherein the 3-10 membered heterocycle is optionally substituted with one or more oxo, or both R.sup.e together with the N to which they are attached are taken together to form a 3-10 membered heterocyclyl, wherein the 3-10 membered heterocyclyl is optionally substituted with one or more halo, oxo, —OH, NH.sub.2, —NH—S(O).sub.2—R.sup.a, or —S(O).sub.2—R.sup.a, [2350] (vi) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl, —OH, oxo or —S(O).sub.2R.sup.a, [2351] (vii) —S(O)—N(C.sub.1-6alkyl)-(C.sub.1-6alkyl), [2352] (viii) —CN; [2353] (ix) —(CH.sub.2).sub.qOH, wherein q is an integer from 0-6; or [2354] (x) —C(O)—C.sub.1-6alkyl [2355] (xi) —P(O)(C.sub.1-6alkyl).sub.2 [2356] or [2357] (2) L.sup.3 is absent; and [2358] one of X.sup.1 and X.sup.2 is N or C(R.sup.5); and [2359] the other of X.sup.1 and X.sup.2 is N or C that is taken together with R.sup.4, and the atoms to which they are attached, to form a 5-10 membered heterocyclyl or a 5-20 membered heteroaryl, wherein [2360] the 5-10 membered heterocyclyl is optionally substituted with one or more R.sup.b, wherein R.sup.b is, independently at each occurrence, selected from the group consisting of halo, oxo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10cycloalkyl, and 3-10 membered heterocyclyl, wherein [2361] the C.sub.1-6alkyl of R.sup.b is optionally substituted with one or more halo, OH, —S(O).sub.2—C.sub.1-6alkyl, or C.sub.3-10cycloalkyl, and wherein the C.sub.3-10cycloalkyl of the C.sub.1-6alkyl of R.sup.b is further optionally substituted with one or more C.sub.1-6alkyl or —OH and [2362] the C.sub.3-10cycloalkyl of R.sup.b is optionally substituted with one or more —OH, C.sub.3-10 cycloalkyl, or C.sub.1-6alkyl, and wherein the C.sub.1-6alkyl of the C.sub.3-10cycloalkyl of R.sup.b is further optionally substituted with one or more —OH, and [2363] the 5-20 membered heteroaryl is optionally substituted with one or more R.sup.e, wherein R.sup.e is, independently at each occurrence, selected from the group consisting of halo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10 cycloalkyl, and 3-10 membered heterocyclyl, wherein [2364] the C.sub.1-6alkyl of R.sup.e is optionally substituted with one or more —S(O).sub.2—C.sub.1-6alkyl, [2365] the C.sub.3-10cycloalkyl of R.sup.e is optionally substituted with one or more —OH, or C.sub.1-6alkyl, [2366] and the 3-10 membered heterocyclyl of R.sup.e is optionally substituted with one or more —OH, or C.sub.1-6alkyl, and wherein the C.sub.1-6alkyl of the 3-10 membered heterocyclyl of R.sup.e is further optionally substituted with one or more —OH; [2367] R.sup.a is, independently at each occurrence: [2368] (i) C.sub.1-6alkyl optionally substituted with one or more halo, —OH, —S(O).sub.2—C.sub.1-6alkyl, or —N(C.sub.1-6alkyl)-C(O)—C.sub.1-6alkyl, [2369] (ii) C.sub.3-10cycloalkyl optionally substituted with one or more —OH, —C(O).sub.2—C.sub.1-6alkyl, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, or —C(O)—C.sub.3-10heterocyclyl, or C.sub.1-6alkyl, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH, or [2370] (iii) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl; [2371] R.sup.5 is, independently at each occurrence, H, halo, —CN, 3-10 membered heterocyclyl, C.sub.1-6alkyl, or C.sub.1-6alkoxy, wherein the C.sub.1-6alkyl of R.sup.5 is optionally substituted with one or more halo, or —OH and wherein the C.sub.1-6alkoxy of R.sup.5 is optionally substituted with one or more halo; and [2372] R.sup.6 and R.sup.7 are each independently H or halo; [2373] to give a compound of formula (I'), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable

salt of any of the foregoing.

[2374] In some embodiments, the compound of formula (I) is prepared by a step comprising:

[2375] a) coupling of an alcohol compound of formula (I'—C) with a phenol compound of formula (I'-D), or a heterocyclic variant, under Mitsunobu-type reaction conditions; or [2376] b) reacting an alkyl halide compound of formula (I'—C) with a phenol or amine compound of formula (I'-D), in the presence of a catalyst.

[2377] In some embodiments, the compound of formula (I') is prepared by a step comprising:

[2378] a) coupling of an alcohol compound of formula (I'—C') with a phenol compound of formula (I'-D'), or a heterocyclic variant, under Mitsunobu-type reaction conditions; or [2379] b) reacting an alkyl halide compound of formula (I'—C') with a phenol or amine compound of formula (I'-D'), in the presence of a catalyst.

[2380] In some embodiments, the compound of formula (I) is prepared by a step comprising coupling of an alcohol compound of formula (I'—C) with a phenol compound of formula (I'-D), or a heterocyclic variant, under Mitsunobu-type reaction conditions. In some embodiments, the Mitsunobu-type reaction conditions comprises coupling of an alcohol compound of formula (I'—C) with a phenol compound of formula (I'-D), or a heterocyclic variant, in the presence of triphenylphosphine and diisopropyl azodicarboxylate.

[2381] In some embodiments, the compound of formula (I') is prepared by a step comprising coupling of an alcohol compound of formula (I'—C') with a phenol compound of formula (I'-D'), or a heterocyclic variant, under Mitsunobu-type reaction conditions. In some embodiments, the Mitsunobu-type reaction conditions comprises coupling of an alcohol compound of formula (I'—C') with a phenol compound of formula (I'-D'), or a heterocyclic variant, in the presence of triphenylphosphine and diisopropyl azodicarboxylate.

[2382] In some embodiments, the compound of formula (I) is prepared by a step comprising reacting an alkyl halide compound of formula (I'—C) with a phenol compound of formula (I'-D), in the presence of a base. In some embodiments, the base is silver oxide or potassium carbonate.

[2383] In some embodiments, the compound of formula (I') is prepared by a step comprising reacting an alkyl halide compound of formula (I'-C') with a phenol compound of formula (I'-D'), in the presence of a base. In some embodiments, the base is silver oxide or potassium carbonate.

[2384] It is understood that the present disclosure also provides for methods of preparing compounds of formula (II). For example, compounds of formula (II) may be prepared by reacting a compound of formula (II'—C):

##STR00884##

wherein Y^{sup.2} is halo, —OH or —NH₂; and m, n, p, R^{sup.1}, R^{sup.2}, R^{sup.3}, and L^{sup.1} are as defined for a compound of formula (II), with a compound of formula (II'-D):

##STR00885##

wherein Y^{sup.3} is —OH or —NH(R^{sup.x}), wherein each R^{sup.x} is independently H or C₁₋₆alkyl; L^{sup.3}, R^{sup.4}, X^{sup.1}, X^{sup.2}, X^{sup.3}, and X^{sup.4} are as defined for a compound of formula (II),

to give a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing.

[2385] In further examples, the compound of formula (II), formula (I'), formula (I), or any embodiment or variation thereof, such as a compound of formula (I), (I-A) (I-B), (I-C), (I-D), (I-E), (I-F), (I-G), or (II-A), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing is prepared by methods shown in the examples below.

EXAMPLES

[2386] The following synthetic reaction schemes, which are detailed in the Schemes, General Procedures, and Examples, are merely illustrative of some of the methods by which the compounds of the present disclosure, or an embodiment or aspect thereof, can be synthesized. Various modifications to these synthetic reaction schemes can be made, as will be apparent to those of

ordinary skill in the art.

[2387] The starting materials and the intermediates of the synthetic reaction schemes can be isolated and purified if desired using conventional techniques, including but not limited to, filtration, distillation, crystallization, chromatography, and the like. Such materials can be characterized using conventional means, including physical constants and spectral data.

[2388] Although certain exemplary embodiments are depicted and described herein, the compounds of the present disclosure, or any variation or embodiment thereof, may be prepared using appropriate starting materials according to the methods described generally herein and/or by methods available to one of ordinary skill in the art.

SYNTHETIC EXAMPLES

[2389] As depicted in the Schemes, General Procedures, and Examples below, in certain exemplary embodiments, compounds of formula (I), formula (I), formula (II), or any variation or embodiment thereof, as described elsewhere herein, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, are prepared according to the general procedures. The general methods below, and other methods known to synthetic chemists of ordinary skill in the art, can be applied to all formulae, variations, embodiments, and species described herein.

##STR00886##

[2390] Compounds of formula S1-3 may be prepared according to Scheme 1. Alkylation of amine S1-2 with an alkyl halide such as S1-1 in the presence of an inorganic base such as potassium carbonate and a polar aprotic solvent such as DMF provides compound S1-3. In cases where L3 represents a chiral atom, compound S1-3 may be further purified by chiral SFC to deliver single stereoisomers.

##STR00887##

[2391] Compounds of formula S2-3 may be prepared may also be prepared by coupling a phenol or heterocyclic variant such as S2-1 with an alcohol such as S2-2 in the presence of triphenylphosphine and DIAD in an aprotic solvent such as THF, as depicted in Scheme 2. Alternative Mitsunobu-type reaction conditions with polymer-bound triphenylphosphine and DIAD in toluene may also be used to generate compounds of formula S2-3.

##STR00888##

[2392] As outlined in Scheme 3, compounds of formula S3-3 may be prepared from alkyl chlorides such as S3-1 and amine S3-2 by heating in an aprotic solvent such as DMF in the presence of sodium iodide and an inorganic base such as potassium carbonate.

##STR00889##

[2393] Compounds of formula S4-3 may also be prepared from phenol S4-1 and alkyl halide S4-2 by heating in an aprotic solvent such as DMF in the presence of silver oxide, as shown in Scheme 4. Alternative reaction conditions involving potassium carbonate in DMF may also be used to generate compounds of formula S4-3.

##STR00890##

[2394] Oxindoles such as S5-1 may be elaborated by reaction with an alkyl halide in the presence of an inorganic base such as potassium carbonate in DMF, at elevated temperature. This provides compounds such as S5-2, as depicted in Scheme 5.

##STR00891##

[2395] Scheme 6 depicts an approach to compounds of formula S6-5. Alkylation of oxindole S6-1 with an alkyl halide in the presence of a base such as potassium carbonate generates S6-2. Removal of the Boc group by treatment with a protic acid such as TFA in an aprotic solvent such as DCM generates amine S6-3, which can be further elaborated by coupling with an alkyl halide such as S6-4, according to procedures shown in Scheme 1.

##STR00892##

[2396] An alternative approach to compounds of formula S7-3 involving reductive amination of ketone S7-2 with amine S7-2 under the action of sodium triacetoxyborohydride, titanium

tetraisopropoxide and acetic acid, is depicted in Scheme 7.

##STR00893##

[2397] Scheme 8 depicts an alternative approach to compounds of formula S8-6. Generation of benzylic chloride S8-2 from benzylic alcohol S8-1, followed by introduction of a sulfonyl group, gives rise to compound S8-3. Deprotection of S8-3 with in situ-formed TMSI gives phenol S8-4, which can be coupled with alkyl chloride S8-5 in the presence of potassium carbonate and sodium iodide in acetone at elevated temperature. Single stereoisomers may be obtained by chiral SFC purification of S8-6.

##STR00894## ##STR00895##

[2398] Scheme 9 shows an approach to compounds of formula S9-7. Alkylation of phenol S9-1 with ethylene oxide, followed by S_{sub}.NAr reaction with sodium methanethiolate gives thioether S9-3. Oxidation with mCPBA gives sulfone S9-4. Treatment of S9-4 with methanesulfonyl chloride and a tertiary amine base such as DIPEA gives S9-5, which can be converted to S9-7 under conditions similar to Scheme 1.

##STR00896##

[2399] As shown in Scheme 10, compounds of formula S10-1 may be further elaborated via coupling with a sulfonyl chloride such as methanesulfonyl chloride in the presence of triethylamine in DCM to provide S10-2. Compound S10-1 may also be acylated with an anhydride such as acetic anhydride and triethylamine to generate S10-3.

##STR00897##

[2400] As shown in Scheme 11, compounds of formula S11-1 may be elaborated via coupling with a carbamoyl chloride such as N-methylcarbamoyl chloride in the presence of triethylamine to give a urea of the formula S11-2.

##STR00898##

[2401] An alternative approach to compounds of formula S12-3 is depicted in Scheme 12, via coupling of a mesylate such as S12-1 and amine such as S12-2 in acetonitrile using sodium bicarbonate as base.

##STR00899##

[2402] Scheme 13 depicts the conversion of cis isomer S13-1 to trans isomer S13-3. Mitsunobu coupling with p-nitrobenzoic acid generates ester S13-2, which can undergo hydrolysis with lithium hydroxide to generate the trans isomer S13-3.

##STR00900##

[2403] Elaboration of compounds of formula S14-1 (generated using methods described in Scheme 1) is depicted in Scheme 14. Cross coupling of S14-1 with potassium vinyltrifluoroborate using a palladium catalyst and an inorganic base provides olefin S14-2. Dihydroxylation with an osmium catalyst and NMO gives S14-3. Diol S14-3 may be further modified by treatment with DAST, giving rise to difluoride S14-4 as an isomeric mixture. Single isomer analogs such as S14-5 may be obtained by chiral SFC purification.

##STR00901##

[2404] Iodide S15-1 may be elaborated using photoredox methods outlined in Scheme 15. Coupling of S15-1 and 3-iodooxetane can be achieved with nickel(II) chloride ethylene glycol dimethyl ether complex, ligand S15-2, (Ir[dF(CF_{sub}3)ppy]_{sub}.2(dtbpv))PF_{sub}6, bis(trimethylsilyl)silyl-trimethylsilane, Na_{sub}2CO_{sub}3 and blue LED to generate S15-3.

##STR00902## ##STR00903##

[2405] Conversion of cis-disubstituted cyclobutanol S16-1 to the trans isomer S16-4 is depicted in Scheme 16. Treatment with methanesulfonic anhydride and TEA generates mesylate S16-2. Heating with potassium acetate generates ester S16-3, which can be converted to the alcohol under the action of potassium carbonate in MeOH to give S16-4.

##STR00904##

[2406] Compounds of the formula S17-1 can be converted to the corresponding alcohol by

treatment with a reducing agent such as LiAlH₄, as depicted in Scheme 17. Alternatively, S17-1 can undergo hydrolysis with lithium hydroxide to give acid S17-3. Amide bond formation with an amine such as methylamine under the action of T3P and TEA generates amide S17-4.

##STR00905##

[2407] Compounds of formula S18-1 can be elaborated by Suzuki couplings of boronate esters such as S18-2, a catalyst such as bis(4-(di-tert-butylphosphanyl)-N,N-dimethylaniline) dichloropalladium, and a base such as KOAc, as depicted in Scheme 18.

##STR00906##

[2408] Scheme 19 begins with protection of an oxindole nitrogen using sodium hydride and SEMCl to give S19-2. Coupling with a phenol such as S19-3 gives S19-4. Removal of the SEM group is achieved by heating a mixture of S19-4, potassium carbonate, and NaI in acetone to give compound of formula S19-5.

##STR00907##

[2409] Heterocyclic electrophiles bearing protecting groups such as compound S20-1 may be coupled with amine nucleophiles such as S20-2 under the conditions shown in Scheme 20. Deprotection of a protecting group such as a SEM N—O acetal may be achieved by treatment of S20-3 with a protic acid such as TFA to deliver S20-4.

##STR00908## ##STR00909##

[2410] An approach to compounds of formula S21-9 is shown in Scheme 21. Alkylation of thiol S21-1 followed by oxidation to the sulfone generates compound S21-3. Cyclopropane formation via double alkylation in the presence of potassium carbonate gives S21-4. S_NAr reaction with ethylene glycol and mesylation of S21-5 provides compound S21-6, which can be coupled with amine S21-7 under standard conditions (Scheme 12). Treatment of compound S21-8 with a reducing agent such as LiAlH₄ gives compound S21-9.

##STR00910##

[2411] Compounds of formula S22-2 may be obtained by treatment of compound S22-1 with a base such as sodium hydride and an alkylating agent such as iodomethane, as depicted in Scheme 22.

##STR00911## ##STR00912##

[2412] Primary amides of formula S23-7 can be generated according to Scheme 23. Palladium-catalyzed borylation of pyrimidine S23-1 followed by oxidation gives compound S23-3. Alkylation with 1,2-dibromoethane to provide S23-4 and coupling with amine S23-5 under conditions outlined above (Scheme 12) generates compound S23-6. Conversion of the nitrile to the primary amide S23-7 occurs upon treatment with potassium carbonate and aqueous H₂O₂.

##STR00913##

[2413] Compounds of formula S24-8 can be synthesized according to Scheme 24. Esterification and alkylation of carboxylic acid S24-1 provides alkyl bromide S24-3. Coupling with amine S24-4 under standard conditions (Scheme 12) gives S24-5. Hydrolysis followed by amide bond formation with amine S24-7, using N-methyl imidazole and TCFH, gives S24-8.

##STR00914##

[2414] Compounds of formula S25-3 may also be prepared by reaction of an amine such as S25-1 with a secondary mesylate such as S25-2 in the presence of NaHCO₃ and potassium iodide in DMF at elevated temperature, as depicted in Scheme 25. Compounds of formula S25-3 may be further purified by chiral SFC to generate isomerically pure analogs.

##STR00915##

[2415] Compounds of formula S26-10 may be prepared according to Scheme 26. Lithium-halogen exchange with aryl bromide S26-1, followed by addition to a ketone such as S26-2, gives alcohol S26-3 as a mixture of cis and trans isomers. Reaction with sodium methanesulfinate and TFA gives sulfone S26-4. Removal of benzyl ethers by hydrogenation gives diol S26-5. Selective alkylation of the phenol moiety with bromoethyl acetate gives ester S26-6. Reduction with LiAlH₄ gives alcohol S26-7, which can be converted to S26-8 and coupled with amine S26-9 using procedures

described above (Scheme 12). Further purification of S26-10 to obtain single stereoisomers may be achieved by methods such as reverse phase HPLC or chiral SFC.

##STR00916##

[2416] Compounds of formula S27-5 can be prepared according to Scheme 27. Coupling of acid S27-1 with an amine such as S27-2 under the action of EDCI and HOBt gives amide S27-3. Treatment with a protic acid such as TFA gives amine S27-4. Amine S27-4 may be further elaborated, such as by reaction with methanesulfonic anhydride and TEA, to give compounds such as S27-5.

##STR00917##

[2417] Compounds of formula S28-3 can be prepared according to Scheme 28, using reaction conditions similar to those described above. Coupling of amine S28-1 with azatetrahydronaphthyridinones bearing tertiary alcohol moieties such S28-2 occurs upon heating in an aprotic solvent such as acetonitrile in the presence of an inorganic base such as sodium bicarbonate to give compounds of formula S28-3.

##STR00918##

[2418] Compounds of formula S29-4 may can be prepared according to Scheme 29. Reaction of amine S29-1 with an alkyl bromide bearing a secondary alcohol such as S29-2 occurs upon heating in a aprotic solvent such as acetonitrile in the presence of an inorganic base such as sodium bicarbonate to give compounds of formula S29-3. Further purification using methods such as chiral SFC may be used to generate single isomer analogs such as S29-4.

##STR00919##

[2419] Compounds of formula S30-3 and S30-5, which bear multiple substituents on the pyrimidine moiety, are depicted in Scheme 30. Thus, coupling of the amine HCl salt S30-1 with alkyl halide S30-2 may be achieved upon heating in an aprotic solvent such as DMF, in the presence of potassium carbonate and potassium iodide, to deliver compounds of formula 30-3. Alternatively, coupling under identical reaction conditions as those described above with amine S30-1 and a chiral center bearing intermediate such as S30-4, may give rise to compounds of the formula S30-5. If desired, further purification of S30-5 may be achieved using methods such as chiral SFC.

##STR00920##

[2420] Compounds of formula S31-12 may be prepared according to Scheme 31. Treatment of indazole S31-1 with iodine and potassium hydroxide in a solvent such as DMF provides S31-2. Alkylation of S31-2 with an electrophile such as S31-3 occurs in the presence of an inorganic base such as potassium carbonate in warm acetone to give S31-4. Treatment with a Grignard reagent such as methylmagnesium bromide at -10°C . in a solvent such as DCM gives tertiary alcohol S31-5. Protodeiodination occurs on treatment with Zn metal and acetic acid in methanol to give S31-6. Palladium catalyzed borylation may be conducted in a solvent such as 1,4-dioxane by reaction with Bis(pinacolato)diboron, potassium acetate as base, and a palladium catalyst such as [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium(II) to give S31-7. Reaction with Oxone gives rise to alcohol S31-8. Phenol alkylation occurs on treatment with an electrophile such as bis-tosylate S31-9 to give S31-10. Heating a mixture of S31-10, an amine nucleophile such as S31-11, and an inorganic base such as sodium bicarbonate in a solvent such as acetonitrile gives compounds of formula S31-12.

##STR00921##

[2421] Compounds of formula S32-6 may be prepared according to Scheme 32. Heating a mixture of S32-1, 1,2-dibromoethane, and an inorganic base such as cesium carbonate gives S32-2. Heating a mixture of S32-2, an amine nucleophile such as S32-3, an inorganic base such as potassium bicarbonate, and potassium iodide in a solvent such as DMF gives compounds of formula S32-4. Compounds of formula S32-4 may be further elaborated by S.sub.NAr reaction with an amine such as S32-5, a tertiary amine base such as triethylamine, in DMSO at 100°C . in a microwave reactor

to give compounds of formula S32-6.

##STR00922##

[2422] Compounds of formula S33-12 may be prepared according to Scheme 33. Reaction of diamine derivative S33-1 with a carboxylic acid such as S33-2 gives amide S33-3. Cyclization to benzimidazole S33-4 may be achieved by heating S33-3 in acetic acid. Treatment of S33-4 with an excess of 2-(Trimethylsilyl)ethoxymethyl chloride and sodium hydride in an aprotic solvent such as THF gives S33-5. Palladium catalyzed borylation may be conducted in a solvent such as 1,4-dioxane by reaction with diboron reagent S33-6, potassium acetate as base, and a palladium catalyst such as [1,1'-Bis(diphenylphosphino)ferrocene]dichloropalladium(II) to give S33-7. Treatment with Oxone in a mixed solvent such as THE and water gives S33-8. Alkylation with 1,2-dibromoethane in the presence of an inorganic base such as cesium carbonate gives S33-9. S33-9 may be coupled with amine S33-10 in hot acetonitrile in the presence of a base such as sodium bicarbonate to give S33-11. SEM-protecting group removal is achieved by treatment with a protic acid such as TFA, to give a benzimidazole which may then be converted to the free base by treatment with aqueous ammonia to deliver compounds of formula S33-12.

##STR00923##

[2423] Compounds of formula S34-11 may be prepared according to Scheme 34. Reductive amination of an ortho-fluoro aldehyde such as S34-1 with an amine such as S34-2 with TFA and triethylsilane gives S34-3. S.sub.NAr reaction with an amine such as S34-4 occurs upon heating with DIEA in DMSO to give S34-5. Removal of the Boc-protecting group by treatment with a protic acid such as HCl in an aprotic solvent such as EtOAc gives amine S34-6, which can undergo conversion to the cyclic urea on treatment with CDI to give S34-7. Heating S34-7 in ethylene glycol, copper(I) iodide, and lithium tert-butoxide gives alcohol S34-8. Conversion of the alcohol moiety to an alkyl bromide may be achieved by reaction with carbon tetrabromide and triphenyl phosphine in a solvent such as DCM to give S34-9. S34-9 may be coupled with amine S34-10 in hot acetonitrile in the presence of a base such as sodium bicarbonate to give compounds of formula S34-11.

##STR00924## ##STR00925##

[2424] Compounds of formula S35-9 may be prepared according to Scheme 35. S.sub.NAr reaction between S35-1 and amine S35-2 occurs upon heating with a tertiary amine base such as DIEA in DMSO solvent to give S35-3. Nitro group reduction by heating with iron and ammonium chloride in a mixed solvent such as ethanol and water gives S35-4. S35-4 is then converted to benzimidazole derivative S35-5 by heating in trimethyl orthoformate in the presence of p-toluene sulfonic acid. Heating S35-5 in ethylene glycol in the presence of copper (II) chloride and potassium carbonate gives the primary alcohol S35-6. Conversion to the alkyl iodide by treatment with iodine, imidazole, and triphenyl phosphine provides S35-7. S35-7 may be coupled with amine S35-8 in hot acetonitrile in the presence of a base such as sodium bicarbonate to give compounds of formula S35-9.

##STR00926##

[2425] Compounds of formula S36-13 may be prepared according to Scheme 36. Conversion of aniline S36-1 to corresponding sulfonyl chloride S36-2 occurs upon treatment with sodium nitrite, thionyl chloride, copper(I) chloride, and HCl. Reaction of S36-2 with aqueous ammonia provides sulfonamide S36-3. Reaction with N,N-dimethylformamide dimethyl acetal then generates N-sulfonylformamidine S36-4. Suzuki coupling with a vinyl boronate ester such as S36-5 gives enol ether S36-6. Treatment with HCl leads to hydrolysis of the formamidine and enol ether and cyclization to give S36-7. Hydrogenation of S36-7 with Pd/C as catalyst gives S36-8. C—H oxidation of S36-8 with pinacol borane in the presence of catalyst Bis(1,5-cyclooctadiene)dimethoxydiiridium gives S36-9, which may undergo oxidation upon treatment with H.sub.2O.sub.2 and potassium carbonate to give phenol derivative S36-10. S36-10 undergoes alkylation with 1,2-dibromoethane under conditions described above to deliver S36-11. Coupling

of S36-11 with amine S36-12 under conditions described above gives compounds of formula S36-13.

General Procedure for Intermediate A-1

##STR00927##

Step 1: 1-(4-methanesulfonylphenoxy)propan-2-one

##STR00928##

[2426] A mixture of 4-methylsulfonylphenol (500 mg, 2.90 mmol), 1-chloropropan-2-one (268 mg, 2.90 mmol) and K₂CO₃ (521 mg, 3.77 mmol) in DMF (10 mL) was stirred for 3 h. The reaction mixture was cooled to 0° C. and H₂O (20 mL) was added. The biphasic mixture was extracted with EtOAc (3×20 mL). The combined organic layers were washed with H₂O (20 mL), brine (20 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo to give 1-(4-methylsulfonylphenoxy)propan-2-one (Intermediate A-1), which was taken to the next step without further purification. MS=229.1 [M+H]⁺.

[2427] The following intermediate in Table 2 was prepared according to procedures similar those described for Intermediate A-1 using the appropriate starting materials.

General Procedure for Intermediate A-2

TABLE-US-00003 TABLE 2 Exact Mass Intermediate # Structure IUPAC Name [M + H]⁺ A-2 [00929]  1-methyl-6-(2-oxopropoxy)-1,2,3,4-tetrahydroquinolin-2-one Calc'd 234.1 Found 234.2

General Procedure for Intermediate A-3

##STR00930##

Step 1: 1-(2-bromoethoxy)-4-methanesulfonylbenzene

##STR00931##

[2428] To a solution of 4-(methylsulfonyl)phenol (5.00 g, 29.0 mmol) in DMF (50 mL) were added Cs₂CO₃ (18.9 g, 58.1 mmol) and 1,2-dibromoethane (11.0 mL, 145 mmol). The mixture was stirred at 100° C. for 4 h. After cooling to room temperature, the reaction mixture was diluted with H₂O (250 mL) and extracted with EtOAc (3×50 mL). The combined organic layers were washed with brine (30 mL), dried over Na₂SO₄, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 40 g cartridge, 0-60% EtOAc:petroleum ether). The crude product was triturated with MTBE (10 mL) to give 1-(2-bromoethoxy)-4-methanesulfonylbenzene (Intermediate A-3). ¹H NMR (400 MHz, DMSO-d₆): δ 7.85 (d, J=7.2 Hz, 2H), 7.19 (d, J=6.8 Hz, 2H), 4.44 (t, J=4.2 Hz, 2H), 3.84 (t, J=4.2 Hz, 2H), 3.16 (s, 3H). MS=279.0/281.0 [M+H]⁺.

[2429] The following intermediates in Table 3 were prepared according to procedures similar to steps described for Intermediate A-3 using the appropriate starting materials.

General Procedure for Intermediates A-4 to A-8 & A-83

TABLE-US-00004 TABLE 3 Exact Mass Intermediate # Structure IUPAC Name [M + H]⁺ A-4 [00932]  5-(2-bromoethoxy)-1-methyl-2,3-dihydro-1H-indol-2-one Calc'd 270.0 Found 270.0 A-5 [00933]  5-(2-bromoethoxy)-1-methyl-1H-indazole Calc'd 255.0 Found 255.0 A-6 [00934]  4-(2-bromoethoxy)-2-fluoro-1-methanesulfonylbenzene No charge in LCMS A-7 [00935]  N-[4-(2-bromoethoxy)phenyl]-N-methylmethanesulfonamide Calc'd 308.0 Found 308.0 A-8 [00936]  {[4-(2-bromoethoxy)phenyl]imino}dimethyl-λ⁶-sulfanone Calc'd 292.0 Found 292.0 A-83 [00937]  6-(2-bromoethoxy)-1,2,3,4-tetrahydroquinolin-2-one Calc'd 270.0/ 272.0 Found 270.0/ 271.9

General Procedure for Intermediates A-9 and A-10

##STR00938##

Step 1: 3-methanesulfonylpropyl methanesulfonate

##STR00939##

[2430] To a 0° C. solution of 3-methanesulfonylpropan-1-ol (500 mg, 3.62 mmol) in DCM (5 mL)

were added TEA (1.01 mL, 7.24 mmol) and methanesulfonic anhydride (945 mg, 5.43 mmol). The mixture was warmed to room temperature and stirred for 2 h. The reaction mixture was quenched with H₂O (20 mL) and extracted with DCM (3×10 mL). The combined organic layers were washed with brine (15 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo to give 3-methanesulfonylpropyl methanesulfonate, which was used in the next step without further purification.

Step 2: 4-[(3-methanesulfonylpropyl)sulfanyl]phenol

##STR00940##

[2431] To a mixture of 4-sulfanylphenol (319 mg, 2.52 mmol) and 3-methanesulfonylpropyl methanesulfonate (390 mg, 1.80 mmol) in MeCN (10 mL) was added Cs₂CO₃ (705 mg, 2.16 mmol). The mixture was stirred for 2 h, then was diluted with H₂O (10 mL) and extracted with EtOAc (3×10 mL). The combined organic layers were washed with brine (12 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 4 g cartridge, 0-60% EtOAc:petroleum ether) to give 4-[(3-methanesulfonylpropyl)sulfanyl]phenol. MS=245.1 [M-H]⁺.

Step 3: 4-(3-methanesulfonylpropanesulfonyl)phenol

##STR00941##

[2432] To a solution of 4-[(3-methanesulfonylpropyl)sulfanyl]phenol (370 mg, 1.50 mmol) in THE (3 mL) and H₂O (3 mL) was added NaIO₄ (250 μL, 4.51 mmol). The mixture was stirred at 70° C. for 12 h. After cooling to room temperature, the reaction mixture was quenched with saturated aqueous Na₂SO₃ (20 mL), then extracted with EtOAc (3×10 mL). The combined organic layers were washed with brine (10 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 4 g cartridge, 0-60% EtOAc:petroleum ether) to give 4-(3-methanesulfonylpropanesulfonyl)phenol (Intermediate A-9). MS=277.1 [M-H]⁺.

Step 4: 1-(2-bromoethoxy)-4-(3-methanesulfonylpropanesulfonyl)benzene

##STR00942##

[2433] To a solution of 4-(3-methanesulfonylpropanesulfonyl)phenol (Intermediate A-9) (300 mg, 1.08 mmol) in DMF (3 mL) were added K₂CO₃ (223 mg, 1.62 mmol) and 1,2-dibromoethane (407 μL, 5.39 mmol). The mixture was stirred at 100° C. for 12 h. After cooling to room temperature, the reaction mixture was diluted with H₂O (20 mL) and extracted with EtOAc (3×10 mL). The combined organic layers were washed with brine (8 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by preparative TLC (SiO₂, 1:1 EtOAc:petroleum ether) to give 1-(2-bromoethoxy)-4-(3-methanesulfonylpropanesulfonyl)benzene (Intermediate A-10). ¹H NMR (400 MHz, DMSO-d₆): δ 7.84 (d, J=8.8 Hz, 2H), 7.24 (d, J=8.8 Hz, 2H), 4.47 (t, J=4.8 Hz, 2H), 3.86 (t, J=5.2 Hz, 2H), 3.43 (t, J=5.2 Hz, 2H), 3.22 (t, J=7.6 Hz, 2H), 2.98 (s, 3H), 2.00-1.92 (m, 2H).

General Procedure for Intermediate A-11

##STR00943##

Step 1: 3,5-difluoro-4-methanesulfonylphenol

##STR00944##

[2434] To a solution of 4-bromo-3,5-difluorophenol (1.00 g, 4.78 mmol) and MeSO₂Na (4.88 g, 47.8 mmol) in DMSO (15 mL) were added CuI (91.1 mg, 0.478 mmol), DMEDA (84 mg, 0.956 mmol) and K₃PO₄ (203 mg, 0.956 mmol). The mixture was stirred at 80° C. for 16 h. After cooling to room temperature, the reaction mixture was diluted with H₂O (20 mL) and extracted with EtOAc (3×10 mL). The combined organic layers were washed with brine (10 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-50% EtOAc:petroleum ether) to give the crude product. The product was further purified by reverse phase preparative HPLC (Agela C18 column, 5-35% Methanol: 0.04% HCl in H₂O) to give 3,5-difluoro-4-

methanesulfonylphenol (Intermediate A-11). MS=207.1 [M-H].sup.-.

General Procedure for Intermediates A-12 and A-13

##STR00945##

Step 1: 4-bromo-3-(difluoromethyl)phenol

##STR00946##

[2435] To a 0° C. solution of 2-bromo-5-hydroxybenzaldehyde (17.6 g, 87.5 mmol) in DCM (170 mL) was added DAST (14.1 g, 87.6 mmol) in one portion. The mixture was allowed to warm to room temperature and stirred for 3 h. The reaction mixture was cooled to 0° C., quenched with saturated aqueous NaHCO₃ (150 mL), and extracted with DCM (2×100 mL). The combined organic layers were washed with brine (300 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 120 g cartridge, 8-15% EtOAc:petroleum ether) to give 4-bromo-3-(difluoromethyl)phenol.

Step 2: 3-(difluoromethyl)-4-methanesulfonylphenol

##STR00947##

[2436] A mixture of 4-bromo-3-(difluoromethyl)phenol (3.60 g, 16.1 mmol), sodium methanesulfinate (16.5 g, 161 mmol), NaOH (323 mg, 8.07 mmol), (2S)-pyrrolidine-2-carboxylic acid (929 mg, 8.07 mmol) and copper(I) iodide (3.07 g, 16.1 mmol) in DMSO (40 mL) was degassed and purged with N₂ (3×). The mixture was stirred at 100° C. for 48 h. The reaction mixture was poured into ice water (200 mL) and then extracted with EtOAc (3×150 mL). The combined organic layers were washed with brine (2×300 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 80 g cartridge, 0-30% EtOAc:petroleum ether) to give 3-(difluoromethyl)-4-methanesulfonylphenol (Intermediate A-12). .sup.1H NMR (400 MHz, DMSO-d₆): δ 11.04 (s, 1H), 7.87 (d, J=8.4 Hz, 1H), 7.67-7.40 (m, 1H), 7.20 (s, 1H), 7.11 (d, J=4.2 Hz, 1H), 3.21 (s, 3H).

Step 3: 4-(2-bromoethoxy)-2-(difluoromethyl)-1-methanesulfonylbenzene

##STR00948##

[2437] To a solution of 3-(difluoromethyl)-4-methanesulfonylphenol (300 mg, 1.35 mmol) in 1, 2-dibromoethane (7.5 mL) was added Cs₂CO₃ (1.32 g, 4.05 mmol). The mixture was stirred at 100° C. for 12 h. After cooling to room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 20 g cartridge, 0-30% EtOAc:petroleum ether) to give 4-(2-bromoethoxy)-2-(difluoromethyl)-1-methanesulfonylbenzene (Intermediate A-13). .sup.1H NMR (400 MHz, DMSO-d₆): δ 7.99 (d, J=8.8 Hz, 1H), 7.74-7.55 (m, 1H), 7.47-7.32 (m, 2H), 4.56-4.49 (m, 2H), 3.87-3.82 (m, 2H), 3.26 (s, 3H).

General Procedure for Intermediate A-14

##STR00949##

Step 1: 4-methoxy-N-(propan-2-yl)aniline

##STR00950##

[2438] To a solution of 4-methoxyaniline (500 mg, 4.06 mmol) in DCM (10 mL) was added dropwise acetone (943 mg, 16.2 mmol) and AcOH (975 mg, 16.2 mmol). The mixture was stirred at room temperature for 1 h, and then NaBH(OAc)₃ (1.08 g, 5.08 mmol) was added. The resulting mixture was stirred at room temperature for 2 h. The reaction mixture was diluted with H₂O (10 mL) and extracted with EtOAc (3×10 mL). The combined organic layers were washed with brine (30 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-30% EtOAc:petroleum ether) to give 4-methoxy-N-(propan-2-yl)aniline. .sup.1H NMR (400 MHz, DMSO-d₆): δ 6.72-6.61 (m, 2H), 6.54-6.45 (m, 2H), 4.81 (s, 1H), 3.62 (s, 3H), 3.49-3.39 (m, 1H), 1.08 (d, J=6.4 Hz, 6H). MS=166.2 [M+H].sup.+.

Step 2: 2-chloro-N-(4-methoxyphenyl)-N-(propan-2-yl)acetamide

##STR00951##

[2439] A mixture of 4-methoxy-N-(propan-2-yl)aniline (200 mg, 1.21 mmol), TEA (253 μ L, 1.82 mmol) in DCM (3 mL) was degassed and purged with N₂ (3 $\times$). The mixture was cooled to 0° C. and 2-chloroacetyl chloride (273 mg, 2.42 mmol) was added dropwise. The reaction was allowed to warm to room temperature and stirred for 12 h under N₂ atmosphere. The reaction mixture was quenched with H.sub.2O (15 mL) and extracted with EtOAc (3 $\times$ 10 mL). The combined organic layers were washed with saturated aqueous NaHCO₃ (20 mL) followed by brine (20 mL), dried over Na.sub.2SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Isco 4 g cartridge, 0-40% EtOAc:petroleum ether) to give 2-chloro-N-(4-methoxyphenyl)-N-(propan-2-yl)acetamide. MS=242.1 [M+H].sup.+.

Step 3: 5-hydroxy-1-(propan-2-yl)-2,3-dihydro-1H-indol-2-one

##STR00952##

[2440] To 2-chloro-N-(4-methoxyphenyl)-N-(propan-2-yl)acetamide (1.00 g, 4.14 mmol) was slowly added aluminum chloride (2.21 g, 16.6 mmol). The mixture was stirred at 130° C. for 3 h. After cooling to room temperature, the reaction mixture was quenched with aqueous 1.0 M NaOH (20 mL), diluted with H.sub.2O (10 mL), and extracted with EtOAc (3 $\times$ 30 mL). The combined organic layers were washed with brine (20 mL), dried over Na.sub.2SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Isco 12 g cartridge, 0-40% EtOAc:petroleum ether) to give 5-hydroxy-1-(propan-2-yl)-2,3-dihydro-1H-indol-2-one. MS=192.2 [M+H].sup.+.

Step 4: 5-(2-bromoethoxy)-1-(propan-2-yl)-2,3-dihydro-1H-indol-2-one

##STR00953##

[2441] To a solution of 5-hydroxy-1-(propan-2-yl)-2,3-dihydro-1H-indol-2-one (500 mg, 2.61 mmol) in 1,2-dibromoethane (8 mL) was added Cs.sub.2CO₃ (2.56 g, 7.84 mmol). The mixture was stirred at 80° C. for 12 h. After cooling to room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Isco 12 g cartridge, 0-60% EtOAc:petroleum ether) to give 5-(2-bromoethoxy)-1-(propan-2-yl)-2,3-dihydro-1H-indol-2-one (Intermediate A-14). MS=298.1/300.1 [M+H].sup.+.

General Procedure for Intermediates A-15 & A-16

##STR00954##

Step 1: 4-[(difluoromethyl)sulfanyl]phenol

##STR00955##

[2442] To a solution of 4-sulfanylphenol (300 mg, 2.38 mmol) in DMF (4 mL) and H.sub.2O (1 mL) were added K.sub.2CO₃ (1.97 g, 14.3 mmol) and 2-chloro-2,2-difluoro-acetic acid (302 μ L, 3.57 mmol). The mixture was stirred at 120° C. for 15 h. After cooling to room temperature, the reaction mixture was quenched with H.sub.2O (10 mL) and extracted with EtOAc (3 $\times$ 10 mL). The combined organic layers were dried over Na.sub.2SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-30% EtOAc:petroleum ether) to give 4-[(difluoromethyl)sulfanyl]phenol. MS=177.0 [M+H].sup.+.

Step 2: 4-difluoromethanesulfonylphenol

##STR00956##

[2443] To a 0° C. solution of 4-[(difluoromethyl)sulfanyl]phenol (100 mg, 0.568 mmol) in THE (1 mL) and H.sub.2O (1 mL) was added NaIO₄ (94.4 μ L, 1.70 mmol). The mixture was stirred at 70° C. for 15 h. After cooling to room temperature, the reaction mixture was quenched by the addition of saturated aqueous Na.sub.2SO₃ (20 mL), and then extracted with EtOAc (3 $\times$ 5 mL). The combined organic layers were dried over Na.sub.2SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 4 g

cartridge, 0-30% EtOAc:petroleum ether) to give 4-difluoromethanesulfonylphenol (Intermediate A-15). MS=207.0 [M-H].sup.+.

Step 3: 1-(2-bromoethoxy)-4-difluoromethanesulfonylbenzene

##STR00957##

[2444] To a mixture of 4-(difluoromethylsulfonyl)phenol (500 mg, 2.40 mmol) and 1,2-dibromoethane (906 μ L, 12.0 mmol) in DMF (5 mL) was added K.sub.2CO.sub.3 (664 mg, 4.80 mmol). The mixture was stirred at 80° C. for 16 h. After cooling to room temperature, the reaction mixture was diluted with H.sub.2O (20 mL), and then extracted with EtOAc (3 $\times$ 15 mL). The combined organic layers were washed with brine (20 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-50% EtOAc:petroleum ether) to give 1-(2-bromoethoxy)-4-difluoromethanesulfonylbenzene (Intermediate A-16). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 7.80 (d, J=2.8 Hz, 2H), 7.21-7.04 (m, 3H), 4.95 (t, J=7.2 Hz, 2H), 3.85 (t, J=7.2 Hz, 2H).

General Procedure for Intermediate A-17

##STR00958##

Step 1: 4-methanesulfonyl-3-(trifluoromethyl)phenol

##STR00959##

[2445] A mixture of 4-bromo-3-(trifluoromethyl)phenol (7.00 g, 29.1 mmol), sodium methanesulfinate (29.7 g, 290 mmol), CuI (2.77 g, 14.5 mmol), NaOH (581 mg, 14.5 mmol) and L-Proline (1.67 g, 14.5 mmol) in DMSO (100 mL) was degassed and purged with N₂ (3 $\times$). The mixture was stirred at 100° C. for 3 d under N₂ atmosphere. After cooling to room temperature, the reaction mixture was quenched with H.sub.2O (80 mL) and extracted with EtOAc (3 $\times$ 150 mL). The combined organic layers were washed with brine (2 $\times$ 100 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 80 g cartridge, 0-50% EtOAc:petroleum ether) to give 4-methanesulfonyl-3-(trifluoromethyl)phenol. MS=239.1 [M-H].sup.-.

Step 2: 4-(2-bromoethoxy)-1-methanesulfonyl-2-(trifluoromethyl)benzene

##STR00960##

[2446] To a solution of 4-methylsulfonyl-3-(trifluoromethyl)phenol (500 mg, 2.08 mmol) in DMF (5 mL) were added 1,2-dibromoethane (2.36 mL, 31.2 mmol) and K.sub.2CO.sub.3 (575 mg, 4.16 mmol). The mixture was stirred at 80° C. for 2 h. After cooling to room temperature, the reaction mixture was quenched with H.sub.2O (8 mL) and extracted with EtOAc (3 $\times$ 15 mL). The combined organic layers were washed with brine (2 $\times$ 10 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-25% EtOAc:petroleum ether) to give 4-(2-bromoethoxy)-1-methanesulfonyl-2-(trifluoromethyl)benzene (Intermediate A-17). MS=347.0/349.0 [M+H].sup.+.

[2447] The following intermediates in Table 4 were prepared according to procedures similar those described for Intermediate A-17 using the appropriate starting materials.

General Procedure for Intermediates A-18 to A-22

TABLE-US-00005 TABLE 4 Exact Mass Intermediate # Structure Name [M + H].sup.+ A-18

[00961]  5-(2-bromoethoxy)-1,3-difluoro-2-methanesulfonylbenzene Calc'd

332.0/ 334.0 Found 332.1/ 334.0 [M + NH.sub.4].sup.+ A-19 [00962]  4-(2-

bromoethoxy)-2-chloro- 1-methanesulfonylbenzene Calc'd 312.9/ 314.9 Found 312.9/ 315.0 A-20

[00963]  5-(2-bromoethoxy)-2- methanesulfonyl-1,3- dimethylbenzene Calc'd

307.0/ 309.0 Found 307.1/ 309.0 A-21 [00964]  5-(2-bromoethoxy)-2-

methanesulfonyl-3- methylpyridine Calc'd 294.0/ 296.0 Found 294.0/ 296.0 A-22 [00965]

 4-(2-bromoethoxy)-1- methanesulfonyl-2- methylbenzene Calc'd 293.0/ 295.0

Found 293.0/ 295.0

General Procedure for Intermediate A-23

##STR00966##

Step 1: 1-bromo-2-(difluoromethoxy)-4-methoxybenzene

##STR00967##

[2448] To a solution of 2-bromo-5-methoxyphenol (2.50 g, 12.3 mmol) in DMF (36 mL) and H.sub.2O (9 mL) were added K.sub.2CO.sub.3 (10.2 g, 73.9 mmol) and sodium chlorodifluoroacetate (4.69 g, 30.8 mmol). The mixture was stirred at 120° C. for 15 h. After cooling to room temperature, the reaction mixture was diluted with H.sub.2O (30 mL) and extracted with EtOAc (3×30 mL). The combined organic layers were washed with brine (30 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Isco 40 g cartridge, 0-7% EtOAc:petroleum ether) to give 1-bromo-2-(difluoromethoxy)-4-methoxybenzene.

Step 2: 2-(difluoromethoxy)-1-methanesulfonyl-4-methoxybenzene

##STR00968##

[2449] A mixture of 1-bromo-2-(difluoromethoxy)-4-methoxybenzene (3.00 g, 11.9 mmol), sodium methanesulfinate (6.05 g, 59.3 mmol), L-Proline (409 mg, 3.56 mmol), sodium hydroxide (142 mg, 3.56 mmol) and copper (I) iodide (1.13 g, 5.93 mmol) in DMSO (30 mL) was degassed and purged with N₂ (3×). The mixture was stirred at 100° C. for 12 h under N₂ atmosphere. After cooling to room temperature, the reaction mixture was quenched with H.sub.2O (80 mL) and extracted with EtOAc (2×100 mL). The combined organic layers were washed with brine (40 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Isco 80 g cartridge, 0-23% EtOAc:petroleum ether) to give 2-(difluoromethoxy)-1-methanesulfonyl-4-methoxybenzene. MS=253.2 [M+H].sup.+.

Step 3: 3-(difluoromethoxy)-4-methanesulfonylphenol

##STR00969##

[2450] A mixture of 2,2,4,6,6-pentamethylheptane-4-thiol (2.70 g, 13.3 mmol) and t-BuOK (1.56 g, 13.9 mmol) in DMF (14 mL) was degassed and purged with N₂ (3×). The mixture was cooled to 0° C. and a solution of 2-(difluoromethoxy)-1-methanesulfonyl-4-methoxybenzene (1.4 g, 5.55 mmol) in DMF (14 mL) was added dropwise. The resulting mixture was stirred at 50° C. for 2 h. After cooling to room temperature, the reaction mixture was quenched with H.sub.2O (50 mL) and extracted with EtOAc (3×20 mL). The combined organic layers were washed with brine (10 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Isco 40 g cartridge, 0-33% EtOAc:petroleum ether) to give 3-(difluoromethoxy)-4-methanesulfonylphenol. MS=239.2 [M+H].sup.+.

Step 4: 4-(2-bromoethoxy)-2-(difluoromethoxy)-1-methanesulfonylbenzene

##STR00970##

[2451] To a mixture of 3-(difluoromethoxy)-4-methanesulfonylphenol (370 mg, 1.55 mmol) and 1,2-dibromoethane (3.00 mL, 39.7 mmol) in DMF (3 mL) was added Cs.sub.2CO.sub.3 (1.01 g, 3.11 mmol). After stirring at room temperature for 12 h, the mixture was diluted with H.sub.2O (15 mL) and extracted with DCM (3×10 mL). The combined organic layers were washed with brine (5 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Isco 4 g cartridge, 0-30% EtOAc:petroleum ether) to give 4-(2-bromoethoxy)-2-(difluoromethoxy)-1-methanesulfonylbenzene (Intermediate A-23). MS=362.1/364.0 [M+NH.sub.4].sup.+.

General Procedure for Intermediate A-24

##STR00971##

Step 1: methyl 2-[(4-fluorophenyl)sulfanyl]acetate

##STR00972##

[2452] To a solution of 4-fluorobenzene-1-thiol (1.25 mL, 11.7 mmol) in DMF (15 mL) were added K.sub.2CO.sub.3 (3.23 g, 23.4 mmol) and methyl 2-bromoacetate (1.33 mL, 14.0 mmol). The mixture was stirred at room temperature for 5 h. After cooling to 0° C., the reaction mixture was

quenched with H.sub.2O (15 mL) and extracted with EtOAc (2×15 mL). The combined organic layers were washed with brine (2×15 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 25 g cartridge, 1-20% EtOAc:petroleum ether) to give methyl 2-[(4-fluorophenyl)sulfonyl]acetate. MS=201.0 [M+H].sup.+.

Step 2: methyl 2-(4-fluorobenzenesulfonyl)acetate

##STR00973##

[2453] To a solution of methyl 2-[(4-fluorophenyl)sulfonyl]acetate (2.20 g, 11.0 mmol) in MeCN (30 mL) and H.sub.2O (3 mL) was added Oxone (8.11 g, 13.2 mmol). After stirring at room temperature for 16 h, the mixture cooled to 0° C. and quenched with saturated aqueous Na.sub.2S.sub.2O.sub.3 (40 mL) and extracted with EtOAc (2×20 mL). The combined organic layers were washed with brine (2×20 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 25 g cartridge, 15-45% EtOAc:petroleum ether) to give methyl 2-(4-fluorobenzenesulfonyl)acetate. MS=233.0 [M+H].sup.+.

Step 3: methyl 1-(4-fluorobenzenesulfonyl)cyclopropane-1-carboxylate

##STR00974##

[2454] To a solution of methyl 2-(4-fluorobenzenesulfonyl)acetate (1.10 g, 4.74 mmol) in DMF (10 mL) were added K.sub.2CO.sub.3 (1.64 g, 11.8 mmol) and 1,2-dibromoethane (429 µL, 5.68 mmol). The mixture was stirred at 60° C. for 6 h. The reaction mixture was cooled to 0° C., quenched with H.sub.2O (20 mL), and extracted with EtOAc (2×15 mL). The combined organic layers were washed with brine (2×15 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 15-45% EtOAc:petroleum ether) to give methyl 1-(4-fluorobenzenesulfonyl)cyclopropane-1-carboxylate. MS=259.0 [M+H].sup.+.

Step 4: methyl 1-[4-(2-hydroxyethoxy)benzenesulfonyl]cyclopropane-1-carboxylate

##STR00975##

[2455] To a solution of methyl 1-(4-fluorobenzenesulfonyl)cyclopropane-1-carboxylate (1.50 g, 5.81 mmol) in DMSO (15 mL) were added Cs.sub.2CO.sub.3 (5.68 g, 17.4 mmol) and ethylene glycol (6.50 mL, 116 mmol). The mixture was stirred at 100° C. for 5 h. The hydrolysis product of the ester was detected. After cooling to room temperature, Mel (1.08 mL, 17.4 mmol) was added to the mixture. The reaction mixture was stirred at room temperature for 1 h and was then quenched with H.sub.2O (30 mL) and extracted with EtOAc (2×15 mL). The combined organic layers were washed with brine (2×15 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 45-80% EtOAc:petroleum ether) to give methyl 1-[4-(2-hydroxyethoxy)benzenesulfonyl]cyclopropane-1-carboxylate. MS=301.1 [M+H].sup.+.

Step 5: methyl 1-{4-[2-(methanesulfonyloxy)ethoxy]benzenesulfonyl}cyclopropane-1-carboxylate

##STR00976##

[2456] To a 0° C. solution of methyl 1-[4-(2-hydroxyethoxy)benzenesulfonyl]cyclopropane-1-carboxylate (570 mg, 1.90 mmol) in DCM (10 mL) were added TEA (528 µL, 3.80 mmol) and methylsulfonyl methanesulfonate (397 mg, 2.28 mmol). The mixture was stirred at 0° C. for 3 h, and then quenched with H.sub.2O (10 mL). 4 M aqueous HCl (0.1 mL) was added, and the mixture was extracted with EtOAc (8 mL×2). The combined organic layers were washed with brine (2×8 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give methyl 1-{4-[2-(methanesulfonyloxy)ethoxy]benzenesulfonyl}cyclopropane-1-carboxylate (Intermediate A-24), which was used in the subsequent step without further purification. MS=378.9 [M+H].sup.+.

General Procedure for Intermediates A-25 & A-85

##STR00977##

Step 1: 4-[(2-hydroxyethyl)sulfonyl]phenol

##STR00978##

[2457] A solution of 4-bromophenol (10.0 g, 57.8 mmol), 2-sulfanylethanol (5.24 mL, 75.1 mmol), XantPhos (6.69 g, 11.6 mmol), Pd.sub.2(dba).sub.3 (5.29 g, 5.78 mmol) and DIEA (25.2 mL, 145 mmol) in 1,4-dioxane (10 mL) was degassed and purged with N₂ (3×). The mixture was stirred at 100° C. for 12 h under N₂ atmosphere. After cooling to room temperature, the reaction mixture was quenched with H.sub.2O (20 mL) and extracted with EtOAc (2×50 mL). The combined organic layers were washed with brine (20 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 20 g cartridge, 0-50% EtOAc:petroleum ether) to give 4-[(2-hydroxyethyl)sulfanyl]phenol.

Step 2: 4-(2-hydroxyethanesulfonyl)phenol

##STR00979##

[2458] A solution of 4-[(2-hydroxyethyl)sulfanyl]phenol (4.80 g, 28.2 mmol), NaIO.sub.4 (4.69 mL, 84.6 mmol) in THE (50 mL) and H.sub.2O (50 mL) was degassed and purged with N₂ (3×). The mixture was stirred at 70° C. for 12 h under N₂ atmosphere. After cooling to room temperature, the reaction mixture was quenched with saturated aqueous Na.sub.2S.sub.2O.sub.3 (100 mL) and extracted with EtOAc (2×70 mL). The combined organic layers were washed with brine (70 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give 4-(2-hydroxyethanesulfonyl)phenol (Intermediate A-85), which was used in the subsequent step without further purification. MS=203.0 [M+H].sup.+.

Step 3: 4-(2-chloroethanesulfonyl)phenol

##STR00980##

[2459] To a 0° C. mixture of 4-(2-hydroxyethanesulfonyl)phenol (600 mg, 2.97 mmol) and pyridine (478 µL, 5.93 mmol) in DCM (6 mL) was added SOCl.sub.2 (645 µL, 8.90 mmol). The mixture was stirred at 35° C. for 16 h. After cooling to room temperature, the reaction mixture was quenched with H.sub.2O (20 mL) and extracted with DCM (3×10 mL). The combined organic layers were washed with brine (30 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-100% EtOAc:petroleum ether) to give 4-(2-chloroethanesulfonyl)phenol.

Step 4: 4-[2-(methylamino)ethanesulfonyl]phenol

##STR00981##

[2460] A mixture of 4-(2-chloroethanesulfonyl)phenol (250 mg, 1.13 mmol), 2.0 M methylamine in MeOH (7.93 mL, 15.9 mmol), and KI (75 mg, 0.45 mmol) in THF (2 mL) was stirred at 60° C. for 16 h under N₂ atmosphere. After cooling to room temperature, the mixture was filtered, and the filtrate was concentrated in vacuo to give 4-[2-(methylamino)ethanesulfonyl]phenol, which was used in the subsequent step without further purification. MS=216.1 [M+H].sup.+.

Step 5: N-[2-(4-hydroxybenzenesulfonyl)ethyl]-N-methylacetamide

##STR00982##

[2461] A mixture of 4-[2-(methylamino)ethanesulfonyl]phenol (340 mg, 1.58 mmol) and Ac.sub.2O (147 µL, 1.58 mmol) in THE (5 mL) was stirred for 16 h. The mixture was filtered and the filtered cake was concentrated under in vacuo to give N-[2-(4-hydroxybenzenesulfonyl)ethyl]-N-methylacetamide (Intermediate A-25), which was used in the subsequent step without further purification. MS=258.2 [M+H].sup.+.

General Procedure for Intermediate A-26

##STR00983##

Step 1: 5-chloro-2-(oxetan-3-ylsulfanyl)pyridine

##STR00984##

[2462] To a solution of 5-chloropyridine-2-thiol (4.50 g, 30.9 mmol) in DMF (50 mL) were added K.sub.2CO.sub.3 (8.54 g, 61.8 mmol) and 3-iodooxetane (6.82 g, 37.1 mmol). The mixture was stirred at room temperature for 2 h. The reaction mixture was quenched with H.sub.2O (100 mL) and extracted with EtOAc (3×50 mL). The combined organic layers were washed with brine (3×50

mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 80 g cartridge, 0-15% EtOAc:petroleum ether) to give 5-chloro-2-(oxetan-3-ylsulfanyl)pyridine. MS=202.0 [M+H].sup.+.

Step 2: 5-chloro-2-(oxetane-3-sulfonyl)pyridine

##STR00985##

[2463] To a 0° C. solution of 5-chloro-2-(oxetan-3-ylsulfanyl)pyridine (3.00 g, 14.9 mmol) in DCM (40 mL) was added m-CPBA (6.04 g, 85% purity, 29.8 mmol). The mixture was allowed to warm to room temperature and stirred for 2 h. The reaction mixture was quenched with saturated aqueous Na.sub.2SO.sub.3 (20 mL), diluted with H.sub.2O (100 mL), and extracted with DCM (3×50 mL). The combined organic layers were washed with saturated aqueous Na.sub.2SO.sub.3 (2×50 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude product was triturated with EtOAc (5 mL) and isolated by filtration to give 5-chloro-2-(oxetane-3-sulfonyl)pyridine. MS=234.0 [M+H].sup.+.

Step 3: 5-chloro-2-[(3-methyloxetan-3-yl)sulfonyl]pyridine

##STR00986##

[2464] To a -78° C. solution of 5-chloro-2-(oxetane-3-sulfonyl)pyridine (500 mg, 2.14 mmol) in THE (8 mL) was added 1.0 M LiHMDS in THE (4.28 mL, 4.28 mmol). After stirring for 30 min, Mel (160 µL, 2.57 mmol) was added. The mixture was allowed to warm to room temperature and stirred for 30 min. The reaction mixture was quenched with saturated aqueous NH.sub.4Cl (20 mL) and extracted with EtOAc (3×10 mL). The combined organic layers were washed with brine (2×10 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-15% EtOAc:petroleum ether) to give 5-chloro-2-[(3-methyloxetan-3-yl)sulfonyl]pyridine. MS=248.0 [M+H].sup.+.

Step 4: 2-({6-[(3-methyloxetan-3-yl)sulfonyl]pyridin-3-yl}oxy)ethan-1-ol

##STR00987##

[2465] To a solution of 5-chloro-2-[(3-methyloxetan-3-yl)sulfonyl]pyridine (350 mg, 1.41 mmol) in DMSO (2 mL) were added ethylene glycol (2.00 mL, 35.8 mmol) and Cs.sub.2CO.sub.3 (921 mg, 2.83 mmol). The mixture was stirred at 100° C. for 7 h. After cooling to 0° C., the reaction mixture was quenched with H.sub.2O (20 mL) and extracted with EtOAc (6×10 mL). The combined organic layers were washed with brine (2×15 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 4 g cartridge, 0-70% EtOAc:petroleum ether) to give 2-({6-[(3-methyloxetan-3-yl)sulfonyl]pyridin-3-yl}oxy)ethan-1-ol. MS=274.1 [M+H].sup.+.

Step 5: 2-({6-[(3-methyloxetan-3-yl)sulfonyl]pyridin-3-yl}oxy)ethyl methanesulfonate

##STR00988##

[2466] To a solution of 2-({6-[(3-methyloxetan-3-yl)sulfonyl]pyridin-3-yl}oxy)ethan-1-ol (130 mg, 0.476 mmol) in DCM (3 mL) was added TEA (199 µL, 1.43 mmol) and methanesulfonic anhydride (166 mg, 0.951 mmol). The mixture was stirred at room temperature for 5 h, and then diluted with H.sub.2O (10 mL) and extracted with DCM (3×5 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered and concentrated in vacuo to give 2-({6-[(3-methyloxetan-3-yl)sulfonyl]pyridin-3-yl}oxy)ethyl methanesulfonate (Intermediate A-26), which was used in the subsequent step without further purification. MS=352.1 [M+H].sup.+.

[2467] The following intermediate in Table 5 was prepared using procedures similar to those described for Intermediate A-26, using the appropriate starting materials.

General Procedure for Intermediate A-27

TABLE-US-00006 TABLE 5 Exact Mass Intermediate # Structure Name [M + H].sup.+ A-27

[00989] embedded image 2-({4-[(3-methyloxetan-3-yl)sulfonyl]phenoxy}ethyl

methanesulfonate) Calc'd 351.1 Found 350.9

General Procedure for Intermediate A-28

##STR00990##

Step 1: 1-(methanesulfonylmethyl)-4-methoxybenzene

##STR00991##

[2468] A mixture of 1-(chloromethyl)-4-methoxybenzene (8.70 mL, 63.9 mmol), sodium methanesulfinate (9.13 g, 89.4 mmol) and NaI (28.7 mg, 0.192 mmol) in DMF (80 mL) was stirred at 80° C. for 3 h. After cooling to 0° C., the reaction mixture was quenched with H.sub.2O (200 mL) and extracted with EtOAc (3×100 mL). The combined organic layers were washed with brine (2×100 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was triturated with EtOAc (50 mL) to give 1-(methanesulfonylmethyl)-4-methoxybenzene, which was used in the subsequent step without further purification.

Step 2: 1-(1-methanesulfonylethenyl)-4-methoxybenzene

##STR00992##

[2469] A mixture of 1-(methanesulfonylmethyl)-4-methoxybenzene (9.00 g, 44.9 mmol), Cs.sub.2CO.sub.3 (43.9 g, 135 mmol), TBAI (166 mg, 0.449 mmol) and HCHO (12.4 mL, 449 mmol) in toluene (150 mL) was stirred at 70° C. for 3 h. After cooling to 0° C., the reaction mixture was quenched with H.sub.2O (200 mL) and extracted with EtOAc (2×100 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 120 g cartridge, 0-30% EtOAc:petroleum ether) to give 1-(1-methanesulfonylethenyl)-4-methoxybenzene. MS=213.1 [M+H].sup.+.

Step 3: 1-(1-methanesulfonylcyclopropyl)-4-methoxybenzene

##STR00993##

[2470] To a solution of trimethylsulfoxonium iodide (523 mg, 2.37 mmol) in DMSO (5 mL) was added NaH (84.4 mg, 60 wt % in mineral oil, 2.11 mmol). After stirring at room temperature for 1 h, 1-(1-methanesulfonylethenyl)-4-methoxybenzene (0.28 g, 1.32 mmol) was added. The mixture was stirred for 16 h, then cooled to 0° C., quenched with saturated aqueous NH.sub.4Cl (10 mL), and extracted with EtOAc (3×5 mL). The combined organic layers were washed with brine (2×10 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 4 g cartridge, 0-50% EtOAc:petroleum ether) to give 1-(1-methanesulfonylcyclopropyl)-4-methoxybenzene. MS=244.2 [M+NH.sub.4].sup.+.

Step 4: 4-(1-methanesulfonylcyclopropyl)phenol

##STR00994##

[2471] To a 0° C. solution of 1-(1-methanesulfonylcyclopropyl)-4-methoxybenzene (1.20 g, 5.30 mmol) in DCM (20 mL) was added BBr.sub.3 (1.53 mL, 15.9 mmol). The mixture was allowed to warm to room temperature and stirred for 3 h. After cooling to 0° C., the reaction mixture was quenched with H.sub.2O (30 mL). The pH of the 0° C. solution was adjusted to pH=7 with the dropwise addition of saturated aqueous NaHCO.sub.3. The biphasic mixture was extracted with DCM (3×20 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered and concentrated in vacuo to give 4-(1-methanesulfonylcyclopropyl)phenol, which was used in the subsequent step without further purification.

Step 5: 1-(2-bromoethoxy)-4-(1-methanesulfonylcyclopropyl)benzene

##STR00995##

[2472] To a solution of 4-(1-methanesulfonylcyclopropyl)phenol (200 mg, 0.942 mmol) in 1,2-dibromoethane (4.27 mL, 56.5 mmol) was added Cs.sub.2CO.sub.3 (614 mg, 1.88 mmol). The mixture was stirred at 100° C. for 4 h. After cooling to room temperature, the reaction mixture was quenched with H.sub.2O (10 mL) and extracted with DCM (3×5 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 4 g cartridge, 0-30% EtOAc:petroleum ether) to give 1-(2-bromoethoxy)-4-(1-methanesulfonylcyclopropyl)benzene

(Intermediate A-28). MS=336.0/338.1 [M+NH.sub.4].sup.+.

General Procedure for Intermediates A-29 & A-30

##STR00996##

Step 1: 5-(benzyloxy)-2-bromo-1,3-difluorobenzene

##STR00997##

[2473] To a solution of 4-bromo-3,5-difluorophenol (10.0 g, 47.8 mmol) in MeCN (100 mL) were added BnBr (6.82 mL, 57.4 mmol) and K.sub.2CO.sub.3 (19.8 g, 144 mmol). The mixture was stirred at 60° C. for 12 h. After cooling to room temperature, the mixture was poured into saturated aqueous NaHCO.sub.3 (400 mL) and extracted with EtOAc (3×200 mL). The combined organic layers were washed with brine (2×200 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 80 g cartridge, 0-1% EtOAc:petroleum ether) to give 5-(benzyloxy)-2-bromo-1,3-difluorobenzene (Intermediate A-29). .sup.1H NMR (400 MHz, CDCl.sub.3): δ 7.42-7.35 (m, 5H), 6.64-6.60 (m, 2H), 5.04 (s, 2H).

Step 2: 1-[4-(benzyloxy)-2,6-difluorophenyl]ethan-1-ol

##STR00998##

[2474] To a 0° C. solution of 5-(benzyloxy)-2-bromo-1,3-difluorobenzene (12.0 g, 40.1 mmol) in THE (24 mL) was added 1.3 M i-PrMgCl.Math.LiCl in THE (34.0 mL, 44.1 mmol). The mixture was allowed to warm to room temperature and stirred for 1 h. The mixture was cooled to 0° C. and 5.0 M acetaldehyde in THF (8.82 mL, 44.1 mmol) was added into the reaction mixture. After stirring at 0° C. for 30 min, the mixture was warmed to room temperature and stirred for another 30 min. The reaction mixture was quenched with saturated aqueous NH.sub.4Cl (300 mL) and extracted with EtOAc (3×300 mL). The combined organic layers were washed with brine (2×300 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 80 g cartridge, 0-35% EtOAc:petroleum ether) to give 1-[4-(benzyloxy)-2,6-difluorophenyl]ethan-1-ol. MS=247.2 [M-OH].sup.+.

Step 3: 5-(benzyloxy)-2-(1-chloroethyl)-1,3-difluorobenzene

##STR00999##

[2475] To a 0° C. solution of 1-[4-(benzyloxy)-2,6-difluorophenyl]ethan-1-ol (5.8 g, 21.9 mmol) in DCM (60 mL) was added SOCl.sub.2 (3.18 mL, 43.9 mmol). The mixture was stirred at room temperature for 4 h, and then concentrated in vacuo to give 5-(benzyloxy)-2-(1-chloroethyl)-1,3-difluorobenzene, which was used in the subsequent step without further purification.

Step 4: 5-(benzyloxy)-1,3-difluoro-2-(1-methanesulfonylethyl)benzene

##STR01000##

[2476] To a solution of 5-(benzyloxy)-2-(1-chloroethyl)-1,3-difluorobenzene (6.20 g, 21.9 mmol) in DMF (60 mL) was added sodium methanesulfinate (4.48 g, 43.9 mmol). The mixture was stirred at 80° C. for 12 h. After cooling to room temperature, the reaction mixture was quenched with H.sub.2O (100 mL) and extracted with EtOAc (3×100 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 80 g cartridge, 0-50% EtOAc:petroleum ether) to give 5-(benzyloxy)-1,3-difluoro-2-(1-methanesulfonylethyl)benzene.

Step 5: 3,5-difluoro-4-(1-methanesulfonylethyl)phenol

##STR01001##

[2477] To a solution of 5-(benzyloxy)-1,3-difluoro-2-(1-methanesulfonylethyl)benzene (3.00 g, 9.19 mmol) in MeOH (30 mL) was added Pd/C (0.50 g, 10 wt %, 0.47 mmol) and Pd(OH).sub.2/C (0.50 g, 20 wt %, 0.70 mmol). The mixture was stirred at 40° C. for 16 h under an atmosphere of H.sub.2 (50 psi). After cooling to room temperature, the reaction mixture was filtered through Celite and the filtrate was concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 80 g cartridge, 0-100% EtOAc:petroleum ether) to give 3,5-

difluoro-4-(1-methanesulfonylethyl)phenol. MS=254.1 [M+NH.sub.4].sup.+.

Step 6: 5-(2-bromoethoxy)-1,3-difluoro-2-(1-methanesulfonylethyl)benzene

##STR01002##

[2478] To a solution of 3,5-difluoro-4-(1-methanesulfonylethyl)phenol (0.43 g, 1.82 mmol) in 1,2-dibromoethane (10 mL, 132 mmol) was added Cs.sub.2CO.sub.3 (1.19 g, 3.64 mmol). The mixture was stirred at 100° C. for 12 h. After cooling to room temperature, the reaction mixture was filtered through Celite and the filtrate was concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 20 g cartridge, 0-50% EtOAc:petroleum ether) to give 5-(2-bromoethoxy)-1,3-difluoro-2-(1-methanesulfonylethyl)benzene (Intermediate A-30).

MS=360.1/362.1 [M+NH.sub.4].sup.+.

[2479] The following intermediate in Table 6 was prepared using procedures similar to those described for Intermediate A-30, using the appropriate starting materials.

General Procedure for Intermediate A-31

TABLE-US-00007 TABLE 6 Intermediate # Structure Name Exact Mass [M + H].sup.+ A-31

[01003]  embedded image 1-(2-bromoethoxy)-4-(1-methanesulfonylethyl)benzene Calc'd

227.0/229.0 Found 227.1/229.1 [M - CH.sub.3O.sub.2S].sup.+

General Procedure for Intermediate A-32

##STR01004## ##STR01005##

Step 1: 3-[4-(benzyloxy)-2,6-difluorophenyl]oxetan-3-ol

##STR01006##

[2480] To a 0° C. solution of 5-(benzyloxy)-2-bromo-1,3-difluorobenzene (Intermediate A-29, 14.0 g, 46.8 mmol) in THE (150 mL) was added 2.0 M i-PrMgCl in THE (25.7 mL, 51.4 mmol). After stirring at 0° C. for 30 min, oxetan-3-one (4.05 g, 56.2 mmol) was added. The mixture was stirred at 0° C. for another 30 min, and then quenched with H.sub.2O (200 mL) and extracted with EtOAc (3×100 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 80 g cartridge, 1:10 EtOAc:petroleum ether) to give 3-[4-(benzyloxy)-2,6-difluorophenyl]oxetan-3-ol. MS=315.2 [M+Na].sup.+.

Step 2: 3-[4-(benzyloxy)-2,6-difluorophenyl]-3-[(triphenylmethyl)sulfanyl]oxetane

##STR01007##

[2481] To a solution of 3-[4-(benzyloxy)-2,6-difluorophenyl]oxetan-3-ol (20.0 g, 68.4 mmol) and triphenylmethanethiol (75.7 g, 274 mmol) in CHCl.sub.3 (200 mL) was added

[bis(trifluoromethylsulfonyl)amino]lithium (2.16 g, 7.53 mmol) and tetrabutylammonium hexafluorophosphate (1.46 g, 3.76 mmol). The mixture was stirred at 40° C. for 2 h. After cooling to room temperature, the reaction mixture was quenched with H.sub.2O (300 mL) and extracted with DCM (3×150 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 330 g cartridge, 0-40% EtOAc:petroleum ether) to give 3-[4-(benzyloxy)-2,6-difluorophenyl]-3-[(triphenylmethyl)sulfanyl]oxetane.

Step 3: 3-[4-(benzyloxy)-2,6-difluorophenyl]oxetane-3-thiol

##STR01008##

[2482] To a solution of 3-[4-(benzyloxy)-2,6-difluorophenyl]-3-[(triphenylmethyl)sulfanyl]oxetane (400 mg, 0.726 mmol) in DCM (2 mL) was added TFA (4 mL) and Et.sub.3SiH (290 µL, 1.82 mmol). The mixture was stirred at room temperature for 2 h. The reaction mixture was quenched with H.sub.2O (10 mL) and extracted with DCM (3×5 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-10% EtOAc:petroleum ether) to give 3-[4-(benzyloxy)-2,6-difluorophenyl]oxetane-3-thiol. MS=307.1 [M-H].sup.-.

Step 4: 3-[4-(benzyloxy)-2,6-difluorophenyl]-3-(methylsulfonyl)oxetane

##STR01009##

[2483] To a solution of 3-[4-(benzyloxy)-2,6-difluorophenyl]oxetane-3-thiol (2.50 g, 8.11 mmol) in DMF (25 mL) was added K₂CO₃ (1.68 g, 12.2 mmol) and MeI (2.52 mL, 40.5 mmol). The mixture was stirred at room temperature for 1 h. After cooling to 0° C., the reaction mixture was quenched with H₂O (50 mL) and extracted with DCM (3×30 mL). The combined organic layers were washed with brine (3×30 mL) dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 80 g cartridge, 0-15% EtOAc:petroleum ether) to give 3-[4-(benzyloxy)-2,6-difluorophenyl]-3-(methylsulfonyl)oxetane. MS=321.0 [M-H]⁺.

Step 5: 3-[4-(benzyloxy)-2,6-difluorophenyl]-3-methanesulfonyloxetane

##STR01010##

[2484] To a solution of 3-[4-(benzyloxy)-2,6-difluorophenyl]-3-(methylsulfonyl)oxetane (2.00 g, 6.20 mmol) in DCM (30 mL) was added m-CPBA (3.78 g, 85% purity, 18.6 mmol). The reaction mixture was diluted with H₂O (50 mL) and extracted with DCM (3×20 mL). The combined organic layers were washed with saturated aqueous Na₂SO₃ (2×20 mL) dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was triturated with EtOAc and then dried in vacuo to give 3-[4-(benzyloxy)-2,6-difluorophenyl]-3-methanesulfonyloxetane, which was used in the subsequent step without further purification. ¹H NMR (400 MHz, DMSO-d₆): δ 7.46-7.35 (m, 5H), 6.95-6.92 (m, 2H), 5.16-5.10 (m, 6H), 3.13 (s, 3H).

Step 6: 3,5-difluoro-4-(3-methanesulfonyloxetan-3-yl)phenol

##STR01011##

[2485] To a solution of 3-[4-(benzyloxy)-2,6-difluorophenyl]-3-methanesulfonyloxetane (600 mg, 1.69 mmol) in EtOAc (20 mL) under Argon atmosphere was added Pd/C (1.00 g, 10 wt %, 0.943 mmol). The mixture was stirred at room temperature for 3 h under an atmosphere of H₂ (15 psi). The reaction mixture was filtered through Celite and the filtrate was concentrated in vacuo to give 3,5-difluoro-4-(3-methanesulfonyloxetan-3-yl)phenol, which was used in the subsequent step without further purification. ¹H NMR (400 MHz, DMSO-d₆, 9/10 H): δ 6.54-6.51 (m, 2H), 5.13-5.08 (m, 4H), 3.09 (s, 3H).

Step 7: 3-[4-(2-bromoethoxy)-2,6-difluorophenyl]-3-methanesulfonyloxetane

##STR01012##

[2486] To a solution of 3,5-difluoro-4-(3-methanesulfonyloxetan-3-yl)phenol (200 mg, 0.757 mmol) in 1,2-dibromoethane (2.00 mL, 26.5 mmol) was added Cs₂CO₃ (493 mg, 1.51 mmol). The mixture was stirred at 80° C. for 3 h. After cooling to 0° C., the reaction mixture was quenched with H₂O (10 mL) and extracted with DCM (3×5 mL). The combined organic layers were dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 4 g cartridge, 0-40% EtOAc:petroleum ether) to give 3-[4-(2-bromoethoxy)-2,6-difluorophenyl]-3-methanesulfonyloxetane (Intermediate A-32). MS=388.0/390.0 [M+NH₄]⁺.

[2487] The following intermediates in Table 7 were prepared using procedures similar to those described for Intermediate A-32, using the appropriate starting materials.

General Procedure for Intermediates A-33 & A-34

TABLE-US-00008 TABLE 7 Intermediate # Structure Name Exact Mass [M + H]⁺ A-33

[01013]  3-[4-(2-bromoethoxy)phenyl]-3-methanesulfonyloxetane Calc'd 352.0/354.0 Found 352.1/354.0 [M + NH₄]⁺ A-34 [01014]  3-[4-(2-bromoethoxy)-2-(trifluoromethyl)phenyl]-3-methanesulfonyloxetane Calc'd 420.0/422.0 Found 420.0/422.0 [M + NH₄]⁺

General Procedure for Intermediates A-35, A-36, & A-84

##STR01015## ##STR01016##

Step 1: 3-[4-(benzyloxy)phenyl]oxetane-3-ol

##STR01017##

[2488] To a -78°C . solution of 1-(benzyloxy)-4-bromobenzene (3.00 g, 11.4 mmol) in THE (20 mL) was added 2.5 M n-BuLi in THE (7.30 mL, 18.3 mmol). After stirring for 1 h at -78°C ., oxetan-3-one (986 mg, 13.7 mmol) was added. The mixture was warmed to room temperature and stirred for 3 h. The reaction mixture cooled to 0°C ., quenched with H.sub.2O (30 mL), and extracted with EtOAc (3 $\times$ 20 mL). The combined organic layers were washed with brine (40 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 45 g cartridge, 0-50% EtOAc:petroleum ether) to give 3-[4-(benzyloxy)phenyl]oxetan-3-ol (Intermediate A-35). MS=279.1 [M+Na].sup.+.

Step 2: 3-[4-(benzyloxy)phenyl]-3-[(triphenylmethyl)sulfanyl]oxetane

##STR01018##

[2489] To a solution of 3-[4-(benzyloxy)phenyl]oxetan-3-ol (7.40 g, 28.9 mmol) and triphenylmethanethiol (39.9 g, 144 mmol) in CHCl.sub.3 (80 mL) were added [bis(trifluoromethylsulfonyl)amino]lithium (911 mg, 3.18 mmol) and tetrabutylammonium hexafluorophosphate (615 mg, 1.59 mmol). The mixture was stirred at 40°C . for 30 min. After cooling to room temperature, the reaction mixture was quenched with H.sub.2O (80 mL) and extracted with DCM (3 $\times$ 50 mL). The combined organic layers were washed with brine (100 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 120 g cartridge, 0-40% EtOAc:petroleum ether) to give 3-[4-(benzyloxy)phenyl]-3-[(triphenylmethyl)sulfanyl]oxetane. MS=537.3 [M+Na].sup.+.

Step 3: 3-[4-(benzyloxy)phenyl]oxetane-3-thiol

##STR01019##

[2490] To a solution of 3-[4-(benzyloxy)phenyl]-3-[(triphenylmethyl)sulfanyl]oxetane (10.0 g, 19.4 mmol) in DCM (40 mL) was added TFA (60.0 mL, 810 mmol) and Et.sub.3SiH (7.76 mL, 48.6 mmol). The mixture was stirred at room temperature for 3 h. The reaction mixture was quenched with H.sub.2O (80 mL) and extracted with DCM (3 $\times$ 50 mL). The combined organic layers were washed with brine (100 mL), dried over Na.sub.2SO.sub.4, filtered and concentrated under reduced pressure to give 3-(4-benzyloxyphenyl)oxetane-3-thiol, which was used in the subsequent step without further purification. MS=273.1 [M+H].sup.+.

Step 4: 3-[4-(benzyloxy)phenyl]-3-(methylsulfanyl)oxetane

##STR01020##

[2491] To a solution of 3-(4-benzyloxyphenyl)oxetane-3-thiol (5.00 g, 18.4 mmol) in DMF (40 mL) was added MeI (13.0 g, 91.8 mmol) and K.sub.2CO.sub.3 (3.81 g, 27.5 mmol). The mixture was stirred at room temperature for 1 h. The reaction mixture was quenched with H.sub.2O (80 mL) and extracted with EtOAc (3 $\times$ 50 mL). The combined organic layers were washed with brine (2 $\times$ 50 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 80 g cartridge, 0-30% EtOAc:petroleum ether) to give 3-[4-(benzyloxy)phenyl]-3-(methylsulfanyl)oxetane. .sup.1H NMR (400 MHz, CDCl.sub.3): δ 7.46-7.39 (m, 5H), 7.11 (d, J=8.8 Hz, 2H), 6.98 (d, J=8.8 Hz, 2H), 5.18 (d, J=6.4 Hz, 2H), 5.08 (s, 2H), 4.90 (d, J=6.4 Hz, 2H), 2.04 (s, 3H).

Step 5: 3-[4-(benzyloxy)phenyl]-3-methanesulfonyloxetane

##STR01021##

[2492] To a 0°C . solution of 3-[4-(benzyloxy)phenyl]-3-(methylsulfanyl)oxetane (3.25 g, 11.4 mmol) in DCM (40 mL) was added m-CPBA (6.91 g, 85% purity, 34.04 mmol). The mixture was stirred at room temperature for 3 h. The reaction mixture was quenched with saturated aqueous Na.sub.2SO.sub.3 (20 mL), diluted with H.sub.2O (60 mL), and extracted with DCM (3 $\times$ 40 mL). The combined organic layers were with saturated aqueous NaHCO.sub.3 (3 $\times$ 50 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The resulting residue was triturated with EtOAc, the solids were collected via filtration, and the filter cake was dried in vacuo to give 3-[4-(benzyloxy)phenyl]-3-methanesulfonyloxetane. MS=336.2 [M+NH.sub.4].sup.+.

Step 6: 4-(3-methanesulfonyloxetan-3-yl)phenol

##STR01022##

[2493] To a solution of 3-[4-(benzyloxy)phenyl]-3-methanesulfonyloxetane (1.60 g, 5.03 mmol) in EtOAc (100 mL) under N₂ atmosphere was added Pd/C (2.00 g, 10 wt %, 1.89 mmol). The suspension was degassed under vacuum and purged with H₂ several times. The mixture was stirred under H₂ (15 psi) at room temperature for 10 h. The reaction mixture was filtered through Celite and the filtrate was concentrated in vacuo to give 4-(3-methanesulfonyloxetan-3-yl)phenol (Intermediate A-84), which was used in the subsequent step without further purification. MS=246.1 [M+NH₄]⁺.

Step 7: 1-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]propan-2-one

##STR01023##

[2494] To a solution of 4-(3-methylsulfonyloxetan-3-yl)phenol (500 mg, 2.19 mmol) in DMF (5 mL) was added K₂CO₃ (605 mg, 4.38 mmol) and 1-chloropropan-2-one (304 mg, 3.29 mmol). The mixture was stirred at room temperature for 1 h. The reaction mixture was cooled to 0° C. and quenched with H₂O (15 mL), and then extracted with EtOAc (2×20 mL). The combined organic layers were washed with brine (2×30 mL), dried over Na₂SO₄, filtered and concentrated in vacuo to give 1-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]propan-2-one, which was used in the subsequent step without further purification. MS=302.2 [M+NH₄]⁺.

Step 8: 1-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]propan-2-ol

##STR01024##

[2495] To a solution of 1-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]propan-2-one (600 mg, 2.11 mmol) in THE (15 mL) and EtOH (2 mL) was added NaBH₄ (120 mg, 3.17 mmol). The mixture was stirred at room temperature for 30 min. The reaction mixture was cooled to 0° C., quenched with H₂O (15 mL), and the solution was adjusted to pH=6-7 with the dropwise addition of 4.0 M aqueous HCl. The mixture was extracted with EtOAc (2×10 mL). The combined organic layers were washed with brine (10 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo to give 1-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]propan-2-ol, which was used in the subsequent step without further purification. MS=304.1 [M+NH₄]⁺.

Step 9: 1-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]propan-2-yl methanesulfonate

##STR01025##

[2496] To a solution of 1-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]propan-2-ol (570 mg, 1.99 mmol) in DCM (15 mL) was added TEA (0.831 mL, 5.97 mmol) and methanesulfonyl anhydride (694 mg, 3.98 mmol). The mixture was stirred at room temperature for 1 h. The reaction mixture was quenched with H₂O (15 mL), and the solution was adjusted to pH=6 with the dropwise addition of 4.0 M aqueous HCl. The mixture was extracted with DCM (3×10 mL). The combined organic layers were washed with brine (2×20 mL), dried over Na₂SO₄, filtered and concentrated in vacuo to give 1-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]propan-2-yl methanesulfonate, which was used in the subsequent step without further purification (Intermediate A-36). MS=382.1 [M+NH₄]⁺.

General Procedure for Intermediate A-37

##STR01026##

Step 1: 3-[4-(benzyloxy)phenyl]-3-(ethanesulfonyl)oxetane

##STR01027##

[2497] To a 0° C. solution of 3-[4-(benzyloxy)phenyl]oxetan-3-ol (Intermediate A-35, 400 mg, 1.56 mmol) in DCM (5 mL) was added sodium ethanesulfinate (906 mg, 7.80 mmol) and TFA (2.31 mL, 31.2 mmol). The mixture was stirred at room temperature for 3 h. The reaction mixture was quenched with H₂O (30 mL). The resulting precipitate was collected by filtration and dried in vacuo to give 3-[4-(benzyloxy)phenyl]-3-(ethanesulfonyl)oxetane. MS=350.1 [M+NH₄]⁺.

Step 2: 4-[3-(ethanesulfonyl)oxetan-3-yl]phenol

##STR01028##

[2498] To a solution of 3-[4-(benzyloxy)phenyl]-3-(ethanesulfonyl)oxetane (400 mg, 1.20 mmol) in EtOAc (20 mL) under N₂ atmosphere was added Pd/C (1.00 g, 10 wt %, 0.943 mmol). The suspension was degassed under vacuum and purged with H₂ (3×). The mixture was stirred under H₂ (15 psi) at room temperature for 3 h. The reaction mixture was filtered through Celite and the filtrate was concentrated in vacuo to give 4-[3-(ethanesulfonyl)oxetan-3-yl]phenol, which was used in the subsequent step without further purification.

Step 3: 3-[4-(2-bromoethoxy)phenyl]-3-(ethanesulfonyl)oxetane

##STR01029##

[2499] To a solution of 4-[3-(ethanesulfonyl)oxetan-3-yl]phenol (150 mg, 0.619 mmol) in MeCN (4 mL) were added K₂CO₃ (428 mg, 3.10 mmol) and 1,2-dibromoethane (1.87 mL, 24.8 mmol). The mixture was stirred at 65° C. for 10 h. After cooling to room temperature, the reaction mixture was filtered, washed with MeCN (20 mL), and the filtrate was concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 4 g cartridge, 0-45% EtOAc:petroleum ether) to give 3-[4-(2-bromoethoxy)phenyl]-3-(ethanesulfonyl)oxetane (Intermediate A-37). MS=366.0/368.0 [M+NH₄]⁺.

General Procedure for Intermediate A-38

##STR01030##

Step 1: 1-{[(3-bromopropyl)sulfanyl]methyl}-4-methoxybenzene

##STR01031##

[2500] To a solution of (4-methoxyphenyl)methanethiol (9.01 mL, 64.8 mmol) and 1,3-dibromopropane (33.1 mL, 324 mmol) in MeOH (50 mL) was added NaOMe (5.25 g, 97.3 mmol). The reaction mixture stirred at room temperature for 16 h. After cooling to 0° C., the reaction mixture was quenched with H₂O (200 mL) and extracted with EtOAc (3×50 mL). The combined organic layers were washed with brine (100 mL), dried over (Na₂SO₄), filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 120 g cartridge, 0-10% EtOAc:petroleum ether) to give 1-{[(3-bromopropyl)sulfanyl]methyl}-4-methoxybenzene.

Step 2: 1-[(3-bromopropanesulfonyl)methyl]-4-methoxybenzene

##STR01032##

[2501] To a 0° C. solution of 1-{[(3-bromopropyl)sulfanyl]methyl}-4-methoxybenzene (5.00 g, 18.2 mmol) in DCM (100 mL) was added m-CPBA (9.22 g, 85% purity, 45.4 mmol). The mixture was stirred at 0° C. for 2 h and was then quenched with saturated Na₂SO₃ (50 mL). The organic layer was separated, washed with saturated NaHCO₃ (2×100 mL), brine (100 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 120 g cartridge, 0-50% EtOAc:petroleum ether) to give 1-[(3-bromopropanesulfonyl)methyl]-4-methoxybenzene.

Step 3: 2-(4-methoxyphenyl)-1λ.SUP.6.-thiolane-1,1-dione

##STR01033##

[2502] To a 0° C. solution of 1-[(3-bromopropanesulfonyl)methyl]-4-methoxybenzene (3.00 g, 9.77 mmol) in DMA (30 mL) was added NaH (898 mg, 60 wt % in mineral oil, 22.5 mmol). The mixture was allowed to warm to room temperature and stirred for 16 h. The reaction mixture was quenched with ice water (20 mL), diluted with H₂O (10 mL) and extracted with EtOAc (3×30 mL). The combined organic layers were washed with brine (30 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 25 g cartridge, 0-100% EtOAc:petroleum ether) to give 2-(4-methoxyphenyl)-1λ.sup.6-thiolane-1,1-dione. MS=227.2 [M+H]⁺.

Step 4: 2-(4-hydroxyphenyl)-1λ.SUP.6.-thiolane-1,1-dione

##STR01034##

[2503] A solution of 2-(4-methoxyphenyl)-1 λ .sup.6-thiolane-1,1-dione (1.00 g, 4.42 mmol) in HBr in H.sub.2O (10 mL, 45 wt %) was stirred at 80° C. for 16 h. After cooling to room temperature, the reaction mixture was concentrated in vacuo. The residue was diluted with H.sub.2O (20 mL) and extracted with EtOAc (3 $\times$ 10 mL). The combined organic layers were washed with brine (20 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 25 g cartridge, 0-100% EtOAc:petroleum ether) to give 2-(4-hydroxyphenyl)-1 λ .sup.6-thiolane-1,1-dione. MS=213.1 [M+H].sup.+.

Step 5: 2-[4-(2-bromoethoxy)phenyl]-1 λ .SUP.6.-thiolane-1,1-dione

##STR01035##

[2504] To a mixture of 2-(4-hydroxyphenyl)-1 λ .sup.6-thiolane-1,1-dione (660 mg, 3.11 mmol) and 1,2-dibromoethane (2.35 mL, 31.1 mmol) in DMF (5 mL) was added Cs.sub.2CO.sub.3 (3.04 g, 9.33 mmol). The reaction mixture was stirred at 100° C. for 16 h. After cooling to room temperature, the mixture was diluted with H.sub.2O (30 mL) and extracted with EtOAc (3 $\times$ 20 mL). The combined organic layers were washed with brine (50 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 25 g cartridge, 0-100% EtOAc:petroleum ether) to give 2-(4-hydroxyphenyl)-1 λ .sup.6-thiolane-1,1-dione (Intermediate A-38). MS=319.0/321.0 [M+H].sup.+.

General Procedure for Intermediate A-39

##STR01036##

Step 1: 5-(benzyloxy)-2-bromobenzonitrile

##STR01037##

[2505] To a solution of 2-bromo-5-hydroxybenzonitrile (15.0 g, 75.8 mmol) in MeCN (160 mL) were added benzyl bromide (10.8 mL, 90.9 mmol) and K.sub.2CO.sub.3 (15.7 g, 114 mmol). The mixture was stirred at 60° C. for 15 h. After cooling to room temperature, the reaction mixture was filtered to remove solids. The filtrate was diluted with water (60 mL) and extracted with EtOAc (3 $\times$ 80 mL). The combined organic layers were washed with brine (100 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was triturated with petroleum ether and dried in vacuo to give 5-(benzyloxy)-2-bromobenzonitrile, which was used in the subsequent step without further purification.

Step 2: 5-(benzyloxy)-2-(3-hydroxyoxetan-3-yl)benzonitrile

##STR01038##

[2506] To a 0° C. solution of 5-(benzyloxy)-2-bromobenzonitrile (10.0 g, 34.7 mmol) in THE (120 mL) was slowly added 1.3 M i-PrMgCl.Math.LiCl in THE (34.71 mL, 45.1 mmol). After stirring for 10 min, oxetan-3-one (3.50 g, 48.59 mmol) was added to the mixture, and stirring was continued for another 2 h. The reaction mixture was quenched with saturated aqueous NH.sub.4Cl (80 mL) and stirred for 30 min, and then was extracted with EtOAc (3 $\times$ 80 mL). The combined organic layers were washed with brine (100 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 120 g cartridge, 0-20% EtOAc:petroleum ether) to give 5-(benzyloxy)-2-(3-hydroxyoxetan-3-yl)benzonitrile. MS=282.2 [M+H].sup.+.

Step 3: 5-(benzyloxy)-3H-spiro[2-benzofuran-1,3'-oxetan]-3-one

##STR01039##

[2507] To a 0° C. solution of 5-(benzyloxy)-2-(3-hydroxyoxetan-3-yl)benzonitrile (1.50 g, 5.33 mmol) in DCM (20 mL) was added sodium methanesulfinate (2.72 g, 26.7 mmol) followed by the dropwise addition of TFA (3.95 mL, 53.3 mmol). The mixture was stirred at 40° C. for 15 h. After cooling to room temperature, the reaction mixture was quenched with H.sub.2O (20 mL) and extracted with EtOAc (3 $\times$ 15 mL). The combined organic layers were washed with brine (20 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 20 g cartridge, 0-25% EtOAc:petroleum ether)

to give 5-(benzyloxy)-3H-spiro[2-benzofuran-1,3'-oxetan]-3-one. MS=283.2 [M+H].sup.+.

Step 4: 5-hydroxy-3H-spiro[2-benzofuran-1,3'-oxetan]-3-one

##STR01040##

[2508] To a solution of 5-(benzyloxy)-3H-spiro[2-benzofuran-1,3'-oxetan]-3-one (900 mg, 3.19 mmol) in MeOH (25 mL) was added Pd/C (200 mg, 10 wt %, 0.189 mmol). The mixture was stirred at room temperature for 15 h under H.sub.2 (15 psi). The reaction mixture was filtered and concentrated in vacuo to give 5-(benzyloxy)-3H-spiro[2-benzofuran-1,3'-oxetan]-3-one, which was used in the subsequent step without further purification. MS=193.2 [M+H].sup.+.

Step 5: 5-(2-bromoethoxy)-3H-spiro[2-benzofuran-1,3'-oxetan]-3-one

##STR01041##

[2509] To a solution of 5-(benzyloxy)-3H-spiro[2-benzofuran-1,3'-oxetan]-3-one (420 mg, 2.19 mmol) in MeCN (5 mL) were added K.sub.2CO.sub.3 (1.51 g, 10.9 mmol) and 1,2-dibromoethane (6.60 mL, 87.4 mmol). The mixture was stirred at 65° C. for 15 h. After cooling to room temperature, the reaction mixture was quenched with H.sub.2O (20 mL) and extracted with EtOAc (3×20 mL). The combined organic layers were washed with brine (20 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-33% EtOAc:petroleum ether) to give 5-(2-bromoethoxy)-3H-spiro[2-benzofuran-1,3'-oxetan]-3-one (Intermediate A-39). MS=299.0/301.1 [M+H].sup.+.

General Procedure for Intermediate A-40

##STR01042##

Step 1: 1-[4-(benzyloxy)phenyl]cyclobutan-1-ol

##STR01043##

[2510] To a -78° C. solution of 1-(benzyloxy)-4-bromobenzene (5.00 g, 19.0 mmol) in THE (50 mL) was added 2.5 M n-BuLi in THE (12.16 mL, 30.4 mmol). After stirring at -78° C. for 30 min, cyclobutanone (1.70 mL, 22.8 mmol) was added. The mixture was allowed to warm to room temperature and stirred for 1 h. The reaction mixture was cooled to 0° C., quenched with saturated aqueous NH.sub.4Cl (50 mL), and then extracted with EtOAc (3×70 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 40 g cartridge, 0-15% EtOAc:petroleum ether) to give 1-[4-(benzyloxy)phenyl]cyclobutan-1-ol. .sup.1H NMR (400 MHz, DMSO-d.sub.6, 17/18 H): δ 7.46-7.38 (m, 7H), 6.99 (d, J=8.8 Hz, 2H), 5.09 (s, 2H), 2.57-2.54 (m, 2H), 2.40-2.35 (m, 2H), 2.00-1.98 (m, 1H), 1.69-1.64 (m, 1H).

Step 2: 1-(benzyloxy)-4-(1-methanesulfonylcyclobutyl)benzene

##STR01044##

[2511] To a 0° C. solution of 1-[4-(benzyloxy)phenyl]cyclobutan-1-ol (3.80 g, 14.9 mmol) and sodium methanesulfinate (10.7 g, 105 mmol) in DCM (40 mL) was added TFA (22.1 mL, 299 mmol). The mixture was allowed to warm to room temperature and stirred for 3 h. The reaction mixture was cooled to 0° C. and quenched with H.sub.2O (50 mL). The resulting biphasic mixture was adjusted to pH=6-7 with saturated aqueous Na.sub.2CO.sub.3 and then extracted with DCM (3×50 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was triturated with 1:1 petroleum ether:EtOAc. The mixture was filtered, the filter cake was washed with EtOAc, and dried in vacuo to give 1-(benzyloxy)-4-(1-methanesulfonylcyclobutyl)benzene, which was used in the subsequent step without further purification. .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 7.46-7.34 (m, 7H), 7.04 (d, J=8.8 Hz, 2H), 5.12 (s, 2H), 2.94-2.91 (m, 2H), 2.61-2.55 (m, 5H), 2.09-2.07 (m, 1H), 1.89-1.86 (m, 1H).

Step 3: 4-(1-methanesulfonylcyclobutyl)phenol

##STR01045##

[2512] To a solution of 1-(benzyloxy)-4-(1-methanesulfonylcyclobutyl)benzene (2.60 g, 8.22 mmol) in EtOAc (100 mL) under a N₂ environment was added Pd/C (2.00 g, 10 wt %, 1.89 mmol).

The suspension was degassed under vacuum and purged with H.sub.2 (3×). The mixture was stirred under H.sub.2 (15 psi) at room temperature for 5 h. The reaction mixture was filtered through Celite, the filter cake was washed with EtOAc, and the filtrate was concentrated in vacuo to give 1-(benzyloxy)-4-(1-methanesulfonylcyclobutyl)benzene, which was used in the subsequent step without further purification. .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 9.64 (s, 1H), 7.20 (d, J=6.8 Hz, 2H), 6.77 (d, J=6.8 Hz, 2H), 2.91-2.87 (m, 2H), 2.58 (s, 3H), 2.56-2.52 (m, 2H), 2.08-2.05 (m, 1H), 1.88-1.84 (s, 1H).

Step 4: 1-(2-bromoethoxy)-4-(1-methanesulfonylcyclobutyl)benzene

##STR01046##

[2513] A mixture of 1-(benzyloxy)-4-(1-methanesulfonylcyclobutyl)benzene (500 mg, 2.21 mmol), 1,2-dibromoethane (6.67 mL, 88.4 mmol) and K.sub.2CO.sub.3 (1.53 g, 11.1 mmol) in MeCN (10 mL) was stirred at 65° C. for 16 h. After cooling to room temperature, the reaction mixture was filtered and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-30% EtOAc:petroleum ether) to give 1-(2-bromoethoxy)-4-(1-methanesulfonylcyclobutyl)benzene (Intermediate A-40). MS=350.1/352.1 [M+NH.sub.4.sup.+].

[2514] The following intermediate in Table 8 was prepared according to procedures similar those described for Intermediate A-40 using the appropriate starting materials.

General Procedure for Intermediate A-41

TABLE-US-00009 TABLE 8 Intermediate # Structure Name Exact Mass [M + H].sup.+ A-41

[01047]  4-[4-(2-bromoethoxy)phenyl]-4-methanesulfonyloxane Calc'd 380.1/382.1 Found 380.1/382.1 [M + NH.sub.4].sup.+

General Procedure for Intermediate A-42

##STR01048## ##STR01049##

Step 1: 6-[4-(benzyloxy)phenyl]-2-oxaspiro[3.3]heptan-6-ol

##STR01050##

[2515] To a -78° C. solution of 1-(benzyloxy)-4-bromobenzene (2.68 g, 10.2 mmol) in THE (10 mL) was added 2.5 M n-BuLi in THE (4.74 mL, 11.9 mmol). After stirring for 20 min, a solution of 2-oxaspiro[3.3]heptan-6-one (950 mg, 8.47 mmol) in THE (3 mL) was added. The mixture was stirred at -78° C. for 30 min. The reaction mixture was warmed to 0° C., quenched with saturated aqueous NH.sub.4Cl (15 mL), and then extracted with EtOAc (2×20 mL). The combined organic layers were washed with brine (2×30 mL), dried over Na.sub.2SO.sub.4, filtered and concentrated under reduced pressure to give 6-[4-(benzyloxy)phenyl]-2-oxaspiro[3.3]heptan-6-ol. MS=279.1 [M-OH].sup.+.

Step 2: 6-[4-(benzyloxy)phenyl]-6-methanesulfonyl-2-oxaspiro[3.3]heptane

##STR01051##

[2516] To a solution of 6-[4-(benzyloxy)phenyl]-2-oxaspiro[3.3]heptan-6-ol (2.00 g, 6.75 mmol) in DCM (30 mL) were added sodium methanesulfinate (3.44 g, 33.7 mmol), and TFA (9.99 mL, 135 mmol). The mixture was stirred at room temperature for 3 h. The reaction mixture was cooled to 0° C., quenched with H.sub.2O (50 mL), and then extracted with DCM (2×15 mL). The combined organic layers were washed with brine (30 mL×2), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 20 g cartridge, 0-50% EtOAc:petroleum ether) to give 6-[4-(benzyloxy)phenyl]-6-methanesulfonyl-2-oxaspiro[3.3]heptane. MS=359.1 [M+H].sup.+.

Step 3: 4-{6-methanesulfonyl-2-oxaspiro[3.3]heptan-6-yl}phenol

##STR01052##

[2517] To a solution of 6-[4-(benzyloxy)phenyl]-6-methanesulfonyl-2-oxaspiro[3.3]heptane (400 mg, 1.12 mmol) in EtOAc (10 mL) under N2 atmosphere was added Pd/C (200 mg, 10 wt %, 1.89 mmol). The suspension was degassed under vacuum and purged with H.sub.2 (3×). The mixture was stirred under H.sub.2 (15 psi) at room temperature for 3 h. The mixture was filtered through

Celite and the filtrate was concentrated in vacuo to give 4-{6-methanesulfonyl-2-oxaspiro[3.3]heptan-6-yl}phenol, which was used in the subsequent step without further purification.

Step 4: methyl 2-(4-{6-methanesulfonyl-2-oxaspiro[3.3]heptan-6-yl}phenoxy)acetate

##STR01053##

[2518] A mixture of 4-{6-methanesulfonyl-2-oxaspiro[3.3]heptan-6-yl}phenol (240 mg, 0.894 mmol), methyl 2-bromoacetate (0.169 mL, 1.79 mmol), and K.sub.2CO.sub.3 (309 mg, 2.24 mmol) in DMF (5 mL) was degassed and purged with N₂ (3×). The mixture was stirred at 50° C. for 3 h under N₂ atmosphere. The reaction mixture was cooled to 0° C., quenched with H.sub.2O (15 mL), and extracted with EtOAc (3×10 mL). The combined organic layers were washed with brine (2×15 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give methyl 2-(4-{6-methanesulfonyl-2-oxaspiro[3.3]heptan-6-yl}phenoxy)acetate. MS=341.1 [M+H].sup.+.

Step 5: 2-(4-{6-methanesulfonyl-2-oxaspiro[3.3]heptan-6-yl}phenoxy)ethan-1-ol

##STR01054##

[2519] To a 0° C. solution of methyl 2-(4-{6-methanesulfonyl-2-oxaspiro[3.3]heptan-6-yl}phenoxy)acetate (300 mg, 0.881 mmol) in THE (10 mL) under a N₂ environment was added LiAlH.sub.4 (66.9 mg, 1.76 mmol). The mixture was stirred at 0° C. for 30 min, and then was quenched with H.sub.2O (15 mL). The resulting biphasic mixture was adjusted to pH=6-7 with 2.0 M aqueous HCl solution, and then was extracted with EtOAc (3×15 mL). The combined organic layers were washed with brine (2×20 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give 2-(4-{6-methanesulfonyl-2-oxaspiro[3.3]heptan-6-yl}phenoxy)ethan-1-ol, which was used in the subsequent step without further purification. MS=313.1 [M+H].sup.+.

Step 6: 2-(4-{6-methanesulfonyl-2-oxaspiro[3.3]heptan-6-yl}phenoxy)ethyl methanesulfonate

##STR01055##

[2520] To a solution of 2-(4-{6-methanesulfonyl-2-oxaspiro[3.3]heptan-6-yl}phenoxy)ethan-1-ol (280 mg, 0.896 mmol) in DCM (5 mL) was added TEA (0.250 mL, 1.79 mmol) and methanesulfonic anhydride (234 mg, 1.34 mmol). The mixture was stirred at room temperature for 1 h, and then was quenched with H.sub.2O (10 mL). The resulting biphasic mixture was adjusted to pH=6-7 with 2.0 M aqueous HCl solution, and then was extracted with DCM (2×10 mL). The combined organic layers were washed with brine (2×20 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give 2-(4-{6-methanesulfonyl-2-oxaspiro[3.3]heptan-6-yl}phenoxy)ethyl methanesulfonate (Intermediate A-42), which was used in the subsequent step without further purification. MS=391.1 [M+H].sup.+.

General Procedure for Intermediate A-43

##STR01056##

Step 1: 5-(benzyloxy)-2-chloropyrimidine

##STR01057##

[2521] To a solution of 2-chloropyrimidin-5-ol (5.00 g, 38.3 mmol) and benzyl bromide (5.46 mL, 46.0 mmol) in MeCN (50 mL) was added K.sub.2CO.sub.3 (7.94 g, 57.5 mmol). The mixture was stirred at 60° C. for 2 h. After cooling to room temperature, the mixture was filtered, and the filtrate was concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 40 g cartridge, 0-15% EtOAc:petroleum ether) to give 5-(benzyloxy)-2-chloropyrimidine. MS=221.1 [M+H].sup.+.

Step 2: 5-(benzyloxy)-2-[(3-methyloxetan-3-yl)oxy]pyrimidine

##STR01058##

[2522] To a solution of 3-methyloxetan-3-ol (1.20 g, 13.6 mmol) and 5-benzyloxy-2-chloropyrimidine (1.00 g, 4.53 mmol) in DMF (20 mL) was added Cs.sub.2CO.sub.3 (4.43 g, 13.6 mmol). The mixture was stirred at room temperature for 6 h. Solids were removed by filtration, and the filtrate was concentrated under reduced pressure. The crude residue was purified by normal phase silica gel chromatography (Biotage 20 g cartridge, 0-15% EtOAc:petroleum ether) to give 5-

(benzyloxy)-2-[(3-methyloxetan-3-yl)oxy]pyrimidine. MS=273.1 [M+H].sup.+.

Step 3: 2-[(3-methyloxetan-3-yl)oxy]pyrimidin-5-ol

##STR01059##

[2523] To a suspension of Pd/C (150 mg, 10 wt %, 0.14 mmol) in MeOH (15 mL) was added 5-(benzyloxy)-2-[(3-methyloxetan-3-yl)oxy]pyrimidine (600 mg, 2.20 mmol) under an atmosphere of N₂. The suspension was degassed and purged with H₂ (3×). The mixture was stirred at room temperature under H₂ (15 psi) for 3 h. The mixture was filtered through a pad of Celite, and the filter cake was washed with MeOH. The filtrate was concentrated in vacuo to give 2-[(3-methyloxetan-3-yl)oxy]pyrimidin-5-ol. MS=183.1 [M+H].sup.+.

Step 4: 5-(2-bromoethoxy)-2-[(3-methyloxetan-3-yl)oxy]pyrimidine

##STR01060##

[2524] To a solution of 2-[(3-methyloxetan-3-yl)oxy]pyrimidin-5-ol (370 mg, 2.03 mmol) and 1,2-dibromoethane (6.13 mL, 81.2 mmol) in MeCN (8 mL) was added K₂CO₃ (1.40 g, 10.2 mmol). The mixture was stirred at 80° C. for 16 h. After cooling to room temperature, the mixture was concentrated in vacuo. The residue was diluted with H₂O (30 mL) and extracted with EtOAc (2×20 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-15% EtOAc:petroleum ether) to give 2-[(3-methyloxetan-3-yl)oxy]pyrimidin-5-ol (Intermediate A-43). MS=289.0/291.0 [M+H].sup.+.

General Procedure for Intermediate A-44

##STR01061##

Step 1: N-(2,6-difluoro-4-methoxyphenyl)methanesulfonamide

##STR01062##

[2525] To a 0° C. mixture of 2,6-difluoro-4-methoxyaniline (3.00 g, 18.9 mmol) and TEA (3.15 mL, 22.6 mmol) in DCM (30 mL) was added methanesulfonic anhydride (8.21 g, 47.1 mmol) dropwise. The mixture was stirred at room temperature for 10 h. The reaction mixture was quenched with H₂O (30 mL) and extracted with EtOAc (3×50 mL). The combined organic layers were dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 40 g cartridge, 0-50% EtOAc:petroleum ether) to give N-(2,6-difluoro-4-methoxyphenyl)methanesulfonamide. MS=236.1 [M-H].sup.-.

Step 2: N-(2,6-difluoro-4-methoxyphenyl)-N-methylmethanesulfonamide

##STR01063##

[2526] To a 0° C. solution of N-(2,6-difluoro-4-methoxyphenyl)methanesulfonamide (1.00 g, 4.22 mmol) in THE (10 mL) was added NaH (184 mg, 60 wt % in mineral oil, 4.59 mmol). After stirring at 0° C. for 15 min, MeI (2.62 mL, 42.2 mmol) was added. The mixture was stirred at room temperature for 5 h. The reaction mixture was cooled to 0° C., quenched with H₂O (30 mL), and extracted with EtOAc (3×50 mL). The combined organic layers were dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 20 g cartridge, 0-30% EtOAc:petroleum ether) to give N-(2,6-difluoro-4-methoxyphenyl)-N-methylmethanesulfonamide. MS=252.1 [M+H].sup.+.

Step 3: N-(2,6-difluoro-4-hydroxyphenyl)-N-methylmethanesulfonamide

##STR01064##

[2527] To a 0° C. solution of N-(2,6-difluoro-4-methoxyphenyl)-N-methylmethanesulfonamide (786 mg, 3.13 mmol) in DCM (8 mL) was added BBr₃ (0.904 mL, 9.39 mmol) dropwise. The mixture was stirred at room temperature for 6 h. The reaction mixture was cooled to 0° C. and quenched with H₂O (10 mL). The resulting biphasic mixture was concentrated in vacuo to remove DCM and the remaining aqueous phase was adjusted to pH=8-9 with saturated aqueous NaHCO₃. The resulting solid was isolated via filtration and washed with MTBE. The filter cake was dried in vacuo to give N-(2,6-difluoro-4-hydroxyphenyl)-N-methylmethanesulfonamide.

MS=238.1 [M+H].sup.+.

Step 4: N-[4-(2-bromoethoxy)-2,6-difluorophenyl]-N-methylmethanesulfonamide

##STR01065##

[2528] To a solution of N-(2,6-difluoro-4-hydroxyphenyl)-N-methylmethanesulfonamide (710 mg, 2.99 mmol) in 1,2-dibromoethane (10 mL) was added Cs.sub.2CO.sub.3 (1.95 g, 5.99 mmol). The mixture was stirred at 100° C. for 4 h. The reaction mixture was cooled to 0° C., quenched with H.sub.2O (20 mL), and extracted with EtOAc (3×15 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 20 g cartridge, 0-30% EtOAc:petroleum ether) to give N-[4-(2-bromoethoxy)-2,6-difluorophenyl]-N-methylmethanesulfonamide (Intermediate A-44). MS=361.1/363.1 [M+NH.sub.4].sup.+.

General Procedure for Intermediate A-45

##STR01066##

Step 1: 1-methanesulfinyl-4-methoxybenzene

##STR01067##

[2529] To a 0° C. solution of 1-methoxy-4-(methylsulfonyl)benzene (15.3 mL, 110 mmol) in THE (160 mL) and H.sub.2O (160 mL) was added NaIO.sub.4 (12.2 mL, 220 mmol) portionwise. The mixture was allowed to warm to room temperature and stirred for 16 h, and then solids were removed by filtration. The filtrate was cooled to 0° C., quenched with saturated aqueous Na.sub.2SO.sub.3 solution (200 mL), and extracted with EtOAc (5×200 mL). The combined organic layers were washed with saturated aqueous Na.sub.2SO.sub.3 solution (2×200 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give 1-methanesulfinyl-4-methoxybenzene, which was used in the subsequent step without further purification. MS=171.1 [M+H].sup.+.

Step 2: imino(4-methoxyphenyl)methyl-λ.SUP.6.-sulfanone

##STR01068##

[2530] To a 0° C. solution of 1-methanesulfinyl-4-methoxybenzene (110 mmol) and PhI(OAc).sub.2 (15 g, 470 mmol) in MeOH (200 mL) was added ammonium carbamate (13.8 g, 176 mmol). The mixture was stirred at 0° C. for 1 h, and then was concentrated in vacuo. The residue was diluted with H.sub.2O (100 mL) and adjusted to pH=6-7 with the dropwise addition of saturated aqueous Na.sub.2CO.sub.3. Solids were removed by filtration, and the filtrate was extracted with EtOAc (5×200 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 330 g cartridge, 80-100% EtOAc:petroleum ether) to give imino(4-methoxyphenyl)methyl-λ.sup.6-sulfanone. MS=186.1 [M+H].sup.+.

Step 3: [(4-methoxyphenyl)(methyl)oxo-λ.SUP.6.-sulfanylidene](methyl)amine

##STR01069##

[2531] A solution of imino(4-methoxyphenyl)methyl-λ.sup.6-sulfanone (10.0 g, 54.0 mmol) and formaldehyde aqueous solution (29.7 mL, 37 wt %, 1.08 mol) in formic acid (100 mL) was stirred at 100° C. for 30 h. The mixture was cooled to 0° C. and then diluted with H.sub.2O (120 mL). The solution was adjusted to pH=6-7 with the dropwise addition of 5.0 M aqueous NaOH. The mixture was extracted with EtOAc (3×120 mL). The combined organic layers were washed with brine (80 mL), dried over anhydrous Na.sub.2SO.sub.4, filtered and concentrated in vacuo to give [(4-methoxyphenyl)(methyl)oxo-λ.sup.6-sulfanylidene](methyl)amine, which was used in the subsequent step without further purification. MS=200.2 [M+H].sup.+.

Step 4: 4-[methyl(methylimino)oxo-λ.SUP.6.-sulfanyl]phenol

##STR01070##

[2532] To a 0° C. solution of [(4-methoxyphenyl)(methyl)oxo-λ.sup.6-sulfanylidene](methyl)amine (11.0 g, 55.2 mmol) in DCM (120 mL) was added BBr.sub.3 (21.3 mL, 221 mmol). The mixture was stirred at room temperature for 2 h, then was cooled to 0° C. and quenched by the addition of

H.sub.2O (15 mL). The solution was adjusted to pH=6-8 with the dropwise addition of saturated aqueous NaHCO₃. The resulting solids were removed by filtration and the filter cake was washed with THF (3×200 mL). The resulting filtrate was concentrated in vacuo, and the resulting residue was triturated with a 1:1 mixture of petroleum ether and EtOAc, then dried in vacuo to give 4-[methyl(methylimino)oxo-λ⁶-sulfanyl]phenol. MS=186.0 [M+H]⁺.

Step 5: {[4-(2-bromoethoxy)phenyl](methyl)oxo-λ⁶-sulfanylidene}(methyl)amine
##STR01071##

[2533] To a mixture of 4-[methyl(methylimino)oxo-λ⁶-sulfanyl]phenol (900 mg, 4.86 mmol) and 1,2-dibromoethane (10.0 mL, 133 mmol) was added Cs₂CO₃ (3.17 g, 9.72 mmol). The mixture was stirred at 100° C. for 16 h. After cooling to room temperature, solids were removed via filtration and the filtrate was concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 20 g cartridge, 0-50% EtOAc:petroleum ether) to give {[4-(2-bromoethoxy)phenyl](methyl)oxo-λ⁶-sulfanylidene}(methyl)amine (Intermediate A-45). MS=291.9/293.9 [M+H]⁺.

General Procedure for Intermediate A-46

##STR01072##

Step 1: 4-(2-bromoethoxy)-2-(trifluoromethyl) aniline

##STR01073##

[2534] To a solution 4-amino-3-(trifluoromethyl)phenol (3.00 g, 16.9 mmol) in 1,2-dibromoethane (30 mL, 399 mmol) was added Cs₂CO₃ (16.6 g, 50.8 mmol). The mixture was stirred at 100° C. for 12 h. After cooling to room temperature, the mixture was filtered and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 40 g cartridge, 0-50% EtOAc:petroleum ether) to give 4-(2-bromoethoxy)-2-(trifluoromethyl) aniline. MS=284.0/286.0 [M+H]⁺.

Step 2: N-[4-(2-bromoethoxy)-2-(trifluoromethyl)phenyl]-N-methanesulfonylmethanesulfonamide
##STR01074##

[2535] To a solution of 4-(2-bromoethoxy)-2-(trifluoromethyl) aniline (600 mg, 2.11 mmol) in DCM (10 mL) was added methanesulfonic anhydride (736 mg, 4.22 mmol), TEA (0.882 mL, 6.34 mmol) and DMAP (25.8 mg, 0.211 mmol). The mixture was stirred at 40° C. for 16 h. After cooling to temperature, the reaction mixture was quenched with H₂O (20 mL), and then extracted with EtOAc (3×20 mL). The combined organic layers were washed with brine (25 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-50% EtOAc:petroleum ether) to give N-[4-(2-bromoethoxy)-2-(trifluoromethyl)phenyl]-N-methanesulfonylmethanesulfonamide. Step 3: N-[4-(2-bromoethoxy)-2-(trifluoromethyl)phenyl]methanesulfonamide

##STR01075##

[2536] To a solution of N-[4-(2-bromoethoxy)-2-(trifluoromethyl)phenyl]-N-methanesulfonylmethanesulfonamide (521 mg, 1.18 mmol) in THE (6 mL) was added a solution of NaOH (237 mg, 5.92 mmol) in H₂O (1.5 mL). The mixture was stirred at room temperature for 16 h. The reaction mixture was quenched with H₂O (10 mL), and then extracted with EtOAc (3×10 mL). The combined organic layers were washed with brine (15 mL), dried over Na₂SO₄, filtered and concentrated in vacuo to give N-[4-(2-bromoethoxy)-2-(trifluoromethyl)phenyl]methanesulfonamide (Intermediate A-46). MS=359.9/362.0 [M-H]⁻.

General Procedure for Intermediate A-47

##STR01076##

Step 1: 2-[4-(2-bromoethoxy)phenyl]-2-methylpropanenitrile

##STR01077##

[2537] To a solution of 2-(4-hydroxyphenyl)-2-methylpropanenitrile (500 mg, 3.10 mmol) in 1,2-dibromoethane (8.0 mL, 106 mmol) was added Cs₂CO₃ (3.03 g, 9.31 mmol). The mixture was stirred at 100° C. for 12 h. After cooling to room temperature, the mixture was filtered

to remove solids, and the resulting filtrate was concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 20 g cartridge, 0-50% EtOAc:petroleum ether) to give 2-[4-(2-bromoethoxy)phenyl]-2-methylpropanenitrile (Intermediate A-47). MS=241.1/243.0 [M-CN].sup.+.

General Procedure for Intermediates A-48 & A-49

##STR01078## ##STR01079##

Step 1: 2-bromo-4-fluorobenzene-1-sulfonamide

##STR01080##

[2538] A solution of 2-bromo-4-fluorobenzene-1-sulfonyl chloride (1.07 mL, 7.31 mmol) in MeOH (30 mL) was cooled to 0° C. and NH₃ was bubbled into the reaction mixture over a period of 15 min. The reaction mixture was warmed to room temperature and stirred for 16 h. The mixture was concentrated in vacuo and diluted with H₂O (100 mL). The reaction mixture was extracted with EtOAc (2×100 mL). The combined organic layers were dried over Na₂SO₄, filtered, and concentrated in vacuo give 2-bromo-4-fluorobenzene-1-sulfonamide, which was used in the subsequent step without further purification.

Step 2: N'-(2-bromo-4-fluorobenzenesulfonyl)-N,N-dimethylmethanimidamide

##STR01081##

[2539] To a mixture of 2-bromo-4-fluorobenzene-1-sulfonamide (1.00 g, 3.94 mmol) in MeCN (10 mL) was added a solution N,N-dimethylformamide dimethyl acetal (0.627 mL, 4.72 mmol) in MeCN (1 mL). The mixture was stirred at room temperature for 1 h, and was then concentrated in vacuo to give N-(2-bromo-4-fluorobenzenesulfonyl)-N,N-dimethylmethanimidamide, which was used in the subsequent step without further purification. MS=309.0/311.0 [M+H].sup.+.

Step 3: N'-{2-[(1E)-2-ethoxyethenyl]-4-fluorobenzenesulfonyl}-N,N-dimethylmethanimidamide

##STR01082##

[2540] A solution of 2-[(1E)-2-ethoxyethenyl]-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (895 mg, 4.52 mmol), N'-(2-bromo-4-fluorobenzenesulfonyl)-N,N-dimethylmethanimidamide (700 mg, 2.26 mmol), K₂CO₃ (937 mg, 6.78 mmol), Pd(PPh₃)₄ (261 mg, 0.226 mmol), and PCy₃ (73.3 µL, 0.226 mmol) in 1,4-dioxane (5 mL) and H₂O (0.5 mL) was purged with N₂ (2×) and heated via microwave for 1 h at 100° C. After cooling to room temperature, the mixture was extracted with EtOAc (3×10 mL). The combined organic layers were dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 10 g cartridge, 0-30% EtOAc:petroleum ether) to give N'-{2-[(1E)-2-ethoxyethenyl]-4-fluorobenzenesulfonyl}-N,N-dimethylmethanimidamide. MS=301.1 [M+H].sup.+.

Step 4: 6-fluoro-2H-1λ.SUP.6.,2-benzothiazine-1,1-dione

##STR01083##

[2541] A solution of N'-{2-[(1E)-2-ethoxyethenyl]-4-fluorobenzenesulfonyl}-N,N-dimethylmethanimidamide (500 mg, 1.66 mmol) in 6.0 M aqueous HCl (2.77 mL, 16.6 mmol) and THE (5 mL) was stirred at 50° C. for 16 h. After cooling to room temperature, the mixture was concentrated in vacuo. The residue was extracted with EtOAc (3×5 mL). The combined organic layers were dried over Na₂SO₄, filtered, and concentrated in vacuo to give 6-fluoro-2H-1λ.sup.6,2-benzothiazine-1,1-dione, which was used in the subsequent step without further purification. MS=198.1 [M-H].sup.-.

Step 5: 6-fluoro-3,4-dihydro-2H-1λ.SUP.6.,2-benzothiazine-1,1-dione

##STR01084##

[2542] To a mixture of Pd/C (10.0 mg, 10 wt %, 0.0094 mmol) in MeOH (3 mL) was added 6-fluoro-2H-1λ.sup.6,2-benzothiazine-1,1-dione (50.0 mg, 0.251 mmol). The mixture was purged with H₂ (2×). The mixture was stirred at room temperature for 16 h under an atmosphere of H₂ (15 psi). The reaction mixture filtered through celite, and the filtrate was concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 4 g

cartridge, 0-50% EtOAc:petroleum ether) to give 6-fluoro-3,4-dihydro-2H-1λ.sup.6,2-benzothiazine-1,1-dione (Intermediate A-48). MS=200.0 [M-H].sup.-.

Step 6: 6-fluoro-2-methyl-3,4-dihydro-2H-1λ.SUP.6.,2-benzothiazine-1,1-dione

##STR01085##

[2543] To a solution of 6-fluoro-3,4-dihydro-2H-1λ.sup.6,2-benzothiazine-1,1-dione (500 mg, 2.48 mmol) and Cs.sub.2CO.sub.3 (891 mg, 2.73 mmol) in DMF (10 mL) was added Mel (1.55 mL, 24.9 mmol). The mixture was stirred at room temperature for 3 h and was then diluted with H.sub.2O (10 mL). The mixture was extracted with EtOAc (3×5 mL). The combined organic layers were washed with brine (2×5 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give 6-fluoro-2-methyl-3,4-dihydro-2H-1λ.sup.6,2-benzothiazine-1,1-dione, which was used in the subsequent step without further purification. MS=216.1 [M+H].sup.+.

Step 7: 6-(2-hydroxyethoxy)-2-methyl-3,4-dihydro-2H-1λ.SUP.6.,2-benzothiazine-1,1-dione

##STR01086##

[2544] A solution of 6-fluoro-2-methyl-3,4-dihydro-2H-1λ.sup.6,2-benzothiazine-1,1-dione (700 mg, 3.25 mmol), ethylene glycol (1.82 mL, 32.5 mmol) and Cs.sub.2CO.sub.3 (3.18 g, 9.76 mmol) in DMF (3 mL) was stirred at 100° C. for 5 h. After cooling to room temperature, the mixture was extracted with EtOAc (3×5 mL). The combined organic layers were washed with brine (2×5 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 4 g cartridge, 0-60% EtOAc:petroleum ether) to give 6-(2-hydroxyethoxy)-2-methyl-3,4-dihydro-2H-1λ.sup.6,2-benzothiazine-1,1-dione. MS=258.0 [M+H].sup.+.

Step 8: 2-[(2-methyl-1,1-dioxo-3,4-dihydro-2H-1λ.SUP.6.,2-benzothiazin-6-yl)oxy]ethyl methanesulfonate

##STR01087##

[2545] To a 0° C. solution of 6-(2-hydroxyethoxy)-2-methyl-3,4-dihydro-2H-1λ.sup.6,2-benzothiazine-1,1-dione (180 mg, 0.700 mmol) and TEA (0.146 mL, 1.05 mmol) in DCM (5 mL) was added methanesulfonic anhydride (146 mg, 0.839 mmol). After stirring at 0° C. for 1 h, the reaction mixture was quenched with H.sub.2O (3 mL). The solution was adjusted to pH=6 with the dropwise addition of 2.0 M aqueous HCl. The mixture was extracted with DCM (3×5 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give 2-[(2-methyl-1,1-dioxo-3,4-dihydro-2H-1λ.sup.6,2-benzothiazin-6-yl)oxy]ethyl methanesulfonate, which was used in the subsequent step without further purification (Intermediate A-49). MS=336.1 [M+H].sup.+.

General Procedure for Intermediate A-50

##STR01088##

Step 1: 6-(2-hydroxyethoxy)-3,4-dihydro-2H-1λ.SUP.6.,2-benzothiazine-1,1-dione

##STR01089##

[2546] A solution of 6-fluoro-3,4-dihydro-2H-1λ.sup.6,2-benzothiazine-1,1-dione (Intermediate A-48, 2.50 g, 12.4 mmol), ethylene glycol (3.47 mL, 62.1 mmol) and t-BuOK (6.97 g, 62.1 mmol) in NMP (15 mL) was stirred at 100° C. for 16 h. After cooling to room temperature, the reaction was quenched with H.sub.2O (30 mL) and extracted with EtOAc (3×20 mL). The combined organic layers were washed with brine (20 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 20 g cartridge, 0-100% EtOAc:petroleum ether) to give 6-(2-hydroxyethoxy)-3,4-dihydro-2H-1λ.sup.6,2-benzothiazine-1,1-dione. MS=242.1 [M-H].sup.-.

Step 2: 2-[(1,1-dioxo-3,4-dihydro-2H-1λ.SUP.6.,2-benzothiazin-6-yl)oxy]ethyl methanesulfonate

##STR01090##

[2547] To a 0° C. solution of 6-(2-hydroxyethoxy)-3,4-dihydro-2H-1λ.sup.6,2-benzothiazine-1,1-dione (400 mg, 1.64 mmol) and TEA (0.343 mL, 2.47 mmol) in DCM (5 mL) was added methanesulfonic anhydride (344 mg, 1.97 mmol). After stirring at 0° C. for 1 h, the reaction

mixture was quenched with H.sub.2O (3 mL). The solution was adjusted to pH=6 with the dropwise addition of 2.0 M aqueous HCl. The mixture was extracted with DCM (3×5 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 4 g cartridge, 0-80% EtOAc:petroleum ether) to give 2-[(1,1-dioxo-3,4-dihydro-2H-1λ.sup.6,2-benzothiazin-6-yl)oxy]ethyl methanesulfonate (Intermediate A-50). MS=320.1 [M-H].sup.+.

General Procedure for Intermediate A-51

##STR01091##

Step 1: N-(4-methoxy-2-methylphenyl)methanesulfonamide

##STR01092##

[2548] To a 0° C. solution of 4-methoxy-2-methylaniline (7.41 mL, 58.3 mmol) and TEA (12.2 mL, 87.4 mmol) in DCM (50 mL) was added methanesulfonic anhydride (11.2 g, 64.2 mmol). After stirring at room temperature for 1 h, the reaction mixture was quenched with H.sub.2O (35 mL). The solution was adjusted to pH=6 with the dropwise addition of 1.0 M aqueous HCl. The mixture was extracted with DCM (3×25 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude product was triturated with EtOAc, and the solid was isolated via filtration then concentrated in vacuo to give N-(4-methoxy-2-methylphenyl)methanesulfonamide. MS=214.2 [M-H].sup.-.

Step 2: N-(4-methoxy-2-methylphenyl)-N-methylmethanesulfonamide

##STR01093##

[2549] A solution of N-(4-methoxy-2-methylphenyl)methanesulfonamide (10.0 g, 46.5 mmol), MeI (4.34 mL, 69.68 mmol) and K.sub.2CO.sub.3 (12.8 g, 92.9 mmol) in DMF (30 mL) was stirred at for 2 h. The reaction mixture was quenched with H.sub.2O (15 mL) and extracted with EtOAc (3×30 mL). The combined organic layers were washed with brine (2×20 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give N-(4-methoxy-2-methylphenyl)-N-methylmethanesulfonamide, which was used in the subsequent step without further purification.

Step 3: N-[2-(bromomethyl)-4-methoxyphenyl]-N-methylmethanesulfonamide

##STR01094##

[2550] A solution N-(4-methoxy-2-methylphenyl)-N-methylmethanesulfonamide (5.00 g, 21.8 mmol), NBS (4.66 g, 26.2 mmol) and AIBN (71.6 mg, 0.436 mmol) in CCl.sub.4 (50 mL) was stirred at 90° C. for 2 h. After cooling to room temperature, the reaction mixture was diluted with H.sub.2O (50 mL) and extracted with EtOAc (2×50 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 40 g cartridge, 0-80% EtOAc:petroleum ether) to give N-[2-(bromomethyl)-4-methoxyphenyl]-N-methylmethanesulfonamide. MS=308.0/310.0 [M+H].sup.+.

Step 4: 6-methoxy-1-methyl-3,4-dihydro-1H-2λ.SUP.6,1-benzothiazine-2,2-dione

##STR01095##

[2551] To a 0° C. solution of N-[2-(bromomethyl)-4-methoxyphenyl]-N-methylmethanesulfonamide (2.60 g, 8.44 mmol) in DMA (15 mL) was added NaH (371 mg, 60 wt % in mineral oil, 9.28 mmol) in portions. The mixture was stirred at 75° C. for 4 h. After cooling to room temperature, the reaction was quenched with H.sub.2O (20 mL). The mixture was extracted with EtOAc (3×15 mL). The combined organic layers were washed with brine (2×15 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-30% EtOAc:petroleum ether) to give 6-methoxy-1-methyl-3,4-dihydro-1H-2λ.sup.6,1-benzothiazine-2,2-dione. MS=228.1 [M+H].sup.+.

Step 5: 6-hydroxy-1-methyl-3,4-dihydro-1H-2.SUP.6,1-benzothiazine-2,2-dione

##STR01096##

[2552] To a -20° C. solution of 6-methoxy-1-methyl-3,4-dihydro-1H-2λ.sup.6,1-benzothiazine-

2,2-dione (1.20 g, 5.28 mmol) in DCM (15 mL) was added BBr.sub.3 (1.53 mL, 15.8 mmol) dropwise. The mixture was warmed to 0° C. and stirred for 1 h. The reaction mixture was quenched by the addition of H.sub.2O (15 mL), warmed to room temperature, and then adjusted to pH=8 with saturated aqueous NaHCO.sub.3. The mixture was extracted with DCM (3×20 mL). The combined organic layers were washed with brine (15 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 10 g cartridge, 0-30% EtOAc:petroleum ether) to give 6-hydroxy-1-methyl-3,4-dihydro-1H-2λ.sup.6,1-benzothiazine-2,2-dione. MS=212.1 [M-H].sup.-.

Step 6: 6-(2-bromoethoxy)-1-methyl-3,4-dihydro-1H-2λ.SUP.6.,1-benzothiazine-2,2-dione
##STR01097##

[2553] A solution of 1,2-dibromoethane (1.59 mL, 21.1 mmol), 6-hydroxy-1-methyl-3,4-dihydro-1H-2λ.sup.6,1-benzothiazine-2,2-dione (900 mg, 4.22 mmol) and Cs.sub.2CO.sub.3 (2.06 g, 6.33 mmol) in DMF (2 mL) was stirred at 100° C. for 16 h. After cooling to room temperature, the reaction mixture was diluted with H.sub.2O (15 mL) and extracted with EtOAc (3×10 mL). The combined organic layers were washed with brine (2×10 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 10 g cartridge, 0-30% EtOAc:petroleum ether) to give 6-(2-bromoethoxy)-1-methyl-3,4-dihydro-1H-2λ.sup.6,1-benzothiazine-2,2-dione (Intermediate A-51). MS=319.9/321.9 [M+H].sup.+.

General Procedure for Intermediate A-52

##STR01098##

Step 1: 5-hydroxy-1λ.SUP.6.-benzothiophene-1,1-dione

##STR01099##

[2554] To a 0° C. solution of Oxone (1.84 g, 3.00 mmol) in H.sub.2O (15 mL) was added a solution of 1-benzothiophen-5-ol (300 mg, 2.00 mmol) in MeOH (15 mL) dropwise. The mixture was stirred at room temperature for 6 h. The reaction mixture was concentrated in vacuo, diluted with H.sub.2O (10 mL), and then extracted with EtOAc (3×20 mL). The combined organic layers were washed with brine (30 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-50% EtOAc:petroleum ether) to give 5-hydroxy-1λ.sup.6-benzothiophene-1,1-dione. MS=181.1 [M-H].sup.-.

Step 2: 5-hydroxy-2,3-dihydro-1λ.SUP.6.-benzothiophene-1,1-dione

##STR01100##

[2555] To a solution of 5-hydroxy-1λ.sup.6-benzothiophene-1,1-dione (270 mg, 1.48 mmol) in EtOAc (30 mL) was added Pd/C (27 mg, 10 wt %, 0.025 mmol) under N₂ atmosphere. The mixture was purged with H.sub.2 (3×). The mixture was stirred at room temperature for 3 h under an atmosphere of H.sub.2 (15 psi). The reaction mixture filtered through Celite, and the filtrate was concentrated in vacuo to give 5-hydroxy-2,3-dihydro-1λ.sup.6-benzothiophene-1,1-dione (Intermediate A-52), which was used in the subsequent step without further purification. MS=183.1 [M-H].sup.-.

General Procedure for Intermediate A-53

##STR01101##

Step 1: 5-methoxy-2-methyl-2,3-dihydro-1H-isoindol-1-one

##STR01102##

[2556] To a solution of 5-methoxy-2,3-dihydro-1H-isoindol-1-one (750 mg, 4.60 mmol,) in DMF (15 mL) were added Cs.sub.2CO.sub.3 (2.25 g, 6.89 mmol) and MeI (0.343 mL, 5.52 mmol). The mixture was stirred at room temperature for 16 h. The mixture was filtered to remove solids, and the resulting filtrate was concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 25 g cartridge, 50-100% EtOAc:petroleum ether) to give 5-methoxy-2-methyl-2,3-dihydro-1H-isoindol-1-one. MS=178.0 [M+H].sup.+.

Step 2: 5-hydroxy-2-methyl-2,3-dihydro-1H-isoindol-1-one

##STR01103##

[2557] To a 0° C. solution of 5-methoxy-2-methyl-2,3-dihydro-1H-isoindol-1-one (800 mg, 4.51 mmol) in DCM (10 mL) was added BBr.sub.3 (1.31 mL, 13.5 mmol). The mixture was stirred at room temperature for 2 h. The reaction mixture was cooled to 0° C. and quenched by the addition of MeOH (5 mL), and then concentrated in vacuo. The crude residue was triturated with EtOAc to give 5-hydroxy-2-methyl-2,3-dihydro-1H-isoindol-1-one, which was taken to the next step without further purification. MS=164.0 [M+H].sup.+.

Step 3: 5-(2-bromoethoxy)-2-methyl-2,3-dihydro-1H-isoindol-1-one

##STR01104##

[2558] To a solution of 5-hydroxy-2-methyl-2,3-dihydro-1H-isoindol-1-one (1.25 g, 7.66 mmol) in 1,2-dibromoethane (12.5 mL, 166 mmol) was added Cs.sub.2CO.sub.3 (4.99 g, 15.3 mmol). The mixture was stirred at 100° C. for 16 h. After cooling to room temperature, the reaction mixture was concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 25 g cartridge, 70-100% EtOAc:petroleum ether) to give 5-(2-bromoethoxy)-2-methyl-2,3-dihydro-1H-isoindol-1-one (Intermediate A-53). MS=270.0/272.0 [M+H].sup.+.

General Procedure for Intermediates A-54 & A-55

##STR01105##

Step 1: 2-methanesulfonyl-1,2,3,4-tetrahydroisoquinolin-6-ol

##STR01106##

[2559] A mixture of 1,2,3,4-tetrahydroisoquinolin-6-ol (500 mg, 3.35 mmol), methanesulfonic anhydride (1.17 g, 6.70 mmol), and TEA (0.933 mL, 6.70 mmol) in DCM (10 mL) was degassed and purged with N₂ (3×). After stirring at room temperature for 3 h under N₂ atmosphere, the reaction mixture was quenched with H.sub.2O (5 mL) and extracted with DCM (3×10 mL). The combined organic layers were washed with brine (2×20 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-50% EtOAc:petroleum ether) to give 2-methanesulfonyl-1,2,3,4-tetrahydroisoquinolin-6-ol (Intermediate A-54). MS=228.1 [M+H].sup.+.

Step 2: 6-(2-bromoethoxy)-2-methanesulfonyl-1,2,3,4-tetrahydroisoquinoline

##STR01107##

[2560] To a solution of 2-methanesulfonyl-1,2,3,4-tetrahydroisoquinolin-6-ol (100 mg, 0.440 mmol) in 1,2-dibromoethane (5 mL) was added Cs.sub.2CO.sub.3 (430 mg, 1.32 mmol). The mixture was stirred at 100° C. for 5 h. After cooling to room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 4 g cartridge, 0-100% EtOAc:petroleum ether) to give 6-(2-bromoethoxy)-2-methanesulfonyl-1,2,3,4-tetrahydroisoquinoline (Intermediate A-55). MS=334.1/336.1 [M+H].sup.+.

[2561] The following intermediate in Table 9 was prepared using procedures similar to Step 1 described for Intermediate A-54, using the appropriate starting materials.

General Procedure for Intermediate A-56

TABLE-US-00010 TABLE 9 Exact Mass Intermediate # Structure Name [M + H].sup.+ A-56

[01108]  embedded image 1-(6-hydroxy-1,2,3,4-tetrahydroisoquinolin-2-yl)ethan-1-one Calc'd 192.1 Found 192.2

General Procedure for Intermediate A-57

##STR01109##

Step 1: 8-hydroxy-1H,2H,3H,5H,6H,10bH-pyrrolo[2,1-a]isoquinolin-3-one

##STR01110##

[2562] A mixture of 3-(2-aminoethyl)phenol (1.00 g, 5.76 mmol, HCl salt), methyl 4-oxobutanoate (1.00 g, 8.64 mmol), and sodium ascorbate (1.14 g, 5.76 mmol) in a solution of 0.3 M aqueous

potassium phosphate buffer (200 mL) and MeCN (200 mL) was stirred at 60° C. for 18 h under an atmosphere of N₂. 1.0 M aqueous Na₂CO₃ (2.0 mL, 2.0 mmol) was added, and the mixture was stirred for another 4 h at 60° C. After cooling to room temperature, the mixture was extracted with EtOAc (3×300 mL). The combined organic layers were washed with brine (200 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 40 g cartridge, 0-100% EtOAc:petroleum ether to 0-10% MeOH:EtOAc) to give 8-hydroxy-1H,2H,3H,5H,6H,10bH-pyrrolo[2,1-a]isoquinolin-3-one. MS=204.1 [M+H]⁺.

Step 2: 8-(2-bromoethoxy)-1H,2H,3H,5H,6H,10bH-pyrrolo[2,1-a]isoquinolin-3-one

##STR01111##

[2563] To a solution of 8-hydroxy-1H,2H,3H,5H,6H,10bH-pyrrolo[2,1-a]isoquinolin-3-one (300 mg, 1.48 mmol) in 1,2-dibromoethane (5.00 mL, 66.3 mmol) was added Cs₂CO₃ (962 mg, 2.95 mmol). The mixture was stirred at 100° C. for 32 h. After cooling to room temperature, the reaction mixture was quenched with H₂O (10 mL) and extracted with EtOAc (4×8 mL). The combined organic layers were washed with brine (2×8 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 25 g cartridge, 70-100% EtOAc:petroleum ether) to give 8-(2-bromoethoxy)-1H,2H,3H,5H,6H,10bH-pyrrolo[2,1-a]isoquinolin-3-one (Intermediate A-57). MS=310.0/312.0 [M+H]⁺.

General Procedure for Intermediate A-58

##STR01112##

Step 1: 2-[(6-methoxy-1,2,3,4-tetrahydroisoquinolin-1-yl)methyl]-2,3-dihydro-1H-isoindole-1,3-dione

##STR01113##

[2564] A mixture of 2-(3-methoxyphenyl)ethan-1-amine (4.85 mL, 33.1 mmol), 2-(1,3-dioxo-2,3-dihydro-1H-isoindol-2-yl)acetaldehyde (7.51 g, 39.7 mmol), and TFA (19.6 mL, 265 mmol) in toluene (50 mL) heated via microwave for 30 min at 140° C. under N₂ atmosphere. After cooling to room temperature, the mixture was diluted with H₂O (60 mL) and extracted with EtOAc (3×50 mL). The combined organic layers were washed with brine (2×50 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 80 g cartridge, 0-50% EtOAc:petroleum ether) to give 2-[(6-methoxy-1,2,3,4-tetrahydroisoquinolin-1-yl)methyl]-2,3-dihydro-1H-isoindole-1,3-dione. MS=323.2 [M+H]⁺.

Step 2: 1-(6-methoxy-1,2,3,4-tetrahydroisoquinolin-1-yl)methanamine

##STR01114##

[2565] A mixture of 2-[(6-methoxy-1,2,3,4-tetrahydroisoquinolin-1-yl)methyl]-2,3-dihydro-1H-isoindole-1,3-dione (5.00 g, 15.5 mmol), H₂NNH₂·H₂O (2.66 mL, 85% purity, 46.53 mmol) in EtOH (50 mL) was degassed and purged with N₂ (3×). The mixture was stirred at 60° C. for 5 h under N₂ atmosphere. After cooling to room temperature, the mixture was filtered to remove solids, and the resulting filtrate was concentrated in vacuo to give 1-(6-methoxy-1,2,3,4-tetrahydroisoquinolin-1-yl)methanamine, which was used in the subsequent step without further purification. MS=193.2 [M+H]⁺.

Step 3: 8-methoxy-1H,2H,3H,5H,6H,10bH-imidazo[4,3-a]isoquinolin-3-one

##STR01115##

[2566] To a solution of 1-(6-methoxy-1,2,3,4-tetrahydroisoquinolin-1-yl)methanamine (1.00 g, 5.20 mmol) in THE (20 mL) were added CDI (4.22 g, 26.0 mmol) and TEA (0.724 mL, 5.20 mmol). A total of 5 identical reactions were set up using the described procedure. The mixtures were stirred at 80° C. for 12 h. After cooling to room temperature, the 5 reaction mixtures were combined into one solution and diluted with H₂O (50 mL) and extracted with EtOAc (3×50 mL). The combined organic layers were washed with brine (2×50 mL), dried over

Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 80 g cartridge, 0-60% EtOAc:petroleum ether) to give 8-methoxy-1H,2H,3H,5H,6H,10bH-imidazo[4,3-a]isoquinolin-3-one. MS=219.2 [M+H].sup.+.

Step 4: 8-hydroxy-1H,2H,3H,5H,6H,10bH-imidazo[4,3-a]isoquinolin-3-one

##STR01116##

[2567] To a 0° C. solution of 8-methoxy-1H,2H,3H,5H,6H,10bH-imidazo[4,3-a]isoquinolin-3-one (1.00 g, 4.58 mmol) in DCM (10 mL) was added BBr.sub.3 (0.883 mL, 9.16 mmol) dropwise. The mixture was allowed to warm to room temperature and stirred for 2 h. The mixture was quenched with H.sub.2O (20 mL), then filtered, and the filtrate was concentrated in vacuo to give 8-methoxy-1H,2H,3H,5H,6H,10bH-imidazo[4,3-a]isoquinolin-3-one, which was used in the subsequent step without further purification. MS=205.1 [M+H].sup.+.

Step 5: 8-(2-bromoethoxy)-1H,2H,3H,5H,6H,10bH-imidazo[4,3-a]isoquinolin-3-one

##STR01117##

[2568] A mixture of 8-methoxy-1H,2H,3H,5H,6H,10bH-imidazo[4,3-a]isoquinolin-3-one (700 mg, 3.43 mmol) and Cs.sub.2CO.sub.3 (3.19 g, 9.79 mmol) in 1,2-dibromoethane (15 mL, 197 mmol) was degassed and purged with N.sub.2 (3×). The mixture was stirred at 100° C. for 12 h under N.sub.2 atmosphere. After cooling to room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 40 g cartridge, 0-100% EtOAc:petroleum ether) to give 8-(2-bromoethoxy)-1H,2H,3H,5H,6H,10bH-imidazo[4,3-a]isoquinolin-3-one (Intermediate A-58). MS=311.1/313.1 [M+H].sup.+.

General Procedure for Intermediate A-59

##STR01118##

Step 1: 6-methoxy-3-methyl-2,3-dihydro-1,3-benzoxazol-2-one

##STR01119##

[2569] To a solution of 6-methoxy-2,3-dihydro-1,3-benzoxazol-2-one (1.80 g, 10.9 mmol) and K.sub.2CO.sub.3 (3.01 g, 21.8 mmol) in DMF (18 mL) was added MeI (1.36 mL, 21.8 mmol). The mixture was stirred at room temperature for 1 h and was then quenched H.sub.2O (15 mL). The mixture was extracted with EtOAc (3×20 mL). The combined organic layers were washed with brine (2×20 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 10 g cartridge, 0-20% EtOAc:petroleum ether) to give 6-methoxy-3-methyl-2,3-dihydro-1,3-benzoxazol-2-one. MS=180.0 [M+H].sup.+.

Step 2: 6-hydroxy-3-methyl-2,3-dihydro-1,3-benzoxazol-2-one

##STR01120##

[2570] To a -20° C. solution of 6-methoxy-3-methyl-2,3-dihydro-1,3-benzoxazol-2-one (1.50 g, 8.37 mmol) in DCM (3 mL) was added BBr.sub.3 (2.42 mL, 25.1 mmol) dropwise. The mixture was warmed to 0° C. and stirred for 1 h, then poured into H.sub.2O (15 mL). The resulting solids were isolated by filtration. The filter cake was washed with H.sub.2O and then dried in vacuo to provide the first fraction of desired product. The filtrate was extracted with DCM (3×20 mL). The combined organic layers were washed with brine (15 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was washed with MeCN (2 mL), isolated by filtration, and the filter cake was dried in vacuo to give the second fraction of 6-hydroxy-3-methyl-2,3-dihydro-1,3-benzoxazol-2-one, which was combined with the first fraction and used in the subsequent step without further purification. MS=166.3 [M+H].sup.+.

Step 3: 6-(2-bromoethoxy)-3-methyl-2,3-dihydro-1,3-benzoxazol-2-one

##STR01121##

[2571] A solution of 1,2-dibromoethane (2.97 mL, 39.4 mmol), 6-hydroxy-3-methyl-2,3-dihydro-1,3-benzoxazol-2-one (1.30 g, 7.87 mmol) and Cs.sub.2CO.sub.3 (3.85 g, 11.8 mmol) in DMF (2 mL) was stirred at 100° C. for 16 h. After cooling to room temperature, the mixture was quenched

with H.sub.2O (15 mL). The mixture was extracted with EtOAc (3×15 mL). The combined organic layers were washed with brine (2×10 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 20 g cartridge, 0-30% EtOAc:petroleum ether) to give 6-(2-bromoethoxy)-3-methyl-2,3-dihydro-1,3-benzoxazol-2-one (Intermediate A-59). MS=271.9/273.8 [M+H].sup.+.

General Procedure for Intermediate A-60

##STR01122##

Step 1: 5-bromo-7-fluoro-1-methyl-2,3-dihydro-1H-indole-2,3-dione

##STR01123##

[2572] A solution of 7-fluoro-1-methyl-2,3-dihydro-1H-indole-2,3-dione (4.00 g, 22.3 mmol) and NBS (5.17 g, 29.0 mmol) in DMF (45 mL) was stirred at 80° C. for 2 h under N.sub.2. After cooling to room temperature, the reaction mixture was poured into H.sub.2O (100 mL) and a solid crashed out of solution. The solid was isolated via filtration and dried in vacuo to give 5-bromo-7-fluoro-1-methyl-2,3-dihydro-1H-indole-2,3-dione, which was used in the subsequent step without further purification. MS=258.0/260.0 [M+H].sup.+.

Step 2: 7-fluoro-1-methyl-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-2,3-dihydro-1H-indole-2,3-dione

##STR01124##

[2573] To a solution of 5-bromo-7-fluoro-1-methyl-2,3-dihydro-1H-indole-2,3-dione (3.98 g, 15.4 mmol) and bis(pinacolato)diboron (5.88 g, 23.1 mmol) in 1,4-dioxane (40 mL) was added KOAc (3.03 g, 30.9 mmol) and Pd(dppf)Cl.sub.2 (1.13 g, 1.54 mmol). The mixture was stirred at 90° C. for 15 h under an atmosphere of N.sub.2. After cooling to room temperature, the mixture was diluted with H.sub.2O (50 mL). The mixture was extracted with EtOAc (3×30 mL). The combined organic layers were washed with brine (2×30 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 80 g cartridge, 0-15% EtOAc:petroleum ether) to give 7-fluoro-1-methyl-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-2,3-dihydro-1H-indole-2,3-dione. MS=306.1 [M+H].sup.+.

Step 3: 7-fluoro-5-hydroxy-1-methyl-2,3-dihydro-1H-indole-2,3-dione

##STR01125##

[2574] To a solution of 7-fluoro-1-methyl-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-2,3-dihydro-1H-indole-2,3-dione (2.80 g, 9.18 mmol) in DCM (35 mL) were added K.sub.2CO.sub.3 (2.54 g, 18.4 mmol) and H.sub.2O.sub.2 in H.sub.2O (2.20 mL, 30 wt %, 22.94 mmol). The mixture was stirred at room temperature for 1 h. The reaction mixture was cooled to 0° C., quenched with saturated Na.sub.2SO.sub.3 (10 mL), and extracted with DCM (3×10 mL). The combined organic layers were washed with brine (5 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 4 g cartridge, 0-35% EtOAc:petroleum ether) to give 7-fluoro-5-hydroxy-1-methyl-2,3-dihydro-1H-indole-2,3-dione. MS=196.1 [M+H].sup.+.

Step 4: 7-fluoro-5-hydroxy-1-methyl-2,3-dihydro-1H-indol-2-one

##STR01126##

[2575] To a solution of 7-fluoro-5-hydroxy-1-methyl-2,3-dihydro-1H-indole-2,3-dione (600 mg, 3.07 mmol) in DMSO (2.5 mL) was added NH.sub.2NH.sub.2. H.sub.2O (2.46 mL, 85% purity, 43.0 mmol). The mixture was stirred at 120° C. for 3 h. After cooling to 0° C., the reaction mixture was quenched with H.sub.2O (8 mL) and extracted with EtOAc (3×7 mL). The combined organic layers were washed with brine (5 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-29% EtOAc:petroleum ether) to give 7-fluoro-5-hydroxy-1-methyl-2,3-dihydro-1H-indol-2-one. MS=180.1 [M-H].sup.-.

Step 5: 5-(2-bromoethoxy)-7-fluoro-1-methyl-2,3-dihydro-1H-indol-2-one

##STR01127##

[2576] A solution of 7-fluoro-5-hydroxy-1-methyl-2,3-dihydro-1H-indol-2-one (250 mg, 1.38 mmol) and Cs.sub.2CO.sub.3 (1.12 g, 3.45 mmol) in 1,2-dibromoethane (6.00 mL, 79.5 mmol) was stirred at 100° C. for 15 h. After cooling to room temperature, the reaction mixture was quenched with H.sub.2O (18 mL) and extracted with EtOAc (3×12 mL). The combined organic layers were washed with brine (5 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 4 g cartridge, 0-17% EtOAc:petroleum ether) to give 5-(2-bromoethoxy)-7-fluoro-1-methyl-2,3-dihydro-1H-indol-2-one (Intermediate A-60). MS=287.9/289.9 [M+H].sup.+.

General Procedure for Intermediate A-61

##STR01128##

Step 1: 5-(2-hydroxyethoxy)-2,3-dihydro-1H-isoindol-1-one

##STR01129##

[2577] To a mixture of 5-fluoro-2,3-dihydro-1H-isoindol-1-one (750 mg, 4.96 mmol) and ethylene glycol (4.00 mL, 71.53 mmol) in NMP (4 mL) was added t-BuOK (1.67 g, 14.8 mmol). The mixture was stirred at 100° C. for 15 h. After cooling to room temperature, the reaction mixture was diluted with NMP (2 mL), and solids were removed by filtration. The filtrate was concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Waters Xbridge BEH C.sub.18 column, 1-30% MeCN:10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 5-(2-hydroxyethoxy)-2,3-dihydro-1H-isoindol-1-one. MS=194.2 [M+H].sup.+.

Step 2: 2-[(1-oxo-2,3-dihydro-1H-isoindol-5-yl)oxy]ethyl methanesulfonate

##STR01130##

[2578] To a 0° C. mixture of 5-(2-hydroxyethoxy)-2,3-dihydro-1H-isoindol-1-one (100 mg, 0.518 mmol) and TEA (0.144 mL, 1.04 mmol) in DCM (2 mL) was added methanesulfonic anhydride (451 mg, 2.59 mmol) portionwise. The reaction mixture was warmed to room temperature and stirred for 15 h. The reaction mixture was then quenched with H.sub.2O (10 mL) and extracted with EtOAc (3×10 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give 2-[(1-oxo-2,3-dihydro-1H-isoindol-5-yl)oxy]ethyl methanesulfonate (Intermediate A-61), which was used in the subsequent step without further purification. MS=272.1 [M+H].sup.+.

General Procedure for Intermediate A-62

##STR01131##

Step 1: 5-methoxy-1-[trans-3-(benzyloxy)cyclobutyl]-1H-indazole

##STR01132##

[2579] To a mixture of 5-methoxy-1H-indazole (665 mg, 4.49 mmol) and (3-benzyloxycyclobutyl) methanesulfonate (1.15 g, 4.49 mmol) in DMF (15 mL) was added Cs.sub.2CO.sub.3 (2.92 g, 8.98 mmol). The mixture was stirred at 80° C. for 15 h. After cooling to room temperature, the reaction mixture was quenched with H.sub.2O (45 mL) and extracted with EtOAc (3×50 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 80 g cartridge, 0-50% EtOAc:petroleum ether). The first eluting product, desired 5-methoxy-1-[trans-3-(benzyloxy)cyclobutyl]-1H-indazole: .sup.1H NMR (400 MHz, CDCl.sub.3): δ 7.94 (s, 1H), 7.40-7.35 (m, 6H), 7.08-7.04 (m, 2H), 4.67-4.59 (m, 1H), 4.53 (s, 2H), 4.06-4.00 (m, 1H), 3.86 (s, 3H), 2.93-2.83 (m, 4H). MS=309.3 [M+H].sup.+.

The second eluting product, desired 5-methoxy-1-[cis-3-(benzyloxy)cyclobutyl]-1H-indazole: .sup.1H NMR (400 MHz, CDCl.sub.3): δ 7.95 (s, 1H), 7.41-7.29 (m, 6H), 7.07-7.04 (m, 2H), 5.29-5.22 (m, 1H), 4.54-4.50 (m, 3H), 3.86 (s, 3H), 2.96-2.92 (m, 2H), 2.73-2.67 (m, 2H).

The third eluting product, undesired 5-methoxy-2-[(trans)-3-(benzyloxy)cyclobutyl]-2H-indazole: .sup.1H NMR (400 MHz, CDCl.sub.3): δ 7.81 (s, 1H), 7.64 (d, J=9.6 Hz, 1H), 7.39-7.37 (m, 5H), 7.00 (dd, J=9.6, 2.4 Hz, 1H), 6.87 (d, J=2.4 Hz, 1H), 5.21-5.17 (m, 1H), 4.56-4.52 (m, 1H), 4.51 (s, 2H), 3.84 (s, 3H), 2.99-2.92 (m, 2H), 2.77-2.72 (m, 2H).

The fourth eluting product, undesired 5-methoxy-2-[(cis)-3-(benzyloxy)cyclobutyl]-2H-indazole:

.sup.1H NMR (400 MHz, CDCl₃): δ 7.88 (s, 1H), 7.62 (d, J=9.2 Hz, 1H), 7.38-7.33 (m, 5H), 7.99 (dd, J=9.2, 2.4 Hz, 1H), 6.87 (d, J=2.4 Hz, 1H), 4.67-4.62 (m, 1H), 4.52 (s, 2H), 4.06-3.99 (m, 1H), 3.84 (s, 3H), 3.01-2.98 (m, 2H), 2.70-2.67 (m, 2H).

Step 2: 1-[trans-3-hydroxycyclobutyl]-1H-indazol-5-ol

##STR01133##

[2580] To a -10° C. solution of 5-methoxy-1-[trans-3-(benzyloxy)cyclobutyl]-1H-indazole (250 mg, 0.811 mmol) in DCM (2.5 mL) was added BBr₃ (156 µL, 1.62 mmol). The mixture was stirred at -10° C. for 4 h, and then quenched with MeOH (10 mL). The solution was placed in a 0° C. ice bath and adjusted to pH=7-8 with the dropwise addition of saturated aqueous NaHCO₃. The biphasic mixture was concentrated in vacuo, and the remaining aqueous layer was extracted with EtOAc (3×20 mL). The combined organic layers were washed with brine (2×40 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-100% EtOAc:petroleum ether) to give 1-[trans-3-hydroxycyclobutyl]-1H-indazol-5-ol. MS=205.2 [M+H]⁺.

Step 3: trans-3-[5-(2-bromoethoxy)-1H-indazol-1-yl]cyclobutan-1-ol

##STR01134##

[2581] To a solution of 1-[trans-3-hydroxycyclobutyl]-1H-indazol-5-ol (100 mg, 0.490 mmol) in 1,2-dibromoethane (10 mL, 133 mmol) was added Cs₂CO₃ (479 mg, 1.47 mmol). The mixture was stirred at 100° C. for 15 h. After cooling to room temperature, the reaction mixture was quenched with H₂O (30 mL) and extracted with EtOAc (3×15 mL). The combined organic layers were dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 25 g cartridge, 0-100% EtOAc:petroleum ether) to give trans-3-[5-(2-bromoethoxy)-1H-indazol-1-yl]cyclobutan-1-ol (Intermediate A-62). MS=311.2/313.2 [M+H]⁺.

[2582] The following intermediate in Table 10 was prepared using procedures similar to those described for Intermediate A-62, using the appropriate starting materials.

General Procedures for Intermediate A-63

TABLE-US-00011 TABLE 10 Intermediate # Structure Name Exact Mass [M + H]⁺ A-63
[01135]  embedded image cis-3-[5-(2-bromoethoxy)-1H-indazol-1-yl]cyclobutan-1-ol Calc'd 311.0/313.0 Found 311.2/313.2

General Procedure for Intermediate A-64

##STR01136##

Step 1: 6-(benzyloxy)-1,2,3,4-tetrahydroquinolin-2-one

##STR01137##

[2583] To a 0° C. solution of 6-hydroxy-1,2,3,4-tetrahydroquinolin-2-one (1.00 g, 6.13 mmol) in EtOH (10 mL) were added NaOH (490 mg, 12.3 mmol) and benzyl chloride (2.12 mL, 18.4 mmol). The mixture was stirred at room temperature for 15 h, was then poured into H₂O (20 mL), and filtered to give a filter cake. The filter cake was washed with EtOH, triturated with MTBE, then dried in vacuo to give 6-(benzyloxy)-1,2,3,4-tetrahydroquinolin-2-one, which was taken to the next step without further purification. MS=254.2 [M+H]⁺.

Step 2: 6-(benzyloxy)-1-(propan-2-yl)-1,2,3,4-tetrahydroquinolin-2-one

##STR01138##

[2584] To a 0° C. solution of 6-(benzyloxy)-1,2,3,4-tetrahydroquinolin-2-one (1.50 g, 5.92 mmol) in DMA (20 mL) under N₂ atmosphere was added NaH (616 mg, 60 wt % in mineral oil, 15.4 mmol). After 30 min, 2-iodopropane (2.37 mL, 23.7 mmol) was added. The reaction mixture was warmed to room temperature, and then heated to 35° C. and stirred for 15.5 h under an atmosphere of N₂. After cooling to 0° C., the reaction mixture was quenched with H₂O (30 mL) and extracted with EtOAc (3×20 mL). The combined organic layers were washed with brine (5 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 20 g cartridge, 0-27% EtOAc:petroleum ether)

to give 6-(benzyloxy)-1-(propan-2-yl)-1,2,3,4-tetrahydroquinolin-2-one. MS=296.1 [M+H].sup.+.

Step 3: 6-hydroxy-1-(propan-2-yl)-1,2,3,4-tetrahydroquinolin-2-one

##STR01139##

[2585] To a solution of 6-(benzyloxy)-1-(propan-2-yl)-1,2,3,4-tetrahydroquinolin-2-one (1.30 g, 4.40 mmol) in MeOH (25 mL) was added Pd/C (0.30 g, 10 wt %, 0.28 mmol). The mixture was stirred at 40° C. for 16 h under an atmosphere of H.sub.2 (40 psi). After cooling to room temperature, the reaction mixture was filtered through Celite and the filtrate was concentrated in vacuo to give 6-hydroxy-1-(propan-2-yl)-1,2,3,4-tetrahydroquinolin-2-one, which was used in the subsequent step without further purification. MS=206.0 [M+H].sup.+.

Step 4: 6-(2-bromoethoxy)-1-(propan-2-yl)-1,2,3,4-tetrahydroquinolin-2-one

##STR01140##

[2586] To a solution of 6-hydroxy-1-(propan-2-yl)-1,2,3,4-tetrahydroquinolin-2-one (810 mg, 3.95 mmol) in 1,2-dibromoethane (12 mL, 159 mmol) was added Cs.sub.2CO.sub.3 (2.57 g, 7.89 mmol). The mixture was stirred at 100° C. for 15 h. After cooling to room temperature, the reaction mixture was quenched with H.sub.2O (20 mL) and extracted with EtOAc (3×20 mL). The combined organic layers were washed with brine (2×25 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-20% EtOAc:petroleum ether) to give 6-(2-bromoethoxy)-1-(propan-2-yl)-1,2,3,4-tetrahydroquinolin-2-one (Intermediate A-64). MS=312.0/314.0 [M+H].sup.+.

[2587] The following intermediates in Table 11 were prepared according to procedures similar those described for Intermediate A-64 using the appropriate starting materials.

General Procedure for Intermediates A-65 to A-68

TABLE-US-00012 TABLE 11 Intermediate # Structure Name Exact Mass [M + H].sup.+ A-65

[01141]  6-(2-bromoethoxy)-1-(2-hydroxyethyl)-1,2,3,4-tetrahydroquinolin-2-one Calc'd 314.0/316.0 Found 313.9/315.9 A-66 [01142]  6-(2-bromoethoxy)-1-(oxetan-3-yl)-1,2,3,4-tetrahydroquinolin-2-one Calc'd 326.0/328.0 Found 326.1/328.1 A-67 [01143]  6-(2-bromoethoxy)-1-(2,2-difluoroethyl)-1,2,3,4-tetrahydroquinolin-2-one Calc'd 334.0/336.0 Found 334.1/336.1 A-68 [01144]  N-[4-(2-bromoethoxy)phenyl]-N-(2-hydroxyethyl)methanesulfonamide Calc'd 338.0/340.0 Found 338.1/340.1

General Procedure for Intermediate A-69

##STR01145##

Step 1: 6-(benzyloxy)-1-[(cis)-3-(benzyloxy)cyclobutyl]-1,2,3,4-tetrahydroquinolin-2-one

##STR01146##

[2588] To a solution of 6-(benzyloxy)-1,2,3,4-tetrahydroquinolin-2-one (300 mg, 1.18 mmol) in DMF (4 mL) was added Cs.sub.2CO.sub.3 (1.54 g, 4.74 mmol) and 3-(benzyloxy)cyclobutyl methanesulfonate (910 mg, 3.55 mmol). The mixture was stirred at 100° C. for 12 h. After cooling to room temperature, the reaction mixture was poured into H.sub.2O (10 mL) and extracted with EtOAc (3×10 mL). The combined organic layers were washed with brine (20 mL), filtered, dried over Na.sub.2SO.sub.4, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 20 g cartridge, 0-15% EtOAc:petroleum ether). The first eluting isomer, minor product 6-(benzyloxy)-1-[(trans)-3-(benzyloxy)cyclobutyl]-1,2,3,4-tetrahydroquinolin-2-one: .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 7.44-7.33 (m, 10H), 6.93 (d, J=2.4 Hz, 1H), 6.92-6.80 (m, 2H), 5.05 (s, 2H), 4.62-4.56 (m, 1H), 4.40 (s, 2H), 4.14-4.11 (m, 1H), 2.76-3.73 (m, 2H), 2.53-2.50 (m, 2H), 2.44-2.37 (m, 4H). The second eluting isomer, desired major product 6-(benzyloxy)-1-[(cis)-3-(benzyloxy)cyclobutyl]-1,2,3,4-tetrahydroquinolin-2-one: .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 7.45-7.39 (m, 4H), 7.33-7.28 (m, 6H), 6.94 (d, J=2.4 Hz, 1H), 6.93-6.77 (m, 2H), 5.06 (s, 2H), 4.36 (s, 2H), 3.93-3.88 (m, 1H), 3.80-3.77 (m, 1H), 2.88-2.86 (m, 2H), 2.77-2.73 (m, 2H), 2.40-2.37 (m, 2H), 1.95-1.89 (m, 2H).

Step 2: 6-hydroxy-1-[(cis)-3-hydroxycyclobutyl]-1,2,3,4-tetrahydroquinolin-2-one

##STR01147##

[2589] To a solution of 6-(benzyloxy)-1-[(cis)-3-(benzyloxy)cyclobutyl]-1,2,3,4-tetrahydroquinolin-2-one (400 mg, 0.967 mmol) in MeOH (20 mL) under N.sub.2 atmosphere was added Pd/C (50 mg, 10 wt %, 0.47 mmol) The mixture was degassed and purged with H.sub.2 (3×), then stirred under H.sub.2 (50 Psi) atmosphere at room temperature for 12 h. The reaction mixture was filtered through Celite and concentrated under in vacuo to give 6-hydroxy-1-[(cis)-3-hydroxycyclobutyl]-1,2,3,4-tetrahydroquinolin-2-one, which was used in the subsequent step without further purification. MS=234.3 [M+H].sup.+.

Step 3: 6-(2-bromoethoxy)-1-[(cis)-3-hydroxycyclobutyl]-1,2,3,4-tetrahydroquinolin-2-one

##STR01148##

[2590] To a solution of 6-hydroxy-1-[(cis)-3-hydroxycyclobutyl]-1,2,3,4-tetrahydroquinolin-2-one (200 mg, 0.857 mmol) in 1,2-dibromoethane (5 mL, 133 mmol) was added Cs.sub.2CO.sub.3 (558 mg, 1.71 mmol). The mixture was stirred at 100° C. for 12 h. After cooling to room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo. The residue was diluted with H.sub.2O (10 mL) and extracted with DCM (3×10 mL). The combined organic layers were washed with brine (20 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Isco 4 g cartridge, 0-47% EtOAc:petroleum ether) to give 6-(2-bromoethoxy)-1-[(cis)-3-hydroxycyclobutyl]-1,2,3,4-tetrahydroquinolin-2-one (Intermediate A-69). MS=340.1/342.1 [M+H].sup.+.

General Procedure for Intermediates A-70 & A-71

##STR01149##

Step 1: 6-hydroxy-1-methyl-1,2,3,4-tetrahydroquinolin-2-one

##STR01150##

[2591] To a 0° C. solution of 6-methoxy-1-methyl-1,2,3,4-tetrahydroquinolin-2-one (150 mg, 0.784 mmol) in DCM (5 mL) was added BBr.sub.3 (378 µL, 3.92 mmol) dropwise. The mixture was stirred at 40° C. for 4 h. After cooling to 0° C., the reaction mixture was quenched with H.sub.2O (10 mL) and adjusted to pH=8 with saturated aqueous Na.sub.2CO.sub.3. The biphasic mixture was extracted with EtOAc (5×5 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 20 g cartridge, 80-100% EtOAc:petroleum ether) to give 6-hydroxy-1-methyl-1,2,3,4-tetrahydroquinolin-2-one (Intermediate A-70). MS=178.1 [M+H].sup.+.

Step 2: 6-(2-bromoethoxy)-1-methyl-1,2,3,4-tetrahydroquinolin-2-one

##STR01151##

[2592] To a solution of 6-hydroxy-1-methyl-3,4-dihydroquinolin-2-one (0.300 g, 1.69 mmol) in 1,2-dibromoethane (5.11 mL, 67.7 mmol) was added Cs.sub.2CO.sub.3 (1.10 g, 3.39 mmol). The mixture was stirred at 100° C. for 4 h. After cooling to 0° C., the reaction mixture was quenched with H.sub.2O (20 mL) and extracted with DCM (3×10 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 4 g cartridge, 0-25% EtOAc:petroleum ether) to give 6-(2-bromoethoxy)-1-methyl-1,2,3,4-tetrahydroquinolin-2-one (Intermediate A-71). MS=284.0/286.1 [M+H].sup.+.

General Procedure for Intermediate A-72

##STR01152##

Step 1: 6-bromo-1-methyl-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR01153##

[2593] To a 0° C. solution of 6-bromo-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (739 mg, 3.26 mmol) in DMF (11 mL) was added NaH (156 mg, 60 wt %, 3.91 mmol). The mixture was stirred at for 1 h and then MeI (693 mg, 4.88 mmol) was added. The reaction mixture was warmed to room temperature and stirred for 1 h. The reaction mixture was then quenched with H.sub.2O (30 mL)

and extracted with EtOAc (3×20 mL). The combined organic layers were washed with brine (30 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 25 g cartridge, 0-100% EtOAc/hexanes) to give 6-bromo-1-methyl-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one. ¹H NMR (500 MHz, CDCl₃): δ 8.31 (d, J=2.3 Hz, 1H), 7.59 (dd, J=2.3, 1.1 Hz, 1H), 3.45 (s, 3H), 2.91 (t, J=7.5 Hz, 2H), 2.72 (t, J=7.5 Hz, 2H).

Step 2: 6-hydroxy-1-methyl-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR01154##

[2594] A solution of 6-bromo-1-methyl-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (450 mg, 1.87 mmol), potassium acetate (366 mg, 3.73 mmol), bis(pinacolato)diboron (569 mg, 2.24 mmol) in 1,4-dioxane (9.3 mL) was sparged with nitrogen for 10 min. Pd(dppf)Cl₂ (68 mg, 0.093 mmol) was then added, and the reaction was heated to 90° C. for 2 h. The reaction mixture was cooled to room temperature, filtered over Celite, and washed with EtOAc (30 mL). The filtrate was concentrated in vacuo. The crude residue was purified by normal phase chromatography (Biotage 25 g cartridge, 10% EtOAc:hexanes). The resulting residue was dissolved in THF (25 mL) and H₂O (10 mL), and 1 M aqueous NaOH (5 mL) was added. The solution was cooled to 0° C. and H₂O (423 mg, 30 wt %, 3.7 mmol) was added dropwise over 3 min. The reaction mixture was allowed to warm to room temperature and stirred for 16 h. The mixture was adjusted to pH=5 with 1.0 M aqueous HCl. The reaction mixture was extracted with EtOAc (3×50 mL). The combined organic layers were washed with brine (30 mL) and dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 25 g cartridge, 10% MeOH:DCM) to give 6-hydroxy-1-methyl-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one. ¹H NMR (500 MHz, DMSO-d₆): δ 9.57 (s, 1H), 7.77 (d, J=2.7 Hz, 1H), 7.09 (d, J=2.8 Hz, 1H), 3.26 (s, 3H), 2.81 (t, J=7.5 Hz, 2H), 2.62-2.36 (m, 2H). MS=179.1 [M+H]⁺.

Step 3: 6-(2-bromoethoxy)-1-methyl-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR01155##

[2595] To a solution of 6-hydroxy-1-methyl-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (221 mg, 1.24 mmol) in 1,2-dibromoethane (4.1 mL, 53 mmol) was added Cs₂CO₃ (1.2 g, 3.7 mmol) and the reaction was stirred at 100° C. for 6 h. The reaction was cooled to room temperature and diluted with EtOAc (10 mL). The reaction mixture was filtered over Celite and washed with EtOAc (30 mL). The reaction mixture was then concentrated in vacuo to afford 6-(2-bromoethoxy)-1-methyl-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (Intermediate A-72) which was used in the subsequent step without further purification. ¹H NMR (500 MHz, CDCl₃): δ 7.94 (d, J=2.8 Hz, 1H), 7.11 (d, J=2.8 Hz, 1H), 4.33 (t, J=6.1 Hz, 2H), 3.65 (t, J=6.1 Hz, 2H), 3.45 (s, 3H), 2.88 (t, J=7.6 Hz, 2H), 2.69 (t, J=7.6 Hz, 2H).

General Procedure for Intermediate A-73

##STR01156##

Step 1: 1-(6-bromo-1,2,3,4-tetrahydro-1,8-naphthyridin-1-yl)ethan-1-one

##STR01157##

[2596] To a solution of 6-bromo-1,2,3,4-tetrahydro-1,8-naphthyridine (500 mg, 2.35 mmol) in DCM (12 mL) was added pyridine (757 μL, 9.39 mmol), acetic anhydride (660 μL, 7.05 mmol) and DMAP (5.0 mg, 0.041 mmol). The mixture was heated to 40° C. and stirred for 16 h, and then cooled to room temperature. The reaction mixture was then quenched with H₂O (30 mL) and extracted with EtOAc (3×20 mL). The combined organic layers were washed with brine (30 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 25 g cartridge, 0-100% EtOAc:Hexanes) to give a mixture of 1-(6-bromo-1,2,3,4-tetrahydro-1,8-naphthyridin-1-yl)ethan-1-one and unreacted 6-bromo-1,2,3,4-tetrahydro-1,8-naphthyridine. The mixture was redissolved in DCM (11.7 mL) and TEA (326 μL, 2.35 mmol) was added followed by acetyl chloride (167 μL, 2.35 mmol). After

stirring for 16 h, the reaction mixture was quenched with H.sub.2O (30 mL) and extracted with EtOAc (3×20 mL). The combined organic layers were washed with brine (30 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 25 g cartridge, 0-100% EtOAc:Hexanes) to afford 1-(6-bromo-1,2,3,4-tetrahydro-1,8-naphthyridin-1-yl)ethan-1-one.

Step 2: 1-(6-hydroxy-1,2,3,4-tetrahydro-1,8-naphthyridin-1-yl)ethan-1-one

##STR01158##

[2597] A solution of 1-(6-bromo-1,2,3,4-tetrahydro-1,8-naphthyridin-1-yl)ethan-1-one (435 mg, 1.71 mmol), potassium acetate (335 mg, 3.41 mmol), and bis(pinacolato)diboron (520 mg 2.05 mmol) in 1,4-dioxane (8.5 mL) was sparged with N.sub.2 for 10 min. Pd(dppf)Cl.sub.2 (62 mg, 0.085 mmol) was added, and the reaction was heated at 90° C. for 2 h. The reaction mixture was cooled to room temperature, filtered over Celite and washed with EtOAc (30 mL), then concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 25 g cartridge, 10% EtOAc:Hexanes). The resulting residue was dissolved in THF (25 mL) and H.sub.2O (10 mL), and 1.0 M aqueous NaOH (5 mL) was added. The solution was cooled to 0° C. and H.sub.2O.sub.2 in H.sub.2O (387 mg, 30 wt %, 3.4 mmol) was added dropwise over 3 min. The reaction mixture was allowed to warm room temperature and stirred for 16 h. The mixture was adjusted to pH=5 with 1.0 M aqueous HCl. The reaction mixture was extracted with EtOAc (3×50 mL). The combined organic layers were washed with brine (30 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase chromatography (Biotage 25 g cartridge, 10% MeOH:DCM) to give 1-(6-hydroxy-1,2,3,4-tetrahydro-1,8-naphthyridin-1-yl)ethan-1-one. .sup.1H NMR (500 MHz, CDCl.sub.3, 11/12 H): δ 7.92 (d, J=2.8 Hz, 1H), 7.04 (d, J=2.8 Hz, 1H), 3.89 (t, J=6.3 Hz, 2H), 2.75 (t, J=6.6 Hz, 2H), 2.41 (s, 3H), 2.00-1.85 (m, 2H). MS=193.1 [M+H].sup.+.

Step 3: 1-[6-(2-bromoethoxy)-1,2,3,4-tetrahydro-1,8-naphthyridin-1-yl]ethan-1-one

##STR01159##

[2598] To a solution of 1-(6-hydroxy-1,2,3,4-tetrahydro-1,8-naphthyridin-1-yl)ethan-1-one (124 mg, 0.645 mmol) in 1,2-dibromoethane (2.1 mL, 27.8 mmol) was added Cs.sub.2CO.sub.3 (631 mg, 1.94 mmol), and the reaction was stirred at 70° C. for 6 h. The reaction was cooled to room temperature and diluted with EtOAc (10 mL). The reaction mixture was filtered over Celite and washed with EtOAc (30 mL). The reaction mixture was then concentrated in vacuo to afford 1-[6-(2-bromoethoxy)-1,2,3,4-tetrahydro-1,8-naphthyridin-1-yl]ethan-1-one (Intermediate A-73) which was used without further purification. .sup.1H NMR (500 MHz, CDCl.sub.3): δ 7.96 (d, J=2.9 Hz, 1H), 7.08 (d, J=2.9 Hz, 1H), 4.35 (t, J=6.1 Hz, 2H), 3.90 (t, J=6.2 Hz, 2H), 3.67 (t, J=6.1 Hz, 2H), 2.78 (t, J=6.7 Hz, 2H), 2.43 (s, 3H), 1.98-1.93 (m, 2H).

General Procedure for Intermediate A-74

##STR01160##

Step 1: 6-bromo-3-methyl-1H,2H,3H-imidazo[4,5-b]pyridin-2-one

##STR01161##

[2599] To a solution of 5-bromo-N2-methylpyridine-2,3-diamine (1.00 g, 4.95 mmol) in DMF (15 mL) was added CDI (3.21 g, 19.8 mmol). The mixture was stirred at room temperature for 16 h. The reaction mixture was quenched with H.sub.2O (20 mL) and extracted with EtOAc (2×20 mL). The combined organic layers were washed with brine (2×10 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 20 g cartridge, 0-100% EtOAc:Hexanes) to give 6-bromo-3-methyl-1H,2H,3H-imidazo[4,5-b]pyridin-2-one. MS=228.1/230.1 [M+H].sup.+.

Step 2: 6-bromo-3-methyl-1-{[2-(trimethylsilyl)ethoxy]methyl}-1H,2H,3H-imidazo[4,5-b]pyridin-2-one

##STR01162##

[2600] To a 0° C. solution of 6-bromo-3-methyl-1H,2H,3H-imidazo[4,5-b]pyridin-2-one (500 mg,

2.19 mmol) in DMA (8 mL) was added NaH (263 mg, 60 wt % in mineral oil, 6.58 mmol) portionwise. The mixture was stirred at 0° C. for 30 min and then 2-(trimethylsilyl)ethoxymethyl chloride (0.582 mL, 3.29 mmol) was added dropwise. The mixture was allowed to warm to room temperature and was stirred for an additional 4.5 h. The reaction mixture was cooled to 0° C. and then quenched with saturated aqueous NH₄Cl (5 mL), diluted with H₂O (5 mL), and extracted with EtOAc (3×5 mL). The combined organic layers were washed with brine (5 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 10 g cartridge, 0-100% EtOAc:Hexanes) to give 6-bromo-3-methyl-1-{[2-(trimethylsilyl)ethoxy]methyl}-1H,2H,3H-imidazo[4,5-b]pyridin-2-one. MS=358.1/360.1 [M+H].sup.+.

Step 3: 3-methyl-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1-{[2-(trimethylsilyl)ethoxy]methyl}-1H,2H,3H-imidazo[4,5-b]pyridin-2-one
##STR01163##

[2601] A mixture of 6-bromo-3-methyl-1-{[2-(trimethylsilyl)ethoxy]methyl}-1H,2H,3H-imidazo[4,5-b]pyridin-2-one (1.80 g, 5.02 mmol), bis(pinacolato)diboron (1.53 g, 6.03 mmol), Pd(dppf)Cl₂ (368 mg, 0.502 mmol), and KOAc (986 mg, 10.1 mmol) in 1,4-dioxane (25 mL) was degassed and purged with N₂ (3×). The mixture was heated at 80° C. for 12 h. After cooling to room temperature, the reaction mixture was quenched with H₂O (20 mL) and extracted with EtOAc (2×30 mL). The combined organic layers were washed with brine (2×20 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 20 g cartridge, 0-100% EtOAc:petroleum ether) to give 3-methyl-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1-{[2-(trimethylsilyl)ethoxy]methyl}-1H,2H,3H-imidazo[4,5-b]pyridin-2-one. MS=406.3 [M+H].sup.+.

Step 4: 6-hydroxy-3-methyl-1-{[2-(trimethylsilyl)ethoxy]methyl}-1H,2H,3H-imidazo[4,5-b]pyridin-2-one
##STR01164##

[2602] To a 0° C. mixture of 3-methyl-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1-{[2-(trimethylsilyl)ethoxy]methyl}-1H,2H,3H-imidazo[4,5-b]pyridin-2-one (2.00 g, 4.93 mmol) in DCM (30 mL) was added H₂O₂ in H₂O (1.19 mL, 30 wt %, 12.3 mmol) and K₂CO₃ (1.36 g, 9.87 mmol). The mixture was stirred at room temperature for 10 h. The reaction mixture was cooled to 0° C., quenched with saturated aqueous Na₂SO₃ (40 mL), and extracted with EtOAc (2×50 mL). The combined organic layers were washed with saturated aqueous Na₂SO₃ (40 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 20 g cartridge, 0-100% EtOAc:petroleum ether) to give 6-hydroxy-3-methyl-1-{[2-(trimethylsilyl)ethoxy]methyl}-1H,2H,3H-imidazo[4,5-b]pyridin-2-one. MS=296.2 [M+H].sup.+.

Step 5: 6-(2-bromoethoxy)-3-methyl-1-{[2-(trimethylsilyl)ethoxy]methyl}-1H,2H,3H-imidazo[4,5-b]pyridin-2-one
##STR01165##

[2603] To a mixture of 6-hydroxy-3-methyl-1-{[2-(trimethylsilyl)ethoxy]methyl}-1H,2H,3H-imidazo[4,5-b]pyridin-2-one (600 mg, 2.03 mmol) and 1,2-dibromoethane (1.53 mL, 20.3 mmol) in DMF (5 mL) was added Cs₂CO₃ (1.32 g, 4.06 mmol). The mixture was stirred at 100° C. for 10 h. After cooling to room temperature, the reaction mixture was quenched by the addition of H₂O (10 mL) and extracted with EtOAc (2×10 mL). The combined organic layers were washed with brine (10 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 10 g cartridge, 0-100% EtOAc:petroleum ether) to give 6-(2-bromoethoxy)-3-methyl-1-{[2-(trimethylsilyl)ethoxy]methyl}-1H,2H,3H-imidazo[4,5-b]pyridin-2-one (Intermediate A-74). MS=402.2/404.1 [M+H].sup.+.

General Procedure for Intermediate A-75

##STR01166##

Step 1: 2-(Chloromethyl)-5-methoxypyrimidine

##STR01167##

[2604] To a solution of (5-methoxypyrimidin-2-yl)methanol (4.00 g, 28.5 mmol) in DCM (143 mL) under N.sub.2 atmosphere was added thionyl chloride (6.21 mL, 64.2 mmol). The reaction mixture was heated at 40° C. for 2 h. After cooling to room temperature, the reaction was quenched with H.sub.2O (100 mL) and extracted with EtOAc (3×100 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo give 2-(chloromethyl)-5-methoxypyrimidine, which was used in the subsequent step without further purification.

Step 2: -(Methanesulfonylmethyl)-5-methoxypyrimidine

##STR01168##

[2605] To a microwave vial equipped with a stir bar was added 2-(chloromethyl)-5-methoxypyrimidine (1.00 g, 6.31 mmol) and sodium methanesulfinate (0.837 g, 8.20 mmol) followed by H.sub.2O (12.6 mL). The vial was sealed and heated at 120° C. in a microwave at 40 psi for 30 min. The reaction mixture was cooled to 0° C. the resulting solid was isolated by filtration and dried in vacuo to give 2-(methanesulfonylmethyl)-5-methoxypyrimidine, which was used in the subsequent step without further purification. MS=203.0 [M+H].sup.+.

Step 3: 2-(1-Methanesulfonylethyl)-5-methoxypyrimidine

##STR01169##

[2606] To a solution of 2-(methanesulfonylmethyl)-5-methoxypyrimidine (1.25 g, 6.18 mmol) in DMF (29.4 mL) at 0° C. was added NaH (0.494 g, 60 wt % in mineral oil, 12.4 mmol). The reaction mixture was allowed to warm to room temperature and stirred for 10 min. MeI (0.366 mL, 5.87 mmol) was added, and the mixture was allowed to stir at room temperature for 1 h. The reaction mixture was quenched with H.sub.2O (50 mL) and extracted with EtOAc (3×50 mL). The combined organic layers were combined and washed with 5% aqueous LiCl solution (100 mL), dried over Na.sub.2SO.sub.4, and concentrated in vacuo. Purification by normal phase silica gel chromatography (0-100% EtOAc in Hexanes) provided 2-(1-methanesulfonylethyl)-5-methoxypyrimidine. MS=217.12 [M+H].sup.+.

Step 4: 2-(1-Methanesulfonylethyl)pyrimidin-5-ol

##STR01170##

[2607] To a solution of 2-(1-methanesulfonylethyl)-5-methoxypyrimidine (500 mg, 2.31 mmol) in DCM (11.6 mL) at 0° C. was added a solution of 1.0 M BBr.sub.3 in DCM (11.6 mL, 11.6 mmol). The mixture was allowed to warm to room temperature and stirred for 5 h. The reaction mixture was concentrated in vacuo, then re-dissolved in DCM and concentrated in vacuo (2×). The residue was quenched with MeOH (30 mL), and then concentrated in vacuo to provide 5-(2-bromoethoxy)-2-(1-methanesulfonylethyl)pyrimidine, which was used in the subsequent step without further purification. MS=202.95 [M+H].sup.+.

Step 5: 5-(2-Bromoethoxy)-2-(1-methanesulfonylethyl)pyrimidine

##STR01171##

[2608] To a solution of 2-(1-methanesulfonylethyl)pyrimidin-5-ol (500 mg, 2.47 mmol) in 1,2-dibromoethane (8.24 mL, 93.9 mmol) and DMF (0.5 mL) was added Cs.sub.2CO.sub.3 (2.42 g, 7.42 mmol). The reaction mixture was heated at 100° C. for 6 h, then allowed to cool to room temperature. The reaction mixture was then diluted with 20 mL EtOAc, filtered over Celite and concentrated in vacuo to afford crude 5-(2-bromoethoxy)-2-(1-methanesulfonylethyl)pyrimidine (Intermediate A-75), which was used in the subsequent step without further purification. MS=309.0/311.0 [M+H].sup.+.

General Procedure for Intermediate A-76

##STR01172##

Step 1: 5-Bromo-2-(chloromethyl)pyrimidine

##STR01173##

[2609] To a solution of (5-bromopyrimidin-2-yl)methanol (5.00 g, 26.5 mmol) in DCM (132 mL) under N₂ atmosphere was added thionyl chloride (5.76 mL, 79.4 mmol). The reaction mixture was heated at 40° C. for 2 h. After cooling to room temperature, the mixture was quenched with H₂O (100 mL) and extracted with EtOAc (3×100 mL). The combined organic layers were dried over Na₂SO₄, filtered, then concentrated in vacuo to give 5-bromo-2-(chloromethyl)pyrimidine, which was used in the subsequent step without further purification. MS=207.0/209.0 [M+H]⁺.

Step 2: 5-Bromo-2-(methanesulfonylmethyl)pyrimidine

##STR01174##

[2610] To a microwave vial equipped with a stir bar were added 5-bromo-2-(chloromethyl)pyrimidine (2.00 g, 9.64 mmol), sodium methanesulfinate (1.28 g, 12.5 mmol), and then H₂O (19 mL). The vial was sealed and heated at 120° C. in a microwave at 40 psi for 30 min. The reaction mixture was then cooled to 0° C. and the resulting solid was isolated by filtration and dried in vacuo to give 2-(methanesulfonylmethyl)-5-methoxypyrimidine, which was used in the subsequent step without further purification. MS=251.0/253.0 [M+H]⁺.

Step 3: 5-Bromo-2-(1-methanesulfonylcyclopropyl)pyrimidine

##STR01175##

[2611] To a solution of 5-bromo-2-(methanesulfonylmethyl)pyrimidine (250 mg, 1.00 mmol) in DMF (5.0 mL) at 0° C. was added NaH (0.119 g, 60 wt % in mineral oil, 2.99 mmol). The reaction mixture was allowed to warm to room temperature for 10 min, then 1,2-dibromoethane (0.561 g, 2.99 mmol) was added. The mixture was allowed to stir for 1 h, then was diluted with H₂O (20 mL). The aqueous layer was extracted with EtOAc (3×20 mL). The combined organic layers were washed with 5% aqueous LiCl solution (50 mL), dried over Na₂SO₄, and concentrated in vacuo. Purification by normal phase silica gel chromatography (0-100% EtOAc:hexanes) gave 5-bromo-2-(1-methanesulfonylcyclopropyl)pyrimidine. ¹H NMR (500 MHz, CDCl₃): δ 8.74 (s, 2H), 3.36 (s, 3H), 2.03-1.98 (m, 2H), 1.79-1.75 (m, 2H). MS=277.0/279.0 [M+H]⁺.

Step 4: 2-(1-Methanesulfonylcyclopropyl)-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyrimidine

##STR01176##

[2612] To a vial equipped with a stir bar were added 5-bromo-2-(1-methanesulfonylcyclopropyl)pyrimidine (500 mg, 1.80 mmol), bis(pinacolato)diboron (687 mg, 2.71 mmol), potassium acetate (619 mg, 6.31 mmol), 1,1'-bis(diphenylphosphino)ferrocene palladium dichloride (66 mg, 0.09 mmol), and dry 1,4-dioxane (9.0 mL). The vial was sealed and evacuated until gas evolution from the solution was observed. The vial was then filled with N₂ and evacuated. This process was repeated three times, and the sealed vial was then heated at 100° C. for 16 h. The reaction mixture was allowed to cool to room temperature and was then diluted with EtOAc (20 mL). The mixture was filtered through Celite, and the filtrate was concentrated in vacuo to provide 2-(1-methanesulfonylcyclopropyl)-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyrimidine, which was used in the subsequent step without further purification.

Step 5: 2-(1-Methanesulfonylcyclopropyl)pyrimidin-5-ol

##STR01177##

[2613] To a vial containing a solution of 2-(1-methanesulfonylcyclopropyl)-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyrimidine (300 mg, 0.93 mmol) in MeOH (3.1 mL) was added 30% hydrogen peroxide in H₂O (0.284 mL, 2.78 mmol) dropwise. The mixture was allowed to stir for 2 h. The reaction mixture was concentrated in vacuo, diluted with MeOH, and concentrated in vacuo. The resulting mixture was diluted with MeOH and concentrated to give 2-(1-methanesulfonylcyclopropyl)pyrimidin-5-ol, which was used in the subsequent step without further purification. MS=215.04 [M+H]⁺.

Step 6: 5-(2-Bromoethoxy)-2-(1-methanesulfonylcyclopropyl)pyrimidine

##STR01178##

[2614] A vial was charged with 2-(1-methanesulfonylcyclopropyl)pyrimidin-5-ol (200 mg, 0.934 mmol) and dissolved in 1,2-dibromoethane (3.1 mL, 41 mmol) and DMF (0.5 mL).

Cs.sub.2CO.sub.3 (0.912 g, 2.801 mmol) was added, and the reaction mixture was heated at 100° C. for 6 h. After cooling to room temperature, the reaction mixture was then filtered over Celite and concentrated in vacuo to afford 5-(2-bromoethoxy)-2-(1-methanesulfonylcyclopropyl)pyrimidine (Intermediate A-76), which was carried onto the next step without further purification.

MS=321.1/323.1 [M+H].sup.+.

General Procedure for Intermediate A-77

##STR01179##

Step 1: 6-bromo-1-(2-hydroxyethyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR01180##

[2615] To a 0° C. solution of 6-bromo-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (4.00 g, 17.6 mmol) in DMA (80 mL) under N.sub.2 atmosphere was added NaH (3.52 g, 60 wt % in mineral oil, 88.1 mmol). After stirring for 15 min, 2-bromoethanol (22.0 g, 176 mmol) was added dropwise.

The reaction mixture was heated to 100° C. and stirred for 16 h. The reaction mixture was cooled to 0° C., quenched with H.sub.2O (40 mL), and extracted with EtOAc (3×50 mL). The combined organic layers were washed with brine (2×40 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 40 g cartridge, 0-45% EtOAc:petroleum ether) to give 6-bromo-1-(2-hydroxyethyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one. MS=271.0/273.0 [M+H].sup.+.

Step 2: 1-(2-hydroxyethyl)-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR01181##

[2616] To a mixture of 6-bromo-1-(2-hydroxyethyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (2.00 g, 7.38 mmol) and bis(pinacolato)diboron (2.25 g, 8.85 mmol) in 1,4-dioxane (30 mL) was added KOAc (1.45 g, 14.8 mmol). The mixture was degassed and purged with N.sub.2 (3×). Pd(dppf)Cl.sub.2 (540 mg, 0.738 mmol) was added, then the mixture was heated to 80° C. and stirred for 16 h under N.sub.2 atmosphere. After cooling to room temperature, the reaction mixture was filtered, and the filtrate was concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 20 g cartridge, 0-30% EtOAc:petroleum ether) to give 1-(2-hydroxyethyl)-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one. MS=319.2.0 [M+H].sup.+.

Step 3: 6-hydroxy-1-(2-hydroxyethyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR01182##

[2617] To a 0° C. solution of 1-(2-hydroxyethyl)-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (1.60 g, 5.03 mmol) in acetone (10 mL) and H.sub.2O (10 mL) was added Oxone (3.71 g, 6.03 mmol). The mixture was stirred at room temperature for 2 h. The reaction mixture was cooled to 0° C., quenched with saturated aqueous Na.sub.2SO.sub.3 (40 mL), and extracted with EtOAc (2×50 mL). The combined organic layers were washed with saturated aqueous Na.sub.2SO.sub.3 (50 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give 6-hydroxy-1-(2-hydroxyethyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one, which was used in the subsequent step without further purification. MS=209.1 [M+H].sup.+.

Step 4: 6-(2-bromoethoxy)-1-(2-hydroxyethyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR01183##

[2618] A mixture of 6-hydroxy-1-(2-hydroxyethyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (200 mg, 0.961 mmol), 1,2-dibromoethane (1.45 mL, 19.2 mmol) and K.sub.2CO.sub.3 (664 mg, 4.80 mmol) in MeCN (3 mL) was stirred at 80° C. for 16 h. After cooling to room temperature, the reaction mixture was filtered, and the filtrate was concentrated in vacuo. The crude residue was

purified by normal phase silica gel chromatography (Biotage 20 g cartridge, 0-30% EtOAc:petroleum ether) to give 6-(2-bromoethoxy)-1-(2-hydroxyethyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (Intermediate A-77). MS=315.1/317.1 [M+H].sup.+.

General Procedure for Intermediate A-78

##STR01184##

Step 1: 5-bromo-2-(methanesulfonylmethyl)pyridine

##STR01185##

[2619] To a 0° C. solution of DMSO (11.5 mL, 142 mmol) in THE (100 mL) under N.sub.2 atmosphere was added 1.0 M NaHMDS in THE (142 mL, 142 mmol) dropwise. The mixture was stirred at 0° C. for 1 h, then a solution of 5-bromo-2-fluoropyridine (2.92 mL, 28.4 mmol) in THE (50 mL) was added dropwise, and the mixture was stirred for another 1 h. The reaction mixture was quenched with H.sub.2O (40 mL) and then extracted with EtOAc (3×50 mL). The combined organic layers were washed with brine (2×50 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. Purification by reverse phase preparative HPLC (Agela C.sub.18 column, 20-50% MeCN in water with 0.1% NH.sub.4OH modifier) afforded 5-bromo-2-(methanesulfonylmethyl)pyridine. MS=250.0/252.0 [M+H].sup.+.

Step 2: 5-bromo-2-(2-methanesulfonylpropan-2-yl)pyridine

##STR01186##

[2620] To a 0° C. solution of 5-bromo-2-(methanesulfonylmethyl)pyridine (3.00 g, 12.0 mmol) in DMA (30 mL) under N.sub.2 atmosphere was added NaH (1.06 g, 60 wt % in mineral oil, 26.4 mmol) portionwise. After stirring at 0° C. for 1 h, a solution of MeI (2.24 mL, 36.0 mmol, 2.24 mL) in DMA (30 mL) was added dropwise. The mixture was allowed to warm to room temperature and stirred for another 12 h. The mixture was cooled to 0° C., quenched with H.sub.2O (40 mL), and extracted with EtOAc (3×50 mL). The combined organic layers were washed with brine (2×50 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 40 g cartridge, 0-40% EtOAc:petroleum ether) to give 5-bromo-2-(2-methanesulfonylpropan-2-yl)pyridine. MS=278.1/280.1 [M+H].sup.+.

Step 3: 2-(2-methanesulfonylpropan-2-yl)-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyridine

##STR01187##

[2621] To a mixture of 5-bromo-2-(2-methanesulfonylpropan-2-yl)pyridine (2.40 g, 8.63 mmol), bis(pinacolato)diboron (3.29 g, 12.9 mmol) in 1,4-dioxane (10 mL) was added KOAc (1.69 g, 17.3 mmol). The mixture was degassed and purged with N.sub.2 (3×), and then the Pd(dppf)Cl.sub.2 (316 mg, 0.431 mmol) was added. The mixture was heated to 80° C. and stirred for 12 h under N.sub.2 atmosphere. The reactions mixture was cooled to room temperature, was diluted with H.sub.2O (40 mL), and extracted with EtOAc (3×50 mL). The combined organic layers washed with brine (2×50 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 40 g cartridge, 0-40% EtOAc:petroleum ether) to give 2-(2-methanesulfonylpropan-2-yl)-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyridine. MS=326.2 [M+H].sup.+.

Step 4: 6-(2-methanesulfonylpropan-2-yl)pyridin-3-ol

##STR01188##

[2622] To a 0° C. solution of 2-(2-methanesulfonylpropan-2-yl)-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyridine (900 mg, 2.77 mmol) in acetone (10 mL) and H.sub.2O (10 mL) was added Oxone (3.40 g, 5.53 mmol). The mixture was allowed to warm to room temperature and stirred for 2 h. The reaction mixture was cooled to 0° C. and quenched with saturated aqueous Na.sub.2SO.sub.3 (50 mL), then extracted with EtOAc (3×50 mL). The combined organic layers were washed with brine (2×50 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 40 g cartridge, 0-40% EtOAc:petroleum ether) to give 6-(2-methanesulfonylpropan-2-yl)pyridin-3-ol.

MS=216.1 [M+H].sup.+.

Step 5: methyl 2-{{[6-(2-methanesulfonylpropan-2-yl)pyridin-3-yl]oxy}acetate

##STR01189##

[2623] To a solution of 6-(2-methanesulfonylpropan-2-yl)pyridin-3-ol (300 mg, 1.39 mmol) in DMF (6 mL) were added K.sub.2CO.sub.3 (385 mg, 2.79 mmol) and methyl 2-bromoacetate (255 mg, 1.67 mmol). The mixture was stirred at room temperature. The mixture was quenched with H.sub.2O (20 mL) and then extracted with EtOAc (2×30 mL). The combined organic layers were washed with brine (2×20 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 40 g cartridge, 0-40% EtOAc:petroleum ether) to give methyl 2-{{[6-(2-methanesulfonylpropan-2-yl)pyridin-3-yl]oxy}acetate. MS=288.1 [M+H].sup.+.

Step 6: 2-{{[6-(2-methanesulfonylpropan-2-yl)pyridin-3-yl]oxy}ethan-1-ol

##STR01190##

[2624] To a 0° C. solution of methyl 2-{{[6-(2-methanesulfonylpropan-2-yl)pyridin-3-yl]oxy}acetate (400 mg, 1.39 mmol) in THE (8 mL) under N.sub.2 atmosphere was added LiAlH.sub.4 (79.3 mg, 2.09 mmol). The mixture was stirred at 0° C. for 2 h. The mixture was quenched with H.sub.2O (20 mL) and then extracted with EtOAc (3×30 mL). The combined organic layers were washed with brine (2×20 mL), dried with Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give methyl 2-{{[6-(2-methanesulfonylpropan-2-yl)pyridin-3-yl]oxy}acetate, which was used in the subsequent step without further purification. MS=260.1 [M+H].sup.+.

Step 7: 2-{{[6-(2-methanesulfonylpropan-2-yl)pyridin-3-yl]oxy}ethyl methanesulfonate

##STR01191##

[2625] To a 0° C. solution of methyl 2-{{[6-(2-methanesulfonylpropan-2-yl)pyridin-3-yl]oxy}acetate (361 mg, 1.39 mmol) in DCM (5 mL) was added methanesulfonic anhydride (364 mg, 2.09 mmol) and TEA (388 µL, 2.78 mmol). The mixture was warmed to room temperature and stirred for 2 h. The mixture was quenched with H.sub.2O (10 mL) and then extracted with EtOAc (3×20 mL). The combined organic layers were washed with brine (2×10 mL), dried with Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give 2-{{[6-(2-methanesulfonylpropan-2-yl)pyridin-3-yl]oxy}ethyl methanesulfonate (Intermediate A-78). MS=338.1 [M+H].sup.+.

General Procedure for Intermediate A-79

##STR01192##

Step 1: 6-bromo-1-(3-oxocyclobutyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR01193##

[2626] To a solution of 6-bromo-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (250 mg, 1.10 mmol) in DMF (5 mL) under N.sub.2 atmosphere was added K.sub.2CO.sub.3 (380 mg, 2.75 mmol). The mixture was stirred at 50° C. for 4 h, and then a solution of 3-bromocyclobutanone (197 mg, 1.32 mmol) in DMF (5 mL) was added to the reaction mixture. The mixture was stirred at 50° C. for an additional 1 h. After cooling to room temperature, the mixture was concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-50% EtOAc:petroleum ether) to give 6-bromo-1-(3-oxocyclobutyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one. MS=295.1/297.1 [M+H].sup.+.

Step 2: 1-(3-oxocyclobutyl)-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR01194##

[2627] A mixture of 6-bromo-1-(3-oxocyclobutyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (1.20 g, 4.07 mmol), bis(pinacolato)diboron (1.24 g, 4.88 mmol), KOAc (798 mg, 8.13 mmol) and Pd(dppf)Cl.sub.2 (149 mg, 0.203 mmol) in 1,4-dioxane (60 mL) was degassed and purged with N.sub.2 (3×). The mixture was stirred at 80° C. for 12 h under N.sub.2 atmosphere. After cooling to room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo to

provide 1-(3-oxocyclobutyl)-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one, which was used into the next step without further purification. MS=343.2 [M+H].sup.+.

Step 3: 6-hydroxy-1-(3-oxocyclobutyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR01195##

[2628] To a 0° C. solution of 1-(3-oxocyclobutyl)-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (1.00 g, 2.92 mmol) in acetone (30 mL) and H.sub.2O (30 mL) was added Oxone (3.59 g, 5.84 mmol). The mixture was stirred at room temperature for 2 h. The mixture was quenched with H.sub.2O (20 mL) and extracted with EtOAc (3×20 mL). The combined organic layers were washed with brine (30 mL), dried with Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-100% EtOAc:petroleum ether) to give 6-hydroxy-1-(3-oxocyclobutyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one. MS=233.2 [M+H].sup.+.

Step 4: 6-hydroxy-1-[(cis)-3-hydroxycyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR01196##

[2629] To a 0° C. solution of 6-hydroxy-1-(3-oxocyclobutyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (500 mg, 2.15 mmol) in THE (15 mL) under N.sub.2 atmosphere was added NaBH.sub.4 (163 mg, 4.31 mmol). The mixture was stirred at room temperature for 2 h, and then was quenched with saturated aqueous NaHCO.sub.3 solution (10 mL). The aqueous phase was extracted with EtOAc (3×20 mL). The combined organic layers were washed with brine (2×30 mL), dried with Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-50% EtOAc:petroleum ether) to give 6-hydroxy-1-[(cis)-3-hydroxycyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one. MS=235.2 [M+H].sup.+.

Step 5: 6-(2-bromoethoxy)-1-[(cis)-3-hydroxycyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR01197##

[2630] To a solution of 6-hydroxy-1-[(cis)-3-hydroxycyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (300 mg, 1.28 mmol) in MeCN (4 mL) was added K.sub.2CO.sub.3 (531 mg, 3.84 mmol) and 1,2-dibromoethane (3.86 mL, 51.23 mmol). The mixture was stirred at 80° C. for 12 h. After cooling to room temperature, the reaction mixture was concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-50% EtOAc:petroleum ether) to give 6-(2-bromoethoxy)-1-[(cis)-3-hydroxycyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (Intermediate A-79). MS=341.1/343.1 [M+H].sup.+.

General Procedure for Intermediate A-80

##STR01198##

Step 1: 5-(2-bromoethoxy)-1H-indazole

##STR01199##

[2631] A solution of 1H-indazol-5-ol (1.00 g, 7.46 mmol), 1,2-dibromoethane (11.3 mL, 149 mmol), and K.sub.2CO.sub.3 (1.24 g, 8.95 mmol) in DMF (10 mL) was stirred at 80° C. for 16 h. After cooling to room temperature, the mixture was diluted with H.sub.2O (10 mL), then extracted with EtOAc (3×10 mL). The combined organic layers were washed with brine (30 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 10 g cartridge, 0-100% EtOAc:petroleum ether) to give 5-(2-bromoethoxy)-1H-indazole (Intermediate A-80). MS=241.1/243.1 [M+H].sup.+.

General Procedure for Intermediate A-81

##STR01200##

Step 1: 6-hydroxy-N-methyl-1,2,3,4-tetrahydroisoquinoline-2-carboxamide

##STR01201##

[2632] To a solution of 1,2,3,4-tetrahydroisoquinolin-6-ol (500 mg, 3.35 mmol) in DCM (7 mL) was added TEA (847 mg, 8.38 mmol) followed by the dropwise addition of N-methylcarbamoyl chloride (783 mg, 8.38 mmol). The mixture was stirred at room temperature for 12 h. The reaction mixture was diluted with H₂O (10 mL) and extracted with EtOAc (3×10 mL). The combined organic layers were washed with brine (20 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 4 g cartridge, 0-100% EtOAc:petroleum ether) to give 6-hydroxy-N-methyl-1,2,3,4-tetrahydroisoquinoline-2-carboxamide (Intermediate A-81). MS=207.2 [M+H]⁺.

General Procedure for Intermediate A-82

##STR01202##

Step 1: 4-methoxy-N1-methylbenzene-1,2-diamine

##STR01203##

[2633] To a solution of 4-methoxy-N-methyl-2-nitroaniline (500 mg, 2.74 mmol) in MeOH (10 mL) under N₂ atmosphere was added Pd/C (150 mg, 10 wt %, 0.141 mmol). The suspension was degassed under vacuum and purged with H₂ (3×). After stirring under an atmosphere of H₂ (15 psi) at room temperature for 12 h, solids were removed by filtration through Celite and the filtrate was concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 20 g cartridge, 0-50% EtOAc:petroleum ether) to give 4-methoxy-N1-methylbenzene-1,2-diamine. MS=153.2 [M+H]⁺.

Step 2: 5-methoxy-1-methyl-2,3-dihydro-1H-1,3-benzodiazol-2-one

##STR01204##

[2634] To a solution of 4-methoxy-N1-methylbenzene-1,2-diamine (150 mg, 0.986 mmol) in THF (20 mL) were added TEA (99.0 mg, 0.986 mmol) and CDI (799 mg, 4.93 mmol). The mixture was stirred at 80° C. for 12 h under N₂ atmosphere. After cooling to room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-100% EtOAc:petroleum ether) to give 5-methoxy-1-methyl-2,3-dihydro-1H-1,3-benzodiazol-2-one. ¹H NMR (400 MHz, DMSO-d₆): δ 10.72 (s, 1H), 6.96 (d, J=8.4 Hz, 1H), 6.61-6.56 (m, 2H), 3.71 (s, 3H), 3.23 (s, 3H).

Step 3: 5-hydroxy-1-methyl-2,3-dihydro-1H-1,3-benzodiazol-2-one

##STR01205##

[2635] To a 0° C. solution of 5-methoxy-1-methyl-2,3-dihydro-1H-1,3-benzodiazol-2-one (216 mg, 1.21 mmol) in DCM (5 mL) was added BBr₃ (1.52 g, 6.06 mmol) dropwise. The mixture was stirred at 40° C. for 4 h. After cooling to room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 4 g cartridge, 0-100% EtOAc:petroleum ether) to give 5-methoxy-1-methyl-2,3-dihydro-1H-1,3-benzodiazol-2-one.

Step 4: 5-(2-bromoethoxy)-1-methyl-2,3-dihydro-1H-1,3-benzodiazol-2-one

##STR01206##

[2636] A mixture of 5-methoxy-1-methyl-2,3-dihydro-1H-1,3-benzodiazol-2-one (100 mg, 0.609 mmol) and Cs₂CO₃ (595 mg, 1.83 mmol) in 1,2-dibromoethane (5 mL, 66 mmol) under N₂ atmosphere was stirred at 80° C. for 12 h. After cooling to room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 4 g cartridge, 0-100% EtOAc:petroleum ether) to give 5-(2-bromoethoxy)-1-methyl-2,3-dihydro-1H-1,3-benzodiazol-2-one (Intermediate A-82). MS=271.0/273.0 [M+H]⁺.

General Procedure for Intermediate A-86

##STR01207##

Step 1: 5-bromo-2-{1-[(tert-butyldimethylsilyl)oxy]ethenyl}pyrimidine

##STR01208##

[2637] To a solution of 1-(5-bromopyrimidin-2-yl)ethan-1-one (500 mg, 2.49 mmol) in DCM (10 mL) were added TEA (378 mg, 3.73 mmol) and TBSOTf (988 mg, 3.73 mmol). The mixture was stirred at 0° C. for 0.5 h. The reaction mixture was quenched with H.sub.2O (10 mL) and then extracted with DCM (3×10 mL). The combined organic layers were washed with brine (20 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-20% EtOAc:Petroleum ether) to give 5-bromo-2-{1-[(tert-butyldimethylsilyl)oxy]ethenyl}pyrimidine. ¹H NMR (400 MHz, DMSO-d.sub.6): δ 8.99 (s, 2H), 5.80 (s, 1H), 4.89 (s, 1H), 0.96 (s, 9H), 0.16 (s, 6H).

Step 2: 5-bromo-2-{1-[(tert-butyldimethylsilyl)oxy]ethyl}pyrimidine

##STR01209##

[2638] To a solution of 5-bromo-2-{1-[(tert-butyldimethylsilyl)oxy]ethenyl}pyrimidine (300 mg, 0.952 mmol) in MeOH (30 mL) under N.sub.2 atmosphere was added Rh(PPh.sub.3).sub.3Cl (178 mg, 0.190 mmol). The suspension was degassed under vacuum and purged with H.sub.2 several times. The mixture was stirred under H.sub.2 (50 psi) at 50° C. for 4 h. After cooling to room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-10% EtOAc:Petroleum ether) to give 5-bromo-2-{1-[(tert-butyldimethylsilyl)oxy]ethyl}pyrimidine. MS=317.1/319.1 [M+H].sup.+.

Step 3: 2-{1-[(tert-butyldimethylsilyl)oxy]ethyl}-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyrimidine

##STR01210##

[2639] A mixture of 5-bromo-2-{1-[(tert-butyldimethylsilyl)oxy]ethyl}pyrimidine (0.25 g, 0.788 mmol), bis(pinacolato)diboron (240 mg, 0.945 mmol), KOAc (232 mg, 2.36 mmol), and Pd(dppf)Cl.sub.2 (58 mg, 78.8 μmol) in 1,4-dioxane (15 mL) was degassed and purged with N.sub.2 (3×). The mixture was stirred at 90° C. for 5 h under N.sub.2 atmosphere. After cooling to room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo to give 2-{1-[(tert-butyldimethylsilyl)oxy]ethyl}-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyrimidine, which was used in the subsequent step without further purification.

Step 4: 2-{1-[(tert-butyldimethylsilyl)oxy]ethyl}pyrimidin-5-ol

##STR01211##

[2640] To a 0° C. solution of 2-{1-[(tert-butyldimethylsilyl)oxy]ethyl}-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyrimidine (300 mg, 0.823 mmol) in MeCN (10 mL) and H.sub.2O (10 mL) was added Oxone (607 mg, 0.988 mmol). The mixture was stirred at 0° C. for 1 h, then was quenched with saturated aqueous Na.sub.2SO.sub.3 (5 mL) and extracted with DCM (3×10 mL). The combined organic layers were washed with brine (15 mL), saturated aqueous Na.sub.2SO.sub.3 (15 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-100% EtOAc:Petroleum ether) to give 2-{1-[(tert-butyldimethylsilyl)oxy]ethyl}pyrimidin-5-ol. MS=255.2 [M+H].sup.+.

Step 5: 5-(2-bromoethoxy)-2-{1-[(tert-butyldimethylsilyl)oxy]ethyl}pyrimidine

##STR01212##

[2641] To a solution of 2-{1-[(tert-butyldimethylsilyl)oxy]ethyl}pyrimidin-5-ol (80.0 mg, 0.314 mmol) in MeCN (3 mL) was added K.sub.2CO.sub.3 (218 mg, 1.57 mmol) and 1,2-dibromoethane (2.36 g, 12.6 mmol). The mixture was stirred at 65° C. for 16 h. After cooling to room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 4 g cartridge, 0-30% EtOAc:Petroleum ether) to give 5-(2-bromoethoxy)-2-{1-[(tert-butyldimethylsilyl)oxy]ethyl}pyrimidine. 5-(2-bromoethoxy)-2-(1-((tert-butyldimethylsilyl)oxy)ethyl)pyrimidine. MS=361.1/363.1 [M+H].sup.+.

Step 6: 1-[5-(2-bromoethoxy)pyrimidin-2-yl]ethan-1-ol

##STR01213##

[2642] To a solution of 5-(2-bromoethoxy)-2-(1-((tert-butyldimethylsilyl)oxy)ethyl)pyrimidine (130 mg, 0.360 mmol) in EtOAc (3 mL) was added 4.0 M HCl in EtOAc (10 mL, 40.0 mmol). The mixture was stirred at room temperature for 5 h. The mixture was concentrated in vacuo to give 1-[5-(2-bromoethoxy)pyrimidin-2-yl]ethan-1-ol (Intermediate A-86), which was used in the subsequent steps without further purification. MS=247.0/249.0 [M+H].sup.+.

General Procedure for Intermediate A-87

##STR01214##

Step 1: methyl 2-(5-bromo-3-fluoropyridin-2-yl)-2-methanesulfonylacetate

##STR01215##

[2643] To a solution of 5-bromo-2,3-difluoropyridine (5.00 g, 25.8 mmol) in DMSO (55 mL) was added Cs.sub.2CO.sub.3 (16.8 g, 51.6 mmol) and methyl 2-methylsulfonylacetate (4.71 g, 30.9 mmol). The mixture was stirred at 55° C. for 30 h. After cooling to room temperature, the reaction mixture was quenched with H.sub.2O (50 mL) and extracted with EtOAc (3×30 mL). The combined organic layers were washed with brine (10 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 40 g cartridge, 0-25% EtOAc:Petroleum ether) to give methyl 2-(5-bromo-3-fluoropyridin-2-yl)-2-methanesulfonylacetate. MS=325.8/327.8 [M+H].sup.+.

Step 2: 5-bromo-3-fluoro-2-(methanesulfonylmethyl)pyridine

##STR01216##

[2644] To a solution of methyl 2-(5-bromo-3-fluoropyridin-2-yl)-2-methanesulfonylacetate (5.30 g, 16.3 mmol) in H.sub.2O (26 mL) and AcOH (26 mL) was added H.sub.2SO.sub.4 (5.10 mL, 95.8 mmol). The mixture was stirred at 80° C. for 15 h. After cooling to room temperature, the reaction mixture was adjusted to pH=4.0 with saturated aqueous Na.sub.2CO.sub.3. The mixture was extracted with EtOAc (3×25 mL), the combined organic layers were washed with brine (10 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give crude 5-bromo-3-fluoro-2-(methanesulfonylmethyl)pyridine, which was used in the subsequent step without further purification. MS=267.8/269.8 [M+H].sup.+.

Step 3: 5-bromo-3-fluoro-2-(1-methanesulfonylcyclopropyl)pyridine

##STR01217##

[2645] To a solution of 5-bromo-3-fluoro-2-(methanesulfonylmethyl)pyridine (3.30 g, 12.3 mmol) in THE (35 mL) were added KOH (3.45 g, 61.5 mmol), 1,2-dibromoethane (13.9 g, 73.9 mmol) and TBAB (793 mg, 2.46 mmol). The mixture was stirred at 50° C. for 24 h. After cooling to room temperature, the reaction mixture was quenched with H.sub.2O (30 mL) and extracted with EtOAc (3×25 mL). The combined organic layers were washed with brine (10 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 20 g cartridge, 0-30% EtOAc:Petroleum ether) to give 5-bromo-3-fluoro-2-(1-methanesulfonylcyclopropyl)pyridine. MS=293.9/295.9 [M+H].sup.+.

Step 4: 3-fluoro-2-(1-methanesulfonylcyclopropyl)-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyridine

##STR01218##

[2646] To a solution of 5-bromo-3-fluoro-2-(1-methanesulfonylcyclopropyl)pyridine (500 mg, 1.70 mmol) in 1,4-dioxane (6 mL) under N.sub.2 atmosphere was added bis(pinacolato)diboron (648 mg, 2.55 mmol), KOAc (334 mg, 3.40 mmol) and Pd(dppf)Cl.sub.2 (125 mg, 0.170 mmol). The mixture was stirred at 90° C. for 15 h under N.sub.2 atmosphere. After cooling to room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 20 g cartridge, 0-55% EtOAc:Petroleum ether) to give 3-fluoro-2-(1-methanesulfonylcyclopropyl)-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyridine.

Step 5: 5-fluoro-6-(1-methanesulfonylcyclopropyl)pyridin-3-ol

##STR01219##

[2647] To a solution of 3-fluoro-2-(1-methanesulfonylcyclopropyl)-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyridine (400 mg, 1.17 mmol) in H.sub.2O (3 mL) and THE (3 mL) was added Oxone (721 mg, 1.17 mmol). The mixture was stirred at room temperature for 2 h. The reaction mixture was diluted with H.sub.2O (15 mL) and extracted with EtOAc (3×15 mL). The combined organic layers were washed with brine (20 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-70% EtOAc:Petroleum ether) to give 5-fluoro-6-(1-methanesulfonylcyclopropyl)pyridin-3-ol. MS=232.0 [M+H].sup.+.

Step 6: 5-(2-bromoethoxy)-3-fluoro-2-(1-methanesulfonylcyclopropyl)pyridine

##STR01220##

[2648] To a solution of 5-fluoro-6-(1-methanesulfonylcyclopropyl)pyridin-3-ol (140 mg, 0.605 mmol) in MeCN (2 mL) were added 1,2-dibromoethane (4.55 g, 24.2 mmol) and K.sub.2CO.sub.3 (419 mg, 3.03 mmol). The mixture was stirred at 65° C. for 15 h. After cooling to room temperature, the reaction mixture was quenched with H.sub.2O (10 mL) and extracted with EtOAc (3×10 mL). The combined organic layers were washed with brine (20 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 4 g cartridge, 0-40% EtOAc:Petroleum ether) to give 5-(2-bromoethoxy)-3-fluoro-2-(1-methanesulfonylcyclopropyl)pyridine (Intermediate A-87).

MS=337.9/339.9 [M+H].sup.+.

General Procedure for Intermediate A-88

##STR01221##

Step 1: 3-(methylsulfanyl)oxetane-3-carbonitrile

##STR01222##

[2649] To a -78° C. solution of oxetane-3-carbonitrile (10.0 g, 12.4 mmol) and (methyldisulfanyl)methane (21.6 mL, 241 mmol) in THF (100 mL) under N.sub.2 atmosphere was added 2.0 M LDA in THF (151 mL, 302 mmol) dropwise. The mixture was warmed to room temperature and stirred for 3 h. The reaction mixture was cooled to 0° C. and then quenched with H.sub.2O (200 mL) and extracted with EtOAc (3×100 mL). The combined organic layers were washed with brine (20 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 120 g cartridge, 0-25% EtOAc:Petroleum ether) to give 3-(methylsulfanyl)oxetane-3-carbonitrile. .sup.1H NMR (400 MHz, CDCl.sub.3): δ 5.13 (d, J=6.8 Hz, 2H), 4.64 (d, J=6.8 Hz, 2H), 2.31 (s, 3H).

Step 2: 3-(methylsulfanyl)oxetane-3-carboximidamide

##STR01223##

[2650] To a 0° C. solution of 3-(methylsulfanyl)oxetane-3-carbonitrile (2.50 g, 19.4 mmol) in THE (30 mL) under N.sub.2 atmosphere was added 1.0 M LiHMDS in THE (96.8 mL, 96.8 mmol). The mixture was allowed to warm to room temperature and stirred for 15 h, then was quenched with AcOH (40 mL). The resulting solids were isolated by filtration and dried in vacuo to give 3-(methylsulfanyl)oxetane-3-carboximidamide, which was used in the subsequent step without further purification. MS=147.2 [M+H].sup.+.

Step 3: 5-bromo-2-[3-(methylsulfanyl)oxetan-3-yl]pyrimidine

##STR01224##

[2651] A mixture of 3-(methylsulfanyl)oxetane-3-carboximidamide (2.50 g, 17.1 mmol) and 2-bromopropanedial (5.16 g, 34.2 mmol) in AcOH (50 mL) was stirred at 80° C. for 8 h. After cooling to room temperature, the reaction mixture was quenched with H.sub.2O (50 mL), and then extracted with EtOAc (2×50 mL). The combined organic layers were washed with saturated aqueous Na.sub.2CO.sub.3 (2×50 mL) and brine (50 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 20 g cartridge, 0-20% EtOAc:Petroleum ether) to give 5-bromo-2-[3-(methylsulfanyl)oxetan-3-yl]pyrimidine. MS=261.1/263.1 [M+H].sup.+.

Step 4: 5-bromo-2-(3-methanesulfonyloxetan-3-yl)pyrimidine

##STR01225##

[2652] To a solution of 5-bromo-2-[3-(methylsulfonyl)oxetan-3-yl]pyrimidine (150 mg, 0.573 mmol) in DCM (15 mL) was added m-CPBA (350 mg, 1.72 mmol, 85% purity). The mixture was stirred at room temperature for 2 h. The reaction mixture was quenched with saturated aqueous Na.sub.2SO.sub.3 (20 mL), diluted with H.sub.2O (10 mL), and extracted with DCM (3×15 mL). The combined organic layers were washed with saturated aqueous NaHCO.sub.3 (2×20 mL) and brine (20 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-50% EtOAc:Petroleum ether) to give 5-bromo-2-(3-methanesulfonyloxetan-3-yl)pyrimidine. MS=293.0/295.0 [M+H].sup.+.

Step 5: 2-(3-methanesulfonyloxetan-3-yl)-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyrimidine

##STR01226##

[2653] A mixture of 5-bromo-2-(3-methanesulfonyloxetan-3-yl)pyrimidine (140 mg, 0.476 mmol), bis(pinacolato)diboron (182 mg, 0.718 mmol), KOAc (141 mg, 1.43 mmol), and Pd(dppf)Cl.sub.2 (35 mg, 47.8 µmol) in 1,4-dioxane (15 mL) was degassed and purged with N.sub.2 (3×). The mixture was stirred at 90° C. for 15 h under N.sub.2 atmosphere. After cooling to room temperature, solids were removed by filtration, and the filtrate was concentrated in vacuo to give 2-(3-methanesulfonyloxetan-3-yl)-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyrimidine, which was used in the subsequent step without further purification. MS=341.2 [M+H].sup.+

Step 6: 2-(3-methanesulfonyloxetan-3-yl)pyrimidin-5-ol

##STR01227##

[2654] To a 0° C. solution of 2-(3-methanesulfonyloxetan-3-yl)-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyrimidine (220 mg, 0.646 mmol) in DCM (20 mL) were added K.sub.2CO.sub.3 (179 mg, 0.118 mmol) and 30% H.sub.2O.sub.2 in H.sub.2O (0.25 mL, 2.59 mmol). The mixture was stirred at room temperature for 2 h, then was quenched with saturated aqueous Na.sub.2S.sub.2O.sub.3 (5 mL). The mixture was extracted with DCM (2×20 mL), the combined organic layers were washed with brine (20 mL), dried over Na.sub.2SO.sub.4, filtered and concentrated in vacuo give 2-(3-methanesulfonyloxetan-3-yl)pyrimidin-5-ol, which was used in the subsequent step without further purification. MS=229.0 [M-H].sup.+.

Step 7: 5-(2-bromoethoxy)-2-(3-methanesulfonyloxetan-3-yl)pyrimidine

##STR01228##

[2655] To a solution of 2-(3-methanesulfonyloxetan-3-yl)pyrimidin-5-ol (220 mg, 0.957 mmol) in MeCN (10 mL) was added K.sub.2CO.sub.3 (660 mg, 4.78 mmol) and 1,2-dibromoethane (7.18 g, 37.5 mmol). The mixture was stirred at 80° C. for 2 h. After cooling to room temperature, solids were removed by filtration and the filtrate was concentrated under reduced pressure. The crude residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-60% EtOAc:Petroleum ether) to give 5-(2-bromoethoxy)-2-(3-methanesulfonyloxetan-3-yl)pyrimidine (Intermediate A-88). MS=336.9/338.9 [M+H].sup.+

General Procedure for Intermediate A-89

##STR01229##

Step 1: 6-bromo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR01230##

[2656] To a 0° C. solution of 6-bromo-1-(3-oxocyclobutyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (Procedure for Intermediate A-79, Step 1, 1.00 g, 3.39 mmol) in THE (20 mL) was added 3.0 M MeMgBr in Et.sub.2O (1.24 mL, 3.72 mmol). The mixture was stirred at 0° C. for 2 h, and then was quenched with saturated aqueous NH.sub.4Cl solution (100 mL). The biphasic mixture was extracted with EtOAc (3×50 mL), and the combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by flash

silica gel chromatography (Biotage 12 g cartridge, 0-48% EtOAc:Petroleum ether) to give 6-bromo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one. ¹H NMR (400 MHz, DMSO-d₆): δ 8.30 (d, J=2.4 Hz, 1H), 7.90 (d, J=2.4 Hz, 1H), 4.83 (s, 1H), 4.20-4.15 (m, 1H), 2.82 (t, J=7.2 Hz, 2H), 2.58-2.52 (m, 2H), 2.49-2.42 (m, 2H), 2.33-2.25 (m, 2H), 1.25 (s, 3H). MS=311.0/313.0 [M+H]⁺.

Step 2: 1-[(cis)-3-hydroxy-3-methylcyclobutyl]-6-(4,4,5,5-tetramethyl-1,3,2 dioxaborolan-2-yl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR01231##

[2657] To a solution of 6-bromo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (1.00 g, 3.21 mmol) in 1,4-dioxane (20 mL) were added bis(pinacolato)diboron (1.06 g, 4.18 mmol), Pd(dppf)Cl₂·CH₂Cl₂ (210 mg, 257 μmol) and KOAc (788 mg, 8.03 mmol). The mixture was stirred at 90° C. for 12 h under N₂ atmosphere, and then cooled to room temperature. The mixture was diluted with H₂O (50 mL) and extracted with EtOAc (3×50 mL). The combined organic layers were washed with brine (50 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-58% EtOAc:Petroleum ether) to give 1-[(cis)-3-hydroxy-3-methylcyclobutyl]-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one. MS=359.1 [M+H]⁺.

Step 3: 6-hydroxy-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR01232##

[2658] To a solution of 1-[(cis)-3-hydroxy-3-methylcyclobutyl]-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (610 mg, 1.70 mmol) in acetone (10 mL) and H₂O (5 mL) was added Oxone (1.57 g, 2.55 mmol). The mixture was stirred for 2 h, and then was diluted with H₂O (10 mL) and extracted with EtOAc (20 mL×3). The combined organic layers were washed with brine (50 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-85% EtOAc:Petroleum ether) to give 6-hydroxy-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one. MS=249.0 [M+H]⁺.

Step 4: 6-(2-bromoethoxy)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR01233##

[2659] To a solution of 6-hydroxy-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (400 mg, 1.61 mmol) and 1,2-dibromoethane (1.82 mL, 24.2 mmol) in MeCN (10 mL) was added K₂CO₃ (1.11 g, 8.06 mmol). The mixture was stirred at 65° C. for 12 h. After cooling to room temperature, the mixture was diluted with H₂O (10 mL) and extracted with EtOAc (3×10 mL). The combined organic layers were washed with brine (20 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-70% EtOAc:Petroleum ether) to give 6-(2-bromoethoxy)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (Intermediate A-89). ¹H NMR (400 MHz, DMSO-d₆): δ 7.93 (d, J=2.8 Hz, 1H), 7.39 (d, J=2.8 Hz, 1H), 4.82 (s, 1H), 4.36 (t, J=5.6 Hz, 2H), 4.25-4.21 (m, 1H), 3.80 (t, J=5.6 Hz, 2H), 2.79 (t, J=6.8 Hz, 2H), 2.51-2.49 (m, 2H), 2.47-2.41 (m, 2H), 2.34-2.33 (m, 2H), 1.25 (s, 3H). MS=355.0/357.0 [M+H]⁺.

[2660] The following intermediates in Table 11.1 were prepared according to procedures similar those described for Intermediate A-89 using the appropriate starting materials.

General Procedure for Intermediates A-90 to A-91

TABLE-US-00013 TABLE 11.1 Intermediate # Structure Name Exact Mass [M + H]⁺ A-90

[01234]  6-(2-bromoethoxy)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one Calc'd 357.0/357.0 Found 357.1/357.1 A-91 [01235]

embedded image 6-(2-bromoethoxy)-3-[(cis)-3-hydroxy-3-methylcyclobutyl]-2H,3H-[1,3]oxazolo[4,5-b]pyridin-2-one Calc'd 343.0/343.0 Found 343.0/343.0

General Procedure for Intermediate A-92

##STR01236##

Step 1: 5-bromo-2-fluoro-1-nitro-3-(trifluoromethyl)benzene

##STR01237##

[2661] To concentrated H₂SO₄ (15 mL) at 0° C. was added a solution of 2-fluoro-1-nitro-3-(trifluoromethyl)benzene (5.00 g, 23.0 mmol) in TFA (10 mL). NBS (5.11 g, 28.7 mol) was added to the 0° C. mixture in several portions. The mixture was stirred at 60° C. for 16 h. The reaction mixture was cooled to room temperature, and then poured into ice water (200 mL). The mixture was extracted with petroleum ether (2×80 mL). The combined organic layers were washed with saturated aqueous NaHCO₃ until pH=8-9. The organic layer was separated, washed with brine (50 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo to give 5-bromo-2-fluoro-1-nitro-3-(trifluoromethyl)benzene. ¹H NMR (400 MHz, DMSO-d₆): δ 8.70 (dd, J=6.0 Hz, 2.4 Hz, 1H), 8.46 (dd, J=5.6 Hz, 2.4 Hz, 1H).

Step 2: (cis)-3-{[4-bromo-2-nitro-6-(trifluoromethyl)phenyl]amino}-1-methylcyclobutan-1-ol

##STR01238##

[2662] A mixture of 5-bromo-2-fluoro-1-nitro-3-(trifluoromethyl)benzene (6.50 g, 22.6 mmol), DIEA (11.2 mL, 67.7 mmol) and (cis)-3-amino-1-methylcyclobutan-1-ol (3.42 g, 24.8 mmol, HCl salt) in THE (35 mL) and MeCN (35 mL) was stirred at 50° C. for 2 h. The reaction mixture was cooled to room temperature and concentrated in vacuo. The residue was diluted with H₂O (150 mL) and extracted with MTBE (2×80 mL). The combined organic layers were washed with brine (80 mL), dried over Na₂SO₄, filtered and concentrated in vacuo to give (cis)-3-{[4-bromo-2-nitro-6-(trifluoromethyl)phenyl]amino}-1-methylcyclobutan-1-ol, which was used in the subsequent step without further purification. MS=368.9/370.9 [M+H]⁺.

Step 3: (cis)-3-{[2-amino-4-bromo-6-(trifluoromethyl)phenyl]amino}-1-methylcyclobutan-1-ol

##STR01239##

[2663] To a three-neck round-bottom flask equipped with a magnetic stir bar and a thermometer was added H₂O (170 mL) and sodium dithionite (32.1 g, 184 mmol). To the mixture was added a solution of (cis)-3-{[4-bromo-2-nitro-6-(trifluoromethyl)phenyl]amino}-1-methylcyclobutan-1-ol (17.0 g, 46.1 mmol) in MeOH (170 mL) dropwise. The mixture was stirred at room temperature for 1 h, and then 12 M aqueous HCl (35 mL) was added. The mixture was stirred at 60° C. for 1 h. After cooling to room temperature, the mixture was concentrated in vacuo to remove MeOH, and the residue was diluted with H₂O (150 mL). The aqueous layer was adjusted to pH>7 by addition of solid Na₂CO₃. The mixture was extracted with MTBE (2×200 mL). The combined organic layers were washed with brine (2×100 mL), dried over Na₂SO₄, filtered and concentrated in vacuo to give (cis)-3-{[2-amino-4-bromo-6-(trifluoromethyl)phenyl]amino}-1-methylcyclobutan-1-ol, which was used in the subsequent step without further purification. MS=339.0/341.0 [M+H]⁺.

Step 4: (cis)-3-[5-bromo-7-(trifluoromethyl)-1H-1,3-benzodiazol-1-yl]-1-methylcyclobutan-1-ol

##STR01240##

[2664] A mixture of (cis)-3-{[2-amino-4-bromo-6-(trifluoromethyl)phenyl]amino}-1-methylcyclobutan-1-ol (13.0 g, 38.3 mmol), trimethoxymethane (6.10 g, 57.5 mmol), and TsOH.Math.H₂O (729 mg, 3.83 mmol) in THE (130 mL) was stirred at 50° C. for 1 h. The mixture was cooled to room temperature and concentrated in vacuo to remove most of the THF. The residual solution was diluted with saturated aqueous NaHCO₃ solution (200 mL) and extracted with MTBE (2×100 mL). The combined organic layers were dried over Na₂SO₄, filtered, and concentrated in vacuo. The residue was triturated in MTBE (20 mL) for 30 min. The mixture was filtered to collect the solid to afford (cis)-3-[5-bromo-7-(trifluoromethyl)-1H-1,3-benzodiazol-1-yl]-1-methylcyclobutan-1-ol. MS=349.0/350.9

[M+H].sup.+.

Step 5: (cis)-3-[5-(5,5-dimethyl-1,3,2-dioxaborinan-2-yl)-7-(trifluoromethyl)-1H-1,3-benzodiazol-1-yl]-1-methylcyclobutan-1-ol

##STR01241##

[2665] A mixture of (cis)-3-[5-bromo-7-(trifluoromethyl)-1H-1,3-benzodiazol-1-yl]-1-methylcyclobutan-1-ol (6.00 g, 17.2 mmol), 2-(5,5-dimethyl-1,3,2-dioxaborinan-2-yl)-5,5-dimethyl-1,3,2-dioxaborinane (5.82 g, 25.8 mmol), KOAc (5.06 g, 51.6 mmol) and Pd(dppf)Cl.sub.2 (130 mg, 0.172 mmol) in 1,4-dioxane (60 mL) was purged with N.sub.2 (3×) and then stirred at 100° C. for 16 h. After cooling to room temperature, the reaction mixture was filtered and the filtrate was concentrated in vacuo to give (cis)-3-[5-(5,5-dimethyl-1,3,2-dioxaborinan-2-yl)-7-(trifluoromethyl)-1H-1,3-benzodiazol-1-yl]-1-methylcyclobutan-1-ol, which was used in the subsequent step without further purification.

Step 6: 1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-ol

##STR01242##

[2666] To a 0° C. solution of (cis)-3-[5-(5,5-dimethyl-1,3,2-dioxaborinan-2-yl)-7-(trifluoromethyl)-1H-1,3-benzodiazol-1-yl]-1-methylcyclobutan-1-ol (9.00 g, 23.6 mmol) in THE (50 mL) and H.sub.2O (50 mL) was added Oxone (14.5 g, 23.6 mmol). The mixture was stirred at room temperature for 2 h. The mixture was cooled to 0° C. and quenched by addition of saturated aqueous Na.sub.2SO.sub.3 (60 mL). The mixture was adjusted to pH=7 by addition of saturated aqueous NaHCO.sub.3 and extracted with EtOAc (2×60 mL). The combined organic phases were washed with brine (30 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was triturated with MTBE (20 mL) to give 1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-ol (Intermediate A-92), which was used in the subsequent step without further purification. MS=287.1 [M+H].sup.+.

General Procedure for Intermediate A-95

##STR01243##

Step 1: methyl 2-[5-bromo-3-(trifluoromethyl)pyridin-2-yl]-2-methanesulfonylacetate

##STR01244##

[2667] To a solution of 5-bromo-2-chloro-3-(trifluoromethyl)pyridine (7.50 g, 28.8 mmol) in DMF (70 mL) was added Cs.sub.2CO.sub.3 (23.5 g, 72.0 mmol) and methyl 2-methylsulfonylacetate (6.57 g, 43.2 mmol). The mixture was stirred at 120° C. for 2 h. After cooling to 0° C., the reaction mixture was quenched by addition of H.sub.2O (70 mL), and then extracted with EtOAc (5×50 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 220 g cartridge, 0-20% EtOAc/Petroleum ether) to give methyl 2-[5-bromo-3-(trifluoromethyl)pyridin-2-yl]-2-methanesulfonylacetate. MS=376.1/378.0 [M+H].sup.+.

Step 2: methyl 2-[5-bromo-3-(trifluoromethyl)pyridin-2-yl]-2-methanesulfonylpropanoate

##STR01245##

[2668] To a 0° C. solution of methyl 2-[5-bromo-3-(trifluoromethyl)pyridin-2-yl]-2-methanesulfonylacetate (14.0 g, 37.2 mmol) and K.sub.2CO.sub.3 (10.3 g, 74.4 mmol) in DMF (140 mL) was added MeI (10.6 g, 74.4 mmol) dropwise via syringe. The mixture was stirred at 40° C. for 4 h. The reaction mixture was quenched by addition of ice water (100 mL) and stirred below 10° C. for 10 min, and then extracted with EtOAc (3×150 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 220 g cartridge, 0-30% EtOAc/Petroleum ether) to give methyl 2-[5-bromo-3-(trifluoromethyl)pyridin-2-yl]-2-methanesulfonylpropanoate. MS=390.1/392.1 [M+H].sup.+.

Step 3: 5-bromo-2-(1-methanesulfonylethyl)-3-(trifluoromethyl)pyridine

##STR01246##

[2669] To a solution of methyl 2-[5-bromo-3-(trifluoromethyl)pyridin-2-yl]-2-

methanesulfonylpropanoate (13.0 g, 33.3 mmol) in H.sub.2O (20 mL) and MeOH (50 mL) was added a solution of NaOH (13.3 g, 333 mmol) in H.sub.2O (30 mL). The mixture was stirred at 75° C. for 6 h. After cooling to room temperature, the reaction mixture was filtered, and the filter cake was washed with H.sub.2O (3×30 mL). The filter cake was dried in vacuo to give 5-bromo-2-(1-methanesulfonylethyl)-3-(trifluoromethyl)pyridine, which was taken to the next step without further purification MS=332.1/334.1 [M+H].sup.+.

Step 4: 5-bromo-2-(1-ethoxy-2-methanesulfonylpropan-2-yl)-3-(trifluoromethyl)pyridine
##STR01247##

[2670] To a round-bottom flask equipped with a magnetic stir bar and thermometer under N.sub.2 atmosphere was added 5-bromo-2-(1-methanesulfonylethyl)-3-(trifluoromethyl)pyridine (7.20 g, 21.7 mmol) and DMA (420 mL). The mixture was cooled to 0° C. and NaH (3.48 g, 86.7 mmol, 60% in mineral oil) was added slowly. The resulting mixture was stirred for 3 h at room temperature under N.sub.2 atmosphere. Chloromethoxyethane (10.26 g, 108 mmol) was then slowly added to the reaction mixture. The reaction mixture was stirred at room temperature for 2 h, then was cooled to 10° C. and quenched by addition of saturated aqueous NH.sub.4Cl solution (150 mL). The mixture was extracted with EtOAc (3×100 mL), then the combined organic layers were washed with brine (150 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 80 g cartridge, 0-25% EtOAc/Petroleum ether) to give 5-bromo-2-(1-ethoxy-2-methanesulfonylpropan-2-yl)-3-(trifluoromethyl)pyridine. MS=389.9/391.9 [M+H].sup.+.

Step 5: 2-[5-bromo-3-(trifluoromethyl)pyridin-2-yl]-2-methanesulfonylpropan-1-ol
##STR01248##

[2671] To a round-bottom flask equipped with a magnetic stir bar and thermometer were added 5-bromo-2-(1-ethoxy-2-methanesulfonylpropan-2-yl)-3-(trifluoromethyl)pyridine (3.90 g, 9.99 mmol) and DCM (150 mL). The mixture was cooled to 0° C. and BBr.sub.3 (12.5 g, 50.0 mmol) was added dropwise via syringe. The resulting mixture was stirred for 6 h at room temperature. The reaction mixture was then cooled to 10° C. and quenched by addition of saturated aqueous NaHCO.sub.3 solution (150 mL) and then extracted with EtOAc (3×75 mL). The combined organic layers were washed with brine (150 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 40 g cartridge, 0-25% EtOAc/Petroleum ether) to give 2-[5-bromo-3-(trifluoromethyl)pyridine-2-yl]-2-methanesulfonylpropan-1-ol. MS=361.9/363.9 [M+H].sup.+.

Step 6: 2-methanesulfonyl-2-[5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-3-(trifluoromethyl)pyridine-2-yl]propan-1-ol
##STR01249##

[2672] A mixture of 2-[5-bromo-3-(trifluoromethyl)pyridine-2-yl]-2-methanesulfonylpropan-1-ol (1.20 g, 3.31 mmol), bis(pinacolato)diboron (1.01 g, 3.98 mmol), KOAc (650 mg, 6.63 mmol), and Pd(dppf)Cl.sub.2 (242 mg, 331 µmol) in 1,4-dioxane (16 mL) was degassed and purged with N.sub.2 (3×). The mixture was then stirred at 90° C. for 16 h under N.sub.2 atmosphere. After cooling to room temperature, the mixture was diluted with H.sub.2O (20 mL) and extracted with EtOAc (3×20 mL). The combined organic layer was dried over Na.sub.2SO.sub.4, filtered and concentrated in vacuo to give 2-methanesulfonyl-2-[5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-3-(trifluoromethyl)pyridin-2-yl]propan-1-ol, which was taken to the next step without further purification. MS=328.1 [M-C.sub.6H.sub.10+H].sup.+.

Step 7: 6-(1-hydroxy-2-methanesulfonylpropan-2-yl)-5-(trifluoromethyl)pyridin-3-ol
##STR01250##

[2673] To a solution of 2-methanesulfonyl-2-[5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-3-(trifluoromethyl)pyridin-2-yl]propan-1-ol (2.00 g, 4.88 mmol) in acetone (15 mL) and H.sub.2O (15 mL) was added Oxone (6.00 g, 9.78 mmol). The mixture was stirred at room temperature for 2 h. The reaction mixture was cooled to 0° C., quenched by addition of saturated aqueous

Na.sub.2SO.sub.3 solution (50 mL), then stirred for 30 min. The mixture was diluted with H.sub.2O (20 mL) and extracted with EtOAc (3×50 mL). The combined organic layer was dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-50% EtOAc/Petroleum ether) to give 6-(1-hydroxy-2-methanesulfonylpropan-2-yl)-5-(trifluoromethyl)pyridin-3-ol (Intermediate A-95). MS=300.0 [M+H].sup.+.

General Procedure for Intermediate A-96

##STR01251##

Step 1: 5-bromo-2-(chloromethyl)pyrimidine

##STR01252##

[2674] To a solution of (5-bromopyrimidin-2-yl)methanol (4.00 g, 21.2 mmol) in DCM (200 mL) was added SOCl.sub.2 (20 mL) slowly via syringe. The mixture was stirred at room temperature for 3 h. The reaction mixture was concentrated under reduced pressure and the resulting residue was quenched with H.sub.2O (80 mL). The mixture was adjusted to pH=8 with saturated aqueous NaHCO.sub.3 and extracted with EtOAc (3×100 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give 5-bromo-2-(chloromethyl)pyrimidine, which was taken to the next step without further purification. .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 9.05 (s, 2H), 4.81 (s, 2H).

Step 2: 5-bromo-2-[(dimethylphosphoryl)methyl]pyrimidine

##STR01253##

[2675] To a 250 mL three-necked round bottom flask equipped with magnetic stirrer, addition funnel, and thermometer under N.sub.2 atmosphere were added dimethylphosphine oxide (1.66 g, 21.2 mmol) and THE (140 mL). To the mixture was added 1 M NaHMDS in THE (23.1 mL, 23.1 mmol) portionwise while keeping the temperature of the reaction mixture between 15-20° C. The mixture was stirred at 15° C. for 1 h under N.sub.2, and then a solution of 5-bromo-2-(chloromethyl)pyrimidine (4.00 g, 19.3 mmol) in THF (30 mL) was added dropwise. The resulting mixture was heated to 60° C. and stirred for 15 h under N.sub.2 atmosphere. The reaction mixture was cooled to 10° C. and quenched by addition of saturated NH.sub.4Cl solution (100 mL) and stirred for 10 min, and then extracted with EtOAc (3×80 mL). The combined organic layers were washed with brine (100 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by reverse phase preparative HPLC (Welch Xtimate Cis column, 0-20% MeCN:10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 5-bromo-2-[(dimethylphosphoryl)methyl]pyrimidine. MS=248.9/250.9 [M+H].sup.+.

Step 3: 5-bromo-2-[1-(dimethylphosphoryl)ethenyl]pyrimidine

##STR01254##

[2676] To a solution of 5-bromo-2-(dimethylphosphorylmethyl)pyrimidine (1.00 g, 4.02 mmol) and N,N,N',N'-tetramethylmethanediamine (821 mg, 8.03 mmol) in DMF (8 mL) was added Ac.sub.2O (1.64 g, 16.1 mmol). The mixture was stirred at 60° C. for 2 h. After cooling to room temperature, the reaction was filtered. The residue was purified by reverse phase preparative HPLC (Phenomenex Luna C.sub.18 column, 5-35% MeCN:10 mM TFA in H.sub.2O) to give 5-bromo-2-[1-(dimethylphosphoryl)ethenyl]pyrimidine. MS=260.9/262.9 [M+H].sup.+.

Step 4: 5-bromo-2-[1-(dimethylphosphoryl)cyclopropyl]pyrimidine

##STR01255##

[2677] To a solution of trimethylsulfoxonium iodide (2.02 g, 9.19 mmol) in DMSO (6 mL) was added t-BuOK (904 mg, 8.04 mmol). The mixture was stirred at 50° C. for 30 min, and then a solution of 5-bromo-2-[1-(dimethylphosphoryl)ethenyl]pyrimidine (600 mg, 2.30 mmol) in DMSO (1 mL) was added dropwise. The mixture was stirred at 50° C. for 30 min. After cooling to room temperature, the reaction was filtered, and the filtrate was concentrated in vacuo. The residue was purified by reverse phase preparative HPLC (Phenomenex Luna C.sub.18 column, 1-40% MeCN:10 mM TFA in H.sub.2O) to give 5-bromo-2-[1-

(dimethylphosphoryl)cyclopropyl]pyrimidine. MS=274.9/276.9 [M+H].sup.+.

Step 5: 2-[1-(dimethylphosphoryl)cyclopropyl]pyrimidin-5-ol

##STR01256##

[2678] A mixture of 5-bromo-2-[1-(dimethylphosphoryl)cyclopropyl]pyrimidine (400 mg, 1.45 mmol), di-tert-butyl-[2,3,4,5-tetramethyl-6-(2,4,6-triisopropylphenyl)phenyl]phosphane (69.9 mg, 145 µmol), Pd.sub.2(dba).sub.3 (26.6 mg, 29.1 µmol) and KOH (245 mg, 4.36 mmol) in 1,4-dioxane (5 mL) and H.sub.2O (2 mL) was degassed and purged with N.sub.2 (3×). The mixture was stirred at 100° C. for 16 h under N.sub.2 atmosphere. After cooling to room temperature, the reaction mixture was filtered, and the filtrate was concentrated in vacuo. The residue was purified by reverse phase preparative HPLC (Phenomenex Luna C.sub.18 column, 1-25% MeCN:10 mM TFA in H.sub.2O) to give 2-[1-(dimethylphosphoryl)cyclopropyl]pyrimidin-5-ol (Intermediate A-96). MS=213.2 [M+H].sup.+.

General Procedure for Intermediate A-97

##STR01257##

Step 1: 5-bromo-2-vinylpyrimidine

##STR01258##

[2679] A mixture of 5-bromo-2-iodo-pyrimidine (5.00 g, 17.6 mmol), potassium vinyltrifluoroborate (2.59 g, 19.3 mmol), Pd(dppf)Cl.sub.2 (1.28 g, 1.76 mmol), and K.sub.2CO.sub.3 (6.06 g, 43.9 mmol) in 1,4-dioxane (75 mL) and H.sub.2O (25 mL) was degassed and purged with N.sub.2 (3×). The mixture was stirred at 90° C. for 12 h under N.sub.2 atmosphere. After cooling to room temperature, the reaction mixture was diluted with H.sub.2O (60 mL) and extracted with EtOAc (3×20 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 40 g cartridge, 0-15% EtOAc/Petroleum ether) to give 5-bromo-2-vinylpyrimidine. MS=185.1/187.1 [M+H].sup.+.

Step 2: 1-(5-bromopyrimidin-2-yl)ethane-1,2-diol

##STR01259##

[2680] To a 0° C. solution of 5-bromo-2-vinylpyrimidine (1.80 g, 9.73 mmol) in THE (20 mL) and H.sub.2O (20 mL) was added K.sub.2OsO.sub.4.Math.2H.sub.2O (358 mg, 97 µmol) and NMO (2.28 g, 19.5 mmol). The mixture was stirred at room temperature for 12 h. The reaction mixture was diluted with H.sub.2O (30 mL) and extracted with EtOAc (3×20 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by reverse phase preparative HPLC (Welch Xtimate C.sub.18 column, 0-20% MeCN:10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 1-(5-bromopyrimidin-2-yl)ethane-1,2-diol. MS=219.1/221.1 [M+H].sup.+.

Step 3: 5-bromo-2-(2,2-dimethyl-1,3-dioxolan-4-yl)pyrimidine

##STR01260##

[2681] To a solution of 1-(5-bromopyrimidin-2-yl)ethane-1,2-diol (860 mg, 3.93 mmol) in acetone (9 mL) was added 2,2-dimethoxypropane (8.18 g, 78.5 mmol) and TsOH (67.6 mg, 0.393 mmol). The mixture was stirred at room temperature for 12 h. The reaction mixture was diluted with H.sub.2O (50 mL) and extracted with EtOAc (4×20 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-70% EtOAc/Petroleum ether) to give 5-bromo-2-(2,2-dimethyl-1,3-dioxolan-4-yl)pyrimidine. MS=259.0/261.0 [M+H].sup.+.

Step 4: 2-(2,2-dimethyl-1,3-dioxolan-4-yl)pyrimidin-5-ol

##STR01261##

[2682] A mixture of 5-bromo-2-(2,2-dimethyl-1,3-dioxolan-4-yl)pyrimidine (200 mg, 0.772 mmol), Pd.sub.2(dba).sub.3 (14.0 mg, 15.4 µmol), di-tert-butyl-[2,3,4,5-tetramethyl-6-(2,4,6-triisopropylphenyl)phenyl]phosphane (37.0 mg, 77.2 µmol) and KOH (130 mg, 2.32 mmol) in H.sub.2O (2 mL) and 1,4-dioxane (5 mL) was degassed and purged with N.sub.2 (3×). The mixture

was stirred at 100° C. for 16 h under N.sub.2 atmosphere. After cooling to room temperature, the reaction mixture was filtered and the filtrate was concentrated in vacuo to give 2-(2,2-dimethyl-1,3-dioxolan-4-yl)pyrimidin-5-ol (Intermediate A-97), which was taken to the next step without further purification. MS=197.1 [M+H].sup.+.

General Procedure for Intermediate A-98

##STR01262##

Step 1: (cis)-3-{6-bromo-3H-imidazo[4,5-b]pyridine-3-yl}-1-methylcyclobutan-1-ol

##STR01263##

[2683] To a solution of (cis)-3-[(3-amino-5-bromopyridin-2-yl)amino]-1-methylcyclobutan-1-ol (Intermediate A-93, 750 mg, 2.76 mmol) in CH(OMe).sub.3 (10 mL) was added

TsOH.Math.H.sub.2O (52.4 mg, 0.276 mmol). The mixture was stirred at 85° C. for 2 h. After cooling to room temperature, the mixture was diluted with H.sub.2O (30 mL) and extracted with EtOAc (2×20 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give (cis)-3-{6-bromo-3H-imidazo[4,5-b]pyridin-3-yl}-1-methylcyclobutan-1-ol, which was taken to the next step without further purification.

MS=282.0/284.0 [M+H].sup.+.

Step 2: (cis)-1-methyl-3-[6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-3H-imidazo[4,5-b]pyridin-3-yl]cyclobutan-1-ol

##STR01264##

[2684] To a solution of (cis)-3-{6-bromo-3H-imidazo[4,5-b]pyridin-3-yl}-1-methylcyclobutan-1-ol (500 mg, 1.77 mmol) in 1,4-dioxane (4 mL) were added bis(pinacolato)diboron (540 mg, 2.13 mmol), KOAc (348 mg, 3.54 mmol) and Pd(dppf)Cl.sub.2 (130 mg, 177 µmol). The mixture was degassed and purged with N.sub.2 (3×), then stirred at 80° C. for 16 h under N.sub.2 atmosphere.

After cooling to room temperature, the mixture was filtered and concentrated in vacuo to give (cis)-1-methyl-3-[6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-3H-imidazo[4,5-b]pyridin-3-yl]cyclobutan-1-ol, which was taken to the next step without further purification. MS=247.9

[M-C.sub.6H.sub.10+H].sup.+.

Step 3: 3-[(cis)-3-hydroxy-3-methylcyclobutyl]-3H-imidazo[4,5-b]pyridin-6-ol

##STR01265##

[2685] To a solution of (cis)-1-methyl-3-[6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-3H-imidazo[4,5-b]pyridin-3-yl]cyclobutan-1-ol (300 mg, 0.911 mmol) in THE (2 mL) and H.sub.2O (2 mL) was added Oxone (560 mg, 911 µmol). The mixture was stirred at room temperature for 2 h. The mixture was concentrated in vacuo to give 3-[(cis)-3-hydroxy-3-methylcyclobutyl]-3H-imidazo[4,5-b]pyridin-6-ol (Intermediate A-98), which was taken to the next step without further purification. MS=220.2 [M+H].sup.+.

General Procedure for Intermediate A-99

##STR01266##

Step 1: 1-[5-bromo-3-(trifluoromethyl)pyridin-2-yl]-3-methylazetidin-3-ol

##STR01267##

[2686] To a solution of 5-bromo-2-chloro-3-(trifluoromethyl)pyridine (1.00 g, 3.84 mmol) and 3-methylazetidin-3-ol (522 mg, 4.22 mmol, HCl salt) in DMF (10 mL) was added K.sub.2CO.sub.3 (1.86 g, 13.4 mmol). The mixture was stirred at 100° C. for 16 h. After cooling to room temperature, the reaction mixture was concentrated in vacuo. The residue was diluted with H.sub.2O (20 mL) and extracted with EtOAc (3×15 mL). The combined organic layers were washed with brine (10 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-50% EtOAc/Petroleum ether) to give 1-[5-bromo-3-(trifluoromethyl)pyridin-2-yl]-3-methylazetidin-3-ol. MS=311.0/313.0 [M+H].sup.+.

Step 2: 6-(3-hydroxy-3-methylazetidin-1-yl)-5-(trifluoromethyl)pyridin-3-ol

##STR01268##

[2687] A mixture of 1-[5-bromo-3-(trifluoromethyl)pyridin-2-yl]-3-methylazetidin-3-ol (900 mg, 2.89 mmol), Pd.sub.2(dba).sub.3 (53.0 mg, 57.9 μ mol), t-BuBrettphos (1.40 g, 2.89 mmol) and KOH (487 mg, 8.68 mmol) in 1,4-dioxane (15 mL) and H.sub.2O (6 mL) was degassed and purged with N.sub.2 (3 $\times$). The mixture was stirred at 100 $^{\circ}$ C. for 16 h under N.sub.2 atmosphere. The reaction mixture was allowed to cool to room temperature, then was concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-100% EtOAc/Petroleum ether) to give 6-(3-hydroxy-3-methylazetidin-1-yl)-5-(trifluoromethyl)pyridin-3-ol (Intermediate A-99). MS=249.2 [M+H].sup.+.

General Procedure for Intermediate A-100

##STR01269##

Step 1: ethyl 2-(5-bromopyrimidin-2-yl)-2-methylpropanoate

##STR01270##

[2688] To a three-neck round-bottom flask equipped with a magnetic stir bar and thermometer under N.sub.2 atmosphere was added ethyl 2-(5-bromopyrimidin-2-yl)acetate (2.00 g, 8.16 mmol) and THF (20 mL). The mixture was cooled to -78 $^{\circ}$ C. and 1.0 M LiHMDS in THF (8.98 mL, 8.98 mmol) was added dropwise. After stirring at -78 $^{\circ}$ C. for 1 h, MeI (2.42 g, 17.1 mmol) was added in one portion. The mixture was warmed up to room temperature and stirred for 2 h under N.sub.2 atmosphere. The reaction mixture was cooled to ~0-5 $^{\circ}$ C. and quenched by addition of saturated aqueous NH.sub.4Cl (40 mL), then diluted with H.sub.2O (20 mL) and extracted with EtOAc (3 $\times$ 25 mL). The combined organic layers were washed with brine (2 $\times$ 25 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give ethyl 2-(5-bromopyrimidin-2-yl)-2-methylpropanoate, which was taken to the next step without further purification. MS=273.2/275.2 [M+H].sup.+.

Step 2: 2-(5-bromopyrimidin-2-yl)-2-methylpropan-1-ol

##STR01271##

[2689] To a three-neck round-bottom flask equipped with a magnetic stir bar and thermometer under N.sub.2 atmosphere were added ethyl 2-(5-bromopyrimidin-2-yl)-2-methylpropanoate (2.60 g, 9.52 mmol) and DCM (30 mL). The mixture was cooled to 0 $^{\circ}$ C. and 1.0 M DIBAL-H in THE (19.0 mL, 19.0 mmol) was added dropwise. The mixture was stirred at 0 $^{\circ}$ C. for 3 h under N.sub.2 atmosphere. The reaction mixture was quenched by addition of H.sub.2O (60 mL), adjusted to pH=3-4 by dropwise addition of aqueous 3.0 M HCl, then extracted with DCM (3 $\times$ 30 mL). The combined organic layers were washed with brine (2 $\times$ 30 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 20 g cartridge, 20-60% EtOAc/Petroleum ether) to give 2-(5-bromopyrimidin-2-yl)-2-methylpropan-1-ol. MS=231.0/233.0 [M+H].sup.+.

Step 3: 2-methyl-2-[5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyrimidin-2-yl]propan-1-ol

##STR01272##

[2690] A mixture of 2-(5-bromopyrimidin-2-yl)-2-methylpropan-1-ol (800 mg, 3.46 mmol), bis(pinacolato)diboron (1.32 g, 5.19 mmol), KOAc (1.02 g, 10.4 mmol) and Pd(dppf)Cl.sub.2 (253 mg, 346 μ mol) in 1,4-dioxane (10 mL) was degassed and purged with N.sub.2 (3 $\times$), and then the mixture was stirred at 80 $^{\circ}$ C. for 16 h under N.sub.2 atmosphere. After cooling to room temperature, the mixture was filtered, the filtrate was concentrated in vacuo to give 2-methyl-2-[5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyrimidin-2-yl]propan-1-ol, which was taken to the next step without further purification. MS=197.2 [M-C.sub.6H.sub.10+H].sup.+.

Step 4: 2-(1-hydroxy-2-methylpropan-2-yl)pyrimidin-5-ol

##STR01273##

[2691] To a three-neck round-bottom flask equipped with a magnetic stir bar and thermometer under a N.sub.2 atmosphere were added 2-methyl-2-[5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyrimidin-2-yl]propan-1-ol (960 mg, 3.45 mmol), K.sub.2CO.sub.3 (1.19 g, 8.63 mmol) and DCM (10 mL). The mixture was cooled to 0 $^{\circ}$ C. and 30% H.sub.2O.sub.2 in H.sub.2O (1.16 mL,

12.1 mmol) was added in portions. The mixture was stirred at 0° C. for 2 h under N.sub.2 atmosphere. The reaction mixture was quenched by addition of saturated aqueous Na.sub.2SO.sub.3 solution (20 mL), and then the reaction mixture was concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Waters Xbridge OBD C.sub.18 column, 20% MeCN:10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 2-(1-hydroxy-2-methylpropan-2-yl)pyrimidin-5-ol (Intermediate A-100). MS=169.2 [M+H].sup.+.

General Procedure for Intermediate A-101

##STR01274##

Step 1: methyl 1-(hydroxymethyl)cyclopropanecarboximide

##STR01275##

[2692] HCl gas was bubbled into a solution of 1-(hydroxymethyl)cyclopropanecarbonitrile (4.00 g, 41.2 mmol) in MeOH (40 mL) at -78° C. for 20 min. The mixture was then stirred at room temperature for 15 h. The mixture was concentrated in vacuo to give methyl 1-(hydroxymethyl)cyclopropanecarboximide, which was taken to the next step without further purification. MS=130.0 [M+H].sup.+.

Step 2: 1-(hydroxymethyl)cyclopropanecarboxamidine

##STR01276##

[2693] NH.sub.3 gas was bubbled into a solution of methyl 1-(hydroxymethyl)cyclopropanecarboximide (6.00 g, 46.5 mmol) in MeOH (60 mL) at -78° C. for 10 min. The mixture was then stirred at room temperature for 15 h. The mixture was concentrated in vacuo to give 1-(hydroxymethyl)cyclopropanecarboxamidine, which was taken to the next step without further purification. MS=115.2 [M+H].sup.+.

Step 3: [1-(pyrimidin-2-yl)cyclopropyl]methanol

##STR01277##

[2694] A mixture of 1-(hydroxymethyl)cyclopropanecarboxamidine (5.50 g, 48.2 mmol), (E)-3-(dimethylamino)prop-2-enal (4.54 g, 45.8 mmol) and K.sub.2CO.sub.3 (13.3 g, 96.4 mmol) in EtOH (50 mL) was stirred at 85° C. for 15 h. The mixture was concentrated in vacuo. The residue was diluted with water (60 mL) at 0° C., and then extracted with EtOAc (4×20 mL). The combined organic layers were washed with brine (50 mL×2), dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (Biotage 40 g cartridge, 0-50% EtOAc/Petroleum ether) to give [1-(pyrimidin-2-yl)cyclopropyl]methanol. MS=151.2 [M+H].sup.+.

Step 4: [1-(5-bromopyrimidin-2-yl)cyclopropyl]methyl acetate

##STR01278##

[2695] To a solution of (1-pyrimidin-2-ylcyclopropyl)methanol (1.00 g, 6.66 mmol) in HOAc (10 mL) was added NaOAc (655 mg, 7.99 mmol). The mixture was heated to 80° C., Br.sub.2 (1.60 g, 9.99 mmol) was added, and the reaction was stirred at 80° C. for 3 h. After cooling to 0° C., the reaction mixture was quenched by addition of H.sub.2O (20 mL). The mixture was adjusted to pH=8 by slow addition of solid NaOH, and then extracted with EtOAc (3×15 mL). The combined organic layers were washed with brine (2×30 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-20% EtOAc/Petroleum ether) to give [1-(5-bromopyrimidin-2-yl)cyclopropyl]methyl acetate. MS=271.0/272.9 [M+H].sup.+.

Step 5: {1-[5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyrimidin-2-yl]cyclopropyl}methyl acetate

##STR01279##

[2696] A mixture of [1-(5-bromopyrimidin-2-yl)cyclopropyl]methyl acetate (500 mg, 1.84 mmol), bis(pinacolato)diboron (609 mg, 2.40 mmol), KOAc (543 mg, 5.53 mmol), and Pd(dppf)Cl.sub.2 (67.5 mg, 92.2 µmol) in 1,4-dioxane (15 mL) was degassed and purged with (N.sub.2 3×), and then the mixture was stirred at 80° C. for 15 h under N.sub.2 atmosphere. The mixture was filtered, and

the filtrate was concentrated in vacuo to give {1-[5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyrimidin-2-yl]cyclopropyl}methyl acetate, which was taken to the next step without further purification. MS=319.3 [M+H].sup.+.

Step 6: [1-(5-hydroxypyrimidin-2-yl)cyclopropyl]methyl acetate

##STR01280##

[2697] To a solution of {1-[5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyrimidin-2-yl]cyclopropyl}methyl acetate (800 mg, 2.51 mmol) in MeCN (10 mL) and H.sub.2O (10 mL) was added Oxone (1.85 g, 3.01 mmol). The mixture was stirred at room temperature for 2 h. After cooling to 0° C., the reaction mixture was quenched by addition of saturated aqueous Na.sub.2SO.sub.3 solution (15 mL), and then extracted with EtOAc (3×15 mL). The combined organic layers were washed with brine (2×30 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-50% EtOAc/Petroleum ether) to give [1-(5-hydroxypyrimidin-2-yl)cyclopropyl]methyl acetate MS=209.2 [M+H].sup.+.

Step 7: {1-[5-(2-bromoethoxy)pyrimidin-2-yl]cyclopropyl}methanol

##STR01281##

[2698] To a solution of [1-(5-hydroxypyrimidin-2-yl)cyclopropyl]methyl acetate (400 mg, 1.92 mmol) in MeCN (5 mL) was added K.sub.2CO.sub.3 (1.06 g, 7.68 mmol) and 1,2-dibromoethane (5.79 mL, 76.8 mmol). The mixture was stirred at 65° C. for 15 h. The mixture was cooled to room temperature and MeOH (10 mL) and additional K.sub.2CO.sub.3 (1.06 g, 7.68 mmol) was added, and the reaction was stirred for 2 h. The mixture was filtered, and the filtrate was concentrated to provide [{1-[5-(2-bromoethoxy)pyrimidin-2-yl]cyclopropyl}methanol (Intermediate 101), which was used in the subsequent step without further purification. MS=273.2/275.1 [M+H].sup.+.

General Procedure for Intermediate A-102

##STR01282##

Step 1: 1-bromo-4-[(dimethylphosphoryl)methyl]benzene

##STR01283##

[2699] In a three-neck round-bottom flask with a magnetic stir bar and thermometer, 1.0 M NaHMDS in THF (96.0 mL, 96.0 mmol) was added dropwise to a 0° C. solution of methylphosphonoylmethane (6.87 g, 88.2 mmol) in THF (200 mL). The mixture was stirred at 0° C. for 15 min, and then a solution of 1-bromo-4-(bromomethyl)benzene (20.0 g, 80.0 mmol) in THF (25 mL) was added dropwise. The resulting mixture was stirred at room temperature for 2 h. The mixture was quenched with H.sub.2O (200 mL) and extracted with DCM (5×50 mL). The combined organic layers were dried over Na.sub.2SO.sub.4 and filtered, and the solvent was removed in vacuo. The residue was purified by flash silica gel chromatography (Biotage 80 g cartridge, 0-10% MeOH/EtOAc) to give 1-bromo-4-[(dimethylphosphoryl)methyl]benzene. MS=247.0/249.0 [M+H].sup.+.

Step 2: 1-bromo-4-[1-(dimethylphosphoryl)cyclopropyl]benzene

##STR01284##

[2700] To a three-necked round-bottom flask equipped with a magnetic stir bar and thermometer under N.sub.2 atmosphere were added 1-bromo-4-(dimethylphosphorylmethyl)benzene (3.00 g, 12.1 mmol) and freshly distilled THE (40 mL). The mixture was cooled to -70° C. and 2.0 M LDA in THE (13.4 mL, 13.4 mmol) was added dropwise. The mixture was stirred for 10 min at -70° C., then 1,2-dibromoethane (3.42 g, 18.1 mmol) was added, and the mixture was stirred at room temperature for 6 h. The reaction was cooled to 0° C., quenched with H.sub.2O (15 mL), and then stirred at 0° C. for 10 min. The mixture was extracted with EtOAc (3×50 mL). The combined organic layers were dried over Na.sub.2SO.sub.4 and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Phenomenex Luna C.sub.18 column, 25-55% MeCN:10 mM TFA in H.sub.2O) to give 1-bromo-4-[1-(dimethylphosphoryl)cyclopropyl]benzene. MS=273.1/275.1 [M+H].sup.+.

Step 3: 4-[1-(dimethylphosphoryl)cyclopropyl]phenol

##STR01285##

[2701] A mixture of 1-bromo-4-(1-dimethylphosphorylcyclopropyl)benzene (350 mg, 1.28 mmol), di-tert-butyl-[2,3,4,5-tetramethyl-6-(2,4,6-triisopropylphenyl)phenyl]phosphane (61.6 mg, 128 μ mol), Pd.sub.2(dba).sub.3 (23.5 mg, 25.6 μ mol) and KOH (216 mg, 3.84 mmol) in 1,4-dioxane (5 mL) and H.sub.2O (2 mL) was degassed and purged with N.sub.2 (3 $\times$). The mixture was stirred at 100° C. for 16 h. After cooling to room temperature, the reaction mixture was filtered, and the filtrate was concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Phenomenex Luna C.sub.18 column, 1-45% MeCN:10 mM TFA in H.sub.2O) to give 4-[1-(dimethylphosphoryl)cyclopropyl]phenol (Intermediate A-102). MS=211.0 [M+H].sup.+.

General Procedure for Intermediate A-103

##STR01286##

Step 1: 4-methoxy-1-(methylsulfanyl)-2-(trifluoromethyl)benzene

##STR01287##

[2702] A mixture of 1-fluoro-4-methoxy-2-(trifluoromethyl)benzene (10.0 g, 51.5 mmol) and sodium methanethiolate (5.42 g, 77.3 mmol) in DMF (100 mL) was stirred at 50° C. for 16 h. After cooling to 0° C., the reaction mixture was quenched by addition of H.sub.2O (150 mL) and then extracted with EtOAc (3 $\times$ 150 mL). The combined organic layers were washed with H.sub.2O (3 $\times$ 150 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure to give 4-methoxy-1-(methylsulfanyl)-2-(trifluoromethyl)benzene, which was taken to the next step without further purification. .sup.1H NMR (400 MHz, CDCl.sub.3): δ 7.43 (d, J=8.4 Hz, 1H), 7.20 (d, J=2.8 Hz, 1H), 7.03 (dd, J=8.4, 2.4 Hz, 1H), 3.84 (s, 3H), 2.46 (s, 3H).

Step 2: imino[4-methoxy-2-(trifluoromethyl)phenyl]methyl- λ .SUP.6.-sulfanone

##STR01288##

[2703] To a 0° C. solution of 4-methoxy-1-(methylsulfanyl)-2-(trifluoromethyl)benzene (10.5 g, 47.3 mmol) and NH.sub.2CO.sub.2NH.sub.4 (14.8 g, 189 mmol) in MeOH (150 mL) was added PhI(OAc).sub.2 (38.1 g, 118 mmol). The mixture was stirred at room temperature for 16 h. The reaction mixture was quenched by addition of H.sub.2O (300 mL) and then concentrated under reduced pressure to remove MeOH. The mixture was extracted with EtOAc (3 $\times$ 150 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (Biotage 120 g cartridge, 0-70% EtOAc/Petroleum ether) to give imino[4-methoxy-2-(trifluoromethyl)phenyl]methyl- λ .sup.6-sulfanone. .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 8.22 (d, J=8.8 Hz, 1H), 7.47-7.36 (m, 2H), 4.43 (s, 1H), 3.91 (s, 3H), 3.07 (s, 3H).

Step 3: {[4-methoxy-2-(trifluoromethyl)phenyl](methyl)oxo- λ .SUP.6.-sulfanylidene}(methyl)amine

##STR01289##

[2704] To a solution of imino[4-methoxy-2-(trifluoromethyl)phenyl]methyl- λ .sup.6-sulfanone (4.00 g, 15.8 mmol) in HCOOH (40 mL) was added 37% HCHO in H.sub.2O (6.41 g, 79.0 mmol). The mixture was stirred at 100° C. for 16 h. After cooling to 0° C., the reaction mixture was quenched by addition of H.sub.2O (100 mL), and then adjusted to pH=7 by dropwise addition of saturated aqueous Na.sub.2CO.sub.3 solution. The aqueous phase was extracted with EtOAc (3 $\times$ 100 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered and concentrated under reduced pressure to give {[4-methoxy-2-(trifluoromethyl)phenyl](methyl)oxo- λ .sup.6-sulfanylidene}(methyl)amine, which was taken to the next step without further purification. MS=268.1 [M+H].sup.+.

Step 4: 4-[methyl(methylimino)oxo- λ .SUP.6.-sulfanyl]-3-(trifluoromethyl)phenol

##STR01290##

[2705] To a 0° C. solution of {[4-methoxy-2-(trifluoromethyl)phenyl](methyl)oxo- λ .sup.6-sulfanylidene}(methyl)amine (2.50 g, 9.35 mmol) in DCM (30 mL) was added BBr.sub.3 (7.21

mL, 74.8 mmol). The mixture was stirred at room temperature for 3 h, then was quenched by slow addition of MeOH (200 mL). The mixture was stirred at room temperature for 10 min, then concentrated in vacuo. The residue was diluted with H.sub.2O (100 mL) and extracted with EtOAc (6×100 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure to give 4-[methyl(methylimino)oxo-λ.sup.6-sulfanyl]-3-(trifluoromethyl)phenol, which was taken to the next step without further purification. MS=254.1 [M+H].sup.+.

Step 5: {[4-(2-bromoethoxy)-2-(trifluoromethyl)phenyl](methyl)oxo-λ.SUP.6.-sulfanylidene} (methyl)amine

##STR01291##

[2706] A mixture of 4-[methyl(methylimino)oxo-λ.sup.6-sulfanyl]-3-(trifluoromethyl)phenol (2.00 g, 7.90 mmol), 1,2-dibromoethane (59.4 g, 316 mmol) and K.sub.2CO.sub.3 (8.73 g, 63.2 mmol) in MeCN (20 mL) was stirred at 65° C. for 16 h. After cooling to room temperature, the reaction mixture was quenched by addition of H.sub.2O (30 mL) and extracted with EtOAc (3×50 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (Biotage 20 g cartridge, 0-90% EtOAc/Petroleum ether) to give {[4-(2-bromoethoxy)-2-(trifluoromethyl)phenyl] (methyl)oxo-λ.sup.6-sulfanylidene}(methyl)amine (Intermediate A-103). MS=360.0/362.0 [M+H].sup.+.

General Procedure for Intermediate A-104

##STR01292##

Step 1: 4-(1-methanesulfonylpiperidin-4-yl)phenyl methanesulfonate

##STR01293##

[2707] To a 0° C. solution of 4-(4-piperidyl)phenol hydrochloride (1.00 g, 4.68 mmol) and TEA (1.42 g, 14.0 mmol) in DCM (10 mL) was added Ms.sub.2O (1.63 g, 9.36 mmol) in portions. The mixture was stirred at room temperature for 1 h. The reaction mixture was quenched by addition of H.sub.2O (15 mL) and extracted with DCM (3×15 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered and concentrated under reduced pressure to give 4-(1-methanesulfonylpiperidin-4-yl)phenyl methanesulfonate, which was taken to the next step without further purification.

Step 2: 4-(1-methanesulfonylpiperidin-4-yl)phenol

##STR01294##

[2708] To a solution of [4-(1-methylsulfonyl-4-piperidyl)phenyl]methanesulfonate (1.30 g, 3.90 mmol) in MeOH (15 mL) and H.sub.2O (10 mL) was added NaOH (1.19 g, 29.6 mmol). The mixture was stirred at 80° C. for 1 h. After cooling to room temperature, the reaction mixture was concentrated under reduced pressure to remove MeOH and then adjusted to pH=1 by dropwise addition of 3.0 M aqueous HCl. The resulting precipitated solid was collected by filtration and dried in vacuo to give 4-(1-methanesulfonylpiperidin-4-yl)phenol, which was taken to the next step without further purification. MS=256.2 [M+H].sup.+.

Step 3: 4-[4-(2-bromoethoxy)phenyl]-1-methanesulfonylpiperidine

##STR01295##

[2709] A mixture of 4-(1-methanesulfonylpiperidin-4-yl)phenol (600 mg, 2.35 mmol), 1,2-dibromoethane (17.7 g, 94.0 mmol) and K.sub.2CO.sub.3 (1.95 g, 14.1 mmol) in MeCN (10 mL) was stirred at 65° C. for 16 h. The reaction mixture was cooled to room temperature and quenched by addition of H.sub.2O (15 mL), then extracted with EtOAc (3×20 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-65% EtOAc/Petroleum ether) to give 4-[4-(2-bromoethoxy)phenyl]-1-methanesulfonylpiperidine (Intermediate A-104). MS=364.1/362.1 [M+H].sup.+.

General Procedure for Intermediates A-105 & A-106

##STR01296##

Step 1: 6-bromo-1-(2-hydroxy-2-methylpropyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR01297##

[2710] A mixture of 6-bromo-3,4-dihydro-1H-1,8-naphthyridin-2-one (2.00 g, 8.81 mmol), 2,2-dimethyloxirane (1.27 g, 17.6 mmol), and Cs.sub.2CO.sub.3 (4.30 g, 13.2 mmol) in DMF (30 mL) was stirred at 80° C. for 16 h. After cooling to 0° C., the reaction mixture was quenched by addition of H.sub.2O (100 mL) and extracted with EtOAc (3×50 mL). The combined organic layers were washed with saturated brine (2×50 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-30% EtOAc/Petroleum ether) to give 6-bromo-1-(2-hydroxy-2-methylpropyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one. MS=298.9/300.9 [M+H].sup.+.

Step 2: 1-(2-hydroxy-2-methylpropyl)-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR01298##

[2711] A mixture of 6-bromo-1-(2-hydroxy-2-methylpropyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (900 mg, 3.01 mmol), bis(pinacolato)diboron (917 mg, 3.61 mmol), KOAc (591 mg, 6.02 mmol) and Pd(dppf)Cl.sub.2.Math.CH.sub.2Cl.sub.2 (246 mg, 301 µmol) in 1,4-dioxane (10 mL) was degassed and purged with N.sub.2 (3×), and then the mixture was stirred at 80° C. for 16 h under N.sub.2 atmosphere. The reaction mixture was filtered, and the filtrate was concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-50% EtOAc/Petroleum ether) to give 1-(2-hydroxy-2-methylpropyl)-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one. MS=347.3 [M+H].sup.+.

Step 3: 6-hydroxy-1-(2-hydroxy-2-methylpropyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR01299##

[2712] To a 0° C. solution of 1-(2-hydroxy-2-methylpropyl)-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (800 mg, 2.31 mmol) and K.sub.2CO.sub.3 (639 mg, 4.62 mmol) in DCM (10 mL) was added 30% H.sub.2O.sub.2 in H.sub.2O (0.577 mL, 6.00 mmol). The mixture was stirred at room temperature for 16 h. After cooling to 0° C., the reaction mixture was quenched by addition of saturated aqueous Na.sub.2SO.sub.3 solution (20 mL). The mixture was diluted with H.sub.2O (20 mL) and adjusted to pH=6 by dropwise addition of 4.0 M aqueous HCl solution. The mixture was extracted with DCM (3×10 mL), and the combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-100% EtOAc/Petroleum ether) to give a crude product. The crude product was further triturated with petroleum ether (10 mL) at room temperature for 10 min and then filtered to give 6-hydroxy-1-(2-hydroxy-2-methylpropyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (Intermediate A-105). MS=237.2 [M+H].sup.+.

Step 4: 6-(2-bromoethoxy)-1-(2-hydroxy-2-methylpropyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR01300##

[2713] To a solution of 6-hydroxy-1-(2-hydroxy-2-methylpropyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (190 mg, 804 µmol) in MeCN (0.5 mL) was added K.sub.2CO.sub.3 (111 mg, 804 µmol) and 1,2-dibromoethane (6.04 g, 32.2 mmol). The mixture was stirred at 80° C. for 16 h. After cooling to room temperature, the reaction mixture was filtered, and the filtrate was concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (Biotage 4 g cartridge, 0-100% EtOAc/Petroleum ether) to give 6-(2-bromoethoxy)-1-(2-hydroxy-2-methylpropyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (Intermediate A-106). MS=343.1/345.1 [M+H].sup.+.

General Procedure for Intermediate A-107

##STR01301##

Step 1: 3-[4-(benzyloxy)phenyl]-3-(propan-2-ylsulfanyl)oxetane

##STR01302##

[2714] To a solution of 3-[4-(benzyloxy)phenyl]oxetane-3-thiol (Step 3, Intermediate A-36, 1.00 g, 3.67 mmol) in DMF (10 mL) was added K₂CO₃ (761 mg, 5.51 mmol) and 2-iodopropane (1.87 g, 11.0 mmol). The mixture was stirred at room temperature for 2 h. After cooling to 0° C., the reaction mixture was quenched by addition of water (10 mL) and then extracted with EtOAc (2×10 mL). The combined organic layers were washed with brine (2×15 mL), dried over Na₂SO₄, filtered, and concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-20% EtOAc/Petroleum ether) to give 3-[4-(benzyloxy)phenyl]-3-(propan-2-ylsulfanyl)oxetane. ¹H NMR (400 MHz, DMSO-*d*₆): δ 7.44-7.31 (m, 5H), 7.24 (d, J=8.8 Hz, 2H), 7.01 (d, J=8.4 Hz, 2H), 5.09 (s, 2H), 5.05 (d, J=6.4 Hz, 2H), 4.74 (d, J=6.4 Hz, 2H), 2.44-2.39 (m, 1H), 1.02 (d, J=6.8 Hz, 6H).

Step 2: 3-[4-(benzyloxy)phenyl]-3-(propane-2-sulfonyl)oxetane

##STR01303##

[2715] To a 0° C. solution of 3-(4-benzyloxyphenyl)-3-isopropylsulfanyl-oxetane (1.00 g, 3.18 mmol) in DCM (20 mL) was added *m*-CPBA (1.94 g, 85% purity, 9.54 mmol). The mixture was stirred at room temperature for 2 h. After cooling to 0° C., the reaction mixture was quenched by addition of saturated aqueous Na₂SO₃ solution (20 mL), and then diluted with H₂O (10 mL) and extracted with DCM (3×15 mL). The combined organic layers were washed with saturated aqueous NaHCO₃, dried over Na₂SO₄, filtered, and concentrated under reduced pressure to give 3-(3-[4-(benzyloxy)phenyl]-3-(propane-2-sulfonyl)oxetane, which was taken to the next step without further purification. MS=364.1 [M+NH₄]⁺.

Step 3: 4-[3-(propane-2-sulfonyl)oxetan-3-yl]phenol

##STR01304##

[2716] To a solution of 3-[4-(benzyloxy)phenyl]-3-(propane-2-sulfonyl)oxetane (1.00 g, 2.89 mmol) in EtOAc (20 mL) under Ar atmosphere was added Pd/C (2.00 g, 10 wt %, 1.89 mmol). The suspension was degassed in vacuo and purged with H₂. The mixture was stirred under H₂ (15 psi) at room temperature for 2 h. The mixture was filtered, and the filtrate was concentrated in vacuo to give 4-[3-(propane-2-sulfonyl)oxetan-3-yl]phenol, which was taken to the next step without further purification. MS=274.1 [M+NH₄]⁺.

Step 4: 3-[4-(2-bromoethoxy)phenyl]-3-(propane-2-sulfonyl)oxetane

##STR01305##

[2717] To a solution of 4-[3-(propane-2-sulfonyl)oxetan-3-yl]phenol (650 mg, 2.54 mmol) in MeCN (5 mL) was added K₂CO₃ (1.75 g, 12.7 mmol) and 1,2-dibromoethane (19.1 g, 101 mmol). The mixture was stirred at 65° C. for 15 h. After cooling to room temperature, the mixture was filtered, and the filtrate was concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-50% EtOAc/Petroleum ether) to give 3-[4-(2-bromoethoxy)phenyl]-3-(propane-2-sulfonyl)oxetane (Intermediate A-107). MS=380.0/382.0 [M+NH₄]⁺.

General Procedure for Intermediate A-108

##STR01306##

Step 1: 3-[5-bromo-7-(trifluoromethyl)-2H-indazol-2-yl]cyclobutan-1-one

##STR01307##

[2718] To a mixture of 5-bromo-7-(trifluoromethyl)-1H-indazole (950 mg, 3.58 mmol) and K₂CO₃ (991 mg, 7.17 mmol) in DMF (9 mL) at room temperature was added 3-bromocyclobutan-1-one (641 mg, 4.30 mmol) dropwise. The mixture was stirred at room temperature for 1 h. The mixture was then poured into ice water (100 mL) and extracted with EtOAc (3×50 mL). The combined organic layers were dried over Na₂SO₄, filtered and

concentrated. The crude product was purified by silica gel chromatography (Biotage 50 g cartridge, 0-10% EtOAc/DCM) to give 3-[5-bromo-7-(trifluoromethyl)-2H-indazol-2-yl]cyclobutan-1-one (2.sup.nd eluting isomer). MS=333.0/335.0 [M+H].sup.+.

Step 2: (cis)-3-[5-bromo-7-(trifluoromethyl)-2H-indazol-2-yl]-1-methylcyclobutan-1-ol

##STR01308##

[2719] To a solution of 3-[5-bromo-7-(trifluoromethyl)-2H-indazol-2-yl]cyclobutan-1-one (995 mg, 2.99 mmol) in THE (10 mL) at 0° C. was added 3.0 M MeMgBr in Et.sub.2O (1.29 mL, 3.88 mmol) dropwise. The resulting mixture was stirred at 0° C. for 1 h. The mixture was quenched by addition of saturated NH.sub.4Cl solution (30 mL) and extracted with EtOAc (3×30 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered and concentrated. The residue was purified by flash silica gel chromatography twice (Biotage 50 g cartridge, 0-50% EtOAc/DCM) to give (cis)-3-[5-bromo-7-(trifluoromethyl)-2H-indazol-2-yl]-1-methylcyclobutan-1-ol. MS=349.1/351.1 [M+H].sup.+.

Step 3: (cis)-1-methyl-3-[5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-7-(trifluoromethyl)-2H-indazol-2-yl]cyclobutan-1-ol

##STR01309##

[2720] A mixture of (cis)-3-[5-bromo-7-(trifluoromethyl)-2H-indazol-2-yl]-1-methylcyclobutan-1-ol (800 mg, 2.29 mmol), bis(pinacolato)diboron (756 mg, 2.98 mmol), Pd(dppf)Cl.sub.2 dichloromethane complex (93.6 mg, 0.115 mmol) and KOAc (450 mg, 4.58 mmol) in 1,4-dioxane (23 mL) was sparged with N.sub.2 gas and then stirred at 90° C. for 16 h. The reaction mixture was cooled to room temperature and diluted with EtOAc, then filtered over a celite plug. The resulting filtrate was concentrated and used in the subsequent step without further purification. MS=397.3 [M+H].sup.+.

Step 4: 2-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-2H-indazol-5-ol

##STR01310##

[2721] The crude residue from the previous step was dissolved in MeOH (11 mL) and cooled to 0° C. A solution of 30% H.sub.2O.sub.2 in H.sub.2O (0.585 mL, 5.72 mmol) was added dropwise. The mixture was then stirred at room temperature for 3 h. The mixture was concentrated under reduced pressure and purified by silica gel chromatography (Biotage 50 g cartridge, 0-50% EtOAc/DCM) to give 2-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-2H-indazol-5-ol (Intermediate A-108). MS=287.1 [M+H].sup.+.

General Procedure for Intermediate A-109

##STR01311##

Step 1: 1-(5-bromo-2-fluoropyridin-3-yl)-2,2,2-trifluoroethan-1-ol

##STR01312##

[2722] To a 0° C. mixture of 5-bromo-2-fluoro-pyridine-3-carbaldehyde (10.0 g, 49.0 mmol) and TMSCF.sub.3 (8.36 g, 58.8 mmol) in THF (100 mL) was added dropwise 1.0 M TBAF in THF (9.80 mL, 9.80 mmol). The mixture was stirred at room temperature for 1 h, then was diluted with brine (150 mL) and extracted with EtOAc (2×80 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 120 g cartridge, 0-8% EtOAc/Petroleum ether) to provide 1-(5-bromo-2-fluoropyridin-3-yl)-2,2,2-trifluoroethan-1-ol. MS=273.8/275.8 [M+H].sup.+.

Step 2: 1-(5-bromo-2-fluoropyridin-3-yl)-2,2,2-trifluoroethan-1-one

##STR01313##

[2723] To a 0° C. mixture of 1-(5-bromo-2-fluoropyridin-3-yl)-2,2,2-trifluoroethan-1-ol (12.5 g, 45.6 mmol) in DCM (150 mL) was added Dess-Martin periodinane (23.2 g, 54.7 mmol). The mixture was stirred at room temperature for 16 h, then was quenched by addition of saturated aqueous Na.sub.2SO.sub.3 solution (200 mL) and extracted with DCM (2×80 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 120 g cartridge, 0-25% EtOAc/Petroleum

ether) to provide 1-(5-bromo-2-fluoropyridin-3-yl)-2,2,2-trifluoroethan-1-one. MS=289.8/291.8 [M+H].sub.2O+H].sup.+.

Step 3: 1-(5-bromo-2-([(cis)-3-hydroxy-3-methylcyclobutyl]amino})pyridin-3-yl)-2,2,2-trifluoroethan-1-one

##STR01314##

[2724] To a mixture of 1-(5-bromo-2-fluoropyridin-3-yl)-2,2,2-trifluoroethan-1-one (4.00 g, 14.7 mmol) and (cis)-3-amino-1-methylcyclobutan-1-ol (2.02 g, 14.7 mmol, HCl salt) in DMSO (35 mL) was added DIPEA (7.68 mL, 44.1 mmol). The mixture was stirred at 90° C. for 4 h. After cooling to room temperature, the mixture was diluted with H.sub.2O (150 mL) and extracted with EtOAc (2×80 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 80 g cartridge, 0-25% EtOAc/Petroleum ether) to provide 1-(5-bromo-2-([(cis)-3-hydroxy-3-methylcyclobutyl]amino})pyridin-3-yl)-2,2,2-trifluoroethan-1-one. MS=352.9/354.8 [M+H].sup.+.

Step 4: 6-bromo-1-([(cis)-3-hydroxy-3-methylcyclobutyl]-4-(trifluoromethyl)-1,2-dihydro-1,8-naphthyridin-2-one

##STR01315##

[2725] To a 0° C. mixture of ethyl 2-diethoxyphosphorylacetate (2.22 mL, 11.2 mmol) in THE (40 mL) was added t-BuOK (2.10 g, 18.7 mmol). The mixture was stirred at room temperature for 1 h, then cooled to 0° C. A solution of 1-(5-bromo-2-([(cis)-3-hydroxy-3-methylcyclobutyl]amino})pyridin-3-yl)-2,2,2-trifluoroethan-1-one (3.30 g, 9.34 mmol) in THE (25 mL) was added dropwise to the 0° C. mixture. After stirring at 0° C. for 2 h, the mixture was quenched by addition of saturated aqueous NH.sub.4Cl solution (60 mL). The organic layer was separated, and the aqueous layer was extracted with EtOAc (2×40 mL). The combined organic layers were concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 40 g cartridge, 0-17% EtOAc/Petroleum ether) to provide 6-bromo-1-([(cis)-3-hydroxy-3-methylcyclobutyl]-4-(trifluoromethyl)-1,2-dihydro-1,8-naphthyridin-2-one. MS=377.0/378.9 [M+H].sup.+.

Step 5: 1-([(cis)-3-hydroxy-3-methylcyclobutyl]-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-4-(trifluoromethyl)-1,2-dihydro-1,8-naphthyridin-2-one

##STR01316##

[2726] A mixture of 6-bromo-1-([(cis)-3-hydroxy-3-methylcyclobutyl]-4-(trifluoromethyl)-1,2-dihydro-1,8-naphthyridin-2-one (1.30 g, 3.45 mmol), bis(pinacolato)diboron (1.75 g, 6.89 mmol), KOAc (846 mg, 8.62 mmol) and Pd(dppf)Cl.sub.2 (126 mg, 172 µmol) in 1,4-dioxane (20 mL) was purged with N.sub.2 (3×) and stirred at 90° C. for 16 h. After cooling to room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo to provide 1-([(cis)-3-hydroxy-3-methylcyclobutyl]-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-4-(trifluoromethyl)-1,2-dihydro-1,8-naphthyridin-2-one, which was used in the subsequent step without further purification.

Step 6: 6-hydroxy-1-([(cis)-3-hydroxy-3-methylcyclobutyl]-4-(trifluoromethyl)-1,2-dihydro-1,8-naphthyridin-2-one

##STR01317##

[2727] To a 0° C. mixture of 1-([(cis)-3-hydroxy-3-methylcyclobutyl]-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-4-(trifluoromethyl)-1,2-dihydro-1,8-naphthyridin-2-one (1.50 g, 3.54 mmol) in a THE (20 mL) and H.sub.2O (10 mL) was added Oxone (2.61 g, 4.24 mmol). The mixture was stirred at room temperature for 1 h. The reaction mixture was cooled to 0-10° C., quenched by addition of saturated Na.sub.2SO.sub.3 solution (60 mL) at 0-10° C., and stirred for 30 min. The mixture was extracted with EtOAc (2×40 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 20 g cartridge, 0-15% EtOAc/Petroleum ether) to provide 6-hydroxy-1-([(cis)-3-hydroxy-3-methylcyclobutyl]-4-(trifluoromethyl)-1,2-dihydro-1,8-naphthyridin-2-one

(Intermediate A-109). MS=315.0 [M+H].sup.+.

General Procedure for Intermediate A-110

##STR01318## ##STR01319##

Step 1: 2-fluoro-5-nitro-3-(trifluoromethyl)benzaldehyde

##STR01320##

[2728] To a 250 mL three-neck round bottom flask equipped with a magnetic stir bar and thermometer containing a solution of 2-fluoro-3-(trifluoromethyl)benzaldehyde (7.19 mL, 52.1 mmol) in H.sub.2SO.sub.4 (50 mL) at -5-0° C. was added HNO.sub.3 (21.1 mL, 468 mmol) dropwise. The mixture was stirred at 0° C. for 30 min. The reaction mixture was poured into ice water (80 mL), adjusted to pH=7 by addition of solid NaHCO.sub.3, and extracted with EtOAc (3×50 mL). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 40 g cartridge, 0-3% EtOAc/Petroleum ether) to provide 2-fluoro-5-nitro-3-(trifluoromethyl)benzaldehyde. .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.43 (s, 1H), 8.96 (dd, J=5.2, 2.8 Hz, 1H), 8.77 (dd, J=5.6, 2.8 Hz, 1H).

Step 2: ethyl (2E)-3-[2-fluoro-5-nitro-3-(trifluoromethyl)phenyl]prop-2-enoate

##STR01321##

[2729] To a 0° C. solution of ethyl 2-(diethoxyphosphoryl)acetate (13.8 mL, 69.6 mmol) in THE (300 mL) was added t-BuOK (8.52 g, 75.9 mmol) in portions. The mixture was stirred at 0° C. for 1 h. A solution of 2-fluoro-5-nitro-3-(trifluoromethyl)benzaldehyde (15.0 g, 63.3 mmol) in THE (30 mL) was added to the 0° C. mixture, and reaction was allowed to warm to room temperature and stirred for 2 h. The reaction mixture was quenched by addition of H.sub.2O (100 mL) and extracted with EtOAc (3×100 mL). The combined organic layers were washed with brine (80 mL), dried over anhydrous Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 80 g cartridge, 0-10% EtOAc/Petroleum ether) to provide ethyl (2E)-3-[2-fluoro-5-nitro-3-(trifluoromethyl)phenyl]prop-2-enoate. .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 8.66 (dd, J=5.6, 2.8 Hz, 1H), 8.53 (dd, J=5.6, 2.8 Hz, 1H), 7.82 (d, J=16.0 Hz, 1H), 6.72 (d, J=16.0 Hz, 1H), 4.33 (q, J=7.2 Hz, 2H), 1.37 (t, J=7.2 Hz, 3H).

Step 3: ethyl 3-[2-fluoro-5-nitro-3-(trifluoromethyl)phenyl]propanoate

##STR01322##

[2730] To a solution of ethyl (2E)-3-[2-fluoro-5-nitro-3-(trifluoromethyl)phenyl]prop-2-enoate (8.00 g, 26.0 mmol) in EtOH (200 mL) was added chlorotris(triphenylphosphine)rhodium(I) (1.20 g, 1.30 mmol). The suspension was degassed and purged with H.sub.2 (3×). The mixture was stirred under H.sub.2 (15 psi) at room temperature for 1 h. Solids were removed by filtration and the filtrate was concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-8% EtOAc/Petroleum ether) to provide ethyl 3-[2-fluoro-5-nitro-3-(trifluoromethyl)phenyl]propanoate. .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 8.42 (s, 1H), 8.41 (s 1H), 4.19-4.13 (m, 2H), 3.13 (t, J=7.2 Hz, 2H), 2.73 (t, J=7.2 Hz, 2H), 1.26 (t, J=7.2 Hz, 3H).

Step 4: ethyl 3-(5-nitro-2-{[(cis)-3-hydroxy-3-methylcyclobutyl]amino}-3-(trifluoromethyl)phenyl)propanoate

##STR01323##

[2731] To a solution of ethyl 3-[2-fluoro-5-nitro-3-(trifluoromethyl)phenyl]propanoate (7.00 g, 22.6 mmol) in DMSO (100 mL) was added (cis)-3-amino-1-methylcyclobutan-1-ol (3.43 g, 24.9 mmol, HCl salt) and DIEA (11.8 mL, 67.9 mmol). The mixture was stirred at 80° C. for 12 h. After cooling to room temperature, the reaction mixture was quenched by addition of H.sub.2O (80 mL) and extracted with EtOAc (3×80 mL). The combined organic layers were washed with brine (150 mL), dried over anhydrous Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 40 g cartridge, 0-10% EtOAc/Petroleum ether) to provide ethyl 3-(5-nitro-2-{[(cis)-3-hydroxy-3-methylcyclobutyl]amino}-3-

(trifluoromethyl)phenyl)propanoate. MS=391.1 [M+H].sup.+.

Step 5: 6-nitro-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-8-(trifluoromethyl)-1,2,3,4-tetrahydroquinolin-2-one

##STR01324##

[2732] To a solution of ethyl 3-(5-nitro-2-[(cis)-3-hydroxy-3-methylcyclobutyl]amino)-3-(trifluoromethyl)phenyl)propanoate (7.00 g, 17.9 mmol) in DMF (80 mL) and MeOH (10 mL) was added K.sub.2CO.sub.3 (7.44 g, 53.8 mmol). The mixture was stirred at 80° C. for 12 h. After cooling to room temperature, the reaction mixture was quenched by addition of H.sub.2O (100 mL) and extracted with EtOAc (3×100 mL). The combined organic layers were washed with brine (15 mL), dried over anhydrous Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 40 g cartridge, 10-27% EtOAc/Petroleum ether) to provide 6-nitro-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-8-(trifluoromethyl)-1,2,3,4-tetrahydroquinolin-2-one. MS=345.1 [M+H].sup.+.

Step 6: 6-amino-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-8-(trifluoromethyl)-1,2,3,4-tetrahydroquinolin-2-one

##STR01325##

[2733] To a solution of 6-nitro-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-8-(trifluoromethyl)-1,2,3,4-tetrahydroquinolin-2-one (3.00 g, 8.71 mmol) in EtOAc (60 mL) under N.sub.2 atmosphere was added Pd/C (244 mg, 10 wt %, 0.231 mmol). The suspension was degassed and purged with H.sub.2 (3×). The mixture was stirred under H.sub.2 (15 psi) at room temperature for 12 h. Solids were removed by filtration, and the filtrate was concentrated in vacuo to provide 6-amino-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-8-(trifluoromethyl)-1,2,3,4-tetrahydroquinolin-2-one, which was used in the subsequent step without further purification. MS=315.1 [M+H].sup.+.

Step 7: 1-[(cis)-3-hydroxy-3-methylcyclobutyl]-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-8-(trifluoromethyl)-1,2,3,4-tetrahydroquinolin-2-one

##STR01326##

[2734] To a solution of 6-amino-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-8-(trifluoromethyl)-1,2,3,4-tetrahydroquinolin-2-one (1.00 g, 3.18 mmol) in MeCN (10 mL) under N.sub.2 atmosphere was added bis(pinacolato)diboron (970 mg, 3.82 mmol) and tert-butyl nitrite (492 mg, 4.77 mmol). The mixture was stirred at 80° C. for 2 h. After cooling to room temperature, the reaction mixture was quenched by addition of H.sub.2O (20 mL) and extracted with EtOAc (3×50 mL). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give 1-[(cis)-3-hydroxy-3-methylcyclobutyl]-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-8-(trifluoromethyl)-1,2,3,4-tetrahydroquinolin-2-one, which was used in the subsequent step without further purification. MS=426.1 [M+H].sup.+.

Step 8: 6-hydroxy-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-8-(trifluoromethyl)-1,2,3,4-tetrahydroquinolin-2-one

##STR01327##

[2735] To a solution of 1-[(cis)-3-hydroxy-3-methylcyclobutyl]-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-8-(trifluoromethyl)-1,2,3,4-tetrahydroquinolin-2-one (800 mg, 1.88 mmol) in THE (9 mL) and H.sub.2O (3 mL) was added Oxone (1.16 g, 1.88 mmol). The mixture was stirred at 0° C. for 1 h. The reaction mixture was quenched by addition of saturated aqueous Na.sub.2SO.sub.3 (20 mL) and extracted with EtOAc (3×20 mL). The combined organic layers were washed with brine (20 mL), dried over anhydrous Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give 6-hydroxy-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-8-(trifluoromethyl)-1,2,3,4-tetrahydroquinolin-2-one (Intermediate A-110), which was used in the subsequent step without further purification. MS=316.0 [M+H].sup.+.

General Procedure for Intermediate A-111

##STR01328## ##STR01329##

Step 1: 1-(5-bromo-2-fluoropyridin-3-yl)-2,2-difluoroethan-1-one

##STR01330##

[2736] To a -78°C . mixture of 5-bromo-2-fluoropyridine (2.92 mL, 28.4 mmol) in THF (45 mL) under N.sub.2 atmosphere was added 2.0 M LDA in THF (17.1 mL, 34.1 mmol). The mixture was stirred at -78°C . for 30 min under N.sub.2 atmosphere, then a solution of ethyl 2,2-difluoroacetate (5.29 g, 42.6 mmol) in THF (5 mL) was added dropwise. The mixture was stirred at -78°C . for another 2 h under N.sub.2 atmosphere, then was warmed to 0°C . The reaction mixture was quenched by addition of saturated aqueous NH.sub.4Cl solution (50 mL) and then extracted with EtOAc (3 $\times$ 100 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 40 g cartridge, 0-20% EtOAc/Petroleum ether) to provide 1-(5-bromo-2-fluoropyridin-3-yl)-2,2-difluoroethan-1-one. MS=272.0/273.9 [M+H+H.sub.2O].sup.+.

Step 2: 1-(5-bromo-2-([(cis)-3-hydroxy-3-methylcyclobutyl]amino)pyridin-3-yl)-2,2-difluoroethan-1-one

##STR01331##

[2737] To a solution of 1-(5-bromo-2-fluoropyridin-3-yl)-2,2-difluoroethan-1-one (1.80 g, 7.09 mmol) and (cis)-3-amino-1-methylcyclobutan-1-ol (1.07 g, 7.80 mmol, HCl salt) in DMSO (20 mL) was added DIEA (4.94 mL, 28.4 mmol). The mixture was stirred at 50°C . for 2 h, then was cooled to 0°C ., quenched by addition of H.sub.2O (20 mL), and extracted with EtOAc (3 $\times$ 20 mL). The combined organic layers were washed with brine (50 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 20 g cartridge, 0-30% EtOAc/Petroleum ether) to provide 1-(5-bromo-2-([(cis)-3-hydroxy-3-methylcyclobutyl]amino)pyridin-3-yl)-2,2-difluoroethan-1-one. MS=335.0/336.9 [M+H].sup.+.

Step 3: ethyl 3-(5-bromo-2-([(cis)-3-hydroxy-3-methylcyclobutyl]amino)pyridin-3-yl)-4,4-difluorobut-2-enoate

##STR01332##

[2738] To a 0°C . mixture of ethyl 2-diethoxyphosphorylacetate (803 mg, 3.58 mmol) in THE (10 mL) was added t-BuOK (670 mg, 5.97 mmol). The mixture was allowed to warm to room temperature and stirred for 1 h, then cooled to 0°C . and a solution of 1-(5-bromo-2-([(cis)-3-hydroxy-3-methylcyclobutyl]amino)pyridin-3-yl)-2,2-difluoroethan-1-one (1.00 g, 2.98 mmol) in THE (3 mL) was added dropwise. Then the mixture was stirred at 0°C . for 2 h, then was quenched by addition of H.sub.2O (15 mL) and extracted with EtOAc (3 $\times$ 15 mL). The combined organic layers were washed with brine (50 mL), dried over Na.sub.2SO.sub.4, filtered and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 20 g cartridge, 0-40% EtOAc/Petroleum ether) to provide ethyl 3-(5-bromo-2-([(cis)-3-hydroxy-3-methylcyclobutyl]amino)pyridin-3-yl)-4,4-difluorobut-2-enoate. MS=405.0/407.0 [M+H].sup.+.

Step 4: 6-bromo-4-(difluoromethyl)-1-([(cis)-3-hydroxy-3-methylcyclobutyl]-1,2-dihydro-1,8-naphthyridin-2-one

##STR01333##

[2739] To a mixture of ethyl 3-(5-bromo-2-([(cis)-3-hydroxy-3-methylcyclobutyl]amino)pyridin-3-yl)-4,4-difluorobut-2-enoate (730 mg, 1.80 mmol) in DMSO (10 mL) was added K.sub.2CO.sub.3 (747 mg, 5.40 mmol). The mixture was stirred at 80°C . for 2. After cooling to room temperature, the reaction mixture was quenched by addition of H.sub.2O (20 mL) and extracted with EtOAc (3 $\times$ 20 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-35% EtOAc/Petroleum ether) to provide 6-bromo-4-(difluoromethyl)-1-([(cis)-3-hydroxy-3-methylcyclobutyl]-1,2-dihydro-1,8-naphthyridin-2-one. MS=359.0/361.0 [M+H].sup.+.

Step 5: 4-(difluoromethyl)-1-([(cis)-3-hydroxy-3-methylcyclobutyl]-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2-dihydro-1,8-naphthyridin-2-one

##STR01334##

[2740] A mixture of 6-bromo-4-(difluoromethyl)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1,2-dihydro-1,8-naphthyridin-2-one (300 mg, 835 μ mol), bis(pinacolato)diboron (318 mg, 1.25 mmol), KOAc (205 mg, 2.09 mmol) and Pd(dppf)Cl.sub.2.Math.CH.sub.2Cl.sub.2 (34.1 mg, 41.8 μ mol) in 1,4-dioxane (4 mL) was degassed and purged with N.sub.2 (3 $\times$), and then stirred at 90° C. for 12 h under N.sub.2 atmosphere. After cooling to room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo to give 4-(difluoromethyl)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2-dihydro-1,8-naphthyridin-2-one, which was used in the subsequent step without further purification.

Step 6: 4-(difluoromethyl)-6-hydroxy-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1,2-dihydro-1,8-naphthyridin-2-one

##STR01335##

[2741] To a 0° C. solution of 4-(difluoromethyl)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2-dihydro-1,8-naphthyridin-2-one (400 mg, 985 μ mol) in THE (5 mL) and H.sub.2O (2 mL) was added Oxone (424 mg, 689 μ mol). The mixture was stirred at 0° C. for 1 h, and then was quenched with saturated aqueous Na.sub.2SO.sub.3 solution (20 mL) and extracted with EtOAc (3 $\times$ 30 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 4 g cartridge, 0-40% EtOAc/Petroleum ether) to provide 4-(difluoromethyl)-6-hydroxy-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1,2-dihydro-1,8-naphthyridin-2-one (Intermediate A-111). MS=297.0 [M+H].sup.+.

General Procedure for Intermediate A-112

##STR01336##

Step 1: 3-(5-bromopyrimidin-2-yl)oxetan-3-ol

##STR01337##

[2742] To a -78° C. mixture of 5-bromo-2-iodo-pyrimidine (5.00 g, 17.6 mmol) in toluene (40 mL) under N.sub.2 atmosphere was added 2.5 M n-BuLi in n-hexane (7.37 mL, 18.4 mmol) dropwise. The mixture was stirred at -78° C. for 30 min under N.sub.2 atmosphere and then a solution of oxetan-3-one (1.52 g, 21.1 mmol) in toluene (20 mL) was added dropwise. The resulting mixture was stirred at -78° C. for 30 min and then stirred at room temperature for 1 h. The reaction mixture was cooled to 0° C., quenched by addition of saturated aqueous NH.sub.4Cl solution (50 mL), then extracted with EtOAc (3 $\times$ 50 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 40 g cartridge, 100% EtOAc) to provide 3-(5-bromopyrimidin-2-yl)oxetan-3-ol. MS=231.0/233.0 [M+H].sup.+.

Step 2: 2-(3-hydroxyoxetan-3-yl)pyrimidin-5-ol

##STR01338##

[2743] A mixture of 3-(5-bromopyrimidin-2-yl)oxetan-3-ol (1.00 g, 4.33 mmol), Pd.sub.2(dba).sub.3 (79.3 mg, 86.6 μ mol), KOH (729 mg, 13.0 mmol) and ditert-butyl-[2,3,4,5-tetramethyl-6-(2,4,6-triisopropylphenyl)phenyl]phosphane (208 mg, 433 μ mol) in 1,4-dioxane (10 mL) and H.sub.2O (5 mL) was degassed and purged with N.sub.2 (3 $\times$) and then stirred at 100° C. for 4 h under N.sub.2 atmosphere. The reaction mixture was cooled to room temperature, quenched by addition of H.sub.2O (20 mL), and then extracted with EtOAc (3 $\times$ 20 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (Biotage 25 g cartridge, 0-100% EtOAc/Petroleum ether) to provide 2-(3-hydroxyoxetan-3-yl)pyrimidin-5-ol (Intermediate A-112). MS=169.1 [M+H].sup.+.

[2744] The following intermediates in Table 11.2 were prepared according to procedures analogous to those described for Intermediate A-112 using the appropriate starting materials or common intermediates.

General Procedure for Intermediate A-113

TABLE-US-00014 TABLE 11.2 Inter- IUPAC Exact Mass mediate Structure Name [M + H].sup.+ A-113 [01339]  embedded image 2-(1- hydroxycyclo- butyl)pyrimidin- 5-ol Calc'd 167.1 Found 167.1

General Procedure for Intermediate A-114

##STR01340##

Step 1: (cis)-1-methyl-3-{[2-nitro-6-(trifluoromethyl)phenyl]amino}cyclobutan-1-ol

##STR01341##

[2745] A mixture of 2-fluoro-1-nitro-3-(trifluoromethyl)benzene (1.50 g, 7.17 mmol), DIEA (3.71 g, 28.7 mmol) and (cis)-3-amino-1-methylcyclobutan-1-ol (1.09 g, 7.89 mmol, HCl salt) in DMSO (15 mL) was stirred at 50° C. for 2 h. After cooling to room temperature, the reaction mixture was poured into H.sub.2O (50 mL). The mixture was extracted with EtOAc (3×60 mL). The combined organic layers were washed with brine (2×40 mL), dried over Na.sub.2SO.sub.4, filtered and concentrated under reduced pressure to give (cis)-1-methyl-3-{[2-nitro-6-(trifluoromethyl)phenyl]amino}cyclobutan-1-ol, which was taken to the next step without further purification. MS=290.9 [M+H].sup.+.

Step 2: (cis)-3-amino-1-methylcyclobutan-1-ol

##STR01342##

[2746] To a mixture of Pd/C (4.00 g, 10 wt %, 3.78 mmol) in MeOH (100 mL) was added (cis)-1-methyl-3-{[2-nitro-6-(trifluoromethyl)phenyl]amino}cyclobutan-1-ol (6.00 g, 20.67 mmol). The mixture was stirred at room temperature for 2 h under H.sub.2 (15 psi) atmosphere. The mixture was filtered and the filtrate was concentrated in vacuo to give (cis)-3-amino-1-methylcyclobutan-1-ol, which was taken to the next step without further purification MS=261.0 [M+H].sup.+.

Step 3: (cis)-1-methyl-3-[7-(trifluoromethyl)-1H-1,2,3-benzotriazol-1-yl]cyclobutan-1-ol

##STR01343##

[2747] To a 0° C. mixture of 3-[2-amino-6-(trifluoromethyl)anilino]-1-methyl-cyclobutanol (2.00 g, 7.68 mmol) in AcOH (4 mL) and MeCN (20 mL) was added NaNO.sub.2 (1.06 g, 15.4 mmol). The mixture was allowed to warm to room temperature and stirred for 16 h. The mixture was diluted with saturated NaHCO.sub.3 solution (100 mL) and extracted with EtOAc (2×50 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give (cis)-1-methyl-3-[7-(trifluoromethyl)-1H-1,2,3-benzotriazol-1-yl]cyclobutan-1-ol, which was used in the subsequent step without further purification. MS=272.0 [M+H].sup.+.

Step 4: (cis)-1-methyl-3-[5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-7-(trifluoromethyl)-1H-1,2,3-benzotriazol-1-yl]cyclobutan-1-ol

##STR01344##

[2748] A mixture of (cis)-1-methyl-3-[7-(trifluoromethyl)-1H-1,2,3-benzotriazol-1-yl]cyclobutan-1-ol (2.30 g, 8.48 mmol), pinacolborane (12.3 mL, 84.8 mmol), 4,4-di-tert-butyl-2,2'-dipyridine (228 mg, 0.848 mmol) and (1,5-cyclooctadiene)(methoxy)iridium(I) dimer (281 mg, 0.424 mmol) in THE (30 mL) was degassed with N.sub.2 for 10 min, and then stirred at 80° C. for 16 h under N.sub.2 atmosphere. After cooling to room temperature, the reaction was filtered and the filtrate was concentrated in vacuo to give (cis)-1-methyl-3-[5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-7-(trifluoromethyl)-1H-1,2,3-benzotriazol-1-yl]cyclobutan-1-ol, which was used in the next step without further purification.

Step 5: 1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,2,3-benzotriazol-5-ol

##STR01345##

[2749] To a 0° C. solution of (cis)-1-methyl-3-[5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-7-(trifluoromethyl)-1H-1,2,3-benzotriazol-1-yl]cyclobutan-1-ol (3.3 g, 8.31 mmol) in THE (20 mL) and H.sub.2O (10 mL) was added Oxone (15.3 g, 24.9 mmol). The mixture was stirred at 0° C. for 2 h. The mixture was quenched with saturated aqueous Na.sub.2SO.sub.3 solution. The mixture was extracted with chloroform (3×30 mL). The combined organic layers were washed with brine

(2×50 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 40 g cartridge, 0-60% EtOAc/Petroleum ether) to provide 1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,2,3-benzotriazol-5-ol (Intermediate A-114). MS=288.0 [M+H].sup.+.

General Procedure for Intermediate A-115

##STR01346##

Step 1: 2-{3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl}-5-methoxypyrimidine

##STR01347##

[2750] In a glove box, to a vial equipped with a magnetic stir bar was added 2-bromo-5-methoxypyrimidine (1.00 g, 5.30 mmol) and (3-bromo-1-methyl-cyclobutoxy)-tert-butyl-dimethylsilane (7.40 g, 26.5 mmol) in DMA (80 mL). To the mixture was added manganese (3.49 g, 63.5 mmol), pyridine-2-carboxamide hydrochloride (2.50 g, 15.9 mmol), TBAI (1.35 g, 3.65 mmol) and dichloro(dimethoxyethane)nickel (872 mg, 3.97 mmol). The reaction was purged with Ar for 10 min, and then stirred at 80° C. for 12 h. After cooling to room temperature, the reaction mixture was quenched with H.sub.2O (50 mL) and filtered to remove solids. The filtrate was extracted with EtOAc (3×50 mL). The combined organic layers were washed with brine (50 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 40 g cartridge, 0-25% EtOAc/Petroleum ether) to provide 2-{3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl}-5-methoxypyrimidine. MS=309.4 [M+H].sup.+.

Step 2: 2-(3-hydroxy-3-methylcyclobutyl)pyrimidin-5-ol

##STR01348##

[2751] To a solution of 2-{3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl}-5-methoxypyrimidine (350 mg, 1.13 mmol) in H.sub.2O (7 mL) was added HBr (7.00 mL, 51.6 mmol). The mixture was stirred at 100° C. for 30 h. After cooling to room temperature, the reaction mixture was quenched by addition of saturated aqueous NaHCO.sub.3 solution (15 mL), then extracted with EtOAc (3×20 mL). The combined organic layers were concentrated in vacuo. The residue was purified by reverse phase preparative HPLC (Phenomenex Luna C.sub.18 column, 1-20% MeCN:10 mM TFA in H.sub.2O) to provide 2-(3-hydroxy-3-methylcyclobutyl)pyrimidin-5-ol (Intermediate A-115). MS=181.0 [M+H].sup.+.

General Procedure for Intermediate A-116

##STR01349##

Step 1: 3-{5-bromo-1H-pyrazolo[3,4-b]pyridin-1-yl}cyclobutan-1-one

##STR01350##

[2752] To a solution of 5-bromo-1H-pyrazolo[3,4-b]pyridine (5.00 g, 25.3 mmol) in DMF (45 mL) was added Na.sub.2CO.sub.3 (5.35 g, 50.5 mmol). The mixture was stirred at room temperature for 10 min, and then a solution 3-bromocyclobutanone (8.28 g, 55.6 mmol) in DMF (5 mL) was added dropwise by syringe pump over 2 h. The resulting mixture was stirred at room temperature for 3 h, then was quenched by addition of H.sub.2O (50 mL), and extracted with EtOAc (3×50 mL). The combined organic layers were washed with brine (2×50 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 80 g cartridge, 0-100% EtOAc/Petroleum ether) to provide 3-{5-bromo-1H-pyrazolo[3,4-b]pyridin-1-yl}cyclobutan-1-one (1.sup.st eluting isomer). MS=266.0/268.0 [M+H].sup.+.

Step 2: 3-{5-bromo-1H-pyrazolo[3,4-b]pyridin-1-yl}-1-methylcyclobutan-1-ol

##STR01351##

[2753] To a 0° C. solution of 3-(5-bromopyrazolo[3,4-b]pyridin-1-yl)cyclobutanone (400 mg, 1.50 mmol) in THE (4 mL) under N.sub.2 atmosphere was added 3.0 M MeMgBr in THE (0.55 mL, 1.65 mmol) dropwise. The mixture was stirred at 0° C. for 3 h under N.sub.2 atmosphere. The reaction mixture was quenched by addition of saturated aqueous NH.sub.4Cl solution (15 mL) and extracted with EtOAc (3×10 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography

(Biotage 40 g cartridge, 0-25% EtOAc/Petroleum ether) to provide 3-{5-bromo-1H-pyrazolo[3,4-b]pyridin-1-yl}-1-methylcyclobutan-1-ol. MS=282.0/284.0 [M+H].sup.+.

Step 3: 1-methyl-3-[5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1H-pyrazolo[3,4-b]pyridin-1-yl]cyclobutan-1-ol

##STR01352##

[2754] A mixture of 3-{5-bromo-1H-pyrazolo[3,4-b]pyridin-1-yl}-1-methylcyclobutan-1-ol (170 mg, 0.603 mmol), bis(pinacolato)diboron (184 mg, 0.723 mmol), KOAc (148 mg, 1.51 mmol), Pd(dppf)Cl.sub.2.Math.CH.sub.2Cl.sub.2 (49.1 mg, 60.3 µmol) in 1,4-dioxane (3.3 mL) was degassed and purged with N.sub.2 (3×), and then the mixture was stirred at 90° C. for 12 h under N.sub.2 atmosphere. The reaction mixture was allowed to cool to room temperature, then was diluted with H.sub.2O (10 mL) and extracted with EtOAc (3×10 mL). The combined organic layers were washed with brine (20 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to provide 1-methyl-3-[5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1H-pyrazolo[3,4-b]pyridin-1-yl]cyclobutan-1-ol, which was used in the subsequent step without further purification.

MS=330.2 [M+H].sup.+.

Step 4. 1-(3-hydroxy-3-methylcyclobutyl)-1H-pyrazolo[3,4-b]pyridin-5-ol

##STR01353##

[2755] To a 0° C. solution of 1-methyl-3-[5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1H-pyrazolo[3,4-b]pyridin-1-yl]cyclobutan-1-ol (330 mg, 1.00 mmol) in THF (5 mL) and H.sub.2O (2.5 mL) was added Oxone (616 mg, 1.00 mmol). The mixture was stirred at 0° C. for 1 h, then was quenched by addition of saturated aqueous Na.sub.2SO.sub.3 solution. The mixture was extracted with chloroform (3×30 mL). The combined organic layers were washed with brine (50 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 4 g cartridge, 0-60% EtOAc/Petroleum ether) to provide 1-(3-hydroxy-3-methylcyclobutyl)-1H-pyrazolo[3,4-b]pyridin-5-ol (Intermediate A-116). MS=220.1 [M+H].sup.+.

General Procedure for Intermediate A-117

##STR01354##

Step 1: 3-nitro-2-{[(cis)-3-hydroxy-3-methylcyclobutyl]amino}benzonitrile

##STR01355##

[2756] To a solution of 2-fluoro-3-nitro-benzonitrile (2.50 g, 15.1 mmol) in DMSO (25 mL) was added DIEA (7.86 mL, 45.2 mmol) and (cis)-3-amino-1-methylcyclobutan-1-ol (1.67 g, 16.6 mmol). The mixture was stirred at 60° C. for 1 h. After cooling to room temperature, the mixture was diluted with H.sub.2O (30 mL), filtered, and concentrated in vacuo to give 3-nitro-2-{[(cis)-3-hydroxy-3-methylcyclobutyl]amino}benzonitrile, which was used in the subsequent step without further purification. MS=248.3 [M+H].sup.+.

Step 2: 5-bromo-3-nitro-2-{[(cis)-3-hydroxy-3-methylcyclobutyl]amino}benzonitrile

##STR01356##

[2757] To a solution of 3-nitro-2-{[(cis)-3-hydroxy-3-methylcyclobutyl]amino}benzonitrile (2.50 g, 10.1 mmol) in DMF (25 mL) was added NBS (3.60 g, 20.2 mmol). The mixture was stirred at 60° C. for 16 h. After cooling to room temperature, the mixture was diluted with H.sub.2O (30 mL) and extracted with EtOAc (2×20 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 40 g cartridge, 0-15% EtOAc/Petroleum ether) to provide 5-bromo-3-nitro-2-{[(cis)-3-hydroxy-3-methylcyclobutyl]amino}benzonitrile.

Step 3: 5-bromo-3-nitro-2-{[(cis)-3-hydroxy-3-methylcyclobutyl]amino}benzonitrile

##STR01357##

[2758] To a solution of 5-bromo-3-nitro-2-{[(cis)-3-hydroxy-3-methylcyclobutyl]amino}benzonitrile (2.00 g, 6.13 mmol) in EtOH (20 mL) and H.sub.2O (10 mL) was added Fe (3.42 g, 61.3 mmol) and NH.sub.4Cl (4.92 g, 92.0 mmol). The mixture was stirred at

60° C. for 2 h. After cooling to room temperature, solids were removed by filtration. The filtrate was diluted with H.sub.2O (30 mL) and extracted with EtOAc (2×20 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give 5-bromo-3-nitro-2-[[*(cis)*-3-hydroxy-3-methylcyclobutyl]amino]benzonitrile, which was used in the subsequent step without further purification. MS=295.9/297.9 [M+H].sup.+.

Step 4: 5-bromo-1-[[*(cis)*-3-hydroxy-3-methylcyclobutyl]-1H-1,3-benzodiazole-7-carbonitrile
##STR01358##

[2759] To a solution of 5-bromo-3-nitro-2-[[*(cis)*-3-hydroxy-3-methylcyclobutyl]amino]benzonitrile (1.80 g, 6.08 mmol) in MeOH (15 mL) was added triethyl orthoformate (3.03 mL, 18.23 mmol) and TsOH.Math.H.sub.2O (116 mg, 0.608 mmol). The mixture was stirred at room temperature for 16 h. The mixture was diluted with H.sub.2O (30 mL) and extracted with EtOAc (2×20 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 40 g cartridge, 0-70% EtOAc/Petroleum ether) to provide 5-bromo-1-[[*(cis)*-3-hydroxy-3-methylcyclobutyl]-1H-1,3-benzodiazole-7-carbonitrile. MS=305.8/307.9 [M+H].sup.+.

Step 5: 1-[[*(cis)*-3-hydroxy-3-methylcyclobutyl]-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1H-1,3-benzodiazole-7-carbonitrile
##STR01359##

[2760] To a solution of 5-bromo-1-[[*(cis)*-3-hydroxy-3-methylcyclobutyl]-1H-1,3-benzodiazole-7-carbonitrile (1.15 g, 3.76 mmol) in 1,4-dioxane (20 mL) was added bis(pinacolato)diboron (4.77 g, 18.8 mmol), KOAc (737 mg, 7.51 mmol), and Pd(dppf)Cl.sub.2 (275 mg, 376 µmol). The mixture was degassed and purged with N.sub.2 for 10 min, and then stirred at 80° C. for 16 h under N.sub.2 atmosphere. After cooling to room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo to give 1-[[*(cis)*-3-hydroxy-3-methylcyclobutyl]-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1H-1,3-benzodiazole-7-carbonitrile, which was used in the subsequent step without further purification.

Step 6: 5-hydroxy-1-[[*(cis)*-3-hydroxy-3-methylcyclobutyl]-1H-1,3-benzodiazole-7-carbonitrile
##STR01360##

[2761] To a 0° C. solution of 1-[[*(cis)*-3-hydroxy-3-methylcyclobutyl]-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1H-1,3-benzodiazole-7-carbonitrile (1.30 g, 3.68 mmol) in THE (10 mL) and H.sub.2O (10 mL) was added Oxone (2.26 g, 3.68 mmol). The mixture was stirred at room temperature for 2 h, then was diluted with H.sub.2O (30 mL) and extracted with EtOAc (2×20 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 20 g cartridge, 0-80% EtOAc/Petroleum ether) to provide 5-hydroxy-1-[[*(cis)*-3-hydroxy-3-methylcyclobutyl]-1H-1,3-benzodiazole-7-carbonitrile (Intermediate A-117). MS=244.0 [M+H].sup.+.

General Procedure for Intermediate A-118

##STR01361##

Step 1: (*cis*)-3-{6-bromo-3H-[1,2,3]triazolo[4,5-b]pyridin-3-yl}-1-methylcyclobutan-1-ol
##STR01362##

[2762] To a solution of (*cis*)-3-[(3-amino-5-bromopyridin-2-yl)amino]-1-methylcyclobutan-1-ol (Intermediate A-93, 1.00 g, 3.67 mmol) in AcOH (10 mL) and MeCN (3 mL) was added NaNO.sub.2 (507 mg, 7.35 mmol). The mixture was stirred at room temperature for 3 h. The mixture was diluted with H.sub.2O (30 mL) and extracted with EtOAc (2×20 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give (*cis*)-3-{6-bromo-3H-[1,2,3]triazolo[4,5-b]pyridin-3-yl}-1-methylcyclobutan-1-ol, which was used in the subsequent step without further purification. MS=282.8/284.9 [M+H].sup.+.

Step 2: (*cis*)-1-methyl-3-[6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-3H-[1,2,3]triazolo[4,5-b]pyridin-3-yl]cyclobutan-1-ol

##STR01363##

[2763] A mixture of (cis)-3-{6-bromo-3H-[1,2,3]triazolo[4,5-b]pyridin-3-yl}-1-methylcyclobutan-1-ol (400 mg, 1.41 mmol), bis(pinacolato)diboron (431 mg, 1.70 mmol), KOAc (277 mg, 2.83 mmol) and Pd(dppf)Cl.sub.2 (103 mg, 141 μ mol) in 1,4-dioxane (10 mL) was degassed and purged with N.sub.2 for 10 min, and then stirred at 85° C. for 16 h. Solids were removed by filtration and the filtrate was concentrated in vacuo to give (cis)-1-methyl-3-[6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-3H-[1,2,3]triazolo[4,5-b]pyridin-3-yl]cyclobutan-1-ol, which was taken to the next step without further purification.

Step 3: 3-[(cis)-3-hydroxy-3-methylcyclobutyl]-3H-[1,2,3]triazolo[4,5-b]pyridin-6-ol

##STR01364##

[2764] To a solution of (cis)-1-methyl-3-[6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-3H-[1,2,3]triazolo[4,5-b]pyridin-3-yl]cyclobutan-1-ol (400 mg, 1.21 mmol) in THE (5 mL) and H.sub.2O (5 mL) was added Oxone (745 mg, 1.21 mmol). The mixture was stirred at room temperature for 1 h, and then was concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-85% EtOAc/Petroleum ether) to provide 3-[(cis)-3-hydroxy-3-methylcyclobutyl]-3H-[1,2,3]triazolo[4,5-b]pyridin-6-ol (Intermediate A-118).

MS=221.2 [M+H].sup.+.

General Procedure for Intermediate A-119

##STR01365##

Step 1: (cis)-3-(5-bromo-2-methyl-7-(trifluoromethyl)-1H-benzo[d]imidazol-1-yl)-1-methylcyclobutan-1-ol

##STR01366##

[2765] To a solution of (cis)-3-{[2-amino-4-bromo-6-(trifluoromethyl)phenyl]amino}-1-methylcyclobutan-1-ol (General procedure for Intermediate A-92, Step 3, 2.00 g, 5.90 mmol) and 1,1,1-trimethoxyethane (2.13 g, 17.7 mmol) in THF (30 mL) was added TsOH.Math.H.sub.2O (112 mg, 589 μ mol). The mixture was stirred at 60° C. for 16 h. The mixture was diluted with H.sub.2O (30 mL) and extracted with EtOAc (3 $\times$ 40 mL). The combined organic layers were washed with brine (50 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (Biotage 40 g cartridge, 0-50% EtOAc/Petroleum ether) to give (cis)-3-(5-bromo-2-methyl-7-(trifluoromethyl)-1H-benzo[d]imidazol-1-yl)-1-methylcyclobutan-1-ol. MS=363.0/365.0 [M+H].sup.+.

Step 2: (cis)-3-(5-(5,5-dimethyl-1,3,2-dioxaborinan-2-yl)-2-methyl-7-(trifluoromethyl)-1H-benzo[d]imidazole-1-yl)-1-methylcyclobutan-1-ol

##STR01367##

[2766] A mixture of (cis)-3-(5-bromo-2-methyl-7-(trifluoromethyl)-1H-benzo[d]imidazole-1-yl)-1-methylcyclobutan-1-ol (2.00 g, 5.51 mmol), 2-(5,5-dimethyl-1,3,2-dioxaborinan-2-yl)-5,5-dimethyl-1,3,2-dioxaborinane (2.49 g, 11.0 mmol), Pd(dppf)Cl.sub.2 (403 mg, 551 μ mol) and KOAc (1.62 g, 16.5 mmol) in 1,4-dioxane (30 mL) was degassed and purged with N.sub.2 (3 $\times$), and then the mixture was stirred at 95° C. for 16 h under N.sub.2 atmosphere. After cooling to room temperature, the mixture was diluted with H.sub.2O (30 mL) and extracted with EtOAc (3 $\times$ 40 mL). The combined organic layers were washed with brine (50 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (Biotage 40 g cartridge, 70-85% EtOAc/Petroleum ether) to give (cis)-3-(5-(5,5-dimethyl-1,3,2-dioxaborinan-2-yl)-2-methyl-7-(trifluoromethyl)-1H-benzo[d]imidazol-1-yl)-1-methylcyclobutan-1-ol. MS=329.1 [M-C.sub.5H.sub.8+H].sup.+.

Step 3: 1-((cis)-3-hydroxy-3-methylcyclobutyl)-2-methyl-7-(trifluoromethyl)-1H-benzo[d]imidazol-5-ol

##STR01368##

[2767] To a solution of (cis)-3-(5-(5,5-dimethyl-1,3,2-dioxaborinan-2-yl)-2-methyl-7-(trifluoromethyl)-1H-benzo[d]imidazol-1-yl)-1-methylcyclobutan-1-ol (2.00 g, 6.10 mmol) in THE

(30 mL) and H.sub.2O (10 mL) was added Oxone (3.75 g, 6.10 mmol). The mixture was stirred at room temperature for 1 h. The reaction mixture was quenched with saturated aqueous Na.sub.2SO.sub.3 solution (30 mL) and extracted with EtOAc (3×30 mL). The combined organic layers were washed with brine (50 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (Biotage 40 g cartridge, 0-100% EtOAc/Petroleum ether) to give 1-((cis)-3-hydroxy-3-methylcyclobutyl)-2-methyl-7-(trifluoromethyl)-1H-benzo[d]imidazole-5-ol (Intermediate A-119). MS=301.1 [M+H].sup.+.

General Procedure for Intermediate A-120

##STR01369## ##STR01370##

Step 1: (cis)-3-((2-bromo-6-nitrophenyl)amino)-1-methylcyclobutan-1-ol

##STR01371##

[2768] To a solution of 1-bromo-2-fluoro-3-nitro-benzene (11.0 g, 50.0 mmol) and (cis)-3-amino-1-methylcyclobutan-1-ol (8.26 g, 60.0 mmol, HCl salt) in DMSO (60 mL) was added DIEA (34.9 mL, 200 mmol). The mixture was stirred at 50° C. for 2 h. The reaction mixture was then quenched by addition of H.sub.2O (150 mL) and extracted with EtOAc (3×150 mL). The combined organic layers were washed with H.sub.2O (3×100 mL), dried over Na.sub.2SO.sub.4, filtered and concentrated under reduced pressure to give (cis)-3-[(2-bromo-6-nitrophenyl)amino]-1-methylcyclobutan-1-ol, which was taken to the next step without further purification.

MS=301.1/303.1 [M+H].sup.+.

Step 2: (cis)-3-((2-amino-6-bromophenyl)amino)-1-methylcyclobutan-1-ol

##STR01372##

[2769] To a solution of (cis)-3-[(2-bromo-6-nitrophenyl)amino]-1-methylcyclobutan-1-ol (16.4 g, 54.5 mmol) in MeOH (120 mL) and 12 M aqueous HCl (30 mL, 360 mmol) was added SnCl.sub.2 (31.0 g, 163 mmol). The mixture was stirred at room temperature for 2 h. The reaction mixture was adjusted to pH=8 with 10% aqueous NaOH solution and then extracted with EtOAc (3×250 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (Biotage 120 g cartridge, 0-45% EtOAc/Petroleum ether) to give (cis)-3-[(2-amino-6-bromophenyl)amino]-1-methylcyclobutan-1-ol. MS=271.2/273.2 [M+H].sup.+.

Step 3: (cis)-3-(7-bromo-1H-1,3-benzodiazol-1-yl)-1-methylcyclobutan-1-ol

##STR01373##

[2770] To a solution of (cis)-3-[(2-amino-6-bromophenyl)amino]-1-methylcyclobutan-1-ol (13.8 g, 50.7 mmol) in THE (130 mL) was added TsOH.Math.H.sub.2O (1.93 g, 10.1 mmol) and trimethyl orthoformate (11.0 mL, 101 mmol). The mixture was stirred at 60° C. for 2 h. After cooling to room temperature, the reaction mixture was quenched by addition of H.sub.2O (300 mL) and extracted with EtOAc (3×200 mL). The combined organic layers were washed with saturated aqueous NaHCO.sub.3 (2×200 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude product was triturated with MTBE (40 mL), then the filter cake was isolated by filtration and dried in vacuo to give (cis)-3-(7-bromo-1H-1,3-benzodiazol-1-yl)-1-methylcyclobutan-1-ol. MS=281.2/283.2 [M+H].sup.+.

Step 4: 7-bromo-1-[(cis)-3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl]-1H-1,3-benzodiazole

##STR01374##

[2771] To a solution of (cis)-3-(7-bromo-1H-1,3-benzodiazol-1-yl)-1-methylcyclobutan-1-ol (12.5 g, 44.4 mmol) in DCM (150 mL) was added TBSCl (20.1 g, 133 mmol) and imidazole (18.2 g, 267 mmol). The mixture was stirred at room temperature for 16 h. The reaction mixture was quenched by addition of H.sub.2O (200 mL), and then extracted with DCM (2×200 mL). The combined organic layers were washed with saturated aqueous NaHCO.sub.3 solution (2×200 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel

chromatography (Biotage 120 g cartridge, 0-8% EtOAc/Petroleum ether) to give 7-bromo-1-[(cis)-3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl]-1H-1,3-benzodiazole. MS=395.3/397.3. [M+H].sup.+.

Step 5: 7-ethenyl-1-[(cis)-3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl]-1H-1,3-benzodiazole

##STR01375##

[2772] A mixture of 7-bromo-1-[(cis)-3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl]-1H-1,3-benzodiazole (16.9 g, 42.7 mmol), potassium vinyltrifluoroborate (17.2 g, 128 mmol), K.sub.2CO.sub.3 (17.7 g, 128 mmol) and Pd(dppf)Cl.sub.2 (3.13 g, 4.27 mmol) in 1,4-dioxane (140 mL) and H.sub.2O (35 mL) was degassed and purged with N.sub.2 (3×), and then the mixture was stirred at 100° C. for 16 h under N.sub.2 atmosphere. The reaction mixture was concentrated in vacuo, then was diluted with H.sub.2O (200 mL) and extracted with EtOAc (3×200 mL). The combined organic layers were washed with brine (2×200 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 220 g cartridge, 0-20% EtOAc/Petroleum ether) to give 7-ethenyl-1-[(cis)-3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl]-1H-1,3-benzodiazole. MS=343.2 [M+H].sup.+.

Step 6: 1-[(cis)-3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl]-1H-1,3-benzodiazole-7-carbaldehyde

##STR01376##

[2773] Ozone was bubbled into a -78° C. solution of 7-ethenyl-1-[(cis)-3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl]-1H-1,3-benzodiazole (3.00 g, 8.76 mmol) in DCM (40 mL) for 20 min. The resulting solution was then purged with O₂ for an additional 10 min before being treated with TEA (886 mg, 8.76 mmol), and then the mixture was allowed to warm to room temperature for 10 min. The reaction mixture was adjusted to pH=5 with 3.0 M aqueous HCl solution, and then extracted with DCM (3×50 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 40 g cartridge, 0-15% EtOAc/Petroleum ether) to give 1-[(cis)-3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl]-1H-1,3-benzodiazole-7-carbaldehyde. MS=345.3. [M+H].sup.+.

Step 7: 7-(difluoromethyl)-1-[(cis)-3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl]-1H-1,3-benzodiazole

##STR01377##

[2774] To a 0° C. solution of 1-[(cis)-3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl]-1H-1,3-benzodiazole-7-carbaldehyde (2.54 g, 7.37 mmol) in DCM (30 mL) was added DAST (4.16 g, 25.8 mmol). The mixture was stirred at room temperature for 16 h. The reaction mixture was cooled to 0° C., quenched by addition of H.sub.2O (50 mL), and then extracted with DCM (2×50 mL). The combined organic layers were washed with saturated aqueous NaHCO.sub.3 solution (3×50 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (Biotage 40 g cartridge, 0-15% EtOAc/Petroleum ether) to give 7-(difluoromethyl)-1-[(cis)-3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl]-1H-1,3-benzodiazole. MS=367.3 [M+H].sup.+.

Step 8: 7-(difluoromethyl)-1-[(cis)-3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl]-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1H-1,3-benzodiazole

##STR01378##

[2775] To a solution of 7-(difluoromethyl)-1-[(cis)-3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl]-1H-1,3-benzodiazole (500 mg, 1.36 mmol) in THE (10 mL) was added pinacolborane (2.62 g, 20.5 mmol), 4,4'-di-tert-butyl-2,2'-dipyridyl (73.2 mg, 273 μmol) and (1,5-cyclooctadiene)(methoxy)iridium(I) dimer (91.6 mg, 136 μmol). The mixture was degassed and purged with N.sub.2 (3×), and then stirred at 80° C. for 32 h. After cooling to room temperature, the reaction was filtered and the filtrate was concentrated in vacuo to give 7-(difluoromethyl)-1-

[(cis)-3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl]-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1H-1,3-benzodiazole, which was taken to the next step without further purification. MS=493.2 [M+H].sup.+.

Step 9: 7-(difluoromethyl)-1-[(cis)-3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl]-1H-1,3-benzodiazol-5-ol

##STR01379##

[2776] To a 0° C. solution of 7-(difluoromethyl)-1-[(cis)-3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl]-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1H-1,3-benzodiazole (700 mg, 1.42 mmol) in H.sub.2O (5 mL) and THE (5 mL) was added Oxone (1.31 g, 2.13 mmol). The mixture was then stirred at room temperature for 2 h. The reaction mixture was quenched by addition of saturated aqueous Na.sub.2SO.sub.3 solution (100 mL) and then extracted with EtOAc (3×50 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure. The residue was purified by reverse phase preparative HPLC (Phenomenex Luna C.sub.18 column, 40-70% MeCN:10 mM TFA in H.sub.2O) to give 7-(difluoromethyl)-1-[(cis)-3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl]-1H-1,3-benzodiazol-5-ol. MS=383.2 [M+H].sup.+.

Step 10: 7-(difluoromethyl)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-1,3-benzodiazol-5-ol

##STR01380##

[2777] To a solution of 7-(difluoromethyl)-1-[(cis)-3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl]-1H-1,3-benzodiazol-5-ol (50.0 mg, 65.4 µmol) in 1,4-dioxane (0.5 mL) was added 4.0 M HCl in 1,4-dioxane (2.0 mL, 8.0 mmol). The mixture was stirred at room temperature for 2 h, then was concentrated in vacuo to give 7-(difluoromethyl)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-1,3-benzodiazol-5-ol (Intermediate A-120), which was taken to the next step without further purification. MS=269.1 [M+H].sup.+.

General Procedure for Intermediate A-121

##STR01381## ##STR01382##

Step 1: 1-(difluoromethoxy)-2-fluoro-3-nitro-benzene

##STR01383##

[2778] To a mixture of 2-fluoro-3-nitro-phenol (2.00 g, 12.7 mmol) and Cs.sub.2CO.sub.3 (12.4 g, 38.2 mmol) in DMF (20 mL) and H.sub.2O (4 mL) was added sodium chlorodifluoroacetate (19.4 g, 127 mmol). The mixture was stirred at 100° C. for 6 h. After cooling to room temperature, the reaction mixture was quenched by addition of H.sub.2O (30 mL) and extracted with EtOAc (30 mL×3). The combined organic layers were washed with brine (2×30 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (Biotage 20 g cartridge, 0-15% EtOAc/Petroleum ether) to give 1-(difluoromethoxy)-2-fluoro-3-nitro-benzene. MS=208.0 [M+H].sup.+.

Step 2: (cis)-3-{[2-(difluoromethoxy)-6-nitrophenyl]amino}-1-methylcyclobutan-1-ol

##STR01384##

[2779] To a solution of 1-(difluoromethoxy)-2-fluoro-3-nitro-benzene (1.20 g, 5.79 mmol) in DMSO (12 mL) was added DIEA (3.00 g, 23.2 mmol) and (cis)-3-amino-1-methylcyclobutan-1-ol (877 mg, 6.37 mmol, HCl salt). The mixture was stirred at 50° C. for 2 h. After cooling to room temperature, the reaction mixture was diluted with H.sub.2O (30 mL) and extracted with EtOAc (3×30 mL). The combined organic layers were washed with brine (3×5 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure to give (cis)-3-{[2-(difluoromethoxy)-6-nitrophenyl]amino}-1-methylcyclobutan-1-ol, which was taken to the next step without further purification. MS=288.9 [M+H].sup.+.

Step 3: (cis)-3-{[4-bromo-2-(difluoromethoxy)-6-nitrophenyl]amino}-1-methylcyclobutan-1-ol

##STR01385##

[2780] To a solution of (cis)-3-{[2-(difluoromethoxy)-6-nitrophenyl]amino}-1-methylcyclobutan-1-ol (1.80 g, 6.24 mmol) in DMF (40 mL) was added NBS (2.22 g, 12.5 mmol). The mixture was

stirred at 60° C. for 12 h. After cooling to room temperature, the reaction mixture was quenched by addition of H.sub.2O (120 mL) and extracted with EtOAc (3×50 mL). The combined organic layers were washed with brine (2×50 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-15% EtOAc/Petroleum ether) to give (cis)-3-{[4-bromo-2-(difluoromethoxy)-6-nitrophenyl]amino}-1-methylcyclobutan-1-ol. MS=367.0/369.0 [M+H].sup.+.

Step 4: (cis)-3-{[2-amino-4-bromo-6-(difluoromethoxy)phenyl]amino}-1-methylcyclobutan-1-ol
##STR01386##

[2781] To a solution of K.sub.2CO.sub.3 (1.88 g, 13.6 mmol) in H.sub.2O (5 mL) was added Na.sub.2S.sub.2O.sub.4 (1.42 g, 8.17 mmol) followed by (cis)-3-{[4-bromo-2-(difluoromethoxy)-6-nitrophenyl]amino}-1-methylcyclobutan-1-ol (1.00 g, 2.72 mmol) in MeOH (5 mL). The mixture was stirred at room temperature for 1 h. The reaction mixture was quenched by addition of H.sub.2O (50 mL) and extracted with EtOAc (3×50 mL). The combined organic layers were washed with brine (20 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give (cis)-3-{[2-amino-4-bromo-6-(difluoromethoxy)phenyl]amino}-1-methylcyclobutan-1-ol, which was taken to the next step without further purification. MS=337.0/339.0 [M+H].sup.+.

Step 5: (cis)-3-[5-bromo-7-(difluoromethoxy)-1H-1,3-benzodiazol-1-yl]-1-methylcyclobutan-1-ol
##STR01387##

[2782] To a solution of (cis)-3-{[2-amino-4-bromo-6-(difluoromethoxy)phenyl]amino}-1-methylcyclobutan-1-ol (720 mg, 2.14 mmol) in THE (11 mL) was added TsOH.Math.H.sub.2O (81.2 mg, 427 µmol) and trimethyl orthoformate (453 mg, 4.27 mmol). The mixture was stirred at 70° C. for 2 h. After cooling to room temperature, the reaction mixture was quenched by addition of H.sub.2O (20 mL) and extracted with EtOAc (3×20 mL). The combined organic layers were washed with brine (10 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The resulting solid was triturated with MTBE (20 mL) for 10 min, then the filter cake was isolated by filtration and dried in vacuo to give (cis)-3-[5-bromo-7-(difluoromethoxy)-1H-1,3-benzodiazol-1-yl]-1-methylcyclobutan-1-ol, which was taken to the next step without further purification. MS=347.0/348.9 [M+H].sup.+.

Step 6: (cis)-3-[7-(difluoromethoxy)-5-(5,5-dimethyl-1,3,2-dioxaborinan-2-yl)-1H-1,3-benzodiazol-1-yl]-1-methylcyclobutan-1-ol

##STR01388##

[2783] A mixture of (cis)-3-[5-bromo-7-(difluoromethoxy)-1H-1,3-benzodiazol-1-yl]-1-methylcyclobutan-1-ol (1.50 g, 4.32 mmol), 2-(5,5-dimethyl-1,3,2-dioxaborinan-2-yl)-5,5-dimethyl-1,3,2-dioxaborinane (2.93 g, 12.9 mmol), KOAc (1.06 g, 10.8 mmol) and Pd(dppf)Cl.sub.2.Math.CH.sub.2C.sub.12 (353 mg, 432 µmol) in 1,4-dioxane (25 mL) was degassed and purged with N.sub.2 (3×), and then the mixture was stirred at 90° C. for 12 h under N.sub.2 atmosphere. After cooling to room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo to give (cis)-3-[7-(difluoromethoxy)-5-(5,5-dimethyl-1,3,2-dioxaborinan-2-yl)-1H-1,3-benzodiazol-1-yl]-1-methylcyclobutan-1-ol, which was taken to the next step without further purification. MS=313.1 [M-C.sub.5H.sub.8+H].sup.+.

Step 7: 7-(difluoromethoxy)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-1,3-benzodiazol-5-ol
##STR01389##

[2784] To a 0° C. solution of (cis)-3-[7-(difluoromethoxy)-5-(5,5-dimethyl-1,3,2-dioxaborinan-2-yl)-1H-1,3-benzodiazol-1-yl]-1-methylcyclobutan-1-ol (2.00 g, 5.26 mmol) in MeOH (25 mL) was added dropwise 30% H.sub.2O.sub.2 in H.sub.2O (10.2 mL, 106 mmol). The resulting mixture was then stirred at room temperature for 1 h. The mixture was quenched with saturated aqueous Na.sub.2SO.sub.3 solution (20 mL) and extracted with EtOAc (3×20 mL). The combined organic layers were dried over Na.sub.2SO.sub.4 filtered and concentrated in vacuo to give 7-(difluoromethoxy)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-1,3-benzodiazol-5-ol (Intermediate A-121), which was taken to the next step without further purification. MS=285.1 [M+H].sup.+.

General Procedure for Intermediate A-122

##STR01390##

Step 1: (cis)-3-{6-bromo-2-methyl-3H-imidazo[4,5-b]pyridin-3-yl}-1-methylcyclobutan-1-ol

##STR01391##

[2785] A mixture of (cis)-3-[(3-amino-5-bromopyridin-2-yl)amino]-1-methylcyclobutan-1-ol (Intermediate A-93, 630 mg, 2.32 mmol), pTSA.Math.H.sub.2O (220 mg, 1.16 mmol) and triethyl orthoacetate (5.89 mL, 46.3 mmol) was stirred at 85° C. for 2 h. The mixture was cooled to room temperature, then diluted with H.sub.2O (20 mL), and extracted with EtOAc (3×20 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered and concentrated. The crude product was purified by silica gel chromatography (Biotage 50 g cartridge, 0-100% EtOAc/hexane) to give (cis)-3-{6-bromo-2-methylimidazo[4,5-b]pyridin-3-yl}-1-methylcyclobutan-1-ol.

MS=296.2/298.2 [M+H].sup.+.

Step 2: (cis)-1-methyl-3-[2-methyl-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-3H-imidazo[4,5-b]pyridin-3-yl]cyclobutan-1-ol

##STR01392##

[2786] A mixture of (cis)-3-{6-bromo-2-methylimidazo[4,5-b]pyridin-3-yl}-1-methylcyclobutan-1-ol (250 mg, 0.844 mmol), bis(pinacolato)diboron (257 mg, 1.01 mmol.),

Pd(dppf)Cl.sub.2.Math.CH.sub.2Cl.sub.2 (34 mg, 0.042 mmol) and KOAc (166 mg, 1.69 mmol) in 1,4-dioxane (8 mL) was purged with nitrogen for 10 min at room temperature, then stirred at 85° C. for 16 h. The reaction mixture was cooled to room temperature and filtered through a Celite pad. The filtrate was concentrated in vacuo to give (cis)-1-methyl-3-[2-methyl-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)imidazo[4,5-b]pyridin-3-yl]cyclobutan-1-ol, which was taken to the next step without further purification. MS=344.0 [M+H].sup.+.

Step 3: 2-methyl-3-[(cis)-3-hydroxy-3-methylcyclobutyl]-3H-imidazo[4,5-b]pyridin-6-ol

##STR01393##

[2787] To a mixture of (cis)-1-methyl-3-[2-methyl-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)imidazo[4,5-b]pyridin-3-yl]cyclobutan-1-ol (0.290 g, 0.844 mmol) in MeOH (4 mL) at 0° C. was added 30% H.sub.2O.sub.2 in H.sub.2O (0.216 mL, 2.11 mmol) dropwise. The mixture was warmed to room temperature and stirred for 2 h. The mixture was then concentrated in vacuo to give 2-methyl-3-[(cis)-3-hydroxy-3-methylcyclobutyl]imidazo[4,5-b]pyridin-6-ol (Intermediate A-122), which was taken to the next step without further purification. MS=234.2 [M+H].sup.+.

General Procedure for Intermediate A-123

##STR01394##

Step 1: 5-bromo-3-iodo-7-(trifluoromethyl)-1H-indazole

##STR01395##

[2788] To a 0° C. solution of 5-bromo-7-(trifluoromethyl)-1H-indazole (4.50 g, 17.0 mmol) in DMA (40 mL) was added 12 (3.42 mL, 17.0 mmol) and KOH (2.86 g, 50.9 mmol). The mixture was stirred at room temperature for 6 h. The mixture was diluted with H.sub.2O (40 mL) and extracted with EtOAc (2×30 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage, 40 g cartridge, 0-25% EtOAc/Petroleum ether) to give 5-bromo-3-iodo-7-(trifluoromethyl)-1H-indazole. MS=390.6/392.6 [M+H].sup.+.

Step 2: 3-[5-bromo-3-iodo-7-(trifluoromethyl)-1H-indazol-1-yl]cyclobutan-1-one

##STR01396##

[2789] To a solution of 5-bromo-3-iodo-7-(trifluoromethyl)-1H-indazole (3.40 g, 8.70 mmol) in acetone (30 mL) was added K.sub.2CO.sub.3 (3.61 g, 26.1 mmol) and 3-bromocyclobutanone (3.89 g, 26.1 mmol). The mixture was stirred at 30° C. for 12 h. The mixture was diluted with H.sub.2O (30 mL) and extracted with EtOAc (2×20 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 20 g cartridge, 0-20% EtOAc/Petroleum ether) to provide 3-(5-

bromo-3-iodo-7-(trifluoromethyl)-1H-indazol-1-yl)cyclobutan-1-one as the first eluting isomer. MS=458.7/460.6 [M+H].sup.+.

Step 3: (cis)-3-[5-bromo-3-iodo-7-(trifluoromethyl)-1H-indazol-1-yl]-1-methylcyclobutan-1-ol
##STR01397##

[2790] To a -10° C. solution of 3-(5-bromo-3-iodo-7-(trifluoromethyl)-1H-indazol-1-yl)cyclobutan-1-one (1.20 g, 2.62 mmol) in DCM (20 mL) under N.sub.2 atmosphere was added 3.0 M MeMgBr in 2-Me-THF (871 µL, 2.62 mmol). The mixture was stirred at room temperature for 1 h under N.sub.2 atmosphere. The reaction mixture was cooled to 0° C. and quenched by addition of H.sub.2O (5 mL), and then extracted with EtOAc (3×10 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Isco 20 g cartridge, 0-8% EtOAc/Petroleum ether) to provide (cis)-3-[5-bromo-3-iodo-7-(trifluoromethyl)-1H-indazol-1-yl]-1-methylcyclobutan-1-ol. MS=474.9/476.9 [M+H].sup.+.

Step 4: (cis)-3-[5-bromo-7-(trifluoromethyl)-1H-indazol-1-yl]-1-methylcyclobutan-1-ol
##STR01398##

[2791] To a 0° C. mixture of (cis)-3-(5-bromo-3-iodo-7-(trifluoromethyl)-1H-indazol-1-yl)-1-methylcyclobutan-1-ol (650 mg, 1.37 mmol) in HOAc (7 mL) and MeOH (7 mL) was added Zn (930 mg, 14.2 mmol) portion wise. The mixture was stirred at 0° C. for 30 min, then allowed to warm to room temperature and stirred for another 30 min. The mixture was filtered, and the filtrate was poured into H.sub.2O (50 mL) and extracted with EtOAc (2×50 mL). The combined organic layers were washed with brine (50 mL), dried over Na.sub.2SO.sub.4, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (SepaFlash 20 g cartridge, 0-20% EtOAc/Petroleum ether) to provide (cis)-3-[5-bromo-7-(trifluoromethyl)-1H-indazol-1-yl]-1-methylcyclobutan-1-ol. MS=348.9/350.9 [M+H].sup.+.

Step 5: (cis)-1-methyl-3-[5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-7-(trifluoromethyl)-1H-indazol-1-yl]cyclobutan-1-ol

##STR01399##

[2792] A mixture of (cis)-3-[5-bromo-7-(trifluoromethyl)-1H-indazol-1-yl]-1-methylcyclobutan-1-ol (380 mg, 1.09 mmol), bis(pinacolato)diboron (553 mg, 2.18 mmol), Pd(dppf)Cl.sub.2 (79.6 mg, 109 µmol), and KOAc (320 mg, 3.27 mmol) in 1,4-dioxane (10 mL) was degassed and purged with N.sub.2 (3×), and then the mixture was stirred at 90° C. for 2 h under N.sub.2 atmosphere. After cooling to room temperature, the mixture was filtered, and the filtrate was concentrated in vacuo to give (cis)-1-methyl-3-[5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-7-(trifluoromethyl)-1H-indazol-1-yl]cyclobutan-1-ol. MS=397.1 [M+H].sup.+.

Step 6: 1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-indazol-5-ol

##STR01400##

[2793] A 0° C. mixture of (cis)-1-methyl-3-[5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-7-(trifluoromethyl)-1H-indazol-1-yl]cyclobutan-1-ol (431 mg, 1.09 mmol) and Oxone (1.34 g, 2.18 mmol) in THE (10 mL) and H.sub.2O (10 mL) was stirred for 1 h. The 0° C. mixture was quenched with saturated aqueous Na.sub.2SO.sub.3 solution (20 mL) and extracted with EtOAc (2×20 mL). The combined organic layers were washed with brine (20 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure to provide 1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-indazol-5-ol, which was used in the subsequent step without further purification. MS=287.1 [M+H].sup.+.

Step 7: 2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-indazol-5-yl}oxy)ethyl 4-methylbenzene-1-sulfonate

##STR01401##

[2794] A mixture of 1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-indazol-5-ol (250 mg, 873 mol), 1,2-bis(tosyloxy)ethane (1.29 g, 3.49 mmol) and Cs.sub.2CO.sub.3 (854 mg, 2.62 mmol) in DMF (12 mL) was stirred at 50° C. for 4 h. After cooling to room temperature, the

reaction was quenched with ice water (30 mL) and extracted with EtOAc (2×40 mL). The combined organic layers were washed with brine (100 mL), dried over Na₂SO₄, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (SepaFlash 20 g cartridge, 0-60% EtOAc/Petroleum ether) to provide 2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-indazol-5-yl}oxy)ethyl 4-methylbenzene-1-sulfonate (Intermediate A-123). MS=485.0 [M+H]⁺.sup.+.
General Procedure for Intermediate A-124

##STR01402##

Step 1: (cis)-3-[(4-bromo-2-fluoro-6-nitrophenyl)amino]-1-methylcyclobutan-1-ol

##STR01403##

[2795] To a solution of 5-bromo-1,2-difluoro-3-nitro-benzene (12.0 g, 50.4 mmol) in MeCN (120 mL) was added (cis)-3-amino-1-methylcyclobutan-1-ol (6.94 g, 50.4 mmol, HCl salt) and DIEA (26.4 mL, 151 mmol). The mixture was stirred at 50° C. for 12 h. After cooling to room temperature, the reaction mixture was concentrated in vacuo. The residue was diluted with H₂O (60 mL) and extracted with EtOAc (4×80 mL). The combined organic layers were washed with brine (100 mL), dried over anhydrous Na₂SO₄, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Sepaflash 120 g cartridge, 0-15% EtOAc/Petroleum ether) to provide (cis)-3-[(4-bromo-2-fluoro-6-nitrophenyl)amino]-1-methylcyclobutan-1-ol. MS=319.0/320.9 [M+H]⁺.sup.+.
Step 2: (cis)-3-[(2-amino-4-bromo-6-fluorophenyl)amino]-1-methylcyclobutan-1-ol

##STR01404##

[2796] To a solution of (cis)-3-[(4-bromo-2-fluoro-6-nitrophenyl)amino]-1-methylcyclobutan-1-ol (2.00 g, 6.27 mmol) in EtOH (20 mL) and H₂O (10 mL) was added Fe (1.05 g, 18.8 mmol) and NH₄Cl (1.68 g, 31.3 mmol). The mixture was stirred at 80° C. for 2 h. After cooling to room temperature, solids were removed by filtration and the filtrate was extracted with EtOAc (3×20 mL). The combined organic layers were washed with brine (50 mL), dried over anhydrous Na₂SO₄, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Sepaflash 40 g cartridge, 0-10% EtOAc/Petroleum ether) to provide (cis)-3-[(2-amino-4-bromo-6-fluorophenyl)amino]-1-methylcyclobutan-1-ol. MS=289.1/291.1 [M+H]⁺.sup.+.
Step 3: (cis)-3-(5-bromo-7-fluoro-1H-1,3-benzodiazol-1-yl)-1-methylcyclobutan-1-ol

##STR01405##

[2797] To a solution of trimethyl orthoformate (2.27 mL, 20.8 mmol) and (cis)-3-[(2-amino-4-bromo-6-fluorophenyl)amino]-1-methylcyclobutan-1-ol (5.00 g, 17.3 mmol) in THE (50 mL) was added TsOH.Math.H₂O (329 mg, 1.73 mmol). After slowly warming to 50° C., the mixture was stirred at 50° C. for 5 h. The reaction mixture was cooled to room temperature, quenched with H₂O (30 mL), and adjusted to pH=8 by dropwise addition of saturated aqueous NaHCO₃ solution. The mixture was extracted with EtOAc (3×30 mL). The combined organic layers were washed with brine (50 mL), dried over Na₂SO₄, filtered, and concentrated under reduced pressure to give (cis)-3-(5-bromo-7-fluoro-1H-1,3-benzodiazol-1-yl)-1-methylcyclobutan-1-ol, which was used in the subsequent step without further purification. MS=299.0/301.1 [M+H]⁺.sup.+.
Step 4: (cis)-3-[5-(5,5-dimethyl-1,3,2-dioxaborinan-2-yl)-7-fluoro-1H-1,3-benzodiazol-1-yl]-1-methylcyclobutan-1-ol

##STR01406##

[2798] To a solution of (cis)-3-(5-bromo-7-fluoro-1H-benzo[d]imidazol-1-yl)-1-methylcyclobutan-1-ol (4.00 g, 13.4 mmol) and 2-(5,5-dimethyl-1,3,2-dioxaborinan-2-yl)-5,5-dimethyl-1,3,2-dioxaborinane (3.62 g, 16.1 mmol) in 1,4-dioxane (60 mL) under N₂ atmosphere was added KOAc (3.28 g, 33.4 mmol) and Pd(dppf)Cl₂ (978 mg, 1.34 mmol). The mixture was stirred at 80° C. for 16 h under N₂ atmosphere. The reaction mixture was allowed to cool to room temperature, then solids were removed by filtration and the filtrate was concentrated in vacuo to give (cis)-3-[5-(5,5-dimethyl-1,3,2-dioxaborinan-2-yl)-7-fluoro-1H-1,3-benzodiazol-1-yl]-1-

methylcyclobutan-1-ol, which was used in the subsequent step without further purification.

MS=333.1 [M+H].sup.+.

Step 5: 7-fluoro-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-1,3-benzodiazol-5-ol

##STR01407##

[2799] To a 0° C. solution of (cis)-3-[5-(5,5-dimethyl-1,3,2-dioxaborinan-2-yl)-7-fluoro-1H-1,3-benzodiazol-1-yl]-1-methylcyclobutan-1-ol (3.00 g, 9.03 mmol) and H.sub.2O (10 mL) in THE (30 mL) was added Oxone (2.78 g, 4.52 mmol) portion wise, and then the mixture was stirred at room temperature for 2 h. The reaction mixture was cooled to 0° C. then quenched by addition of saturated aqueous Na.sub.2SO.sub.3 solution (30 mL) and extracted with EtOAc (3×20 mL). The combined organic layers were washed with brine (50 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give 7-fluoro-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-1,3-benzodiazol-5-ol (Intermediate A-124). MS=237.2 [M+H].sup.+.

General Procedure for Intermediate B-1

##STR01408##

Step 1: tert-Butyl 5-ethenyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate

##STR01409##

[2800] To a solution of tert-butyl 5-bromo-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate (555 mg, 1.45 mmol) in DMF (7.3 mL) was added tributyl(vinyl)tin (640 µL, 2.2 mmol). The mixture was sparged with N.sub.2 for 10 min and tetrakis(triphenylphosphine)palladium (168 mg, 0.15 mmol) was added. The reaction mixture was flushed with N.sub.2, sealed, and heated to 80° C. After 16 h the reaction mixture was removed from heat and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 25 g cartridge, 0-100% EtOAc:hexanes) to afford tert-butyl 5-ethenyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate. .sup.1H NMR (500 MHz, CDCl.sub.3): δ 8.93 (s, 1H), 7.39 (s, 1H), 7.31 (d, J=8.0, 1H), 6.94 (d, J=8.0 Hz, 1H), 6.72 (dd, J=17.5, 10.8 Hz, 1H), 5.69 (d, J=17.6 Hz, 1H), 5.22 (d, J=10.9 Hz, 1H), 3.98-3.74 (m, 4H), 1.99-1.78 (m, 4H), 1.56 (s, 9H).

Step 2: tert-Butyl 5-formyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate

##STR01410##

[2801] To a solution of tert-butyl 5-ethenyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate (311 mg, 0.95 mmol) in 1,4-dioxane (8 mL) and water (4.5 mL) was added sodium periodate (405 mg, 1.89 mmol) followed by OsO.sub.4 (500 µL, 2.5% w/w in t-BuOH, 0.38 mmol). The reaction mixture was stirred for 3 h at room temperature and then diluted with EtOAc (30 mL) and quenched with saturated aqueous Na.sub.2SO.sub.3 (50 mL). The aqueous layer was then extracted with EtOAc (3×30 mL). The combined organic layers were washed with brine (50 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. Purification by normal phase silica gel chromatography (Biotage 25 g cartridge, 80% EtOAc:hexanes) afforded tert-butyl 5-formyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate. .sup.1H NMR (500 MHz, DMSO-d.sub.6): δ 9.93 (s, 1H), 8.34 (br s, 1H), 7.86 (s, 1H), 7.81 (d, J=7.9 Hz, 1H), 7.08 (d, J=8.0 Hz, 1H), 3.86 (dd, J=6.7, 4.9 Hz, 4H), 1.98-1.71 (m, 4H), 1.54 (s, 9H). MS=231.1

[M-C.sub.4H.sub.8+H].sup.+.

Step 3: tert-Butyl 5-(difluoromethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate

##STR01411##

[2802] To a solution of tert-butyl 5-formyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate (71.0 mg, 0.215 mmol) in DCM (1.4 mL) was added DAST (85 µL, 0.64 mmol) and the reaction was stirred at room temperature 16 h. The reaction mixture was then quenched by the addition of saturated aqueous NaHCO.sub.3. The reaction mixture was extracted with EtOAc (3×30 mL). The combined organic layers were washed with brine (30 mL), dried over Na.sub.2SO.sub.4, filtered and concentrated. The crude residue was purified by normal phase silica gel chromatography (Biotage 10 g cartridge, 40% EtOAc:hexanes) to afford tert-butyl 5-

(difluoromethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate. ¹H NMR (500 MHz, CDCl₃): δ 8.08 (s, 1H), 7.35 (s, 1H), 7.32-7.30 (m, 1H), 6.90 (d, J=8.0 Hz, 1H), 6.55 (t, J=56.6 Hz, 1H), 3.76 (dd, J=8.2, 4.3 Hz, 4H), 1.83-1.73 (m, 4H), 1.44 (s, 9H). MS=297.0 [M-C_{sub}.4H_{sub}.8+H]_{sup}+

Step 4: 5-(Difluoromethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR01412##

[2803] tert-Butyl 5-(difluoromethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate (32 mg, 0.091 mmol) was dissolved in a 4.0 M HCl in dioxane solution (300 µL, 1.2 mmol) and stirred at room temperature for 1 h. The reaction mixture was then concentrated in vacuo to afford 5-(difluoromethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-1), which was used without further purification. MS=253.1 [M+H]_{sup}+

General Procedure for Intermediate B-2

##STR01413##

Step 1: tert-butyl 5-bromo-2-oxo-1-{[2-(trimethylsilyl)ethoxy]methyl}-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate

##STR01414##

[2804] To a solution of tert-butyl 5-bromo-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate (1.55 g, 4.06 mmol) in DMF (21 mL) was added sodium hydride (195 mg, 60 wt % in mineral oil, 4.9 mmol). After one hour, SEMCl (935 µL, 5.3 mmol) was added dropwise over 30 sec and the reaction mixture was heated to 40° C. After 48 h the reaction was removed from heat and diluted with EtOAc (50 mL) and quenched with H_{sub}.2O (100 mL). The aqueous layer was then extracted with EtOAc (3×50 mL). The combined organic layers were then washed with brine (150 mL), dried over Na_{sub}.2SO_{sub}.4, filtered, and concentrated in vacuo. The crude reaction mixture was then purified by normal phase chromatography (Biotage 50 g cartridge 20% EtOAc:hexanes) to afford tert-butyl 5-bromo-2-oxo-1-{[2-(trimethylsilyl)ethoxy]methyl}-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate. ¹H NMR (500 MHz, CDCl₃): δ 7.46 (app d, J=1.9 Hz, 2H), 7.02-6.96 (m, 1H), 5.17 (s, 2H), 3.85 (m, 4H), 3.69-3.44 (m, 2H), 2.00-1.74 (m, 4H), 1.55 (s, 9H), 1.05-0.82 (m, 2H), 0.00 (s, 9H).

Step 2: tert-butyl 2-oxo-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1-{[2-(trimethylsilyl)ethoxy]methyl}-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate

##STR01415##

[2805] A solution tert-butyl 5-bromo-2-oxo-1-{[2-(trimethylsilyl)ethoxy]methyl}-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate (1.53 g, 3.1 mmol), KOAc (604 mg, 6.1 mmol), bis(pinacolato)diboron (937 mg, 2.2 mmol) in 1,4-dioxane (9.3 mL) was sparged with N_{sub}.2 for 10 min. Pd(dppf)Cl_{sub}.2 (113 mg, 0.15 mmol) was then added, and the reaction was heated to 90° C. for 2 h. The reaction mixture was then cooled to room temperature, filtered over Celite, washed with EtOAc (30 mL), and concentrated in vacuo. The crude material was then purified by normal phase chromatography (Biotage 25 g cartridge, 10% EtOAc:hexanes) to give tert-butyl 2-oxo-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1-{[2-(trimethylsilyl)ethoxy]methyl}-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate.

Step 3. Tert-butyl 5-hydroxy-2-oxo-1-{[2-(trimethylsilyl)ethoxy]methyl}-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate

##STR01416##

[2806] tert-Butyl 5-hydroxy-2-oxo-1-{[2-(trimethylsilyl)ethoxy]methyl}-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate (3.1 mmol) was dissolved in MeOH (20.5 mL). The solution was cooled to 0° C. and 30% aqueous hydrogen peroxide solution (1.05 mL, 9.3 mmol) was added dropwise over 3 min. The reaction mixture was then warmed to room temperature and stirred for 16 h. The reaction mixture was then extracted with EtOAc (3×50 mL). The combined organic layers were washed with brine (30 mL), dried over Na_{sub}.2SO_{sub}.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase chromatography (Biotage 25 g cartridge,

40% EtOAc in hexanes) to afford tert-butyl 5-hydroxy-2-oxo-1-{[2-(trimethylsilyl)ethoxy]methyl}-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate. ¹H NMR (500 MHz, CDCl₃, 35/36 H): δ 6.96 (d, J=8.4 Hz, 1H), 6.91 (s, 1H), 6.82 (d, J=8.4, 1H), 5.16 (s, 2H), 3.96-3.71 (m, 4H), 3.67-3.49 (m, 2H), 1.87-1.85 (m, 2H), 1.77-1.75 (m, 2H), 1.56 (s, 9H), 1.00-0.91 (m, 2H), 0.00 (s, 9H).

Step 4: tert-Butyl 5-(difluoromethoxy)-2-oxo-1-{[2-(trimethylsilyl)ethoxy]methyl}-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate

##STR01417##

[2807] A solution of tert-butyl 5-hydroxy-2-oxo-1-{[2-(trimethylsilyl)ethoxy]methyl}-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate (295 mg, 0.66 mmol) in MeCN (2 mL) and H₂O (2 mL) was cooled to 0° C. and KOH (738 mg, 13.1 mmol) was added. The mixture was stirred at 0° C. for 20 min, then diethyl bromodifluoromethyl phosphonate (316 mg, 1.18 mmol) was added. After 2 h, the reaction mixture was acidified to pH=3 by addition of 1.0 M aqueous HCl. The reaction mixture was extracted with EtOAc (3×30 mL). The combined organic layers were washed with brine (30 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. Purification by normal phase chromatography (Biotage 10 g cartridge, 30% EtOAc:hexanes) gave tert-butyl 5-(difluoromethoxy)-2-oxo-1-{[2-(trimethylsilyl)ethoxy]methyl}-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate. ¹H NMR (500 MHz, CDCl₃): δ 7.17-7.04 (m, 3H), 6.50 (t, J=73.8 Hz, 1H), 5.18 (s, 2H), 3.98-3.78 (m, 4H), 3.68-3.46 (m, 2H), 1.98-1.74 (m, 4H), 1.54 (s, 9H), 1.06-0.84 (m, 2H), 0.00 (s, 9H).

Step 5: 5-(Difluoromethoxy)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR01418##

[2808] To a solution of tert-butyl 5-(difluoromethoxy)-2-oxo-1-{[2-(trimethylsilyl)ethoxy]methyl}-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate (139 mg, 0.28 mmol) in DCM (1.4 mL) was added TFA (530 μL, 5.6 mmol), then was stirred at room temperature for 2 h. The reaction mixture was then concentrated in vacuo and dissolved in MeOH (1.5 mL) and ammonium hydroxide (500 μL). After 5 min, the reaction mixture was concentrated and triturated with MTBE (5 mL). The solid was then filtered and dried in vacuo to afford 5-(difluoromethoxy)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-2), which was used in the subsequent step without further purification. MS=269.2 [M+H]⁺.

General Procedure for Intermediates B-3 to B-5

##STR01419##

Step 1: tert-butyl 5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate

##STR01420##

[2809] To a solution of tert-butyl 4-formylpiperidine-1-carboxylate (15.0 g, 70.3 mmol) and (4-chlorophenyl)hydrazine (15.1 g, 84.4 mmol, HCl salt) in CHCl₃ (200 mL) was added TFA (15.6 mL, 211 mmol). The mixture was stirred at 70° C. for 1 h, then cooled to 0° C. and m-CPBA (35.7 g, 85% purity, 176 mmol) was added. The mixture was warmed to room temperature and stirred for 1 h. The reaction mixture was cooled to 0° C. then quenched with saturated aqueous Na₂CO₃ (300 mL) and extracted with EtOAc (3×200 mL). The combined organic layers were washed with saturated aqueous Na₂SO₄ (150 mL) and brine (3×100 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 330 g cartridge, 0-100% EtOAc:petroleum ether) to give tert-butyl 5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate (Intermediate B-3). MS=281.1 [M-C₄H₈+H]⁺.

Step 2: 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR01421##

[2810] A solution of tert-butyl 5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate (5.00 g, 14.9 mmol) and 4.0 M HCl in EtOAc (50 mL) was stirred for 1 h. The mixture was partially concentrated under reduced pressure to give a slurry. The suspension was filtered to

isolate a solid, which was triturated with EtOAc (5 mL) and dried in vacuo to give 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, HCl salt). MS=237.1 [M+H].sup.+.

Step 3: 5-chloro-1'-(2-chloroethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR01422##

[2811] A 0° C. solution of 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (3.50 g, 12.8 mmol, HCl salt) and 2-chloroacetaldehyde in H.sub.2O (2.68 mL, 40 wt %, 16.6 mmol) in DCM (50 mL) was stirred for 10 min, then NaBH(OAc).sub.3 (6.09 g, 28.7 mmol) was added. The mixture was stirred at 0° C. for 1 h and was then quenched with H.sub.2O (30 mL). The resulting biphasic mixture was adjusted to pH=9 with saturated aqueous Na.sub.2CO.sub.3 and was extracted with DCM (3×15 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered and concentrated in vacuo to give 5-chloro-1'-(2-chloroethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-5). ¹H NMR (400 MHz, DMSO-d.sub.6): δ 10.50 (s, 1H), 7.52 (br s, 1H), 7.25 (dd, J=8.4, 2.0 Hz, 1H), 6.84 (d, J=8.0 Hz, 1H), 3.75-3.72 (m, 2H), 2.90-2.56 (m, 6H), 1.79-1.70 (m, 4H). MS=299.1 [M+H].sup.+.

General Procedure for Intermediates B-6 & B-7

##STR01423##

Step 1: tert-butyl 1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate

##STR01424##

[2812] A mixture of tert-butyl 1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate (10.0 g, 33.1 mmol), Cs.sub.2CO.sub.3 (21.6 g, 66.1 mmol), and MeI (20.6 mL, 331 mmol) in DMF (100 mL) was degassed and purged with N.sub.2 (3×). The mixture was stirred at 30° C. for 12 h under N.sub.2 atmosphere and was then filtered and concentrated in vacuo to give tert-butyl 1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate, which was taken to the next step without further purification. MS=317.1 [M+H].sup.+.

Step 2: 1-methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR01425##

[2813] To a solution of tert-butyl 1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate (33.1 mmol) in DCM (100 mL) was added 4.0 M HCl in 1,4-dioxane (40 mL, 160 mmol). The mixture was stirred for 12 h. The reaction mixture was adjusted to pH=8 with the dropwise addition of TEA. The solution was filtered and concentrated in vacuo to give 1-methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one, which was taken to the next step without further purification (Intermediate B-6). MS=217.1 [M+H].sup.+.

Step 3: 1'-(2-hydroxyethyl)-1-methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR01426##

[2814] To a solution of 1-methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (33.1 mmol) and 2-bromoethanol (3.15 mL, 44.4 mmol) in MeCN (80 mL) was added KI (614 mg, 3.70 mmol) and K.sub.2CO.sub.3 (10.2 g, 74.0 mmol). The mixture was stirred at 50° C. for 12 h. After cooling to room temperature, the reaction mixture was filtered and concentrated in vacuo to give 1'-(2-hydroxyethyl)-1-methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-7), which was taken to the next steps without purification. MS=261.2 [M+H].sup.+.

General Procedure for Intermediates B-8 & B-9

##STR01427##

Step 1: tert-butyl 5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate

##STR01428##

[2815] To a solution of tert-butyl 5-bromo-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate (6.00 g, 15.7 mmol), Zn(CN).sub.2 (5.54 g, 47.2 mmol), DPPF (436 mg, 0.787 mmol) in DMF (120 mL) was added Pd(dba).sub.2 (905 mg, 1.57 mmol). The mixture was degassed and purged with N.sub.2 (3×), and then the mixture was stirred at 120° C. for 12 h. After cooling to room temperature, the reaction mixture was quenched by the addition of H.sub.2O (50 mL) and extracted with EtOAc (3×50 mL). The combined organic layers were washed with brine (3×50

mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 80 g cartridge, 0-50% EtOAc:petroleum ether) to give tert-butyl 5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate (Intermediate B-8). MS=272.2 [M-C.sub.4H.sub.8+H].sup.+.

Step 2: 2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile

##STR01429##

[2816] To a solution of tert-butyl 5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate (3.80 g, 11.6 mmol) in EtOAc (40 mL) was added 4.0 M HCl in EtOAc (10 mL, 40 mmol). The mixture was stirred for 2 h, and the resulting solid was isolated via filtration to give 2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (Intermediate B-9, HCl salt), which was used in subsequent steps without further purification. .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 11.19 (s, 1H), 9.33 (s, 2H), 7.70 (s, 1H), 7.60 (d, J=8.4 Hz, 1H), 7.04 (d, J=8.4 Hz, 1H), 3.43-3.41 (m, 2H), 3.40-3.39 (m, 2H), 2.13-2.07 (m, 2H), 1.99-1.92 (m, 2H). MS=228.3 [M+H].sup.+.

General Procedure for Intermediate B-10

##STR01430##

Step 1: 1'-(2-chloroethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR01431##

[2817] A mixture of 1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (1.00 g, 4.19 mmol, HCl salt), 2-chloroacetaldehyde in H.sub.2O (1.35 mL, 40 wt %, 8.38 mmol) and TEA (1.75 mL, 12.8 mmol) in DCM (10 mL) was stirred for 2 h, and then NaBH(OAc).sub.3 (1.78 g, 8.38 mmol) was added. After stirring for 30 min, the reaction mixture was quenched by the addition of H.sub.2O (10 mL) and extracted with DCM (3 $\times$ 10 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Isco 10 g cartridge, 0-100% EtOAc:petroleum ether) to give 1'-(2-chloroethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-10). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.36 (s, 1H), 7.47-7.45 (m, 1H), 7.19 (app t, J=7.6 Hz, 1H), 6.95 (app t, J=7.6 Hz, 1H), 6.85-6.83 (m, 1H), 3.77-3.74 (m, 2H), 2.97-2.67 (m, 6H), 1.79-1.64 (m, 4H). MS=265.1 [M+H].sup.+.

General Procedure for Intermediate B-11

##STR01432##

Step 1: tert-butyl 5-chloro-1-(2-hydroxyethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate

##STR01433##

[2818] To a solution of tert-butyl 5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate (Intermediate B-3, 500 mg, 1.48 mmol) and 2-bromoethanol (0.738 mL, 10.4 mmol) in DMF (5 mL) was added K.sub.2CO.sub.3 (1.03 g, 7.42 mmol). The mixture was stirred at 100° C. for 12 h. After cooling to room temperature, the reaction mixture was diluted with H.sub.2O (10 mL) and extracted with EtOAc (3 $\times$ 30 mL). The combined organic layers were washed with brine (40 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-100% EtOAc:petroleum ether) to give tert-butyl 5-chloro-1-(2-hydroxyethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate. MS=325.1 [M-C.sub.4H.sub.8+H].sup.+.

Step 2: 5-chloro-1-(2-hydroxyethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR01434##

[2819] A solution of tert-butyl 5-chloro-1-(2-hydroxyethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate (480 mg, 1.26 mmol) in 4.0 M HCl in 1,4-dioxane (2.0 mL, 8.0 mmol) was stirred at room temperature for 1 h. The mixture was concentrated in vacuo to give 5-chloro-1-(2-hydroxyethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-11), which was used in the subsequent step without further purification. MS=281.2 [M+H].sup.+.

[2820] The following intermediates in Table 12 were prepared according to procedures similar

those described for Intermediate B-11 using the appropriate starting materials.

General Procedure for Intermediates B-12 to B-15

TABLE-US-00015 TABLE 12 Exact Mass Intermediate # Structure Name [M + H].sup.+ B-12 [01435]  embedded image 1-(2-(2-hydroxyethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile Calc'd 272.1 Found 272.2 B-13 [01436]  embedded image 2-oxo-1-(2,2,2-trifluoroethyl)-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile Calc'd 310.1 Found 310.1 B-14 [01437]  embedded image 5-chloro-1-(2,2,2-trifluoroethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 319.1 Found 319.1 B-15 [01438]  embedded image 1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile Calc'd 242.1 Found 242.2

General Procedure for Intermediate B-16

##STR01439##

Step 1: tert-butyl 5-chloro-2-oxo-1-(3-oxocyclobutyl)-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate

##STR01440##

[2821] To a solution of tert-butyl 5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate (Intermediate B-3, 1.00 g, 2.97 mmol) and 3-bromocyclobutanone (1.33 g, 8.91 mmol) in DMF (13 mL) was added K.sub.2CO.sub.3 (820 mg, 5.94 mmol). The mixture was stirred at 50° C. for 4 h. The reaction mixture was quenched with H.sub.2O (30 mL), and then extracted with EtOAc (2×30 mL). The combined organic layers were washed with brine (2×30 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 40 g cartridge, 0-30% EtOAc:Petroleum ether) to give tert-butyl 5-chloro-2-oxo-1-(3-oxocyclobutyl)-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate. MS=349.0 [M-C.sub.4H.sub.8+H].sup.+.

Step 2: tert-butyl 5-chloro-1-(3-hydroxycyclobutyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate

##STR01441##

[2822] To a 0° C. solution of tert-butyl 5-chloro-2-oxo-1-(3-oxocyclobutyl)-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate (500 mg, 1.23 mmol) in THE (10 mL) was added NaBH.sub.4 (160 mg, 4.23 mmol). The mixture was allowed to warm to room temperature and stirred for 30 min. The reaction mixture was quenched with H.sub.2O (20 mL) and then extracted with EtOAc (2×20 mL). The combined organic layers were washed with brine (2×20 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 20 g cartridge, 20-30% EtOAc:petroleum ether) to give tert-butyl 5-chloro-1-(3-hydroxycyclobutyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate. .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 7.60 (d, J=2.0 Hz, 1H), 7.35-7.28 (m, 2H), 5.32 (d, J=6.8 Hz, 1H), 4.22-4.14 (m, 1H), 3.99-3.91 (m, 1H), 3.72-3.63 (m, 4H), 2.66-2.52 (m, 4H), 1.77-1.62 (m, 4H), 1.44 (s, 9H).

Step 3: 5-chloro-1-(3-hydroxycyclobutyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR01442##

[2823] To a solution of tert-butyl 5-chloro-1-(3-hydroxycyclobutyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate (400 mg, 0.983 mmol) in DCM (5 mL) was added TFA (1.50 g, 13.5 mmol). Then the mixture was stirred at room temperature for 1 h. The mixture was concentrated in vacuo, then diluted with H.sub.2O (5 mL), and adjusted to pH=7-8 via dropwise addition of saturated aqueous NaHCO.sub.3. The mixture was extracted with EtOAc (3×15 mL). The combined organic layers were washed with brine (20 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give 5-chloro-1-(3-hydroxycyclobutyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-16), which was used in the subsequent step without further purification. MS=307.2 [M+H].sup.+.

General Procedure for Intermediates B-17 and B-18

##STR01443##

Step 1: 5-chloro-1'-[(2S)-1-hydroxypropan-2-yl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one and 5-chloro-1'-[(2R)-2-hydroxypropyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR01444##

[2824] To a solution of 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 500 mg, 2.11 mmol) in EtOH (7 mL) was added (2R)-2-methyloxirane (0.592 mL, 8.45 mmol). The mixture was stirred at room temperature for 20 h. The reaction was concentrated in vacuo to give a mixture of 5-chloro-1'-[(2S)-1-hydroxypropan-2-yl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one and 5-chloro-1'-[(2R)-2-hydroxypropyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediates B-17 and B-18). MS=295.2 [M+H].sup.+.

[2825] The following intermediates in Table 13 were prepared according to procedures similar those described for Intermediates B-17 and B-18 using the appropriate starting materials.

General Procedure for Intermediates B-19 to B-22

TABLE-US-00016 TABLE 13 Intermediate Inter- Exact Starting mediate Mass Materials #
Structure Name [M + H].sup.+ Used B-19 and B-20 [01445]  + [01446]  (S)-1'-[(1-hydroxypropan-2-yl)-2-oxospiro[indoline-3,4'-piperidine]-5-carbonitrile and 1'-[(2R)-2-hydroxypropyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile Calc'd 286.1 Found 286.2 B-9 B-21 and B-22 [01447]  + [01448]  1'-[(2S)-1-hydroxypropan-2-yl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one and 1'-[(2R)-2-hydroxypropyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 261.1 Found 261.3 n/a

General Procedure for Intermediate B-23

##STR01449##

Step 1: tert-butyl 5-(difluoromethyl)-1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate

##STR01450##

[2826] To a 0° C. solution of tert-butyl 5-(difluoromethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate (Procedure for Intermediate B-1, Step 3, 4.50 g, 12.8 mmol) in THF (50 mL) was added sodium hydride (613 mg, 60 wt % in mineral oil, 15.3 mmol). After stirring for 30 min, iodomethane (2.18 g, 15.32 mmol) was added to the mixture. The mixture was allowed to warm to room temperature and stirred for 12 h. The reaction mixture was quenched with saturated aqueous NH₄Cl (50 mL) and extracted with EtOAc (3×50 mL). The combined organic layers were dried over Na₂SO₄, filtered, and concentrated under reduced pressure to give tert-butyl 5-(difluoromethyl)-1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate, which was used in the subsequent step without further purification. MS=311.1

[M-C₄H₈+H].sup.+.

Step 2: 5-(difluoromethyl)-1-methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR01451##

[2827] To a 0° C. solution of tert-butyl 5-(difluoromethyl)-1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate (4.00 g, 10.9 mmol) in EtOAc (20 mL) was added 4.0 M HCl in EtOAc (20.0 mL, 80.0 mmol). The mixture was allowed to warm to room temperature and stirred for 2 h. The resulting solids were isolated by filtration and dried in vacuo to give 5-(difluoromethyl)-1-methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-23, HCl salt), which was used in the subsequent steps without further purification. MS=267.1 [M+H].sup.+.

General Procedure for Intermediate B-24

##STR01452##

Step 1: tert-butyl 5-chloro-7-iodo-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate

##STR01453##

[2828] To a solution of 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 2.40 g, 10.1 mmol) in MeOH (20 mL) was added ICl (2.59 mL, 50.7 mmol). The mixture was

stirred at 75° C. for 36 h. After cooling to room temperature, the reaction mixture was quenched with saturated aqueous Na.sub.2S.sub.2O.sub.3 (20 mL). The mixture was concentrated under reduced pressure to remove MeOH. The aqueous solution was adjusted to pH=10 with saturated aqueous NaHCO.sub.3, and then tert-butoxycarbonyl tert-butyl carbonate (2.21 g, 10.1 mmol) was added. The mixture was stirred at room temperature for 2 h. The reaction mixture was extracted with EtOAc (3×20 mL). The combined organic layers were washed with brine (30 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 25 g cartridge, 0-10% MeOH:DCM) to give tert-butyl 5-chloro-7-iodo-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate.

Step 2: 5-chloro-7-iodo-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR01454##

[2829] To a solution of tert-butyl 5-chloro-7-iodo-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate (369 mg, 0.797 mmol) in MeOH (3 mL) was added 4.0 M HCl in MeOH (15 mL, 60 mmol). The mixture was stirred at room temperature for 2 h. The reaction mixture was concentrated in vacuo to give 5-chloro-7-iodo-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-24). MS=363.0 [M+H].sup.+.

General Procedure for Intermediate C-1

##STR01455##

Step 1: tert-butyl 6-(2-bromoethoxy)-1,2,3,4-tetrahydroquinoline-1-carboxylate

##STR01456##

[2830] To a vial holding tert-butyl 6-hydroxy-1,2,3,4-tetrahydroquinoline-1-carboxylate (0.500 g, 2.01 mmol) were added toluene (20 mL), bromoethanol (0.501 g, 4.01 mmol), then polymer-bound PPh.sub.3 (1.34 g, 3.00 mmol/g, 4.01 mmol). The reaction mixture was cooled to 0° C. and DIAD (0.608 g, 3.01 mmol) was added dropwise. The mixture was allowed to warm to room temperature and stirred for 16 h. The solids were removed by filtration and washed with EtOAc (25 mL). The filtrate was concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 25 g cartridge, 0-10% EtOAc:hexanes) to give tert-butyl 6-(2-bromoethoxy)-1,2,3,4-tetrahydroquinoline-1-carboxylate. MS=300.0

[M-C.sub.4H.sub.8+H].sup.+.

Step 2: tert-butyl 6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-1,2,3,4-tetrahydroquinoline-1-carboxylate

##STR01457##

[2831] A solution of tert-butyl 6-(2-bromoethoxy)-1,2,3,4-tetrahydroquinoline-1-carboxylate (200 mg, 0.561 mmol), 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 161 mg, 0.589 mmol, HCl salt) and K.sub.2CO.sub.3 (0.233 g, 1.68 mmol) in DMF (5.6 mL) was heated at 80° C. for 2 h. After cooling to room temperature, the reaction mixture was diluted with H.sub.2O (25 mL) and extracted with EtOAc (2×25 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 25 g cartridge, 0-15% MeOH:DCM) to give tert-butyl 6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-1,2,3,4-tetrahydroquinoline-1-carboxylate. MS=512.2 [M+H].sup.+.

Step 3: 5-chloro-1'-[2-(1,2,3,4-tetrahydroquinolin-6-yloxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR01458##

[2832] To a solution tert-butyl 6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-1,2,3,4-tetrahydroquinoline-1-carboxylate (177 mg, 0.346 mmol) in 1,4-dioxane (2 mL) was added 4.0 M HCl in 1,4-dioxane (2.16 mL, 8.64 mmol). The mixture was stirred for 1 h. The mixture was concentrated in vacuo to give 5-chloro-1'-[2-(1,2,3,4-tetrahydroquinolin-6-yloxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate C-1, HCl salt), which was used in the subsequent step without further purification. MS=412.1 [M+H].sup.+.

[2833] The following intermediates in Table 14 were prepared according to procedures similar to steps 1-3 described for Intermediate C-1 using the appropriate starting materials.

General Procedure for Intermediates C-2 to C-5

TABLE-US-00017 TABLE 14 Intermediate Inter- Exact Starting mediate Mass Materials #

Structure Name [M + H].sup.+ Used C-2 [01459]  embedded image 5-chloro-1'-[2-(2,3-dihydro-1H-isoindol-5-yloxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (HCl salt) Calc'd 398.2 Found 398.1 B-4 C-3 [01460]  embedded image 5-chloro-1'-[2-(1,2,3,4-tetrahydroisoquinolin-6-yloxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (HCl salt) Calc'd 412.2 Found 412.2 B-4 C-4 [01461]  embedded image 2-oxo-1'-[2-(1,2,3,4-tetrahydroquinolin-6-yloxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (HCl salt) Calc'd 403.2 Found 403.2 B-9 C-5 [01462]  embedded image 1-methyl-1'-[2-(1,2,3,4-tetrahydroisoquinolin-6-yloxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (HCl salt) Calc'd 392.2 Found 392.1 B-6

General Procedure for Intermediate D-1

##STR01463##

Step 1: 3-(bromomethyl)-1-methylcyclobutan-1-ol

##STR01464##

[2834] To a 0° C. solution of 3-(bromomethyl)cyclobutanone (3.00 g, 18.4 mmol) in THF (30 mL) was added 3.0 M MeMgBr in 2-MeTHF (6.8 mL, 20.4 mmol). The mixture was stirred at room temperature for 1 h. The reaction mixture was quenched by addition of saturated aqueous NH.sub.4Cl (30 mL), and then extracted with 3:1 DCM/MeOH (3×50 mL). The combined organic layers were washed with brine (100 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give 3-(bromomethyl)-1-methylcyclobutan-1-ol (Intermediate D-1), which was used in the subsequent steps without further purification. .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ=4.13 (s, 1H), 3.52 (d, J=7.6 Hz, 2H), 2.17-2.11 (m, 1H), 2.08-1.99 (m, 2H), 1.72-1.66 (m, 2H), 1.20 (s, 3H).

Example 1

5-Chloro-1'-(2-([2-(1-methanesulfonyl)ethyl]pyrimidin-5-yl)oxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 1) 5-chloro-1'-[2-([2-((1S) or (1R)-1-methanesulfonyl)ethyl]pyrimidin-5-yl)oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 2) and 5-chloro-1'-[2-([2-((1R) or (1S)-1-methanesulfonyl)ethyl]pyrimidin-5-yl)oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 3)

##STR01465##

Step 1: 5-Chloro-1'-(2-([2-(1-methanesulfonyl)ethyl]pyrimidin-5-yl)oxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR01466##

[2835] To a vial was added 5-(2-bromoethoxy)-2-(1-methanesulfonyl)pyrimidine (Intermediate A-75, 221 mg, 0.714 mmol), 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 150 mg, 0.55 mmol, HCl salt), followed by DMF (2.7 mL). K.sub.2CO.sub.3 (0.19 g, 1.37 mmol) was then added, and the reaction mixture was heated to 60° C. After 2 h, the reaction was removed from heat and diluted with EtOAc, then filtered over Celite. The filtrate was concentrated, and the residue was diluted in a 1:1 mixture of MeCN and H.sub.2O (3 mL). Purification by reverse phase preparative HPLC (Phenomenex Kinetex C.sub.18 column, 5-40% MeCN in water with 0.1% formic acid modifier) afforded 5-chloro-1'-(2-([2-(1-methanesulfonyl)ethyl]pyrimidin-5-yl)oxy)ethyl)-1H-spiro[indole-3,4'-piperidin]-2-one (Compound 1). .sup.1H NMR (500 MHz, DMSO-d.sub.6): δ 10.42 (s, 1H), 8.58 (s, 2H), 7.44 (d, J=2.1 Hz, 1H), 7.17 (dd, J=8.3, 2.1 Hz, 1H), 6.78 (d, J=8.3 Hz, 1H), 4.58 (q, J=7.2 Hz, 1H), 4.29-4.27 (m, 2H), 2.94 (s, 3H), 2.91-2.79 (m, 4H), 2.66-2.64 (m, 2H), 1.71-1.69 (m, 2H), 1.66-1.61 (m, 5H). MS=465.03 [M+H].sup.+.

Step 2: 5-chloro-1'-[2-([2-((1S) or (1R)-1-methanesulfonyl)ethyl]pyrimidin-5-yl)oxy)ethyl]-1,2-

dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 2) and 5-chloro-1'-[2-({2-[(1R) or (1S)-1-methanesulfonylethyl]pyrimidin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 3)

##STR01467##

[2836] 5-Chloro-1'-(2-{[2-(1-methanesulfonylethyl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one was separated by preparative chiral SFC (Daicel Chiralpak AD-3, 55% isopropanol with 0.1% NH₄OH in CO₂). The first eluting enantiomer of the title compound, 5-chloro-1'-[2-({2-[(1S) or (1R)-1-methanesulfonylethyl]pyrimidin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 2): ¹H NMR (400 MHz, DMSO-d₆): δ 10.45 (s, 1H), 8.61 (s, 2H), 7.47 (d, J=1.6 Hz, 1H), 7.20 (dd, J=8.4, 2.0 Hz, 1H), 6.81 (d, J=8.4 Hz, 1H), 4.62 (q, J=7.2 Hz, 1H), 4.31 (t, J=5.6 Hz, 2H), 2.98 (s, 3H), 2.93-2.82 (m, 4H), 2.71-2.62 (m, 2H), 1.78-1.71 (m, 2H), 1.70-1.63 (m, 5H). MS=465.1 [M+H]⁺. The second eluting enantiomer of the title compound, 5-chloro-1'-[2-({2-[(1R) or (1S)-1-methanesulfonylethyl]pyrimidin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 3): ¹H NMR (400 MHz, DMSO-d₆): δ 10.50 (s, 1H), 8.66 (s, 2H), 7.52 (s, 1H), 7.29-7.22 (m, 1H), 6.86 (d, J=8.3 Hz, 1H), 4.67 (q, J=7.0 Hz, 1H), 4.36 (br s, 2H), 3.03 (s, 3H), 2.98-2.87 (m, 4H), 2.76-2.67 (m, 2H), 1.79-1.78 (m, 2H), 1.72-1.70 (m, 5H). MS=465.1 [M+H]⁺.

Example 2

5-chloro-1'-(2-{[2-(1-methanesulfonylcyclopropyl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 4)

##STR01468##

[2837] To a solution of 5-(2-bromoethoxy)-2-(1-methanesulfonylcyclopropyl)pyrimidine (Intermediate A-76, 300 mg, 0.93 mmol) and 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 0.306 g, 1.12 mmol, HCl salt) in DMF (4.7 mL) was added K₂CO₃ (0.387 g, 2.80 mmol). The reaction mixture was heated to 60° C. for 2 h. The reaction was allowed to cool to room temperature, diluted in H₂O (20 mL) and extracted with EtOAc (3×30 mL). The combined organic layers were dried over Na₂SO₄, filtered, and concentrated in vacuo. Purification by reverse phase preparative HPLC (Phenomenex Kinetex C₁₈ column, 5-40% MeCN in water with 0.1% formic acid modifier) gave 5-chloro-1'-(2-{[2-(1-methanesulfonylcyclopropyl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 4). ¹H NMR (500 MHz, DMSO-d₆): δ 10.49 (s, 1H), 8.61 (s, 2H), 7.50 (d, J=2.1 Hz, 1H), 7.23 (dd, J=8.2, 2.1 Hz, 1H), 6.84 (d, J=8.3 Hz, 1H), 4.34 (t, J=5.6 Hz, 2H), 3.36 (s, 3H), 2.96-2.85 (m, 4H), 2.73-2.68 (m, 2H), 1.80-1.67 (i, 6H), 1.60-1.58 (i, 2H). MS=477.12 [M+H]⁺.

[2838] The following compounds in Table 15 were prepared according to procedures analogous to those described for Compound 4 using the appropriate starting materials.

TABLE-US-00018 TABLE 15 Exact Inter- IUPAC Mass mediates # Structure Name [M + H]⁺ Used HNMR Spectra 5 [01469]  5-(difluoro-methyl)-1'-(2-{[2-(1-methanesulfonyl-cyclopropyl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 493.2 Found 493.2 A-76 and B-1 ¹H NMR (500 MHz, CDCl₃, 25/26 H): δ 8.34 (s, 2H), 7.69 (s, 1H), 7.39 (s, 1H), 7.34-7.25 (m, 1H), 6.87 (d, J = 8.0 Hz, 1H), 6.54 (t, J = 56.5 Hz, 1H), 4.35 (t, J = 5.1 Hz, 2H), 3.30 (t, J = 11.8 Hz, 2H), 3.26 (s, 3H), 3.14 (t, J = 5.1 Hz, 2H), 3.06 (d, J = 11.4 Hz, 1H), 2.22 (s, 2H), 1.94-1.82 (m, 4H), 1.72-1.57 (m, 2H). 6 [01470]

 5-(difluoro-methoxy)-1'-(2-{[2-(1-methanesulfonyl-cyclopropyl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 509.2 Found 509.1 A-76 and B-2 ¹H NMR (500 MHz, DMSO-d₆): δ 10.39 (s, 1H), 8.55 (s, 2H), 8.07 (s, 1H), 7.03 (t, J = 74.7 Hz, 1H), 6.95 (d, J = 8.4, 1H), 6.79 (d, J = 8.4 Hz, 1H), 4.30 (t, J = 5.5 Hz, 2H), 3.29 (br s, 3H), 2.99-2.86 (m, 4H), 2.75-2.66 (m, 2H), 1.77-1.73 (m, 2H), 1.68-1.65 (m, 4H), 1.54-1.51 (m, 2H).

Example 3

5,7-dichloro-1'-[2-(4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 7)

##STR01471##

Step 1: tert-butyl 5,7-dichloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate

##STR01472##

[2839] To a solution of (2,4-dichlorophenyl)hydrazine (2.60 g, 12.2 mmol, HCl salt) in CHCl₃ (30 mL) was added TEA (1.30 mL, 9.36 mmol), tert-butyl 4-formylpiperidine-1-carboxylate (2.00 g, 9.36 mmol) and TFA (2.80 mL, 28.1 mmol). The mixture was stirred at 70° C. for 17 h. The mixture was cooled to 0° C. and m-CPBA (1.14 g, 85% purity, 5.63 mmol) was added in portions. Then the mixture was stirred at room temperature for 1 h. The reaction mixture was quenched with saturated aqueous Na₂CO₃ (60 mL) and extracted with EtOAc (3×40 mL). The combined organic layers were washed with saturated aqueous Na₂SO₄ (50 mL) and brine (2×50 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 25 g cartridge, 0-40% EtOAc:petroleum ether). The residue was further purified by reverse phase preparative HPLC (Phenomenex Luna C₁₈ column, 40-75% MeCN: 10 mM TFA in H₂O) to give tert-butyl 5,7-dichloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate. MS=315.0 [M-C₄H₈+H]⁺.

Step 2: 5,7-dichloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR01473##

[2840] To a solution of tert-butyl 5,7-dichloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1-carboxylate (280 mg, 0.75 mmol) in EtOAc (5 mL) was added 4.0 M HCl in EtOAc (10 mL, 40 mmol). The mixture was stirred at room temperature for 1 h. The mixture was concentrated under reduced pressure to give 5,7-dichloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one. MS=271.0/273.0 [M+H]⁺.

Step 3: 5,7-dichloro-1'-[2-(4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 7)

##STR01474##

[2841] A mixture of 5,7-dichloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (100 mg, 0.325 mmol, HCl salt), 1-(2-bromoethoxy)-4-methylsulfonyl-benzene (Intermediate A-3, 77.1 mg, 0.276 mmol) and K₂CO₃ (89.9 mg, 650 μmol) in MeCN (4 mL) was stirred at 70° C. for 3 h. After cooling to the reaction to room temperature, the reaction mixture was diluted with H₂O (10 mL) and extracted with EtOAc (2×15 mL). The combined organic layers were washed with brine (2×15 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Waters Xbridge BEH C₁₈ column, 30-60% MeCN:10 mM NH₄HCO₃ in H₂O) to give 5,7-dichloro-1'-[2-(4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 7): ¹H NMR (400 MHz, DMSO-d₆): δ 10.90 (br s, 1H), 7.85-7.83 (d, J=8.8 Hz, 2H), 7.53 (s, 1H), 7.43-7.42 (m, 1H), 7.19 (d, J=8.8 Hz, 2H), 4.25 (t, J=5.6 Hz, 2H), 3.16 (s, 3H), 2.92-2.86 (m, 4H), 2.70-2.74 (m, 2H), 1.80-1.78 (m, 4H). MS=469.1 [M+H]⁺.

[2842] The following compounds in Table 16 were prepared according to procedures similar to steps 1-3 described for Compound 7 using the appropriate starting materials.

TABLE-US-00019 TABLE 16 Inter- Exact Mass mediate # Structure IUPAC Name [M + H]⁺ Used 8 [01475]

 5-chloro-1'-[2-(4-methanesulfonylphenoxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-7-carbonitrile Calc'd 460.1 Found 460.1 A-3 9 [01476]

Example 4

5-chloro-1'-[(2S)-1-(4-methanesulfonylphenoxy)propan-2-yl]-1,2-dihydrospiro[indole-3,4'-

piperidin]-2-one (Compound 10) and 5-chloro-1'-[(2S)-2-(4-methanesulfonylphenoxy)propyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 11)

##STR01477##

[2843] To a mixture of 5-chloro-1'-[(2S)-1-hydroxypropan-2-yl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one and 5-chloro-1'-[(2R)-2-hydroxypropyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediates B-17 and B-18, 200 mg, 0.678 mmol) and 4-methylsulfonylphenol (175 mg, 1.02 mmol) in THE (1 mL) was added PPh.sub.3 (267 mg, 1.02 mmol). The mixture was cooled to 0° C., and DIAD (0.198 mL, 1.02 mmol) was added dropwise. The mixture was stirred at 25° C. for 3 h. The reaction mixture was cooled to 0° C. and quenched with H.sub.2O (10 mL), then extracted with EtOAc (3×10 mL). The combined organic layers were washed with brine (8 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-100% EtOAc:petroleum ether). The residue was further purified by reverse phase preparative HPLC (Waters Xbridge BEH C.sub.18 column, 30-50% MeCN:10 mM NH.sub.4HCO.sub.3 in H.sub.2O). The first eluting product, 5-chloro-1'-[(2S)-1-(4-methanesulfonylphenoxy)propan-2-yl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 10): .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.49 (s, 1H), 7.85 (d, J=8.8 Hz, 2H), 7.44 (s, 1H), 7.24-7.21 (m, 3H), 6.84 (d, J=8.0 Hz, 1H), 4.26-4.22 (m, 1H), 4.10-4.06 (m, 1H), 3.16 (s, 3H), 3.14-3.12 (m, 1H), 3.02-3.00 (m, 2H), 2.84-2.74 (m, 2H), 1.77-1.66 (m, 4H), 1.16 (d, J=6.4 Hz, 3H), MS=449.1 [M+H].sup.+.

The second eluting product, 5-chloro-1'-[(2S)-2-(4-methanesulfonylphenoxy)propyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 11): .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.49 (s, 1H), 7.83 (d, J=8.8 Hz, 2H), 7.47 (s, 1H), 7.24-7.19 (m, 3H), 6.83 (d, J=8.4 Hz, 1H), 4.89-4.85 (m, 1H), 3.16 (s, 3H), 2.90-2.88 (m, 2H), 2.78-2.76 (m, 1H), 2.70-2.63 (m, 3H), 1.75-1.64 (m, 4H), 1.30 (d, J=6.0 Hz, 3H). MS=449.1 [M+H].sup.+.

[2844] The following compounds in Table 17 were prepared according to procedures similar to steps described for Compound 10 and Compound 11 using the appropriate starting materials.

TABLE-US-00020	TABLE 17	Exact Step	2	Inter-	Mass Step	2	Elution mediates	#	Structure	IUPAC
Name	[M + H].sup.+	Column	Order	Used	12	[01478]		1'-[(2S)-1-(4-methanesulfonylphenoxy)propan-2-yl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one	Calc'd 415.1	
Found	415.2	Waters Xbridge BEH C.sub.18 column	1	sup.st	B-21 & B-22	13	[01479]		1'-[(2S)-2-(4-methanesulfonylphenoxy)propyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one	Calc'd 415.1
Found	415.1	Waters Xbridge BEH C.sub.18 column	2	sup.nd	B-21 & B-22					

Example 5

5-chloro-1'-(2-{[1-(3-methanesulfonylpropyl)-1H-indazol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 14)

##STR01480##

Step 1: 1-(3-methanesulfonylpropyl)-5-methoxy-1H-indazole

##STR01481##

[2845] A mixture of 5-methoxy-1H-indazole (200 mg, 1.35 mmol), 3-methylsulfonylpropyl methanesulfonate (292 mg, 1.35 mmol) and Cs.sub.2CO.sub.3 (880 mg, 2.70 mmol) in DMF (5 mL) was stirred at 80° C. for 2 h. The reaction mixture was cooled to 0° C. and quenched by the addition of H.sub.2O (20 mL), then extracted with EtOAc (3×10 mL). The combined organic layers were washed with brine (2×10 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by preparative TLC (SiO.sub.2, 100% EtOAc). The lower polarity (higher R.sub.f) product, desired 1-(3-methanesulfonylpropyl)-5-methoxy-1H-indazole: .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 7.96 (d, J=0.8 Hz, 1H), 7.59 (d, J=9.2 Hz, 1H), 7.18 (d, J=2.4 Hz, 1H), 7.06 (dd, J=9.2, 2.4 Hz, 1H), 4.50 (t, J=7.2 Hz, 2H), 3.78 (s, 3H), 3.08 (dd, J=10.4, 5.6 Hz, 2H), 2.96 (s, 3H), 2.26-2.18 (m, 2H). MS=269.1 [M+H].sup.+.

The higher polarity product, byproduct 2-(3-methanesulfonylpropyl)-5-methoxy-2H-indazole: .sup.1H NMR (400

MHz, DMSO-d₆): δ 8.21 (s, 1H), 7.51 (d, J=9.2 Hz, 1H), 7.00 (d, J=2.4 Hz, 1H), 6.90 (dd, J=9.2, 2.8 Hz, 1H), 4.50 (t, J=6.8 Hz, 2H), 3.75 (s, 3H), 3.11-3.07 (m, 2H), 2.98 (s, 3H), 2.36-2.28 (m, 2H). MS=269.1 [M+H].sup.+.

Step 2: 1-(3-methanesulfonylpropyl)-1H-indazol-5-ol

##STR01482##

[2846] To a 0° C. solution of 1-(3-methanesulfonylpropyl)-5-methoxy-1H-indazole (150 mg, 0.559 mmol) in DCM (8 mL) was added BBr.sub.3 (215 µL, 2.24 mmol) dropwise. The mixture was stirred at room temperature for 10 h. The reaction mixture was cooled to 0° C. and quenched by the addition of H.sub.2O (10 mL), then adjusted to pH=9 with saturated aqueous NaHCO.sub.3 and extracted with DCM (3×5 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo give 1-(3-methanesulfonylpropyl)-1H-indazol-5-ol. MS=255.1 [M+H].sup.+.

Step 3: 5-chloro-1'-(2-{[1-(3-methanesulfonylpropyl)-1H-indazol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 14)

##STR01483##

[2847] A mixture of 1-(3-methanesulfonylpropyl)-1H-indazol-5-ol (30 mg, 0.118 mmol), 5-chloro-1'-(2-chloroethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-5, 52.9 mg, 0.177 mmol), NaI (0.8 mg, 5.90 µmol) and K.sub.2CO.sub.3 (16.3 mg, 0.118 mmol) in acetone (1 mL) was stirred at 50° C. for 16 h. After cooling to the reaction to room temperature, the mixture was diluted with H.sub.2O (20 mL) and extracted with EtOAc (3×10 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Waters Xbridge BEH C.sub.18 column, 25-60% MeCN:10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 5-chloro-1'-(2-{[1-(3-methanesulfonylpropyl)-1H-indazol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 14): .sup.1H NMR (400 MHz, DMSO-d₆): δ 10.50 (s, 1H), 7.96 (s, 1H), 7.60 (d, J=9.2 Hz, 1H), 7.51 (d, J=2.0 Hz, 1H), 7.24-7.22 (m, 2H), 7.10 (d, J=8.8 Hz, 1H), 6.84 (d, J=8.0 Hz, 1H), 4.50 (t, J=6.8 Hz, 2H), 4.16 (t, J=6.4 Hz, 2H), 3.08 (t, J=8.0 Hz, 2H), 2.96 (s, 3H), 2.94-2.87 (m, 4H), 2.72-2.70 (m, 2H), 2.22-2.20 (m, 2H), 1.79-1.70 (m, 4H). MS=517.2 [M+H].sup.+.

[2848] The following compounds in Table 18 were prepared according to procedures similar to steps 1-3 described for Compound 14 using the appropriate starting materials. Step 1 separation of indazole regioisomers was performed using normal phase SiO.sub.2 column chromatography.

TABLE-US-00021
TABLE 18
Exact Mass Step 1 Elution Intermediate # Structure IUPAC Name [M + H].sup.+ Order Used
15 [01484]  5-chloro-1'-(2-{[1-(3-methanesulfonylpropyl)-1H-indazol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 453.1 Found 453.2
First; lower polarity product
16 [01485]  5-chloro-1'-(2-{[1-(propan-2-yl)-1H-indazol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 439.1 Found 439.3
First; lower polarity product

Example 6

5-chloro-1'-[2-(4-{[(cis)-3-hydroxycyclobutyl]sulfonyl}phenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 17)

##STR01486##

Step 1: 3-[(4-hydroxyphenyl)sulfanyl]cyclobutan-1-one

##STR01487##

[2849] To a solution of 4-sulfanylphenol (200 mg, 1.59 mmol) and 3-bromocyclobutanone (236 mg, 1.59 mmol) in acetone (2 mL) was added K.sub.2CO.sub.3 (219 mg, 1.59 mmol) and NaI (237 mg, 1.59 mmol). The mixture was stirred at 50° C. for 3 h. After cooling to room temperature, the reaction mixture was diluted with H.sub.2O (10 mL) and extracted with EtOAc (3×5 mL). The combined organic layers were washed with brine (10 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by preparative TLC (SiO.sub.2, 1:2 EtOAc:petroleum ether) to give 3-[(4-hydroxyphenyl)sulfanyl]cyclobutan-1-one. MS=195.0

[M+H].sup.+.

Step 2: 3-(4-hydroxybenzenesulfonyl)cyclobutan-1-one

##STR01488##

[2850] To a 0° C. solution of 3-[(4-hydroxyphenyl)sulfonyl]cyclobutan-1-one (50.0 mg, 0.257 mmol) in THE (2 mL) and H.sub.2O (1 mL) was added Oxone (316 mg, 0.514 mmol). The mixture was stirred at room temperature for 1 h. The reaction mixture was cooled to 0° C. and quenched by the addition of saturated aqueous Na.sub.2S.sub.2O.sub.3 (3 mL), then diluted with H.sub.2O (5 mL) and extracted with EtOAc (3×5 mL). The combined organic layers were washed with brine (5 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by preparative TLC (SiO.sub.2, 1:1 EtOAc:petroleum ether) to give 3-(4-hydroxybenzenesulfonyl)cyclobutan-1-one. MS=225.0 [M-H].sup.+.

Step 3: Preparation of 4-{[(cis)-3-hydroxycyclobutyl]sulfonyl}phenol

##STR01489##

[2851] To a 0° C. solution of 3-[(4-hydroxyphenyl)sulfonyl]cyclobutan-1-one (75.0 mg, 0.331 mmol) in THE (5 mL) was added NaBH.sub.4 (25 mg, 0.662 mmol). The mixture was stirred at room temperature for 1 h. The reaction mixture was quenched with saturated aqueous NH.sub.4Cl (5 mL), diluted with H.sub.2O (5 mL), and extracted with EtOAc (3×5 mL). The combined organic layers were washed with brine (5 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give crude 4-{[(cis)-3-hydroxycyclobutyl]sulfonyl}phenol, which was carried onto the next step without additional purification. MS=229.1 [M+H].sup.+.

Step 4: 5-chloro-1'-[2-(4-{[(cis)-3-hydroxycyclobutyl]sulfonyl}phenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 17)

##STR01490##

[2852] To a solution of 4-{[(cis)-3-hydroxycyclobutyl]sulfonyl}phenol (100 mg, 0.438 mmol) and 5-chloro-1'-(2-chloroethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-5, 196 mg, 0.657 mmol) in DMF (3 mL) was added Ag.sub.2O (203 mg, 0.876 mmol). The mixture was stirred at 50° C. for 16 h. After cooling to room temperature, the reaction mixture was diluted with H.sub.2O (10 mL) and extracted with EtOAc (3×5 mL). The combined organic layers were washed with brine (5 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by reverse phase preparative HPLC (Waters Xbridge BEH C.sub.18 column, 25-55% MeCN: 10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 5-chloro-1'-[2-(4-{[(cis)-3-hydroxycyclobutyl]sulfonyl}phenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 17). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.49 (s, 1H), 7.73 (d, J=8.8 Hz, 2H), 7.50 (d, J=1.6 Hz, 1H), 7.24-7.21 (m, 1H), 7.19 (d, J=9.2 Hz, 2H), 6.84 (d, J=8.4 Hz, 1H), 5.44 (d, J=6.4 Hz, 1H), 4.24 (t, J=6.4 Hz, 2H), 3.99-3.91 (m, 1H), 3.53-3.45 (m, 1H), 2.95-2.86 (m, 4H), 2.71-2.66 (m, 2H), 2.32-2.26 (m, 2H), 2.15-2.03 (m, 2H), 1.80-1.68 (m, 4H). MS=491.1 [M+H].sup.+.

Example 7

5-chloro-1-(2-hydroxyethyl)-1'-[2-(4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 18)

##STR01491##

[2853] To a solution of 5-chloro-1'-[2-(4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 69, 100 mg, 0.230 mmol) in DMF (1.0 mL) was added K.sub.2CO.sub.3 (159 mg, 1.15 mmol) and 2-bromoethanol (54.6 µL, 0.768 mmol). The mixture was stirred at 80° C. for 12 h. After cooling to room temperature, the mixture was filtered and concentrated in vacuo. The residue was purified by reverse phase preparative HPLC (Waters Xbridge BEH C.sub.18 column, 20-50% MeCN: 10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 5-chloro-1-(2-hydroxyethyl)-1'-[2-(4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 18). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 7.84 (d, J=8.8 Hz, 2H), 7.54 (d, J=2.4 Hz, 1H), 7.32 (dd, J=8.4, 2.0 Hz, 1H), 7.20 (d, J=8.8 Hz, 2H), 7.10 (d, J=8.4

Hz, 1H), 4.82 (t, J=6.0 Hz, 1H), 4.26 (t, J=6.0 Hz, 2H), 3.73-3.68 (m, 2H), 3.57-3.54 (m, 2H), 3.16 (s, 3H), 2.98-2.86 (m, 4H), 2.76-2.69 (m, 2H), 1.77 (t, J=5.2 Hz, 4H). MS=479.1 [M+H].sup.+.

[2854] The following compounds in Table 19 were prepared according to procedures similar to steps described for Compound 18 using the appropriate starting materials.

TABLE-US-00022 TABLE 19 Exact Mass Intermediated # Structure IUPAC Name [M + H].sup.+
Used 19 [01492]

 embedded image 1'-[2-(4-methanesulfonylphenoxy) ethyl]-1-[(cis)-3-hydroxycyclobutyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 471.1 Found 471.2

Compound 73 20 [01493]  embedded image 5-chloro-1'-[2-(4-methanesulfonylphenoxy) ethyl]-1-(2-methoxyethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 493.2 Found 493.2

Compound 69 21 [01494]  embedded image 1'-[2-(4-methanesulfonylphenoxy) ethyl]-1-(2-methoxyethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 459.2 Found 459.2

Compound 73

Example 8

5-chloro-1'-(2-(4-(methylsulfonyl)phenoxy)ethyl)-1-(oxetan-3-yl)spiro[indoline-3,4'-piperidin]-2-one (Compound 22)

##STR01495##

Step 1: tert-butyl 5-chloro-1-(oxetan-3-yl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate

##STR01496##

[2855] To a solution of tert-butyl 5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate (Intermediate B-3, 300 mg, 0.891 mmol) in DMF (5 mL) were added K.sub.2CO.sub.3 (369 mg, 2.67 mmol) and 3-iodooxetane (655 mg, 3.56 mmol). The mixture was stirred at 100° C. for 12 h. After cooling to room temperature, the reaction mixture was diluted with H.sub.2O (10 mL) and extracted with EtOAc (3×15 mL). The combined organic layers were washed with brine (10 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-18%

EtOAc:petroleum ether) to give tert-butyl 5-chloro-1-(oxetan-3-yl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate. MS=393.2 [M+H].sup.+.

Step 2: 5-chloro-1-(oxetan-3-yl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR01497##

[2856] A solution of tert-butyl 5-chloro-1-(oxetan-3-yl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate (200 mg, 509 mmol) in DCM (4 mL) and TFA (0.5 mL) was stirred at room temperature for 1 h. The reaction mixture was concentrated under reduced pressure to give 5-chloro-1-(oxetan-3-yl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one. MS=293.1 [M+H].sup.+.

Step 3: 5-chloro-1'-(2-(4-(methylsulfonyl)phenoxy)ethyl)-1-(oxetan-3-yl)spiro[indoline-3,4'-piperidin]-2-one (Compound 22)

##STR01498##

[2857] To a solution of 5-chloro-1-(oxetan-3-yl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (150 mg, 0.369 mmol, TFA salt) in MeCN (3 mL) were added K.sub.2CO.sub.3 (102 mg, 0.737 mmol) and 1-(2-bromoethoxy)-4-methanesulfonylbenzene (Intermediate A-3, 154 mg, 0.553 mmol). The mixture was stirred at 80° C. for 12 h. After cooling to room temperature, the reaction mixture was filtered and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Waters Xbridge BEH C.sub.18 column, 30-55% MeCN: 10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 5-chloro-1'-(2-(4-(methylsulfonyl)phenoxy)ethyl)-1-(oxetan-3-yl)spiro[indoline-3,4'-piperidin]-2-one (Compound 22). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 7.84 (d, J=8.8 Hz, 2H), 7.61 (s, 1H), 7.42-7.34 (m, 2H), 7.19 (d, J=8.8 Hz, 2H), 5.43-5.35 (m, 1H), 5.00-4.87 (m, 4H), 4.25 (t, J=5.6 Hz, 2H), 3.16 (s, 3H), 2.94-2.88 (m, 4H), 2.75-2.19 (m, 2H), 1.85-1.75 (m, 4H). MS=491.1 [M+H].sup.+.

[2858] The following compounds in Table 20 were prepared according to procedures similar to steps 1-3 described for Compound 22 using the appropriate starting materials.

TABLE-US-00023 TABLE 20 Exact Mass Intermediate # Structure IUPAC Name [M + H].sup.+
Used 23 [01499]  5-chloro-1'-{2-[4-(3-methanesulfonylpropanesulfonyl)phenoxy]ethyl}-1-methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 555.1 Found 555.2 A-10 and B-3 24 [01500]  5-bromo-1-(2-hydroxyethyl)-1'-[2-(4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 523.1 Found 523.0 A-3 25 [01501]  1'-[2-(4-methanesulfonylphenoxy)ethyl]-1-(2-methoxyethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile Calc'd 484.2 Found 484.2 A-3 and B-8

Example 9

1-(2-(4-(methylsulfonyl)phenoxy)ethyl)spiro[azepane-4,3'-indolin]-2'-one (Compound 26)

##STR01502##

Step 1: tert-butyl N-[2-(2-oxo-2,3-dihydro-1H-indol-3-yl)ethyl]carbamate

##STR01503##

[2859] To a solution of 2-(1H-indol-3-yl)ethan-1-amine (15 g, 93.6 mmol) in AcOH (90 mL) was added a solution of 12.0 M aqueous HCl (45 mL, 540 mmol) in DMSO (15 mL) dropwise. The mixture was stirred at room temperature for 3 h. The reaction mixture was diluted with H₂O (200 mL) and adjusted to pH=8 with the addition of solid Na₂CO₃. Boc₂O (21.8 g, 99.9 mmol) was added, and the mixture was stirred at room temperature for 16 h. The reaction mixture was filtered to give a crude product, which was isolated as a solid. The crude product was triturated with 4:1 petroleum ether:EtOAc and filtered to give tert-butyl N-[2-(2-oxo-2,3-dihydro-1H-indol-3-yl)ethyl]carbamate. MS=177.1 [M-C₅H₈O₂+H].sup.+.

Step 2: tert-butyl N-{2-[2-oxo-3-(prop-2-en-1-yl)-2,3-dihydro-1H-indol-3-yl]ethyl}carbamate

##STR01504##

[2860] To a 0° C. solution of tert-butyl N-[2-(2-oxo-2,3-dihydro-1H-indol-3-yl)ethyl]carbamate (4.50 g, 16.3 mmol) in DMA (60 mL) was added NaH (782 mg, 60 wt % in mineral oil, 19.5 mmol). After stirring for 30 min, 3-bromoprop-1-ene (1.97 g, 16.3 mmol) was added dropwise. The mixture was stirred at room temperature for 2.5 h. The reaction mixture was cooled to 0° C. and quenched by the addition of H₂O (20 mL), and then was extracted with EtOAc (3×20 mL). The combined organic layers were washed with brine (2×20 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 40 g cartridge, 0-18% EtOAc:petroleum ether) to give tert-butyl N-{2-[2-oxo-3-(prop-2-en-1-yl)-2,3-dihydro-1H-indol-3-yl]ethyl}carbamate. MS=217.1 [M-C₅H₈O₂+H].sup.+.

Step 3: tert-butyl N-{2-[3-(3-hydroxypropyl)-2-oxo-2,3-dihydro-1H-indol-3-yl]ethyl}carbamate

##STR01505##

[2861] To a 0° C. solution of tert-butyl N-{2-[2-oxo-3-(prop-2-en-1-yl)-2,3-dihydro-1H-indol-3-yl]ethyl}carbamate (4.10 g, 13.0 mmol) in THE (50 mL) was added 1.0 M BH₃.Math.THF in THE (38.9 mL, 38.9 mmol) dropwise. The mixture was stirred at 0° C. for 2 h under N₂ atmosphere. The reaction mixture was quenched by the addition of H₂O (30 mL) at 0° C., and then NaOH (1.04 g, 25.9 mmol) and H₂O in H₂O (5.27 mL, 30 wt %, 54.9 mmol) was added. The mixture was stirred at room temperature for 14 h. The reaction mixture was cooled to 0° C. and quenched by the addition of saturated aqueous Na₂SO₃ (20 mL) and extracted with EtOAc (3×30 mL). The combined organic layers were washed with saturated aqueous Na₂SO₃ (2×20 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 40 g cartridge, 0-100% EtOAc:petroleum ether) to give tert-butyl N-{2-[3-(3-hydroxypropyl)-2-oxo-2,3-dihydro-1H-indol-3-yl]ethyl}carbamate. MS=235.2 [M-C₅H₈O₂+H].sup.+.

Step 4: tert-butyl N-{2-[2-oxo-3-(3-oxopropyl)-2,3-dihydro-1H-indol-3-yl]ethyl}carbamate

##STR01506##

[2862] To a -78° C. solution of (COCl)₂ (825 µL, 9.42 mmol) in DCM (50 mL) was added

DMSO (981 μ L, 12.6 mmol) dropwise followed by tert-butyl N-{2-[3-(3-hydroxypropyl)-2-oxo-2,3-dihydro-1H-indol-3-yl]ethyl}carbamate (2.10 g, 6.28 mmol). After 30 min, TEA (2.62 mL, 18.8 mmol) was added. The mixture was warmed up to room temperature and stirred for 30 min. The reaction mixture was diluted with H₂O (20 mL) and extracted with DCM (3 $\times$ 10 mL). The combined organic layers were dried over Na₂SO₄, filtered and concentrated in vacuo to give crude tert-butyl N-{2-[2-oxo-3-(3-oxopropyl)-2,3-dihydro-1H-indol-3-yl]ethyl}carbamate, which was used in the subsequent step without further purification. MS=233.1

[M-C₅H₈O₂+H]⁺.

Step 5: 3-[3-(2-aminoethyl)-2-oxo-2,3-dihydro-1H-indol-3-yl]propanal

##STR01507##

[2863] To a solution of tert-butyl N-{2-[2-oxo-3-(3-oxopropyl)-2,3-dihydro-1H-indol-3-yl]ethyl}carbamate (600 mg, 1.81 mmol) in DCM (20 mL) was added TFA (10.0 mL, 135 mmol). The mixture was stirred at room temperature for 3 h. The reaction mixture was concentrated in vacuo to give 3-[3-(2-aminoethyl)-2-oxo-2,3-dihydro-1H-indol-3-yl]propanal, which was used in the subsequent step without further purification. MS=233.1 [M+H]⁺.

Step 6: 1',2'-dihydrospiro[azepane-4,3'-indol]-2'-one

##STR01508##

[2864] A solution of 3-[3-(2-aminoethyl)-2-oxo-2,3-dihydro-1H-indol-3-yl]propanal (1.81 mmol, TFA salt) in MeOH (20 mL) was adjusted to pH=6 by the dropwise addition of TEA. Then NaBH₃CN (210 mg, 3.35 mmol) was added, and the mixture was stirred at room temperature for 16 h. The reaction mixture cooled to 0° C. and was quenched by the addition H₂O (10 mL) and was then concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Waters Xbridge BEH C₁₈ column, 1-30% MeCN: 10 mM NH₄HCO₃ in H₂O) to give 1',2'-dihydrospiro[azepane-4,3'-indol]-2'-one. MS=217.1 [M+H]⁺.

Step 7: 1-(2-(4-(methylsulfonyl)phenoxy)ethyl)spiro[azepane-4,3'-indolin]-2'-one (Compound 26)

##STR01509##

[2865] A 70° C. solution of spiro[azepane-4,3'-indoline]-2'-one (50.0 mg, 0.231 mmol), 1-(2-bromoethoxy)-4-methanesulfonylbenzene (Intermediate A-3, 64.5 mg, 0.231 mmol) and K₂CO₃ (63.9 mg, 0.462 mmol) in MeCN (5 mL) was stirred for 10 h. After cooling to room temperature, the reaction mixture was concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Waters Welch Xtimate C₁₈ column, 5-25% MeCN: 0.04% HCl in H₂O) to give 1-(2-(4-(methylsulfonyl)phenoxy)ethyl)spiro[azepane-4,3'-indolin]-2'-one (Compound 26, HCl salt). ¹H NMR (400 MHz, DMSO-d₆): δ 11.08 (s, 1H), 10.52-10.46 (m, 1H), 7.91-7.88 (m, 2H), 7.52-7.40 (m, 1H), 7.25 (d, J=8.8 Hz, 2H), 7.23-7.18 (m, 1H), 7.00 (t, J=7.6 Hz, 1H), 6.86 (d, J=7.6 Hz, 1H), 4.59-4.54 (m, 2H), 3.92-3.80 (m, 1H), 3.67-3.60 (m, 4H), 3.36-3.34 (m, 1H), 3.17 (s, 3H), 2.43-2.38 (m, 1H), 2.22-1.88 (m, 5H).

MS=415.2 [M+H]⁺.

Example 10

5-chloro-1'-[2-[(1-methyl-2-oxo-3,4-dihydroquinolin-6-yl)oxy]ethyl]spiro[indoline-3,4'-piperidine]-2-one (Compound 27)

##STR01510##

[2866] To a solution of 6-hydroxy-1-methyl-1,2,3,4-tetrahydroquinolin-2-one (Intermediate A-70, 50.0 mg, 0.282 mmol) and 5-chloro-1'-(2-chloroethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-5, 169 mg, 0.564 mmol) in DMF (1 mL) was added K₂CO₃ (78.9 mg, 0.564 mmol). The mixture was stirred at 50° C. for 12 h under N₂ atmosphere, cooled to room temperature, and then filtered. The filtrate was purified by reverse phase preparative HPLC (Phenomenex Luna C₁₈ column, 5-25% MeCN: 0.04% HCl in H₂O) to give 5-chloro-1'-[2-[(1-methyl-2-oxo-3,4-dihydroquinolin-6-yl)oxy]ethyl]spiro[indoline-3,4'-piperidine]-2-one (Compound 27). ¹H NMR (400 MHz, DMSO-d₆): δ 10.53-10.49 (m, 1H), 7.50 (s, 1H), 7.23 (d, J=8.4 Hz, 1H), 7.00 (d, J=8.8 Hz, 1H), 6.88-6.83 (m, 3H), 4.10 (t, J=5.6 Hz, 2H), 3.22 (s,

3H), 2.91-2.78 (m, 6H), 2.67-2.64 (m, 2H), 2.44-2.42 (m, 2H), 1.83-1.66 (m, 4H). MS=440.3 [M+H].sup.+.

Example 11

1'-(1-(4-(methylsulfonyl)phenoxy)propan-2-yl)spiro[indoline-3,4'-piperidin]-2-one (Compound 28)
##STR01511##

[2867] A 50° C. mixture of 1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (150 mg, 0.742 mmol), 1-(4-methanesulfonylphenoxy)propan-2-one (Intermediate A-1, 169 mg, 0.742 mmol), Ti(i-PrO).sub.4 (211 mg, 0.742 mmol) and HOAc (4.0 mg, 0.074 mmol) in DCE (5 mL) was stirred for 30 min. After cooling to 0° C., NaBH(OAc).sub.3 (236 mg, 1.11 mmol) was added. The mixture was stirred at 50° C. for 16 h. After cooling to room temperature, the reaction mixture was quenched with addition of H.sub.2O (10 mL). The mixture was filtered, and the filtrate was extracted with EtOAc (3×20 mL). The combined organic layers were washed with brine (30 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Phenomenex Gemini NX-C.sub.18, 10-50% MeCN:10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 1'-(1-(4-(methylsulfonyl)phenoxy)propan-2-yl)spiro[indoline-3,4'-piperidin]-2-one (Compound 28). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.34 (s, 1H), 7.87-7.84 (m, 2H), 7.43-7.41 (m, 1H), 7.23-7.15 (m, 3H), 6.94-6.92 (m, 1H), 6.85-6.83 (m, 1H), 4.27-4.23 (m, 1H), 4.09-4.05 (m, 1H), 3.16 (s, 3H), 3.15-3.13 (m, 1H), 3.00-2.97 (m, 2H), 2.86-2.78 (m, 2H), 1.79-1.77 (m, 2H), 1.61-1.58 (m, 2H), 1.16 (d, J=6.4 Hz, 3H). MS=415.0 [M+H].sup.+.

Example 12

1-methyl-1'-{2-[(3-methyl-4-oxo-3,4-dihydroquinazolin-6-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 29)

##STR01512##

[2868] 1'-(2-hydroxyethyl)-1-methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-7, 130 mg, 0.50 mmol) and 6-hydroxy-3-methyl-3,4-dihydroquinazolin-4-one (88 mg, 0.50 mmol) were added to a 40 mL vial, followed by toluene (2.00 mL), polymer-bound PPh.sub.3 (167 mg, 78.6% PPh.sub.3 loading by weight, 0.500 mmol) and DIAD (120 mg, 0.600 mmol). The mixture was stirred at room temperature for 12 h, then was filtered and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Phenomenex Gemini NX-C.sub.18, 0-50% MeCN: 10 mM NH.sub.4OH in H.sub.2O) to give 1'-(1-(4-(methylsulfonyl)phenoxy)propan-2-yl)spiro[indoline-3,4'-piperidin]-2-one (Compound 29). .sup.1H NMR (400 MHz, CDCl.sub.3): δ 7.96 (s, 1H), 7.71 (d, J=2.85 Hz, 1H), 7.65 (d, J=8.99 Hz, 1H), 7.39 (dd, J=8.88, 2.74 Hz, 2H), 7.27-7.31 (m, 1H), 7.00-7.12 (m, 1H), 6.84 (d, J=7.67 Hz, 1H), 4.23-4.42 (m, 2H), 3.60 (s, 3H), 3.20 (s, 3H), 3.00-3.19 (m, 4H), 2.81-3.00 (m, 2H), 1.81-2.06 (m, 4H). MS=419.2 [M+H].sup.+.

[2869] The following compounds in Table 21 were prepared according to procedures similar to those described for Compound 29 using the appropriate starting materials.

TABLE-US-00024 TABLE 21 Exact Mass Intermediate # Structure IUPAC Name [M + H].sup.+

Used 30 [01513]  1-methyl-1'-{2-[(2-methyl-1-oxo-2,3-dihydro-1H-indol-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 406.2 Found 406.2 B-7 31

[01514]  1-methyl-1'-{2-[4-(1-methyl-1H-pyrazol-4-yl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 417.2 Found 417.2 B-7 32 [01515]

 5-(2-{1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2,3-dihydro-1λ.sup.6-benzothiophene-1,1-dione Calc'd 427.1 Found 427.2 B-7 33

[01516]  1-methyl-1'-{2-[(1-methyl-2-oxo-2,3-dihydro-1H-indol-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 406.2 Found 406.2 B-7 34

[01517]  1-methyl-1'-[2-(quinoxalin-6-yloxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 389.1 Found 389.2 B-7 35 [01518]  1-methyl-1'-{2-[4-(5-methyl-1,2,4-oxadiazol-3-yl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

one Calc'd 419.2 Found 419.2 B-7 36 [01519]  1'-{2-[(1-acetyl- 1,2,3,4-tetrahydroquinolin-6-yl)oxy]ethyl}-1-methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one
Calc'd 434.2 Found 434.3 B-7 37 [01520]  1-methyl-1'-{2-[4-(2-oxopyrrolidin-1-yl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one
Calc'd 420.2 Found 420.2 B-7

Example 13

5-chloro-1'-{2-[4-(1-methanesulfonylethyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 38), 5-chloro-1'-(2-{4-[(1S) or (1R)-1-methanesulfonylethyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 39), and 5-chloro-1'-(2-{4-[(1R) or (1S)-1-methanesulfonylethyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 40)

##STR01521##

Step 1: 1-(1-chloroethyl)-4-methoxybenzene

##STR01522##

[2870] To a 0° C. solution of 1-(4-methoxyphenyl)ethan-1-ol (5.00 g, 32.9 mmol) in DCM (50 mL) was added SOCl₂ (2.86 mL, 39.4 mmol). The mixture was allowed to warm to room temperature and stirred for 3 h. The reaction mixture was concentrated in vacuo to give 1-(1-chloroethyl)-4-methoxybenzene, which was used in the next step without further purification.

Step 2: 1-(1-methanesulfonylethyl)-4-methoxybenzene

##STR01523##

[2871] A mixture of 1-(1-chloroethyl)-4-methoxybenzene (6.00 g, 35.1 mmol) and sodium methanesulfinate (7.18 g, 70.3 mmol) in DMF (60 mL) was stirred at 80° C. for 16 h. The mixture was cooled to 0° C. and quenched by the addition of H₂O (50 mL) and extracted with EtOAc (3×50 mL). The combined organic layers were washed with H₂O (50 mL), brine (50 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 80 g cartridge, 0-50% EtOAc:petroleum ether) to give 1-(1-methanesulfonylethyl)-4-methoxybenzene. MS=135.1 [M-CH₃SO₂]⁺.

Step 3: 4-(1-methanesulfonylethyl)phenol

##STR01524##

[2872] To a solution of 1-(1-methanesulfonylethyl)-4-methoxybenzene (500 mg, 2.33 mmol) in MeCN (5 mL) was added NaI (1.40 g, 9.33 mmol) and TMSCl (1.18 mL, 9.33 mmol). The mixture was stirred at 80° C. for 16 h. After cooling to room temperature, the reaction mixture was quenched by the addition of H₂O (20 mL) and extracted with EtOAc (3×20 mL). The combined organic layers were washed with saturated aqueous Na₂S₂O₃ (60 mL), brine (60 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 25-50% EtOAc:petroleum ether) to give 4-(1-methanesulfonylethyl)phenol. MS=218.2

[M+NH₄]⁺.

Step 4: 5-chloro-1'-{2-[4-(1-methanesulfonylethyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 38)

##STR01525##

[2873] To a mixture of 5-chloro-1'-(2-chloroethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-5, 269 mg, 0.899 mmol) in acetone (10 mL) was added NaI (4.5 mg, 0.030 mmol), K₂CO₃ (82.8 mg, 0.599 mmol) and 4-(1-methanesulfonylethyl)phenol (120 mg, 0.599 mmol). The mixture was stirred at 50° C. for 12 h. After cooling to room temperature, the reaction mixture was diluted with H₂O (10 mL) and extracted with EtOAc (3×10 mL). The combined organic layers were washed with brine (10 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Phenomenex Gemini NX-C₁₈, 20-50% MeCN: 10 mM NH₄HCO₃ in H₂O) to give 5-chloro-1'-{2-[4-(1-methanesulfonylethyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-

piperidin]-2-one (Compound 38). ¹H NMR (400 MHz, DMSO-d₆): δ 10.49 (s, 1H), 7.51 (s, 1H), 7.36 (d, J=8.8 Hz, 2H), 7.23 (m, 1H), 6.99 (d, J=8.8 Hz, 2H), 6.84 (d, J=8.4 Hz, 1H), 4.47 (m, 1H), 4.14 (t, J=5.4 Hz, 2H), 2.91-2.84 (m, 4H), 2.77 (s, 3H), 2.73-2.65 (m, 1H), 1.83-1.67 (m, 5H), 1.59 (d, J=7.2 Hz, 3H). MS=463.3 [M+H].sup.+.

Step 5: 5-chloro-1'-(2-{4-[(1S) or (1R)-1-methanesulfonylethyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 39) and 5-chloro-1'-(2-{4-[(1R) or (1S)-1-methanesulfonylethyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 40)

##STR01526##

[2874] 5-chloro-1'-(2-[4-(1-methanesulfonylethyl)phenoxy]ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 38, 45 mg, 0.060 mmol) was separated by preparative chiral SFC (Chiralcel OD-3 column, 60% ethanol with 0.1% NH₄OH in CO₂). The first eluting enantiomer of the title compound, 5-chloro-1'-(2-{4-[(1S) or (1R)-1-methanesulfonylethyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 39): ¹H NMR (400 MHz, DMSO-d₆): δ 10.50 (s, 1H), 7.51 (d, J=2.0 Hz, 1H), 7.36 (d, J=8.8 Hz, 2H), 7.23 (dd, J=8.4, 2.2 Hz, 1H), 6.99 (d, J=8.8 Hz, 2H), 6.84 (d, J=8.4 Hz, 1H), 4.47 (q, J=7.2 Hz, 1H), 4.14 (t, J=5.6 Hz, 2H), 2.92-2.89 (m, 2H), 2.86-2.83 (m, 2H), 2.77 (s, 3H), 2.71-2.68 (m, 2H), 1.82-1.67 (m, 4H), 1.59 (d, J=7.2 Hz, 3H). MS=463.2 [M+H].sup.+.

The second eluting enantiomer of the title compound, 5-chloro-1'-(2-{4-[(1R) or (1S)-1-methanesulfonylethyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 40): ¹H NMR (400 MHz, DMSO-d₆): δ 10.49 (s, 1H), 7.51 (d, J=2.0 Hz, 1H), 7.36 (d, J=8.8 Hz, 2H), 7.23 (dd, J=8.4, 2.2 Hz, 1H), 6.99 (d, J=9.0 Hz, 2H), 6.84 (d, J=8.0 Hz, 1H), 4.47 (q, J=7.2 Hz, 1H), 4.14 (t, J=5.6 Hz, 2H), 2.95-2.88 (m, 2H), 2.85 (t, J=5.6 Hz, 2H), 2.77 (s, 3H), 2.74-2.68 (m, 2H), 1.82-1.68 (m, 4H), 1.60 (d, J=7.2 Hz, 3H). MS=463.2 [M+H].sup.+.

Example 14

5-chloro-1'-(2-[4-(3-methanesulfonylpropanesulfonyl)phenoxy]ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 41)

##STR01527##

[2875] To a solution of 4-(3-methanesulfonylpropanesulfonyl)phenol (Intermediate A-9, 60.0 mg, 0.216 mmol) in DMF (2 mL) was added K₂CO₃ (59.6 mg, 0.431 mmol) and 5-chloro-1'-(2-chloroethyl)-1,2-dihydrospiro [indole-3,4'-piperidin]-2-one (Intermediate B-5, 129 mg, 0.431 mmol). The mixture was stirred at 50° C. for 12 h. The reaction mixture was filtered and the filtrate was purified by reverse phase preparative HPLC (Waters Xbridge BEH C₁₈ column, 20-55% MeCN: 10 mM NH₄HCO₃ in H₂O) to give 5-chloro-1'-(2-[4-(3-methanesulfonylpropanesulfonyl)phenoxy]ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 41). ¹H NMR (400 MHz, DMSO-d₆): δ 10.48 (s, 1H), 7.82 (d, J=8.8 Hz, 2H), 7.51 (d, J=2.0 Hz, 1H), 7.25-7.21 (m, 3H), 6.84 (d, J=8.4 Hz, 1H), 4.26 (t, J=5.6 Hz, 2H), 3.44-3.37 (m, 2H), 3.24-3.18 (m, 2H), 2.97 (s, 3H), 2.94-2.84 (m, 4H), 2.75-2.70 (m, 2H), 1.99-1.94 (m, 2H), 1.77-1.73 (m, 4H). MS=541.2 [M+H].sup.+.

[2876] The following compounds in Table 22 were prepared according to procedures similar to those described for Compound 41 using the appropriate starting materials

TABLE-US-00025 TABLE 22 Exact Mass Intermediates # Structure IUPAC Name [M + H].sup.+

Used 42 [01528]  1'-(2-(3,5-difluoro-4-methanesulfonylphenoxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 437.1 Found 437.0 A-11 and B-10 43 [01529]

 5-chloro-1'-(2-[4-(cyclopropanesulfonyl)phenoxy]ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 461.1 Found 461.1 B-5 44 [01530]

 1'-(2-(4-difluoromethanesulfonylphenoxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 437.1 Found 437.2 A-15 and B-10

Example 15

4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)benzamide (Compound

45)

##STR01531##

Step 1: 4-(2-bromoethoxy)benzamide

##STR01532##

[2877] A mixture of 4-hydroxybenzamide (200 mg, 1.46 mmol), Cs.sub.2CO.sub.3 (950 mg, 2.92 mmol), and 1,2-dibromoethane (2.01 mL, 23.3 mmol) was stirred at 110° C. for 18 h in a sealed vial. After cooling to room temperature, the reaction mixture was diluted with DCM and filtered. The filter cake was then washed with acetone (3×5 mL) and the filtrate was concentrated in vacuo to give 4-(2-bromoethoxy)benzamide, which was taken to the next step without further purification. .sup.1H NMR (500 MHz, CDCl.sub.3): δ 7.81-7.76 (m, 2H), 6.95 (dd, J=8.8, 2.2 Hz, 2H), 6.17-5.19 (m, 2H), 4.34 (t, J=2.0 Hz, 2H), 3.66 (t, J=2.1 Hz, 2H). MS=243.9 [M+H].sup.+.

Step 2: 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)benzamide (Compound 45)

##STR01533##

[2878] A mixture of 4-(2-bromoethoxy)benzamide, (36.0 mg, 0.15 mmol), 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 40 mg, 0.15 mmol, HCl salt), and potassium carbonate (61 mg, 0.44 mmol) was taken up in DMF (0.6 mL). The mixture was stirred at 80° C. for 3 h. After cooling to room temperature, the reaction mixture was diluted with H.sub.2O (0.15 mL) and formic acid (3 drops) was added. The mixture was stirred at room temperature for an additional 10 min and solids were removed by filtration. The filtrate was purified reverse phase preparative HPLC (Phenomenex Kinetex C.sub.18 column, 5-30% MeCN:H.sub.2O with 0.1% formic acid modifier) to give 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)benzamide (Compound 45). .sup.1H NMR (500 MHz, DMSO-d.sub.6): δ 10.52 (s, 1H), 7.84-7.78 (m, 3H), 7.51 (s, 1H), 7.25 (d, J=8.3 Hz, 1H), 7.17 (s, 1H), 7.02 (d, J=8.3 Hz, 2H), 6.85 (d, J=8.3 Hz, 1H), 4.23 (s, 2H), 3.15-2.66 (m, 6H), 2.13-1.60 (m, 4H). MS=400.1 [M+H].sup.+.

[2879] The following compounds in Table 23 were prepared according to procedures analogous to steps 1-2 described for Compound 45 using the appropriate starting materials.

TABLE-US-00026 TABLE 23 Exact Mass Intermediate # Structure IUPAC Name [M + H].sup.+

Used 46 [01534]  4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-fluorobenzamide Calc'd 418.1 Found 418.1 B-4 47 [01535]

 5-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)pyridine-2-carboxamide Calc'd 401.1 Found 401.2 B-4 48 [01536] 

5-chloro-1'-[2-(2-fluoro-4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 453.1 Found 453.0 B-4 49 [01537] 

4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-methylbenzamide Calc'd 414.2 Found 414.2 B-4

Example 16

5-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-methanesulfonylbenzonitrile (Compound 50)

##STR01538##

Step 1: 2-fluoro-5-(2-hydroxyethoxy)benzonitrile

##STR01539##

[2880] A mixture of 2-fluoro-5-hydroxybenzonitrile (1.00 g, 7.29 mmol), K.sub.2CO.sub.3 (3.02 g, 21.9 mmol), and ethylene carbonate (20.6 mL, 24.1 mmol) was taken up in DMF (10 mL). The mixture was stirred at 110° C. for 18 hours. After cooling to room temperature, the reaction mixture was filtered to remove solids and concentrated in vacuo. The residue was taken up in EtOAc (15 mL), and then washed with H.sub.2O (5 mL) and brine (5 mL). The organic layer was dried over Na.sub.2SO.sub.4, filtered, and concentrated to give 2-fluoro-5-(2-hydroxyethoxy)benzonitrile, which was taken to the next step without further purification. .sup.1H NMR (500 MHz,

CDCl₃.sub.3): δ 7.18-7.11 (m, 2H), 7.10 (dd, J=5.2, 2.5 Hz, 1H), 4.10-4.04 (m, 2H), 3.98 (t, J=4.4 Hz, 2H), 1.98 (s, 1H).

Step 2: 5-(2-hydroxyethoxy)-2-(methylsulfanyl)benzonitrile

##STR01540##

[2881] A mixture of 2-fluoro-5-(2-hydroxyethoxy)benzonitrile (875 mg, 4.83 mmol), aqueous sodium methanethiolate (6.77 g, 15 wt %, 14.5 mmol), and DMF (6.7 mL) was stirred at 60° C. for 18 h. After cooling to room temperature, the reaction mixture was diluted with brine (10 mL) and extracted EtOAc (3×10 mL). The combined organic layers were dried over Na₂SO₄, filtered, and concentrated in vacuo to give 5-(2-hydroxyethoxy)-2-(methylsulfanyl)benzonitrile, which was taken to the next step without further purification. ¹H NMR (500 MHz, CDCl₃, 10/11 H): δ 7.36 (d, J=8.8 Hz, 1H), 7.15 (d, J=2.8 Hz, 1H), 7.11 (dd, J=8.8, 2.8 Hz, 1H), 4.08 (dd, J=5.1, 3.8 Hz, 2H), 3.98 (t, J=4.4 Hz, 2H), 2.52 (s, 3H).

Step 3: 5-(2-hydroxyethoxy)-2-methanesulfonylbenzonitrile

##STR01541##

[2882] To a 0° C. suspension of 5-(2-hydroxyethoxy)-2-(methylsulfanyl)benzonitrile (169 mg, 0.808 mmol) in DCM (3 mL) was added m-CPBA (432 mg, 2.50 mmol) in a single portion. The mixture was allowed to slowly warm to room temperature and stirred for 18 h. The reaction mixture was diluted with saturated aqueous NaHCO₃ (10 mL) and allowed to stir for an additional 15 min. The organic layer was separated, and the aqueous layer was extracted with EtOAc (3×10 mL). The combined organic layers were dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 10 g cartridge, 0-100% EtOAc:Hexanes) to give 5-(2-hydroxyethoxy)-2-methanesulfonylbenzonitrile. ¹H NMR (500 MHz, CDCl₃, 10/11 H): δ 8.12 (d, J=8.9 Hz, 1H), 7.43 (s, 1H), 7.30 (dd, J=9.0, 2.6 Hz, 1H), 4.24 (t, J=4.4 Hz, 2H), 4.07 (t, J=4.5 Hz, 2H), 3.27 (s, 3H).

Step 4: 2-(3-cyano-4-methanesulfonylphenoxy)ethyl methanesulfonate

##STR01542##

[2883] Methanesulfonyl chloride (70 µL, 0.82 mmol) was added dropwise via syringe to a 0° C. solution of 5-(2-hydroxyethoxy)-2-methanesulfonylbenzonitrile (132 mg, 0.548 mmol) and DIEA (0.19 mL, 0.82 mmol) in DCM (2 mL). The mixture was allowed to slowly warm to room temperature and stirred for 3 h. The reaction mixture was diluted with saturated aqueous NH₄Cl (5 mL) and stirred for an additional 15 min. The organic layer was separated, and the aqueous layer was extracted with DCM (3×2 mL). The combined organic layers were dried over Na₂SO₄, filtered, and concentrated in vacuo to give 2-(3-cyano-4-methanesulfonylphenoxy)ethyl methanesulfonate, which was taken to the next step without further purification.

Step 5: 5-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-methanesulfonylbenzonitrile (Compound 50)

##STR01543##

[2884] A mixture of 2-(3-cyano-4-methanesulfonylphenoxy)ethyl methanesulfonate (51.0 mg, 0.16 mmol), 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 40.0 mg, 0.146 mmol, HCl salt) and K₂CO₃ (50.6 mg, 0.366 mmol), in DMF (0.6 mL) was stirred at 80° C. for 3 h. After cooling to room temperature, the reaction mixture was diluted with H₂O (0.15 mL) and purified by reverse phase preparative HPLC (Phenomenex Kinetex C₁₈ column, 10-30% MeCN:H₂O with 0.1% formic acid modifier) to give 5-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-methanesulfonylbenzonitrile (Compound 50). ¹H NMR (500 MHz, DMSO-d₆): δ 10.49 (s, 1H), 8.02 (d, J=8.9 Hz, 1H), 7.83 (d, J=2.6 Hz, 1H), 7.58-7.44 (m, 2H), 7.23 (dd, J=8.2, 2.0 Hz, 1H), 6.84 (d, J=8.2 Hz, 1H), 4.34 (t, J=5.6 Hz, 2H), 3.11 (br s, 3H), 2.97-2.83 (m, 4H), 2.74-2.65 (m, 2H), 1.83-1.74 (m, 2H), 1.74-1.61 (m, 2H). MS=460.0 [M+H]⁺.

Example 17

4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-methylbenzamide (Compound 51)

##STR01544##

Step 1: 5-hydroxy-N,N-dimethylpyridine-2-carboxamide

##STR01545##

[2885] To a mixture of 5-hydroxypyridine-2-carboxylic acid (500 mg, 3.59 mmol) and dimethylamine hydrochloride (440 mg, 5.40 mmol) in DMF (2.5 mL) was added DIEA (1.88 mL, 10.8 mmol). The mixture was stirred for 10 min, then HATU (1.64 g, 4.3 mmol) was added in a single portion. After stirring for 18 h, the reaction mixture was diluted with brine (5 mL) and filtered to remove solids. The filtrate was extracted with EtOAc (3×5 mL). The combined organic layers were dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 10 g cartridge, 0-20% MeOH:DCM) to give 5-hydroxy-N,N-dimethylpyridine-2-carboxamide. ¹H NMR (500 MHz, CDCl₃, 9/10 H): δ 8.07 (d, J=2.8 Hz, 1H), 7.41 (d, J=8.5 Hz, 1H), 7.08 (dd, J=8.5, 2.8 Hz, 1H), 3.14-3.07 (m, 6H).

Step 2: 5-(2-bromoethoxy)-N,N-dimethylpyridine-2-carboxamide

##STR01546##

[2886] A mixture of 5-hydroxy-N,N-dimethylpyridine-2-carboxamide (158 mg, 0.951 mmol) and Cs₂CO₃ (620 mg, 1.90 mmol) in 1,2-dibromoethane (1.24 mL, 14.3 mmol) was stirred at 110° C. for 18 h in a sealed vial. After cooling to room temperature, the reaction mixture was diluted with DCM and filtered to remove solids. The filter cake was then washed with DCM (3×5 mL) and concentrated under reduced pressure to give 5-(2-bromoethoxy)-N,N-dimethylpyridine-2-carboxamide, which was taken to the next step without further purification. ¹H NMR (500 MHz, DMSO-d₆): δ 8.30 (s, 1H), 7.60-7.53 (m, 1H), 7.53-7.48 (m, 1H), 4.46 (t, J=4.5 Hz, 2H), 3.84 (t, J=4.3 Hz, 2H), 2.99 (s, 6H).

Step 3: 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-methylbenzamide (Compound 51)

##STR01547##

[2887] A mixture of 5-(2-bromoethoxy)-N,N-dimethylpyridine-2-carboxamide (25.0 mg, 0.0915 mmol), 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 25.0 mg, 0.0915 mmol, HCl salt), K₂CO₃ (31.6 mg, 0.229 mmol), and KI (15.0 mg, 0.0904 mmol) in DMF (0.6 mL) was stirred at 75° C. for 3 h. After cooling to room temperature, the reaction mixture was diluted with H₂O (0.15 mL) and purified by reverse phase preparative HPLC (Phenomenex Kinetex C₁₈ column, 10-30% MeCN:H₂O with 0.1% formic acid modifier) to give 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-methylbenzamide (Compound 51). ¹H NMR (500 MHz, DMSO-d₆): δ 10.49 (s, 1H), 8.29 (d, J=2.7 Hz, 1H), 7.57 (d, J=8.6 Hz, 1H), 7.54-7.48 (m, 2H), 7.23 (dd, J=8.2, 2.1 Hz, 1H), 6.84 (d, J=8.3 Hz, 1H), 4.27 (t, J=5.7 Hz, 2H), 3.00 (s, 3H), 2.99 (s, 3H), 2.97-2.90 (m, 2H), 2.88 (t, J=5.7 Hz, 2H), 2.75-2.66 (m, 2H), 1.86-1.75 (m, 2H), 1.75-1.64 (m, 2H). MS=429.1

[M+H]⁺.

Example 18

4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-methylbenzamide (Compound 52)

##STR01548##

Step 1: 4-(2-hydroxyethoxy)-N-methylbenzamide

##STR01549##

[2888] To a mixture of 4-(2-hydroxyethoxy)benzoic acid (500 mg, 2.75 mmol) and methylamine hydrochloride (278 mg, 4.12 mmol) in a solution of 10:1 THF/DMF (5.5 mL) was added DIEA (1.43 mL, 8.2 mmol). The mixture was stirred at room temperature for 10 min. HATU (1.25 g, 3.3

mmol) was added in a single portion and stirring was continued for 18 h. The reaction mixture was diluted with brine (5 mL) and extracted with EtOAc (3×5 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 10 g cartridge, 0-10% MeOH:DCM) to give 4-(2-hydroxyethoxy)-N-methylbenzamide. MS=196.1 [M+H].sup.+.

Step 2: 2-[4-(methylcarbamoyl)phenoxy]ethyl methanesulfonate

##STR01550##

[2889] Methanesulfonyl chloride (0.45 mL, 2.9 mmol) was added dropwise to a 0° C. solution of 4-(2-hydroxyethoxy)-N-methylbenzamide (380 mg, 1.95 mmol) and DIEA (0.300 mL, 3.41 mmol) in DCM (4 mL). The mixture was allowed to slowly warm to room temperature and stirring was continued 16 h. The reaction mixture was diluted with saturated aqueous NH.sub.4Cl (5 mL) and allowed to stir an additional 15 min. The organic layer was separated, and the aqueous layer was extracted with DCM (3×2 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 10 g cartridge, 0-10% MeOH:DCM) to give 2-[4-(methylcarbamoyl)phenoxy]ethyl methanesulfonate. MS=274.1 [M+H].sup.+.

Step 3: 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-methylbenzamide (Compound 52)

##STR01551##

[2890] A mixture of 2-[4-(methylcarbamoyl)phenoxy]ethyl methanesulfonate (30.0 mg, 0.110 mmol), 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 30.0 mg, 0.110 mmol, HCl salt), K.sub.2CO.sub.3 (37.9 mg, 0.274 mmol), and KI (18.2 mg, 0.110 mmol) in DMF (0.6 mL) was stirred at 75° C. for 5 h. After cooling to room temperature, the reaction mixture was diluted with H.sub.2O (0.2 mL) and 5 drops of formic acid was added. The mixture was purified by reverse phase preparative HPLC (Phenomenex Kinetex C.sub.18 column, 10-30% MeCN:H.sub.2O with 0.1% formic acid modifier) to give 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-methylbenzamide (Compound 52). .sup.1H NMR (500 MHz, DMSO-d.sub.6): δ 10.53 (s, 1H), 8.35-8.19 (m, 1H), 7.80 (d, J=10.0, 2H), 7.52 (s, 1H), 7.24 (dd, J=8.3, 2.1 Hz, 1H), 7.02 (d, J=10.0, 2H), 6.85 (d, J=8.3 Hz, 1H), 4.23 (t, J=5.6 Hz, 2H), 3.04-3.00 (m, 4H) 2.83 (br s, 2H), 2.76 (d, J=4.4 Hz, 3H), 1.99-1.58 (m, 4H). MS=414.2 [M+H].sup.+.

Example 19

5-chloro-1'-{2-[(1-methanesulfonyl-1,2,3,4-tetrahydroquinolin-6-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 53)

##STR01552##

[2891] To a 0° C. solution of tert-butyl 6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-1,2,3,4-tetrahydroquinoline-1-carboxylate (Intermediate C-1, 84.0 g, 0.173 mmol, HCl salt) in DCM (1.73 mL) was added TEA (0.121 mL, 0.865 mmol), followed by methanesulfonyl chloride (20.1 μL, 0.260 mmol) dropwise. The mixture was allowed to warm to room temperature and stirred for 2 h. The reaction mixture was diluted with H.sub.2O (10 mL) and extracted with DCM (2×20 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Phenomenex Kinetex C.sub.18 column, 0-50% MeCN:H.sub.2O with 0.1% formic acid modifier) to give 5-chloro-1'-{2-[(1-methanesulfonyl-1,2,3,4-tetrahydroquinolin-6-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 53). .sup.1H NMR (500 MHz, DMSO-d.sub.6): δ 10.49 (s, 1H), 7.51 (d, J=2.0 Hz, 1H), 7.43 (d, J=8.6 Hz, 1H), 7.24 (d, J=8.2 Hz, 1H), 6.85 (d, J=8.3 Hz, 1H), 6.80 (d, J=8.9 Hz, 2H), 4.11 (t, J=5.7 Hz, 2H), 3.66 (t, J=4.9 Hz, 2H), 2.97-2.89 (m, 5H), 2.86 (t, J=5.8 Hz, 2H), 2.78 (t, J=6.7 Hz, 2H), 2.75-2.67 (m, 2H), 1.96-1.86 (m, 2H), 1.83-1.70 (m, 4H). MS=490.2 [M+H].sup.+.

[2892] The following compounds in Table 24 were prepared according to procedures analogous to

those described for Compound 53 using the appropriate starting materials.

TABLE-US-00027 TABLE 24 Exact Mass Intermediates # Structure IUPAC Name [M + H].sup.+
Used 54 [01553]  embedded image 5-chloro-1'-{2-[(2-methanesulfonyl-1,2,3,4-tetrahydroisoquinolin-6-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 476.1 Found 476.1
C-2 55 [01554]  embedded image 1'-{2-[(2-methanesulfonyl-1,2,3,4-tetrahydroisoquinolin-6-yl)oxy]ethyl}-1-methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 470.2 Found 470.2
C-5

Example 20

1'-{2-[(1-acetyl-1,2,3,4-tetrahydroquinolin-6-yl)oxy]ethyl}-5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 56)

##STR01555##

[2893] To a 0° C. solution of tert-butyl 6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-1,2,3,4-tetrahydroquinoline-1-carboxylate (Intermediate C-1, 84.0 mg, 0.173 mmol, HCl salt) in DCM (1.73 mL) was added TEA (0.121 mL, 0.865 mmol), followed by acetic anhydride (23.9 µL, 0.260 mmol) dropwise. The mixture was stirred for 30 min and was then diluted with H.sub.2O (10 mL) and extracted with DCM (2×20 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Phenomenex Kinetex C.sub.18 column, 0-40% MeCN:H.sub.2O with 0.100 formic acid modifier) to give 1'-{2-[(1-acetyl-1,2,3,4-tetrahydroquinolin-6-yl)oxy]ethyl}-5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 56). .sup.1H NMR (500 MHz, DMSO-d.sub.6): δ 10.49 (s, 1H), 8.15 (s, 1H), 7.51 (d, J=2.1 Hz, 1H), 7.24 (dd, J=8.2, 2.1 Hz, 1H), 6.85 (d, J=8.3 Hz, 1H), 6.83-6.75 (m, 2H), 4.12 (t, J=5.8 Hz, 2H), 3.64 (t, J=6.4 Hz, 2H), 2.98-2.89 (m, 2H), 2.86 (t, J=5.8 Hz, 2H), 2.75-2.66 (m, 4H), 2.11 (s, 3H), 1.88-1.68 (m, 6H). MS=454.1 [M+H].sup.+.

[2894] The following compounds in Table 25 were prepared according to procedures analogous to those described for Compound 56 using the appropriate starting materials.

TABLE-US-00028 TABLE 25 Exact Mass Intermediates # Structure IUPAC Name [M + H].sup.+
Used 57 [01556]  embedded image 1'-{2-[(2-acetyl-2,3-dihydro-1H-isoindol-5-yl)oxy]ethyl}-5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 440.2 Found 440.1
C-2 58 [01557]  embedded image 1'-{2-[(2-acetyl-1,2,3,4-tetrahydroisoquinolin-6-yl)oxy]ethyl}-1-methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 434.2 Found 434.1
C-5 59 [01558]  embedded image 1'-{2-[(1-acetyl-1,2,3,4-tetrahydroquinolin-6-yl)oxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile Calc'd 445.2 Found 445.2
C-4

Example 21

6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-methyl-1,2,3,4-tetrahydroquinoline-1-carboxamide (Compound 60)

##STR01559##

[2895] To a 0° C. mixture of 5-chloro-1'-[2-(1,2,3,4-tetrahydroquinolin-6-yloxy)ethyl]-1H-spiro[indole-3,4'-piperidin]-2-one (Intermediate C-1, 0.118 g, 0.244 mmol) in DCM (1.6 mL) was added TEA (0.134 mL, 0.976 mmol), followed by N-methylcarbonyl chloride (0.034 g, 0.366 mmol). The mixture was allowed to warm to room temperature and stirred for 2 h. The reaction was quenched with saturated aqueous NaHCO.sub.3 (20 mL) and extracted with DCM (2×20 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Phenomenex Kinetex C.sub.18 column, 0-40% MeCN:H.sub.2O with 0.1% formic acid modifier) to give 6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-methyl-1,2,3,4-tetrahydroquinoline-1-carboxamide (Compound 60). .sup.1H NMR (500 MHz, DMSO-d.sub.6): δ 10.49 (s, 1H), 7.52 (s, 1H), 7.30 (d, J=8.6 Hz, 1H), 7.27-7.21 (m, 1H), 6.85 (dd, J=8.3, 1.5 Hz, 1H), 6.73 (d, J=8.2 Hz, 2H), 6.40 (q, J=4.6 Hz, 1H), 4.09 (t, J=5.8 Hz, 2H), 3.52 (t, J=6.2 Hz, 2H), 2.98-2.89 (m, 2H), 2.85 (t, J=5.8 Hz, 2H), 2.75-2.68 (m, 2H), 2.66 (t, J=6.6 Hz, 2H), 2.62 (d, J=1.5 Hz, 3H), 1.86-1.76 (m,

4H), 1.75-1.68 (m, 2H). MS=469.1 [M+H].sup.+.

[2896] The following compound in Table 26 was prepared according to procedures analogous to those described for Compound 60 using the appropriate starting materials.

TABLE-US-00029 TABLE 26 Exact Mass Intermediates # Structure IUPAC Name [M + H].sup.+
Used 61 [01560]  5-(2-(5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl)ethoxy)-N-methyl-2,3-dihydro-1H-isoindole-2-carboxamide Calc'd 455.2
Found 455.2 C-2

Example 22

5-chloro-1'-[2-(quinoxalin-6-yloxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one
(Compound 62)

##STR01561##

[2897] A mixture of 5-chloro-1'-(2-chloroethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-5, 40.0 mg, 0.134 mmol), quinoxalin-6-ol (23.4 mg, 0.161 mmol) and K.sub.2CO.sub.3 (27.7 mg, 0.201 mmol) in DMF (1.3 mL) was heated at 80° C. for 1 h. After cooling to room temperature, the mixture was diluted with H.sub.2O (10 mL) and extracted with EtOAc (2×20 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Phenomenex Kinetex C.sub.18 column, 0-30% MeCN:H.sub.2O with 0.1% formic acid modifier) to give 5-chloro-1'-[2-(quinoxalin-6-yloxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 62). .sup.1H NMR (500 MHz, DMSO-d.sub.6): δ 10.50 (s, 1H), 8.87 (d, J=2.4 Hz, 1H), 8.78 (d, J=2.4 Hz, 1H), 8.00 (dd, J=9.2, 2.5 Hz, 1H), 7.58-7.49 (m, 3H), 7.23 (d, J=8.3 Hz, 1H), 6.85 (dd, J=8.3, 2.4 Hz, 1H), 4.37 (t, J=5.3 Hz, 2H), 2.98-2.91 (m, 4H), 2.78 (t, J=9.2 Hz, 2H), 1.93-1.67 (m, 4H). MS=409.1 [M+H].sup.+.

[2898] The following compound in Table 27 was prepared according to procedures analogous to those described for Compound 62 using the appropriate starting materials.

TABLE-US-00030 TABLE 27 Exact Mass Intermediates # Structure IUPAC Name [M + H].sup.+
Used 63 [01562]  5-chloro-1'-[2-(3-fluoro-4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 453.1 Found 453.1 B-5 64 [01563]
 5-chloro-1'-[2-[(6-methanesulfonylpyridin-3-yl)oxy]ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 436.1 Found 436.1 B-5

Example 23

5-chloro-1'-[2-[(1-methyl-2-oxo-2,3-dihydro-1H-indol-5-yl)oxy]ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 65)

##STR01564##

[2899] A mixture of 5-(2-bromoethoxy)-1-methyl-2,3-dihydro-1H-indol-2-one (Intermediate A-4, 20 mg, 0.074 mmol), 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 22 mg, 0.081 mmol) and K.sub.2CO.sub.3 (31 mg, 0.22 mmol) in DMF (0.5 mL) was heated at 60° C. for 3 h. After cooling to room temperature, the mixture was diluted with H.sub.2O (5 mL) and extracted with EtOAc (2×10 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Phenomenex Kinetex Cis column, 0-40% MeCN:H.sub.2O with 0.100 formic acid modifier) to give 5-chloro-1'-[2-[(1-methyl-2-oxo-2,3-dihydro-1H-indol-5-yl)oxy]ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 65). H NMR (500 MHz, CD.sub.3CN): δ 8.34 (s, 1H), 7.36 (d, J=2.2 Hz, 1H), 7.12 (dd, J=8.3, 2.1 Hz, 1H), 6.87-6.83 (m, 1H), 6.82-6.75 (m, 2H), 6.70 (d, J=8.5 Hz, 1H), 4.04 (t, J=5.6 Hz, 2H), 3.34 (s, 2H), 3.02 (s, 3H), 3.00-2.91 (m, 2H), 2.85 (t, J=5.6 Hz, 2H), 2.78-2.70 (m, 2H), 1.82-1.70 (m, 4H). MS=426.1 [M+H].

[2900] The following compound in Table 28 was prepared according to procedures analogous to those described for Compound 65 using the appropriate starting materials.

TABLE-US-00031 TABLE 28 Exact Mass Intermediates # Structure IUPAC Name [M + H].sup.+
Used 66 [01565]  1'-[2-(4-methanesulfonylphenoxy)ethyl]-5-methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

dihydrospiro[indole- 3,4'-piperidin]-2-one Calc'd 415.2 Found 415.1 A-3 67 [01566]
 embedded image 1'-[2-(4- methanesulfonylphenoxy) ethyl]-1-methyl- 1,2- dihydrospiro[indole- 3,4'-piperidin]-2-one Calc'd 415.2 Found 415.1 A-3 and B-6 68 [01567]
 embedded image 5-chloro-1'-{2-[(1- methyl-1H-indazol-5- yl)oxy]ethyl}-1,2- dihydrospiro[indole- 3,4'-piperidin]-2-one Calc'd 411.2 Found 411.1 A-5 and B-4 69 [01568]
 embedded image 5-chloro-1'-[2-(4- methanesulfonylphenoxy) ethyl]-1,2- dihydrospiro[indole- 3,4'-piperidin]-2-one Calc'd 435.1 Found 435.0 A-3 and B-4 70 [01569]
 embedded image 1'-{2-[(1-methyl-1H- indazol-5- yl)oxy]ethyl}-1,2- dihydrospiro[indole- 3,4'-piperidin]-2-one Calc'd 377.2 Found 377.1 A-5 71 [01570]
 embedded image 1'-[2-(4- methanesulfonylphenoxy) ethyl]-2-oxo-1,2- dihydrospiro[indole- 3,4'-piperidine]-5- carbonitrile Calc'd 426.1 Found 426.1 A-3 and B-9 72 [01571]
 embedded image 5-bromo-1'-[2-(4- methanesulfonylphenoxy) ethyl]-1,2- dihydrospiro[indole- 3,4'-piperidin]-2-one Calc'd 479.1 Found 479.1 A-3 73 [01572]
 embedded image 1'-[2-(4- methanesulfonylphenoxy) ethyl]-1,2- dihydrospiro[indole- 3,4'-piperidin]-2-one Calc'd 401.2 Found 401.2 A-3 74 [01573]
 embedded image 1'-[2-(3-fluoro-4- methanesulfonylphenoxy) ethyl]-2-oxo-1,2- dihydrospiro[indole- 3,4'-piperidine]-5- carbonitrile Calc'd 444.1 Found 444.1 A-6 and B-9 75 [01574]
 embedded image N-[4-(2-{5-cyano-2- oxo- 1,2- dihydrospiro[indole- 3,4'-piperidin]-1'- yl}ethoxy)phenyl]-N- methylmethanesulfonamide Calc'd 455.2 Found 455.2 A-7 and B-9 76 [01575]
 embedded image N-[4-(2-{5-chloro-2- oxo- 1,2- dihydrospiro[indole- 3,4'-piperidin]-1'- yl}ethoxy)phenyl]-N- methylmethanesulfonamide Calc'd 464.1 Found 464.2 A-7 and B-4 77 [01576]
 embedded image 1'-[2-(4- {[dimethyl(oxo)- λ.sup.6- sulfanylidene]amino} phenoxy)ethyl]-2-oxo- 1,2- dihydrospiro[indole- 3,4'-piperidine]-5- carbonitrile Calc'd 439.2 Found 439.2 A-8 and B-9 78 [01577]
 embedded image 5-chloro-1'-{2- [(8- methyl-7-oxo-5,6,7,8- tetrahydro-1,8- naphthyridin-3- yl)oxy]ethyl}-1,2- dihydrospiro[indole- 3,4'-piperidin]-2-one Calc'd 441.2 Found 441.1 A-72 and B-4 79 [01578]
 embedded image 1'-{2- [(8-methyl-7- oxo-5,6,7,8- tetrahydro-1,8- naphthyridin-3- yl)oxy]ethyl}-2-oxo- 1,2- dihydrospiro[indole- 3,4'-piperidine]-5- carbonitrile Calc'd 432.2 Found 432.2 A-72 and B-9 80 [01579]
 embedded image 1'-{2-[(8-acetyl- 5,6,7,8-tetrahydro-1,8- naphthyridin-3- yl)oxy]ethyl}-5- chloro-1,2- dihydrospiro[indole- 3,4'-piperidin]-2-one Calc'd 455.2 Found 455.1 A-73 and B-4 81 [01580]
 embedded image 1'-{2-[(8-acetyl- 5,6,7,8-tetrahydro-1,8- naphthyridin-3- yl)oxy]ethyl }-2-oxo- 1,2- dihydrospiro[indole- 3,4'-piperidine]-5- carbonitrile Calc'd 446.2 Found 446.2 A-73 and B-9 82 [01581]
 embedded image 5-chloro-1'-{2-[(2- oxo-1,2,3,4- tetrahydroquinolin-6- yl)oxy]ethyl}-1,2- dihydrospiro[indole- 3,4'-piperidin]-2-one Calc'd 426.2 Found 426.2 A-83 and B-4

Example 24

2-oxo-1'-{2-[(1,3,3-trimethyl-2-oxo-2,3-dihydro-1H-indol-5-yl)oxy]ethyl}-1,2- dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (Compound 83)

##STR01582##

Step 1: 5-(2-hydroxyethoxy)-1,3,3-trimethyl-2,3-dihydro-1H-indol-2-one

##STR01583##

[2901] A mixture of 5-hydroxy-1,3,3-trimethyl-2,3-dihydro-1H-indol-2-one (100 mg, 0.523 mmol), ethylene carbonate (69.7 μ L, 1.05 mmol) and K.sub.2CO.sub.3 (0.145 g, 1.05 mmol) in DMF (5 mL) was heated at 90° C. for 3 h. After cooling to room temperature, the mixture was diluted with H.sub.2O (10 mL) and extracted with EtOAc (2 $\times$ 20 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 10 g cartridge, 0-100% EtOAc:hexanes) to give 5-(2-hydroxyethoxy)-1,3,3-trimethyl-2,3-dihydro-1H-indol-2-one. MS=236.1 [M+H].sup.+.

Step 2: 2-[(1,3,3-trimethyl-2-oxo-2,3-dihydro-1H-indol-5-yl)oxy]ethyl methanesulfonate

##STR01584##

[2902] To a solution of 5-(2-hydroxyethoxy)-1,3,3-trimethyl-2,3-dihydro-1H-indol-2-one (65.0 mg, 0.276 mmol) and TEA (57.9 μ L, 0.414 mmol) in DCM (2 mL) was added methanesulfonyl chloride

(25.6 μ L, 0.332 mmol) The mixture was stirred for 30 min, then quenched with H.sub.2O (5 mL) and extracted with DCM (2 $\times$ 10 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo and used in the subsequent step without further purification. MS=314.1 [M+H].sup.+.

Step 3: 2-oxo-1'-{2-[(1,3,3-trimethyl-2-oxo-2,3-dihydro-1H-indol-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (Compound 83)

##STR01585##

[2903] A mixture of 2-[(1,3,3-trimethyl-2-oxo-2,3-dihydro-1H-indol-5-yl)oxy]ethyl methanesulfonate (84.0 mg, 0.268 mmol), 2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (Intermediate B-9, 64.0 mg, 0.241 mmol, HCl salt) and K.sub.2CO.sub.3 (93.0 mg, 0.670 mmol) in DMF (1.2 mL) was heated at 60° C. for 3 h. After cooling to room temperature, the mixture was diluted with H.sub.2O (10 mL) and extracted with EtOAc (2 $\times$ 20 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Phenomenex Kinetex C.sub.18 column, 0-30% MeCN:H.sub.2O with 0.1% formic acid modifier) to give 2-oxo-1'-{2-[(1,3,3-trimethyl-2-oxoindol-5-yl)oxy]ethyl}-1H-spiro[indole-3,4'-piperidine]-5-carbonitrile (Compound 83). .sup.1H NMR (500 MHz, DMSO-d.sub.6): δ 10.87 (s, 1H), 7.96 (s, 1H), 7.68 (d, J=8.1 Hz, 1H), 7.09 (d, J=2.5 Hz, 1H), 7.00 (d, J=8.1 Hz, 1H), 6.94-6.84 (m, 2H), 4.11 (t, J=5.8 Hz, 2H), 3.10 (s, 3H), 2.96-2.83 (m, 4H), 2.81-2.70 (m, 2H), 1.85-1.71 (m, 4H), 1.26 (s, 6H). MS=445.2 [M+H].sup.+.

Example 25

1'-{2-[3,5-difluoro-4-(3-methanesulfonyloxetan-3-yl)phenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (Compound 84)

##STR01586##

[2904] To a solution of 3-[4-(2-bromoethoxy)-2,6-difluorophenyl]-3-methanesulfonyloxetane (Intermediate A-32, 60.0 mg, 0.162 mmol) in MeCN (1.00 mL) was added NaHCO.sub.3(40.7 mg, 0.485 mmol) and 2-oxospiro[indoline-3,4'-piperidine]-5-carbonitrile (Intermediate B-9, 46.89 mg, 177.8 μ mol, HCl salt). The mixture was stirred at 80° C. for 16 h. After cooling to room temperature, the reaction mixture was filtered and washed with MeCN. The filtrate was concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Waters Xbridge BEH C.sub.18 column, 25-55% MeCN:H.sub.2O with 10 mM NH.sub.4HCO.sub.3 modifier) to give 1'-{2-[3,5-difluoro-4-(3-methanesulfonyloxetan-3-yl)phenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (Compound 84). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.86 (s, 1H), 7.96 (s, 1H), 7.67 (d, J=8.4 Hz, 1H), 6.98 (d, J=8.4 Hz, 1H), 6.90 (d, J=10.8 Hz, 2H), 5.14 (dd, J=14.4, 8.0 Hz, 4H), 4.21 (t, J=5.2 Hz, 2H), 3.13 (s, 3H), 2.89-2.85 (m, 4H), 2.74-2.71 (m, 2H), 1.81-1.72 (m, 4H). MS=518.2 [M+H].sup.+.

[2905] The following compounds in Table 29 were prepared according to procedures analogous to those described for Compound 84 using the appropriate starting materials. In cases where step 1 was followed by chiral SFC purification to separate isomers, chiral column conditions and elution order are specified.

TABLE-US-00032 TABLE 29 Chiral Exact Inter- Column, Mass mediates If App- Elution #
Structure IUPAC Name [M + H].sup.+ Used licable Order 85 [01587]
5-chloro-1'-[2-((1-[(cis)-3-hydroxycyclobutyl]-1H-indazol-5-yl)oxy)ethyl]-1,2-dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 467.2 Found 467.3 A-63 and B-4 n/a n/a 86 [01588]

5-chloro-1'-[2-((1-[(trans)-3-hydroxycyclobutyl]-1H-indazol-5-yl)oxy)ethyl]-1,2-dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 467.2 Found 467.3 A-62 and B-4 n/a n/a 87 [01589]
5-chloro-1'-[2-(2H-indazol-5-yloxy)ethyl]-1,2-dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 397.1 Found 397.1 A-80 and B-4 n/a n/a 88 [01590]
5-chloro-1'-{2-[(1-methyl-2-oxo-2,3-dihydro-1H-1,3-benzodiazol-5-yl)oxy]ethyl}-1,2-dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 427.1 Found 427.2 A-82 and B-4 n/a n/a 89 [01591]

phenoxy)ethyl]-1,2- dihydrospiro [indole-3,4'- piperidine]- 2-one Calc'd 471.1 Found 471.1 A-18 and B-4 n/a n/a 90 [01592]  1'-[2-(3,5- difluoro-4- methanesulfonylphenoxy)ethyl]-2- oxo-1,2- dihydrospiro [indole-3,4'- piperidine]- 5-carbonitrile Calc'd 462.1 Found 462.0 A-18 and B-9 n/a n/a 91 [01593]  methyl 1-[4-(2-{5- chloro-2-oxo-1,2- dihydrospiro [indole-3,4'- piperidin]- 1'-yl}ethoxy)benzene sulfonyl] cyclopropane-1- carboxylate Calc'd 519.1 Found 519.1 A-24 and B-4 n/a n/a 92 [01594]  5-chloro-1'-(2-{4- [(3-methyloxetan- 3- yl)sulfonyl] phenoxy}ethyl)- 1,2- dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 491.1 Found 491.2 A-27 and B-4 n/a n/a 93 [01595]  1'-(2-{4-[(3-methyloxetan-3- yl)sulfonyl] phenoxy}ethyl)- 2-oxo-1,2- dihydrospiro [indole-3,4'- piperidine]- 5-carbonitrile Calc'd 482.2 Found 482.2 A-27 and B-9 n/a n/a 94 [01596]  5-chloro-1'-{2-[(2- methanesulfonyl- 1,2,3,4- tetrahydro- isoquinolin-6- yl)oxy]ethyl}-1,2- dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 490.1 Found 490.2 A-55 and B-4 n/a n/a 95 [01597]  1'-{2-[(1-methyl- 2-oxo-1,2,3,4- tetrahydroquinolin- 6-yl)oxy]ethyl}- 2-oxo-1,2- dihydrospiro [indole-3,4'- piperidine]- 5-carbonitrile Calc'd 431.2 Found 431.3 A-71 and B-9 n/a n/a 96 [01598]  2-oxo-1'-(2-{[2- oxo-1-(propan-2- yl)-1,2,3,4- tetrahydroquinolin- 6-yl]oxy}ethyl)- 1,2- dihydrospiro [indole-3,4'- piperidine]- 5-carbonitrile Calc'd 459.2 Found 459.1 A-64 and B-9 n/a n/a 97 [01599]  5-chloro-1'-(2-{[2- oxo-1-(propan-2- yl)-1,2,3,4- tetrahydroquinolin- 6-yl]oxy}ethyl)- 1,2- dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 468.2 Found 468.2 A-64 and B-4 n/a n/a 98 [01600]  N-[4-(2-{5-chloro- 2-oxo-1,2- dihydrospiro [indole-3,4'- piperidin]- 1'-yl}ethoxy) phenyl]-N-(2- hydroxyethyl) methane- sulfonamide Calc'd 494.1 Found 494.1 A-68 and B-4 n/a n/a 99 [01601]  5-chloro-1'-(2-{[1- (oxetan-3-yl)-2- oxo-1,2,3,4- tetrahydroquinolin- 6-yl]oxy}ethyl)- 1,2- dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 482.2 Found 482.2 A-66 and B-4 n/a n/a 100 [01602]  1'-{2-[3- (difluoromethyl)- 4- methane- sulfonylphenoxy] ethyl}-2- oxo-1,2- dihydrospiro [indole-3,4'- piperidine]- 5-carbonitrile Calc'd 476.1 Found 476.2 A-13 and B-9 n/a n/a 101 [01603]  1'-{2-[4-(1- methane- sulfonyl- cyclopropyl) phenox ylethyl]-2-oxo- 1,2- dihydrospiro [indole-3,4'- piperidine]- 5-carbonitrile Calc'd 466.2 Found 466.2 A-28 and B-9 n/a n/a 102 [01604]  5-chloro-1'-{2-[4- (1- methanesulfonyl- cyclopropyl) phenoxy]ethyl}- 1,2- dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 475.1 Found 475.2 A-28 and B-4 n/a n/a 103 [01605]  6-(2-{5-chloro-2- oxo-1,2- dihydrospiro [indole-3,4'- piperidin]- 1'-yl}ethoxy)-1- methyl-3,4- dihydro-1H-2λ.sup.6,1- benzothiazine-2,2- dione Calc'd 476.1 Found 476.2 A-51 and B-4 n/a n/a 104 [01606]  1'-{2-[(1-methyl- 2,2-dioxo-3,4- dihydro-1H-226,1- benzothiazin- 6- yl)oxy]ethyl}-2- oxo-1,2- dihydrospiro [indole-3,4'- piperidine]- 5-carbonitrile Calc'd 467.2 Found 467.2 A-51 and B-9 n/a n/a 105 [01607]  1'-(2-{[1-(2,2- difluoroethyl)-2- oxo-1,2,3,4- tetrahydroquinolin- 6-yl]oxy}ethyl)- 2-oxo-1,2- dihydrospiro [indole-3,4'- piperidine]- 5-carbonitrile Calc'd 481.2 Found 481.3 A-67 and B-9 n/a n/a 106 [01608]  5-chloro-1'-(2-{[1- (2,2- difluoroethyl)-2- oxo-1,2,3,4- tetrahydroquinolin- 6-yl]oxy}ethyl)- 1,2- dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 490.2 Found 490.3 A-67 and B-4 n/a n/a 107 [01609]  1'-{2-[4- methanesulfonyl- 3- (trifluoromethyl) phenoxy]ethyl}- 2-oxo- 1,2- dihydrospiro [indole-3,4'- piperidine]- 5-carbonitrile Calc'd 494.1 Found 494.1 A-17 and B-9 n/a n/a 108 [01610]  5-chloro-1'-{2-[4- methanesulfonyl- 3- (trifluoromethyl) phenoxy]ethyl}- 1,2- dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 503.1 Found 503.1 A-17 and B-4 n/a n/a 109 [01611]  1'-{2-[4-(3- methanesulfonyl- oxetan-3- yl)phenoxy] ethyl}-2-oxo-1,2- dihydrospiro [indole-3,4'- piperidine]- 5-carbonitrile Calc'd 482.2 Found 482.1 A-33 and B-9 n/a n/a 110 [01612]  5-chloro-1'-[2-(4- methanesulfonyl- 3- methylphenoxy) ethyl]-1,2- dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 449.1 Found 449.1 A-22 and B-4 n/a n/a 111 [01613]  1'-[2-(4- methanesulfonyl- 3- methylphenoxy) ethyl]-2-oxo-1,2- dihydrospiro [indole-3,4'- piperidine]- 5-carbonitrile Calc'd 440.2 Found 440.2 A-22 and B-9 n/a n/a 112 [01614]  5-chloro-1'-[2-(4-

difluoromethane- sulfonylphenoxy) ethyl]-1,2- dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 471.1 Found 471.1 A-16 and B-4 n/a n/a 113 [01615]  1'-[2-(4- difluoromethane-sulfonylphenoxy) ethyl]-2-oxo-1,2- dihydrospiro [indole-3,4'- piperidine]- 5-carbonitrile Calc'd 462.1 Found 462.2 A-16 and B-9 n/a n/a 114 [01616]  5-chloro-1'-[2-({2- oxo-1- [(cis)-3- hydroxycyclobutyl]- 1,2,3,4- tetrahydroquinolin- 6-yl}oxy)ethyl]- 1,2- dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 496.2 Found 496.2 A-69 and B-4 n/a n/a 115 [01617]  5-chloro-1'-[2-({3- oxo- 1H,2H,3H,5H,6H, 10bH- imidazo[4,3- a]isoquinolin-8- yl}oxy)ethyl]-1,2- dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 467.2 Found 467.2 A-58 and B-4 n/a n/a 116 [01618]  (S) or (R)-5- chloro-1'-[2-({3- oxo- 1H,2H,3H,5H,6H, 10bH- imidazo[4,3- a]isoquinolin-8- yl}oxy)ethyl]-1,2- dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 467.2 Found 467.2 A-58 and B-4 Daicel Chiralpa k AD-3 Second 117 [01619]  (R) or (S)-5- chloro-1'-[2-({3- oxo- 1H,2H,3H,5H,6H, 10bH- imidazo[4,3- a]isoquinolin-8- yl}oxy)ethyl]-1,2- dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 467.2 Found 467.2 A-58 and B-4 Daicel Chiralpak AD-3 First 118 [01620]  2-oxo-1'-[2-({3- oxo- 1H,2H,3H,5H,6H, 10bH- imidazo[4,3- a]isoquinolin-8- yl}oxy)ethyl]-1,2- dihydrospiro [indole-3,4'- piperidine]- 5-carbonitrile Calc'd 458.2 Found 458.3 A-58 and B-9 n/a n/a 119 [01621]  (S) or (R)-2-oxo- 1'-[2-({3-oxo- 1H,2H,3H,5H,6H, 10bH- imidazo[4,3- a]isoquinolin-8- yl}oxy)ethyl]-1,2- dihydrospiro [indole-3,4'- piperidine]- 5- carbonitrile Calc'd 458.2 Found 458.3 A-58 and B-9 Daicel Chiralpak AD-3 First 120 [01622]  (R) or (S)-2-oxo- 1'-[2-({3-oxo- 1H,2H,3H,5H,6H, 10bH- imidazo[4,3- a]isoquinolin-8- yl}oxy)ethyl]-1,2- dihydrospiro [indole-3,4'- piperidine]- 5-carbonitrile Calc'd 458.2 Found 458.3 A-58 and B-9 Daicel Chiralpak AD-3 Second 121 [01623]  5-chloro-1'-[2-({3- oxo- 1H,2H,3H,5H,6H, 10bH-pyrrolo[2,1- a]isoquinolin-8- yl}oxy)ethyl]-1,2- dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 466.2 Found 466.1 A-57 and B-4 n/a n/a 122 [01624]  (S) or (R)-5- chloro-1'-[2-({3- oxo- 1H,2H,3H,5H,6H, 10bH- pyrrolo[2,1- a]isoquinolin-8- yl}oxy)ethyl]-1,2- dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 466.2 Found 466.1 A-57 and B-4 Daicel Chiralpak AD-3 First 123 [01625]  (R) or (S)-5- chloro-1'-[2-({3- oxo- 1H,2H,3H,5H,6H, 10bH-pyrrolo[2,1- a]isoquinolin-8- yl}oxy)ethyl]-1,2- dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 466.2 Found 466.1 A-57 and B-4 Daicel Chiralpak AD-3 Second 124 [01626]  2-oxo-1'-[2-({3- oxo- 1H,2H,3H,5H,6H, 10bH-pyrrolo[2,1- a]isoquinolin-8- yl}oxy)ethyl]-1,2- dihydrospiro[indol e-3,4'- piperidine]- 5-carbonitrile Calc'd 457.2 Found 457.1 A-57 and B-9 n/a n/a 125 [01627]  (S) or (R)-2-oxo- 1'-[2-({3-oxo- 1H,2H,3H,5H,6H, 10bH-pyrrolo[2,1- a]isoquinolin-8- yl}oxy)ethyl]-1,2- dihydrospiro [indole-3,4'- piperidine]- 5-carbonitrile Calc'd 457.2 Found 457.1 A-57 and B-9 Daicel Chiralpak AD-3 n/a 126 [01628]  (R) or (S)-2-oxo- 1'-[2-({3-oxo- 1H,2H,3H,5H,6H, 10bH-pyrrolo[2,1- a]isoquinolin-8- yl}oxy)ethyl]-1,2- dihydrospiro [indole-3,4'- piperidine]- 5-carbonitrile Calc'd 457.2 Found 457.1 A-57 and B-9 Daicel Chiralpak AD-3 n/a 127 [01629]  N-[4-(2-{5-chloro- 2-oxo-1,2- dihydrospiro [indole-3,4'- piperidin]- 1'-yl}ethoxy)-2,6- difluorophenyl]- N- methylmethane- sulfonamide Calc'd 500.1 Found 500.2 A-44 and B-4 n/a n/a 128 [01630]  N-[4-(2-{5-cyano- 2-oxo-1,2- dihydrospiro [indole-3,4'- piperidin]- 1'-yl}ethoxy)-2,6- difluorophenyl]- N- methylmethane- sulfonamide Calc'd 491.1 Found 491.2 A-44 and B-9 n/a n/a 129 [01631]  5-chloro-1'-[2-(4- methanesulfonyl- 3,5- dimethylphenoxy) ethyl]-1,2- dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 463.1 Found 463.1 A-20 and B-4 n/a n/a 130 [01632]  1'-[2-(4- methanesulfonyl- 3,5- dimethylphenoxy) ethyl]-2-oxo-1,2- dihydrospiro [indole-3,4'- piperidine]- 5-carbonitrile Calc'd 454.2 Found 454.2 A-20 and B-9 n/a n/a 131 [01633]  5-chloro-1'-{2-[(3- methyl-2-oxo-2,3- dihydro-1,3- benzoxazol-6- yl)oxy]ethyl}-1,2- dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 428.1 Found 428.1 A-59 and B-4 n/a n/a 132 [01634]  1'-{2-[(3-methyl- 2-oxo-2,3- dihydro-1,3- benzoxazol-6- yl)oxy]ethyl}-2- oxo-1,2- dihydrospiro [indole-3,4'- piperidine]- 5-carbonitrile

Calc'd 419.2 Found 419.2 A-59 and B-9 n/a n/a 133 [01635]  embedded image 1'-{2-[4-(1-methanesulfonyl-ethyl)phenoxy]ethyl}-2-oxo-1,2-dihydrospiro [indole-3,4'-piperidine]-5-carbonitrile Calc'd 454.2 Found 454.2 A-31 and B-9 n/a n/a 134 [01636]  embedded image (S) or (R)-1'-{2-[4-(1-methanesulfonyl-ethyl)phenoxy]ethyl}-2-oxo-1,2-dihydrospiro [indole-3,4'-piperidine]-5-carbonitrile Calc'd 454.2 Found 454.2 A-31 and B-9 Daicel Chiralpak OD-3 First 135 [01637]  embedded image (R) or (S)-1'-{2-[4-(1-methanesulfonyl-ethyl)phenoxy]ethyl}-2-oxo-1,2-dihydrospiro [indole-3,4'-piperidine]-5-carbonitrile Calc'd 454.2 Found 454.2 A-31 and B-9 Daicel Chiralpak OD-3 Second 136 [01638]  embedded image 5-chloro-1'-{2-[3,5-difluoro-4-(3-methanesulfonyl-oxetan-3-yl)phenoxy]ethyl}-1,2-dihydrospiro [indole-3,4'-piperidin]-2-one Calc'd 527.1 Found 527.2 A-32 and B-4 n/a n/a 137 [01639]  embedded image 1'-{2-[3,5-difluoro-4-(1-methanesulfonyl-ethyl)phenoxy]ethyl}-2-oxo-1,2-dihydrospiro [indole-3,4'-piperidine]-5-carbonitrile Calc'd 490.2 Found 490.2 A-30 and B-9 n/a n/a 138 [01640]  embedded image (S) or (R)-1'-{2-[3,5-difluoro-4-(1-methanesulfonyl-ethyl)phenoxy]ethyl}-2-oxo-1,2-dihydrospiro [indole-3,4'-piperidine]-5-carbonitrile Calc'd 490.2 Found 490.2 A-30 and B-9 Daicel Chiralpak OD-3 First 139 [01641]  embedded image (R) or (S)-1'-{2-[3,5-difluoro-4-(1-methanesulfonyl-ethyl)phenoxy]ethyl}-2-oxo-1,2-dihydrospiro [indole-3,4'-piperidine]-5-carbonitrile Calc'd 490.2 Found 490.2 A-30 and B-9 Daicel Chiralpak OD-3 Second 140 [01642]  embedded image 1'-{2-[(6-methanesulfonyl-5-methylpyridin-3-yl)oxy]ethyl}-2-oxo-1,2-dihydrospiro [indole-3,4'-piperidine]-5-carbonitrile Calc'd 441.2 Found 441.1 A-21 and B-9 n/a n/a 141 [01643]  embedded image 5-chloro-1'-{2-[3,5-difluoro-4-(1-methanesulfonyl-ethyl)phenoxy]ethyl}-1,2-dihydrospiro [indole-3,4'-piperidin]-2-one Calc'd 499.1 Found 499.2 A-30 and B-4 n/a n/a 142 [01644]  embedded image (S) or (R)-5-chloro-1'-{2-[3,5-difluoro-4-(1-methanesulfonyl-ethyl)phenoxy]ethyl}-1,2-dihydrospiro [indole-3,4'-piperidin]-2-one Calc'd 499.1 Found 499.2 A-30 and B-4 Daicel Chiralpak OD-3 First 143 [01645]  embedded image (R) or (S)-5-chloro-1'-{2-[3,5-difluoro-4-(1-methanesulfonyl-ethyl)phenoxy]ethyl}-1,2-dihydrospiro [indole-3,4'-piperidin]-2-one Calc'd 499.1 Found 499.2 A-30 and B-4 Daicel Chiralpak OD-3 Second 144 [01646]  embedded image 2-oxo-1'-(2-{[2-oxo-1-(propan-2-yl)-2,3-dihydro-1H-indol-5-yl]oxy}ethyl)-1,2-dihydrospiro [indole-3,4'-piperidine]-5-carbonitrile Calc'd 445.2 Found 445.3 A-14 and B-9 n/a n/a 145 [01647]  embedded image 5-chloro-1'-(2-{[2-oxo-1-(propan-2-yl)-2,3-dihydro-1H-indol-5-yl]oxy}ethyl)-1,2-dihydrospiro [indole-3,4'-piperidin]-2-one Calc'd 454.2 Found 454.2 A-14 and B-4 n/a n/a 146 [01648]  embedded image 5-chloro-1'-(2-{[2-methyl-1-oxo-2,3-dihydro-1H-isoindol-5-yl]oxy}ethyl)-1,2-dihydrospiro [indole-3,4'-piperidin]-2-one Calc'd 426.2 Found 426.1 A-53 and B-4 n/a n/a 147 [01649]  embedded image 5-chloro-1'-(2-{4-[methyl (methylinino) oxo-λ^{sup}.6-sulfanyl]phenoxy}ethyl)-1,2-dihydrospiro [indole-3,4'-piperidin]-2-one Calc'd 448.1 Found 448.0 A-45 and B-4 n/a n/a 148 [01650]  embedded image (S) or (R)-5-chloro-1'-(2-{4-[methyl (methylinino) oxo-λ^{sup}.6-sulfanyl]phenoxy}ethyl)-1,2-dihydrospiro [indole-3,4'-piperidin]-2-one Calc'd 448.1 Found 448.0 A-45 and B-4 Daicel Chiralpak ID-3 First 149 [01651]  embedded image (R) or (S)-5-chloro-1'-(2-{4-[methyl (methylinino) oxo-λ^{sup}.6-sulfanyl]phenoxy}ethyl)-1,2-dihydrospiro [indole-3,4'-piperidin]-2-one Calc'd 448.1 Found 448.0 A-45 and B-4 Daicel Chiralpak ID-3 Second 150 [01652]  embedded image 1'-(2-{4-[methyl (methylinino) oxo-λ^{sup}.6-sulfanyl]phenoxy}ethyl)-2-oxo-1,2-dihydrospiro [indole-3,4'-piperidine]-5-carbonitrile Calc'd 439.2 Found 439.0 A-45 and B-9 n/a n/a 151 [01653]  embedded image (S) or (R)-1'-(2-{4-[methyl (methylinino) oxo-λ^{sup}.6-sulfanyl]phenoxy}ethyl)-2-oxo-1,2-dihydrospiro [indole-3,4'-piperidine]-5-carbonitrile Calc'd 439.2 Found 439.0 A-45 and B-9 Daicel Chiralpak ID-3 Second 152 [01654]  embedded image (R) or (S)-1'-(2-{4-[methyl (methylinino) oxo-λ^{sup}.6-sulfanyl]phenoxy}ethyl)-2-oxo-1,2-dihydrospiro [indole-3,4'-piperidine]-5-carbonitrile Calc'd 439.2 Found 439.0 A-45 and B-9 Daicel Chiralpak ID-3 First 153 [01655]  embedded image 6-(2-{5-chloro-2-oxo-1,2-dihydrospiro [indole-3,4'-piperidin]-1'-yl}ethoxy)-2-methyl-3,4-dihydro-2H-126,2-benzothiazine-1,1-dione Calc'd 476.1 Found 476.1 A-49 and B-4 n/a n/a 154 [01656]

 embedded image 1'-{2-[(2-methyl- 1,1-dioxo-3,4- dihydro-2H-126,2- benzothiazin-6-yl)oxy]ethyl}-2- oxo-1,2- dihydrospiro [indole-3,4'- piperidine]- 5-carbonitrile Calc'd 467.2 Found 467.2 A-49 and B-9 n/a n/a 155 [01657]  embedded image 5-chloro-1'-[2-(3- chloro-4-methanesulfonyl- phenoxy)ethyl]-1,2- dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 469.1 Found 469.2 A-19 and B-4 n/a n/a 156 [01658]  embedded image 1'-[2-(3-chloro-4-methanesulfonyl- phenoxy)ethyl]-2- oxo-1,2- dihydrospiro [indole-3,4'- piperidine]- 5-carbonitrile Calc'd 460.1 Found 460.2 A-19 and B-9 n/a n/a 157 [01659]  embedded image 1'-{2-[4-(1-cyano-1-methylethyl) phenoxy]ethyl}- 2-oxo-1,2- dihydrospiro [indole-3,4'- piperidine]- 5-carbonitrile Calc'd 415.2 Found 415.2 A-47 and B-9 n/a n/a 158 [01660]  embedded image 5-chloro-1'-[2-({6-[(3-methyloxetan- 3-yl)sulfonyl] pyridin-3-yl} oxy)ethyl]- 1,2- dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 492.1 Found 492.1 A-26 and B-4 n/a n/a 159 [01661]  embedded image 1-(2- hydroxyethyl)-1'- {2-[4- methanesulfonyl- 3-(trifluoromethyl) phenoxy]ethyl}- 2-oxo-1,2- dihydrospiro [indole-3,4'- piperidine]- 5-carbonitrile Calc'd 538.2 Found 538.2 A-17 and B-12 n/a n/a 160 [01662]  embedded image 5-chloro-1-(2- hydroxyethyl)-1'- {2-[4-(3- methanesulfonyl- oxetan-3- yl)phenoxy] ethyl}-1,2- dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 535.2 Found 535.2 A-33 and B-11 n/a n/a 161 [01663]  embedded image 5-chloro-1'-{2-[(6- methanesulfonyl- 5-methylpyridin- 3-yl)oxy]ethyl}- 1,2- dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 450.1 Found 450.1 A-21 and B-4 n/a n/a 162 [01664]  embedded image 5-chloro-1'-(2-{{1- (2- hydroxyethyl)- 2-oxo-1,2,3,4- tetrahydroquinolin- 6-yl}oxy}ethyl)- 1,2- dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 470.2 Found 470.1 A-65 and B-4 n/a n/a 163 [01665]  embedded image 5-chloro-1'-{2-[3- (difluoromethoxy)- 4-methanesulfonyl- phenoxy]ethyl}- 1,2- dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 501.1 Found 501.2 A-23 and B-4 n/a n/a 164 [01666]  embedded image 1'-{2-[3- (difluoromethoxy)- 4-methanesulfonyl- phenoxy]ethyl}-2- oxo-1,2- dihydrospiro [indole-3,4'- piperidine]- 5-carbonitrile Calc'd 492.1 Found 492.2 A-23 and B-9 n/a n/a 165 [01667]  embedded image 5-chloro-1'-{2-[4- (3- methanesulfonyl- oxetan-3-y1)-3- (trifluoromethyl) phenoxy]ethyl}- 1,2- dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 559.1 Found 559.2 A-34 and B-4 n/a n/a 166 [01668]  embedded image 1'-{2-[4-(3- methanesulfonyl- oxetan-3-y1)-3- (trifluoromethyl) phenoxy]ethyl}-2- oxo-1,2- dihydrospiro [indole-3,4'- piperidine]- 5-carbonitrile Calc'd 550.2 Found 550.2 A-34 and B-9 n/a n/a 167 [01669]  embedded image 6-(2-{5-chloro-2- oxo-1,2- dihydrospiro [indole-3,4'- piperidin]- 1'-yl}ethoxy)-3,4- dihydro-2H-126,2- benzothiazine-1,1- dione Calc'd 462.1 Found 462.0 A-50 and B-4 n/a n/a 168 [01670]  embedded image 1'-{2-[(1, 1-dioxo- 3,4-dihydro-2H- 1λ.sup.6,2- benzothiazin-6- yl)oxy]ethyl}-2- oxo-1,2- dihydrospiro [indole-3,4'- piperidine]- 5-carbonitrile Calc'd 453.2 Found 453.2 A-50 and B-9 n/a n/a 169 [01671]  embedded image 1'-{2-[(7-fluoro-1- methyl-2-oxo-2,3- dihydro-1H-indol- 5-yl)oxy]ethyl}-2- oxo-1,2- dihydrospiro [indole-3,4'- piperidine]- 5-carbonitrile Calc'd 435.2 Found 435.1 A-60 and B-9 n/a n/a 170 [01672]  embedded image 2-oxo-1'-{2-[(1- oxo-2,3-dihydro- 1H-isoindol-5- yl)oxy]ethyl}-1,2- dihydrospiro [indole-3,4'- piperidine]- 5-carbonitrile Calc'd 403.2 Found 403.3 A-61 and B-9 n/a n/a 171 [01673]  embedded image 5-chloro-1'-{2-[(1- oxo-2,3-dihydro- 1H-isoindol-5- yl)oxy]ethyl}-1,2- dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 412.1 Found 412.2 A-61 and B-4 n/a n/a 172 [01674]  embedded image 5-chloro-1'-[2-({2- [(3-methyloxetan- 3-yl)oxy]pyrimidin- 5-y1}oxy)ethyl]- 1,2- dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 445.2 Found 445.2 A-43 and B-4 n/a n/a 173 [01675]  embedded image 1'-[2-({2-[(3- methyloxetan-3- yl)oxy]pyrimidin- 5-y1}oxy)ethyl]-2- oxo-1,2- dihydrospiro[indol e-3,4'-piperidine]- 5-carbonitrile Calc'd 436.2 Found 436.2 A-43 and B-9 n/a n/a 174 [01676]  embedded image 1'-(2-{4-[3- (ethanesulfonyl) oxetan-3- yl]phenoxy} ethyl)-2-oxo-1,2- dihydrospiro [indole-3,4'- piperidine]- 5-carbonitrile Calc'd 496.2 Found 496.0 A-37 and B-9 n/a n/a 175 [01677]  embedded image 2-[4- (2-{5-chloro- 2-oxo-1,2- dihydrospiro [indole-3,4'- piperidin]- 1'-yl}ethoxy) phenyl]- 1λ.sup.6- thiolane-1,1- dione Calc'd 475.1 Found 475.2 A-38 and B-4 n/a n/a 176 [01678]  embedded image (S) or (R)-2-[4-(2- {5-chloro-2-oxo- 1,2-dihydrospiro [indole-3,4'- piperidin]-1'- yl}ethoxy)

phenyl]- 1λ.sup.6-thiolane-1,1- dione Calc'd 475.1 Found 475.2 A-38 and B-4 Regis(S, S)Whelk - O1 First 177 [01679]  embedded image (R) or (S)-2-[4-(2- {5-chloro-2-oxo- 1,2- dihydrospiro [indole-3,4'- piperidin]- 1'-yl}ethoxy) phenyl]- 1λ.sup.6-thiolane-1,1- dione Calc'd 475.1 Found 475.2 A-38 and B-4 Regis(S, S)Whelk -O1 Second 178 [01680]  embedded image 1'-{2-[4-(1,1-dioxo-1λ.sup.6-thiolan- 2-yl)phenoxy] ethyl}-2-oxo-1,2- dihydrospiro [indole-3,4'- piperidine]- 5- carbonitrile Calc'd 466.2 Found 466.2 A-38 and B-9 n/a n/a 179 [01681]  embedded image 5-chloro-1'-(2-{3- oxo-3H-spiro[2- benzofuran-1,3'- oxetan]-5- yloxy}ethyl)-1,2- dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 455.1 Found 455.0 A-38 and B-9 n/a n/a 180 [01682]  embedded image 5-chloro-1'-{2-[4- (1- methanesulfonyl- cyclobutyl) phenoxy] ethyl}-1,2- dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 489.2 Found 489.1 A-40 and B-4 n/a n/a 181 [01683]  embedded image 1'-{2-[4-(1- methanesulfonyl- cyclobutyl) phenoxy] ethyl}-2-oxo-1,2- dihydrospiro [indole-3,4'- piperidine]- 5-carbonitrile Calc'd 480.2 Found 480.2 A-40 and B-9 n/a n/a 182 [01684]  embedded image 5-chloro-1'-[2-(4- {6- methanesulfonyl- 2- oxaspiro[3.3] heptan-6- yl}phenoxy) ethyl]-1,2- dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 531.2 Found 531.1 A-42 and B-4 n/a n/a 183 [01685]  embedded image 1'-[2-(4-{6- methanesulfonyl- 2- oxaspiro[3.3] heptan-6- yl}phenoxy) ethyl]-2-oxo-1,2- dihydrospiro [indole-3,4'- piperidine]- 5- carbonitrile Calc'd 522.2 Found 522.1 A-42 and B-9 n/a n/a 184 [01686]  embedded image 1'-{2-[4-(3- methanesulfonyl- oxetan-3- yl)phenoxy]ethyl}- 1-(2,2,2- trifluoroethyl)-1,2- dihydrospiro [indole-3,4'- piperidine]- 5-carbonitrile Calc'd 564.2 Found 564.2 A-33 and B-13 n/a n/a 185 [01687]  embedded image 5-chloro-1'-{2-[4- (3- methanesulfonyl- oxetan-3- yl)phenoxy]ethyl}- 1-(2,2,2- trifluoroethyl)-1,2- dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 573.1 Found 573.2 A-33 and B-14 n/a n/a 186 [01688]  embedded image N-(4-{2-[5-chloro- 1-(2- hydroxyethyl)-2- oxo-1,2- dihydrospiro [indole-3,4'- piperidin]- 1'-yl]ethoxy}-2- (trifluoromethyl) phenyl)methane- sulfonamide Calc'd 562.1 Found 562.1 A-46 and B-11 n/a n/a 187 [01689]  embedded image 1'-{2-[4-(4- methanesulfonyl- oxan-4- yl)phenoxy] ethyl}-2-oxo-1,2- dihydrospiro [indole- 3,4'-piperidine]- 5-carbonitrile Calc'd 510.2 Found 510.2 A-41 and B-9 n/a n/a 188 [01690]  embedded image 5-chloro-1'-{2-[4- (4- methanesulfonyl- oxan-4- yl)phenoxy] ethyl}-1,2- dihydrospiro [indole- 3,4'-piperidin]- 2-one Calc'd 519.2 Found 519.2 A-41 and B-4 n/a n/a 189 [01691]  embedded image 1'-{2-[3,5- difluoro-4-(3- methanesulfonyl- oxetan-3- yl)phenoxy]ethyl}- 1-methyl-2-oxo- 1,2- dihydrospiro [indole- 3,4'-piperidine]- 5-carbonitrile Calc'd 532.1 Found 532.2 A-32 and B-15 n/a n/a 190 [01692]  embedded image 5-chloro-1'-[2- ({7- oxo-8-[(cis)-3- hydroxycyclobutyl 1-5,6,7,8- tetrahydro-1,8- naphthyridin-3- yl]oxy}ethyl)-1,2- dihydrospiro [indole- 3,4'-piperidin]- 2-one Calc'd 497.2 Found 497.2 A-79 and B-4 n/a n/a 191 [01693]  embedded image 1'-(2-{[6-(2- methanesulfonyl- propan-2- yl)pyridin- 3- yl]oxy}ethyl)-2- oxo-1,2- dihydrospiro [indole-3,4'- piperidine]- 5-carbonitrile Calc'd 469.2 Found 469.2 A-78 and B-9 n/a n/a 192 [01694]  embedded image 5-chloro-1'-(2-{[8- (2-hydroxyethyl)- 7-oxo-5,6,7,8- tetrahydro-1,8- naphthyridin-3- yloxy}ethyl)-1,2- dihydrospiro [indole-3,4'- piperidin]- 2-one Calc'd 471.2 Found 471.2 A-77 and B-4 n/a n/a

Example 26

N-{2-[4-(2-{5-chloro-2-oxo-1,2-dihydrosiro[indole-3,4'-piperidin]1'-yl}ethoxy)benzenesulfonyl]ethyl}-N-methylacetamide (Compound 193)

##STR01695##

[2906] A mixture of 5-chloro-1'-(2-chloroethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-5, 211 mg, 0.707 mmol), N-[2-(4-hydroxybenzenesulfonyl)ethyl]-N-methylacetamide (Intermediate A-25, 140 mg, 0.544 mmol) and K.sub.2CO.sub.3 (150 mg, 1.09 mmol) in DMF (4 mL) was stirred at 50° C. for 16 h. After cooling to room temperature, the mixture was filtered to remove solids. The filtrate was purified by reverse phase preparative HPLC (Phenomenex Gemini-NX C.sub.18 column, 20-50% MeCN: 10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give N-{2-[4-(2-{5-chloro-2-oxo-1,2-dihydrosiro[indole-3,4'-piperidin]-1'-yl}ethoxy)benzenesulfonyl]ethyl}I-N-methylacetamide (Compound 193). .sup.1H NMR (400

MHz, DMSO-d₆.sub.6): δ 10.48 (s, 1H), 7.84-7.80 (m, 2H), 7.51 (s, 1H), 7.24-7.20 (m, 3H), 6.84 (d, J=8.0, 1H), 4.26 (t, J=5.6 Hz, 2H), 3.65-3.46 (m, 4H), 2.93-2.87 (m, 6H), 2.70-2.67 (m, 3H), 1.93 (s, 1H), 1.86 (s, 2H), 1.77-1.74 (m, 4H). MS=520.2 [M+H].sup.+.

[2907] The following compounds in Table 30 were prepared according to procedures analogous to those described for Compound 193 using the appropriate starting materials.

TABLE-US-00033 TABLE 30 Exact Mass Intermediates # Structure IUPAC Name [M + H].sup.+
Used 194 [01696]  5-chloro-1'-{2-[(2-acetyl-1,2,3,4-tetrahydroisoquinolin-6-yl)oxy]ethyl}-5-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 454.2 Found 454.2 A-56 and B-5 195 [01697]  5-chloro-1'-{2-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 491.1 Found 491.1 A-84 and B-5 196 [01698]  5-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2,3-dihydro-1λ.sup.6-benzothiophene-1,1-dione Calc'd 447.1 Found 447.2 A-52 and B-5 197 [01699]  5-chloro-1'-{2-[3-(difluoromethyl)-4-methanesulfonyl-phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 485.1 Found 485.1 A-12 and B-5 198 [01700]  5-chloro-1'-{2-[4-(2-hydroxyethanesulfonyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 465.1 Found 465.1 A-85 and B-5

Example 27

5-chloro-1'-{1-[(1-methyl-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl)oxy]propan-2-yl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 199), chloro-1'-[(2S) or (2R)-1-[(1-methyl-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl)oxy]propan-2-yl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 200), and chloro-1'-[(2R) or (2S)-1-[(1-methyl-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl)oxy]propan-2-yl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 201)

##STR01701##

Step 1: 5-chloro-1'-{1-[(1-methyl-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl)oxy]propan-2-yl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 199)

##STR01702##

[2908] A solution of 1-methyl-6-(2-oxopropoxy)-1,2,3,4-tetrahydroquinolin-2-one (Intermediate A-2, 400 mg, 1.71 mmol), 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 406 mg, 1.71 mmol), Ti(i-PrO)₄ (506 μL, 1.71 mmol) and AcOH (9.81 μL, 0.171 mmol) in 1,2-dichloroethane (5 mL) was stirred at 50° C. for 2 h. The mixture was cooled to 0° C. and NaBH(OAc)₃ (545 mg, 2.57 mmol) was added in portionwise. The mixture was stirred at 50° C. for 12 hours. After cooling to 0° C., the reaction mixture was quenched with H₂O (5 mL), and then filtered. The filtrate was extracted with EtOAc (3×10 mL). The combined organic layers were washed with brine (8 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The residue was purified by reverse phase preparative HPLC (Waters Xbridge Prep OBD C₁₈ column, 30-60% MeCN: 10 mM NH₄HCO₃ in H₂O) to give 5-chloro-1'-{1-[(1-methyl-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl)oxy]propan-2-yl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 199). ¹H NMR (400 MHz, DMSO-d₆.sub.6): δ 10.48 (s, 1H), 7.45 (s, 1H), 7.23 (d, J=6.0 Hz, 1H), 7.01 (d, J=8.8 Hz, 1H), 6.91-6.83 (m, 3H), 4.12-4.08 (m, 1H), 3.96-3.93 (m, 1H), 3.22 (s, 3H), 3.10-2.98 (m, 3H), 2.85-2.81 (m, 3H), 2.79-2.77 (m, 1H), 2.53-2.51 (m, 2H), 1.80-1.76 (m, 2H), 1.68-1.65 (m, 2H), 1.14 (d, J=6.8 Hz, 3H). MS=454.2 [M+H].sup.+.

Step 2: chloro-1'-[(2S) or (2R)-1-[(1-methyl-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl)oxy]propan-2-yl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 200) and chloro-1'-[(2R) or (2S)-1-[(1-methyl-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl)oxy]propan-2-yl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 201)

##STR01703##

[2909] 5-chloro-1'-{1-[(1-methyl-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl)oxy]propan-2-yl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one was separated by preparative chiral SFC (Daicel Chiralpak AD-3, 42% ethanol with 0.1% NH₄OH in CO₂). The first eluting enantiomer

of the title compound, chloro-1'-[(2S) or (2R)-1-[(1-methyl-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl)oxy]propan-2-yl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 200): ¹H NMR (400 MHz, DMSO-d₆): δ 10.49 (s, 1H), 7.45 (s, 1H), 7.23 (d, J=8.0 Hz, 1H), 7.01 (d, J=8.8 Hz, 1H), 6.90-6.83 (m, 3H), 4.12-4.08 (m, 1H), 3.96-3.92 (m, 1H), 3.22 (s, 3H), 3.10-2.98 (m, 3H), 2.85-2.81 (m, 3H), 2.77-2.73 (m, 1H), 2.53-2.51 (m, 2H), 1.80-1.76 (m, 2H), 1.68-1.65 (m, 2H), 1.14 (d, J=6.8 Hz, 3H). MS=454.2 [M+H]⁺. The second eluting enantiomer of the title compound, chloro-1'-[(2R) or (2S)-1-[(1-methyl-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl)oxy]propan-2-yl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 201): ¹H NMR (400 MHz, DMSO-d₆): δ 10.49 (s, 1H), 7.45 (s, 1H), 7.23 (d, J=8.4 Hz, 1H), 7.01 (d, J=8.8 Hz, 1H), 6.91-6.83 (m, 3H), 4.12-4.08 (m, 1H), 3.95-3.93 (m, 1H), 3.22 (s, 3H), 3.10-2.98 (m, 3H), 2.85-2.81 (m, 3H), 2.77-2.73 (m, 1H), 2.53-2.51 (m, 2H), 1.78-1.76 (m, 2H), 1.68-1.65 (m, 2H), 1.14 (d, J=6.8 Hz, 3H). MS=454.2 [M+H]⁺.

[2910] The following compounds in Table 31 were prepared according to procedures analogous to those described for Compounds 199-201 using the appropriate starting materials.

TABLE-US-00034 TABLE 31 Exact Chiral Column, Mass Intermediates If Elution # Structure IUPAC Name [M + H]⁺ Used Applicable Order 202 [01704]  embedded image 1'-{1-[(1-methyl-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl)oxy]propan-2-yl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile Calc'd 445.2 Found 445.2 A-2 and B-9 n/a n/a 203 [01705]  embedded image (S) or (R)-1'-{1-[(1-methyl-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl)oxy]propan-2-yl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile Calc'd 445.2 Found 445.2 A-2 and B-9 Daicel Chiralpak AD-3 First 204 [01706]  embedded image (R) or (S)-1'-{1-[(1-methyl-2-oxo-1,2,3,4-tetrahydroquinolin-6-yl)oxy]propan-2-yl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile Calc'd 445.2 Found 445.2 A-2 and B-9 Daicel Chiralpak AD-3 Second

Example 28

5-chloro-1'-[1-(4-methanesulfonylphenoxy)propan-2-yl]-1-[3-hydroxycyclobutyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 205), 5-chloro-1'-[(2S) or (2R)-1-(4-methanesulfonylphenoxy)propan-2-yl]-1-[(cis) or (trans)-3-hydroxycyclobutyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 206), 5-chloro-1'-[(2S) or (2R)-1-(4-methanesulfonylphenoxy)propan-2-yl]-1-[(trans) or (cis)-3-hydroxycyclobutyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 207), 5-chloro-1'-[(2R) or (2S)-1-(4-methanesulfonylphenoxy)propan-2-yl]-1-[(cis) or (trans)-3-hydroxycyclobutyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 208), and 5-chloro-1'-[(2R) or (2S)-1-(4-methanesulfonylphenoxy)propan-2-yl]-1-[(trans) or (cis)-3-hydroxycyclobutyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 209)

##STR01707##

Step 1: 5-chloro-1'-[1-(4-methanesulfonylphenoxy)propan-2-yl]-1-[3-hydroxycyclobutyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 205)

##STR01708##

[2911] A mixture of 5-chloro-1-(3-hydroxycyclobutyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-16, 250 mg, 0.815 mmol), 1-(4-methanesulfonylphenoxy)propan-2-one (Intermediate A-1, 465 mg, 2.04 mmol), tetrakisopropoxytitanium (231 mg, 0.815 mmol) and AcOH (3.0 mg, 40.7 μmol) in DCE (5 mL) was stirred at room temperature for 1 h. NaBH(OAc)₃ (259 mg, 1.22 mmol) was added into the mixture, which was then stirred at 50° C. for 16 h. The reaction mixture was cooled to 0° C. and then quenched with H₂O (20 mL). The mixture was extracted with EtOAc (3×15 mL). The combined organic layers were washed with brine (20 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The residue was purified by reverse phase preparative HPLC (Waters Xbridge BEH C₁₈ column, 25-55% MeCN: 10 mM NH₄HCO₃ in H₂O) to give 5-chloro-1'-[1-(4-methanesulfonylphenoxy)propan-2-yl]-1-[3-hydroxycyclobutyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one as a mixture of

diastereomers (Compound 205). ^{sup}.1H NMR (400 MHz, DMSO-d.sub.6, 30/31 H): δ 7.85 (d, J=8.8 Hz, 2H), 7.49 (d, J=2.0 Hz, 1H), 7.35-7.21 (m, 4H), 5.31 (d, J=6.8 Hz, 1H), 4.26-4.06 (m, 3H), 3.95-3.90 (m, 1H), 3.16-3.12 (m, 4H), 3.06-2.99 (m, 2H), 2.87-2.71 (m, 2H), 2.64-2.58 (m, 3H), 1.77-1.66 (m, 4H), 1.16 (d, J=6.8 Hz, 3H). MS=519.1 [M+H].^{sup}+

Step 2: 5-chloro-1'-[(2S) or (2R)-1-(4-methanesulfonylphenoxy)propan-2-yl]-1[(cis) or (trans)-3-hydroxycyclobutyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 206) and 5-chloro-1'-[(2R) or (2S)-1-(4-methanesulfonylphenoxy)propan-2-yl]-1-[(cis) or (trans)-3-hydroxycyclobutyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 208)

##STR01709##

[2912] 5-chloro-1'-[1-(4-methanesulfonylphenoxy)propan-2-yl]-1-[3-hydroxycyclobutyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 205) was separated by preparative chiral SFC (Daicel Chiralcel IG-3, 60% EtOH with 0.1% NH.sub.4OH in CO.sub.2). The first eluting peak contained a mixture of isomers that underwent further chiral separation as described in step 3. The second eluting peak, 5-chloro-1'-[(2S) or (2R)-1-(4-methanesulfonylphenoxy)propan-2-yl]-1-[(cis) or (trans)-3-hydroxycyclobutyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 206): ^{sup}.1H NMR (400 MHz, DMSO-d.sub.6): δ 7.85 (d, J=8.8 Hz, 2H), 7.48 (d, J=2.0 Hz, 1H), 7.30 (dd, J=8.4, 2.4 Hz, 1H), 7.22 (d, J=8.8 Hz, 2H), 7.12 (d, J=8.8 Hz, 1H), 4.93-4.85 (m, 1H), 4.44-4.41 (m, 1H), 4.27-4.22 (m, 1H), 4.10-4.06 (m, 1H), 3.16-3.12 (m, 4H), 3.06-3.00 (m, 2H), 2.92-2.88 (m, 2H), 2.84-2.70 (m, 3H), 2.22-2.15 (m, 2H), 1.77-1.65 (m, 4H), 1.16 (d, J=6.8 Hz, 3H). MS=519.1 [M+H].^{sup}+. The third eluting peak, 5-chloro-1'-[(2R) or (2S)-1-(4-methanesulfonylphenoxy)propan-2-yl]-1-[(cis) or (trans)-3-hydroxycyclobutyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 208): ^{sup}.1H NMR (400 MHz, DMSO-d.sub.6, 30/31 H): δ 7.85 (d, J=8.8 Hz, 2H), 7.49 (d, J=2.0 Hz, 1H), 7.34-7.21 (m, 4H), 5.30 (d, J=6.4 Hz, 1H), 4.26-4.06 (m, 3H), 3.97-3.89 (m, 1H), 3.16-3.12 (m, 4H), 3.06-3.00 (m, 2H), 2.84-2.72 (m, 2H), 2.65-2.55 (m, 3H), 1.77-1.66 (m, 4H), 1.17-1.15 (m, 3H). MS=519.1 [M+H].^{sup}+. Step 3: 5-chloro-1'-[(2R) or (2S)-1-(4-methanesulfonylphenoxy)propan-2-yl]-1-[(trans) or (cis)-3-hydroxycyclobutyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 209) and 5-chloro-1'-[(2S) or (2R)-1-(4-methanesulfonylphenoxy)propan-2-yl]-1-[(trans) or (cis)-3-hydroxycyclobutyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 207)

##STR01710##

[2913] The second eluting peak from Step 2 was further separated by preparative chiral SFC (Daicel Chiralcel OJ-3, 30% MeOH with 0.1% NH.sub.4OH in CO.sub.2). The first eluting isomer, 5-chloro-1'-[(2R) or (2S)-1-(4-methanesulfonylphenoxy)propan-2-yl]-1-[(trans) or (cis)-3-hydroxycyclobutyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 209): ^{sup}.1H NMR (400 MHz, DMSO-d.sub.6, 30/31 H): δ 7.85 (d, J=9.2 Hz, 2H), 7.49 (s, 1H), 7.34-7.21 (m, 4H), 5.30 (d, J=6.8 Hz, 1H), 4.26-4.06 (m, 3H), 3.98-3.89 (m, 1H), 3.16-3.11 (m, 4H), 3.04-3.02 (m, 2H), 2.84-2.72 (m, 2H), 2.67-2.60 (m, 3H), 1.84-1.64 (m, 4H), 1.17-1.15 (m, 3H). MS=519.1 [M+H].^{sup}+. The second eluting isomer, 5-chloro-1'-[(2S) or (2R)-1-(4-methanesulfonylphenoxy)propan-2-yl]-1-[(trans) or (cis)-3-hydroxycyclobutyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 207), ^{sup}.1H NMR (400 MHz, DMSO-d.sub.6): δ 7.85 (d, J=8.8 Hz, 2H), 7.49 (d, J=2.0 Hz, 1H), 7.30 (dd, J=8.4, 2.0 Hz, 1H), 7.22 (d, J=8.8 Hz, 2H), 7.12 (d, J=8.4 Hz, 1H), 5.16 (d, J=4.4 Hz, 1H), 4.93-4.85 (m, 1H), 4.44-4.42 (m, 1H), 4.26-4.22 (m, 1H), 4.10-4.06 (m, 1H), 3.16-3.12 (m, 4H), 3.06-3.00 (m, 2H), 2.92-2.72 (m, 4H), 2.22-2.16 (m, 2H), 1.77-1.66 (m, 4H), 1.16 (d, J=6.8 Hz, 3H). MS=519.1 [M+H].^{sup}+

Example 29A

5-chloro-1'-[(2S)-1-[3-(difluoromethyl)-4-methanesulfonylphenoxy]propan-2-yl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 210)

##STR01711##

[2914] To a solution containing a mixture of 5-chloro-1'-[(2S)-1-hydroxypropan-2-yl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one and 5-chloro-1'-[(2R)-2-hydroxypropyl]-1,2-

dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediates B-17 and B-18, 136 mg, 0.461 mmol), 3-(difluoromethyl)-4-methanesulfonylphenol (Intermediate A-12, 154 mg, 0.692 mmol) in THE (4 mL) was added PPh.sub.3 (242 mg, 0.923 mmol). The mixture was cooled to 0° C. and DIAD (179 µL, 0.923 mmol) was added dropwise. The mixture was warmed to room temperature and stirred for 15 h. The reaction mixture was quenched with H.sub.2O (8 mL) and extracted with EtOAc (3×10 mL). The combined organic layers were washed with brine (5 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 4 g cartridge, 0-90% EtOAc:petroleum ether). The crude product was further purified by reverse phase preparative HPLC (Waters Xbridge BEH C.sub.18 column, 25-55% MeCN: 10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give the desired product 5-chloro-1'-[(2S)-1-[3-(difluoromethyl)-4-methanesulfonylphenoxy]propan-2-yl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 210) as the first eluting isomer. ¹H NMR (400 MHz, DMSO-d.sub.6): δ 10.49 (s, 1H), 8.00 (d, J=8.8 Hz, 1H), 7.59 (t, J=54.4 Hz, 1H), 7.44-7.37 (m, 3H), 7.23 (dd, J=8.0, 2.0 Hz, 1H), 6.84 (d, J=8.4 Hz, 1H), 4.31 (dd, J=10.0, 6.4 Hz, 1H), 4.16 (dd, J=10.0, 5.6 Hz, 1H), 3.26 (s, 3H), 3.18-3.13 (m, 1H), 3.03-3.01 (m, 2H), 2.85-2.73 (m, 2H), 1.80-1.75 (m, 2H), 1.67-1.63 (m, 2H), 1.16 (d, J=6.8 Hz, 3H). MS=499.1 [M+H].⁺. [2915] The following compounds in Table 32 were prepared according to procedures analogous to those described for Compound 210 using the appropriate starting materials.

TABLE-US-00035 TABLE 32 Exact Mass Intermediates Elution # Structure IUPAC Name [M + H].⁺ Used Order 211 [01712]  5-chloro-1'-[(2S)-1-[(1-methyl-1H-indazol-5-yl)oxy]propan-2-yl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 425.2 Found 425.2 B-17 and B-18 First eluting isomer 212 [01713]  1'-[(2S)-1-[3-(difluoromethyl)-4-methanesulfonylphenoxy]propan-2-yl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile Calc'd 490.2 Found 490.2 A-13, B-19, and B-20 First eluting isomer 213 [01714]  1'-[(2S)-1-(3,5-difluoro-4-methanesulfonylphenoxy)propan-2-yl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile Calc'd 476.1 Found 476.1 A-11, B-19, and B-20 First eluting isomer

Example 29B

N-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]methanesulfonamide (Compound 214)

##STR01715##

Step 1: 1-(2-bromoethoxy)-4-nitrobenzene

##STR01716##

[2916] To a solution of 4-nitrophenol (5.00 g, 35.9 mmol) in 1,2-dibromoethane (30 mL) was added Cs.sub.2CO.sub.3 (35.1 g, 108 mmol). The mixture was stirred at 100° C. for 12 h. After cooling to room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-50% EtOAc:petroleum ether) to give 1-(2-bromoethoxy)-4-nitrobenzene.

MS=246.0/248.0 [M+H].⁺.

Step 2: 5-chloro-1'-[2-(4-nitrophenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR01717##

[2917] To a solution of 1-(2-bromoethoxy)-4-nitrobenzene (300 mg, 1.22 mmol) in MeCN (3 mL) was added NaHCO.sub.3 (307 mg, 3.66 mmol) and 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 433 mg, 1.58 mmol, HCl salt). The mixture was stirred at 80° C. for 12 h. After cooling to room temperature, the reaction mixture was diluted with H.sub.2O (8 mL) and extracted with EtOAc (3×8 mL). The combined organic layers were washed with brine (10 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-30% EtOAc:petroleum ether) to give 5-chloro-1'-[2-(4-nitrophenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one. MS=402.2 [M+H].⁺.

Step 3: 1'-[2-(4-aminophenoxy)ethyl]-5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one
##STR01718##

[2918] A mixture of 5-chloro-1'-[2-(4-nitrophenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (200 mg, 498 μ mol), Fe (139 mg, 2.49 mmol), and NH₄Cl (79.9 mg, 1.49 mmol) in EtOH (6 mL) and H₂O (2 mL) was stirred at 70° C. for 2 h under a N₂ atmosphere. After cooling to room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo to give 1'-[2-(4-aminophenoxy)ethyl]-5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one, which was used in the subsequent step without further purification. MS=372.2 [M+H]⁺.

Step 4: N-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]methanesulfonamide (Compound 214)
##STR01719##

[2919] To a solution of 1'-[2-(4-aminophenoxy)ethyl]-5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (110 mg, 296 μ mol), and TEA (82.4 μ L, 0.52 mmol) in DCM (3 mL) was added methanesulfonic anhydride (103. mg, 0.592 mmol). The mixture was stirred at room temperature for 2 h under N₂ atmosphere. The reaction mixture was quenched with saturated aqueous NaHCO₃ (10 mL) and extracted with EtOAc (3 $\times$ 15 mL). The combined organic layers were washed with brine (20 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The residue was purified by reverse phase preparative HPLC (Phenomenex C₁₈ column, 20-50% MeCN: 10 mM NH₄HCO₃ in H₂O) to give N-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]methanesulfonamide (Compound 214). MS=450.2 [M+H]⁺.

Example 30

5-chloro-1'-[2-(4-{1[(trans)-3-hydroxycyclobutyl]sulfonyl}phenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 215)
##STR01720##

Step 1: (trans)-3-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)benzenesulfonyl]cyclobutyl 4-nitrobenzoate
##STR01721##

[2920] To a 0° C. solution of 5-chloro-1'-[2-(4-{[(cis)-3-hydroxycyclobutyl]sulfonyl}phenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 17, 13 mg, 26.5 μ mol), 4-nitrobenzoic acid (5.00 mg, 31.8 μ mol) and PPh₃ (27 mg, 106 μ mol) in THE (2 mL) was added DIAD (20.6 μ L, 0.106 mmol). The mixture was stirred at 40° C. for 16 h. After cooling to room temperature, the reaction mixture was diluted with H₂O (10 mL) and extracted with EtOAc (3 $\times$ 5 mL). The combined organic layers were washed with brine (5 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by preparative TLC (SiO₂, DCM:MeOH=10:1) to give (trans)-3-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)benzenesulfonyl]cyclobutyl 4-nitrobenzoate. MS=640.3 [M+H]⁺.

Step 2: 5-chloro-1'-[2-(4-{[(trans)-3-hydroxycyclobutyl]sulfonyl}phenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 215)
##STR01722##

[2921] To a 0° C. solution of (trans)-3-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)benzenesulfonyl]cyclobutyl 4-nitrobenzoate (10.0 mg, 15.6 μ mol) in THE (2 mL) was added a solution of LiOH.Math.H₂O (1.97 mg, 46.9 μ mol) in H₂O (0.5 mL). The mixture was stirred at room temperature for 2 h. The reaction mixture was diluted with H₂O (10 mL) and extracted with EtOAc (3 $\times$ 5 mL). The combined organic layers were washed with brine (5 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The residue was purified by reverse phase preparative HPLC (Waters Xbridge BEH C₁₈ column, 20-50% MeCN: 10 mM NH₄HCO₃ in H₂O) to give 5-chloro-1'-[2-(4-{[(trans)-3-

hydroxycyclobutyl)sulfonyl}phenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 215). ¹H NMR (400 MHz, DMSO-d₆): δ 10.50 (s, 1H), 7.80 (d, J=8.8 Hz, 2H), 7.51 (d, J=2.0 Hz, 1H), 7.25-7.22 (m, 1H), 7.20 (d, J=8.8 Hz, 2H), 6.84 (d, J=8.4 Hz, 1H), 5.36-5.33 (m, 1H), 4.27-4.21 (m, 3H), 3.89-3.81 (m, 1H), 2.95-2.87 (m, 4H), 2.72-2.67 (m, 4H), 2.18-2.11 (m, 2H), 1.81-1.68 (m, 4H). MS=491.1 [M+H]⁺.

Example 31

5-chloro-1'-{2-[3-(1,2-dihydroxyethyl)-4-methanesulfonylphenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 216)

5-chloro-1'-{2-[3-(1,2-difluoroethyl)-4-methanesulfonylphenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 217)

5-chloro-1'-(2-{3-[(1S or 1R)-1,2-difluoroethyl]-4-methanesulfonylphenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 218) and

5-chloro-1'-(2-{3-[(1R or 1S)-1,2-difluoroethyl]-4-methanesulfonylphenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 219)

##STR01723##

Step 1: 3-iodo-4-methanesulfonylphenol

##STR01724##

[2922] To a -20° C. solution of 2-iodo-1-methanesulfonyl-4-methoxybenzene (2.00 g, 6.41 mmol) in DCM (30 mL) under N₂ atmosphere was added BBr₃ (1.85 mL, 19.2 mmol) dropwise. The mixture was stirred at room temperature for 6 h. The reaction mixture was cooled to 0° C., quenched with H₂O (30 mL), and extracted with DCM (3×20 mL). The combined organic layers were washed with brine (15 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo to give 3-iodo-4-methanesulfonylphenol, which was used in the subsequent step without further purification. MS=299.0 [M+H]⁺.

Step 2: 4-(2-bromoethoxy)-2-iodo-1-methanesulfonylbenzene

##STR01725##

[2923] To a solution of 3-iodo-4-methanesulfonylphenol (1.90 g, 6.37 mmol) in DMF (5 mL) were added 1,2-dibromoethane (7.21 mL, 95.6 mmol) and K₂CO₃ (2.20 g, 15.9 mmol). The mixture was stirred at 80° C. for 5 h. After cooling to room temperature, the reaction mixture was quenched with H₂O (15 mL) and extracted with EtOAc (3×20 mL). The combined organic layers were washed with brine (2×15 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 20 g cartridge, 0-35% EtOAc:petroleum ether) to give 4-(2-bromoethoxy)-2-iodo-1-methanesulfonylbenzene. MS=404.9/406.9 [M+H]⁺.

Step 3: 5-chloro-1'-[2-(3-iodo-4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR01726##

[2924] To a solution of 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 1.38 g, 5.04 mmol, HCl salt) in MeCN (20 mL) was added NaHCO₃ (705 mg, 8.39 mmol) and 4-(2-bromoethoxy)-2-iodo-1-methanesulfonylbenzene (1.70 g, 4.20 mmol). The mixture was stirred at 80° C. for 12 h. After cooling to room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 20 g cartridge, 0-100% EtOAc:petroleum ether) to give 5-chloro-1'-[2-(3-iodo-4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one.

MS=561.1 [M+H]⁺.

Step 4: 5-chloro-1'-[2-(3-ethenyl-4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR01727##

[2925] A solution of potassium vinyltrifluoroborate (1.15 g, 8.56 mmol), 5-chloro-1'-[2-(3-iodo-4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (1.60 g, 2.85 mmol),

K.sub.2CO.sub.3 (1.18 g, 8.56 mmol), and Pd(dppf)Cl.sub.2.Math.CH.sub.2Cl.sub.2 (233 mg, 0.286 mmol) in 1,4-dioxane (15 mL) and H.sub.2O (2 mL) was degassed and purged with N.sub.2 (3×). The mixture was stirred at 100° C. for 12 h under N.sub.2 atmosphere. After cooling to room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-100% EtOAc:petroleum ether) to give 5-chloro-1'-[2-(3-ethenyl-4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one. MS=461.2 [M+H].sup.+.

Step 5: 5-chloro-1'-{2-[3-(1,2-dihydroxyethyl)-4-methanesulfonylphenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 216)

##STR01728##

[2926] To a 0° C. solution of 5-chloro-1'-[2-(3-ethenyl-4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (1.10 g, 2.39 mmol) in THE (8 mL) and H.sub.2O (2 mL) was added NMO (756 µL, 7.16 mmol) and K.sub.2OsO.sub.4.Math.2H.sub.2O (87.9 mg, 0.238 mmol) dropwise. The mixture was stirred at room temperature for 12 h. The reaction mixture was cooled to 0° C., quenched with H.sub.2O (15 mL), and then extracted with EtOAc (3×15 mL). The combined organic layers were washed with brine (15 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 20 g cartridge, 0-100% EtOAc:petroleum ether). The crude product was further purified by reverse phase preparative HPLC (Waters Xbridge BEH C.sub.18 column, 15-45% MeCN: 10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 5-chloro-1'-{2-[3-(1,2-dihydroxyethyl)-4-methanesulfonylphenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 216). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.49 (s, 1H), 7.82 (d, J=8.8 Hz, 1H), 7.51 (s, 1H), 7.27-7.22 (m, 2H), 7.09 (d, J=8.8 Hz, 1H), 6.84 (d, J=8.4 Hz, 1H), 5.50 (d, J=4.8 Hz, 1H), 5.41-5.38 (m, 1H), 4.85 (t, J=5.6 Hz, 1H), 4.25-4.21 (m, 2H), 3.55-3.53 (m, 1H), 3.49-3.46 (m, 1H), 3.20 (s, 3H), 2.95-2.87 (m, 4H), 2.72-2.70 (m, 2H), 1.81-1.73 (m, 4H). MS=495.1 [M+H].sup.+.

Step 6: 5-chloro-1'-{2-[3-(1,2-difluoroethyl)-4-methanesulfonylphenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 217)

##STR01729##

[2927] To a -30° C. solution of 5-chloro-1'-{2-[3-(1,2-dihydroxyethyl)-4-methanesulfonylphenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 216, 300 mg, 0.606 mmol) in THE (5 mL) was added DAST (400 µL, 3.03 mmol). The mixture was allowed to warm to room temperature and stirred for 10 h. The reaction mixture was cooled to 0° C. and quenched with H.sub.2O (10 mL) at 0° C., and then extracted with DCM (2×15 mL). The combined organic layers were washed with brine (10 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Waters Xbridge BEH C.sub.18 column, 15-65% MeCN: 10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 5-chloro-1'-{2-[3-(1,2-difluoroethyl)-4-methanesulfonylphenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 217). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.49 (s, 1H), 7.93 (d, J=8.8 Hz, 1H), 7.50 (s, 1H), 7.29-7.22 (m, 3H), 6.84 (d, J=8.4 Hz, 1H), 6.62-6.45 (m, 1H), 4.87-4.75 (m, 2H), 4.32-4.27 (m, 2H), 3.29 (s, 3H), 2.93-2.87 (m, 4H), 2.70-2.67 (m, 2H), 1.78-1.72 (m, 4H). MS=499.1 [M+H].sup.+.

Step 7: 5-chloro-1'-(2-{3-[(1R or 1S)-1,2-difluoroethyl]-4-methanesulfonylphenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 218) and 5-chloro-1'-(2-{3-[(1S or 1R)-1,2-difluoroethyl]-4-methanesulfonylphenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 219)

##STR01730##

[2928] 5-chloro-1'-{2-[3-(1,2-difluoroethyl)-4-methanesulfonylphenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 217, 110 mg, 0.216 mmol) was separated by

preparative chiral SFC (Daicel Chiralcel OJ-3, 25% EtOH with 0.1% NH₄OH in CO₂). 42% ethanol with 0.1% NH₄OH in CO₂). The second eluting enantiomer of the title compound, Compound 218: ¹H NMR (400 MHz, DMSO-d₆): δ 10.49 (s, 1H), 7.93 (d, J=8.8 Hz, 1H), 7.50 (s, 1H), 7.29-7.23 (m, 3H), 6.84 (d, J=8.0 Hz, 1H), 6.62-6.45 (m, 1H), 4.87-4.75 (m, 2H), 4.32-4.27 (m, 2H), 3.29 (s, 3H), 2.93-2.87 (m, 4H), 2.70-2.67 (m, 2H), 1.78-1.68 (m, 4H). MS=499.1 [M+H]⁺. The first eluting enantiomer of the title compound, Compound 219: ¹H NMR (400 MHz, DMSO-d₆): δ 10.49 (s, 1H), 7.93 (d, J=8.8 Hz, 1H), 7.50 (s, 1H), 7.29-7.22 (m, 3H), 6.84 (d, J=8.4 Hz, 1H), 6.62-6.50 (m, 1H), 4.87-4.75 (m, 2H), 4.31-4.27 (m, 2H), 3.29 (s, 3H), 2.93-2.87 (m, 4H), 2.70-2.67 (m, 2H), 1.78-1.72 (m, 4H). MS=499.1 [M+H]⁺.

[2929] The following compounds in Table 33 were prepared according to procedures analogous to those described for Compounds 217-219 using the appropriate starting materials.

TABLE-US-00036 TABLE 33 Exact Inter- Mass mediates Elution Chiral # Structure Name [M + H]⁺ Used Order Column 220 [01731]  1'-{2-[3-(1,2-difluoroethyl)-4-methanesulfonyl-phenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile Calc'd 490.2 Found 490.0 B-9 n/a n/a 221 [01732]  1'-(2-{3-[(1S or 1R)-1,2-difluoroethyl]-4-methanesulfonyl-phenoxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile Calc'd 490.2 Found B-9 Second Daicel chiralcel OD-3 222 [01733]  1'-(2-{3-[(1R or 1S)-1,2-difluoroethyl]-4-methanesulfonyl-phenoxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile Calc'd 490.2 Found B-9 First Daicel Chiralcel OD-3

Example 32

5-chloro-1'-{2-[4-methanesulfonyl-3-(oxetan-3-yl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 223)

##STR01734##

[2930] To a solution of 5-chloro-1'-[2-(3-iodo-4-methanesulfonylphenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Example 31: Step 3, 100 mg, 0.178 mmol) and 3-iodooxetane (328 mg, 1.78 mmol) in DME (3 mL) under N₂ atmosphere was added nickel(II) chloride ethylene glycol dimethyl ether complex (0.20 mg, 0.91 μmol), 4,4'-di-tert-butyl-2,2'-dipyridyl (0.24 mg, 0.89 μmol), Ir[dF(CF₃)ppy]₂(dtbpy)PF₆ (2.0 mg, 1.8 μmol), bis(trimethylsilyl)silyl-trimethyl-silane (55.0 μL, 0.178 mmol) and Na₂CO₃ (37.8 mg, 0.357 mmol). The mixture was stirred at room temperature under 35 W blue LED lights for 15 h. Solids were removed by filtration, and the filtrate was concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Phenomenex C₁₈ column, 20-50% MeCN: 10 mM NH₄HCO₃ in H₂O) to give 5-chloro-1'-{2-[4-methanesulfonyl-3-(oxetan-3-yl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 223). ¹H NMR (400 MHz, DMSO-d₆): δ 10.49 (s, 1H), 7.88 (d, J=8.8 Hz, 1H), 7.51-7.47 (m, 2H), 7.24 (dd, J=8.0, 2.0 Hz, 1H), 7.13 (dd, J=8.8, 2.4 Hz, 1H), 6.85 (d, J=8.4 Hz, 1H), 4.93-4.92 (m, 3H), 4.72-4.69 (m, 2H), 4.32 (t, J=5.6 Hz, 2H), 3.16 (s, 3H), 2.95-2.89 (m, 4H), 2.74-2.70 (m, 2H), 1.82-1.70 (m, 4H). MS=491.0 [M+H]⁺.

Example 33

5-chloro-1'-[2-({2-oxo-1-[(trans)-3-hydroxycyclobutyl]-1,2,3,4-tetrahydroquinolin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 224)

##STR01735##

Step 1: (cis)-3-[6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-oxo-1,2,3,4-tetrahydroquinolin-1-yl]cyclobutyl methanesulfonate

##STR01736##

[2931] To a solution of 5-chloro-1'-[2-({2-oxo-1-[(cis)-3-hydroxycyclobutyl]-1,2,3,4-tetrahydroquinolin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 114, 50.0 mg, 0.101 mmol) in DCM (1 mL) was added TEA (28.1 μL, 0.202 mmol). The mixture was

cooled to 0° C. and methanesulfonic anhydride (26.3 mg, 0.151 mmol) was added dropwise. The mixture was allowed to warm to room temperature and stirred for 2 h. The reaction mixture was cooled to 0° C., quenched with H.sub.2O (8 mL), and then extracted with EtOAc (2×10 mL). The combined organic layers were washed with brine (10 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give (cis)-3-[6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-oxo-1,2,3,4-tetrahydroquinolin-1-yl]cyclobutyl methanesulfonate, which was used in the subsequent step without further purification. MS=574.2 [M+H].sup.+.

Step 2: (trans)-3-[6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-oxo-1,2,3,4-tetrahydroquinolin-1-yl]cyclobutyl acetate

##STR01737##

[2932] To a solution of (cis)-3-[6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-oxo-1,2,3,4-tetrahydroquinolin-1-yl]cyclobutyl methanesulfonate (40.0 mg, 69.9 µmol) in DMF (1 mL) was added KOAc (68.4 mg, 0.697 mmol). The mixture was stirred at 100° C. for 12 h. After cooling to 0° C., the reaction mixture was quenched with H.sub.2O (10 mL), solids were removed by filtration, and the filtrate was extracted with EtOAc (2×10 mL). The combined organic layers were washed with brine (10 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give (trans)-3-[6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-oxo-1,2,3,4-tetrahydroquinolin-1-yl]cyclobutyl acetate, which was used in the subsequent step without further purification. MS=538.3 [M+H].sup.+.

Step 3: 5-chloro-1'-[2-({2-oxo-1-[(trans)-3-hydroxycyclobutyl]-1,2,3,4-tetrahydroquinolin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 224)

##STR01738##

[2933] To a solution of [(trans)-3-[6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-oxo-1,2,3,4-tetrahydroquinolin-1-yl]cyclobutyl acetate (40.0 mg, 74.3 µmol) in MeOH (1 mL) was added K.sub.2CO.sub.3 (20.6 mg, 0.149 mmol). The mixture was stirred at room temperature for 6 h. The reaction mixture was filtered to remove solids, and the filtrate was concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Waters Xbridge BEH C.sub.18 column, 20-50% MeCN: 10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 5-chloro-1'-[2-({2-oxo-1-[(trans)-3-hydroxycyclobutyl]-1,2,3,4-tetrahydroquinolin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 224). MS=496.2 [M+H].sup.+.

Example 34

5-chloro-1'-[2-(4-{[1-(hydroxymethyl)cyclopropyl]sulfonyl}phenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 225)

##STR01739##

[2934] A solution of methyl 1-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)benzenesulfonyl]cyclopropane-1-carboxylate (Compound 91, 180 mg, 0.347 mmol) in THE (3 mL) was degassed and purged with N.sub.2 (3×). The mixture was cooled to 0° C. and 2.0 M LiBH.sub.4 in THE (21.4 mL, 42.8 mmol) was added. After stirring at 0° C. for 3 h, the mixture was allowed to warm to room temperature and stirred for another 13 h. The reaction mixture was cooled to 0° C., quenched with H.sub.2O (5 mL), and then adjusted to pH=6 with 4.0 M aqueous HCl. The combined organic layers were washed with brine (2×6 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Waters Xbridge BEH C.sub.18 column, 25-55% MeCN: 10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 5-chloro-1'-[2-(4-{[1-(hydroxymethyl)cyclopropyl]sulfonyl}phenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 225). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.48 (s, 1H), 7.77 (d, J=8.8 Hz, 2H), 7.50 (d, J=2 Hz, 1H), 7.23 (d, J=8.4 Hz, 1H), 7.17 (d, J=8.8 Hz, 2H), 6.84 (d, J=8.4 Hz, 1H), 4.85 (t, J=6.4 Hz, 1H), 4.25 (t, J=5.6 Hz, 2H), 3.61 (d, J=6.4 Hz, 2H), 2.93-2.87 (m, 4H), 2.72-2.70 (m, 2H), 1.78-1.72 (m, 4H), 1.30 (t, J=4.4 Hz, 2H), 1.01 (t, J=4.8 Hz, 2H). MS=491.1

[M+H].sup.+.

Example 35

1-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)benzenesulfonyl]-N-methylcyclopropane-1-carboxamide (Compound 226)

##STR01740##

Step 1: 1-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)benzenesulfonyl]cyclopropane-1-carboxylic acid

##STR01741##

[2935] To a solution of methyl 1-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)benzenesulfonyl]cyclopropane-1-carboxylate (Compound 91, 1.30 g, 2.50 mmol) in MeOH (8 mL) and H₂O (8 mL) was added LiOH.H₂O (210 mg, 5.01 mmol). The mixture was stirred at room temperature for 3 h. The reaction mixture was concentrated in vacuo, and the residue was adjusted to pH=4 with 1.0 M aqueous HCl. The resulting solid was collected via filtration and dried in vacuo to give 1-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)benzenesulfonyl]cyclopropane-1-carboxylic acid, which was used in the subsequent step without further purification. ¹H NMR (400 MHz, DMSO-d₆, 24/25 H): δ 10.55 (s, 1H), 7.90 (d, J=8.8 Hz, 2H), 7.51 (d, J=1.6 Hz, 1H), 7.26 (dd, J=8.4, 3.0 Hz, 1H), 7.10 (d, J=8.8 Hz, 2H), 6.85 (d, J=8.4 Hz, 1H), 4.28 (t, J=5.2 Hz, 2H), 3.13-3.10 (m, 4H), 2.93-2.90 (m, 2H), 1.86-1.84 (m, 4H), 1.70-1.68 (m, 2H), 1.51-1.49 (m, 2H). MS=505.0 [M+H].sup.+.

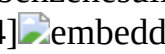
Step 2: 1-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)benzenesulfonyl]-N-methylcyclopropane-1-carboxamide (Compound 226)

##STR01742##

[2936] To a 0° C. solution of 1-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)benzenesulfonyl]cyclopropane-1-carboxylic acid (120 mg, 0.238 mmol) in DCM (2 mL) was added methylamine hydrochloride (160 mg, 2.38 mmol), TEA (198 µL, 1.43 mmol), and then 60% T3P in EtOAc (252 mg, 0.475 mmol). The mixture was allowed to warm to room temperature and stirred for 16 h. The reaction mixture was quenched with H₂O (8 mL) and extracted with EtOAc (2×10 mL). The combined organic layers were washed with brine (2×10 mL), dried over anhydrous Na₂SO₄, filtered and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Phenomenex C₁₈ column, 20-60% MeCN: 10 mM NH₄HCO₃ in H₂O) to give 1-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)benzenesulfonyl]-N-methylcyclopropane-1-carboxamide (Compound 226). ¹H NMR (400 MHz, DMSO-d₆): δ 10.48 (s, 1H), 7.94 (d, J=4.4 Hz, 1H), 7.82-7.80 (m, 2H), 7.51 (d, J=2.4 Hz, 1H), 7.24 (dd, J=8.4, 2.0 Hz, 1H), 7.18-7.16 (m, 2H), 6.85 (d, J=8.0 Hz, 1H), 4.28 (t, J=5.6 Hz, 2H), 2.93-2.87 (m, 4H), 2.72-2.69 (m, 2H), 2.59 (d, J=4.4 Hz, 3H), 1.79-1.74 (m, 4H), 1.59-1.57 (m, 2H), 1.43-1.41 (m, 2H). MS=518.1 [M+H].sup.+.

[2937] The following compounds in Table 34 were prepared according to procedures similar to step 2 described for Compound 226 using the appropriate starting materials.

TABLE-US-00037 TABLE 34 Exact Mass # Structure Name [M + H].sup.+ 227 [01743]

 1-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)benzenesulfonyl]-N,N-dimethylcyclopropane-1-carboxamide Calc'd 532.2 Found 532.1 228 [01744]  1'-[2-(4-{[1-(azetidine-1-carbonyl)cyclopropyl]sulfonyl}phenoxy)ethyl]-5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 544.2 Found 544.0

Example 36

4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-fluoro-N-methylbenzamide (Compound 229)

##STR01745##

Step 1: methyl 4-(2-bromoethoxy)-2-fluorobenzoate

##STR01746##

[2938] To a mixture of methyl 2-fluoro-4-hydroxybenzoate (3.00 g, 17.6 mmol) and 1,2-

dibromomethane (6.65 mL, 88.2 mmol) in DMF (20 mL) was added Cs.sub.2CO.sub.3 (11.5 g, 35.3 mmol). The mixture was stirred at 100° C. for 12 h. After cooling to room temperature, the reaction mixture was quenched with H.sub.2O (20 mL) and extracted with EtOAc (3×20 mL). The combined organic layers were washed with brine (2×20 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 40 g cartridge, 0-100% EtOAc:petroleum ether) to give methyl 4-(2-bromoethoxy)-2-fluorobenzoate. MS=277.1/279.1 [M+H].sup.+.

Step 2: methyl 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-fluorobenzoate

##STR01747##

[2939] To a mixture of 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 2.00 g, 7.32 mmol, HCl salt) and methyl 4-(2-bromoethoxy)-2-fluorobenzoate (2.23 g, 8.05 mmol) in MeCN (30 mL) was added NaHCO.sub.3 (923 mg, 11.0 mmol). The mixture was stirred at 80° C. for 16 h. After cooling to room temperature, the reaction mixture was quenched with H.sub.2O (20 mL) and extracted with EtOAc (3×20 mL). The combined organic layers were washed with brine (2×20 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 20 g cartridge, 0-100% EtOAc:petroleum ether) to give methyl 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-fluorobenzoate. MS=433.2 [M+H].sup.+.

Step 3: 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-fluorobenzoic acid

##STR01748##

[2940] To a solution of methyl 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-fluorobenzoate (1.90 g, 4.39 mmol) in THE (15 mL) and H.sub.2O (5 mL) was added LiOH.Math.H.sub.2O (315 mg, 13.2 mmol). The mixture was stirred at room temperature for 5 h. The reaction mixture was concentrated in vacuo and adjusted to pH=4 with 1.0 M aqueous HCl. The resulting solids were collected via filtration and dried in vacuo to give 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-fluorobenzoic acid, which was used in the subsequent step without further purification. MS=419.1 [M+H].sup.+.

Step 4: 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-fluoro-N-methylbenzamide (Compound 229)

##STR01749##

[2941] To a 0° C. solution of 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-fluorobenzoic acid (150 mg, 0.358 mmol) in DCM (2 mL) was added TEA (150 µL, 1.07 mmol), methylamine hydrochloride (96.7 mg, 1.43 mmol), and then 60% T3P in EtOAc (380 mg, 0.716 mmol) dropwise. The mixture was allowed to warm to room temperature and stirred for 12 h. The reaction mixture was cooled to 0° C., quenched with H.sub.2O (8 mL), and then extracted with EtOAc (2×10 mL). The combined organic layers were washed with brine (10 mL), dried over Na.sub.2SO.sub.4, filtered, and in vacuo. The crude residue was purified by reverse phase preparative HPLC (Waters Xbridge BEH C.sub.18 column, 15-55% MeCN: 10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-fluorobenzoic acid (Compound 229). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.48 (s, 1H), 7.99-7.98 (m, 1H), 7.65-7.63 (m, 1H), 7.51 (s, 1H), 7.23 (dd, J=8.4, 2.0 Hz, 1H), 6.91-6.83 (m, 3H), 4.20 (t, J=5.6 Hz, 2H), 2.87-2.84 (m, 2H), 2.75 (t, J=4.4 Hz, 2H), 2.70 (d, J=4.0 Hz, 3H), 2.69-2.67 (m, 2H), 1.80-1.68 (m, 4H). MS=432.1 [M+H].sup.+.

[2942] The following compounds in Table 35 were prepared according to procedures similar to steps 1-4 described for Compound 229 using the appropriate starting materials. In cases where step 1 was followed by chiral SFC purification to separate isomers, chiral column conditions and elution order are specified.

TABLE-US-00038 TABLE 35 Exact Mass Elution Chiral # Structure Name [M + H].sup.+ Order

Column 230 [01750]  embedded image 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-fluoro-N,N-dimethylbenzamide Calc'd 446.2 Found 446.1 n/a n/a 231 [01751]  embedded image 5-chloro-1'-(2-[3-fluoro-4-(morpholine-4-carbonyl)phenoxy]ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 448.2 Found 488.2 n/a n/a 232 [01752]  embedded image 4-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-fluorobenzoyl]-1λ.sup.6-thiomorpholine-1,1-dione Calc'd 536.1 Found 536.1 n/a n/a 233 [01753]  embedded image 5-chloro-1'-(2-[3-fluoro-4-(3-methanesulfonylazetidine-1-carbonyl)phenoxy]ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 536.1 Found 536.1 n/a n/a 234 [01754]  embedded image 1'-(2-[4-(azetidine-1-carbonyl)-3-fluorophenoxy]ethyl)-5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 458.2 Found 458.1 n/a n/a 235 [01755]  embedded image 5-chloro-1'-(2-[3-fluoro-4-(3-hydroxyazetidine-1-carbonyl)phenoxy]ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 474.2 Found 474.1 n/a n/a 236 [01756]  embedded image 1'-(2-[4-(3-aminoazetidine-1-carbonyl)-3-fluorophenoxy]ethyl)-5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 473.2 Found 473.1 n/a n/a 237 [01757]  embedded image N-{1-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-fluorobenzoyl]azetidin-3-yl}methanesulfonamide Calc'd 551.1 Found 551.2 n/a n/a 238 [01758]  embedded image 4-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-fluorobenzamide Calc'd 409.2 Found 409.2 n/a n/a 239 [01759]  embedded image 6-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-fluorobenzoyl]-2λ.sup.6-thia-6-azaspiro[3.3]heptane-2,2-dione Calc'd 548.1 Found 548.2 n/a n/a 240 [01760]  embedded image 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-(1,1-dioxo-1λ.sup.6-thian-4-yl)-2-fluorobenzamide Calc'd 550.2 Found 550.1 n/a n/a 241 [01761]  embedded image 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-(1,1-dioxo-1λ.sup.6-thiolan-3-yl)-2-fluorobenzamide Calc'd 536.1 Found 536.2 n/a n/a 242 [01762]  embedded image 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-[(3S) or (3R)-1,1-dioxo-1λ.sup.6-thiolan-3-yl]-2-fluorobenzamide Calc'd 536.1 Found 536.2 Daicel Chiralpak OD-3 First 243 [01763]  embedded image 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-[(3R) or (3S)-1,1-dioxo-1λ.sup.6-thiolan-3-yl]-2-fluorobenzamide Calc'd 536.1 Found 536.2 Daicel Chiralpak OD-3 Second

Example 37

5-chloro-1'-(2-[4-(5-methyl-1H-pyrazol-4-yl)phenoxy]ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 244)

##STR01764##

Step 1: 1'-[2-(4-bromophenoxy)ethyl]-5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR01765##

[2943] To a solution of 4-bromophenol (300 mg, 1.73 mmol) in acetone (5 mL) was added K.sub.2CO.sub.3 (240 mg, 1.73 mmol), 5-chloro-1'-(2-chloroethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-5, 778 mg, 2.60 mmol) and NaI (13.0 mg, 86.7 μmol). The mixture was stirred at 50° C. for 16 h. After cooling to room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 4 g cartridge, 0-100% EtOAc:petroleum ether) to give 1'-[2-(4-bromophenoxy)ethyl]-5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one.

MS=435.1/437.1 [M+H].sup.+.

Step 2: 5-chloro-1'-(2-[4-(5-methyl-1H-pyrazol-4-yl)phenoxy]ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 244)

##STR01766##

[2944] To a solution of 1'-[2-(4-bromophenoxy)ethyl]-5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (200 mg, 0.459 mmol) in n-butanol (4 mL) and H.sub.2O (1 mL) was added 3-methyl-4-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1H-pyrazole (287 mg, 1.38 mmol), KOAc

(135 mg, 1.38 mmol), and bis(4-(di-tert-butylphosphanyl)-N,N-dimethylaniline) dichloropalladium (6.50 mg, 9.18 μ mol). The mixture was degassed and purged with N.sub.2 (3 $\times$), and heated via microwave for 2 h at 120° C. After cooling to room temperature, the mixture was filtered through Celite and the filtrate was concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Waters Xbridge Prep OBD C.sub.18 column, 20-50% MeCN: 10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 5-chloro-1'-{2-[4-(5-methyl-1H-pyrazol-4-yl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 244). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 12.52 (s, 1H), 10.49 (s, 1H), 7.82-7.59 (m, 1H), 7.52 (s, 1H), 7.34 (d, J=8.0 Hz, 2H), 7.23 (d, J=6.4 Hz, 1H), 6.98 (d, J=8.8 Hz, 2H), 6.84 (d, J=8.4 Hz, 1H), 4.16-4.13 (m, 2H), 2.87-2.84 (m, 4H), 2.67-2.66 (m, 2H), 2.34-2.29 (m, 3H), 1.79-1.74 (m, 4H). MS=437.2 [M+H].sup.+.

Example 38

6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-methyl-1,2,3,4-tetrahydroisoquinoline-2-carboxamide (Compound 245)

##STR01767##

Step 1: 5-chloro-1'-(2-chloroethyl)-1-{[2-(trimethylsilyl)ethoxy]methyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR01768##

[2945] To a 0° C. solution of 5-chloro-1'-(2-chloroethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-5, 400 mg, 1.34 mmol) in THF (3 mL) was added NaH (160 mg, 60 wt % in mineral oil, 4.01 mmol). The mixture was stirred at 0° C. for 30 min, and then SEM-Cl (355 μ L, 2.01 mmol) was added dropwise. The mixture was allowed to warm to room temperature and stirred for another 3 h. The reaction mixture was cooled to 0° C., quenched with saturated aqueous NH.sub.4Cl (10 mL), and then extracted with EtOAc (3 $\times$ 15 mL). The combined organic layers were washed with brine (10 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-20% EtOAc:petroleum ether) to give 5-chloro-1'-(2-chloroethyl)-1-{[2-(trimethylsilyl)ethoxy]methyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one. MS=429.2 [M+H].sup.+.

Step 2: 6-[2-(5-chloro-2-oxo-1-{[2-(trimethylsilyl)ethoxy]methyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl)ethoxy]-N-methyl-1,2,3,4-tetrahydroisoquinoline-2-carboxamide SEM

##STR01769##

[2946] To a solution of 6-hydroxy-N-methyl-1,2,3,4-tetrahydroisoquinoline-2-carboxamide (Intermediate A-81, 130 mg, 0.630 mmol) in acetone (4 mL) were added K.sub.2CO.sub.3 (87.1 mg, 0.630 mmol), 5-chloro-1'-(2-chloroethyl)-1-{[2-(trimethylsilyl)ethoxy]methyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (406 mg, 0.946 mmol), and NaI (4.72 mg, 31.5 μ mol). The mixture was stirred at 50° C. for 16 h. Solids were removed by filtration, and the filtrate was concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 4 g cartridge, 0-100% EtOAc:petroleum ether) to give 6-[2-(5-chloro-2-oxo-1-{[2-(trimethylsilyl)ethoxy]methyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl)ethoxy]-N-methyl-1,2,3,4-tetrahydroisoquinoline-2-carboxamide. MS=599.4 [M+H].sup.+.

Step 3: 6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-methyl-1,2,3,4-tetrahydroisoquinoline-2-carboxamide (Compound 245)

##STR01770##

[2947] To a solution of 6-[2-(5-chloro-2-oxo-1-{[2-(trimethylsilyl)ethoxy]methyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl)ethoxy]-N-methyl-1,2,3,4-tetrahydroisoquinoline-2-carboxamide (230 mg, 0.384 mmol) in DCM (3 mL) was added TFA (1.50 mL, 20.3 mmol). The mixture was stirred at room temperature for 2 h and was then concentrated in vacuo. The residue was dissolved into MeOH (1.5 mL) and 30% NH.sub.4OH solution in H.sub.2O (1.50 mL) was added. Then the mixture was stirred at room temperature for 2 h and was then concentrated in

vacuo. The crude residue was purified by reverse phase preparative HPLC (Phenomenex C.sub.18 column, 15-45% MeCN: 10 mM NH₄HCO₃ in H₂O) to give 6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-methyl-1,2,3,4-tetrahydroisoquinoline-2-carboxamide (Compound 245). ¹H NMR (400 MHz, DMSO-d₆): δ 10.48 (s, 1H), 7.51 (s, 1H), 7.23 (d, J=10.4 Hz, 1H), 7.02 (d, J=8.4 Hz, 1H), 6.84 (d, J=8.4 Hz, 1H), 6.77 (d, J=7.6 Hz, 2H), 6.43 (s, 1H), 4.38 (s, 2H), 4.11-4.10 (m, 2H), 3.49 (t, J=5.6 Hz, 2H), 2.91-2.85 (m, 2H), 2.84-2.82 (m, 2H), 2.72-2.68 (m, 4H), 2.59 (d, J=4.4 Hz, 3H), 1.78-1.76 (m, 2H), 1.73-1.71 (m, 2H). MS=469.2 [M+H]⁺.

Example 39

6-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-methyl-1,2,3,4-tetrahydroisoquinoline-2-carboxamide (Compound 246)

##STR01771##

Step 1: tert-butyl 6-(2-bromoethoxy)-1,2,3,4-tetrahydroisoquinoline-2-carboxylate

##STR01772##

[2948] To a solution of tert-butyl 6-hydroxy-1,2,3,4-tetrahydroisoquinoline-2-carboxylate (500 mg, 2.01 mmol) in 1,2-dibromoethane (15 mL) was added Cs₂CO₃ (1.96 g, 6.02 mmol). The mixture was stirred at 80° C. for 12 h. The reaction mixture was diluted with H₂O (20 mL) and extracted with EtOAc (3×20 mL). The combined organic layers were washed with brine (30 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Isco 12 g cartridge, 0-40% EtOAc:petroleum ether) to give tert-butyl 6-(2-bromoethoxy)-1,2,3,4-tetrahydroisoquinoline-2-carboxylate.

MS=302.1 [M-C₄H₈+H]⁺.

Step 2: tert-butyl 6-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-1,2,3,4-tetrahydroisoquinoline-2-carboxylate

##STR01773##

[2949] A mixture of tert-butyl 6-(2-bromoethoxy)-1,2,3,4-tetrahydroisoquinoline-2-carboxylate (100 mg, 0.281 mmol), 2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (Intermediate B-9, 74.0 mg, 0.281 mmol, HCl salt) and NaHCO₃ (47.0 mg, 0.561 mmol) in MeCN (5 mL) was stirred at 80° C. for 12 h. The reaction mixture was diluted with H₂O (10 mL) and extracted with EtOAc (3×10 mL). The combined organic layers were washed with brine (20 mL), dried over Na₂SO₄, filtered, and concentrated under reduced pressure to give tert-butyl 6-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-1,2,3,4-tetrahydroisoquinoline-2-carboxylate, which was taken to the next step without further purification.

MS=448.1 [M-C₄H₈+H]⁺.

Step 3: 2-oxo-1'-[2-(1,2,3,4-tetrahydroisoquinolin-6-yloxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile

##STR01774##

[2950] A solution of tert-butyl 6-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-1,2,3,4-tetrahydroisoquinoline-2-carboxylate (120 mg, 0.239 mmol) in 4.0 M HCl in MeOH (3 mL, 12 mmol) was stirred at room temperature for 2 h. The residue was concentrated in vacuo to give 2-oxo-1'-[2-(1,2,3,4-tetrahydroisoquinolin-6-yloxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile, which was used in the subsequent step without further purification.

MS=403.1 [M+H]⁺.

Step 4: 6-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-methyl-1,2,3,4-tetrahydroisoquinoline-2-carboxamide (Compound 246)

##STR01775##

[2951] To a 0° C. solution of 1'-{2-[(2-acetyl-1,2,3,4-tetrahydroisoquinolin-6-yl)oxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (140 mg, 0.348 mmol) in DCM (5 mL) was added TEA (96.8 μL, 0.696 mmol) and N-methylcarbonyl chloride (97.6 mg, 1.04 mmol). The mixture was stirred at 0° C. for 2 h, was quenched with H₂O (1 mL), and then extracted with

DCM (3×5 mL). The combined organic layers were dried over Na.sub.2SO.sub.4 and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Phenomenex C.sub.18 column, 15-45% MeCN: 10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 6-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-N-methyl-1,2,3,4-tetrahydroisoquinoline-2-carboxamide (Compound 246). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.87 (s, 1H), 7.95 (s, 1H), 7.67 (d, J=8.0 Hz, 1H), 7.03-6.98 (m, 2H), 6.79-6.77 (m, 2H), 6.45-6.44 (m, 1H), 4.38 (s, 2H), 4.10-4.08 (m, 2H), 3.49 (t, J=6.0 Hz, 2H), 2.74-2.72 (m, 4H), 2.70-2.65 (m, 4H), 2.59 (d, J=4.4 Hz, 3H), 1.78-1.76 (m, 4H). MS=460.3 [M+H].sup.+.

Example 40

5-chloro-1'-[2-({3-methyl-2-oxo-1H,2H,3H-imidazo[4,5-b]pyridin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 247)

##STR01776##

Step 1: 5-chloro-1'-{2-[(3-methyl-2-oxo-1-{2-(trimethylsilyl)ethoxy)methyl}-1H,2H,3H-imidazo[4,5-b]pyridin-6-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR01777##

[2952] To a mixture of 6-(2-bromoethoxy)-3-methyl-1-{2-(trimethylsilyl)ethoxy)methyl}-1H,2H,3H-imidazo[4,5-b]pyridin-2-one (Intermediate A-74, 150 mg, 0.373 mmol) and 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 112 mg, 0.410 mmol, HCl salt) in MeCN (3 mL) was added NaHCO.sub.3(47.0 mg, 0.559 mmol). The mixture was stirred at 80° C. for 12 h. After cooling to room temperature, the reaction mixture was poured into H.sub.2O (5 mL) and extracted with EtOAc (2×10 mL). The combined organic layers were washed with brine (10 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 4 g cartridge, 0-100% EtOAc:petroleum ether) to give 5-chloro-1'-{2-[(3-methyl-2-oxo-1-{2-(trimethylsilyl)ethoxy)methyl}-1H,2H,3H-imidazo[4,5-b]pyridin-6-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one. MS=558.3 [M+H].sup.+.

Step 2: 5-chloro-1'-[2-({3-methyl-2-oxo-1H,2H,3H-imidazo[4,5-b]pyridin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 247)

##STR01778##

[2953] A solution of 5-chloro-1'-{2-[(3-methyl-2-oxo-1-{2-(trimethylsilyl)ethoxy)methyl}-1H,2H,3H-imidazo[4,5-b]pyridin-6-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (100 mg, 0.179 mmol) in DCM (2 mL) and TFA (0.5 mL) was stirred at room temperature for 1 h. The reaction mixture was concentrated in vacuo, then dissolved in MeOH (3 mL) and 25% NH.sub.4OH solution in H.sub.2O (1 mL) was added. The mixture was stirred at room temperature for 1 h, was diluted with H.sub.2O (5 mL), and then extracted with EtOAc (2×10 mL). The combined organic layers were dried with Na.sub.2SO.sub.4 and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Phenomenex C.sub.18 column, 15-45% MeCN: 10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 5-chloro-1'-[2-({3-methyl-2-oxo-1H,2H,3H-imidazo[4,5-b]pyridin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 247). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 11.03 (s, 1H), 10.48 (s, 1H), 7.71 (d, J=2.4 Hz, 1H), 7.50 (d, J=2.0 Hz, 1H), 7.23 (dd, J=8.0, 2.0 Hz, 1H), 7.04 (d, J=2.4 Hz, 1H), 6.84 (d, J=8.4 Hz, 1H), 4.16 (t, J=5.6 Hz, 2H), 3.26 (s, 3H), 2.95-2.87 (m, 2H), 2.84 (t, J=5.6 Hz, 2H), 2.72-2.65 (m, 2H), 1.84-1.75 (m, 2H), 1.74-1.66 (m, 2H). MS=428.1 [M+H].sup.+.

Example 41

5-chloro-1'-{2-[(6-{[1-(hydroxymethyl)cyclopropyl]sulfonyl}pyridin-3-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one(Compound 248)

##STR01779##

Step 1: methyl 2-[(5-chloropyridin-2-yl)sulfanyl]acetate

##STR01780##

[2954] A mixture of 5-chloropyridine-2-thiol (3.00 g, 20.6 mmol), methyl 2-bromoacetate (2.92 mL, 30.9 mmol), and K₂CO₃ (5.69 g, 41.2 mmol) in DMF (50 mL) was degassed and purged with N₂ (3×). The mixture was stirred at room temperature for 2 h under N₂ atmosphere. The reaction mixture was quenched with H₂O (30 mL) and then extracted with EtOAc (3×50 mL). The combined organic layers were washed with brine (2×20 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 40 g cartridge, 0-5% EtOAc:petroleum ether) to give methyl 2-[(5-chloropyridin-2-yl)sulfanyl]acetate. MS=218.0 [M+H]⁺.

Step 2: methyl 2-[(5-chloropyridin-2-yl)sulfonyl]acetate

##STR01781##

[2955] A mixture of methyl 2-[(5-chloropyridin-2-yl)sulfanyl]acetate (5.70 g, 26.2 mmol), m-CPBA (10.6 g, 85% purity, 52.4 mmol) in DCM (2 mL) was degassed and purged with N₂ (3×). The mixture was stirred at 0° C. for 2 h under N₂ atmosphere. The reaction mixture was quenched with saturated Na₂SO₃ (50 mL) and extracted with EtOAc (3×50 mL). The combined organic layers were washed with brine (2×50 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 40 g cartridge, 0-13% EtOAc:petroleum ether) to give methyl 2-[(5-chloropyridin-2-yl)sulfonyl]acetate. MS=250.0 [M+H]⁺.

Step 3: methyl 1-[(5-chloropyridin-2-yl)sulfonyl]cyclopropane-1-carboxylate

##STR01782##

[2956] A mixture of methyl 2-[(5-chloropyridin-2-yl)sulfonyl]acetate (5.00 g, 20.0 mmol), 1,2-dibromoethane (1.81 mL, 24.0 mmol), and K₂CO₃ (6.92 g, 50.1 mmol) in DMF (50 mL) was degassed and purged with N₂ (3×). The mixture was stirred at 60° C. for 10 h under N₂ atmosphere. After cooling to room temperature, the mixture was diluted with H₂O (50 mL) and extracted with EtOAc (3×50 mL). The combined organic layers were washed with brine (2×50 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 40 g cartridge, 0-60% EtOAc:petroleum ether) to give methyl 1-[(5-chloropyridin-2-yl)sulfonyl]cyclopropane-1-carboxylate. MS=276.0 [M+H]⁺.

Step 4: methyl 1-{[5-(2-hydroxyethoxy)pyridin-2-yl]sulfonyl}cyclopropane-1-carboxylate

##STR01783##

[2957] A mixture of methyl 1-[(5-chloropyridin-2-yl)sulfonyl]cyclopropane-1-carboxylate (5.00 g, 18.1 mmol), ethylene glycol (20.3 mL, 363 mmol), and Cs₂CO₃ (17.7 g, 54.4 mmol) in DMSO (2 mL) was degassed and purged with N₂ (3×). The mixture was stirred at 100° C. for 12 h under N₂ atmosphere. After cooling to room temperature, MeI (3.39 mL, 54.4 mmol) was added, and the mixture was stirred for 12 h. The mixture was quenched with H₂O (50 mL) and extracted with EtOAc (3×50 mL). The combined organic layers were washed with brine (2×30 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 40 g cartridge, 0-60% EtOAc:petroleum ether) to give methyl 1-{[5-(2-hydroxyethoxy)pyridin-2-yl]sulfonyl}cyclopropane-1-carboxylate. MS=302.2 [M+H]⁺.

Step 5: methyl 1-([5-[2-(methanesulfonyloxy)ethoxy]pyridin-2-yl]sulfonyl)cyclopropane-1-carboxylate

##STR01784##

[2958] To a 0° C. mixture of methyl 1-{[5-(2-hydroxyethoxy)pyridin-2-yl]sulfonyl}cyclopropane-1-carboxylate (500 mg, 1.66 mmol) and TEA (0.693 mL, 4.98 mmol) in DCM (2 mL) was added methanesulfonic anhydride (578 mg, 3.32 mmol). The mixture was stirred at 0° C. for 2 h. The mixture was quenched with H₂O (50 mL) and extracted with EtOAc (3×50 mL). The combined organic layers were washed with brine (2×50 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel

chromatography (Biotage 40 g cartridge, 0-60% EtOAc:petroleum ether) to give methyl 1-({5-[2-(methanesulfonyloxy)ethoxy]pyridin-2-yl}sulfonyl)cyclopropane-1-carboxylate. MS=380.1 [M+H].sup.+.

Step 6: methyl 1-{{5-[2-{{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy}pyridin-2-yl}sulfonyl}cyclopropane-1-carboxylate

##STR01785##

[2959] A mixture of methyl 1-({5-[2-(methanesulfonyloxy)ethoxy]pyridin-2-yl}sulfonyl)cyclopropane-1-carboxylate (700 mg, 1.84 mmol), 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 605 mg, 2.21 mmol, HCl salt), and NaHCO₃ (310 mg, 3.69 mmol) in MeCN (10 mL) was degassed and purged with N₂ (3×). The mixture was stirred at 80° C. for 12 h under N₂ atmosphere. After cooling to room temperature, the mixture was quenched with H₂O (50 mL) and then extracted with EtOAc (3×50 mL). The combined organic layers were washed with brine (2×50 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-80% EtOAc:petroleum ether) to give methyl 1-{{5-[2-{{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy}pyridin-2-yl}sulfonyl}cyclopropane-1-carboxylate. MS=520.2 [M+H].sup.+.

Step 7: 5-chloro-1'-{2-[(6-{{1-(hydroxymethyl)cyclopropyl}sulfonyl}pyridin-3-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 248)

##STR01786##

[2960] To a 0° C. solution of methyl 1-{{5-[2-{{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy}pyridin-2-yl}sulfonyl}cyclopropane-1-carboxylate (100 mg, 0.192 mmol) in THE (2 mL) was added LiAlH₄ (14.6 mg, 385 mmol). The reaction mixture allowed to warm to room temperature and stirred for 2 h under N₂ atmosphere. The mixture was quenched with H₂O (50 mL) and extracted with EtOAc (3×50 mL). The combined organic layers were washed with brine (2×50 mL), dried over anhydrous Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Waters Xbridge BEH C₁₈ column, 25-55% MeCN: 10 mM NH₄HCO₃ in H₂O) to give 5-chloro-1'-{2-[(6-{{1-(hydroxymethyl)cyclopropyl}sulfonyl}pyridin-3-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 248). ¹H NMR (400 MHz, DMSO-d₆): δ 10.48 (s, 1H), 8.48 (d, J=2.8 Hz, 1H), 7.97 (d, J=8.8 Hz, 1H), 7.68 (dd, J=8.8, 2.8 Hz, 1H), 7.50 (s, 1H), 7.24-7.22 (m, 1H), 6.84 (d, J=8.4 Hz, 1H), 4.82 (t, J=6 Hz, 1H), 4.33 (t, J=6 Hz, 2H), 3.70 (d, J=6 Hz, 2H), 2.98-2.85 (m, 4H), 2.79-2.68 (m, 2H), 1.83-1.67 (m, 4H), 1.39-1.30 (m, 2H), 1.09-1.02 (m, 2H). MS=492.2 [M+H].sup.+.

Example 42

4-(2-{{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-(difluoromethyl)benzamide (Compound 249)

##STR01787##

Step 1: 4-(benzyloxy)-1-bromo-2-(difluoromethyl)benzene

##STR01788##

[2961] To a 0° C. solution of 5-(benzyloxy)-2-bromobenzaldehyde (3.00 g, 10.3 mmol) in DCM (30 mL) was added DAST (2.72 mL, 20.6 mmol). The mixture was allowed to warm to room temperature and stirred for 12 h. The reaction mixture was cooled to 0° C., quenched with saturated NaHCO₃ aqueous solution (20 mL), diluted with H₂O (10 mL), and then extracted with EtOAc (3×20 mL). The combined organic layers were washed with brine (30 mL), dried with Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 20 g cartridge, 0-20% EtOAc:petroleum ether) to give 4-(benzyloxy)-1-bromo-2-(difluoromethyl)benzene.

Step 2: methyl 4-(benzyloxy)-2-(difluoromethyl)benzoate

##STR01789##

[2962] A mixture of 4-(benzyloxy)-1-bromo-2-(difluoromethyl)benzene (2.20 g, 7.03 mmol), Pd(OAc)₂ (315 mg, 1.41 mmol), and 1,1'-bis(diphenylphosphino)ferrocene (1.56 g, 2.81 mmol) in MeOH (15 mL) and toluene (15 mL) was degassed and purged with N₂ (3×). The suspension was then degassed under vacuum and purged with CO. The mixture was stirred under CO (50 psi) at 80° C. for 12 h. After cooling to room temperature, solids were removed by filtration, and the filtrate was concentrated in vacuo. The residue was taken up in H₂O (40 mL) and extracted with EtOAc (3×30 mL). The combined organic layers were washed with brine (2×12 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Isco 20 g cartridge, 0-23% EtOAc:petroleum ether) to give methyl 4-(benzyloxy)-2-(difluoromethyl)benzoate. MS=293.1 [M+H]⁺.

Step 3: methyl 2-(difluoromethyl)-4-hydroxybenzoate

##STR01790##

[2963] To a solution of methyl 4-(benzyloxy)-2-(difluoromethyl)benzoate (420 mg, 1.44 mmol) in MeOH (10 mL) under N₂ atmosphere was added Pd/C (100 mg, 10 wt %, 0.0940 mmol) and Pd(OH)₂/C (100 mg, 20 wt %, 0.140 mmol). The mixture was stirred at 50° C. for 12 h under H₂ (50 Psi) atmosphere. After cooling to room temperature, the reaction mixture was filtered through Celite and the filtrate was concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Isco 12 g cartridge, 0-30% EtOAc:petroleum ether) to give methyl 2-(difluoromethyl)-4-hydroxybenzoate. MS=203.1 [M+H]⁺.

Step 4: methyl 4-(2-bromoethoxy)-2-(difluoromethyl)benzoate

##STR01791##

[2964] To a solution of methyl 2-(difluoromethyl)-4-hydroxybenzoate (300 mg, 1.48 mmol) in 1,2-dibromoethane (5 mL) was added Cs₂CO₃ (1.45 g, 4.45 mmol). The mixture was stirred at 100° C. for 12 h. After cooling to room temperature, the mixture was filtered, and the filtrate was concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Isco 4 g cartridge, 0-50% EtOAc:petroleum ether) to give methyl 4-(2-bromoethoxy)-2-(difluoromethyl)benzoate.

Step 5: methyl 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-(difluoromethyl)benzoate

##STR01792##

[2965] A mixture of methyl 4-(2-bromoethoxy)-2-(difluoromethyl)benzoate (200 mg, 0.647 mmol), 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 177 mg, 0.647 mmol, HCl salt) and NaHCO₃ (163 mg, 1.94 mmol) in MeCN (4 mL) was degassed and purged with N₂ (3 times). The mixture was stirred at 80° C. for 12 h under N₂ atmosphere. After cooling to room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Isco 12 g cartridge, 0-50% EtOAc:petroleum ether) to give methyl 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-(difluoromethyl)benzoate. MS=465.1 [M+H]⁺.

Step 6: 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-(difluoromethyl)benzoic acid

##STR01793##

[2966] To a solution of methyl 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-(difluoromethyl)benzoate (200 g, 430 mmol) in MeOH (4 mL) was added a solution of LiOH H₂O (144 mg, 3.44 mmol) in H₂O (4 mL). The mixture was stirred at 50° C. for 2 h. After cooling to room temperature, the reaction mixture was concentrated in vacuo to remove MeOH. The residue was adjusted to pH=2-3 with 1.0 M aqueous HCl solution. The resulting solids were isolated by filtration, and then dried in vacuo. The crude product was triturated with H₂O and then dried again in vacuo to give 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-

piperidin]-1'-yl}ethoxy)-2-(difluoromethyl)benzoic acid, which was used in the subsequent step without further purification. MS=451.1 [M+H].sup.+.

Step 7: 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-(difluoromethyl)benzamide (Compound 249)

##STR01794##

[2967] To a 0° C. solution of 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-(difluoromethyl)benzoic acid (200 mg, 0.444 mmol) in DMF (20 mL) was added NH₄Cl (71.2 mg, 1.33 mmol), TEA (216 µL, 1.55 mmol), and then HATU (202 mg, 0.532 mmol). The mixture was allowed to warm to room temperature, stirred for 12 h, and then quenched with H₂O (15 mL). Solids were removed by filtration, and the filtrate was concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Waters Xbridge Prep OBD C₁₈ column, 25-55% MeCN: 10 mM NH₄HCO₃ in H₂O) to give 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-(difluoromethyl)benzamide (Compound 249). ¹H NMR (400 MHz, DMSO-d₆): δ 10.48 (s, 1H), 7.96 (s, 1H), 7.71-7.61 (m, 1H), 7.52-7.32 (m, 3H), 7.25-7.14 (m, 3H), 6.84 (d, J=8.4 Hz, 1H), 4.24 (t, J=5.6 Hz, 2H), 2.97-2.83 (m, 4H), 2.74-2.68 (m, 2H), 1.83-1.66 (m, 4H). MS=450.1 [M+H].sup.+.

[2968] The following compounds in Table 36 were prepared according to procedures similar to steps 1-7 described for Compound 249 using the appropriate starting materials.

TABLE-US-00039 TABLE 36 Exact Mass Intermediate # Structure Name [M + H].sup.+ Used 250 [01795]  4-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-(difluoromethyl)benzamide Calc'd 441.2 Found 441.1 B-9

Example 43

1'-{2-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]ethyl}-1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (Compound 251)

##STR01796##

[2969] To a 0° C. solution of 1'-{2-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]ethyl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (Compound 109, 78.0 mg, 0.162 mmol) in DMA (1 mL) was added NaH (13.0 mg, 60 wt % in mineral oil, 0.324 mmol) portionwise. The mixture was stirred at 0° C. for 30 min, and then MeI (11.1 µL, 0.178 mmol) was added dropwise. The mixture was allowed to warm to room temperature and stirred for 3 h. The reaction mixture was quenched with H₂O (10 mL) and then extracted with EtOAc (3×10 mL). The combined organic layers were washed with brine (10 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Phenomenex C₁₈ column, 20-50% MeCN: 10 mM NH₄HCO₃ in H₂O) to give 1'-{2-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]ethyl}-1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (Compound 251). ¹H NMR (400 MHz, DMSO-d₆): δ 7.99 (d, J=1.2 Hz, 1H), 7.80 (dd, J=8.0, 1.6 Hz, 1H), 7.26 (d, J=8.8 Hz, 2H), 7.20 (d, J=8.0 Hz, 1H), 7.05 (d, J=8.8 Hz, 2H), 5.21 (d, J=7.6 Hz, 2H), 5.01 (d, J=7.6 Hz, 2H), 4.18 (t, J=5.6 Hz, 2H) 3.16 (s, 3H), 2.96-2.88 (m, 4H), 2.86 (s, 3H), 2.79-2.76 (m, 2H), 1.82-1.77 (m, 4H). MS=496.2 [M+H].sup.+.

Example 44

1'-(2-{4-[3-(3-fluoroazetidine-1-carbonyl)oxetan-3-yl]phenoxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (Compound 252)

##STR01797##

Step 1: 2-{3-[4-(benzyloxy)phenyl]oxetan-3-yl}-5-methylfuran

##STR01798##

[2970] To a solution 3-[4-(benzyloxy)phenyl]oxetan-3-ol (Intermediate A-35, 20.0 g, 78.0 mmol) in CHCl₃ (200 mL) was added tetrabutylammonium hexafluorophosphate (1.66 g, 4.29 mmol), [bis(trifluoromethylsulfonyl)amino]lithium (2.46 g, 8.58 mmol), and 2-methylfuran (35.1 mL, 390 mmol). The mixture was stirred at 50° C. for 4 h. After cooling to 0° C., the reaction mixture was

quenched with H.sub.2O (200 mL) and extracted with DCM (3×200 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 220 g cartridge, 0-15% EtOAc:petroleum ether) to give 2-{3-[4-(benzyloxy)phenyl]oxetan-3-yl}-5-methylfuran. MS=321.2 [M+H].sup.+.

Step 2: 3-[4-(benzyloxy)phenyl]oxetane-3-carboxylic acid

##STR01799##

[2971] To a solution of 2-{3-[4-(benzyloxy)phenyl]oxetan-3-yl}-5-methylfuran (11.0 g, 34.3 mmol) in heptane (200 mL), EtOAc (200 mL) and H.sub.2O (400 mL) was added NaIO.sub.4 (51.4 g, 240 mmol) and RuCl.sub.3 (178 mg, 0.858 mmol). The mixture was stirred at room temperature for 16 h. The reaction mixture was cooled to 0° C. and quenched with saturated Na.sub.2SO.sub.3 (200 mL). The mixture was adjusted to pH=8 with saturated NaHCO.sub.3 and washed with EtOAc (3×400 mL). The aqueous layer was adjusted to pH=2-3 with 2.0 M aqueous HCl solution and extracted with EtOAc (3×400 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give 3-[4-(benzyloxy)phenyl]oxetane-3-carboxylic acid, which was used in the subsequent step without further purification. MS=283.0 [M-H].sup.-.

Step 3: methyl 3-[4-(benzyloxy)phenyl]oxetane-3-carboxylate

##STR01800##

[2972] To a solution of 3-[4-(benzyloxy)phenyl]oxetane-3-carboxylic acid (4.00 g, 14.1 mmol) in DMF (50 mL) was added K.sub.2CO.sub.3 (2.92 g, 21.1 mmol) and MeI (4.38 mL, 70.5 mmol). The mixture was stirred at room temperature for 3 h. The reaction mixture was cooled to 0° C., quenched with H.sub.2O (100 mL), and the resulting precipitate was collected by filtration and washed with H.sub.2O to give methyl 3-[4-(benzyloxy)phenyl]oxetane-3-carboxylate, which was used in the subsequent step without further purification. MS=316.2 [M+NH.sub.4].sup.+.

Step 4: methyl 3-(4-hydroxyphenyl)oxetane-3-carboxylate

##STR01801##

[2973] To a solution of methyl 3-[4-(benzyloxy)phenyl]oxetane-3-carboxylate (4.00 g, 13.4 mmol) in EtOAc (100 mL) under N.sub.2 atmosphere was added Pd/C (5.00 g, 10 wt %, 4.70 mmol). The suspension was degassed under vacuum and purged with H.sub.2 (3×). After stirring under an atmosphere of H.sub.2 (15 psi) at room temperature for 4 h, solids were removed by filtration through Celite and the filtrate was concentrated in vacuo to give methyl 3-(4-hydroxyphenyl)oxetane-3-carboxylate, which was used in the subsequent step without further purification. MS=231.1 [M+Na].sup.+.

Step 5: methyl 3-[4-(2-bromoethoxy)phenyl]oxetane-3-carboxylate

##STR01802##

[2974] To a solution of methyl 3-(4-hydroxyphenyl)oxetane-3-carboxylate (2.20 g, 10.6 mmol) in DMF (3 mL) was added 1,2-dibromoethane (17.0 mL, 225 mmol) and Cs.sub.2CO.sub.3 (6.89 g, 21.1 mmol). The mixture was stirred at 100° C. for 6 h. After cooling to 0° C., the reaction mixture was quenched with H.sub.2O (30 mL) and extracted with DCM (3×15 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 45 g cartridge, 0-40% EtOAc:petroleum ether) to give methyl 3-[4-(2-bromoethoxy)phenyl]oxetane-3-carboxylate. MS=332.1/334.1 [M+NH.sub.4].sup.+.

Step 6: methyl 3-[4-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]oxetane-3-carboxylate

##STR01803##

[2975] To a solution of methyl 3-[4-(2-bromoethoxy)phenyl]oxetane-3-carboxylate (2.50 g, 7.93 mmol) in MeCN (30 mL) was added NaHCO.sub.3 (2.00 g, 23.8 mmol) and 2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (Intermediate B-9, 2.09 g, 7.93 mmol, HCl salt).

The mixture was stirred at 80° C. for 16 h. After cooling to room temperature, the reaction mixture was filtered and washed with MeCN (40 mL) and EtOAc (20 mL). The filtrate was concentrated in vacuo to give methyl 3-[4-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]oxetane-3-carboxylate, which was used in the subsequent step without further purification. MS=462.2 [M+H].sup.+.

Step 7: 3-[4-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]oxetane-3-carboxylic acid

##STR01804##

[2976] To a solution of methyl 3-[4-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]oxetane-3-carboxylate (3.00 g, 6.50 mmol) in THE (30 mL) and H.sub.2O (10 mL) was added LiOH (467 mg, 19.5 mmol). The mixture was stirred at room temperature for 6 h, and was then concentrated in vacuo to remove THF. The residue was adjusted to pH=2-3 with 2 M aqueous HCl solution. The resulting precipitate was collected by filtration and washed with H.sub.2O (100 mL) to give 3-[4-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]oxetane-3-carboxylic acid, which was used in the subsequent step without further purification. MS=448.3 [M+H].sup.+.

Step 8: 1'-(2-{4-[3-(3-fluoroazetidine-1-carbonyl)oxetan-3-yl]phenoxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (Compound 252)

##STR01805##

[2977] To a solution of 3-[4-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]oxetane-3-carboxylic acid (100 mg, 0.223 mmol) and 3-fluoroazetidine hydrochloride (125 mg, 1.12 mmol) in DCM (2 mL) was added TEA (249 µL, 1.79 mmol). After stirring for 10 min, the reaction was cooled to 0° C. and T3P (199 µL, 0.670 mmol) was added. The mixture was allowed to warm to room temperature and stirred for 16 h. The reaction mixture was diluted with H.sub.2O (10 mL) and extracted with DCM (3×5 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The resulting solid was triturated with MeCN to give 1'-(2-{4-[3-(3-fluoroazetidine-1-carbonyl)oxetan-3-yl]phenoxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (Compound 252).
.sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.86 (s, 1H), 7.95 (s, 1H), 7.67 (d, J=7.6 Hz, 1H), 7.30 (d, J=8.4 Hz, 2H), 7.03 (d, J=8.4 Hz, 2H), 6.99 (d, J=8.0 Hz, 1H), 5.35-5.22 (m, 1H), 5.10-5.06 (m, 2H), 4.72-4.67 (m, 2H), 4.31-4.22 (m, 1H), 4.15 (s, 2H), 4.00-3.92 (m, 2H), 3.66-3.64 (m, 1H), 2.93-2.87 (m, 4H), 2.78-2.72 (m, 2H), 1.78-1.75 (m, 4H). MS=505.2 [M+H].sup.+.

[2978] The following compounds in Table 37 were prepared according to procedures similar to those described for Compound 252 using the appropriate starting materials.

TABLE-US-00040 TABLE 37 Exact Mass # Structure Name [M + H].sup.+ 253 [01806]

 1'-(2-{4-[3-(azetidine-1-carbonyl)oxetan-3-yl]phenoxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile Calc'd 487.2 Found 487.2 254 [01807]

 3-[4-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]-N,N-dimethyloxetane-3-carboxamide Calc'd 475.2 Found 475.3 255 [01808]

 3-[4-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]-N-methyloxetane-3-carboxamide Calc'd 461.2 Found 461.3

Example 45

5-chloro-1'-(2-[(7-fluoro-2-oxo-2,3-dihydro-1H-indol-5-yl)oxy]ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 256)

##STR01809##

Step 1: 5-bromo-7-fluoro-2,3-dihydro-1H-indole-2,3-dione

##STR01810##

[2979] A mixture of 7-fluoro-2,3-dihydro-1H-indole-2,3-dione (5.00 g, 30.3 mmol) and NBS (7.01 g, 39.4 mmol) in DMF (35 mL) was stirred at 80° C. for 2 h. After cooling to room temperature, the mixture was poured into H.sub.2O (150 mL) and filtered to collect the solid. The filter cake was

washed with H.sub.2O (3×50 mL). The crude product was triturated with MTBE and dried in vacuo to give 5-bromo-7-fluoro-2,3-dihydro-1H-indole-2,3-dione, which was used in the subsequent step without further purification. MS=243.9/245.9 [M+H].sup.+.

Step 2: 5-bromo-7-fluoro-1-{{2-(trimethylsilyl)ethoxy}methyl}-2,3-dihydro-1H-indole-2,3-dione
##STR01811##

[2980] To a 0° C. solution of 5-bromo-7-fluoro-2,3-dihydro-1H-indole-2,3-dione (5.00 g, 20.5 mmol) in DMA (50 mL) under N.sub.2 atmosphere was added NaH (983 mg, 60 wt % in mineral oil, 24.6 mmol). The mixture was allowed to warm to room temperature and stirred for 30 min. After cooling to 0° C., 2-(trimethylsilyl)ethoxymethyl chloride (4.71 mL, 26.6 mmol) was added. The mixture was warmed to room temperature and stirred for another 2 h. The mixture was quenched with saturated aqueous NH.sub.4Cl solution (200 mL), and the mixture was extracted with EtOAc (3×70 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 80 g cartridge, 0-15% EtOAc:petroleum ether) to give 5-bromo-7-fluoro-1-{{2-(trimethylsilyl)ethoxy}methyl}-2,3-dihydro-1H-indole-2,3-dione. MS=391.1/393.1 [M+NH.sub.4].sup.+.

Step 3: 7-fluoro-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1-{{2-(trimethylsilyl)ethoxy}methyl}-2,3-dihydro-1H-indole-2,3-dione
##STR01812##

[2981] A mixture of 5-bromo-7-fluoro-1-{{2-(trimethylsilyl)ethoxy}methyl}-2,3-dihydro-1H-indole-2,3-dione (7.00 g, 18.7 mmol), bis(pinacolato)diboron (5.70 g, 22.4 mmol), KOAc (4.59 g, 46.8 mmol) and Pd(dppf)Cl.sub.2.Math.CH.sub.2Cl.sub.2 (1.22 g, 1.50 mmol) in 1,4-dioxane (70 mL) was stirred at 95° C. for 16 h under N.sub.2 atmosphere. After cooling to room temperature, the mixture was poured into H.sub.2O (200 mL) and extracted with EtOAc (2×100 mL). The combined organic layers were washed with brine (80 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 120 g cartridge, 0-8% EtOAc:petroleum ether) to give 7-fluoro-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1-{{2-(trimethylsilyl)ethoxy}methyl}-2,3-dihydro-1H-indole-2,3-dione. MS=439.3 [M+NH.sub.4].sup.+.

Step 4: 7-fluoro-5-hydroxy-1-{{2-(trimethylsilyl)ethoxy}methyl}-2,3-dihydro-1H-indole-2,3-dione
##STR01813##

[2982] To a 0° C. mixture of 7-fluoro-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1-{{2-(trimethylsilyl)ethoxy}methyl}-2,3-dihydro-1H-indole-2,3-dione (7.80 g, 18.5 mmol) in acetone (70 mL) and H.sub.2O (70 mL) was added Oxone (17.1 g, 27.8 mmol) portionwise. The mixture was allowed to warm to room temperature and stirred for 1 h. The mixture was poured into saturated Na.sub.2SO.sub.3 solution (200 mL) and extracted with EtOAc (2×80 mL). The combined organic layers were washed with brine (60 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuum. The crude residue was purified by normal phase silica gel chromatography (Biotage 80 g cartridge, 0-25% EtOAc:petroleum ether) to give 7-fluoro-5-hydroxy-1-{{2-(trimethylsilyl)ethoxy}methyl}-2,3-dihydro-1H-indole-2,3-dione.

Step 5: 7-fluoro-5-hydroxy-1-{{2-(trimethylsilyl)ethoxy}methyl}-2,3-dihydro-1H-indol-2-one
##STR01814##

[2983] To a solution of 7-fluoro-5-hydroxy-1-{{2-(trimethylsilyl)ethoxy}methyl}-2,3-dihydro-1H-indole-2,3-dione (4.30 g, 13.8 mmol) in DMSO (43 mL) was added N.sub.2H.sub.4.Math.H.sub.2O (11.8 mL, 207 mmol). The mixture was stirred at 120° C. for 3 h. The mixture was diluted with H.sub.2O (150 mL) and extracted with EtOAc (2×80 mL). The combined organic layers were washed with brine (60 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 40 g cartridge, 0-20% EtOAc:petroleum ether) to give 7-fluoro-5-hydroxy-1-{{2-(trimethylsilyl)ethoxy}methyl}-2,3-dihydro-1H-indol-2-one. MS=296.1 [M-H].sup.+.

Step 6: 5-(2-bromoethoxy)-7-fluoro-1-{{2-(trimethylsilyl)ethoxy}methyl}-2,3-dihydro-1H-indol-2-one

##STR01815##

[2984] To a 0° C. mixture of 7-fluoro-5-hydroxy-1-{{2-(trimethylsilyl)ethoxy}methyl}-2,3-dihydro-1H-indol-2-one (500 mg, 1.68 mmol), 2-bromoethanol (477 µL, 6.73 mmol), and PPh.sub.3 (882 mg, 3.36 mmol) in THE (10 mL) was added DIAD (654 µL, 3.36 mmol). The mixture was allowed to warm to room temperature and stirred for 16 h. The mixture was diluted with EtOAc (25 mL) and H.sub.2O (10 mL). The organic layer was separated, and washed with brine (40 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 20 g cartridge, 0-10% EtOAc:petroleum ether) to give 5-(2-bromoethoxy)-7-fluoro-1-{{2-(trimethylsilyl)ethoxy}methyl}-2,3-dihydro-1H-indol-2-one.

Step 7: 5-chloro-1'-{{2-[(7-fluoro-2-oxo-1-{{2-(trimethylsilyl)ethoxy}methyl}-2,3-dihydro-1H-indol-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR01816##

[2985] A mixture of 5-(2-bromoethoxy)-7-fluoro-1-{{2-(trimethylsilyl)ethoxy}methyl}-2,3-dihydro-1H-indol-2-one (170 mg, 0.420 mmol), 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 79.6 mg, 0.291 mmol), and NaHCO.sub.3 (106 mg, 1.26 mmol) in MeCN (2 mL) was stirred at 80° C. for 16 h. After cooling to room temperature, the mixture was diluted with EtOAc (25 mL) and H.sub.2O (10 mL). The organic layer was separated, washed with brine (20 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-100% EtOAc:petroleum ether) to give 5-chloro-1'-{{2-[(7-fluoro-2-oxo-1-{{2-(trimethylsilyl)ethoxy}methyl}-2,3-dihydro-1H-indol-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one. MS=560.2 [M+H].sup.+.

Step 8: 5-chloro-1'-{{2-[(7-fluoro-2-oxo-2,3-dihydro-1H-indol-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 256)

##STR01817##

[2986] A solution of 5-chloro-1'-{{2-[(7-fluoro-2-oxo-1-{{2-(trimethylsilyl)ethoxy}methyl}-2,3-dihydro-1H-indol-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (80.0 mg, 0.143 mmol) in 4.0 M HCl in 1,4-dioxane (2.0 mL, 8.0 mmol) was stirred at room temperature for 1 h. The mixture was concentrated in vacuo, and then NH.sub.3.Math.H.sub.2O (0.5 mL) was added dropwise. The mixture was stirred at room temperature for 30 min, and then diluted with H.sub.2O (5 mL). The mixture was extracted with EtOAc (2×5 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Phenomenex C.sub.18 column, 20-50% MeCN: 10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 5-chloro-1'-{{2-[(7-fluoro-2-oxo-2,3-dihydro-1H-indol-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 256). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.67 (s, 1H), 10.51 (s, 1H), 7.53 (s, 1H), 7.26 (d, J=8.0 Hz, 1H), 6.87 (d, J=8.4 Hz, 1H), 6.82-6.80 (m, 2H), 4.11 (t, J=5.6 Hz, 2H), 3.54 (s, 2H), 2.93-2.86 (m, 2H), 2.85-2.82 (m, 2H), 2.73-2.70 (m, 2H), 1.81-1.73 (m, 4H). MS=430.2 [M+H].sup.+.

Example 46

2-[4-(2-{{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy}phenyl)-2-methylpropanamide (Compound 257)

##STR01818##

Step 1: methyl 2-(4-hydroxyphenyl)-2-methylpropanoate

##STR01819##

[2987] To a 0° C. solution of 2-(4-hydroxyphenyl)-2-methylpropanoic acid (900 mg, 4.99 mmol) in MeOH (18 mL) was added SOCl.sub.2 (471 µL, 6.49 mmol). The mixture was stirred at 70° C. for

2 h. After cooling to room temperature, the mixture was concentrated in vacuo. The residue was diluted with H₂O (30 mL), extracted with EtOAc (2×20 mL), dried over Na₂SO₄, filtered, and concentrated to give methyl 2-(4-hydroxyphenyl)-2-methylpropanoate, which was used in the subsequent step without further purification. MS=193.1 [M-H]⁺.

Step 2: methyl 2-[4-(2-bromoethoxy)phenyl]-2-methylpropanoate

##STR01820##

[2988] To a solution of methyl 2-(4-hydroxyphenyl)-2-methylpropanoate (820 mg, 4.22 mmol) and 1,2-dibromoethane (12.7 mL, 169 mmol) in MeCN (16 mL) was added K₂CO₃ (2.92 g, 21.1 mmol). After stirring at 80° C. for 16 h, the mixture was concentrated in vacuum. The residue was diluted with H₂O (30 mL), extracted with EtOAc (2×20 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 40 g cartridge, 0-20% EtOAc:petroleum ether) to give methyl 2-[4-(2-bromoethoxy)phenyl]-2-methylpropanoate. MS=301.0 [M+H]⁺.

Step 3: methyl 2-[4-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]-2-methylpropanoate

##STR01821##

[2989] To a solution of methyl 2-[4-(2-bromoethoxy)phenyl]-2-methylpropanoate (300 mg, 0.996 mmol) and 2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (Intermediate B-9, 272 mg, 1.03 mmol, HCl salt) in MeCN (3 mL) was added NaHCO₃ (251 mg, 2.99 mmol). The mixture was stirred at 80° C. for 16 h. After cooling to room temperature, the residue was diluted with H₂O (50 mL), extracted with EtOAc (2×30 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-75% EtOAc:petroleum ether) to give methyl 2-[4-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]-2-methylpropanoate. MS=448.2 [M+H]⁺.

Step 4: 2-[4-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]-2-methylpropanoic acid

##STR01822##

[2990] To a solution of methyl 2-[4-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]-2-methylpropanoate (120 mg, 0.268 mmol) in THE (2 mL) was added a solution of NaOH (26.8 mg, 0.670 mmol) in H₂O (1 mL). The mixture was stirred at room temperature for 3 h. The reaction mixture was diluted with H₂O (5 mL) and then adjusted pH=3-4 with 4.0 M aqueous HCl solution. The residue was extracted with EtOAc (2×20 mL) and washed with brine (10 mL). The organic combined organic layers were dried over Na₂SO₄, filtered, and concentrated in vacuo to give 2-[4-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]-2-methylpropanoic acid, which was used in the subsequent step without further purification. MS=434.2 [M+H]⁺.

Step 5: 2-[4-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]-2-methylpropanamide (Compound 257)

##STR01823##

[2991] To a solution of 2-[4-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]-2-methylpropanoic acid (80.0 mg, 0.186 mmol) and NH₄Cl (11.9 mg, 0.221 mmol) in THE (0.5 mL) was added DIEA (77.2 µL, 0.443 mmol) and HATU (84.2 mg, 0.221 mmol). The mixture was stirred at room temperature for 1 h. The reaction mixture was diluted with H₂O (5 mL) and extracted with EtOAc (2×10 mL). The combined organic layers were washed with brine (10 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Waters Xbridge Prep OBD C₁₈ column, 20-50% MeCN: 10 mM NH₄HCO₃ in H₂O) to give 2-[4-(2-{5-cyano-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)phenyl]-2-methylpropanamide (Compound 257). ¹H NMR (400 MHz, DMSO-d₆): δ 10.83 (br s, J=1.2 Hz, 1H) 7.93 (d,

J=1.2 Hz, 1H) 7.65 (dd, J=8.0, 1.5 Hz, 1H) 7.19-7.25 (m, 2H) 6.96 (d, J=8.0 Hz, 1H) 6.88 (d, J=8.8 Hz, 2H), 6.78 (br d, J=6.4 Hz, 2H), 4.09 (t, J=6.0 Hz, 2H), 2.90-2.82 (m, 4H), 2.73-2.71 (m, 2H), 1.84-1.66 (m, 4H), 1.38 (s, 6H). MS=433.2 [M+H].sup.+.

Example 47

5-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)pyrimidine-2-carboxamide (Compound 258)

##STR01824##

Step 1: 5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyrimidine-2-carbonitrile

##STR01825##

[2992] To a mixture of 5-bromopyrimidine-2-carbonitrile (1.00 g, 5.43 mmol) and bis(pinacolato)diboron (1.56 g, 6.52 mmol) in dioxane (10 mL) under N.sub.2 atmosphere was added KOAc (1.07 g, 10.9 mmol) and Pd(dppf)Cl.sub.2 (199 mg, 0.272 mmol). The mixture was degassed and purged with N.sub.2 (3×), and then stirred at 80° C. for 12 h. After cooling to room temperature, the reaction mixture was concentrated in vacuo. The residue was diluted with H.sub.2O (20 mL) and extracted with EtOAc (3×30 mL). The combined organic layers were washed with brine (10 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-35% EtOAc:petroleum ether) to give 5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyrimidine-2-carbonitrile. MS=232.2 [M+H].sup.+.

Step 2: 5-hydroxypyrimidine-2-carbonitrile

##STR01826##

[2993] To a 0° C. solution of 5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyrimidine-2-carbonitrile (15.0 g, 64.9 mmol) in acetone (100 mL) and H.sub.2O (100 mL) was added Oxone (47.9 g, 77.9 mmol). The mixture was stirred at room temperature for 1 h and was then quenched with saturated Na.sub.2SO.sub.3 (100 mL) and extracted with EtOAc (3×100 mL). The combined organic layers were washed with brine (100 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 100 g cartridge, 0-65% EtOAc:petroleum ether) to give 5-hydroxypyrimidine-2-carbonitrile. MS=122.0 [M+H].sup.+.

Step 3: 5-(2-bromoethoxy)pyrimidine-2-carbonitrile

##STR01827##

[2994] To a mixture of 5-hydroxypyrimidine-2-carbonitrile (300 mg, 2.48 mmol) and 1,2-dibromoethane (7.48 mL, 99.1 mmol) in MeCN (6 mL) was added K.sub.2CO.sub.3 (1.03 g, 7.43 mmol). The mixture was stirred at 60° C. for 12 h. After cooling to room temperature, the reaction mixture was concentrated in vacuo to remove MeCN. The residue was diluted with H.sub.2O (10 mL) and extracted with EtOAc (3×30 mL). The combined organic layers were washed with brine (10 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 4 g cartridge, 0-60% EtOAc:petroleum ether) to give 5-(2-bromoethoxy)pyrimidine-2-carbonitrile. MS=228.1/230.1 [M+H].sup.+.

Step 4: 5-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)pyrimidine-2-carbonitrile

##STR01828##

[2995] To a mixture of 5-(2-bromoethoxy)pyrimidine-2-carbonitrile (165 mg, 0.724 mmol) and 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 206 mg, 0.868 mmol,) in MeCN (2 mL) was added NaHCO.sub.3 (122 mg, 1.45 mmol). The mixture was stirred at 80° C. for 12 h. After cooling to room temperature, the reaction mixture was concentrated in vacuo to remove MeCN. The residue was diluted with H.sub.2O (10 mL) and extracted with EtOAc (3×30 mL). The combined organic layers were washed with brine (10 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated 5-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-

yl}ethoxy)pyrimidine-2-carbonitrile, which was used in the subsequent step without further purification. MS=384.2 [M+H].sup.+.

Step 5: 5-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)pyrimidine-2-carboxamide (Compound 258)

##STR01829##

[2996] To a 0° C. mixture of 5-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)pyrimidine-2-carbonitrile (160 mg, 0.417 mmol) and K.sub.2CO.sub.3 (115 mg, 0.834 mmol) in DMSO (2 mL) was added H.sub.2O.sub.2 in H.sub.2O (120 µL, 30 wt % 1.25 mmol) dropwise. The mixture was stirred at room temperature for 2 h. The reaction mixture was cooled to 0° C., quenched with saturated aqueous Na.sub.2SO.sub.3 solution (5 mL), diluted with H.sub.2O (5 mL), and then extracted with EtOAc (3×10 mL). The combined organic layers were washed with brine (5 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Waters Xbridge Prep OBD C.sub.18 column, 15-40% MeCN: 10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 5-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)pyrimidine-2-carboxamide (Compound 258).
.sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.50 (s, 1H), 8.68 (s, 2H), 8.05 (s, 1H), 7.65 (s, 1H), 7.52 (d, J=2.00 Hz, 1H), 7.25 (dd, J=10.40, 6.00 Hz, 1H), 6.87 (d, J=8.40 Hz, 1H), 4.41 (t, J=5.60 Hz, 2H), 2.99-2.88 (m, 4H), 2.76-2.70 (m, 2H), 1.84-1.68 (m, 4H). MS=402.2 [M+H].sup.+.

Example 48

4-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2,6-difluorobenzoyl]-1λ.SUP.6.-thiomorpholine-1,1-dione (Compound 259)

##STR01830##

Step 1: methyl 2,6-difluoro-4-hydroxybenzoate

##STR01831##

[2997] To a 0° C. solution of 2,6-difluoro-4-hydroxybenzoic acid (4.50 g, 25.9 mmol) in MeOH (45 mL) and MTBE (45 mL) was added 2.0 M diazomethyl(trimethyl)silane in DCM (14.2 mL, 28.4 mmol) dropwise. The mixture was warmed to room temperature and stirred for 16 h. The reaction mixture was concentrated in vacuo to provide methyl 2,6-difluoro-4-hydroxybenzoate, which was used in the subsequent step without further purification. MS=187.1 [M-H].sup.+.

Step 2: methyl 4-(2-bromoethoxy)-2,6-difluorobenzoate

##STR01832##

[2998] To a solution of methyl 2,6-difluoro-4-hydroxybenzoate (4.38 g, 23.3 mmol) in MeCN (15 mL) was added K.sub.2CO.sub.3 (16.1 g, 116 mmol) and 1,2-dibromoethane (70.3 mL, 931 mmol). The mixture was heated to 80° C. and stirred for 16 h. After cooling room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo to provide methyl 4-(2-bromoethoxy)-2,6-difluorobenzoate, which was used in the subsequent step without further purification. MS=295.2/296.8 [M+H].sup.+.

Step 3: methyl 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2,6-difluorobenzoate

##STR01833##

[2999] To a solution of methyl 4-(2-bromoethoxy)-2,6-difluorobenzoate (3.00 g, 10.2 mmol) in MeCN (60 mL) was added NaHCO.sub.3 (2.56 g, 30.5 mmol) and 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 2.41 g, 8.82 mmol, HCl salt). The mixture was heated to 80° C. and stirred for 16 h. After cooling to room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo. The filtrate was diluted with H.sub.2O and then extracted with a 10:1 solution of DCM:MeOH (3×40 mL). The combined organic layers were washed with brine (40 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Phenomenex Luna C.sub.18 column, 20-50% MeCN: 10 mM TFA in H.sub.2O) to give methyl 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2,6-difluorobenzoate.

MS=451.1 [M+H].sup.+.

Step 4: 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2,6-difluorobenzoic acid

##STR01834##

[3000] To a solution of methyl 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2,6-difluorobenzoate (1.60 g, 3.55 mmol) in THE (32 mL) and H.sub.2O (64 mL) was added LiOH H.sub.2O (596 mg, 14.2 mmol). The mixture was stirred at room temperature for 16 h, and then was concentrated in vacuo. The residue was diluted with EtOAc (50 mL) and H.sub.2O (10 mL) and was adjusted to pH=5-6 with dropwise addition of 1.0 M aqueous HCl. The organic layer was separated, and aqueous phase was extracted with EtOAc (2×40 mL). The combined organic layers were washed with brine (40 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2,6-difluorobenzoic acid, which was used in the subsequent step without further purification. MS=437.0 [M+H].sup.+.

Step 5: 4-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2,6-difluorobenzoyl]-1λ.SUP.6.-thiomorpholine-1,1-dione (Compound 259)

##STR01835##

[3001] To a 0° C. solution of 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2,6-difluorobenzoic acid (1.00 g, 2.29 mmol) in DMF (5 mL) was added 1-methylimidazole (0.547 mL, 6.87 mmol) and 1λ.sup.6-thiomorpholine-1,1-dione (464 mg, 3.43 mmol). After stirring for 10 min, [chloro(dimethylamino)methylene]-dimethyl-ammonium hexafluorophosphate (963 mg, 3.43 mmol) was added. The mixture was allowed to warm to room temperature and stirred for 2 h. The reaction mixture was poured into the H.sub.2O (30 mL) and then extracted with EtOAc (3×40 mL). The combined organic layers were washed with brine (30 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Kromasil C.sub.18 column, 20-60% MeCN: 10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 4-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2,6-difluorobenzoyl]-1λ.sup.6-thiomorpholine-1,1-dione (Compound 259). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.49 (s, 1H), 7.51 (d, J=1.6 Hz, 1H), 7.24 (d, J=8.4, 2.0 Hz, 1H), 6.94 (d, J=9.6 Hz, 2H), 6.85 (d, J=8.4 Hz, 1H), 4.22 (t, J=5.2 Hz, 2H), 4.11-4.05 (m, 2H), 3.79-3.69 (m, 2H), 3.29-3.28 (m, 2H), 3.11-3.10 (m, 2H), 2.91-2.84 (m, 4H), 2.68-2.67 (m, 2H), 1.78-1.70 (m, 4H). MS=554.2 [M+H].sup.+.

Example 49

1'-{2-[4-(1,1-dioxo-1λ.SUP.6.-thiomorpholine-4-carbonyl)-3,5-difluorophenoxy]ethyl}-1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (Compound 260)

##STR01836##

Step 1: methyl 4-(2-{5-cyano-1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2,6-difluorobenzoate

##STR01837##

[3002] To a solution of methyl 4-(2-bromoethoxy)-2,6-difluorobenzoate (Example 48: Step 2, 300 mg, 1.02 mmol) in MeCN (4 mL) was added NaHCO.sub.3 (256 mg, 3.05 mmol) and 1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (Intermediate B-15, 245 mg, 1.02 mmol). The mixture was heated to 80° C. and stirred for 16 h. After cooling to room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo. The crude residue was purified by preparative TLC (SiO.sub.2, 10:1 DCM:MeOH) to give methyl 4-(2-{5-cyano-1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2,6-difluorobenzoate. MS=456.2 [M+H].sup.+.

Step 2: 4-(2-{5-cyano-1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2,6-difluorobenzoic acid

##STR01838##

[3003] To a solution of methyl 4-(2-{5-cyano-1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2,6-difluorobenzoate (290 mg, 0.637 mmol) in THE (3.6 mL) and H.sub.2O (2.0 mL) was added LiOH H.sub.2O (53.4 mg, 1.42 mmol). The mixture was stirred at room temperature for 16 h. The reaction mixture was poured into H.sub.2O (10 mL) and adjusted to pH=4-6 via the dropwise addition of 1.0 M aqueous HCl. The mixture was extracted with EtOAc (3×15 mL). The combined organic layers were washed with brine (20 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give 4-(2-{5-cyano-1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2,6-difluorobenzoic acid, which was used in the subsequent step without further purification. MS=442.2 [M+H].sup.+.

Step 3: 1'-{2-[4-(1,1-dioxo-1λ.SUP.6.-thiomorpholine-4-carbonyl)-3,5-difluorophenoxy]ethyl}-1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (Compound 260)

##STR01839##

[3004] To a solution of 1λ.sup.6-thiomorpholine-1,1-dione (62.2 mg, 0.362 mmol, HCl salt) and 4-(2-{5-cyano-1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2,6-difluorobenzoic acid (80.0 mg, 0.181 mmol) in DMF (2 mL) was added 1-methylimidazole (72.2 μL, 0.906 mmol). After stirring for 10 min, [chloro(dimethylamino)methylene]-dimethylammonium hexafluorophosphate (102 mg, 0.363 mmol) was added and the mixture was stirred at room temperature for 2 h. The reaction mixture was quenched with H.sub.2O (1 mL), solids were removed by filtration, and the filtrate was concentrated in vacuo. The residue was diluted with H.sub.2O (10 mL) and extracted with EtOAc (3×10 mL). The combined organic layers were washed with brine (20 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Phenomenex C.sub.18 column, 15-45% MeCN: 10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 1'-{2-[4-(1,1-dioxo-1λ.sup.6-thiomorpholine-4-carbonyl)-3,5-difluorophenoxy]ethyl}-1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (Compound 260). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 7.99 (s, 1H), 7.80 (dd, J=8.0, 1.6, 1H), 7.20 (d, J=8.4 Hz, 1H), 6.94 (d, J=10 Hz, 2H), 4.22 (t, J=5.4 Hz, 2H), 4.08-4.05 (m, 2H), 3.74-3.71 (m, 2H), 3.35-3.27 (m, 2H), 3.14 (s, 3H), 3.09 (br s, 2H), 2.90-2.87 (m, 4H), 2.85-2.74 (m, 2H), 1.78-1.74 (m, 4H). MS=559.2 [M+H].sup.+.

Example 50

1'-{2-[4-(1,1-dioxo-1λ.SUP.6.-thiomorpholine-4-carbonyl)-3-(trifluoromethyl)phenoxy]ethyl}-1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (Compound 261)

##STR01840##

Step 1: methyl 4-hydroxy-2-(trifluoromethyl)benzoate

##STR01841##

[3005] To a 0° C. solution of 4 4-hydroxy-2-(trifluoromethyl)benzoic acid (3.30 g, 16.0 mmol) in MeOH (50 mL) was added SOCl.sub.2 (11.0 mL, 152 mmol) dropwise. The mixture was heated to 50° C. and stirred for 12 h. The mixture was cooled to 0° C. and quenched with saturated NH.sub.4Cl aqueous solution (100 mL). The aqueous phase was extracted with EtOAc (3×100 mL). The combined organic layers were washed with brine (100 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 40 g cartridge, 0-50% EtOAc:petroleum ether) to give methyl 4-hydroxy-2-(trifluoromethyl)benzoate. MS=221.1 [M+H].sup.+.

Step 2: methyl 4-(2-bromoethoxy)-2-(trifluoromethyl)benzoate

##STR01842##

[3006] To a solution of methyl 4-hydroxy-2-(trifluoromethyl)benzoate (2.50 g, 11.4 mmol) and 1,2-dibromoethane (34.3 mL, 454 mmol) in MeCN (35 mL) was added K.sub.2CO.sub.3 (7.85 g, 56.8 mmol). The mixture was heated to 65° C. and then stirred for 12 h. After cooling to room temperature, the mixture was concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 40 g cartridge, 0-50% EtOAc:petroleum ether) to give methyl 4-(2-bromoethoxy)-2-(trifluoromethyl)benzoate. MS=327.1/329.1 [M+H].sup.+.

Step 3: methyl 4-(2-{5-cyano-1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-(trifluoromethyl)benzoate

##STR01843##

[3007] To a solution of 1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (Intermediate B-15, 154 mg, 0.556 mmol, HCl salt) and methyl 4-(2-bromoethoxy)-2-(trifluoromethyl)benzoate (200 mg, 0.611 mmol) in MeCN (8 mL) was added NaHCO₃ (140 mg, 1.67 mmol). The mixture was heated to 80° C. and stirred for 12 h. After cooling to room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-50% EtOAc:petroleum ether) to give methyl 4-(2-{5-cyano-1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-(trifluoromethyl)benzoate. MS=488.3 [M+H].sup.+

Step 4: 4-(2-{5-cyano-1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-(trifluoromethyl)benzoic acid

##STR01844##

[3008] To a solution of methyl 4-(2-{5-cyano-1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-(trifluoromethyl)benzoate (250 mg, 0.513 mmol) in MeOH (5 mL) and H₂O (5 mL) was added LiOH H₂O (172 mg, 4.10 mmol). The mixture was stirred at room temperature for 2 h. The reaction mixture was concentrated in vacuo to remove MeOH, and the resulting aqueous phase was adjusted to pH=4-6 via dropwise addition of 1.0 M aqueous HCl. The resulting solid was isolated by filtration, washed with MTBE, and dried in vacuo to give 4-(2-{5-cyano-1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-(trifluoromethyl)benzoic acid, which was taken to the subsequent step without further purification. MS=474.2 [M+H].sup.+.

Step 5: 1'-{2-[4-(1,1-dioxo-1λ⁶-thiomorpholine-4-carbonyl)-3-(trifluoromethyl)phenoxy]ethyl}-1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (Compound 261)

##STR01845##

[3009] To a 0° C. solution of 4-(2-{5-cyano-1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-(trifluoromethyl)benzoic acid (100 mg, 0.211 mmol), 1λ⁶-thiomorpholine-1,1-dione (39.9 mg, 0.232 mmol, HCl salt) and DIEA (92.0 μL, 0.528 mmol) in DMF (2 mL) was added HATU (120 mg, 0.317 mmol). The mixture was allowed to warm to room temperature and stirred for 16 h. The reaction mixture was poured into H₂O (5 mL) and then extracted with EtOAc (3×10 mL). The combined organic layers were washed with brine (20 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Waters Xbridge Prep OBD C₁₈ column, 15-45% MeCN: 10 mM NH₄HCO₃ in H₂O) to give 1'-{2-[4-(1,1-dioxo-1λ⁶-thiomorpholine-4-carbonyl)-3-(trifluoromethyl)phenoxy]ethyl}-1-methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (Compound 261). ¹H NMR (400 MHz, DMSO-d₆): δ 8.00 (s, 1H), 7.80 (d, J=9.6 Hz, 1H), 7.68 (d, J=8.4, 1H), 7.38-7.35 (m, 2H), 7.21 (d, J=8.0, 1H), 4.47 (br d, J=13.2, 1H), 4.29-4.26 (m, 2H), 3.59-3.56 (m, 2H), 3.46-3.41 (m, 2H), 3.29-3.28 (m, 1H), 3.20-3.16 (m, 4H), 2.95-2.88 (m, 5H), 3.77-3.74 (m, 2H), 1.84-1.68 (m, 4H). MS=591.3 [M+H].sup.+.

Example 51

1'-{1-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]propan-2-yl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (Compound 262), (S) or (R)-1'-{1-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]propan-2-yl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (Compound 263), and (R) or (S)-1'-{1-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]propan-2-yl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (Compound 264)

##STR01846##

Step 1: 1'-{1-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]propan-2-yl}-2-oxo-1,2-

dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (Compound 262)

##STR01847##

[3010] A mixture of 1-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]propan-2-yl methanesulfonate (Intermediate A-36, 385 mg, 1.06 mmol), 2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (Intermediate B-9, 200 mg, 0.880 mmol), NaHCO₃ (296 mg, 3.52 mmol) and KI (292 mg, 1.76 mmol) in DMF (1 mL) was degassed and purged with N₂ (3×). The mixture was stirred at 90° C. for 15 h under N₂ atmosphere. After cooling to room temperature, the reaction mixture was quenched with H₂O (15 mL) and extracted with EtOAc (3×10 mL). The combined organic layers were washed with brine (20 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Phenomenex Cis column, 20-40% MeCN. 10 mM NH₄HCO₃ in H₂O) to give 1'-{1-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]propan-2-yl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (Compound 262). MS=496.1 [M+H]⁺.

Step 2: (S) or (R)-1'-{1-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]propan-2-yl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (Compound 263), and (R) or (S)-1'-{1-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]propan-2-yl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (Compound 264)

##STR01848##

[3011] 1'-{1-[4-(3-methanesulfonyloxetan-3-yl)phenoxy]propan-2-yl}-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (110 mg, 0.222 mmol) was purified by preparative chiral SFC (Dailcel Chiralpak AD-3, 60% ethanol with 0.1% NH₄OH in CO₂). The first eluting enantiomer of the title compound, Compound 263: ¹H NMR (400 MHz, DMSO-d₆): δ 10.86 (s, 1H), 7.87 (s, 1H), 7.68 (dd, J=8.4, 1.6 Hz, 1H), 7.26 (d, J=8.8 Hz, 2H), 7.05 (d, J=8.8 Hz, 2H), 6.97 (d, J=8.0 Hz, 1H), 5.21 (d, J=7.6 Hz, 2H), 5.00 (d, J=7.6 Hz, 2H), 4.50-4.20 (m, 1H), 4.01-3.98 (m, 1H), 3.14-3.09 (m, 1H), 3.03-3.01 (m, 2H), 2.89-2.84 (m, 4H), 2.79-2.74 (m, 1H), 1.81-1.68 (m, 4H), 1.16 (d, J=7.2 Hz, 3H). MS=496.1 [M+H]⁺. The second eluting enantiomer of the title compound, Compound 264: ¹H NMR (400 MHz, DMSO-d₆): δ 10.87 (s, 1H), 7.87 (s, 1H), 7.68 (dd, J=8.0, 1.6 Hz, 1H), 7.26 (d, J=8.8 Hz, 2H), 7.06 (d, J=8.4 Hz, 2H), 6.98 (d, J=8.0 Hz, 1H), 5.19 (d, J=7.2 Hz, 2H), 5.00 (d, J=7.6 Hz, 2H), 4.21-4.13 (m, 1H), 4.05-3.98 (m, 1H), 3.15-3.13 (m, 1H), 3.04-3.03 (m, 2H), 2.89-2.84 (m, 4H), 2.81-2.77 (m, 1H), 1.82-1.72 (m, 4H), 1.18 (d, J=6.8 Hz, 3H). MS=496.1 [M+H]⁺.

Example 52

5-chloro-1'-{2-[3-(difluoromethyl)-4-methanesulfonylphenoxy]ethyl}-1-(2-hydroxyethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 265)

##STR01849##

[3012] To a solution of 5-chloro-1'-{2-[3-(difluoromethyl)-4-methanesulfonylphenoxy]ethyl}-1-(2-hydroxyethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 197, 100 mg, 0.206 mmol) in DMF (1 mL) was added K₂CO₃ (142 mg, 1.03 mmol) and 2-bromoethanol (155 mg, 1.24 mmol). The mixture was stirred at 80° C. for 12 h. The reaction mixture was diluted with H₂O (10 mL) and extracted with EtOAc (3×10 mL). The combined organic layers were washed with brine (10 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Phenomenex C_{sub}18 column, 30-65% MeCN: 10 mM NH₄HCO₃ in H₂O) to give 5-chloro-1'-{2-[3-(difluoromethyl)-4-methanesulfonylphenoxy]ethyl}-1-(2-hydroxyethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 265). ¹H NMR (400 MHz, DMSO-d₆): δ 7.99 (d, J=8.8 Hz, 1H), 7.59-7.52 (m, 2H), 7.33-7.30 (m, 3H), 7.09 (d, J=8.4 Hz, 1H), 4.82 (t, J=5.6 Hz, 1H), 4.32 (t, J=5.6 Hz, 2H), 3.70 (t, J=5.6 Hz, 2H), 3.59-3.52 (m, 2H), 3.26 (s, 3H), 2.99-2.86 (m, 4H), 2.77-2.69 (m, 2H), 1.76 (s, 4H). MS=529.1 [M+H]⁺.

[3013] The following compound in Table 38 was prepared according to procedures similar to those described for Compound 265 using the appropriate starting materials.

TABLE-US-00041 TABLE 38 Exact Mass Starting # Structure Name [M + H].sup.+ Material 266
[01850]  embedded image 1'-[2-[3-(difluoromethyl)-4-methanesulfonylphenoxy]ethyl]-1-(2-hydroxyethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile Calc'd 520.2 Found 520.3 Compound 100

Example 53

5-chloro-1'-[2-({7-oxo-8-[(trans)-3-hydroxycyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 267)

##STR01851##

Step 1: (cis)-3-[6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-oxo-1,2,3,4-tetrahydro-1,8-naphthyridin-1-yl]cyclobutyl methanesulfonate

##STR01852##

[3014] To a 0° C. solution of 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-hydroxycyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 190, 150 mg, 0.302 mmol) in DCM (5 mL) was added TEA (61.0 mg, 0.604 mmol) and methanesulfonic anhydride (79 mg, 0.453 mmol). The mixture was allowed to warm to room temperature and stirred for 12 h. The mixture was quenched with H.sub.2O (10 mL) and extracted with DCM (3×10 mL). The combined organic layers were washed with brine (30 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give (cis)-3-[6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-oxo-1,2,3,4-tetrahydro-1,8-naphthyridin-1-yl]cyclobutyl methanesulfonate, which was taken onto the next step without further purification. MS=575.2 [M+H].sup.+.

Step 2: (trans)-3-[6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-oxo-1,2,3,4-tetrahydro-1,8-naphthyridin-1-yl]cyclobutyl acetate

##STR01853##

[3015] To a solution of (cis)-3-[6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-oxo-1,2,3,4-tetrahydro-1,8-naphthyridin-1-yl]cyclobutyl methanesulfonate (200 mg, 0.348 mmol) in DMF (5 mL) was added KOAc (341 mg, 3.48 mmol). The mixture was heated to 100° C. and stirred for 12 h. After cooling to room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Waters Xbridge Prep OBD C.sub.18 column, 25-55% MeCN: 10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give (trans)-3-[6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-oxo-1,2,3,4-tetrahydro-1,8-naphthyridin-1-yl]cyclobutyl acetate. MS=539.3 [M+H].sup.+.

Step 3: 5-chloro-1'-[2-({7-oxo-8-[(trans)-3-hydroxycyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 267)

##STR01854##

[3016] To a mixture of [(trans)-3-[6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-oxo-1,2,3,4-tetrahydro-1,8-naphthyridin-1-yl]cyclobutyl acetate (40.0 mg, 0.0742 mmol) in MeOH (2 mL) was added K.sub.2CO.sub.3 (20.5 mg, 0.148 mmol). The mixture was stirred at room temperature for 6 h. Solids were removed by filtration and the filtrate was concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Waters Xbridge BEH C.sub.18 column, 25-55% MeCN: 10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 5-chloro-1'-[2-({7-oxo-8-[(trans)-3-hydroxycyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 267). .sup.1H NMR (400 MHz, CD.sub.3CN): δ 8.40 (br s, 1H), 7.90 (d, J=2.8 Hz, 1H), 7.45 (d, J=2.0 Hz, 1H), 7.24-7.16 (m, 2H), 6.87 (d, J=8.4 Hz, 1H), 5.46 (q, J=8.6 Hz, 1H), 4.47 (br s, 1H), 4.17 (t, J=5.6 Hz, 2H), 3.04-2.84 (m, 7H), 2.83-2.77 (m, 2H), 2.77-2.67 (m, 2H), 2.56-2.48 (m, 2H), 2.27-2.18 (m, 2H), 1.91-1.82 (m, 2H), 1.79-1.70 (m, 2H). MS=497.2 [M+H].sup.+.

Example 54

5-chloro-1'-[2-[4-(3-hydroxy-1-methanesulfonylcyclobutyl)phenoxy]ethyl]-1,2-

dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 268)

5-chloro-1'-(2-{4-[(cis) or (trans)-3-hydroxy-1-methanesulfonylcyclobutyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 269)

5-chloro-1'-(2-{4-[(trans) or (cis)-3-hydroxy-1-methanesulfonylcyclobutyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 270)

##STR01855##

Step 1: 3-(benzyloxy)-1-[4-(benzyloxy)phenyl]cyclobutan-1-ol

##STR01856##

[3017] To a -78° C. mixture of 1-(benzyloxy)-4-bromobenzene (34.0 g, 129 mmol) in THE (250 mL) was added 2.5 M n-BuLi in THE (62.0 mL, 155 mmol) dropwise. After stirring for 30 min, 3-benzyloxycyclobutanone (25.1 g, 142 mmol) in THE (50 mL) was added. The mixture was stirred at -78° C. for 1 h. The reaction mixture was allowed to warm to 0° C. and then quenched with H.sub.2O (200 mL) and saturated aqueous NH.sub.4Cl (200 mL). The mixture was extracted with EtOAc (3 $\times$ 250 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 330 g cartridge, 0-50% EtOAc:petroleum ether) to give 3-(benzyloxy)-1-[4-(benzyloxy)phenyl]cyclobutan-1-ol as a mixture of diastereomers. ¹H NMR (400 MHz, DMSO-d.sub.6): δ 7.48-7.22 (m, 13H), 6.95-6.91 (m, 1H), 5.54 (s, 1H), 5.08 (s, 2H), 4.37 (s, 2H), 3.78-3.67 (m, 1H), 2.78-2.68 (m, 2H), 2.30-2.22 (m, 2H).

Step 2: 1-(benzyloxy)-4-[3-(benzyloxy)-1-methanesulfonylcyclobutyl]benzene

##STR01857##

[3018] To a 0° C. solution of 3-(benzyloxy)-1-[4-(benzyloxy)phenyl]cyclobutan-1-ol (25.0 g, 69.4 mmol) and sodium methanesulfinate (35.4 g, 347 mmol) in DCM (300 mL) was added TFA (39.5 g, 347 mmol). The mixture was stirred at 0° C. for 2 h. The reaction mixture was then quenched with H.sub.2O (300 mL) and extracted with EtOAc (3 $\times$ 150 mL). The combined organic layers were washed with brine (3 $\times$ 300 mL), dried over MgSO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 12 g cartridge, 0-45% EtOAc:petroleum ether) to give 1-(benzyloxy)-4-[3-(benzyloxy)-1-methanesulfonylcyclobutyl]benzene. MS=440.3 [M+NH.sub.4].sup.+.

Step 3: 4-(3-hydroxy-1-methanesulfonylcyclobutyl)phenol

##STR01858##

[3019] To a mixture of Pd/C (8.00 g, 10 wt %, 7.55 mmol) in MeOH (200 mL) was added 1-(benzyloxy)-4-[3-(benzyloxy)-1-methanesulfonylcyclobutyl]benzene (5.00 g, 11.8 mmol). The suspension was degassed under vacuum and purged with H.sub.2 (3 $\times$). After allowing the mixture to stir at room temperature for 16 h under a H.sub.2 atmosphere, the mixture was filtered through Celite. The filtrate was concentrated in vacuo to give 4-(3-hydroxy-1-methanesulfonylcyclobutyl)phenol, which was taken to the next step without further purification. MS=260.2 [M+NH.sub.4].sup.+.

Step 4: methyl 2-[4-(3-hydroxy-1-methanesulfonylcyclobutyl)phenoxy]acetate

##STR01859##

[3020] To a solution 4-(3-hydroxy-1-methanesulfonylcyclobutyl)phenol (3.80 g, 15.7 mmol) and K.sub.2CO.sub.3 (3.25 g, 23.5 mmol) in DMF (30 mL) was added methyl 2-bromoacetate (2.88 g, 18.8 mmol). The mixture was stirred at room temperature for 16 h, and then quenched with H.sub.2O (30 mL). The mixture was extracted with EtOAc (3 $\times$ 15 mL). The combined organic phase was washed with brine (50 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 10 g cartridge, 0-100% EtOAc:petroleum ether) to give methyl 2-[4-(3-hydroxy-1-methanesulfonylcyclobutyl)phenoxy]acetate as a mixture of diastereomers. MS=332.2 [M+NH.sub.4].sup.+.

Step 5: 3-[4-(2-hydroxyethoxy)phenyl]-3-methanesulfonylcyclobutan-1-ol

##STR01860##

[3021] To a 0° C. solution of methyl 2-[4-(3-hydroxy-1-methanesulfonylcyclobutyl)phenoxy]acetate (500 mg, 1.59 mmol) in THE (25 mL) was added LiAlH.sub.4 (151 mg, 3.98 mmol). The mixture was allowed to warm to room temperature and was stirred for 12 h. The mixture was quenched with saturated aqueous NH.sub.4Cl (5 mL) and then extracted with EtOAc (3×20 mL). The combined organic layers were washed with brine (50 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 10 g cartridge, 0-100% EtOAc:petroleum ether) to give 3-[4-(2-hydroxyethoxy)phenyl]-3-methanesulfonylcyclobutan-1-ol. MS=304.2 [M+NH.sub.4].sup.+.

Step 6: 2-[4-(3-hydroxy-1-methanesulfonylcyclobutyl)phenoxy]ethyl methanesulfonate

##STR01861##

[3022] To a solution of 3-[4-(2-hydroxyethoxy)phenyl]-3-methanesulfonylcyclobutan-1-ol (220 mg, 0.77 mmol) in DCM (15 mL) was added TEA (233 mg, 2.30 mmol) and methanesulfonic anhydride (161 mg, 0.92 mmol). The resulting mixture was stirred at room temperature for 12 h. The mixture was quenched with H.sub.2O (20 mL) and extracted with EtOAc (3×20 mL). The combined organic layers were washed with brine (2×20 mL), dried with Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by preparative TLC (SiO.sub.2, 1:1 petroleum ether/EtOAc) to give 2-[4-(3-hydroxy-1-methanesulfonylcyclobutyl)phenoxy]ethyl methanesulfonate as a mixture of diastereomers. MS=382.1 [M+NH.sub.4].sup.+.

Step 7: 5-chloro-1'-{2-[4-(3-hydroxy-1-methanesulfonylcyclobutyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 268)

##STR01862##

[3023] To a solution of 2-[4-(3-hydroxy-1-methanesulfonylcyclobutyl)phenoxy]ethyl methanesulfonate (85.0 mg, 0.23 mmol) in MeCN (5 mL) was added NaHCO.sub.3 (58.0 mg, 0.700 mmol) and 5-chloro-1'-(2-chloroethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 63.7 mg, 0.23 mmol, HCl salt). The mixture was stirred at 80° C. for 12 h. After allowing to cool to room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Waters Xbridge BEH C.sub.18 column, 20-50% MeCN:10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 5-chloro-1'-{2-[4-(3-hydroxy-1-methanesulfonylcyclobutyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 268). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.48 (s, 1H), 7.51-7.47 (m, 2H), 7.24 (dd, J=7.6, 2.0 Hz, 2H), 7.00 (t, J=8.0 Hz, 2H), 6.84 (d, J=8.4 Hz, 1H), 5.49-5.32 (m, 1H), 4.36-4.14 (m, 1H), 4.18-4.17 (m, 2H), 3.28-3.27 (m, 1H), 2.92-2.84 (m, 5H), 2.72-2.68 (m, 3H), 2.63-2.54 (m, 3H), 2.35-2.45 (m, 1H), 1.79-1.71 (m, 4H). MS=505.2 [M+H].sup.+.

Step 8: 5-chloro-1'-(2-{4-[(cis) or (trans)-3-hydroxy-1-methanesulfonylcyclobutyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 269) and 5-chloro-1'-(2-{4-[(trans) or (cis)-3-hydroxy-1-methanesulfonylcyclobutyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 270)

##STR01863##

[3024] 5-chloro-1'-{2-[4-(3-hydroxy-1-methanesulfonylcyclobutyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 268) was separated by preparative chiral SFC (Chiralpak AD-3 column, 50% isopropanol with 0.1% NH.sub.4OH in CO.sub.2). The first eluting isomer of the title compound, 5-chloro-1'-(2-{4-[(trans) or (cis)-3-hydroxy-1-methanesulfonylcyclobutyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 270): .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.49 (s, 1H), 7.55-7.44 (m, 3H), 7.23 (dd, J=8.4, 2.0 Hz, 1H), 7.01 (d, J=8.8 Hz, 2H), 6.84 (d, J=8.4 Hz, 1H), 5.49 (d, J=6.4 Hz, 1H), 4.17 (t, J=5.2 Hz, 2H), 3.90-3.75 (m, 1H), 2.98-2.79 (m, 6H), 2.76-2.66 (m, 4H), 2.54 (s, 3H),

1.86-1.64 (m, 4H). MS=505.2 [M+H].sup.+. The second eluting isomer of the title compound, 5-chloro-1'-(2-{4-[(cis) or (trans)-3-hydroxy-1-methanesulfonylcyclobutyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 269): .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.49 (s, 1H), 7.55-7.44 (m, 3H), 7.23 (dd, J=8.4, 2.0 Hz, 1H), 7.01 (d, J=8.8 Hz, 2H), 6.84 (d, J=8.4 Hz, 1H), 5.49 (d, J=6.4 Hz, 1H), 4.17 (t, J=5.2 Hz, 2H), 3.90-3.75 (m, 1H), 2.98-2.79 (m, 6H), 2.76-2.66 (m, 4H), 2.54 (s, 3H), 1.86-1.64 (m, 4H). MS=505.2 [M+H].sup.+.

Example 55

4-(4-{2-[5-chloro-1-(2-hydroxyethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl]ethoxy}-2-(trifluoromethyl)benzoyl)-1 λ .SUP.6.-thiomorpholine-1,1-dione (Compound 271)
##STR01864##

Step 1: methyl 4-{2-[5-chloro-1-(2-hydroxyethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl]ethoxy}-2-(trifluoromethyl)benzoate

##STR01865##

[3025] To a room temperature solution of methyl 4-(2-bromoethoxy)-2-(trifluoromethyl)benzoate (Example 50, Step 2, 450 mg, 1.38 mmol) in MeCN (5 mL) was added NaHCO.sub.3 (462 mg, 5.50 mmol) and 5-chloro-1-(2-hydroxyethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-11, 524 mg, 1.65 mmol). The mixture was heated to 80° C. and stirred for 16 h. The mixture was allowed to cool to room temperature, then was diluted with H.sub.2O (30 mL), and extracted with EtOAc (2 $\times$ 20 mL). The combined organic layers were washed with brine (30 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 4 g cartridge, 0-35% EtOAc:petroleum ether) to give methyl 4-{2-[5-chloro-1-(2-hydroxyethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl]ethoxy}-2-(trifluoromethyl)benzoate. MS=527.2. [M+H].sup.+.

Step 2: 4-{2-[5-chloro-1-(2-hydroxyethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl]ethoxy}-2-(trifluoromethyl)benzoic acid

##STR01866##

[3026] To a solution of methyl 4-{2-[5-chloro-1-(2-hydroxyethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl]ethoxy}-2-(trifluoromethyl)benzoate (750 mg, 1.42 mmol) in MeOH (5 mL) was added LiOH H.sub.2O (478 mg, 11.4 mmol) in H.sub.2O (1 mL). The mixture was stirred at 50° C. for 12 h, then was concentrated in vacuo. The resulting mixture was adjusted to pH=4 via dropwise addition of 4.0 M aqueous HCl. The resulting solid was isolated by filtration and concentrated in vacuo to give 4-{2-[5-chloro-1-(2-hydroxyethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl]ethoxy}-2-(trifluoromethyl)benzoic acid, which was taken onto the subsequent step without further purification. MS=513.1 [M+H].sup.+.

Step 3: 4-(4-{2-[5-chloro-1-(2-hydroxyethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl]ethoxy}-2-(trifluoromethyl)benzoyl)-1 λ .SUP.6.-thiomorpholine-1,1-dione (Compound 271)

##STR01867##

[3027] To a solution of 1 λ .sup.6.-thiomorpholine-1,1-dione (31 mg, 0.23 mmol) and 4-{2-[5-chloro-1-(2-hydroxyethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl]ethoxy}-2-(trifluoromethyl)benzoic acid (100 mg, 0.19 mmol) in DCM (2 mL) was added HOBt (8.0 mg, 0.058 mmol), EDCI (56.0 mg, 0.292.45 mmol) and TEA (79 mg, 0.78 mmol). The mixture was stirred at room temperature for 12 h, then was concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Phenomenex C.sub.18 column, 20-55% MeCN: 10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 4-(4-{2-[5-chloro-1-(2-hydroxyethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl]ethoxy}-2-(trifluoromethyl)benzoyl)-1 λ .sup.6.-thiomorpholine-1,1-dione (Compound 271). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 7.68 (d, J=8.0 Hz, 1H), 7.53 (d, J=2.0 Hz, 1H), 7.42-7.26 (m, 3H), 7.10 (d, J=8.4 Hz, 1H), 4.83 (t, J=6.0 Hz, 1H), 4.48 (d, J=13.6 Hz, 1H), 4.27 (t, J=6.0 Hz, 2H), 3.69 (t, J=6.0 Hz, 2H), 3.63-3.49 (m, 4H), 3.49-3.39 (m, 2H), 3.29-3.11 (m, 2H), 3.02-2.82 (m, 5H), 2.77-2.68 (m, 2H), 1.90-1.56 (m, 4H).

MS=630.2 [M+H].sup.+.

Example 56

4-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-(trifluoromethyl)benzoyl]-1 λ .SUP.6.-thiomorpholine-1,1-dione (Compound 272)

##STR01868##

Step 1: Methyl 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-(trifluoromethyl)benzoate

##STR01869##

[3028] To a solution of methyl 4-(2-bromoethoxy)-2-(trifluoromethyl)benzoate (Example 50, Step 2, 1.0 g, 3.06 mmol) and 5-chlorospiro[indoline-3,4'-piperidine]-2-one (Intermediate B-4, 759 mg, 2.78 mmol, HCl salt) in MeCN (40 mL) was added NaHCO₃ (700 mg, 8.34 mmol). The mixture was heated to 80° C. and stirred for 12 h. The reaction mixture was allowed to cool to room temperature, then was filtered and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 20 g cartridge, 0-100% EtOAc:petroleum ether) to give methyl 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-(trifluoromethyl)benzoate. MS=483.1 [M+H].sup.+.

Step 2: 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-(trifluoromethyl)benzoic acid

##STR01870##

[3029] To a solution of methyl 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-(trifluoromethyl)benzoate (300 mg, 0.62 mmol) in MeOH (6 mL) and H₂O (6 mL) was added LiOH H₂O (208 mg, 4.97 mmol). The mixture was stirred at room temperature for 2 h. The reaction mixture was concentrated in vacuo, and the aqueous phase was adjusted to pH=4 via dropwise addition of 1.0 M aqueous HCl. The resulting solid was collected by filtration, washed with MTBE, and dried in vacuo to give 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-(trifluoromethyl)benzoic acid which was taken to the next step without further purification. MS=469.2 [M+H].sup.+.

Step 3: 4-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-(trifluoromethyl)benzoyl]-1 λ .SUP.6.-thiomorpholine-1,1-dione (Compound 272)

##STR01871##

[3030] To a 0° C. solution of 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-(trifluoromethyl)benzoic acid (100 mg, 0.21 mmol) and 1,4-thiazinane 1,1-dioxide (34.60 mg, 0.26 mmol) in DMF (2 mL) was added TEA (64.8 mg, 0.64 mmol), EDCI (61.3 mg, 0.32 mmol) and HOBt (5.8 mg, 0.043 mmol). The mixture was allowed to warm to room temperature and stirred for 12 h. The reaction mixture was poured into the H₂O (5 mL) and extracted with EtOAc (3 $\times$ 10 mL). The combined organic layers were washed with brine (20 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Phenomenex C₁₈ column, 10-55% MeCN: 10 mM NH₄HCO₃ in H₂O) to give 4-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-(trifluoromethyl)benzoyl]-1 λ .sup.6-thiomorpholine-1,1-dione (Compound 272). ¹H NMR (400 MHz, DMSO-d₆): δ 10.49 (s, 1H), 7.67 (d, J=8.4 Hz, 1H), 7.50 (s, 1H), 7.38-7.35 (m, 2H), 7.24 (dd, J=8.0, 2.4 Hz, 1H), 6.85 (d, J=8.4 Hz, 1H), 4.49-4.45 (m, 1H), 4.27 (t, J=5.6 Hz, 2H), 3.59-3.56 (m, 2H), 3.46-3.33 (m, 2H), 3.21-3.18 (m, 2H), 2.95-2.86 (m, 5H), 2.70-2.60 (m, 2H), 1.81-1.70 (m, 4H). MS=586.2 [M+H].sup.+.

Example 57

5-chloro-1'-[2-(3-fluoro-4-{6-methanesulfonyl-2,6-diazaspiro[3.3]heptane-2-carbonyl}phenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 273)

##STR01872##

Step 1: tert-butyl 6-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-fluorobenzoyl]-2,6-diazaspiro[3.3]heptane-2-carboxylate

##STR01873##

[3031] To a solution of 4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-fluorobenzoic acid (Example 36, Step 3, 1.15 g, 2.75 mmol) and tert-butyl 2,6-diazaspiro[3.3]heptane-2-carboxylate (950 mg, 3.29 mmol) in DMF (12 mL) was added TEA (1.11 g, 11.0 mmol), HOBt (74.2 mg, 0.549 mmol), then EDCI (1.05 g, 5.49 mmol). The mixture was stirred at room temperature for 16 h. The reaction mixture was quenched with H.sub.2O (30 mL) and then extracted with EtOAc (3×15 mL). The combined organic layers were washed with brine (3×10 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by normal phase silica gel chromatography (Biotage 20 g cartridge, 50-100% EtOAc:petroleum ether) to give tert-butyl 6-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-fluorobenzoyl]-2,6-diazaspiro[3.3]heptane-2-carboxylate. MS=599.2 [M+H].sup.+.

Step 2: 5-chloro-1'-[2-(4-{2,6-diazaspiro[3.3]heptane-2-carbonyl}-3-fluorophenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR01874##

[3032] To a solution of tert-butyl 6-[4-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-fluorobenzoyl]-2,6-diazaspiro[3.3]heptane-2-carboxylate (1.40 g, 2.34 mmol) in DCM (20 mL) was added TFA (4.31 g, 37.8 mmol). The mixture was stirred at room temperature for 3 h. The mixture was concentrated in vacuo to give 5-chloro-1'-[2-(4-{2,6-diazaspiro[3.3]heptane-2-carbonyl}-3-fluorophenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one. MS=499.1 [M+H].sup.+.

Step 3: 5-chloro-1'-[2-(3-fluoro-4-{6-methanesulfonyl-2,6-diazaspiro[3.3]heptane-2-carbonyl}phenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 273)

##STR01875##

[3033] To a 0° C. solution of 5-chloro-1'-[2-(4-{2,6-diazaspiro[3.3]heptane-2-carbonyl}-3-fluorophenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (1.25 g, 2.51 mmol) in DCM (15 mL) was added TEA (1.74 mL, 12.5 mmol) and methanesulfonic anhydride (655 mg, 3.76 mmol). The mixture was stirred at room temperature for 3 h, and then was concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Kromasil C.sub.18 column, 25-55% MeCN:10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 5-chloro-1'-[2-(3-fluoro-4-{6-methanesulfonyl-2,6-diazaspiro[3.3]heptane-2-carbonyl}phenoxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 273). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ=10.49 (s, 1H), 7.50 (d, J=2.0 Hz, 1H), 7.44 (app t, J=8.4 Hz, 1H), 7.23 (dd, J=8.4, 2.0 Hz, 1H), 6.93 (dd, J=12.4, 2.4 Hz, 1H), 6.89-6.83 (m, 2H), 4.20-4.18 (m, 6H), 4.03 (q, J=8.8 Hz, 4H), 2.97 (s, 3H), 2.95-2.84 (m, 4H), 2.72-2.67 (m, 2H), 1.82-1.66 (m, 4H). MS=577.2 [M+H].sup.+.

Example 58

5-(difluoromethyl)-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 274)

##STR01876##

[3034] To a solution of 6-(2-bromoethoxy)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (Intermediate A-89, 100 mg, 0.281 mmol) and 5-(difluoromethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-1, 89.4 mg, 0.309 mmol, HCl salt) in MeCN (3 mL) was added NaHCO.sub.3 (94.5 mg, 1.13 mmol). The mixture was stirred at 80° C. for 16 h. After cooling to room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Waters Xbridge BEH C.sub.18 column, 25-55% MeCN:10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 5-(difluoromethyl)-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 274). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.62 (s, 1H), 7.94 (d, J=2.4 Hz, 1H), 7.66 (s 1H), 7.43-7.39 (m, 2H), 7.10-6.82 (m, 2H), 4.82 (s, 1H), 4.24 (t,

J=8.0 Hz, 1H), 4.19 (t, J=5.6 Hz, 2H), 2.96-2.94 (m, 2H), 2.86 (t, J=5.6 Hz, 2H), 2.80 (t, J=7.2 Hz, 2H), 2.72-2.67 (m, 2H), 2.44-2.42 (m, 4H), 2.34 (t, J=7.2 Hz, 2H), 1.85-1.80 (m, 2H), 1.71-1.66 (m, 2H), 1.25 (s, 3H). MS=527.3 [M+H].sup.+.

[3035] The following compound in Table 38.1 was prepared according to the procedures similar to those described for Compound 274 using the appropriate starting materials.

TABLE-US-00042 TABLE 38.1 Exact Mass Intermediate No. Structure IUPAC Name [M + H].sup.+ Used 325 [01877]  5-(difluoromethoxy)-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 543.2 Found 543.1 A-89 & B-2 329 [01878]  5-chloro-7-iodo-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 637.1 Found 637.0 A-89 & B-24 289 [01879]  5-chloro-1'-[2-({2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H,2H,4H-pyrido[2,3-d][1,3]oxazin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 513.2 Found 513.2 A-90 & B-4 394 [01880]  5-chloro-1'-[2-({2-oxo-3-[(cis)-3-hydroxy-3-methylcyclobutyl]-2H,3H-[1,3]oxazolo[4,5-b]pyridin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 499.2 Found 499.2 A-91 & B-4

Example 59

5-chloro-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 275)
##STR01881##

[3036] To a solution of 6-(2-bromoethoxy)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (Intermediate A-89, 330 mg, 0.929 mmol) and 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 279 mg, 1.02 mmol, HCl salt) in MeCN (10 mL) was added NaHCO₃ (234 mg, 2.79 mmol). The mixture was stirred at 80° C. for 12 h. After cooling to room temperature, the mixture was diluted with H₂O (20 mL) and extracted with EtOAc (3×10 mL). The combined organic layers were washed with brine (20 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Waters Xbridge BEH C₁₈ column, 20-50% MeCN:10 mM NH₄HCO₃ in H₂O) to give 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 275). ¹H NMR (400 MHz, DMSO-d₆): δ 10.48 (s, 1H), 7.94 (d, J=2.8 Hz, 1H), 7.50 (d, J=2.0 Hz, 1H), 7.39 (d, J=2.8 Hz, 1H), 7.23 (dd, J=8.0 Hz, 2.0 Hz, 1H), 6.84 (d, J=8.4 Hz, 1H), 4.82 (s, 1H), 4.30-4.22 (m, 1H), 4.18 (t, J=5.6 Hz, 2H), 2.95-2.89 (m, 2H), 2.87-2.77 (m, 4H), 2.72-2.67 (m, 2H), 2.48-2.46 (m, 2H), 2.46-2.41 (m, 2H), 2.38-2.32 (m, 2H), 1.82-1.67 (m, 4H), 1.25 (s, 3H). MS=511.1 [M+H].sup.+.

Example 60

5-chloro-1'-(2-{[2-(3-methanesulfonyloxetan-3-yl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 276)
##STR01882##

[3037] To a solution of 5-(2-bromoethoxy)-2-(3-methanesulfonyloxetan-3-yl)pyrimidine (Intermediate A-88, 50.0 mg, 0.148 mmol) and 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 38.6 mg, 0.163 mmol) in MeCN (4 mL) was added NaHCO₃ (62.3 mg, 0.744 mmol). The mixture was stirred at 80° C. for 15 h. After cooling to room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Phenomenex C₁₈ column, 15-45% MeCN:10 mM NH₄HCO₃ in H₂O) to give 5-chloro-1'-(2-{[2-(3-methanesulfonyloxetan-3-yl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 276). ¹H NMR (400 MHz, DMSO-d₆): δ 10.49 (s, 1H), 8.71 (s, 2H), 7.50 (d, J=2.0 Hz, 1H), 7.25-7.22 (m, 1H), 6.84 (d, J=8.4 Hz, 1H), 5.20 (d, J=7.6 Hz, 2H), 5.09 (d, J=7.6 Hz, 2H), 4.38 (t,

J=5.6 Hz, 2H), 3.02 (s, 3H), 2.94-2.89 (m, 4H), 2.72-2.70 (m, 2H), 1.78-1.71 (m, 4H). MS=493.1 [M+H].sup.+.

Example 61

5-(difluoromethyl)-1'-{2-[4-(3-hydroxy-1-methanesulfonylcyclobutyl)phenoxy]ethyl}-1-methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 277), 5-(difluoromethyl)-1-methyl-1'-(2-{4-[(cis) or (trans)-3-hydroxy-1-methanesulfonylcyclobutyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 278), and 5-(difluoromethyl)-1-methyl-1'-(2-{4-[(trans) or (cis)-3-hydroxy-1-methanesulfonylcyclobutyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 279)

##STR01883##

Step 1: 5-(difluoromethyl)-1'-{2-[4-(3-hydroxy-1-methanesulfonylcyclobutyl)phenoxy]ethyl}-1-methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 277)

##STR01884##

[3038] To a solution of 2-[4-(3-hydroxy-1-methanesulfonylcyclobutyl)phenoxy]ethyl methanesulfonate (Example 54, Step 6, 100 mg, 0.274 mmol) and 5-(difluoromethyl)-1-methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-23, 100 mg, 0.329 mmol, HCl salt) in MeCN (4 mL) was added NaHCO₃ (92.0 mg, 1.10 mmol). The mixture was stirred at 80° C. for 12 h. After cooling to room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Phenomenex C₁₈ column, 20-50% MeCN:10 mM NH₄HCO₃ in H₂O) to give 5-(difluoromethyl)-1'-{2-[4-(3-hydroxy-1-methanesulfonylcyclobutyl)phenoxy]ethyl}-1-methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 277). MS=535.2 [M+H].sup.+.

Step 2: 5-(difluoromethyl)-1-methyl-1'-(2-{4-[(cis) or (trans)-3-hydroxy-1-methanesulfonylcyclobutyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 278) and 5-(difluoromethyl)-1-methyl-1'-(2-{4-[(trans) or (cis)-3-hydroxy-1-methanesulfonylcyclobutyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 279)

##STR01885##

[3039] 5-(difluoromethyl)-1'-{2-[4-(3-hydroxy-1-methanesulfonylcyclobutyl)phenoxy]ethyl}-1-methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one was purified by preparative chiral SFC (Daicel Chiralpak AD-3, 25-47% isopropanol with 0.1% NH₄OH in CO₂). The first eluting isomer of the title compound, Compound 278: ¹H NMR (400 MHz, DMSO-d₆): δ 7.71 (s, 1H), 7.53 (d, J=7.6 Hz, 1H), 7.48 (d, J=8.8 Hz, 2H), 7.14 (d, J=8.0 Hz, 1H), 7.04-6.99 (m, 2H), 7.16-6.82 (m, 1H), 5.48 (d, J=6.4 Hz, 1H), 4.18 (s, 2H), 3.89-3.79 (m, 1H), 3.15 (s, 3H), 2.97 (s, 2H), 2.91-2.80 (m, 4H), 2.77-2.67 (m, 4H), 2.54 (s, 3H), 1.88-1.67 (m, 4H). MS=535.2 [M+H].sup.+.

The second eluting isomer of the title compound, Compound 279: ¹H NMR (400 MHz, DMSO-d₆): δ 7.71 (s, 1H), 7.53 (d, J=7.6 Hz, 1H), 7.24 (d, J=8.8 Hz, 2H), 7.14 (d, J=8.0 Hz, 1H), 7.01-6.98 (m, 2H), 7.16-6.84 (m, 1H), 5.33 (d, J=6.8 Hz, 1H), 4.39-4.30 (m, 1H), 4.17 (t, J=8.0 Hz, 2H), 3.29-3.23 (m, 2H), 3.15 (s, 3H), 3.00-2.93 (m, 2H), 2.90-2.84 (m, 2H), 2.75-2.68 (m, 2H), 2.64 (s, 3H), 2.44-2.37 (m, 2H), 1.86-1.68 (m, 4H). MS=535.2 [M+H].sup.+.

Example 62

5-(difluoromethyl)-1'-{2-[4-(3-hydroxy-1-methanesulfonylcyclobutyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 280), 5-(difluoromethyl)-1'-(2-{4-[(cis) or (trans)-3-hydroxy-1-methanesulfonylcyclobutyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 281), and 5-(difluoromethyl)-1'-(2-{4-[(trans) or (cis)-3-hydroxy-1-methanesulfonylcyclobutyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 282)

##STR01886##

Step 1: 5-(difluoromethyl)-1'-{2-[4-(3-hydroxy-1-methanesulfonylcyclobutyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 280)

##STR01887##

[3040] To a solution of 2-[4-(3-hydroxy-1-methanesulfonylcyclobutyl)phenoxy]ethyl methanesulfonate (Example 54, Step 6, 202 mg, 0.554 mmol) and 5-(difluoromethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-1, 160 mg, 554 μ mol, HCl salt) in MeCN (3 mL) was added NaHCO₃ (94.0 mg, 1.11 mmol). The mixture was stirred at 80° C. for 16 h. After cooling to room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Waters XBridge C₁₈ column, 20-50% MeCN:10 mM NH₄HCO₃ in H₂O) to give 5-(difluoromethyl)-1'-{2-[4-(3-hydroxy-1-methanesulfonylcyclobutyl)phenoxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 280). MS=521.2 [M+H]⁺.

Step 2: 5-(difluoromethyl)-1'-(2-{4-[(cis) or (trans)-3-hydroxy-1-methanesulfonylcyclobutyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 281), and

5-(difluoromethyl)-1'-(2-{4-[(trans) or (cis)-3-hydroxy-1-methanesulfonylcyclobutyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 282)

##STR01888##

[3041] 5-(Difluoromethyl)-1'-(2-{4-[(cis) or (trans)-3-hydroxy-1-methanesulfonylcyclobutyl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one was purified by preparative chiral SFC (Daicel Chiralpak AD-3, 40% isopropanol with 0.1% NH₄OH in CO₂). The first eluting isomer of the title compound, Compound 281: ¹H NMR (400 MHz, DMSO-d₆): 10.63 (s, 1H), 7.67 (s, 1H), 7.48 (d, J=8.4 Hz, 2H), 7.42 (d, J=8.0 Hz, 1H), 7.09-6.81 (m, 4H), 5.49 (d, J=6.4 Hz, 1H), 4.18 (t, J=4.8 Hz, 2H), 3.90-3.76 (m, 1H), 2.95 (s, 2H), 2.89-2.81 (m, 4H), 2.75-2.67 (m, 4H), 2.54 (s, 3H), 1.87-1.77 (m, 2H), 1.75-1.62 (m, 2H). MS=521.2 [M+H]⁺. The second eluting isomer of the title compound, Compound 282: ¹H NMR (400 MHz, DMSO-d₆): δ 10.63 (s, 1H), 7.67 (s, 1H), 7.42 (d, J=8.0 Hz, 1H), 7.24 (d, J=8.8 Hz, 2H), 7.10-6.82 (m, 4H), 5.33 (d, J=6.8 Hz, 1H), 4.38-4.28 (m, 1H), 4.16 (t, J=5.6 Hz, 2H), 3.32-3.23 (m, 2H), 2.99-2.90 (m, 2H), 2.86 (t, J=4.8 Hz, 2H), 2.70 (t, J=7.6 Hz, 2H), 2.64 (s, 3H), 2.45-2.37 (m, 2H), 1.87-1.78 (m, 2H), 1.70 (s, 2H). MS=521.2 [M+H]⁺.

Example 63

1'-(2-{[5-fluoro-6-(1-methanesulfonylcyclopropyl)pyridin-3-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (Compound 283)

##STR01889##

[3042] To a solution of 5-(2-bromoethoxy)-3-fluoro-2-(1-methanesulfonylcyclopropyl)pyridine (Intermediate A-87, 90.0 mg, 0.266 mmol) and 2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (Intermediate B-9, 78.0 mg, 0.293 mmol, HCl salt) in MeCN (1.5 mL) was added NaHCO₃ (112 mg, 1.33 mmol). The mixture was stirred at 80° C. for 15 h. After cooling to room temperature, solids were removed by filtration and the filtrate was purified by normal phase preparative TLC (SiO₂, EtOAc:MeOH=10:1) to give 1'-(2-{[5-fluoro-6-(1-methanesulfonylcyclopropyl)pyridin-3-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile (Compound 283). ¹H NMR (400 MHz, DMSO-d₆): δ 10.87 (s, 1H), 8.24 (d, J=2.0 Hz, 1H), 7.96 (s, 1H), 7.68-7.54 (m, 2H), 7.00 (d, J=8.0 Hz, 1H), 4.28 (s, 2H), 2.95 (s, 3H), 2.93-2.83 (m, 4H), 2.74 (d, J=6.8 Hz, 2H), 1.84-1.69 (m, 6H), 1.39-1.36 (m, 2H). MS=485.1 [M+H]⁺.

Example 64

5-chloro-1'-(2-{[2-(1-hydroxyethyl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 284), 5-chloro-1'-[2-({2-[(1S) or (1R)-1-hydroxyethyl]pyrimidin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 285) and 5-chloro-1'-[2-({2-[(1R) or (1S)-1-hydroxyethyl]pyrimidin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 286)

##STR01890##

Step 1: 5-chloro-1'-(2-{[2-(1-hydroxyethyl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR01891##

[3043] A mixture of 1-[5-(2-bromoethoxy)pyrimidin-2-yl]ethan-1-ol (Intermediate A-86, 90.0 mg, 0.364 mmol), 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 91.0 mg, 0.382 mmol, HCl salt), and NaHCO₃ (153 mg, 1.82 mmol) in MeCN (5 mL) under N₂ atmosphere was stirred at 80° C. for 15 h. After cooling to room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 4 g cartridge, 0-20% MeOH:EtOAc) to give 5-chloro-1'-(2-{[2-(1-hydroxyethyl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 284). ¹H NMR (400 MHz, DMSO-d₆): δ 10.49 (s, 1H), 8.55 (s, 2H), 7.50 (d, J=2.0 Hz, 1H), 7.23 (dd, J=8.0 Hz, 2.0 Hz, 1H), 6.84 (d, J=8.0 Hz, 1H), 5.10 (d, J=5.6 Hz, 1H), 4.78-4.72 (m, 1H), 4.30 (t, J=5.6 Hz, 2H), 2.92-2.65 (m, 4H), 2.71-2.69 (m, 2H), 1.78-1.70 (m, 4H), 1.38 (d, J=6.8 Hz, 3H). MS=403.2 [M+H]⁺.

Step 2: 5-chloro-1'-[2-({2-[(1S) or (1R)-1-hydroxyethyl]pyrimidin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 285) and 5-chloro-1'-[2-({2-[(1R) or (1S)-1-hydroxyethyl]pyrimidin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 286)

##STR01892##

[3044] 5-chloro-1'-(2-{[2-(1-hydroxyethyl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (100 mg, 0.248 mmol) was purified by preparative chiral SFC (Daicel Chiralpak IC-3, 55% MeOH with 0.1% NH₄OH in CO₂). The first eluting enantiomer of the title compound, Compound 285: ¹H NMR (400 MHz, DMSO-d₆): δ 10.48 (s, 1H), 8.55 (s, 2H), 7.50 (d, J=2.4 Hz, 1H), 7.23 (dd, J=8.4 Hz, 2.0 Hz, 1H), 6.84 (d, J=8.4 Hz, 1H), 5.10 (d, J=5.6 Hz, 1H), 4.80-4.71 (m, 1H), 4.30 (t, J=5.6 Hz, 2H), 2.95-2.85 (m, 4H), 2.72-2.66 (m, 2H), 1.82-1.75 (m, 2H), 1.73-1.66 (m, 2H), 1.39 (d, J=6.4 Hz, 3H). MS=403.2 [M+H]⁺. The second eluting enantiomer of the title compound, Compound 286: ¹H NMR (400 MHz, DMSO-d₆): δ 10.48 (s, 1H), 8.55 (d, J=2.4 Hz, 2H), 7.50 (s, 1H), 7.23 (d, J=8.4 Hz, 1H), 6.84 (dd, J=8.4 Hz, 2.0 Hz, 1H), 5.10 (dd, J=5.6 Hz, 2.0 Hz, 1H), 4.80-4.71 (m, 1H), 4.30 (t, J=4.0 Hz, 2H), 2.94-2.84 (m, 4H), 2.73-2.67 (m, 2H), 1.81-1.66 (m, 4H), 1.39 (d, J=6.8 Hz, 3H). MS=403.2 [M+H]⁺.

Example 65

5-chloro-1'-(2-{[8-(3-hydroxyazetidin-1-yl)-1,7-naphthyridin-3-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 314)

##STR01893##

Step 1: 1-(3-bromo-1,7-naphthyridin-8-yl)azetidin-3-ol

##STR01894##

[3045] A mixture of 3-bromo-8-chloro-1,7-naphthyridine (500 mg, 2.05 mmol), azetidin-3-ol (270 mg, 2.46 mmol), and diisopropylethylamine (663 mg, 5.13 mmol) in DMF (5 mL) was stirred at 50° C. for 16 h. After cooling to room temperature, the mixture was diluted with EtOAc (10 mL), and H₂O (20 mL) was then added. A precipitated solid was collected by filtration, and dried in vacuo to give 1-(3-bromo-1,7-naphthyridin-8-yl)azetidin-3-ol, which was used in the next step without further purification. MS=280.0/282.0 [M+H]⁺.

Step 2: [8-(3-hydroxyazetidin-1-yl)-1,7-naphthyridin-3-yl]boronic acid

##STR01895##

[3046] A mixture of 1-(3-bromo-1,7-naphthyridin-8-yl)azetidin-3-ol (250 mg, 0.892 mmol), bis(pinacolato)diboron (272 mg, 1.07 mmol), Pd(dppf)Cl₂·2Me₃CH₂Cl₂ (36.0 mg, 0.045 mmol) and KOAc (219 mg, 2.21 mmol) in 1,4-dioxane (15 mL) was purged with N₂ for 10 min, then stirred at 90° C. under N₂ for 16 h. The mixture was cooled to room temperature,

filtered over a celite pad, and then washed with EtOAc (30 mL). The filtrate was concentrated in vacuo to give [8-(3-hydroxyazetidin-1-yl)-1,7-naphthyridin-3-yl]boronic acid, which was used in the next step without further purification. MS=246.2 [M+H].sup.+.

Step 3: 8-(3-hydroxyazetidin-1-yl)-1,7-naphthyridin-3-ol

##STR01896##

[3047] [8-(3-Hydroxyazetidin-1-yl)-1,7-naphthyridin-3-yl]boronic acid (0.892 mmol) was dissolved in MeOH (5 mL) and cooled to 0° C. 30% H.sub.2O.sub.2 in H.sub.2O (0.228 mL, 2.23 mmol) was then added dropwise. The resulting mixture was stirred at room temperature for 3 h, then concentrated in vacuo to give 8-(3-hydroxyazetidin-1-yl)-1,7-naphthyridin-3-ol, which was used in the next step without further purification. MS=218.1 [M+H].sup.+.

Step 4: 1-(3-(2-bromoethoxy)-1,7-naphthyridin-8-yl)azetidin-3-ol

##STR01897##

[3048] A mixture of 8-(3-hydroxyazetidin-1-yl)-1,7-naphthyridin-3-ol (220 mg, 1.01 mmol) and Cs.sub.2CO.sub.3 (660 mg, 2.03 mmol) in 1,2-dibromoethane (3.50 mL, 40.5 mmol) and DMF (1 mL) was stirred at 90° C. for 3 h. The mixture was cooled to room temperature and filtered over a celite pad. The filtrate was concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 25 g cartridge, 0-10% MeOH/DCM) to give 1-[3-(2-bromoethoxy)-1,7-naphthyridin-8-yl]azetidin-3-ol. MS=324.1/326.1 [M+H].sup.+.

Step 5: 5-chloro-1'-(2-{[8-(3-hydroxyazetidin-1-yl)-1,7-naphthyridin-3-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 314)

##STR01898##

[3049] A mixture of 1-[3-(2-bromoethoxy)-1,7-naphthyridin-8-yl]azetidin-3-ol (45.0 mg, 0.139 mmol), 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 39.8 mg, 0.146 mmol, HCl salt), K.sub.2CO.sub.3 (48.0 mg, 0.347 mmol) and KI (12.0 mg, 0.069 mmol) in DMF (1.0 mL) was stirred at 60° C. for 16 h. The mixture was then cooled to room temperature and filtered over a celite pad. The filtrate was concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Phenomenex Kinetex C.sub.18 column, 5-50% MeCN in H.sub.2O with 0.1% NH.sub.4OH modifier) to give 5-chloro-1'-(2-{[8-(3-hydroxyazetidin-1-yl)-1,7-naphthyridin-3-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 314). .sup.1H NMR (500 MHz, DMSO-d.sub.6): δ 10.42 (s, 1H), 8.40 (d, J=2.9 Hz, 1H), 7.82 (d, J=5.6 Hz, 1H), 7.54 (d, J=2.9 Hz, 1H), 7.44 (d, J=2.2 Hz, 1H), 7.17 (dd, J=8.2 Hz, 2.1 Hz, 1H), 6.82 (d, J=5.6 Hz, 1H), 6.78 (d, J=8.3 Hz, 1H), 5.53 (d, J=5.6 Hz, 1H), 4.54-4.49 (m, 3H), 4.24 (t, J=5.7 Hz, 2H), 4.03-3.99 (m, 2H), 2.92-2.83 (m, 4H), 2.70-2.61 (m, 2H), 1.78-1.60 (m, 4H). MS=480.2 [M+H].sup.+.

[3050] The following compound in Table 38.2 was prepared according to the procedures similar to those described for Compound 314 using the appropriate starting materials.

TABLE-US-00043 TABLE 38.2 Exact Mass No. Structure IUPAC Name [M + H].sup.+ 306

[01899] 5-chloro-1'-(2-{[8-(3-hydroxy-3-methylazetidin-1-yl)-1,7-naphthyridin-3-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 494.2 Found 494.2 364 [01900] 5-chloro-1'-(2-{[4-(3-hydroxy-3-methylazetidin-1-yl)pyrido[3,2-d]pyrimidin-7-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 495.2 Found 495.2 366 [01901] 5-chloro-1'-(2-{[4-(3-hydroxyazetidin-1-yl)pyrido[3,2-d]pyrimidin-7-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 481.2 Found 481.2

Example 66

5-chloro-1'-(2-{[2-(2-hydroxypropan-2-yl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 305)

##STR01902##

Step 1: 2-(5-(2-bromoethoxy)pyrimidin-2-yl)propan-2-ol

##STR01903##

[3051] A mixture of 2-(2-hydroxypropan-2-yl)pyrimidin-5-ol (120 mg, 0.778 mmol) and Cs.sub.2CO.sub.3 (507 mg, 1.56 mmol) in 1,2-dibromoethane (2.7 mL, 31.5 mmol) and DMF (0.3 mL) was stirred at 90° C. for 3 h. The mixture was cooled to room temperature and filtered over a celite pad, and the filtrate was concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 25 g cartridge, 0-70% EtOAc/Hexane) to give 2-[5-(2-bromoethoxy)pyrimidin-2-yl]propan-2-ol. MS=261.1/263.1 [M+H].sup.+.

Step 2: 5-chloro-1'-(2-{[2-(2-hydroxypropan-2-yl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 305)

##STR01904##

[3052] A mixture of 2-[5-(2-bromoethoxy)pyrimidin-2-yl]propan-2-ol (51.0 mg, 0.195 mmol), 5-chloro-1H-spiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 58.7 mg, 0.215 mmol, HCl salt), K.sub.2CO.sub.3 (67.0 mg, 0.488 mmol), and KI (16 mg, 0.098 mmol) in DMF (2.0 mL) was stirred at 60° C. for 16 h. The mixture was cooled to room temperature and filtered over a celite pad. The filtrate was concentrated in vacuo, and the residue was purified by reverse phase preparative HPLC (5-40% MeCN in H.sub.2O with 0.1% formic acid modifier) to give 5-chloro-1'-(2-{[2-(2-hydroxypropan-2-yl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 305). .sup.1H NMR (500 MHz, DMSO-d.sub.6): δ 10.47 (s, 1H), 8.50 (s, 2H), 7.45-7.41 (m, 1H), 7.19 (dd, J=8.3 Hz, 2.1 Hz, 1H), 6.79 (d, J=8.3 Hz, 1H), 4.88 (s, 1H), 4.32-4.28 (m, 2H), 3.11-2.47 (m, 6H), 2.00-1.48 (m, 4H), 1.41 (s, 6H). MS=417.2 [M+H].sup.+.

Example 67

5-chloro-1'-(2-{[2-(3-hydroxyazetidin-1-yl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 301)

##STR01905##

Step 1: 5-(2-bromoethoxy)-2-chloropyrimidine

##STR01906##

[3053] To a solution of 2-chloropyrimidin-5-ol (1.00 g, 7.66 mmol) in DMF (7.7 mL) was added 1,2-dibromoethane (15.3 mL, 194 mmol) and Cs.sub.2CO.sub.3 (7.49 g, 23.0 mmol). The mixture was stirred at 90° C. for 16 h. The mixture was cooled to room temperature and filtered over a celite pad, and the filtrate was concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 50 g cartridge, 0-100% EtOAc/Hexanes) to give 5-(2-bromoethoxy)-2-chloropyrimidine as a yellow oil. MS=237.0 [M+H].sup.+.

Step 2: 5-chloro-1'-(2-{[2-(2-chloropyrimidin-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR01907##

[3054] To a vial was added 5-(2-bromoethoxy)-2-chloropyrimidine (1.23 g, 3.13 mmol), 5-chloro-1H-spiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 1.34 g, 4.92 mmol, HCl salt), K.sub.2CO.sub.3 (2.15 g, 15.5 mmol), KI (860 mg, 5.18 mmol), and DMF (17.3 mL). The mixture was stirred at 65° C. for 4 h. After cooling to room temperature, the mixture was diluted with H.sub.2O (100 mL). The resulting solids were isolated by filtration and dried in vacuo to give 5-chloro-1'-(2-{[2-(2-chloropyrimidin-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one, which was used in the subsequent step without further purification. MS=393.2 [M+H].sup.+.

Step 3: 5-chloro-1'-(2-{[2-(3-hydroxyazetidin-1-yl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 301)

##STR01908##

[3055] To a microwave vial was added azetidin-3-ol (63.0 mg, 0.575 mmol, HCl salt), 5-chloro-1'-(2-{[2-(2-chloropyrimidin-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (75.0 mg, 0.191 mmol), DMSO (0.38 mL), and TEA (0.058 g, 0.572 mmol). The reaction stirred at 100° C. for 1 h under microwave irradiation (normal absorption). The reaction was cooled to room temperature and diluted with H.sub.2O (1 mL) and MeCN (1 mL). The mixture was purified by reverse phase C.sub.18 chromatography (Biotage SFär C.sub.18 Duo cartridge, 5-50% MeCN in

H.sub.2O with 0.1% NH.sub.4OH modifier) to give 5-chloro-1'-(2-{[2-(3-hydroxyazetidin-1-yl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 301).
.sup.1H NMR (500 MHz, DMSO-d.sub.6): δ 10.48 (s, 1H), 8.21 (s, 2H), 7.50 (d, J=2.2 Hz, 1H), 7.23 (dd, J=8.3 Hz, 2.1 Hz, 1H), 6.84 (d, J=8.2 Hz, 1H), 5.61 (d, J=5.7 Hz, 1H), 4.56-4.49 (m, 1H), 4.18-4.08 (m, 4H), 3.74-3.68 (m, 2H), 2.93-2.86 (m, 2H), 2.80 (t, J=5.7 Hz, 2H), 2.70-2.62 (m, 2H), 1.80-1.66 (m, 4H). MS=430.2 [M+H].sup.+.

[3056] The following compounds in Table 38.4 were prepared according to the procedures similar to those described for Compound 301 using the appropriate starting materials.

TABLE-US-00044 TABLE 38.4 Exact Inter- Mass mediate No. Structure IUPAC Name [M + H].sup.+ Used 304 [01909]  6-[5-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)pyrimidin-2-yl]-2 λ .sup.6-thia-6-azaspiro[3.3]heptane-2,2-dione Calc'd 504.1 Found 504.2 B-4 303 [01910]  5-chloro-1'-(2-{[2-(dimethylamino)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 402.2 Found 402.2 B-4 302 [01911]  5-chloro-1'-(2-{[2-(3-methanesulfonylazetidin-1-yl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 492.1 Found 492.1 B-4 300 [01912]  5-chloro-1'-(2-{[2-(3-hydroxy-3-methylazetidin-1-yl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 444.2 Found 444.2 B-4 395 [01913]  5-(difluoromethyl)-1'-(2-{[2-(3-hydroxy-3-methylazetidin-1-yl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 460.2 Found 460.2 B-1 396 [01914]  5-chloro-1'-(2-{[2-(6-hydroxy-2-azaspiro[3.3]heptan-2-yl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 470.0 Found 470.2 B-4 397 [01915]  5-chloro-1'-(2-{[2-(3-ethyl-3-hydroxyazetidin-1-yl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 458.2 Found 458.2 B-4 Example 68

5-chloro-1'-(2-{[7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-7,8-dihydro-1,8-naphthyridin-3-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 323)

##STR01916##

Step 1: 6-bromo-1-(3-oxocyclobutyl)-1,2-dihydro-1,8-naphthyridin-2-one

##STR01917##

[3057] A mixture of 6-bromo-1-(3-oxocyclobutyl)-1,2-dihydro-1,8-naphthyridin-2-one (5.00 g, 22.2 mmol) and K.sub.2CO.sub.3 (4.61 g, 33.3 mmol) in DMF (50 mL) was stirred at 50° C. for 1 h under N.sub.2 atmosphere. 3-Bromocyclobutanone (4.63 g, 31.1 mmol) was added to the mixture, which was stirred for an additional 13 h. After cooling to 0° C., the reaction mixture was quenched with H.sub.2O (100 mL) and extracted with EtOAc (3 $\times$ 60 mL). The combined organic layers were washed with brine (3 $\times$ 60 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give 6-bromo-1-(3-oxocyclobutyl)-1,2-dihydro-1,8-naphthyridin-2-one, which was taken on to the subsequent step without further purification. MS=293.0/295.0 [M+H].sup.+.

Step 2: 6-bromo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1,2-dihydro-1,8-naphthyridin-2-one

##STR01918##

[3058] To a three-neck round-bottom flask equipped with a magnetic stir bar and thermometer was added 6-bromo-1-(3-oxocyclobutyl)-1,8-naphthyridin-2-one (6.30 g, 21.5 mmol) and THF (65 mL). The mixture was cooled to 0° C. and 3.0 M MeMgBr in 2-MeTHF (7.88 mL, 23.6 mmol) was added dropwise. After stirring at 0° C. for 2 h, the reaction mixture was quenched by the addition of saturated aqueous NH.sub.4Cl solution (100 mL) and extracted with EtOAc (3 $\times$ 50 mL). The combined organic layers were washed with brine (2 $\times$ 50 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 40 g cartridge, 0-100% EtOAc/Petroleum ether). The resulting crude product was triturated with 1:5 EtOAc/Petroleum ether to give 6-bromo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1,2-dihydro-1,8-naphthyridin-2-one. MS=309.1/311.1 [M+H].sup.+.

Step 3: 1-[(cis)-3-hydroxy-3-methylcyclobutyl]-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2-dihydro-1,8-naphthyridin-2-one

##STR01919##

[3059] To a solution of 6-bromo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1,2-dihydro-1,8-naphthyridin-2-one (3.48 g, 11.3 mmol) in 1,4-dioxane (40 mL) was added bis(pinacolato)diboron (4.29 g, 16.9 mmol), KOAc (2.76 g, 28.1 mmol) and Pd(dppf)Cl.sub.2 (412 mg, 0.563 mmol). The mixture was degassed and purged with N.sub.2 (3×), and then the mixture was stirred at 80° C. for 16 h under N.sub.2 atmosphere. After cooling to room temperature, the mixture was filtered, and the filtrate was concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 40 g cartridge, 0-100% EtOAc/Petroleum ether) to give 1-[(cis)-3-hydroxy-3-methylcyclobutyl]-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2-dihydro-1,8-naphthyridin-2-one.

Step 4: 6-hydroxy-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1,2-dihydro-1,8-naphthyridin-2-one

##STR01920##

[3060] To a solution of 1-[(cis)-3-hydroxy-3-methylcyclobutyl]-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2-dihydro-1,8-naphthyridin-2-one (3.10 g, 8.70 mmol) in THE (15 mL) and H.sub.2O (15 mL) was added Oxone (3.74 g, 6.09 mmol). The mixture was stirred at 0° C. for 2 h. The 0° C. reaction mixture was quenched by addition of saturated aqueous Na.sub.2SO.sub.3 (30 mL) and extracted with EtOAc (3×25 mL). The combined organic layers were washed with brine (2×50 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give 6-hydroxy-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1,2-dihydro-1,8-naphthyridin-2-one, which was used in the subsequent step without further purification. MS=247.2 [M+H].sup.+.

Step 5: 6-(2-bromoethoxy)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1,2-dihydro-1,8-naphthyridin-2-one

##STR01921##

[3061] To a solution of 6-hydroxy-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1,2-dihydro-1,8-naphthyridin-2-one (2.40 g, 9.75 mmol) in MeCN (25 mL) was added K.sub.2CO.sub.3 (5.39 g, 39.0 mmol) and 1,2-dibromoethane (29.4 mL, 390 mmol). The mixture was stirred at 65° C. for 16 h. After cooling to room temperature, the mixture was filtered, and the filtrate was concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 40 g cartridge, 0-100% EtOAc/Petroleum ether) to give 6-(2-bromoethoxy)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1,2-dihydro-1,8-naphthyridin-2-one. MS=353.1/355.1 [M+H].sup.+.

Step 6: 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-7,8-dihydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 323)

##STR01922##

[3062] To a solution of 6-(2-bromoethoxy)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1,2-dihydro-1,8-naphthyridin-2-one (1.00 g, 2.83 mmol) in MeCN (12 mL) was added NaHCO.sub.3 (951 mg, 11.3 mmol) and 5-chloro-1H-spiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 851 mg, 3.11 mmol, HCl salt). The mixture was stirred at 80° C. for 16 h. After cooling to room temperature, the mixture was filtered, and the filter cake was triturated with H.sub.2O (10 mL). The filter cake was dried in vacuo to give 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-7,8-dihydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 323). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.48 (s, 1H), 8.40 (d, J=3.2 Hz, 1H), 7.88-7.82 (m, 2H), 7.49 (d, J=2.0 Hz, 1H), 7.23 (dd, J=8.4 Hz, 2.0 Hz, 1H), 6.84 (d, J=8.4 Hz, 1H), 6.64 (d, J=9.6 Hz, 1H), 5.35-5.24 (m, 1H), 4.94 (s, 1H), 4.26 (t, J=5.6 Hz, 2H), 3.00-2.88 (m, 6H), 2.74-2.65 (m, 2H), 2.43-2.36 (m, 2H), 1.82-1.67 (m, 4H), 1.33 (s, 3H). MS=509.1 [M+H].sup.+.

[3063] The following compounds in Table 38.5 were prepared according to the procedures similar to those described for Compound 323 using the appropriate starting materials.

TABLE-US-00045 TABLE 38.5 Exact Mass Intermediate No. Structure IUPAC Name [M + H].sup.+ used 324 [01923]  5-(difluoromethyl)-1'-[2-({7-oxo-8-[(cis)-3-

hydroxy-3- methylcyclobutyl]-7,8- dihydro-1,8-naphthyridin- 3-yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one Calc'd 525.2 Found 525.2 B-1 325 [01924]
 embedded image 5-(difluoromethoxy)-1'-[2- ({7-oxo-8-[(cis)-3- hydroxy-3- methylcyclobutyl]-7,8- dihydro-1,8-naphthyridin- 3-yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one Calc'd 541.2 Found 541.3 B-2

Example 69

5-chloro-1'-[2-({7-oxo-8-[(cis)-3-ethyl-3-hydroxycyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 321), 5-chloro-1'-[12-({7-oxo-8-[(3-ethyl-3-hydroxycyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 318), and 5-chloro-1'-[2-({7-oxo-8-[(trans)-3-ethyl-3-hydroxycyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 290)

##STR01925##

Step 1: 6-bromo-1-(3-ethyl-3-hydroxycyclobutyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR01926##

[3064] To a 0° C. solution of 6-bromo-1-(3-oxocyclobutyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (Procedure for Intermediate A-79, Step 1, 5.00 g, 16.9 mmol) in THE (100 mL) under N.sub.2 atmosphere was added 3.0 M EtMgBr in 2-MeTHF (8.47 mL, 25.4 mmol). The mixture was stirred at room temperature for 1 h. The reaction mixture was quenched with saturated aqueous NH.sub.4Cl solution (80 mL) and extracted with EtOAc (3×50 mL). The combined organic layers were washed with brine (50 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-80% EtOAc/Petroleum ether) to give 6-bromo-1-(3-ethyl-3-hydroxycyclobutyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one. MS=325.1/327.1 [M+H].sup.+.

Step 2: 1-(3-ethyl-3-hydroxycyclobutyl)-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR01927##

[3065] To a solution of 6-bromo-1-(3-ethyl-3-hydroxycyclobutyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (1.40 g, 4.31 mmol) and bis(pinacolato)diboron (1.20 g, 4.74 mmol) in 1,4-dioxane (30 mL) was added KOAc (1.27 g, 12.9 mmol) and Pd(dppf)Cl.sub.2 (352 mg, 0.431 mmol). The mixture was purged with N.sub.2 (3×), and then the mixture was stirred at 80° C. for 16 h under N.sub.2 atmosphere. After cooling to room temperature, the reaction mixture was filtered, and the filtrate was concentrated in vacuo to give 1-(3-ethyl-3-hydroxycyclobutyl)-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one, which was used in the subsequent step without further purification. MS=373.2 [M+H].sup.+.

Step 3: 1-(3-ethyl-3-hydroxycyclobutyl)-6-hydroxy-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR01928##

[3066] To a 0° C. solution of 1-(3-ethyl-3-hydroxycyclobutyl)-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (1.60 g, 4.30 mmol) in acetone (40 mL) and H.sub.2O (5 mL) was added Oxone (2.91 g, 4.73 mmol), and then the mixture was stirred at room temperature for 1 h. The reaction mixture was filtered, and the filtrate was diluted with H.sub.2O (40 mL) and extracted with EtOAc (4×20 mL). The combined organic layers were washed with brine (20 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Isco 12 g cartridge, 0-100% EtOAc/Petroleum ether) to give 1-(3-ethyl-3-hydroxycyclobutyl)-6-hydroxy-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one. MS=263.2 [M+H].sup.+.

Step 4: 6-(2-bromoethoxy)-1-(3-ethyl-3-hydroxycyclobutyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR01929##

[3067] To a solution of 1-(3-ethyl-3-hydroxycyclobutyl)-6-hydroxy-1,2,3,4-tetrahydro-1,8-

naphthyridin-2-one (400 mg, 1.52 mmol) in MeCN (0.5 mL) and 1,2-dibromoethane (4.60 mL, 61.0 mmol) was added K₂CO₃ (422 mg, 3.05 mmol). The mixture was then stirred at 80° C. for 16 h. After cooling to room temperature, the reaction mixture was filtered, and the filtrate was concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 8 g cartridge, 0-100% EtOAc/Petroleum ether) to give 6-(2-bromoethoxy)-1-(3-ethyl-3-hydroxycyclobutyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one. MS=369.1/371.1 [M+H]⁺.
Step 5: 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-ethyl-3-hydroxycyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 321) and 5-chloro-1'-[2-({7-oxo-8-[(3-ethyl-3-hydroxycyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 318)

##STR01930##

[3068] To a solution of 6-(2-bromoethoxy)-1-(3-ethyl-3-hydroxycyclobutyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (250 mg, 0.677 mmol) and 5-chloro-1H-spiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 203 mg, 0.745 mmol, HCl salt) in MeCN (5 mL) was added NaHCO₃ (228 mg, 2.71 mmol). The mixture was stirred at 80° C. for 16 h. After cooling to room temperature, the reaction mixture was filtered, and the filtrate was concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Waters Xbridge OBD C₁₈ column, 25-55% MeCN:10 mM NH₄HCO₃ in H₂O). Baseline separation of the major (cis) and minor (trans) products was not obtained. The first eluting fractions of product provided 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-ethyl-3-hydroxycyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 321): ¹H NMR (400 MHz, DMSO-d₆): δ 10.48 (s, 1H), 7.94 (t, J=2.8 Hz, 1H), 7.50 (d, J=2.0 Hz, 1H), 7.40 (d, J=2.8 Hz, 1H), 7.25-7.22 (m, 1H), 6.84 (d, J=8.0 Hz, 1H), 4.66 (s, 1H), 4.22-4.16 (m, 3H), 2.92-2.78 (m, 6H), 2.69-2.65 (m, 2H), 2.50-2.49 (m, 4H), 2.27-2.26 (m, 2H), 1.78-1.72 (m, 4H), 1.51-1.49 (m, 2H), 0.87 (t, J=7.2 Hz, 3H). MS=525.2 [M+H]⁺. Remaining fractions were a mixture of cis and trans products, providing 5-chloro-1'-[2-({7-oxo-8-[(3-ethyl-3-hydroxycyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 318): ¹H NMR (400 MHz, DMSO-d₆): δ 10.48 (s, 1H), 7.95-7.94 (m, 1H), 7.51 (s, 1H), 7.50-7.38 (m, 1H), 7.25-7.22 (m, 1H), 6.84 (d, J=8.4 Hz, 1H), 5.39-5.28 (m, 0.3H), 4.67-4.61 (m, 1H), 4.24-4.16 (m, 2.6H), 2.92-2.78 (m, 6H), 2.69-2.54 (m, 6H), 2.27-2.12 (m, 2H), 1.78-1.72 (m, 4H), 1.51-1.49 (m, 2H), 0.88-0.81 (m, 3H). MS=525.2 [M+H]⁺.

Step 6: 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-ethyl-3-hydroxycyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 321) and 5-chloro-1'-[2-({7-oxo-8-[(trans)-3-ethyl-3-hydroxycyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 290)

##STR01931##

[3069] 5-chloro-1'-[2-({7-oxo-8-[(3-ethyl-3-hydroxycyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 318, 220 mg, 0.419 mmol) was separated by preparative chiral SFC (Daicel Chiralpak AD-3 column, 60% IPA with 0.1% NH₄OH in CO₂). The first eluting diastereomer of the title compound, 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-ethyl-3-hydroxycyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 321). The second eluting diastereomer of the title compound, 5-chloro-1'-[2-({7-oxo-8-[(trans)-3-ethyl-3-hydroxycyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 290): ¹H NMR (400 MHz, DMSO-d₆): δ 10.48 (s, 1H), 7.95 (d, J=2.8 Hz, 1H), 7.50 (s, 1H), 7.38 (d, J=2.4 Hz, 1H), 7.25-7.22 (m, 1H), 6.84 (d, J=8.0 Hz, 1H), 5.37-5.28 (m, 1H), 4.61 (s, 1H), 4.18 (t, J=5.2 Hz, 2H), 2.92-2.78 (m, 6H), 2.69-2.62 (m, 6H), 2.13-2.12 (m, 2H), 1.76-1.72 (m, 4H), 1.56-1.55 (m, 2H), 0.81 (t, J=7.2 Hz, 3H). MS=525.2 [M+H]⁺.

[3070] The following compounds in Table 38.6 were prepared according to the procedures similar to those described for Compounds 290, 318, & 321 using the appropriate starting materials.

TABLE-US-00046	TABLE 38.6	Chi-	Exact Elu-	ral Mass Com-	tion Col-	No.	Structure	IUPAC
Name	[M + H].sup.+	ments	Order umn	317	[01932]			
5-chloro-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-(propan-2-yl)cyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one	Calc'd 539.2	Found 539.3	Route only					
provided cis isomer	n/a	n/a	299	[01933]				
5-chloro-1'-[2-({8-(3-cyclopropyl-3-hydroxycyclobutyl)-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one	Calc'd 537.2	Found 537.4	Step 5 provided a mixture of isomers	n/a	n/a	398	[01934]	
5-chloro-1'-[2-({7-oxo-8-[(cis)-3-cyclopropyl-3-hydroxycyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one	Calc'd 537.2	Found 537.3	n/a	2nd Daicel Chiral-pak IF				
297	[01935]							
5-chloro-1'-[2-({7-oxo-8-[(trans)-3-cyclopropyl-3-hydroxycyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one	Calc'd 537.2	Found 537.1	n/a	1st Daicel Chiral-pak IF				
Example 70								

5-chloro-1'-[2-({8-[3-(hydroxymethyl)cyclobutyl]-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one and (Compound 326) 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-(hydroxymethyl)cyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 319)

##STR01936##

Step 1: (3-bromocyclobutyl)methanol

##STR01937##

[3071] To a 0° C. solution of methyl 3-bromocyclobutane-1-carboxylate (5.00 g, 25.9 mmol) in THE (50 mL) under N.sub.2 atmosphere was added 4.0 M LiBH.sub.4 in THF (9.71 mL, 38.8 mmol). The mixture was stirred at room temperature for 2 h. The mixture was diluted with saturated aqueous NH.sub.4Cl (150 mL) and extracted with EtOAc (3×100 mL), dried over Na.sub.2SO.sub.4, filtered and concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 20 g cartridge, 0-100% EtOAc/Petroleum ether) to give (3-bromocyclobutyl)methanol. .sup.1H NMR (400 MHz, CDCl.sub.3, 8/9 H): δ 4.60-4.52 (m, 1H) 3.66 (d, J=6.4 Hz, 2H) 2.79-2.76 (m, 1H) 2.61-2.53 (m, 4H)

Step 2: 6-bromo-1-[3-(hydroxymethyl)cyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR01938##

[3072] To a solution of 6-bromo-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (2.00 g, 8.81 mmol) in DMF (25 mL) was added Cs.sub.2CO.sub.3 (5.74 g, 17.6 mmol), NaI (2.64 g, 17.6 mmol) and (3-bromocyclobutyl)methanol (2.91 g, 17.6 mmol). The mixture was stirred at 130° C. for 16 h. After cooling to room temperature, the residue was diluted with H.sub.2O (50 mL), extracted with EtOAc (3×30 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 20 g cartridge, 0-100% EtOAc/Petroleum ether) to give 6-bromo-1-[3-(hydroxymethyl)cyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one. MS=311.0/313.0 [M+H].sup.+.

Step 3: 1-[3-(hydroxymethyl)cyclobutyl]-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR01939##

[3073] A mixture of 6-bromo-1-[3-(hydroxymethyl)cyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (900 mg, 2.89 mmol), KOAc (568 mg, 5.78 mmol), bis(pinacolato)diboron (808 mg, 3.18 mmol) and Pd(dppf)Cl.sub.2 (236 mg, 0.289 mmol) in 1,4-dioxane (10 mL) was degassed and purged with N.sub.2 (3×), and then stirred at 100° C. for 16 h under N.sub.2 atmosphere. After cooling to room temperature, the reaction mixture was filtered and concentrated under in vacuo to give 1-[3-(hydroxymethyl)cyclobutyl]-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-

2-yl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one, which was taken onto the subsequent step without further purification. MS=359.3 [M+H].sup.+.

Step 4: 6-hydroxy-1-[3-(hydroxymethyl)cyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one
##STR01940##

[3074] To a 0° C. solution of 1-[3-(hydroxymethyl)cyclobutyl]-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (800 mg, 1.67 mmol, 75% purity) in THE (10 mL) and H.sub.2O (3 mL) was added Oxone (2.06 g, 3.35 mmol). The mixture was stirred at room temperature for 1 h. The reaction was quenched with saturated aqueous Na.sub.2SO.sub.3 solution (10 mL) and extracted with EtOAc (3×10 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-100% EtOAc/Petroleum ether) to give 6-hydroxy-1-[3-(hydroxymethyl)cyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one. MS=249.0 [M+H].sup.+.

Step 5: 6-(2-bromoethoxy)-1-[3-(hydroxymethyl)cyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR01941##

[3075] To a solution of 6-hydroxy-1-[3-(hydroxymethyl)cyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (300 mg, 1.21 mmol) in MeCN (2 mL) was added K.sub.2CO.sub.3 (334 mg, 2.42 mmol) and 1,2-dibromoethane (2.73 mL, 36.3 mmol). The mixture was stirred at 80° C. for 4 h. After cooling to room temperature, the reaction mixture was concentrated in vacuo to remove MeCN. The residue was diluted with H.sub.2O (10 mL) and extracted with EtOAc (3×10 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 4 g cartridge, 0-100% EtOAc/Petroleum ether) to give 6-(2-bromoethoxy)-1-[3-(hydroxymethyl)cyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one. MS=354.9/356.9 [M+H].sup.+.

Step 6: 5-chloro-1'-[2-({8-[3-(hydroxymethyl)cyclobutyl]-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one and (Compound 326)

##STR01942##

[3076] To a solution of 6-(2-bromoethoxy)-1-[3-(hydroxymethyl)cyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (140 mg, 0.394 mmol) in MeCN (3 mL) was added NaHCO.sub.3 (66.2 mg, 0.788 mmol) and 5-chloro-1H-spiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 129 mg, 0.473 mmol, HCl salt). The mixture was stirred at 80° C. for 4 h. After cooling to room temperature, the reaction mixture was filtered and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Phenomenex C.sub.18 column, 25-45% MeCN with H.sub.2O with 10 mM NH.sub.4HCO.sub.3 modifier) to give 5-chloro-1'-[2-({8-[3-(hydroxymethyl)cyclobutyl]-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one and (Compound 326). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.49 (s, 1H), 7.99-7.90 (m, 1H), 7.50 (d, J=2.0 Hz, 1H), 7.38 (d, J=2.8 Hz, 1H), 7.23 (d, J=8.4 Hz, 1H), 6.84 (d, J=8.4 Hz, 1H), 4.82 (t, J=8.8 Hz, 1H), 4.55-4.28 (m, 1H), 4.18 (t, J=5.6 Hz, 2H), 3.41 (d, J=6.3 Hz, 2H), 2.92 (t, J=11.2 Hz, 2H), 2.86 (t, J=5.6 Hz, 2H), 2.80 (t, J=7.2 Hz, 2H), 2.73-2.67 (m, 2H), 2.48 (s, 2H), 2.36-2.24 (m, 4H), 2.15-2.02 (m, 1H), 1.83-1.67 (m, 4H). MS=511.1 [M+H].sup.+.

Step 7: 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-(hydroxymethyl)cyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 327)

##STR01943##

[3077] 5-chloro-1'-[2-({8-[3-(hydroxymethyl)cyclobutyl]-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one was separated by preparative chiral SFC (Daicel Chiralcel OX column, 6000 MeOH with 0.1% NH.sub.4OH in CO.sub.2) to remove trace trans impurity. The first eluting diastereomer of the title compound, 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-(hydroxymethyl)cyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 319): .sup.1H NMR (400

MHz, DMSO-d₆): δ 10.48 (s, 1H), 7.94 (d, J=2.8 Hz, 1H), 7.50 (s, 1H), 7.38 (d, J=2.4 Hz, 1H), 7.24 (d, J=8.4 Hz, 1H), 6.84 (d, J=8.4 Hz, 1H), 4.83 (t, J=8.8 Hz, 1H), 4.40 (t, J=5.2 Hz, 1H), 4.18 (t, J=5.6 Hz, 2H), 3.41 (t, J=5.6 Hz, 2H), 2.92 (t, J=7.2 Hz, 3H), 2.87-2.77 (m, 4H), 2.74-2.63 (m, 3H), 2.37-2.23 (m, 4H), 2.15-2.04 (m, 1H), 1.86-1.64 (m, 4H). MS=511.2 [M+H].sup.+.

[3078] The following compounds in Table 38.7 were prepared according to the procedures similar to those described for Compounds 326 & 319 using the appropriate starting materials.

TABLE-US-00047 TABLE 38.7 Chi- Exact Inter- ral Mass mediate Elution Col- No. Structure IUPAC Name [M + H].sup.+ Used Order umn 316 [01944]  5-(difluoromethyl)-1'-[2-({8-[3-(hydroxymethyl)cy-clobutyl]-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 527.2 Found 527.2 B-1 n/a n/a 309 [01945]  5-(difluoromethyl)-1'-[2-({7-oxo-8-[(cis) or (trans)-3-(hydroxymethyl)cy-clobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 527.2 Found 527.3 B-1 1st Daicel Chiral-pak IG 310 [01946]  5-(difluoromethyl)-1'-[2-({7-oxo-8-[(trans) or (cis)-3-(hydroxymethyl)cy-clobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 527.2 Found 527.3 B-1 2nd Daicel Chiral-pak IG 327 [01947]  5-chloro-1'-[2-({8-[(3-hydroxy-3-methylcyclobutyl)methyl]-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 525.2 Found 525.1 B-4 & D-1 n/a n/a 313 [01948]  5-chloro-1'-{2-[(7-oxo-8-[(cis) or (trans)-3-hydroxy-3-methylcyclobutyl]-methyl)-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy}ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 525.2 Found 525.3 B-4 & D-1 1st Chiral-pak IE-3 322 [01949]  5-chloro-1'-{2-[(7-oxo-8-[(trans) or (cis)-3-hydroxy-3-methylcyclobutyl]-methyl)-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy}ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 525.2 Found 525.3 B-4 & D-1 2nd Chial-pak IE-3 320 [01950]  5-(difluoromethyl)-1'-[2-({8-[(3-hydroxy-3-methylcyclobutyl)methyl]-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 541.3 Found 541.3 B-1 & D-1 n/a n/a 311 [01951]  5-(difluoromethyl)-1'-{2-[(7-oxo-8-[(cis) or (trans)-3-hydroxy-3-methylcyclobutyl]-methyl)-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy}ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 541.3 Found 541.1 B-1 & D-1 1st Chiral-pak IE-3 312 [01952]  5-(difluoromethyl)-1'-{2-[(7-oxo-8-[(trans) or (cis)-3-hydroxy-3-methylcyclobutyl]-methyl)-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy}ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 541.3 Found 541.1 B-1 & D-1 2nd Chiral-pak IE-3

Example 71

1'-(2-{{6-acetyl-5-(trifluoromethyl)pyridin-3-yl}oxy}ethyl)-5-(difluoromethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 295), 5-(difluoromethyl)-1'-(2-{{6-(1-hydroxyethyl)-5-(trifluoromethyl)pyridin-3-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 296), 5-(difluoromethyl)-1'-[2-({6-[(1R) or (1S)-1-hydroxyethyl]-5-(trifluoromethyl)pyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 288), and 5-(difluoromethyl)-1'-[2-({6-[(1S) or (1R)-1-hydroxyethyl]-5-(trifluoromethyl)pyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 287)

##STR01953##

Step 1: 2-chloro-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-3-(trifluoromethyl)pyridine

##STR01954##

[3079] To a solution of 5-bromo-2-chloro-3-(trifluoromethyl)pyridine (5.00 g, 19.2 mmol) in 1,4-dioxane (70 mL) was added bis(pinacolato)diboron (6.83 g, 26.88 mmol), KOAc (4.71 g, 48.0 mmol), and Pd(dppf)Cl.sub.2 (1.40 g, 1.92 mmol). The mixture was purged with N.sub.2 (3×), and the mixture was stirred at 95° C. for 16 h under N.sub.2 atmosphere. After cooling to room temperature, the reaction mixture was filtered, and the filter cake was washed with EtOAc. The

filtrate was concentrated in vacuo to give 2-chloro-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-3-(trifluoromethyl)pyridine, which was used in the subsequent step without further purification.
Step 2: 6-chloro-5-(trifluoromethyl)pyridin-3-ol

##STR01955##

[3080] To a solution of 2-chloro-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-3-(trifluoromethyl)pyridine (5.00 g, 16.3 mmol) in THE (45 mL) and H₂O (15 mL) was added Oxone (10.00 g, 16.3 mmol). The mixture was stirred at 0° C. for 1 h. The reaction mixture was quenched with saturated aqueous Na₂SO₃ solution (50 mL) and extracted with EtOAc (3×50 mL). The combined organic layers were washed with brine (40 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 80 g cartridge, 0-30% EtOAc/Petroleum ether) to give 6-chloro-5-(trifluoromethyl)pyridin-3-ol. MS=197.9 [M+H]⁺.

Step 3: 5-(2-bromoethoxy)-2-chloro-3-(trifluoromethyl)pyridine

##STR01956##

[3081] To a solution of 6-chloro-5-(trifluoromethyl)pyridin-3-ol (4.00 g, 20.3 mmol) in MeCN (20 mL) was added K₂CO₃ (14.0 g, 101 mmol) and 1,2-dibromoethane (15.3 mL, 202 mmol). The mixture was stirred at 70° C. for 10 h. After cooling to 0° C., the reaction mixture was quenched with H₂O (50 mL) and extracted with DCM (3×50 mL). The combined organic layers were washed with brine (80 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 80 g cartridge, 0-12% EtOAc/Petroleum ether) to give 5-(2-bromoethoxy)-2-chloro-3-(trifluoromethyl)pyridine. MS=303.8/305.8 [M+H]⁺.

Step 4: 5-(2-bromoethoxy)-2-(1-ethoxyethenyl)-3-(trifluoromethyl)pyridine

##STR01957##

[3082] To a solution of 5-(2-bromoethoxy)-2-chloro-3-(trifluoromethyl)pyridine (3.00 g, 9.85 mmol) in 1,4-dioxane (30 mL) was added tributyl(1-ethoxyvinyl)stannane (4.49 mL, 13.3 mmol) and Pd(PPh₃)₂Cl₂ (691 mg, 0.985 mmol). The mixture was degassed and purged with N₂ (3×) and stirred at 100° C. for 16 h. After cooling to room temperature, the mixture was quenched with saturated aqueous KF (50 mL) and stirred for 1 h. The mixture was filtered, and the filtrate was extracted with EtOAc (3×50 mL). The combined organic layers were dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 80 g cartridge, 0-30% EtOAc/Petroleum ether) to give 5-(2-bromoethoxy)-2-(1-ethoxyethenyl)-3-(trifluoromethyl)pyridine. MS=340.1/342.0 [M+H]⁺.

Step 5: 1-[5-(2-bromoethoxy)-3-(trifluoromethyl)pyridin-2-yl]ethan-1-one

##STR01958##

[3083] To a 0° C. solution of 5-(2-bromoethoxy)-2-(1-ethoxyethenyl)-3-(trifluoromethyl)pyridine (3.00 g, 8.82 mmol) in THE (30 mL) was added 3.0 M aqueous HCl (30 mL, 90 mmol). The mixture was stirred at room temperature for 4 h. The mixture was adjusted to pH=7-8 with saturated NaHCO₃ solution and extracted with EtOAc (3×30 mL). The combined organic layers were washed with brine (3×100 mL), dried over Na₂SO₄, and concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 40 g cartridge, 0-80% EtOAc/Petroleum ether) to give 1-[5-(2-bromoethoxy)-3-(trifluoromethyl)pyridin-2-yl]ethan-1-one. MS=312.0/314.0 [M+H]⁺.

Step 6: 1'-(2-{[6-acetyl-5-(trifluoromethyl)pyridin-3-yl]oxy}ethyl)-5-(difluoromethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 295)

##STR01959##

[3084] To a solution of 1-[5-(2-bromoethoxy)-3-(trifluoromethyl)pyridin-2-yl]ethan-1-one (500 mg, 1.60 mmol) and 5-(difluoromethyl)spiro[indoline-3,4'-piperidine]-2-one (Intermediate B-1, 509 mg, 1.76 mmol, HCl salt) in MeCN (10 mL) was added NaHCO₃ (673 mg, 8.01 mmol). The mixture was stirred at 80° C. for 16 h. After cooling to room temperature, the mixture was

filtered, and filtrate was concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Phenomenex Luna C.sub.18 column, 5-40% MeCN with H.sub.2O with 0.04% HCl modifier) to give 1'-(2-{[6-acetyl-5-(trifluoromethyl)pyridin-3-yl]oxy}ethyl)-5-(difluoromethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 295, HCl salt). MS=484.1 [M+H].sup.+.

Step 7: 5-(difluoromethyl)-1'-(2-{[6-(1-hydroxyethyl)-5-(trifluoromethyl)pyridin-3-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 296)

##STR01960##

[3085] To a 0° C. solution of 1'-(2-{[6-acetyl-5-(trifluoromethyl)pyridin-3-yl]oxy}ethyl)-5-(difluoromethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 295, 700 mg, 1.45 mmol) in THE (10 mL) was added NaBH.sub.4 (110 mg, 2.91 mmol). The mixture was stirred at room temperature for 3 h. The mixture was quenched with the dropwise addition of H.sub.2O (10 mL) and extracted with EtOAc (3×10 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Waters Xbridge OBD C.sub.18 column, 30-70% MeCN:10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 5-(difluoromethyl)-1'-(2-{[6-(1-hydroxyethyl)-5-(trifluoromethyl)pyridin-3-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 296). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.62 (s, 1H), 8.61-8.60 (m, 1H), 7.69-7.65 (m, 2H), 7.42 (d, J=8.0 Hz, 1H), 7.09-7.81 (m, 2H), 5.12 (d, J=6.8 Hz, 1H), 4.98-4.95 (m, 1H), 4.34 (t, J=5.6 Hz, 2H), 2.96-2.93 (m, 2H), 2.90-2.87 (m, 2H), 2.72-2.70 (m, 2H), 1.82-1.80 (m, 2H), 1.69-1.67 (m, 2H), 1.38 (d, J=6.4 Hz, 3H). MS=486.1 [M+H].sup.+.

Step 8: 5-(difluoromethyl)-1'-[2-({6-[(1R) or (1S)-1-hydroxyethyl]-5-(trifluoromethyl)pyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 288) and 5-(difluoromethyl)-1'-[2-({6-[(1S) or (1R)-1-hydroxyethyl]-5-(trifluoromethyl)pyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 287)

##STR01961##

[3086] 5-(difluoromethyl)-1'-(2-{[6-(1-hydroxyethyl)-5-(trifluoromethyl)pyridin-3-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 296) was separated by preparative chiral SFC (Daicel Chiralpak IG-3 column, 50% EtOH with 0.1% NH.sub.4OH in CO.sub.2). The first eluting enantiomer of the title compound, 5-(difluoromethyl)-1'-[2-({6-[(1R) or (1S)-1-hydroxyethyl]-5-(trifluoromethyl)pyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 288). The second eluting enantiomer of the title compound, 5-(difluoromethyl)-1'-[2-({6-[(1S) or (1R)-1-hydroxyethyl]-5-(trifluoromethyl)pyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 287).

[3087] 5-(difluoromethyl)-1'-[2-({6-[(1R) or (1S)-1-hydroxyethyl]-5-(trifluoromethyl)pyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 288): .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.62 (s, 1H), 8.61-8.60 (m, 1H), 7.68 (s, 1H), 7.65 (s, 1H), 7.43-7.41 (m, 1H), 7.09-6.81 (m, 2H), 5.12 (d, J=7.6 Hz, 1H), 4.96 (t, J=6.4 Hz, 1H), 4.34 (t, J=5.6 Hz, 2H), 2.95-2.93 (m, 2H), 2.90-2.87 (m, 2H), 2.72-2.70 (m, 2H), 1.82-1.80 (m, 2H), 1.69-1.67 (m, 2H), 1.38 (d, J=6.8 Hz, 3H). MS=486.1 [M+H].sup.+.

5-(difluoromethyl)-1'-[2-({6-[(1S) or (1R)-1-hydroxyethyl]-5-(trifluoromethyl)pyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 287): .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.62 (s, 1H), 8.61-8.60 (m, 1H), 7.68 (s, 1H), 7.65 (s, 1H), 7.43-7.41 (m, 1H), 6.97-6.81 (m, 2H), 5.12 (d, J=6.8 Hz, 1H), 4.96-4.95 (m, 1H), 4.34 (t, J=5.6 Hz, 2H), 2.95-2.93 (m, 2H), 2.90-2.87 (m, 2H), 2.72-2.70 (m, 2H), 1.82-1.80 (m, 2H), 1.69-1.67 (m, 2H), 1.38 (d, J=6.8 Hz, 3H). MS=486.1 [M+H].sup.+.

[3088] The following compounds in Table 38.8 were prepared according to the procedures similar to those described for Compounds 295, 296, 287, & 288 using the appropriate starting materials.

TABLE-US-00048 TABLE 38.8 Chi- Exact Inter- Elu- ral Mass mediate tion Col- No. Structure IUPAC Name [M + H].sup.+ Used Order umn 294 [01962]

(trifluoromethyl)py-ridin-3-yl}oxy}ethyl)-5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 468.1 Found 468.1 B-4 n/a n/a 399 [01963]  5-chloro-1'-(2-([6-(1-hydroxyethyl)-5-(trifluoromethyl)py-ridin-3-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 470.1 Found 470.2 B-4 n/a n/a 292 [01964]  5-chloro-1'-(2-([6-((1S) or (1R)-1-hydroxyethyl)-5-(trifluoromethyl)py-ridin-3-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 470.1 Found 470.1 B-4 1st Daicel Chiral-pak IG-3 291 [01965]  5-chloro-1'-(2-([6-((1R) or (1S)-1-hydroxyethyl)-5-(trifluoromethyl)py-ridin-3-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 470.1 Found 470.1 B-4 2nd Daicel Chiral-pak IG-3 298 [01966]  1'-(2-([6-acetyl-5-(trifluoromethyl)py-ridin-3-yl}oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile Calc'd 459.2 Found 459.2 B-9 n/a n/a 338 [01967]  1'-(2-([6-(1-hydroxyethyl)-5-(trifluoromethyl)py-ridin-3-yl}oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile Calc'd 461.2 Found 461.3 B-9 n/a n/a 376 [01968]  1'-(2-([6-((1S) or (1R)-1-hydroxyethyl)-5-(trifluoromethyl)py-ridin-3-yl}oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile Calc'd 461.2 Found 461.1 B-9 1st Daicel Chiral-pak IG-3 377 [01969]  1'-(2-([6-((1S) or (1R)-1-hydroxyethyl)-5-(trifluoromethyl)py-ridin-3-yl}oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile Calc'd 461.2 Found 461.1 B-9 2nd Daicel Chiral-pak IG-3

Example 72

5-chloro-1'-(2-([8-(3-hydroxy-3-methylcyclobutyl)-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 315), 5-chloro-1'-(2-([7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 275), and 5-chloro-1'-(2-([7-oxo-8-[(trans)-3-hydroxy-3-methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 307)

##STR01970##

Step 1: 3-(benzyloxy)-1-methylcyclobutan-1-ol

##STR01971##

[3089] To a -20°C . solution of 3.0 M MeMgBr in 2-MeTHF (94 mL, 282 mmol) in THE (1.0 L) under N.sub.2 atmosphere was added a solution of 3-(benzyloxy)cyclobutan-1-one (25.0 g, 142 mmol) in THE (200 mL) dropwise. The mixture was stirred at -20°C . for 30 min. The mixture was warmed to 0°C ., quenched with H.sub.2O (300 mL), and stirred at 0°C . for 15 min. The mixture was extracted with EtOAc (3 $\times$ 100 mL). The combined organic layers were washed with brine (300 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give 3-(benzyloxy)-1-methylcyclobutan-1-ol, which was used in the subsequent step without further purification. ¹H NMR (400 MHz, CDCl.sub.3): δ 7.36-7.31 (m, 5H), 4.45 (s, 3H), 3.75-3.72 (m, 1H), 2.48-2.43 (m, 2H), 2.13-2.09 (m, 2H), 1.32 (s, 3H).

Step 2: [3-(benzyloxy)-1-methylcyclobutoxy](tert-butyl)dimethylsilane

##STR01972##

[3090] To a solution of 3-benzyloxy-1-methyl-cyclobutanol (30 g, 156 mmol) in DCM (300 mL) was added imidazole (32.0 g, 468 mmol) and TBSCl (28.7 mL, 234 mmol). The mixture was stirred at room temperature for 16 h. The mixture was filtered, and filtrate was in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 330 g cartridge, 0-5% EtOAc/Petroleum ether) to give [3-(benzyloxy)-1-methylcyclobutoxy](tert-butyl)dimethylsilane. ¹H NMR (400 MHz, DMSO-d.sub.6): δ 7.34-7.29 (m, 5H), 4.42 (s, 2H), 3.70-3.65 (m, 1H), 2.42-2.38 (m, 2H), 2.19-2.17 (m, 2H), 1.30 (s, 3H), 0.90 (s, 9H), 0.09 (s, 6H).

Step 3: 3-[(tert-butyl)dimethylsilyl]oxy]3-methylcyclobutan-1-ol

##STR01973##

[3091] To a solution of [3-(benzyloxy)-1-methylcyclobutoxy](tert-butyl)dimethylsilane (49.0 g,

160 mmol) in MeOH (1.5 L) under N.sub.2 atmosphere was added Pd/C (20.0 g, 10 wt %, 18.9 mmol). The mixture was degassed and purged with H.sub.2 (3×). The mixture was stirred at 40° C. for 16 h under H.sub.2 (15 psi). After cooling to room temperature, the mixture was filtered, and filtrate was concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 330 g cartridge, 0-30% EtOAc/Petroleum ether) to give 3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutan-1-ol.

Step 4: 3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl methanesulfonate

##STR01974##

[3092] To a 0° C. solution of 3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutan-1-ol (10.0 g, 46.2 mmol) in DCM (100 mL) was added TEA (12.9 mL, 92.4 mmol). Methanesulfonic anhydride (12.1 g, 69.3 mmol) was added, and the mixture was stirred at room temperature for 16 h. The mixture was poured into ice water (100 mL) and extracted with DCM (3×100 mL). The combined organic layers were washed with brine (200 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to provide 3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl methanesulfonate, which was used in the subsequent step without further purification. .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 4.75-4.68 (m, 1H), 3.33 (s, 3H), 2.59-2.54 (m, 2H), 2.25-2.24 (m, 2H), 1.30 (s, 3H), 0.85 (s, 9H), 0.07 (s, 6H).

Step 5: 6-bromo-1-{3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl}-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR01975##

[3093] To a solution of 6-bromo-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (1.20 g, 5.29 mmol) and 3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl methanesulfonate (6.23 g, 21.1 mmol) in DMSO (100 mL) was added Cs.sub.2CO.sub.3 (5.17 g, 15.9 mmol). The mixture was stirred at 130° C. for 16 h. The mixture was poured into H.sub.2O (100 mL) and extracted with EtOAc (3×100 mL). The combined organic layers were washed with brine (200 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 120 g cartridge, 0-30% EtOAc/Petroleum ether) to give 6-bromo-1-{3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl}-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one. MS=425.1/427.1 [M+H].sup.+.

Step 6: 1-{3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl}-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR01976##

[3094] To a solution of 6-bromo-1-{3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl}-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (5.00 g, 11.8 mmol) in 1,4-dioxane (100 mL) was added AcOK (2.31 g, 23.5 mmol) and bis(pinacolato)diboron (3.58 g, 14.1 mmol). Pd(dppf)Cl.sub.2 (860 mg, 1.18 mmol) was added, and the mixture was degassed and purged with N.sub.2 (3×). The mixture was stirred at 80° C. for 16 h. After cooling to room temperature, the mixture was filtered, and the filtrate was concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 80 g cartridge, 0-40% EtOAc/Petroleum ether) to give 1-{3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl}-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one. MS=473.4 [M+H].sup.+.

Step 7: 1-{3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl}-6-hydroxy-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR01977##

[3095] To a 0° C. solution of 1-{3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl}-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (7.00 g, 14.8 mmol) in THE (90 mL) and H.sub.2O (30 mL) was added Oxone (13.7 g, 22.2 mmol). The mixture was stirred at room temperature for 3 h. The mixture was poured into ice water (100 mL) and extracted with EtOAc (3×100 mL). The combined organic layers were washed with brine (200 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was

purified by flash silica gel chromatography (Biotage 80 g cartridge, 0-40% EtOAc/Petroleum ether) to give 1-{3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl}-6-hydroxy-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one. MS=363.2 [M+H].sup.+.

Step 8: 6-(2-bromoethoxy)-1-{3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl}-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR01978##

[3096] To a solution of 1-{3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl}-6-hydroxy-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (1.30 g, 3.59 mmol) and 1,2-dibromoethane (8.12 mL, 108 mmol) in MeCN (1.0 mL) was added K.sub.2CO.sub.3 (1.49 g, 10.8 mmol). The mixture was stirred at 80° C. for 16 h. After cooling to room temperature, the mixture was filtered, and the filtrate was concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-80% EtOAc/Petroleum ether) to give 6-(2-bromoethoxy)-1-{3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl}-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one. MS=469.2/471.2 [M+H].sup.+.

Step 9: 6-(2-bromoethoxy)-1-(3-hydroxy-3-methylcyclobutyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one Br OH

##STR01979##

[3097] To a 0° C. solution of 6-(2-bromoethoxy)-1-{3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl}-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (1.00 g, 2.13 mmol) in THE (2 mL) was added 6.0 M HCl in H.sub.2O (8 mL, 48 mmol). The mixture was stirred at room temperature for 2 h, then was concentrated under reduced pressure. The crude residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-80% EtOAc/Petroleum ether) to give 6-(2-bromoethoxy)-1-(3-hydroxy-3-methylcyclobutyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one. MS=355.1/357.1 [M+H].sup.+.

Step 10: 5-chloro-1'-(2-{[8-(3-hydroxy-3-methylcyclobutyl)-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 315)

##STR01980##

[3098] To a solution of 6-(2-bromoethoxy)-1-(3-hydroxy-3-methylcyclobutyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (200 mg, 0.563 mmol) and 5-chloro-1H-spiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 231 mg, 0.845 mmol, HCl salt) in MeCN (5.0 mL) was added NaHCO.sub.3 (331 mg, 3.94 mmol). The mixture was stirred at 80° C. for 16 h. The mixture was filtered, and filtrate was concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Phenomenex Luna C.sub.18 column, 5-35% MeCN with H.sub.2O with 0.04% HCl modifier) to give 5-chloro-1'-(2-{[8-(3-hydroxy-3-methylcyclobutyl)-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 315, HCl salt). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 11.20-11.17 (m, 0.5H), 10.81-10.73 (m, 1H), 10.63 (s, 0.5H), 8.03-8.00 (m, 1.5H), 7.45 (s, 1H), 7.30-7.28 (m, 1H), 7.15 (s, 0.5H), 6.94-6.89 (m, 1H), 5.34-5.25 (m, 1H), 4.51 (s, 2H), 3.83 (s, 1H), 3.82-3.66 (m, 4H), 3.64 (s, 1H), 2.84-2.80 (m, 2H), 2.66-2.65 (m, 2H), 2.64-2.53 (m, 3H), 2.24-2.18 (m, 3H), 2.18-1.94 (m, 2H), 1.31 (s, 3H). MS=511.2 [M+H].sup.+.

Step 11: 5-chloro-1'-(2-{[7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 275) and 5-chloro-1'-(2-{[7-oxo-8-[(trans)-3-hydroxy-3-methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 307)

##STR01981##

[3099] 5-chloro-1'-(2-{[8-(3-hydroxy-3-methylcyclobutyl)-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 315) was separated by preparative chiral SFC (Daicel Chiralpak AD-3 column, 50% IPA with 0.1% NH.sub.4OH in CO.sub.2). The first eluting diastereomer of the title compound, 5-chloro-1'-(2-{[7-oxo-8-[(trans)-3-hydroxy-3-methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-

yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 307). The second eluting diastereomer of the title compound, 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 275, Example 59). 5-chloro-1'-[2-({7-oxo-8-[(trans)-3-hydroxy-3-methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 307): .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.48 (s, 1H), 7.96 (s, 1H), 7.50-7.49 (m, 1H), 7.38-7.37 (m, 1H), 7.25-7.22 (m, 1H), 6.85-6.83 (m, 1H), 5.32-5.27 (m, 1H), 4.78 (s, 1H), 4.19-4.17 (m, 2H), 2.92-2.78 (m, 2H), 2.69-2.67 (m, 4H), 2.66-2.52 (m, 4H), 2.40-2.20 (m, 2H), 2.19-2.18 (m, 2H), 1.78-1.72 (m, 4H), 1.30 (s, 3H). MS=511.2 [M+H].sup.+.

Example 73

5-chloro-1'-(2-{[2-(1-hydroxycyclopropyl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 308)

##STR01982##

Step 1: 5-bromo-2-[1-(ethoxymethoxy)ethenyl]pyrimidine

##STR01983##

[3100] To a three-neck round-bottom flask equipped with a magnetic stir bar and thermometer under N.sub.2 atmosphere was added 1-(5-bromopyrimidin-2-yl)ethan-1-one (5.00 g, 24.9 mmol) and THE (80 mL). The mixture was cooled to -78° C. and then 1.0 M LiHMDS in hexane (37.3 mL, 37.3 mmol) was added dropwise over 5 min. After stirring for 10 min, chloromethoxyethane (2.77 mL, 29.9 mmol) was added dropwise. The mixture was slowly warmed to room temperature and stirred for another 5 h. After cooling to 0° C., the reaction mixture was quenched with saturated aqueous NH.sub.4Cl (80 mL), and then extracted with EtOAc (3×50 mL). The combined organic layers were washed with brine (100 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 80 g cartridge, 0-20% EtOAc/Petroleum ether) to give 5-bromo-2-[1-(ethoxymethoxy)ethenyl]pyrimidine. MS=259.1/261.0 [M+H].sup.+.

Step 2: 5-bromo-2-[1-(ethoxymethoxy)cyclopropyl]pyrimidine

##STR01984##

[3101] To a solution of trimethylsulfoxonium iodide (4.08 g, 18.5 mmol) in DMSO (10 mL) was added t-BuOK (1.82 g, 16.2 mmol). The mixture was stirred at 50° C. for 30 min, and then a solution of 5-bromo-2-[1-(ethoxymethoxy)vinyl]pyrimidine (1.20 g, 4.63 mmol) in DMSO (10 mL) was added dropwise. The mixture was stirred at 50° C. for 30 min. After cooling to 0° C., the reaction mixture was quenched with H.sub.2O (20 mL), and then extracted with EtOAc (2×15 mL). The combined organic layers were washed with brine (2×30 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 20 g cartridge, 0-20% EtOAc/Petroleum ether) to give 5-bromo-2-[1-(ethoxymethoxy)cyclopropyl]pyrimidine, MS=273.1/275.1 [M+H].sup.+.

Step 3: 2-[1-(ethoxymethoxy)cyclopropyl]-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyrimidine

##STR01985##

[3102] A mixture of 5-bromo-2-[1-(ethoxymethoxy)cyclopropyl]pyrimidine (570 mg, 2.09 mmol), bis(pinacolato)diboron (795 mg, 3.13 mmol), KOAc (614 mg, 6.26 mmol), and Pd(dppf)Cl.sub.2 (76.4 mg, 0.104 mmol) in 1,4-dioxane (15 mL) was degassed and purged with N.sub.2 (3×), and then the mixture was stirred at 80° C. for 15 h under N.sub.2 atmosphere. After cooling to room temperature, the mixture was filtered and the filtrate was concentrated in vacuo to give 2-[1-(ethoxymethoxy)cyclopropyl]-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyrimidine, which was used in the subsequent step without further purification.

Step 4: 2-[1-(ethoxymethoxy)cyclopropyl]pyrimidin-5-ol

##STR01986##

[3103] To a solution of 2-[1-(ethoxymethoxy)cyclopropyl]-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyrimidine (700 mg, 2.19 mmol) in MeCN (15 mL) and H₂O (15 mL) was added Oxone (2.02 g, 3.28 mmol). The mixture was stirred for 2 h. The reaction mixture was quenched with saturated aqueous Na₂SO₃ (10 mL) at 0° C., and then extracted with EtOAc (3×15 mL). The combined organic layers were washed with brine (2×20 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-50% EtOAc/Petroleum ether) to give 2-[1-(ethoxymethoxy)cyclopropyl]pyrimidin-5-ol. MS=211.3 [M+H]⁺.

Step 5: 5-(2-bromoethoxy)-2-[1-(ethoxymethoxy)cyclopropyl]pyrimidine

##STR01987##

[3104] To a solution of 2-[1-(ethoxymethoxy)cyclopropyl]pyrimidin-5-ol (600 mg, 2.85 mmol) and 1,2-dibromoethane (8.61 mL, 114 mmol) in MeCN (10 mL) was added K₂CO₃ (1.97 g, 14.3 mmol). The mixture was stirred at 65° C. for 5 h. After cooling to room temperature, the mixture was filtered, and the filtrate was concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-35% EtOAc/Petroleum ether) to give 5-(2-bromoethoxy)-2-[1-(ethoxymethoxy)cyclopropyl]pyrimidine. MS=317.2/319.1 [M+H]⁺.

Step 6: 1-[5-(2-bromoethoxy)pyrimidin-2-yl]cyclopropan-1-ol

##STR01988##

[3105] To a solution of 5-(2-bromoethoxy)-2-[1-(ethoxymethoxy)cyclopropyl]pyrimidine (300 mg, 0.946 mmol) in DCM (15 mL) was added TFA (1.0 mL). The mixture was stirred for 1 h, then was concentrated in vacuo. The residue was cooled to 0° C. and quenched with H₂O (15 mL), and then adjusted to pH=8 with saturated aqueous NaHCO₃. The biphasic mixture was extracted with EtOAc (3×10 mL). The combined organic layers were washed with brine (2×20 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo to give 5-(2-bromoethoxy)-2-[1-(ethoxymethoxy)cyclopropyl]pyrimidine. MS=259.0/261.0 [M+H]⁺.

Step 7: 5-chloro-1'-(2-{[2-(1-hydroxycyclopropyl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 308)

##STR01989##

[3106] A mixture of 1-[5-(2-bromoethoxy)pyrimidin-2-yl]cyclopropanol (100 mg, 0.386 mmol), 5-chloro-1H-spiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 111 mg, 0.405 mmol, HCl salt), and NaHCO₃ (162 mg, 1.93 mmol) in MeCN (4 mL) was degassed and purged with N₂ (3×), and then the mixture was stirred at 80° C. for 15 h under N₂ atmosphere. After cooling to room temperature, the mixture was filtered, and the filtrate was concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Phenomenex Kinetex C₁₈ column, 15-45% MeCN with H₂O with 0.1% NH₄OH modifier) to give 5-chloro-1'-(2-{[2-(1-hydroxycyclopropyl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 308). ¹H NMR (400 MHz, DMSO-*d*₆): δ 10.48 (s, 1H), 8.49 (s, 2H), 7.50 (d, J=2.0 Hz, 1H), 7.23 (dd, J=8.4 Hz, 2.0 Hz, 1H), 6.84 (d, J=8.4 Hz, 1H), 5.90 (s, 1H), 4.28 (t, J=5.6 Hz, 2H), 2.93-2.85 (m, 4H), 2.71-2.69 (m, 2H), 1.78-1.72 (m, 4H), 1.21-1.18 (m, 2H), 1.09-1.06 (m, 2H). MS=415.1 [M+H]⁺.

Example 74

5-chloro-1'-(2-{[2-(7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 293)

##STR01990##

Step 1: tert-butyl 6-bromo-2-oxo-1,2,3,4-tetrahydro-1,8-naphthyridine-1-carboxylate

##STR01991##

[3107] To a solution of 6-bromo-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (10.0 g, 44.0 mmol) in DCM (70 mL) was added Boc₂O (12.1 mL, 52.9 mmol), DIEA (11.5 mL, 66.1 mmol) and DMAP (538 mg, 4.40 mmol). The mixture was stirred for 4 h, then solids were removed by filtration. The filtrate was diluted with H₂O (80 mL) and extracted with EtOAc (2×60 mL).

The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 80 g cartridge, 0-35% EtOAc/Petroleum ether) to give tert-butyl 6-bromo-2-oxo-1,2,3,4-tetrahydro-1,8-naphthyridine-1-carboxylate.

Step 2: tert-butyl 2-oxo-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2,3,4-tetrahydro-1,8-naphthyridine-1-carboxylate

##STR01992##

[3108] A mixture of tert-butyl 6-bromo-2-oxo-1,2,3,4-tetrahydro-1,8-naphthyridine-1-carboxylate (3.00 g, 9.17 mmol), bis(pinacolato)diboron (2.79 g, 11.0 mmol), KOAc (2.25 g, 22.9 mmol), and Pd(dppf)Cl.sub.2 (67.1 mg, 91.7 µmol) in 1,4-dioxane (30 mL) was degassed and purged with N.sub.2 (3×), and then the mixture was stirred at 90° C. for 12 h under N.sub.2 atmosphere. After cooling to room temperature, the mixture was filtered, and the filtrate was concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 25 g cartridge, 0-100% EtOAc/Petroleum ether) to give tert-butyl 2-oxo-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2,3,4-tetrahydro-1,8-naphthyridine-1-carboxylate. MS=375.3 [M+H].sup.+.

Step 3: tert-butyl 6-hydroxy-2-oxo-1,2,3,4-tetrahydro-1,8-naphthyridine-1-carboxylate

##STR01993##

[3109] To a solution of tert-butyl 2-oxo-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2,3,4-tetrahydro-1,8-naphthyridine-1-carboxylate (1.00 g, 2.67 mmol) in H.sub.2O (2 mL) and THE (10 mL) was added Oxone (1.15 g, 1.87 mmol). The mixture was stirred at 0° C. for 1 h, and then was quenched with saturated aqueous Na.sub.2SO.sub.3 (10 mL). The mixture was extracted with EtOAc (3×10 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give tert-butyl 6-hydroxy-2-oxo-1,2,3,4-tetrahydro-1,8-naphthyridine-1-carboxylate, which was used in the subsequent step without further purification. MS=265.1 [M+H].sup.+.

Step 4: tert-butyl 6-(2-bromoethoxy)-2-oxo-1,2,3,4-tetrahydro-1,8-naphthyridine-1-carboxylate

##STR01994##

[3110] To a solution of tert-butyl 6-hydroxy-2-oxo-1,2,3,4-tetrahydro-1,8-naphthyridine-1-carboxylate (900 mg, 3.41 mmol) in MeCN (7 mL) and 1,2-dibromoethane (16 mL) was added K.sub.2CO.sub.3 (2.35 g, 17.0 mmol). The mixture was stirred at 80° C. for 16 h. After cooling to room temperature, the reaction mixture was concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 4 g cartridge, 0-50% EtOAc/Petroleum ether) to give tert-butyl 6-(2-bromoethoxy)-2-oxo-1,2,3,4-tetrahydro-1,8-naphthyridine-1-carboxylate. MS=371.1/373.1[M+H].sup.+.

Step 5: tert-butyl 6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-oxo-1,2,3,4-tetrahydro-1,8-naphthyridine-1-carboxylate

##STR01995##

[3111] To a solution of 5-chloro-1H-spiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 515 mg, 1.89 mmol, HCl salt) in MeCN (10 mL) was added NaHCO.sub.3 (634 mg, 7.54 mmol) and tert-butyl 6-(2-bromoethoxy)-2-oxo-1,2,3,4-tetrahydro-1,8-naphthyridine-1-carboxylate (700 mg, 1.89 mmol). The mixture was stirred at 80° C. for 12 h. After cooling to room temperature, the reaction mixture was filtered and concentrated in vacuo. The residue was purified by preparative TLC (SiO.sub.2, 100% EtOAc) to give tert-butyl 6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-oxo-1,2,3,4-tetrahydro-1,8-naphthyridine-1-carboxylate. MS=527.0 [M+H].sup.+.

Step 6: 5-chloro-1'-{2-[(7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 293)

##STR01996##

[3112] A mixture of tert-butyl 6-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-2-oxo-1,2,3,4-tetrahydro-1,8-naphthyridine-1-carboxylate (330 mg, 0.626 mmol) in 4.0

M HCl in EtOAc (10 mL, 40 mmol) was stirred at room temperature for 2 h. The reaction mixture was concentrated under reduced pressure. The crude residue was purified by reverse phase preparative HPLC (Waters Xbridge BEH C.sub.18 column, 20-50% MeCN:10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 5-chloro-1'-{2-[(7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 293). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 9.67 (s, 1H), 9.46 (s, 1H), 7.02 (d, J=2.8 Hz, 1H), 6.69 (d, J=2.0 Hz, 1H), 6.53 (d, J=2.6 Hz, 1H), 6.42 (dd, J=8.0 Hz, 2.0 Hz, 1H), 6.03 (d, J=8.4 Hz, 1H), 3.33 (t, J=5.6 Hz, 2H), 1.98-2.17 (m, 6H), 1.87-1.82 (m, 2H), 1.66-1.57 (m, 2H), 0.99-0.90 (m, 4H). MS=427.2 [M+H].sup.+.

Example 75

5,7-dichloro-1'-(2-{[2-(1-methanesulfonylcyclopropyl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 400)

##STR01997##

[3113] To a solution of 5-chloro-1'-(2-{[2-(1-methanesulfonylcyclopropyl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 4, 100 mg, 0.21 mmol) and NCS (0.056 g, 0.419 mmol) in MeCN (0.5 mL) was added TMSCl (0.023 g, 0.21 mmol). The reaction was allowed to stir at room temperature overnight. The reaction was diluted with H.sub.2O (1 mL) and DMSO (1 mL), and then purified by reverse phase C18 chromatography (5-50% MeCN in H.sub.2O with 0.1% NH.sub.4OH modifier) to give 5,7-dichloro-1'-(2-{[2-(1-methanesulfonylcyclopropyl)pyrimidin-5-yl]oxy}ethyl)-1H-spiro[indole-3,4'-piperidin]-2-one (Compound 400). .sup.1H NMR (500 MHz, DMSO): δ 10.87 (s, 1H), 8.54 (s, 2H), 7.45 (s, 1H), 7.36 (s, 1H), 4.26 (t, J=5.6 Hz, 2H), 3.29 (s, 3H), 2.86-2.78 (m, 4H), 2.66-2.61 (m, 2H), 1.72-1.65 (m, 6H), 1.54-1.51 (m, 2H). MS=511.1 [M+H].sup.+.

Example 76

5-chloro-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 352)

##STR01998##

Step 1: (cis)-3-[5-(2-bromoethoxy)-7-(trifluoromethyl)-1H-1,3-benzodiazol-1-yl]-1-methylcyclobutan-1-ol

##STR01999##

[3114] To a solution of 1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-ol (Intermediate A-92, 610 mg, 2.13 mmol) in MeCN (12.2 mL) was added K.sub.2CO.sub.3 (1.47 g, 10.7 mmol) and 1,2-dibromoethane (16.0 g, 85.2 mmol). The mixture was stirred at 80° C. for 16 h. After cooling to room temperature, the reaction mixture was quenched by addition of H.sub.2O (50 mL) and extracted with EtOAc (3×50 mL). The combined organic layers were washed with brine (2×20 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-30% MeOH/EtOAc) to give (cis)-3-[5-(2-bromoethoxy)-7-(trifluoromethyl)-1H-1,3-benzodiazol-1-yl]-1-methylcyclobutan-1-ol. MS=393.0/395.0 [M+H].sup.+.

Alternative Conditions for Step 1: (cis)-3-[5-(2-bromoethoxy)-7-(trifluoromethyl)-1H-1,3-benzodiazol-1-yl]-1-methylcyclobutan-1-ol

##STR02000##

[3115] A solution of 1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-ol (Intermediate A-92, 1.00 g, 3.49 mmol), 1,2-dibromoethane (5.27 mL, 69.9 mmol) and Cs.sub.2CO.sub.3 (2.85 g, 8.73 mmol) in i-PrOH (5 mL) was stirred at 60° C. for 16 h. After cooling to room temperature, solids were removed by filtration and the filtrate was concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 20 g cartridge, 50-90% EtOAc/Petroleum ether) to give (cis)-3-[5-(2-bromoethoxy)-7-(trifluoromethyl)-1H-1,3-benzodiazol-1-yl]-1-methylcyclobutan-1-ol. MS=393.0/394.9

[M+H].sup.+.

Step 2: 5-chloro-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR02001##

[3116] To a solution of (cis)-3-[5-(2-bromoethoxy)-7-(trifluoromethyl)-1H-1,3-benzodiazol-1-yl]-1-methylcyclobutan-1-ol (420 mg, 1.07 mmol) in MeCN (7 mL) was added NaHCO₃ (449 mg, 5.34 mmol) and 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 321 mg, 1.17 mmol, HCl salt). The mixture was stirred at 80° C. for 12 h. After cooling to room temperature, the reaction mixture was quenched by addition of H₂O (20 mL) and extracted with EtOAc (3×40 mL). The combined organic layers were washed with brine (40 mL), dried over Na₂SO₄, filtered, and concentrated under reduced pressure. The residue was suspended in MeCN (10 mL) and the mixture was stirred at 80° C. for 1 h. The mixture was cooled to room temperature and filtered to collect the solid. The filter cake was dried in vacuo to give 5-chloro-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 352). ¹H NMR (400 MHz, DMSO-d₆): δ 10.48 (s, 1H), 8.67 (s, 1H), 7.60 (d, J=2.0 Hz, 1H), 7.49 (d, J=2.0 Hz, 1H), 7.26-7.21 (m, 2H), 6.83 (d, J=8.4 Hz, 1H), 5.30 (s, 1H), 4.57 (pent, J=8.0 Hz, 1H), 4.26-4.23 (m, 2H), 2.93-2.87 (m, 4H), 2.72-2.62 (m, 2H), 2.60-2.50 (m, 4H), 1.79-1.68 (m, 4H), 1.33 (s, 3H). MS=549.0 [M+H].sup.+.

Example 77

5-chloro-1'-[2-({3-oxo-4-[(cis)-3-hydroxy-3-methylcyclobutyl]-3H,4H-pyrido[2,3-b]pyrazin-7-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 358)

##STR02002##

Step 1: 7-(2-bromoethoxy)-4-[(cis)-3-hydroxy-3-methylcyclobutyl]-3H,4H-pyrido[2,3-b]pyrazin-3-one

##STR02003##

[3117] To a solution of 7-hydroxy-4-[(cis)-3-hydroxy-3-methylcyclobutyl]-3H,4H-pyrido[2,3-b]pyrazin-3-one (Intermediate A-94, 140 mg, 0.388 mmol, TFA salt) and 1,2-dibromoethane (1.17 mL, 15.5 mmol) in MeCN (4 mL) was added K₂CO₃ (321 mg, 2.33 mmol). The mixture was stirred at 65° C. for 12 h. After cooling to room temperature, the mixture was diluted with H₂O (30 mL) and extracted with EtOAc (2×50 mL). The combined organic layers were washed with brine (50 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by flash silica gel chromatography (Biotage 4 g cartridge, 0-72% EtOAc/Petroleum ether) to give 7-(2-bromoethoxy)-4-[(cis)-3-hydroxy-3-methylcyclobutyl]-3H,4H-pyrido[2,3-b]pyrazin-3-one. MS=354.2/356.1 [M+H].sup.+.

Step 2: 5-chloro-1'-[2-({3-oxo-4-[(cis)-3-hydroxy-3-methylcyclobutyl]-3H,4H-pyrido[2,3-b]pyrazin-7-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 358)

##STR02004##

[3118] To a solution of 5-chloro-1'-[2-({3-oxo-4-[(cis)-3-hydroxy-3-methylcyclobutyl]-3H,4H-pyrido[2,3-b]pyrazin-7-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (130 mg, 0.367 mmol) and 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 95.6 mg, 0.404 mmol, HCl salt) in MeCN (5 mL) was added NaHCO₃ (93.3 mg, 1.10 mmol). The mixture was stirred at 80° C. for 12 h. After cooling to room temperature, the mixture was diluted with H₂O (10 mL) and extracted with EtOAc (3×15 mL). The combined organic layers were washed with brine (15 mL), dried over Na₂SO₄, filtered, and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Waters Xbridge OBD C₁₈ column, 25-55% MeCN:10 mM NH₄HCO₃ in H₂O) to give 5-chloro-1'-[2-({3-oxo-4-[(cis)-3-hydroxy-3-methylcyclobutyl]-3H,4H-pyrido[2,3-b]pyrazin-7-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 358). ¹H NMR (400 MHz, DMSO-d₆): δ 10.54-10.45 (m, 1H), 8.44 (d, J=2.8 Hz, 1H), 8.23 (s, 1H), 7.93 (d, J=3.2 Hz, 1H), 7.50

(d, J=2.0 Hz, 1H), 7.23 (dd, J=8.4, 2.0 Hz, 1H), 6.84 (d, J=8.0 Hz, 1H), 5.21-5.11 (m, 1H), 5.02 (s, 1H), 4.34-4.29 (m, 2H), 3.02-2.96 (m, 2H), 2.95-2.91 (m, 2H), 2.91-2.88 (m, 2H), 2.74-2.68 (m, 2H), 2.44-2.40 (m, 2H), 1.81-1.75 (m, 2H), 1.74-1.67 (m, 2H), 1.33 (s, 3H). MS=510.1 [M+H].sup.+.

[3119] The following compounds in Table 38.9 were prepared according to procedures analogous to those described for Compound 358 using the appropriate starting materials or common intermediates.

TABLE-US-00049 TABLE 38.9 Chiral Exact Inter- separa- Chiral Mass mediate s tion elution No. Structure IUPAC Name [M + H].sup.+ Used column order 401 [02005]  5-chloro-1'-(2- {[6-(1-hydroxy-2- methanesulfonylpro- pan-2-yl)-5- (trifluoromethyl)py- ridin-3-yl]oxy}ethyl)-1,2- dihydrospiro[indole- 3,4'-piperidin]- 2-one Calc'd 562.1 Found 562.0 A-95 and B-4 n/a n/a 402 [02006]  5-chloro-1'-[2- ({6-[(2S or 2R)-1- hydroxy-2- methanesulfonylpro- pan-2-yl]-5- (trifluoromethyl)py- ridin-3- yl}oxy)ethyl]-1,2- dihydrospiro[indole- 3,4'-piperidin]- 2-one Calc'd 562.1 Found 561.9 A-95 and B-4 Daicel Chiralpak AD- 3 1st 403 [02007]  5-chloro-1'-[2- ({6-[(2R or 2S)-1- hydroxy-2- methanesulfonylpro- pan-2-yl]-5- (trifluoromethyl)py- ridin-3- yl}oxy)ethyl]-1,2- dihydrospiro[indole- 3,4'-piperidin]- 2-one Calc'd 562.1 Found 562.0 A-95 and B-4 Daicel Chiralpak AD- 3 2nd 344 [02008]  5-chloro-1'-[2- ({2-[1- (dimethylphospho- ryl)cyclopropyl]py- rimidin-5- yl}oxy)ethyl)-1,2- dihydrospiro[indole- 3,4'-piperidin]- 2-one Calc'd 475.2 Found 475.2 A-96 and B-4 n/a n/a 404 [02009]  2-oxo-1'-[2-({7- oxo-8-[(cis)-3- hydroxy-3- methylcyclobutyl]- 5,6,7,8- tetrahydro-1,8- naphthyridin-3- yl}oxy)ethyl)-1,2- dihydrospiro[indole- 3,4'- piperidine]-5- carbonitrile Calc'd 502.2 FOund 502.2 A-89 and B-9 n/a n/a 367 [02010]  5-chloro-1'-[2- ({3-[(cis)-3- hydroxy-3- methylcyclobutyl]- 3H-imidazo[4,5- b]pyridin-6- yl}oxy)ethyl)-1,2- dihydrospiro[indole- 3,4'- piperidin]- 2-one Calc'd 482.2 Found 482.2 A-98 and B-4 n/a n/a 405 [02011]  5-chloro-1'-(2- {[6-(3-hydroxy-3- methylazetidin-1- yl)-5- (trifluoromethyl)py- ridin-3- yl]oxy}ethyl)-1,2- dihydrospiro[indole- 3,4'-piperidin]- 2-one Calc'd 511.2 Found 511.2 A-99 and B-4 n/a n/a 406 [02012]  2-oxo-1'-[2-({7- oxo-8-[(cis)-3- hydroxy-3- methylcyclobutyl]- 7,8-dihydro-1,8- naphthyridin-3- yl}oxy)ethyl)-1,2- dihydrospiro[indole- 3,4'- piperidine]-5- carbonitrile Calc'd 500.2 Found 500.3 Example 68, Step 4 and B-9 n/a n/a 357 [02013]  5-chloro-1'-[2- ({1-[(cis)-3- hydroxy-3- methylcyclobutyl]- 1H- pyrazolo[3,4- b]pyridin-5- yl}oxy)ethyl)-1,2- dihydrospiro[indole- 3,4'-piperidin]- 2-one Calc'd 482.2 Found 482.3 A-116 and B-4 Daicel Chiralpak AD 2nd 407 [02014]  5-chloro-1'-(2- {[1-(3-hydroxy-3- methylcyclobutyl)- 1H-pyrazolo[3,4- b]pyridin-5- yl]oxy}ethyl)-1,2- dihydrospiro[indole- 3,4'-piperidin]- 2-one Calc'd 482.2 Found 482.1 A-116 and B-4 n/a n/a 408 [02015]  5-chloro-1'-[2- ({1-[(trans)-3- hydroxy-3- methylcyclobutyl]- 1H-pyrazolo[3,4- b]pyridin-5- yl}oxy)ethyl)-1,2- dihydrospiro[indole- 3,4'- piperidin]- 2-one Calc'd 482.2 Found 482.1 A-116 and B-4 Daicel Chiralpak AD 1st 334 [02016]  5-chloro-1'-(2- {[2-(1-hydroxy-2- methylpropan-2- yl)pyrimidin-5- yl]oxy}ethyl)-1,2- dihydrospiro[indole- 3,4'-piperidin]- 2-one Calc'd 431.2 Found 431.2 A-100 and B-4 n/a n/a 409 [02017]  5,7-dichloro-1'- [2-({7-oxo-8- [(cis)-3-hydroxy- 3- methylcyclobutyl]- 5,6,7,8- tetrahydro-1,8- naphthyridin-3- yl}oxy)ethyl)-1,2- dihydrospiro[indole- 3,4'-piperidin]- 2-one Calc'd 545.2 Found 545.2 General Procedure for Interme- diate A- 89, Step 3 and Example 3, Step 2 n/a n/a 333* [02018]  5-chloro-1'-[2- ({2-[1- (hydroxymethyl)cy- clopropyl]pyrimi- din-5- yl}oxy)ethyl)-1,2- dihydrospiro[indole- 3,4'- piperidin]- 2-one Calc'd 429.2 Found 429.1 A-101 and B-4 n/a n/a 343 [02019]  5-chloro-1'-(2-{4- [1- (dimethylphospho- ryl)cyclopropyl]phe- noxy}ethyl)- 1,2- dihydrospiro[indole- 3,4'-piperidin]- 2-one Calc'd 473.2 Found 473.3 A-102 and B-4 n/a n/a

*Prepared according to procedures analogous to Step 2 in procedure for synthesis of Compound 358

Example 78

5-chloro-1'-[2-({1-[(*cis*)-3-hydroxy-3-methylcyclobutyl]-1H-pyrrolo[2,3-*b*]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 349)

Step 1: [5-bromo-1-[(*cis*)-3-[(*tert*-butyldimethylsilyl)oxy]-3-methylcyclobutyl]-1H-pyrrolo[2,3-*b*]pyridine

##STR02020##

[3120] To a mixture of 5-bromo-1H-pyrrolo[2,3-*b*]pyridine (5.00 g, 25.4 mmol) and *tert*-butyldimethyl[(*cis*)-3-bromo-1-methylcyclobutoxy]silane (10.6 g, 38.1 mmol) in DMF (80 mL) was added Cs.sub.2CO.sub.3 (20.7 g, 63.4 mmol). The mixture was stirred at 90° C. for 30 h. After cooling to room temperature, the reaction mixture was diluted with H.sub.2O (200 mL) and extracted with EtOAc (3×50 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (Biotage 120 g cartridge, 0-10% EtOAc/Petroleum ether), and then further purified by reverse phase preparative HPLC (Phenomenex C.sub.18 column, 75-99% MeCN:10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give *tert*-butyldimethyl[(*cis*)-3-bromo-1-methylcyclobutoxy]silane. MS=395.0/397.0 [M+H].sup.+.

Step 2: 1-[(*cis*)-3-[(*tert*-butyldimethylsilyl)oxy]-3-methylcyclobutyl]-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1H-pyrrolo[2,3-*b*]pyridine

##STR02021##

[3121] A mixture of bis(pinacolato)diboron (1.41 g, 5.55 mmol), *tert*-butyldimethyl[(*cis*)-3-bromo-1-methylcyclobutoxy]silane (1.83 g, 4.63 mmol), KOAc (908 mg, 9.26 mmol) and Pd(dppf)Cl.sub.2 (169 mg, 231 μmol) in 1,4-dioxane (20 mL) was degassed and purged with N.sub.2 (3×), and then the mixture was stirred at 80° C. for 15 h under N.sub.2 atmosphere. After cooling to room temperature, the reaction mixture was filtered through a pad of celite and the filtrate was concentrated under reduced pressure to give 1-[(*cis*)-3-[(*tert*-butyldimethylsilyl)oxy]-3-methylcyclobutyl]-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1H-pyrrolo[2,3-*b*]pyridine, which was taken to the next step without further purification. MS=443.2 [M+H].sup.+.

Step 3: 1-[(*cis*)-3-[(*tert*-butyldimethylsilyl)oxy]-3-methylcyclobutyl]-1H-pyrrolo[2,3-*b*]pyridin-5-ol

##STR02022##

[3122] To a 0° C. solution of 1-[(*cis*)-3-[(*tert*-butyldimethylsilyl)oxy]-3-methylcyclobutyl]-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1H-pyrrolo[2,3-*b*]pyridine (2.05 g, 4.63 mmol) in acetone (20 mL) and H.sub.2O (20 mL) was added Oxone (1.42 g, 2.32 mmol). The mixture was stirred at room temperature for 2 h. The reaction mixture was diluted with H.sub.2O (50 mL) and extracted with EtOAc (3×40 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (Biotage 40 g cartridge, 0-50% EtOAc/Petroleum ether) to give 1-[(*cis*)-3-[(*tert*-butyldimethylsilyl)oxy]-3-methylcyclobutyl]-1H-pyrrolo[2,3-*b*]pyridin-5-ol. MS=333.1 [M+H].sup.+.

Step 4: 5-(2-bromoethoxy)-1-[(*cis*)-3-[(*tert*-butyldimethylsilyl)oxy]-3-methylcyclobutyl]-1H-pyrrolo[2,3-*b*]pyridine

##STR02023##

[3123] To a mixture of 1,2-dibromoethane (7.68 g, 40.9 mmol) and 1-[(*cis*)-3-[(*tert*-butyldimethylsilyl)oxy]-3-methylcyclobutyl]-1H-pyrrolo[2,3-*b*]pyridin-5-ol (340 mg, 1.02 mmol) in MeCN (5 mL) was added K.sub.2CO.sub.3 (707 mg, 5.11 mmol). The mixture was stirred at 85° C. for 63 h. After cooling to room temperature, the reaction mixture was diluted with H.sub.2O (10 mL) and extracted with DCM (3×10 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (Biotage 20 g cartridge, 0-100% EtOAc/Petroleum ether) to give 5-(2-bromoethoxy)-1-[(*cis*)-3-[(*tert*-butyldimethylsilyl)oxy]-3-methylcyclobutyl]-1H-pyrrolo[2,3-*b*]pyridine. MS=439.0/441.0 [M+H].sup.+.

Step 5: (cis)-3-[5-(2-bromoethoxy)-1H-pyrrolo[2,3-b]pyridin-1-yl]-1-methylcyclobutan-1-ol
##STR02024##

[3124] A solution of 5-(2-bromoethoxy)-1-[(cis)-3-[(tert-butyldimethylsilyl)oxy]-3-methylcyclobutyl]-1H-pyrrolo[2,3-b]pyridine (485 mg, 1.10 mmol) in 4.0 M HCl in EtOAc (5.0 mL, 20 mmol) was stirred at room temperature for 2 h. The reaction mixture was filtered and the filter cake was washed with Petroleum ether (3×1 mL) and dried under reduced pressure to give (cis)-3-[5-(2-bromoethoxy)-1H-pyrrolo[2,3-b]pyridin-1-yl]-1-methylcyclobutan-1-ol (HCl salt), which was taken to the next step without further purification. MS=325.0/327.0 [M+H].sup.+.

Step 6: 5-chloro-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-pyrrolo[2,3-b]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 349)

##STR02025##

[3125] To a mixture of (cis)-3-[5-(2-bromoethoxy)-1H-pyrrolo[2,3-b]pyridin-1-yl]-1-methylcyclobutan-1-ol (100 mg, 277 µmol, HCl salt) and 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 98.2 mg, 359 µmol, HCl salt) in MeCN (2 mL) was added NaHCO₃ (116 mg, 1.38 mmol). The mixture was stirred at 80° C. for 15 h. After cooling to room temperature, the reaction mixture was concentrated. The residue was purified by reverse phase preparative HPLC (Waters Xbridge BEH C₁₈ column, 30-60% MeCN:10 mM NH₄HCO₃ in H₂O) to give 5-chloro-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-pyrrolo[2,3-b]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 349). MS=481.0 [M+H].sup.+.

[3126] The following compounds in Table 38.10 were prepared according to procedures analogous to those described for Compound 349 using the appropriate starting materials or common intermediates.

TABLE-US-00050 TABLE 38.10 Exact Mass Intermediate No. Structure IUPAC Name [M + H].sup.+ Used 410 [02026]  2-oxo-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-pyrrolo[2,3-b]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile Calc'd 472.2 Found 472.2 B-9

Example 79

5-chloro-1'-[2-({2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H,2H,3H-pyrrolo[2,3-b]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 347)

##STR02027##

[3127] To a round-bottom flask equipped with a magnetic stir bar were added 5-chloro-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-pyrrolo[2,3-b]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 349, 64.0 mg, 133 µmol), t-BuOH (2 mL), H₂O (0.2 mL) and then Py.Math.Br.₃ (128 mg, 399 µmol). The mixture was stirred at room temperature for 3 h. The mixture was diluted with H₂O (3 mL) and extracted with EtOAc (3×5 mL). The combined organic layers were dried over anhydrous Na₂SO₄, filtered, and concentrated under reduced pressure. The residue was dissolved in AcOH (1 mL) and then Zinc (8.70 mg, 133 µmol) was added. The mixture was stirred at room temperature for 4 h. The reaction mixture was cooled to 0° C. and quenched with H₂O (10 mL), and then adjusted to pH=7 by addition of saturated aqueous NaHCO₃ solution. The mixture was extracted with EtOAc (3×5 mL). The combined organic layers were dried over Na₂SO₄, filtered, and concentrated under reduced pressure. The residue was purified by reverse phase preparative HPLC (Waters Xbridge Prep OBD C₁₈ column, 20-55% MeCN:10 mM NH₄HCO₃ in H₂O) to give 5-chloro-1'-[2-({2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H,2H,3H-pyrrolo[2,3-b]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 347). ¹H NMR (400 MHz, DMSO-d₆): δ 10.48 (s, 1H), 7.89 (d, J=2.8 Hz, 1H), 7.50 (d, J=2.0 Hz, 1H), 7.42 (d, J=2.8 Hz, 1H), 7.24 (dd, J=8.0, 2.0 Hz, 1H), 6.84 (d, J=8.0 Hz, 1H), 5.12 (s, 1H), 4.39-4.37 (m, 1H), 4.16 (t, J=5.6 Hz, 2H), 3.56 (s, 2H), 3.01-2.91 (m, 2H), 2.85-2.84 (m, 2H), 2.82-2.83 (m, 2H), 2.68-2.69 (m, 2H), 2.23-2.20 (m, 2H), 1.78-1.72 (m, 4H), 1.30 (s, 3H).

MS=497.2 [M+H].sup.+.

[3128] The following compounds in Table 38.11 were prepared according to procedures analogous to those described for Compound 347 using the appropriate starting materials or common intermediates.

TABLE-US-00051 TABLE 38.11 Exact Inter- Mass mediate # Structure IUPAC Name [M + H].sup.+ used 348 [02028]  5-(difluoromethyl)-1'-[2-({2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H,2H,3H-pyrrolo[2,3-b]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 513.2 Found 513.2 B-1

Example 80

5-chloro-1'-(2-{[6-(2-hydroxypropan-2-yl)-5-(trifluoromethyl)pyridin-3-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 371)

##STR02029##

Step 1: 2-[5-(2-bromoethoxy)-3-(trifluoromethyl)pyridin-2-yl]propan-2-ol

##STR02030##

[3129] To a solution of 1-[5-(2-bromoethoxy)-3-(trifluoromethyl)pyridin-2-yl]ethan-1-one (Example 71, Step 5, 100 mg, 320 μ mol) in THF (2 mL) at 0° C. was added 3.0 M MeMgBr in THE (160 μ L, 480 μ mmol). The mixture was stirred at 0° C. for 2 h, then was quenched by addition of H.sub.2O (1 mL) and extracted with EtOAc (3 $\times$ 1 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered and concentrated under reduced pressure to give 2-[5-(2-bromoethoxy)-3-(trifluoromethyl)pyridin-2-yl]propan-2-ol, which was taken to the next step without further purification. MS=328.0/330.0 [M+H].sup.+.

Step 2: 5-chloro-1'-(2-{[6-(2-hydroxypropan-2-yl)-5-(trifluoromethyl)pyridin-3-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR02031##

[3130] To a solution of 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 93.7 mg, 305 μ mol, HCl salt) in MeCN (5 mL) was added NaHCO.sub.3 (154 mg, 1.83 mmol) and 2-[5-(2-bromoethoxy)-3-(trifluoromethyl)pyridin-2-yl]propan-2-ol (100 mg, 305 μ mol). The mixture was stirred at 80° C. for 16 h. After cooling to room temperature, the mixture was concentrated in vacuo. The residue was purified by reverse phase preparative HPLC (Phenomenex Luna C.sub.18 column, 10-50% MeCN:10 mM TFA in H.sub.2O) to give 5-chloro-1'-(2-{[6-(2-hydroxypropan-2-yl)-5-(trifluoromethyl)pyridin-3-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 371). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.66-10.20 (m, 1H), 8.49 (d, J=2.4 Hz, 1H), 7.69 (d, J=2.4 Hz, 1H), 7.49 (d, J=2.0 Hz, 1H), 7.30-7.18 (m, 1H), 6.85 (d, J=8.4 Hz, 1H), 5.15 (br d, J=1.2 Hz, 1H), 4.31 (t, J=5.6 Hz, 2H), 3.05-2.82 (m, 4H), 2.77-2.64 (m, 2H), 1.86-1.62 (m, 4H), 1.51 (s, 6H). MS=484.3 [M+H].sup.+.

[3131] The following compounds in Table 38.12 were prepared according to procedures analogous to those described for Compound 371 using the appropriate starting materials or common intermediates.

TABLE-US-00052 TABLE 38.12 Exact Mass Intermediate # Structure IUPAC Name [M + H].sup.+ used 411 [02032]  1'-(2-{[6-(2-hydroxypropan-2-yl)-5-(trifluoromethyl)pyridin-3-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile Calc'd 475.2 Found 475.1 B-9

Example 81

5-chloro-1'-(2-[4-(1-methanesulfonylpiperidin-4-yl)phenoxy]ethyl)-1-(.SUP.2.H.SUB.3.)methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 412)

##STR02033##

Step 1: tert-butyl 5-chloro-1-(.SUP.2.H.SUB.3.)methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate

##STR02034##

[3132] To a three-neck round-bottom flask equipped with a thermometer under N.sub.2 atmosphere

were added tert-butyl 5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate (Intermediate B-3, 8.67 g, 25.8 mmol), CD.sub.3I (5.60 g, 38.6 mmol) and THE (100 mL). The mixture was cooled to 0° C. and sodium hydride (3.09 g, 60% in mineral oil, 77.3 mmol) was added in portions. The mixture was stirred at 0° C. for 3 h under N.sub.2 atmosphere. The reaction mixture was maintained at 0-5° C. and quenched by addition of saturated aqueous NH.sub.4Cl (100 mL), and then extracted with EtOAc (3×150 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 120 g cartridge, 0-40% EtOAc/Petroleum ether) to give tert-butyl 5-chloro-1-(.sup.2H.sub.3)methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate. MS=297.9 [M-C.sub.4H.sub.8+H].sup.+.

Step 2: 5-chloro-1-(trideuteriomethyl)spiro[indoline-3,4'-piperidine]-2-one

##STR02035##

[3133] To a mixture of tert-butyl 5-chloro-1-(.sup.2H.sub.3)methyl-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-1'-carboxylate (8.60 g, 24.3 mmol) in EtOAc (10 mL) was added 4.0 M HCl in EtOAc (90 mL, 360 mmol). The mixture was stirred at room temperature for 2 h, then concentrated in vacuo. The residue was triturated with MTBE (45 mL) at room temperature for 1 h, and solids were collected by filtration. The filter cake was dried in vacuo to give 5-chloro-1-(.sup.2H.sub.3)methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (HCl salt), which was taken to the next step without further purification. MS=254.2 [M+H].sup.+.

Step 3: 5-chloro-1'-{2-[4-(1-methanesulfonylpiperidin-4-yl)phenoxy]ethyl}-1-(.SUP.2.H.SUB.3)methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR02036##

[3134] To a solution of 5-chloro-1-(.sup.2H.sub.3)methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (88.1 mg, 304 µmol, HCl salt) in MeCN (3 mL) was added NaHCO.sub.3 (92.8 mg, 1.10 mmol) and 4-[4-(2-bromoethoxy)phenyl]-1-methanesulfonylpiperidine (Intermediate A-104, 100 mg, 276 µmol). The mixture was stirred at 80° C. for 16 h. After cooling to room temperature, the reaction mixture was filtered to remove NaHCO.sub.3. The filtrate was purified by reverse phase preparative HPLC (Waters Xbridge Prep OBD C.sub.18 column, 35-65% MeCN:10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 5-chloro-1'-{2-[4-(1-methanesulfonylpiperidin-4-yl)phenoxy]ethyl}-1-(.sup.2H.sub.3)methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 412). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 7.56 (d, J=2.0 Hz, 1H), 7.35 (dd, J=8.0, 2.0 Hz, 1H), 7.18 (d, J=8.8 Hz, 2H), 7.03 (d, J=8.4 Hz, 1H), 6.90 (d, J=8.4 Hz, 2H), 4.11 (t, J=5.6 Hz, 2H), 3.66 (d, J=12.0 Hz, 2H), 2.97-2.90 (m, 2H), 2.89 (s, 3H), 2.84 (t, J=5.6 Hz, 2H), 2.82-2.75 (m, 2H), 2.74-2.67 (m, 2H), 2.60-2.54 (m, 1H), 1.86-1.70 (m, 6H), 1.68-1.57 (i, 2H). MS=535.3 [M+H].sup.+.

[3135] The following compounds in Table 38.13 were prepared according to procedures analogous to those described for Compound 412 using the appropriate starting materials or common intermediates.

TABLE-US-00053 TABLE 38.13 Exact Mass # Structure IUPAC Name [M + H].sup.+

Intermediate used 413 [02037]  5-chloro-1-(.sup.2H.sub.3)methyl-1'-(2-{4-[methyl(methylimino)oxo-λ.sup.6-sulfanyl]-3-(trifluoromethyl)phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 533.2 Found 533.3 A-103 414 [02038]

 5-chloro-1-(.sup.2H.sub.3)methyl-1'-(2-{4-[3-(propane-2-sulfonyl)oxetan-3-yl]phenoxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 536.2 Found 536.2 A-107 415 [02039]  5-chloro-1'-(2-[4-(1-methanesulfonylcyclobutyl)

phenoxy]ethyl)-1-(.sup.2H.sub.3)methyl-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 506.2 Found 506.3 A-40 416 [02040]  5-chloro-(.sup.2H.sub.3)methyl-1'-(2-[(2-methyl-1-oxo-2,3-dihydro-1H-isoindol-5-yl)oxy]ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 443.2 Found 443.2 A-53 417 [02041]  5-chloro-1'-(2-[[8-(2-hydroxy-2-methylpropyl)-7-oxo-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl]oxy]ethyl)-1-

(.sup.2H.sub.3)methyl-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one Calc'd 516.2 Found 516.3 A-106 418 [02042]  5-chloro-1'-[2-(4- methanesulfonylphenoxy) ethyl]-1-
(.sup.2H.sub.3)methyl-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one Calc'd 452.1 Found 452.2 A-3 419 [02043]  5-chloro-1-(.sup.2H.sub.3)methyl-1'-[2- ({7-oxo-8-[(1r,3s)-3-hydroxy- 3-methylcyclobutyl]-5,6,7,8- tetrahydro-1,8-naphthpyridin- 3-yl}oxy)ethyl]-1,2-
dihydrospiro[indole-3,4'- piperidin]-2-one Calc'd 528.2 Found 528.2 A-89 420 [02044]  5-chloro-1'-[2-({2-[1- (hydroxymethyl)cyclopropyl] pyrimidin-5-yl}oxy)ethyl]-1-
(.sup.2H.sub.3)methyl-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one Calc'd 446.2 Found 446.1 A-101 421 [02045]  6-{2-[5-chloro-1-(.sup.2H.sub.3)methyl-2-oxo-1,2- dihydrospiro[indole-3,4'- piperidin]-1'-yl]ethoxy}-8- fluoro-3,4-dihydro-2H-1λ.sup.6,2-
benzothiazine-1,1-dione Calc'd 497.1 Found 497.2 422 [02046]  5-chloro-1'-(2-{[2-(3- methanesulfonyloxetan-3- yl)pyrimidin-5-yl]oxy}ethyl)-
1-(.sup.2H.sub.3)methyl-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one Calc'd 510.2 Found 510.3 A-88 423 [02047]  5-chloro-1-(.sup.2H.sub.3)methyl-1'-[2- ({7-oxo-8-[(cis)-3-hydroxycyclobutyl]-5,6,7,8- tetrahydro-1,8-naphthyridin- 3-yl}oxy)ethyl]-1,2-
dihydrospiro[indole-3,4'- piperidin]-2-one Calc'd 514.2 Found 514.1 A-79 424 [02048]  5-chloro-1'-(2-{[2-(1- methanesulfonylcyclopropyl) pyrimidin-5-yl]oxy}ethyl)-1-
(.sup.2H.sub.3)methyl-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one Calc'd 494.2 Found 494.1 A-76 425 [02049]  5-chloro-1'-(2-{[2-(1- methanesulfonylethyl)pyrimidin- 5-yl]oxy}ethyl)-1-
(.sup.2H.sub.3)methyl-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one Calc'd 482.2 Found 482.1 A-75

Example 82

5-chloro-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)(1,1,2,2-.SUP.2.H.SUB.4.)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 426)

##STR02050##

Step 1: 6-[2-bromo(1,1,2,2-.SUP.2.H.SUB.4.)ethoxy]-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR02051##

[3136] A mixture of 6-hydroxy-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-3,4-dihydro-1,8-naphthyridin-2-one (General Procedure for Intermediate A-89, Step 3, 100 mg, 0.403 mmol) and Cs.sub.2CO.sub.3 (262 mg, 0.806 mmol) in dibromo(.sup.2H.sub.4)ethane (1.55 g, 8.05 mmol) and DMF (0.5 mL) was stirred at 90° C. for 3 h. The mixture was cooled to room temperature and filtered. The filtrate was concentrated under reduced pressure. The crude product was purified by flash silica gel chromatography (Biotage 10 g cartridge, 0-100% EtOAc/hexane) to give 6-[2-bromo(1,1,2,2-.sup.2H.sub.4)ethoxy]-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-3,4-dihydro-1,8-naphthyridin-2-one. MS=359.2/361.2 [M+H].sup.+.

Step 2: 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)(1,1,2,2-.SUP.2.H.SUB.4.)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR02052##

[3137] A mixture of 6-[2-bromo(1,1,2,2-.sup.2H.sub.4)ethoxy]-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-3,4-dihydro-1,8-naphthyridin-2-one (60.0 mg, 0.167 mmol), 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 50.2 mg, 0.184 mmol), K.sub.2CO.sub.3 (57.7 mg, 0.418 mmol) and KI (14 mg, 0.084 mmol) in DMF (1 mL) was stirred at 60° C. for 3 h. The mixture was cooled to room temperature and filtered. The filtrate was concentrated under reduced pressure. The crude product was purified by reverse phase silica gel chromatography (Biotage 60 g C.sub.18 cartridge, 5-50% MeCN/H.sub.2O with 0.1% NH.sub.4OH) to give 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-5,6-dihydro-1,8-naphthyridin-3-yl}oxy)(1,1,2,2-.sup.2H.sub.4)ethyl]-1H-spiro[indole-3,4'-piperidin]-2-one

(Compound 426). ¹H NMR (500 MHz, DMSO-d₆): δ 10.54 (s, 1H), 7.95 (d, J=2.9 Hz, 1H), 7.52 (s, 1H), 7.40 (d, J=2.9 Hz, 1H), 7.25 (dd, J=8.3, 2.1 Hz, 1H), 6.86 (d, J=8.2 Hz, 1H), 4.82 (s, 1H), 4.24 (pent, J=8.3 Hz, 1H), 3.20-2.67 (m, 4H), 2.54-2.50 (m, 4H), 2.48-2.40 (m, 2H), 2.38-2.30 (m, 2H), 2.00-1.61 (m, 4H), 1.25 (s, 3H). MS=515.2 [M+H].sup.+.

Example 83

5-chloro-1'-[2-({2-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-2H-indazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 427)

##STR02053##

Step 1: (cis)-3-[5-(2-bromoethoxy)-7-(trifluoromethyl)-2H-indazol-2-yl]-1-methylcyclobutan-1-ol

##STR02054##

[3138] A mixture of 2-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-2H-indazol-5-ol (Intermediate A-108, 365 mg, 1.27 mmol) and Cs₂CO₃ (831 mg, 2.55 mmol) in 1,2-dibromoethane (3.3 mL, 38.2 mmol) and DMF (0.5 mL) was stirred at 90° C. for 19 h. The mixture was cooled to room temperature and filtered. The filtrate was concentrated under reduced pressure. The crude product was purified by flash silica gel chromatography (Biotage 25 g cartridge, 0-100% EtOAc/hexane) to give (cis)-3-[5-(2-bromoethoxy)-7-(trifluoromethyl)-2H-indazol-2-yl]-1-methylcyclobutan-1-ol. MS=393.1/395.1 [M+H].sup.+.

Step 2: 5-chloro-1'-[2-({2-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-2H-indazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 427)

##STR02055##

[3139] A mixture of (cis)-3-[5-(2-bromoethoxy)-7-(trifluoromethyl)indazol-2-yl]-1-methylcyclobutan-1-ol (85 mg, 0.216 mmol), 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 65 mg, 0.238 mmol, HCl salt), K₂CO₃ (75 mg, 0.54 mmol) and KI (18 mg, 0.108 mmol) in DMF (1 mL) was stirred at 60° C. for 4 h. The mixture was cooled to room temperature, filtered, washed with EtOAc (3×20 mL). The filtrate was concentrated under reduced pressure. The crude residue was purified by reverse phase preparative HPLC (Phenomenex Kinetex C₁₈ column, 5-40% MeCN in H₂O with 0.1% formic acid) to give 5-chloro-1'-[2-({2-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-2H-indazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 427). ¹H NMR (500 MHz, DMSO-d₆): δ 10.47 (s, 1H), 8.43 (s, 1H), 7.45 (s, 1H), 7.38 (d, J=2.3 Hz, 1H), 7.27 (s, 1H), 7.18 (dd, J=8.3, 2.1 Hz, 1H), 6.79 (d, J=8.2 Hz, 1H), 5.27 (s, 1H), 4.80 (pent, J=8.3 Hz, 1H), 4.19-4.16 (m, 2H), 3.13-2.68 (m, 6H), 2.61-2.48 (m, 4H), 1.90-1.61 (m, 4H), 1.30 (s, 3H). MS=549.1 [M+H].sup.+.

[3140] The following compound in Table 38.14 was prepared according to the procedures analogous to those described for Compound 427 using the appropriate starting materials.

TABLE-US-00054 TABLE 38.14 Exact # Structure IUPAC Name Mass [M + H].sup.+

Intermediate used 428 [02056]  2-oxo-1'-[2-({2-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-2H-indazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile Calc'd 540.2 Found 540.2 A-108 and B-9

Example 84

5-chloro-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-5-(trifluoromethyl)-7,8-dihydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 341)

##STR02057##

Step 1: 6-(2-bromoethoxy)-1-((cis)-3-hydroxy-3-methylcyclobutyl)-4-(trifluoromethyl)-1,8-naphthyridin-2(1H)-one

##STR02058##

[3141] To a mixture of 6-hydroxy-1-((cis)-3-hydroxy-3-methylcyclobutyl)-4-(trifluoromethyl)-1,8-naphthyridin-2(1H)-one (Intermediate A-109, 400 mg, 1.27 mmol) and 1,2-dibromoethane (2.88 mL, 38.2 mmol) in MeCN (5 mL) was added K₂CO₃ (704 mg, 5.09 mmol). The mixture was stirred at 65° C. for 16 h. After cooling to room temperature, the reaction mixture was diluted

with H.sub.2O (20 mL) and extracted with EtOAc (2×20 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-15% EtOAc/Petroleum ether) to give 6-(2-bromoethoxy)-1-((cis)-3-hydroxy-3-methylcyclobutyl)-4-(trifluoromethyl)-1,8-naphthyridin-2(1H)-one. MS=420.9/422.8 [M+H].sup.+.

Step 2: 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-5-(trifluoromethyl)-7,8-dihydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one
##STR02059##

[3142] To a mixture of 6-(2-bromoethoxy)-1-((cis)-3-hydroxy-3-methylcyclobutyl)-4-(trifluoromethyl)-1,8-naphthyridin-2(1H)-one (120 mg, 285 µmol) and 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 77.82 mg, 285 µmol, HCl salt) in MeCN (3 mL) was added NaHCO.sub.3 (120 mg, 1.42 mmol). The mixture was stirred at 65° C. for 16 h. After cooling to room temperature, the reaction was diluted with H.sub.2O (10 mL) and extracted with EtOAc (2×10 mL). The combined organic layers were washed with brine (10 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Waters Xbridge OBD C.sub.18 column, 30-60% MeCN:10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-5-(trifluoromethyl)-7,8-dihydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 341). .sup.1H NMR (400 MHz, MeCN-d.sub.3): δ 8.48 (d, J=2.8 Hz, 1H), 8.39 (br s, 1H), 7.66-7.59 (m, 1H), 7.43 (d, J=2.0 Hz, 1H), 7.20 (dd, J=8.4, 2.0 Hz, 1H), 7.08 (s, 1H), 6.87 (d, J=8.4 Hz, 1H), 5.77-5.65 (m, 1H), 4.28 (t, J=5.6 Hz, 2H), 3.87 (s, 1H), 3.08-2.86 (m, 6H), 2.80-2.67 (m, 2H), 2.61-2.49 (m, 2H), 1.91-1.81 (m, 2H), 1.79-1.69 (m, 2H), 1.37 (s, 3H). MS=577.3 [M+H].sup.+.

[3143] The following compounds in Table 38.15 were prepared according to procedures analogous to those described for Compound 341 using the appropriate starting materials or common intermediates.

TABLE-US-00055 TABLE 38.15 Exact Inter- Chiral Chiral Mass mediates separation elution #
Structure IUPAC Name [M + H].sup.+ Used column order 342 [02060]  5-(difluoromethyl)-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-5-(trifluoromethyl)-7,8-dihydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 593.2 Found 593.3 A-109 and B-1 n/a n/a 354 [02061]  2-oxo-1'-[2-({2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-8-(trifluoromethyl)-1,2,3,4-tetrahydroquinolin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile Calc'd 569.2 Found 569.2 A-110 and B-9 n/a n/a 353 [02062]  5-chloro-1'-[2-({2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-8-(trifluoromethyl)-1,2,3,4-tetrahydroquinolin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 578.2 Found 578.2 A-110 and B-4 n/a n/a 339 [02063]  5-chloro-1'-(2-{{5-(difluoromethyl)-7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-7,8-dihydro-1,8-naphthyridin-3-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 559.2 Found 559.4 A-111 and B-4 n/a n/a 340 [02064]  5-(difluoromethyl)-1'-(2-{{5-(difluoromethyl)-7-oxo-8-[(cis)-3-hydroxy-3-methylcyclobutyl]-7,8-dihydro-1,8-naphthyridin-3-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 575.2 Found 575.4 A-111 and B-1 n/a n/a 429 [02065]  5-chloro-1'-(2-{{2-(3-hydroxyoxetan-3-yl)pyrimidin-5-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 431.1 Found 431.2 A-112 and B-4 n/a n/a 430 [02066]  5-chloro-1'-(2-{{2-(1-hydroxycyclobutyl)pyrimidin-5-yl}oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 429.1 Found 429.1 A-113 and B-4 n/a n/a 431 [02067]  1'-(2-{{6-(3-hydroxy-3-methylazetidin-1-yl)-5-(trifluoromethyl)pyridin-3-yl}oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile Calc'd 502.2 Found 502.3 A-99 and B-9 n/a n/a 432 [02068]  1'-(2-{{2-(1-hydroxy-2-methylpropan-2-yl)pyrimidin-5-yl}oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile

Calc'd 422.2 Found 422.2 A-100 and B-9 n/a n/a 433 [02069]  1'-(2-((1-((cis)-3-hydroxy-3-methylcyclobutyl)-7-(trifluoromethyl)-1H-benzo[d]imidazol-5-yl)oxy)ethyl)-2-oxospiro[indoline-3,4'-piperidine]-5-carbonitrile Calc'd 540.2 Found 540.1 A-92 and B-9 n/a n/a 434 [02070]  5-chloro-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,2,3-benzotriazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 550.2 Found 550.0 A-114 and B-4 n/a n/a 435 [02071]  2-oxo-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,2,3-benzotriazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile Calc'd 541.2 Found 541.0 A-114 and B-9 n/a n/a 373 [02072]  5-chloro-1'-(2-{[2-(3-hydroxy-3-methylcyclobutyl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 443.2 Found 443.0 A-115 and B-4 n/a n/a 393 [02073]  5-chloro-1'-[2-({2-[(cis) or (trans)-3-hydroxy-3-methylcyclobutyl]pyrimidin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 443.2 Found 443.0 A-115 and B-4 Daicel Chiralpak AD-3 1st 392 [02074]  5-chloro-1'-[2-({2-[(trans) or (cis)-3-hydroxy-3-methylcyclobutyl]pyrimidin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 443.2 Found 443.0 A-115 and B-4 Daicel Chiralpak AD-3 2nd 436 [02075]  1'-(2-({[1-(3-hydroxy-3-methylcyclobutyl)-1H-pyrazolo[3,4-b]pyridin-5-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile Calc'd 473.2 Found 473.1 A-116 & B-9 n/a n/a 437 [02076]  2-oxo-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-pyrazolo[3,4-b]pyridin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile Calc'd 473.2 Found 473.1 A-116 & B-9 Daicel Chiralpak OJ-3 1st 438 [02077]  2-oxo-1'-[2-({1-[(trans)-3-hydroxy-3-methylcyclobutyl]-1H-pyrazolo[3,4-b]pyridin-5-yl}oxy)ethy]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile Calc'd 473.2 Found 473.1 A-116 & B-9 Daicel Chiralpak OJ-3 2nd 439 [02078]  5-(2-{5-chloro-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidin]-1'-yl}ethoxy)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-1,3-benzodiazole-7-carbonitrile Calc'd 506.2 Found 506.2 A-117 & B-4 n/a n/a 440 [02079]  5-chloro-1'-[2-({3-[(cis)-3-hydroxy-3-methylcyclobutyl]-3H-[1,2,3]triazolo[4,5-b]pyridin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 483.2 Found 483.2 A-118 & B-4 n/a n/a

Example 85

5-chloro-1'-[2-({2-oxo-3-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H,2H,3H-imidazo[4,5-b]pyridin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 374)

##STR02080##

Step 1: (cis)-1-methyl-3-((3-nitro-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyridin-2-yl)amino)cyclobutan-1-ol

##STR02081##

[3144] A mixture of (cis)-3-((5-bromo-3-nitropyridin-2-yl)amino)-1-methylcyclobutan-1-ol (General procedure for Intermediate A-92, Step 2, 1.00 g, 3.31 mmol), bis(pinacolato)diboron (1.68 g, 6.62 mmol), KOAc (650 mg, 6.62 mmol) and Pd(dppf)Cl.sub.2 (121 mg, 165 µmol) in 1,4-dioxane (15 mL) was degassed and purged with N.sub.2 for 10 min, and then stirred at 85° C. for 16 h under N.sub.2 atmosphere. After cooling to room temperature, the mixture was filtered, and the filtrate was concentrated in vacuo to give (cis)-1-methyl-3-((3-nitro-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyridin-2-yl)amino)cyclobutan-1-ol, which was used in the subsequent step without further purification. MS=267.9 [M-C.sub.6H.sub.10+H].sup.+.

Step 2: 6-(((cis)-3-hydroxy-3-methylcyclobutyl)amino)-5-nitropyridin-3-ol

##STR02082##

[3145] To a solution of 1(cis)-1-methyl-3-((3-nitro-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyridin-2-yl)amino)cyclobutan-1-ol (1.00 g, 2.86 mmol) in THE (10 mL) and H.sub.2O (10 mL) was added Oxone (1.76 g, 2.86 mmol). The mixture was stirred at room temperature for 2 h, then was diluted with H.sub.2O (30 mL) and extracted with EtOAc (2×20 mL). The combined organic

layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-85% EtOAc/Petroleum ether) to give 6-(((cis)-3-hydroxy-3-methylcyclobutyl)amino)-5-nitropyridin-3-ol. MS=239.9 [M+H].sup.+.

Step 3: (cis)-3-((5-(2-bromoethoxy)-3-nitropyridin-2-yl)amino)-1-methylcyclobutan-1-ol
##STR02083##

[3146] To a solution of 6-(((cis)-3-hydroxy-3-methylcyclobutyl)amino)-5-nitropyridin-3-ol (1.00 g, 4.18 mmol) in MeCN (20 mL) was added K.sub.2CO.sub.3 (2.31 g, 16.7 mmol) and 1,2-dibromoethane (6.31 mL, 83.6 mmol). The mixture was stirred at 65° C. for 12 h. After cooling to room temperature, the mixture was diluted with H.sub.2O (10 mL) and extracted with DCM (3×20 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 20 g cartridge, 0-65% EtOAc/Petroleum ether) to give (cis)-3-((5-(2-bromoethoxy)-3-nitropyridin-2-yl)amino)-1-methylcyclobutan-1-ol. MS=346.1/348.1 [M+H].sup.+.

Step 4: (cis)-3-((3-amino-5-(2-bromoethoxy)pyridin-2-yl)amino)-1-methylcyclobutan-1-ol
##STR02084##

[3147] To a solution of (cis)-3-((5-(2-bromoethoxy)-3-nitropyridin-2-yl)amino)-1-methylcyclobutan-1-ol (300 mg, 867 µmol) in EtOH (5 mL) and H.sub.2O (2.5 mL) was added Fe (484 mg, 8.67 mmol) and NH.sub.4Cl (695 mg, 13.0 mmol). The mixture was stirred at 60° C. for 4 h. After cooling to room temperature, the mixture was filtered. The filtrate was diluted with H.sub.2O (20 mL) and extracted with EtOAc (3×20 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to provide (cis)-3-((3-amino-5-(2-bromoethoxy)pyridin-2-yl)amino)-1-methylcyclobutan-1-ol, which was used in the subsequent step without further purification. MS=315.9/317.9 [M+H].sup.+.

Step 5: 6-(2-bromoethoxy)-3-((cis)-3-hydroxy-3-methylcyclobutyl)-1,3-dihydro-2H-imidazo[4,5-b]pyridin-2-one
##STR02085##

[3148] To a solution of (cis)-3-((3-amino-5-(2-bromoethoxy)pyridin-2-yl)amino)-1-methylcyclobutan-1-ol (100 mg, 316 µmol) in THE (4 mL) was added CDI (103 mg, 633 µmol). The mixture was stirred at 60° C. for 16 h. After cooling to room temperature, the mixture was diluted with H.sub.2O (10 mL) and extracted with EtOAc (2×50 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to provide 6-(2-bromoethoxy)-3-((cis)-3-hydroxy-3-methylcyclobutyl)-1,3-dihydro-2H-imidazo[4,5-b]pyridin-2-one, which was used in the subsequent step without further purification. MS=341.9/343.9 [M+H].sup.+.

Step 6: 5-chloro-1'-[2-({2-oxo-3-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H,2H,3H-imidazo[4,5-b]pyridin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 374)
##STR02086##

[3149] To a solution of 6-(2-bromoethoxy)-3-((cis)-3-hydroxy-3-methylcyclobutyl)-1,3-dihydro-2H-imidazo[4,5-b]pyridin-2-one (100 mg, 292 µmol) in MeCN (3 mL) was added NaHCO.sub.3 (98.2 mg, 1.17 mmol) and 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 79.8 mg, 292 µmol, HCl salt). The mixture was stirred at 80° C. for 16 h. After cooling to room temperature, the mixture was concentrated in vacuo to remove MeCN. The residue was dissolved in DMF (1.5 mL) and was filtered to remove solids. The filtrate was purified by reverse phase preparative HPLC (Waters Xbridge OBD C.sub.18 column, 20-50% MeCN:10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 5-chloro-1'-[2-({2-oxo-3-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H,2H,3H-imidazo[4,5-b]pyridin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 374). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 11.04 (br s, 1H), 10.48 (s, 1H), 7.72 (d, J=2.4 Hz, 1H), 7.49 (d, J=1.8 Hz, 1H), 7.27-7.18 (m, 1H), 7.03 (d, J=2.6 Hz, 1H), 6.84 (d, J=8.4 Hz, 1H), 5.16 (s, 1H), 4.48-4.33 (m, 1H), 4.17 (t, J=6.0 Hz, 2H), 3.08-2.98 (m,

2H), 2.96-2.88 (m, 2H), 2.84 (t, J=6.0 Hz, 2H), 2.75-2.62 (m, 2H), 2.32-2.21 (m, 2H), 1.86-1.64 (m, 4H), 1.32 (s, 3H). MS=498.1 [M+H].sup.+.

Example 86

(5-chloro-1'-[2-({2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H,2H,3H,4H-pyrido[2,3-d]pyrimidin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one, Compound 356)

##STR02087##

Step 1: tert-butyl N-[(5-bromo-2-fluoro-3-pyridyl)methyl]carbamate

##STR02088##

[3150] To a mixture of 5-bromo-2-fluoro-pyridine-3-carbaldehyde (5.00 g, 24.5 mmol) and tert-butyl carbamate (5.74 g, 49.0 mmol) in DCM (50 mL) under N.sub.2 atmosphere was added TFA (5.44 mL, 73.5 mmol) and Et.sub.3SiH (19.6 mL, 123 mmol). The mixture was stirred at room temperature for 16 h under N.sub.2. The mixture was quenched by addition of saturated aqueous NaHCO.sub.3 solution (130 mL) and extracted with DCM (2×100 mL). The combined organic layers were concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 80 g cartridge 0-8% EtOAc/Petroleum ether) to provide tert-butyl N-[(5-bromo-2-fluoro-3-pyridyl)methyl]carbamate. MS=304.9/306.9 [M+H].sup.+.

Step 2: tert-butyl N-[(5-bromo-2-{{(cis)-3-hydroxy-3-methylcyclobutyl}amino}pyridin-3-yl)methyl]carbamate

##STR02089##

[3151] A mixture of tert-butyl N-[(5-bromo-2-fluoro-3-pyridyl)methyl]carbamate (3.80 g, 12.5 mmol), (cis)-3-amino-1-methylcyclobutanol (1.89 g, 13.7 mmol, HCl salt) and DIEA (41 mL, 249 mmol) in DMSO (20 mL) was stirred at 90° C. for 16 h under N.sub.2 atmosphere. After cooling to room temperature, the mixture was diluted with H.sub.2O (150 mL), and extracted with EtOAc (80 mL×2). The combined organic layers were concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 40 g cartridge, 0-25% EtOAc/Petroleum ether) to provide tert-butyl N-[(5-bromo-2-{{(cis)-3-hydroxy-3-methylcyclobutyl}amino}pyridin-3-yl)methyl]carbamate. MS=386.0/387.9 [M+H].sup.+.

Step 3: (cis)-3-{{[3-(aminomethyl)-5-bromopyridin-2-yl]amino}-1-methylcyclobutan-1-ol

##STR02090##

[3152] A solution of tert-butyl N-[(5-bromo-2-{{(cis)-3-hydroxy-3-methylcyclobutyl}amino}pyridin-3-yl)methyl]carbamate (1.50 g, 3.88 mmol) in 4.0 M HCl in EtOAc (15 mL, 60 mmol) was stirred at room temperature for 1 h. The mixture was concentrated in vacuo to give (cis)-3-{{[3-(aminomethyl)-5-bromopyridin-2-yl]amino}-1-methylcyclobutan-1-ol, which was used in the subsequent step without further purification. MS=285.9/287.8 [M+H].sup.+.

Step 4: 6-bromo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H,2H,3H,4H-pyrido[2,3-d]pyrimidin-2-one

##STR02091##

[3153] To a mixture of (cis)-3-{{[3-(aminomethyl)-5-bromopyridin-2-yl]amino}-1-methylcyclobutan-1-ol (1.20 g, 4.19 mmol) in THE (15 mL) was added CDI (1.02 g, 6.29 mmol). The mixture was stirred at room temperature for 1 h, then was diluted with brine (15 mL) and extracted with EtOAc (2×20 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered and concentrated in vacuo to give 6-bromo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H,2H,3H,4H-pyrido[2,3-d]pyrimidin-2-one, which was used in the subsequent step without further purification. MS=312.0/313.9 [M+H].sup.+.

Step 5: 6-(2-hydroxyethoxy)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H,2H,3H,4H-pyrido[2,3-d]pyrimidin-2-one

##STR02092##

[3154] A mixture of 6-bromo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H,2H,3H,4H-pyrido[2,3-d]pyrimidin-2-one (350 mg, 1.12 mmol), CuI (85.4 mg, 0.448 mmol), t-BuOLi (359 mg, 4.48 mmol) and ethylene glycol (4.17 mL, 74.5 mmol) was degassed and purged with N.sub.2 for 10

min, and then stirred at 120° C. for 16 h under N.sub.2 atmosphere. After cooling to room temperature, the mixture was filtered and the filtrate was concentrated in vacuo. The filtrate was purified by reverse phase preparative HPLC (Waters Xbridge OBD C.sub.18 column, 1-30% MeCN:10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 6-(2-hydroxyethoxy)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H,2H,3H,4H-pyrido[2,3-d]pyrimidin-2-one. MS=294.0 [M+H].sup.+.

Step 6: 6-(2-bromoethoxy)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H,2H,3H,4H-pyrido[2,3-d]pyrimidin-2-one

##STR02093##

[3155] To a mixture of 6-(2-hydroxyethoxy)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H,2H,3H,4H-pyrido[2,3-d]pyrimidin-2-one (180 mg, 0.614 mmol) in DCM (8 mL) at 0° C. under N.sub.2 atmosphere was added PPh.sub.3 (805 mg, 3.07 mmol), followed by CBr.sub.4 (1.02 g, 3.07 mmol). The mixture was stirred at room temperature for 16 h under N.sub.2 atmosphere, then was concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 4 g cartridge, 0-100% EtOAc/Petroleum ether) to give 6-(2-bromoethoxy)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H,2H,3H,4H-pyrido[2,3-d]pyrimidin-2-one. MS=356.0/357.9 [M+H].sup.+.

Step 7: 5-chloro-1'-[2-({2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H,2H,3H,4H-pyrido[2,3-d]pyrimidin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR02094##

[3156] To a solution of 6-(2-bromoethoxy)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H,2H,3H,4H-pyrido[2,3-d]pyrimidin-2-one (50.0 mg, 0.140 mmol) in MeCN (4 mL) was added NaHCO.sub.3 (47.2 mg, 0.561 mmol) and 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 38.3 mg, 0.140 mmol, HCl salt). The mixture was stirred at 80° C. for 16 h. After cooling to room temperature, the mixture was concentrated in vacuo. The residue was purified by reverse phase preparative HPLC (Waters Xbridge OBD C.sub.18 column, 20-50% MeCN:10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 5-chloro-1'-[2-({2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H,2H,3H,4H-pyrido[2,3-d]pyrimidin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 356). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.48 (br s, 1H), 7.90 (d, J=2.8 Hz, 1H), 7.50 (d, J=2.0 Hz, 1H), 7.34 (d, J=2.8 Hz, 1H), 7.23 (dd, J=8.4, 2.0 Hz, 1H), 7.11 (s, 1H), 6.84 (d, J=8.4 Hz, 1H), 4.81 (s, 1H), 4.19-4.11 (m, 4H), 4.06 (t, J=7.6 Hz, 1H), 2.96-2.87 (m, 2H), 2.83 (t, J=5.6 Hz, 2H), 2.75-2.64 (m, 2H), 2.47-2.40 (m, 2H), 2.34-2.23 (m, 2H), 1.84-1.65 (m, 4H), 1.25 (s, 3H). MS=512.4 [M+H].sup.+.

Example 87

5-chloro-1'-[2-({2-(2,2-dimethyl-1,3-dioxolan-4-yl)pyrimidin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 5-chloro-1'-[2-({2-(1,2-dihydroxyethyl)pyrimidin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one 5-chloro-1'-[2-({2-[(1R or 1S)-1,2-dihydroxyethyl]pyrimidin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one & 5-chloro-1'-[2-({2-[(1S or 1R)-1,2-dihydroxyethyl]pyrimidin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compounds 441, 330, 378, & 379)

##STR02095##

Step 1: 5-bromo-2-vinylpyrimidine

##STR02096##

[3157] A mixture of 5-bromo-2-iodopyrimidine (5.00 g, 17.6 mmol), vinyl potassium trifluoroborate (2.59 g, 19.3 mmol), Pd(dppf)Cl.sub.2 (1.28 g, 1.76 mmol), and K.sub.2CO.sub.3 (6.06 g, 43.9 mmol) in 1,4-dioxane (75 mL) and H.sub.2O (25 mL) was degassed and purged with N.sub.2 (3×), and then the mixture was stirred at 90° C. for 12 h under N.sub.2 atmosphere. After cooling to room temperature, the reaction mixture was diluted with H.sub.2O (60 mL) and extracted with EtOAc (3×50 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 40 g cartridge, 0-15% EtOAc/Petroleum ether) to give 5-bromo-2-vinylpyrimidine.

MS=185.10 [M+H].sup.+.

Step 2: 1-(5-bromopyrimidin-2-yl)ethane-1,2-diol

##STR02097##

[3158] To a 0° C. solution of 5-bromo-2-vinyl-pyrimidine (1.80 g, 9.73 mmol) in THE (20 mL) and H.sub.2O (20 mL) was added K.sub.2O.sub.4.Math..sub.2H.sub.2O (358 mg, 0.973 mmol) and NMO (2.05 mL, 19.5 mmol). The mixture was allowed to warm to room temperature and stirred for 12 h. The reaction mixture was diluted with H.sub.2O (30 mL) and extracted with EtOAc (3×50 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by reverse phase preparative HPLC (Waters Xbridge OBD C.sub.18 column, 25-55% MeCN:10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 1-(5-bromopyrimidin-2-yl)ethane-1,2-diol. MS=219.10 [M+H].sup.+.

Step 3: 5-bromo-2-(2,2-dimethyl-1,3-dioxolan-4-yl)pyrimidine

##STR02098##

[3159] To a solution of 1-(5-bromopyrimidin-2-yl)ethane-1,2-diol (860 mg, 3.93 mmol) in acetone (9 mL) was added 2,2-dimethoxypropane (9.62 mL, 78.53 mmol) and TsOH (67.6 mg, 0.393 mmol). The mixture was stirred at room temperature for 16 h. The reaction mixture was with H.sub.2O (20 mL) and extracted with EtOAc (4×20 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-70% EtOAc/Petroleum ether) to give 5-bromo-2-(2,2-dimethyl-1,3-dioxolan-4-yl)pyrimidine. MS=259.0 [M+H].sup.+.

Step 4: 2-(2,2-dimethyl-1,3-dioxolan-4-yl)pyrimidin-5-ol

##STR02099##

[3160] A mixture of 5-bromo-2-(2,2-dimethyl-1,3-dioxolan-4-yl)pyrimidine (200 mg, 0.772 mmol), Pd(dba).sub.2 (8.88 mg, 15.4 μmol), ditert-butyl-[2,3,4,5-tetramethyl-6-(2,4,6-triisopropylphenyl)phenyl]phosphane (37.1 mg, 77.2 μmol) and KOH (129.92 mg, 2.32 mmol) in H.sub.2O (2 mL) and 1,4-dioxane (5 mL) was degassed and purged with N.sub.2 (3×), and then the mixture was stirred at 100° C. for 16 h under N.sub.2 atmosphere. After cooling to room temperature, the mixture was extracted with EtOAc (3×10 mL). The combined organic layers were washed with brine (2×10 mL), dried over Na.sub.2SO.sub.4, filtered and concentrated in vacuo to give 2-(2,2-dimethyl-1,3-dioxolan-4-yl)pyrimidin-5-ol), which was used in the subsequent step without further purification. MS=197.1 [M+H].sup.+.

Step 5: 5-(2-bromoethoxy)-2-(2,2-dimethyl-1,3-dioxolan-4-yl)pyrimidine

##STR02100##

[3161] To a solution of 2-(2,2-dimethyl-1,3-dioxolan-4-yl)pyrimidin-5-ol (260 mg, 1.33 mmol) in MeCN (3 mL) and 1,2-dibromoethane (5 mL) under N.sub.2 atmosphere was added K.sub.2CO.sub.3 (916 mg, 6.63 mmol). The mixture was stirred at 80° C. for 16 h under N.sub.2 atmosphere. The reaction mixture was diluted with H.sub.2O (30 mL) and extracted with DCM (3×20 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered and concentrated in vacuo to give 5-(2-bromoethoxy)-2-(2,2-dimethyl-1,3-dioxolan-4-yl)pyrimidine, which was used in the subsequent step without further purification. MS=303.2/305.2 [M+H].sup.+.

Step 6: 5-chloro-1'-(2-{[2-(2,2-dimethyl-1,3-dioxolan-4-yl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 441)

##STR02101##

[3162] To a solution of 5-(2-bromoethoxy)-2-(2,2-dimethyl-1,3-dioxolan-4-yl)pyrimidine (120 mg, 396 μmol) in MeCN (2 mL) under N.sub.2 atmosphere was added NaHCO.sub.3 (166 mg, 1.98 mmol) and 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 112 mg, 475 μmol, HCl salt). The mixture was stirred at 80° C. for 12 h under N.sub.2 atmosphere. After cooling to room temperature, the reaction mixture was filtered and the filtrate was concentrated in vacuo to afford 5-chloro-1'-(2-{[2-(2,2-dimethyl-1,3-dioxolan-4-yl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 441), which was used in the subsequent step

without further purification. MS=459.2 [M+H].sup.+.

Step 7: 5-chloro-1'-(2-{[2-(1,2-dihydroxyethyl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 330)

##STR02102##

[3163] To a 0° C. solution of 5-chloro-1'-(2-{[2-(2,2-dimethyl-1,3-dioxolan-4-yl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 441, 160 mg, 349 µmol) in THE (0.5 mL) was added 12 M aqueous HCl (0.20 mL, 2.4 mmol). The mixture was stirred at 0° C. for 2 h. The reaction mixture was filtered, and the filtrate was concentrated in vacuo. The residue was purified by reverse phase preparative HPLC (Waters Xbridge BEH C.sub.18 column, 15-40% MeCN:10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 5-chloro-1'-(2-{[2-(1,2-dihydroxyethyl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 330). MS=419.3 [M+H].sup.+.

Step 8: 5-chloro-1'-[2-({2-[(1R or 1S)-1,2-dihydroxyethyl]pyrimidin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one & 5-chloro-1'-[2-({2-[(1S or 1R)-1,2-dihydroxyethyl]pyrimidin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compounds 378 & 379)

##STR02103##

[3164] 5-chloro-1'-(2-{[2-(1,2-dihydroxyethyl)pyrimidin-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 330) was separated by preparative chiral SFC (Chiralcel IF column, 30-80% Heptane:(4:1 EtOH:MeCN) with 0.1% NH.sub.4OH in CO.sub.2). The first eluting enantiomer of the title compound, 5-chloro-1'-[2-({2-[(1R or 1S)-1,2-dihydroxyethyl]pyrimidin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 379): .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.50 (s, 1H), 8.56 (s, 2H), 7.51 (s, 1H), 7.24 (dd, J=7.6, 2.0 Hz, 1H), 6.85 (d, J=8.4 Hz, 1H), 5.14 (d, J=2.0 Hz, 1H), 4.62-4.59 (m, 2H), 4.30 (t, J=5.6 Hz, 2H), 3.71-3.70 (m, 1H), 3.64-3.63 (m, 1H), 2.92-2.86 (m, 4H), 2.70-2.67 (m, 2H), 1.78-1.72 (m, 4H). MS=419.30 [M+H].sup.+.

The second eluting enantiomer of the title compound, 5-chloro-1'-[2-({2-[(1S or 1R)-1,2-dihydroxyethyl]pyrimidin-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 378): .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.50 (s, 1H), 8.56 (s, 2H), 7.51 (s, 1H), 7.24 (dd, J=8.0, 2.0 Hz, 1H), 6.85 (d, J=8.4 Hz, 1H), 5.14 (d, J=2.0 Hz, 1H), 4.62-4.59 (m, 2H), 4.30 (t, J=5.6 Hz, 2H), 3.71-3.70 (m, 1H), 3.64-3.63 (m, 1H), 2.92-2.86 (m, 4H), 2.70-2.67 (m, 2H), 1.78-1.72 (m, 4H). MS=419.30 [M+H].sup.+.

Example 88

5-chloro-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 351)

##STR02104##

Step 1: (cis)-3-[(4-bromo-2-nitrophenyl)amino]-1-methylcyclobutan-1-ol

##STR02105##

[3165] A mixture of 4-bromo-1-fluoro-2-nitro-benzene (1.96 mL, 15.9 mmol), (cis)-3-amino-1-methylcyclobutan-1-ol (2.41 g, 17.5 mmol, HCl salt) and DIEA (8.31 mL, 47.7 mmol) in DMSO (25 mL) under N.sub.2 atmosphere was stirred at 90° C. for 1 h. After cooling to room temperature, the reaction mixture was quenched by addition of saturated aqueous NH.sub.4Cl solution (150 mL). The mixture was extracted with EtOAc (3×100 mL). The combined organic layers were washed with brine (2×50 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo to give (cis)-3-[(4-bromo-2-nitrophenyl)amino]-1-methylcyclobutan-1-ol, which was used in the subsequent step without further purification. MS=300.9/302.9 [M+H].sup.+.

Step 2: (cis)-3-[(2-amino-4-bromophenyl)amino]-1-methylcyclobutan-1-ol

##STR02106##

[3166] To a solution of (cis)-3-[(4-bromo-2-nitrophenyl)amino]-1-methylcyclobutan-1-ol (300 mg, 996 µmol) in EtOH (5 mL) and H.sub.2O (2.5 mL) was added Fe (556 mg, 9.96 mmol) and NH.sub.4Cl (799 mg, 14.9 mmol). The mixture was stirred at 60° C. for 2 h. After cooling to room

temperature, the reaction mixture was filtered and the filtrate was concentrated in vacuo to afford (cis)-3-[(2-amino-4-bromophenyl)amino]-1-methylcyclobutan-1-ol, which was used in the subsequent step without further purification. MS=270.9/272.9 [M+H].sup.+.

Step 3: (cis)-3-(5-bromo-1H-1,3-benzodiazol-1-yl)-1-methylcyclobutan-1-ol

##STR02107##

[3167] To a solution of (cis)-3-[(2-amino-4-bromophenyl)amino]-1-methylcyclobutan-1-ol (1.00 g, 3.69 mmol) in triethoxymethane (6.13 mL, 36.9 mmol) was added TsOH (63.5 mg, 369 μ mol). The mixture was stirred at 80° C. for 4 h. After cooling to room temperature, the reaction was filtered and the filtrate was concentrated in vacuo to give (cis)-3-(5-bromo-1H-1,3-benzodiazol-1-yl)-1-methylcyclobutan-1-ol, which was used in the subsequent step without further purification.

MS=281.0/283.0 [M+H].sup.+.

Step 4: (cis)-3-[5-(2-hydroxyethoxy)-1H-1,3-benzodiazol-1-yl]-1-methylcyclobutan-1-ol

##STR02108##

[3168] To a solution of (cis)-3-(5-bromo-1H-1,3-benzodiazol-1-yl)-1-methylcyclobutan-1-ol (1.30 g, 4.62 mmol) in ethylene glycol (15 mL) was added CuCl.sub.2 (124 mg, 925 μ mol) and K.sub.2CO.sub.3 (1.92 g, 13.9 mmol). The mixture was degassed and purged with N.sub.2 for 10 min, and then stirred at 130° C. for 6 h. After cooling to room temperature, the reaction mixture was filtered and the filtrate was concentrated in vacuo. The residue was purified by reverse phase preparative HPLC (Phenomenex Luna C.sub.18 column, 1-30% MeCN:10 mM

NH.sub.4HCO.sub.3 in H.sub.2O) to give (cis)-3-[5-(2-hydroxyethoxy)-1H-1,3-benzodiazol-1-yl]-1-methylcyclobutan-1-ol. MS=263.1 [M+H].sup.+.

Step 5: (cis)-3-[5-(2-iodoethoxy)-1H-1,3-benzodiazol-1-yl]-1-methylcyclobutan-1-ol

##STR02109##

[3169] To a solution of (cis)-3-[5-(2-hydroxyethoxy)-1H-1,3-benzodiazol-1-yl]-1-methylcyclobutan-1-ol (200 mg, 762 μ mol) in DCM (5 mL) was added 12 (169 μ L, 839 μ mol), imidazole (104 mg, 1.52 mmol), and PPh.sub.3 (220 mg, 839 μ mol). The mixture was stirred at room temperature for 4 h. The mixture was filtered, and the filtrate was concentrated in vacuo. The residue was purified by preparative TLC (SiO.sub.2, DCM:MeOH=10:1) to give (cis)-3-[5-(2-iodoethoxy)-1H-1,3-benzodiazol-1-yl]-1-methylcyclobutan-1-ol. MS=373.0 [M+H].sup.+.

Step 6: 5-chloro-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 351)

##STR02110##

[3170] To a solution of (cis)-3-[5-(2-iodoethoxy)-1H-1,3-benzodiazol-1-yl]-1-methylcyclobutan-1-ol (40.0 mg, 107 μ mol) in MeCN (2 mL) was added 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 32.3 mg, 118 μ mol, HCl salt) and NaHCO.sub.3 (36.1 mg, 430 μ mol). The mixture was stirred at 80° C. for 12 h. After cooling to room temperature, the reaction mixture was quenched by addition of H.sub.2O (10 mL) and extracted with EtOAc (3 $\times$ 20 mL). The combined organic layers were washed with brine (30 mL), dried over Na.sub.2SO.sub.4, filtered and concentrated in vacuo. The crude residue was purified by reverse phase preparative HPLC (Phenomenex Luna C.sub.18, 5-20% MeCN in H.sub.2O with 0.2% formic acid modifier) to give 5-chloro-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 351, formic acid salt). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.49 (s, 1H), 8.22 (s, 1H), 8.18 (s, 1H), 7.69-7.46 (m, 2H), 7.32-7.13 (m, 2H), 7.01-6.77 (m, 2H), 4.58 (t, J=8.4 Hz, 1H), 4.16 (t, J=5.6 Hz, 2H), 3.00-2.84 (m, 4H), 2.77-2.58 (m, 4H), 2.56-2.51 (m, 2H), 1.90-1.67 (m, 4H), 1.37 (s, 3H). MS=481.3 [M+H].sup.+.

Example 89

5-chloro-1'-[2-({2-methyl-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 442)

##STR02111##

Step 1: (cis)-3-[5-(2-bromoethoxy)-2-methyl-7-(trifluoromethyl)-1H-1,3-benzodiazol-1-yl]-1-

methylcyclobutan-1-ol

##STR02112##

[3171] To a mixture of 2-methyl-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-ol (Intermediate A-119, 1.40 g, 4.66 mmol) and 1,2-dibromoethane (17.5 g, 93.2 mmol) in i-PrOH (10 mL) was added Cs.sub.2CO.sub.3 (3.80 g, 11.7 mmol). The mixture was stirred at 60° C. for 2 h. After cooling to room temperature, the reaction mixture was filtered, and the filtrate was concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (Biotage 20 g cartridge, 0-100% EtOAc/Petroleum ether) to give (cis)-3-[5-(2-bromoethoxy)-2-methyl-7-(trifluoromethyl)-1H-1,3-benzodiazol-1-yl]-1-methylcyclobutan-1-ol. MS=407.1/409.1 [M+H].sup.+.

Step 2: 5-chloro-1'-[2-({2-methyl-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR02113##

[3172] To a mixture of (cis)-3-[5-(2-bromoethoxy)-2-methyl-7-(trifluoromethyl)-1H-1,3-benzodiazol-1-yl]-1-methylcyclobutan-1-ol (150 mg, 368 mmol) and 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 121 mg, 442 mmol, HCl salt) in MeCN (4 mL) was added NaHCO.sub.3 (92.8 mg, 1.11 mmol). The mixture was stirred at 80° C. for 16 h. After cooling to room temperature, the reaction mixture was filtered, and the filtrate was concentrated under reduced pressure. The residue was purified by reverse phase preparative HPLC (Waters Xbridge OBD C.sub.18 column, 30-60% MeCN:10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 5-chloro-1'-[2-({2-methyl-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 442). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.48 (s, 1H), 7.51-7.48 (m, 1H), 7.47-7.43 (m, 1H), 7.23 (dd, J=8.0, 2.0 Hz, 1H), 7.18 (d, J=2.0 Hz, 1H), 6.84 (d, J=8.0 Hz, 1H), 5.43 (s, 1H), 4.81-4.69 (m, 1H), 4.22 (t, J=5.6 Hz, 2H), 2.97-2.90 (m, 2H), 2.90-2.86 (m, 2H), 2.85 (s, 3H), 2.83-2.76 (m, 2H), 2.73-2.70 (m, 2H), 2.46-2.45 (m, 2H), 1.83-1.75 (m, 2H), 1.74-1.66 (m, 2H), 1.34 (s, 3H). MS=563.1 [M+H].sup.+.

[3173] The following intermediates in Table 38.16 were prepared according to procedures analogous to those described for Compound 442 using the appropriate starting materials or common intermediates.

TABLE-US-00056 TABLE 38.16 Exact Mass Intermediates # Structure IUPAC Name [M + H].sup.+ Used 443 [02114]  embedded image 1'-[2-({2-methyl-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile Calc'd 554.2 Found 554.2 A-119 and B-9 444 [02115]  embedded image 5-chloro-1'-[2-[4-(1-methanesulfonylpiperidin-4-yl)phenoxy]ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 518.2 Found 518.2 General Procedure for Intermediate A-104, Step 2 and B-4 445 [02116]  embedded image 5-chloro-1'-(2-{[7-(difluoromethyl)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-1,3-benzodiazol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 531.2 Found 531.2 A-120 and B-4 446 [02117]  embedded image 1'-(2-{[7-(difluoromethyl)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-1,3-benzodiazol-5-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile Calc'd 522.2 Found 522.2 A-120 and B-9 447 [02118]  embedded image 5-chloro-1'-(2-{[7-(difluoromethoxy)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-1,3-benzodiazol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one Calc'd 547.2 Found 547.3 A-121 and B-4 448 [02119]  embedded image 1'-(2-{[7-(difluoromethoxy)-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-1H-1,3-benzodiazol-5-yl]oxy}ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile Calc'd 538.2 Found 538.3 A-121 and B-9 449 [02120]  embedded image 5-chloro-1'-(2-((2-(2-hydroxypropan-2-yl)pyrimidin-5-yl)oxy)ethyl)-7-iodospiro[indoline-3,4'-piperidin]-2-one Calc'd 543.1 Found 543.2 A-100 and B-24 450 [02121]  embedded image 5-(difluoromethyl)-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

methylcyclobutyl]-7- (trifluoromethyl)-1H-1,3- benzodiazol-5-yl}oxy)ethyl]- 1,2- dihydrospiro[indole-3,4'- piperidin]-2-one Calc'd 565.2 Found 565.2 A-92 and B-1 451 [02122]  5-chloro-1'-[2-({2-methyl-3- [(cis)-3-hydroxy-3- methylcyclobutyl]-3H-imidazo[4,5-b]pyridin-6- yl}oxy)ethyl]-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one Calc'd 496.2 Found 496.2 A-122 and B-4 452 [02123]  5-chloro-1'-(2-{{8-(2-hydroxy-2-methylpropyl)-7-oxo-5,6,7,8- tetrahydro-1,8-naphthyridin-3- yl}oxy)ethyl)-1,2- dihydrospiro[indole-3,4'- piperidin]-2-one Calc'd 499.2 Found 499.2 A-105 and B-4 453 [02124]  5-chloro-7-iodo-1'-[2-({1-[(cis)- 3-hydroxy-3-methylcyclobutyl]- 7- (trifluoromethyl)-1H-1,3- benzodiazol-5-yl}oxy)ethyl]- 1,2-dihydrospiro[indole-3,4'- piperidin]-2-one Calc'd 675.1 Found 675.0 A-92 and B-24 454 [02125]  5-chloro-1'-(2-((7-fluoro-1- ((cis)-3-hydroxy-3- methylcyclobutyl)-1H- benzo[d]imidazol-5- yl)oxy)ethyl)spiro[indoline-3,4'- piperidin]-2-one Calc'd 499.2 Found 499.2 A-124 and B-4 455 [02126]  1'-(2-((7-fluoro-1-((cis)-3- hydroxy-3-methylcyclobutyl)- 1H- benzo[d]imidazol-5- yl)oxy)ethyl)-2- oxospiro[indoline-3,4'- piperidine]-5-carbonitrile Calc'd 490.2 Found 490.2 A-124 and B-9

Example 90

5-chloro-1'-[2-({2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-8-(trifluoromethyl)-1,2,3,4-tetrahydroquinazolin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 456)

##STR02127##

Step 1: 2-{{[(cis)-3-hydroxy-3-methylcyclobutyl]amino}-3-(trifluoromethyl)benzonitrile

##STR02128##

[3174] To a solution of (cis)-3-amino-1-methylcyclobutanol (12.0 g, 87.3 mmol, HCl salt) in DMSO (150 mL) was added 2-fluoro-3-(trifluoromethyl)benzonitrile (15.0 g, 79.3 mmol) and DIEA (30.8 g, 238 mmol). The mixture was stirred at 60° C. for 6 h. After cooling to room temperature, the reaction mixture was quenched by addition of H.sub.2O (200 mL) and extracted with EtOAc (3×80 mL). The combined organic layers were washed with brine (150 mL), dried over anhydrous Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 40 g cartridge, 0-10% EtOAc/Petroleum ether) to give 2-{{[(cis)-3-hydroxy-3-methylcyclobutyl]amino}-3-(trifluoromethyl)benzonitrile. MS=271.1 [M+H].sup.+.

Step 2: 5-bromo-2-{{[(cis)-3-hydroxy-3-methylcyclobutyl]amino}-3-(trifluoromethyl)benzonitrile

##STR02129##

[3175] To a solution of 2-{{[(cis)-3-hydroxy-3-methylcyclobutyl]amino}-3-(trifluoromethyl)benzonitrile (12.0 g, 44.4 mmol) in DMF (150 mL) was added NBS (10.3 g, 57.7 mmol). The mixture was stirred at 80° C. for 16 h. After cooling to room temperature, the reaction mixture was quenched by addition of H.sub.2O (100 mL) and extracted with EtOAc (3×100 mL). The combined organic layers were washed with brine (300 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (Biotage 80 g cartridge, 0-15% EtOAc/Petroleum ether) to give 5-bromo-2-{{[(cis)-3-hydroxy-3-methylcyclobutyl]amino}-3-(trifluoromethyl)benzonitrile. MS=349.0/351.0 [M+H].sup.+.

Step 3: 2-{{[(cis)-3-hydroxy-3-methylcyclobutyl]amino}-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-3-(trifluoromethyl)benzonitrile

##STR02130##

[3176] To a solution of 5-bromo-2-{{[(cis)-3-hydroxy-3-methylcyclobutyl]amino}-3-(trifluoromethyl)benzonitrile (9.00 g, 25.8 mmol) in 1,4-dioxane (150 mL) was added KOAc (6.32 g, 64.4 mmol), bis(pinacolato)diboron (6.55 g, 25.8 mmol) and Pd(dppf)Cl.sub.2 (1.89 g, 2.58 mmol). The mixture was degassed and purged with N.sub.2 (3×), then stirred at 80° C. for 16 h. After cooling to room temperature, the reaction mixture was filtered, and the filtrate was

concentrated under reduced pressure to give 2-[[*(cis)*-3-hydroxy-3-methylcyclobutyl]amino]-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-3-(trifluoromethyl)benzonitrile, which was taken to the next step without further purification. MS=397.1 [M+H].sup.+.

Step 4: 5-hydroxy-2-[[*(cis)*-3-hydroxy-3-methylcyclobutyl]amino]-3-(trifluoromethyl)benzonitrile
##STR02131##

[3177] To a solution of 2-[[*(cis)*-3-hydroxy-3-methylcyclobutyl]amino]-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-3-(trifluoromethyl)benzonitrile (15.0 g, 22.7 mmol, 60% purity) in THE (90 mL) and H.sub.2O (30 mL) was added Oxone (13.9 g, 22.7 mmol). The mixture was stirred at room temperature for 2 h. The reaction mixture was quenched by addition of saturated aqueous Na.sub.2SO.sub.3 (100 mL) and extracted with EtOAc (3×200 mL). The combined organic layers were washed with brine (200 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (Biotage 80 g cartridge, 0-37% EtOAc/Petroleum ether) to give 5-hydroxy-2-[[*(cis)*-3-hydroxy-3-methylcyclobutyl]amino]-3-(trifluoromethyl)benzonitrile. MS=287.1 [M+H].sup.+.

Step 5: 5-(2-bromoethoxy)-2-[[*(cis)*-3-hydroxy-3-methylcyclobutyl]amino]-3-(trifluoromethyl)benzonitrile

##STR02132##

[3178] To a solution of 5-hydroxy-2-[[*(cis)*-3-hydroxy-3-methylcyclobutyl]amino]-3-(trifluoromethyl)benzonitrile (6.00 g, 20.9 mmol) in 1,2-dibromoethane (63.3 mL, 838 mmol) and MeCN (20 mL) was added K.sub.2CO.sub.3 (14.5 g, 105 mmol). The mixture was stirred at 80° C. for 12 h. After cooling to room temperature, solids were removed by filtration. H.sub.2O (80 mL) was then added to the filtrate, and the mixture was extracted with DCM (3×60 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (Biotage 40 g cartridge, 0-15% EtOAc/Petroleum ether) to give 5-(2-bromoethoxy)-2-[[*(cis)*-3-hydroxy-3-methylcyclobutyl]amino]-3-(trifluoromethyl)benzonitrile. MS=393.0/395.0 [M+H].sup.+.

Step 6: *(cis)*-3-{[2-(aminomethyl)-4-(2-bromoethoxy)-6-(trifluoromethyl)phenyl]amino}-1-methylcyclobutan-1-ol

##STR02133##

[3179] To a mixture of 5-(2-bromoethoxy)-2-[[*(cis)*-3-hydroxy-3-methylcyclobutyl]amino]-3-(trifluoromethyl)benzonitrile (3.00 g, 7.63 mmol) in THE (50 mL) at 0° C. was added 10 M BH.sub.3.Math.Me.sub.2S in Me.sub.2S (2.29 mL, 22.9 mmol). The mixture was then stirred at 60° C. for 5 h. After cooling to room temperature, the reaction mixture was quenched by addition of MeOH (30 mL) and H.sub.2O (20 mL), then extracted with EtOAc (3×100 mL). The combined organic phases were dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure to give *(cis)*-3-{[2-(aminomethyl)-4-(2-bromoethoxy)-6-(trifluoromethyl)phenyl]amino}-1-methylcyclobutan-1-ol, which was taken to the next step without further purification. MS=397.0/399.0 [M+H].sup.+.

Step 7: 6-(2-bromoethoxy)-1-[[*(cis)*-3-hydroxy-3-methylcyclobutyl]-8-(trifluoromethyl)-1,2,3,4-tetrahydroquinazolin-2-one

##STR02134##

[3180] To a solution of *(cis)*-3-{[2-(aminomethyl)-4-(2-bromoethoxy)-6-(trifluoromethyl)phenyl]amino}-1-methylcyclobutan-1-ol (200 mg, 503 μmol) in THE (5 mL) was added CDI (245 mg, 1.51 mmol). The mixture was stirred at 60° C. for 5 h. After cooling to room temperature, the reaction mixture was quenched with H.sub.2O (10 mL) and extracted with EtOAc (3×10 mL). The combined organic phases were dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure. The residue was purified by preparative TLC (SiO.sub.2, 100% EtOAc) to give 6-(2-bromoethoxy)-1-[[*(cis)*-3-hydroxy-3-methylcyclobutyl]-8-(trifluoromethyl)-1,2,3,4-tetrahydroquinazolin-2-one. MS=423.0/425.0 [M+H].sup.+.

Step 8: 5-chloro-1'-[2-({2-oxo-1-[[*(cis)*-3-hydroxy-3-methylcyclobutyl]-8-(trifluoromethyl)-1,2,3,4-

tetrahydroquinazolin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR02135##

[3181] To a solution of 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 51.6 mg, 189 μ mol, HCl salt) in MeCN (2 mL) was added 6-(2-bromoethoxy)-1-(3-hydroxy-3-methyl-cyclobutyl)-8-(trifluoromethyl)-3,4-dihydroquinazolin-2-one (80.0 mg, 189 μ mol) and NaHCO₃ (63.5 mg, 756 μ mol). The mixture was stirred at 80° C. for 5 h under N₂ atmosphere. The mixture was then diluted with H₂O (10 mL) and extracted with EtOAc (3×10 mL). The combined organic layers were dried over Na₂SO₄, filtered, and concentrated under reduced pressure. The filtrate was purified by reverse phase preparative HPLC (Phenomenex Luna Cis column, 10-30% MeCN:0.04% HCl in H₂O) to give 5-chloro-1'-[2-({2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-8-(trifluoromethyl)-1,2,3,4-tetrahydroquinazolin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 456). H NMR (400 MHz, DMSO-d₆, HCl salt): δ 10.96-10.66 (in, 1.5H), 10.30 (s, 0.5H), 7.99 (s, 1H), 7.35-7.31 (m, 1H), 7.30-7.27 (m, 1H), 7.26-7.15 (m, 2H), 6.91 (dd, J=8.0, 14.0 Hz, 1H), 5.32-4.67 (m, 1H), 4.49 (d, J=4.0 Hz, 2H), 4.13 (s, 2H), 3.87-3.55 (m, 7H), 2.55-2.51 (m, 1H), 2.39-2.28 (m, 2H), 2.24-2.12 (m, 1H), 2.06-1.86 (m, 4H), 1.16 (s, 3H). MS=579.1 [M+H]⁺.

[3182] The following compounds in Table 38.17 were prepared according to procedures similar to steps described for Example 90 using the appropriate starting materials or common intermediates.

TABLE-US-00057 TABLE 38.17 Exact Mass Intermediates # Structure IUPAC Name [M + H]⁺ used 457 [02136]  2-oxo-1'-[2-({2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-8-(trifluoromethyl)-1,2,3,4-tetrahydroquinazolin-6-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile Calc'd 570.2 Found 570.1 B-9

Example 91

5-chloro-1'-[2-({7,7-dimethyl-5-oxo-5H,7H-furo[3,4-b]pyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 458)

##STR02137##

Step 1: 2-chloro-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyridine-3-carbonitrile

##STR02138##

[3183] A mixture of 5-bromo-2-chloro-pyridine-3-carbonitrile (10.0 g, 46.0 mmol), bis(pinacolato)diboron (17.5 g, 69.0 mmol), KOAc (11.3 g, 115 mmol) and Pd(dppf)Cl₂·DCH₂Cl₂ (1.88 g, 2.30 mmol) in 1,4-dioxane (100 mL) was degassed and purged with N₂ (3×), and then the mixture was stirred at 90° C. for 12 h under N₂ atmosphere. After cooling to room temperature, solids were removed by filtration, and then the mixture concentrated in vacuo give 2-chloro-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyridine-3-carbonitrile, which was taken to the next step without further purification.

Step 2: 2-chloro-5-hydroxypyridine-3-carbonitrile

##STR02139##

[3184] To a solution of 2-chloro-5-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)pyridine-3-carbonitrile (1.30 g, 4.91 mmol) in THE (10 mL) and H₂O (4 mL) was added Oxone (2.42 g, 3.93 mmol). The mixture was stirred at 0° C. for 1 h, then was quenched with saturated aqueous Na₂SO₃ solution (10 mL) and extracted with EtOAc (3×10 mL). The combined organic phases were dried over Na₂SO₄, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-30% EtOAc/Petroleum ether) to provide 2-chloro-5-hydroxypyridine-3-carbonitrile. MS=155.1 [M+H]⁺.

Step 3: 5-(2-bromoethoxy)-2-chloropyridine-3-carbonitrile

##STR02140##

[3185] To a solution of 2-chloro-5-hydroxypyridine-3-carbonitrile (550 mg, 3.56 mmol) in MeCN (7 mL) was added K₂CO₃ (2.46 g, 17.8 mmol) and 1,2-dibromoethane (20.1 g, 107 mmol). The mixture was stirred at 80° C. for 12 h. After cooling to room temperature, the solids were removed by filtration and the filtrate was concentrated in vacuo. The residue was purified by

flash silica gel chromatography (Biotage 12 g cartridge, 0-30% EtOAc/Petroleum ether) to provide 5-(2-bromoethoxy)-2-chloropyridine-3-carbonitrile. MS=260.9/262.9 [M+H].sup.+.

Step 4: 5-(2-bromoethoxy)-2-(prop-1-en-2-yl)pyridine-3-carbonitrile

##STR02141##

[3186] A mixture of 5-(2-bromoethoxy)-2-chloropyridine-3-carbonitrile (300 mg, 1.15 mmol), 2-isopropenyl-4,4,5,5-tetramethyl-1,3,2-dioxaborolane (386 mg, 2.29 mmol), Cs.sub.2CO.sub.3 (1.87 g, 5.74 mmol) and Pd(PPh.sub.3)Cl.sub.2 (8.05 mg, 11.5 µmol) in 1,4-dioxane (3 mL) and H.sub.2O (0.5 mL) was degassed and purged with N.sub.2 (3×), and then the mixture was stirred at 100° C. for 2 h under N.sub.2 atmosphere. After cooling to room temperature, the reaction mixture was filtered, and the filtrate was concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-20% EtOAc/Petroleum ether) to give 5-(2-bromoethoxy)-2-(prop-1-en-2-yl)pyridine-3-carbonitrile. MS=267.0/269.0 [M+H].sup.+.

Step 5: 5-(2-bromoethoxy)-2-(2-hydroxypropan-2-yl)pyridine-3-carbonitrile

##STR02142##

[3187] A mixture of 5-(2-bromoethoxy)-2-(prop-1-en-2-yl)pyridine-3-carbonitrile (100 mg, 374 µmol), tris(2,2,6,6-tetramethyl-3,5-heptanedionato)manganese(III) (22.6 mg, 37.4 µmol), and phenylsilane (81.0 mg, 749 µmol) in DCM (0.5 mL) and i-PrOH (10 mL) was degassed and purged with O₂ (3×), and then the mixture was stirred at 0° C. for 2 h under O₂ atmosphere. The reaction mixture was filtered, and the filtrate was concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (Biotage 4 g cartridge, 0-40% EtOAc/Petroleum ether) to give 5-(2-bromoethoxy)-2-(2-hydroxypropan-2-yl)pyridine-3-carbonitrile. MS=285.0/286.9 [M+H].sup.+.

Step 6: 5-chloro-1'-[2-({7,7-dimethyl-5-oxo-5H,7H-furo[3,4-b]pyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR02143##

[3188] To a solution of 5-(2-bromoethoxy)-2-(2-hydroxypropan-2-yl)pyridine-3-carbonitrile (100 mg, 351 µmol) and 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 105 mg, 386 µmol, HCl salt) in MeCN (3 mL) was added NaHCO.sub.3 (88.4 mg, 1.05 mmol). The mixture was stirred at 80° C. for 12 h. After cooling to room temperature, the reaction mixture was filtered, and the filtrate was concentrated under reduced pressure. The filtrate was purified by reverse phase preparative HPLC (Phenomenex Luna C.sub.18 column, 20-50% MeCN:10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 5-chloro-1'-[2-({7,7-dimethyl-5-oxo-5H,7H-furo[3,4-b]pyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 458). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.49 (s, 1H), 8.64 (d, J=2.8 Hz, 1H), 7.78 (d, J=2.4 Hz, 1H), 7.50 (d, J=1.6 Hz, 1H), 7.24 (dd, J=8.0, 2.0 Hz, 1H), 6.85 (d, J=8.4 Hz, 1H), 4.33 (t, J=5.6 Hz, 2H), 2.96-2.86 (m, 4H), 2.74-2.67 (m, 2H), 1.82-1.67 (m, 4H), 1.61 (s, 6H). MS=442.3 [M+H].sup.+.

Example 92

5-chloro-1'-{2-[(2-{6-hydroxy-6-methyl-2-azaspiro[3.3]heptan-2-yl}pyrimidin-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 459)

##STR02144##

Step 1: 5-(2-bromoethoxy)-2-chloropyrimidine

##STR02145##

[3189] To a solution of 2-chloropyrimidin-5-ol (1.00 g, 7.66 mmol) in DMF (7.661 mL) was added 1,2-dibromoethane (15.3 mL) and Cs.sub.2CO.sub.3 (7.49 g, 23.0 mmol). The reaction was heated at 90° C. for 16 h. The reaction was cooled to room temperature, diluted with EtOAc (30 mL), and filtered through a pad of Celite. The filtrate was concentrated, and the residue was purified by flash silica gel chromatography (Biotage 50 g cartridge, 0-100% EtOAc/Hexanes) to provide 5-(2-bromoethoxy)-2-chloropyrimidine. MS=236.96 [M+H].sup.+.

Step 2: 5-chloro-1'-{2-[(2-chloropyrimidin-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR02146##

[3190] To a vial was added 5-(2-bromoethoxy)-2-chloropyrimidine (1.23 g, 5.18 mmol), 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 1.34 g, 4.92 mmol, HCl salt), K.sub.2CO.sub.3 (2.15 g, 15.5 mmol), and KI (860 mg, 5.18 mmol) followed by DMF (17.3 mL). The mixture was heated at 65° C. for 4 h. The reaction was cooled to room temperature and diluted with H.sub.2O (100 mL). The resulting precipitate was collected by filtration and dried in vacuo to provide 5-chloro-1'-{2-[(2-chloropyrimidin-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one, which was used in the subsequent step without further purification. MS=393.2 [M+H].sup.+.

Step 3: 5-chloro-1'-{2-[(2-{6-hydroxy-6-methyl-2-azaspiro[3.3]heptan-2-yl}pyrimidin-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR02147##

[3191] To a solution of 5-chloro-1'-{2-[(2-chloropyrimidin-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (75.0 mg, 0.191 mmol) and 6-methyl-2-azaspiro[3.3]heptan-6-ol (156 mg, 0.954 mmol, HCl salt) in DMSO (0.381 mL) was added triethylamine (0.132 mL, 0.954 mmol). The reaction was heated at 100° C. under microwave irradiation for 1 h. After cooling to room temperature, the mixture was diluted with H.sub.2O (1 mL) and MeCN (1 mL). The resulting solution was purified by reverse phase preparative HPLC (Phenomenex Kinetex C.sub.18 column, 5-50% MeCN in H.sub.2O with 0.1% NH.sub.4OH modifier) to give 5-chloro-1'-{2-[(2-{6-hydroxy-6-methyl-2-azaspiro[3.3]heptan-2-yl}pyrimidin-5-yl)oxy]ethyl}-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 459). .sup.1H NMR (500 MHz, DMSO-d.sub.6, 29/30H): δ 10.41 (s, 1H), 8.15-8.06 (m, 2H), 7.42 (d, J=2.1 Hz, 1H), 7.16 (dd, J=8.2, 2.1 Hz, 1H), 6.77 (d, J=8.2 Hz, 1H), 4.82 (s, 1H), 4.27-4.02 (m, 2H), 3.94-3.77 (m, 4H), 2.87-2.77 (m, 2H), 2.73 (t, J=5.7 Hz, 2H), 2.65-2.55 (m, 2H), 2.16-2.07 (m, 3H), 1.77-1.57 (m, 4H), 1.12 (s, 3H). MS=484.28 [M+H].sup.+.

Example 93

5-chloro-1'-(2-{2'-oxo-1'-[(cis)-3-hydroxy-3-methylcyclobutyl]-1',2'-dihydrospiro[cyclopropane-1,3'-pyrrolo[2,3-b]pyridin]-5'-yloxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 460)

##STR02148##

Step 1: 5'-bromo-1'-(3-oxocyclobutyl)spiro[cyclopropane-1,3'-pyrrolo[2,3-b]pyridin]-2'-one

##STR02149##

[3192] To a solution of 5'-bromo-1',2'-dihydrospiro[cyclopropane-1,3'-pyrrolo[2,3-b]pyridin]-2'-one (600 mg, 2.51 mmol) in DMF (8.37 mL) was added K.sub.2CO.sub.3 (1.04 g, 7.53 mmol) followed by 3-bromocyclobutan-1-one (0.56 g, 3.77 mmol). The reaction was allowed to stir at room temperature for 3 h, then was diluted with H.sub.2O (50 mL) and extracted with EtOAc (3×50 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and then concentrated in vacuo to give 5'-bromo-1'-(3-oxocyclobutyl)-1',2'-dihydrospiro[cyclopropane-1,3'-pyrrolo[2,3-b]pyridin]-2'-one which was used in the subsequent step without further purification. MS=307.02 [M+H].sup.+.

Step 2: 5'-bromo-1'-[(cis)-3-hydroxy-3-methylcyclobutyl]-1',2'-dihydrospiro[cyclopropane-1,3'-pyrrolo[2,3-b]pyridin]-2'-one

##STR02150##

[3193] To a solution of 5'-bromo-1'-(3-oxocyclobutyl)-1',2'-dihydrospiro[cyclopropane-1,3'-pyrrolo[2,3-b]pyridin]-2'-one (700 mg, 2.28 mmol) in THE (11.4 mL) at 0° C. was added 3.0 M methyl magnesium bromide solution in THE (0.836 mL, 2.51 mmol) dropwise. The reaction was allowed to stir at 0° C. for 1 h, then was quenched with saturated aqueous NH.sub.4Cl solution (30 mL), and the mixture was extracted with EtOAc (3×30 mL). The combined organic layers were dried over Na.sub.2SO.sub.4, filtered, and then concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 50 g cartridge, 0-100% EtOAc/Hexanes) to provide 5'-

bromo-1'-[(cis)-3-hydroxy-3-methylcyclobutyl]-1',2'-dihydrospiro[cyclopropane-1,3'-pyrrolo[2,3-b]pyridin]-2'-one MS=323.1 [M+H].sup.+.

Step 3: 1'-[(cis)-3-hydroxy-3-methylcyclobutyl]-5'-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1',2'-dihydrospiro[cyclopropane-1,3'-pyrrolo[2,3-b]pyridin]-2'-one

##STR02151##

[3194] To a solution of 5'-bromo-1'-[(cis)-3-hydroxy-3-methylcyclobutyl]-1',2'-dihydrospiro[cyclopropane-1,3'-pyrrolo[2,3-b]pyridin]-2'-one (480 mg, 1.49 mmol) and bis(pinacolato)diboron (0.566 g, 2.23 mmol) in 1,4-dioxane (4.95 mL) was added Pd(dppf)Cl.sub.2.Math.CH.sub.2Cl.sub.2 (0.121 g, 0.149 mmol) and KOAc (0.437 g, 4.46 mmol). The mixture was sparged with N.sub.2 for 5 min, and then the reaction was heated at 90° C. for 16 h. The reaction was cooled to room temperature, diluted with EtOAc (20 mL), and filtered over Celite. The resulting filtrate was concentrated in vacuo to provide 1'-[(cis)-3-hydroxy-3-methylcyclobutyl]-5'-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1',2'-dihydrospiro[cyclopropane-1,3'-pyrrolo[2,3-b]pyridin]-2'-one, which was used in the subsequent step without further purification. MS=371.3 [M+H].sup.+.

Step 4: 5'-hydroxy-1'-[(cis)-3-hydroxy-3-methylcyclobutyl]-1',2'-dihydrospiro[cyclopropane-1,3'-pyrrolo[2,3-b]pyridin]-2'-one

##STR02152##

[3195] To a solution of 1'-[(cis)-3-hydroxy-3-methylcyclobutyl]-5'-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1',2'-dihydrospiro[cyclopropane-1,3'-pyrrolo[2,3-b]pyridin]-2'-one (550 mg, 1.49 mmol) in MeOH (7.43 mL) was added 30% H.sub.2O.sub.2 in H.sub.2O (0.486 mL, 4.46 mmol) dropwise. The reaction was stirred for 1 h, then was diluted with MeOH (20 mL) and concentrated in vacuo to provide 5'-hydroxy-1'-[(cis)-3-hydroxy-3-methylcyclobutyl]-1',2'-dihydrospiro[cyclopropane-1,3'-pyrrolo[2,3-b]pyridin]-2'-one, which was used in the subsequent step without further purification. MS=261.1 [M+H].sup.+.

Step 5: 5'-(2-bromoethoxy)-1'-[(cis)-3-hydroxy-3-methylcyclobutyl]-1',2'-dihydrospiro[cyclopropane-1,3'-pyrrolo[2,3-b]pyridin]-2'-one

##STR02153##

[3196] To a solution of 5'-hydroxy-1'-[(cis)-3-hydroxy-3-methylcyclobutyl]-1',2'-dihydrospiro[cyclopropane-1,3'-pyrrolo[2,3-b]pyridin]-2'-one (400 mg, 1.54 mmol) in DMF (1.54 mL) was added 1,2-dibromoethane (7.68 mL) and Cs.sub.2CO.sub.3 (1.502 g, 4.61 mmol). The reaction was heated at 90° C. for 4 h, then was cooled to room temperature and diluted with EtOAc (30 mL). The solution was filtered over Celite and the filtrate was concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 25 g cartridge, 0-70% EtOAc/Hexanes) to provide 5'-(2-bromoethoxy)-1'-[(cis)-3-hydroxy-3-methylcyclobutyl]-1',2'-dihydrospiro[cyclopropane-1,3'-pyrrolo[2,3-b]pyridin]-2'-one. MS=367.13 [M+H].sup.+.

Step 6: 5-chloro-1'-(2-{2'-oxo-1'-[(cis)-3-hydroxy-3-methylcyclobutyl]-1',2'-dihydrospiro[cyclopropane-1,3'-pyrrolo[2,3-b]pyridin]-5'-yloxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR02154##

[3197] To a solution of 5'-(2-bromoethoxy)-1'-[(cis)-3-hydroxy-3-methylcyclobutyl]-1',2'-dihydrospiro[cyclopropane-1,3'-pyrrolo[2,3-b]pyridin]-2'-one (100 mg, 0.272 mmol) and 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 74.0 mg, 0.272 mmol, HCl salt) in DMF (1.361 mL) was added K.sub.2CO.sub.3 (113 mg, 0.817 mmol) and KI (45.0 g, 0.272 mmol). The reaction was heated to 60° C. for 3 h. After cooling to room temperature, the mixture was diluted with H.sub.2O (1.5 mL) and MeCN (0.5 mL), and then filtered. The filtrate was purified by reverse phase preparative HPLC (Phenomenex Kinetex C.sub.18 column, 5-50% MeCN in H.sub.2O with 0.1% formic acid modifier) to give 5-chloro-1'-(2-{2'-oxo-1'-[(cis)-3-hydroxy-3-methylcyclobutyl]-1',2'-dihydrospiro[cyclopropane-1,3'-pyrrolo[2,3-b]pyridin]-5'-yloxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 460). .sup.1H NMR (500

MHz, DMSO-d₆): δ 10.50 (s, 1H), 7.88 (d, J=2.7 Hz, 1H), 7.50 (d, J=2.2 Hz, 1H), 7.28 (d, J=2.6 Hz, 1H), 7.24 (dd, J=8.3, 2.1 Hz, 1H), 6.85 (d, J=8.2 Hz, 1H), 5.13 (s, 1H), 4.48-4.38 (m, 1H), 4.16 (t, J=5.7 Hz, 2H), 3.05-2.99 (m, 2H), 2.99-2.91 (m, 2H), 2.91-2.84 (m, 2H), 2.77-2.68 (m, 2H), 2.28-2.21 (m, 2H), 1.83-1.66 (m, 6H), 1.58-1.53 (m, 2H), 1.31 (s, 3H). MS=523.17 [M+H].sup.+.

Example 94

5-chloro-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-(.SUP.2.H.SUB.3.)methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 461)

##STR02155##

Step 1: 6-bromo-1-(3-oxocyclobutyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR02156##

[3198] To a solution of 6-bromo-3,4-dihydro-1H-1,8-naphthyridin-2-one (5.00 g, 22.0 mmol) in DMF (100 mL) was added K₂CO₃ (9.13 g, 66.1 mmol). The mixture was stirred at 60° C. for 1 h, and then 3-bromocyclobutan-1-one (4.92 g, 33.0 mmol) was added dropwise via syringe pump over 20 h. The mixture was poured into H₂O (200 mL) and a solid precipitated. The solid was isolated via filtration, and the filter cake was dried in vacuo to provide 6-bromo-1-(3-oxocyclobutyl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one, which was used in the subsequent step without further purification. MS=295.04 [M+H].sup.+.

Step 2: 6-bromo-1-[(cis)-3-hydroxy-3-(.SUP.2.H.SUB.3.)methylcyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR02157##

[3199] To a solution of 6-bromo-1-(3-oxocyclobutyl)-3,4-dihydro-1,8-naphthyridin-2-one (4.34 g, 14.7 mmol) in THE (73.5 mL) at 0° C. was added 1.0 M CD₃MgI in Et₂O (17.6 mL, 17.6 mmol) dropwise. The reaction was allowed to warm to room temperature and stirred for 1 h. The mixture was quenched with saturated aqueous NH₄Cl solution (100 mL) and extracted with EtOAc (3×100 mL). The combined organic layers were dried over anhydrous Na₂SO₄, then concentrated in vacuo. The residue was taken up in DCM, solids were removed by filtration, and the filtrate was concentrated in vacuo to give 6-bromo-1-[(cis)-3-hydroxy-3-(.sup.2H.sub.3)methylcyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one. MS=314.1 [M+H].sup.+.

Step 3: 1-[(cis)-3-hydroxy-3-(.SUP.2.H.SUB.3.)methylcyclobutyl]-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR02158##

[3200] A solution of 6-bromo-1-[(cis)-3-hydroxy-3-(.sup.2H.sub.3)methylcyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (4.13 g, 12.4 mmol), bis(pinacolato)diboron (4.71 g, 18.5 mmol), Pd(dppf)Cl₂ (0.904 g, 1.24 mmol), and KOAc (3.64 g, 37.1 mmol) in 1,4-dioxane (61.8 mL) was sparged with N₂, and then heated at 90° C. for 16 h. The reaction was cooled to room temperature and diluted with EtOAc (150 mL), then filtered over Celite. The resulting filtrate was concentrated in vacuo to give 1-[(cis)-3-hydroxy-3-(.sup.2H.sub.3)methylcyclobutyl]-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one, which was used in the subsequent step without further purification. MS=362.34 [M+H].sup.+.

Step 4: 6-hydroxy-1-[(cis)-3-hydroxy-3-(.SUP.2.H.SUB.3.)methylcyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR02159##

[3201] To a solution of 1-[(cis)-3-hydroxy-3-(.sup.2H.sub.3)methylcyclobutyl]-6-(4,4,5,5-tetramethyl-1,3,2-dioxaborolan-2-yl)-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (4.46 g, 12.346 mmol) in MeOH (61.7 mL) was added 30% H₂O₂ in H₂O (3.78 mL, 37.0 mmol) dropwise. The reaction was allowed to stir at room temperature for 2 h, then was concentrated in vacuo to give 6-hydroxy-1-[(cis)-3-hydroxy-3-(.sup.2H.sub.3)methylcyclobutyl]-1,2,3,4-

tetrahydro-1,8-naphthyridin-2-one, which was used in the subsequent step without further purification. MS=252.2 [M+H].sup.+.

Step 5: 6-(2-bromoethoxy)-1-[(cis)-3-hydroxy-3-(.SUP.2.H.SUB.3.)methylcyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR02160##

[3202] 6-hydroxy-1-[(cis)-3-hydroxy-3-(.sup.2H.sub.3)methylcyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (1.00 g, 3.98 mmol) was dissolved in DMF (3.98 mL), then Cs.sub.2CO.sub.3 (3.89 g, 11.9 mmol) and 1,2-dibromoethane (13.3 mL) were added and the reaction was heated at 90° C. for 8 h. The reaction was cooled to room temperature, filtered through a pad of Celite, and the filtrate was concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 50 g cartridge, 0-100% EtOAc/Hexanes) to provide 6-(2-bromoethoxy)-1-[(cis)-3-hydroxy-3-(.sup.2H.sub.3)methylcyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one.

MS=358.16 [M+H].sup.+.

Step 6: 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-(.SUP.2.H.SUB.3.)methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR02161##

[3203] To a solution of 6-(2-bromoethoxy)-1-[(cis)-3-hydroxy-3-(.sup.2H.sub.3)methylcyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one (500 mg, 1.40 mmol) and 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 0.381 g, 1.40 mmol, HCl salt) in DMF (2.79 mL) was added K.sub.2CO.sub.3 (0.579 g, 4.19 mmol) and KI (0.232 g, 1.40 mmol). The reaction was heated at 60° C. for 3 h. After cooling to room temperature, the reaction was poured over H.sub.2O (20 mL) and extracted with EtOAc (3×20 mL). The combined organic layers were combined and washed with 5% LiCl solution in H.sub.2O (50 mL). The organic phase was dried over Na.sub.2SO.sub.4, filtered, and concentrated in vacuo. The crude product was purified by reverse phase silica gel chromatography (Biotage 30 g C.sub.18 cartridge, 5-50% MeCN/H.sub.2O with 0.1% NH.sub.4OH) to give 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-(.sup.2H.sub.3)methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 461). .sup.1H NMR (500 MHz, DMSO-d.sub.6): δ 10.48 (s, 1H), 7.94 (d, J=2.9 Hz, 1H), 7.50 (d, J=2.2 Hz, 1H), 7.39 (d, J=2.9 Hz, 1H), 7.23 (dd, J=8.3, 2.1 Hz, 1H), 6.84 (d, J=8.3 Hz, 1H), 4.81 (s, 1H), 4.28-4.20 (m, 1H), 4.18 (t, J=5.7 Hz, 2H), 2.95-2.88 (m, 2H), 2.85 (t, J=5.7 Hz, 2H), 2.82-2.77 (m, 2H), 2.73-2.66 (m, 2H), 2.47-2.41 (m, 3H), 2.37-2.31 (m, 3H), 1.81-1.74 (m, 2H), 1.74-1.67 (m, 2H).

MS=514.27 [M+H].sup.+.

Example 95

5-chloro-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-(.SUP.2.H.SUB.3.)methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)(1,1,2,2-.SUP.2.H.SUB.4.)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 462)

##STR02162##

Step 1: 6-[2-bromo(1,1,2,2-.SUP.2.H.SUB.4.)ethoxy]-1-[(cis)-3-hydroxy-3-(.SUP.2.H.SUB.3.)methylcyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one

##STR02163##

[3204] To a solution of 6-hydroxy-1-[(cis)-3-hydroxy-3-(.sup.2H.sub.3)methylcyclobutyl]-3,4-dihydro-1,8-naphthyridin-2-one (Example 94, Step 4, 3.00 g, 11.9 mmol) in DMF (23.9 mL) was added dibromo(.sup.2H.sub.4)ethane (45.8 g, 239 mmol) and Cs.sub.2CO.sub.3 (11.7 g, 35.8 mmol). The reaction was heated at 90° C. for 8 h. The reaction was cooled to room temperature and diluted with EtOAc (100 mL), filtered through Celite, and the filtrate was concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 100 g cartridge, 0-80% EtOAc/Hexanes) to provide 6-[2-bromo(1,1,2,2-.sup.2H.sub.4)ethoxy]-1-[(cis)-3-hydroxy-3-(.sup.2H.sub.3)methylcyclobutyl]-1,2,3,4-tetrahydro-1,8-naphthyridin-2-one. MS=362.2 [M+H].sup.+.

Step 2: 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-(.SUP.2.H.SUB.3.)methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)(1,1,2,2-.SUP.2.H.SUB.4.)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR02164##

[3205] To a solution of 6-[2-bromo(1,1,2,2-.sup.2H.sub.4)ethoxy]-1-[(cis)-3-hydroxy-3-(.sup.2H.sub.3)methylcyclobutyl]-3,4-dihydro-1,8-naphthyridin-2-one (1.40 g, 3.86 mmol) and 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 1.06 g, 3.86 mmol, HCl salt) in DMF (7.729 mL) was added K.sub.2CO.sub.3 (1.60 g, 11.6 mmol) and KI (0.642 g, 3.86 mmol). The reaction was heated at 60° C. for 3 h. The reaction was cooled to room temperature and diluted with EtOAc (50 mL). The mixture was then filtered through a pad of Celite, and the filtrate concentrated in vacuo. The crude product was purified by reverse phase silica gel chromatography (Biotage 30 g C.sub.18 cartridge, 5-50% MeCN/H.sub.2O with 0.1% NH.sub.4OH) to give 5-chloro-1'-[2-({7-oxo-8-[(cis)-3-hydroxy-3-(.sup.2H.sub.3)methylcyclobutyl]-5,6,7,8-tetrahydro-1,8-naphthyridin-3-yl}oxy)(1,1,2,2-.sup.2H.sub.4)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 462). .sup.1H NMR (500 MHz, DMSO-d.sub.6, 22/24 H): δ 10.43 (s, 1H), 7.87 (d, J=2.9 Hz, 1H), 7.44 (d, J=2.1 Hz, 1H), 7.32 (d, J=2.9 Hz, 1H), 7.17 (dd, J=8.2, 2.1 Hz, 1H), 6.78 (d, J=8.2 Hz, 1H), 4.76 (s, 1H), 4.17 (p, J=8.3 Hz, 1H), 2.92-2.79 (m, 2H), 2.73 (t, J=7.2 Hz, 2H), 2.67-2.58 (m, 2H), 2.41-2.34 (m, 2H), 2.31-2.22 (m, 2H), 1.75-1.59 (m, 4H). MS=518.3

[M+H].sup.+.

Example 97

5-chloro-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl}oxy)(1,1,2,2-.SUP.2.H.SUB.4.)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 463)

##STR02165##

Step 1: (cis)-3-{5-[2-bromo(1,1,2,2-.SUP.2.H.SUB.4.)ethoxy]-7-(trifluoromethyl)-1,3-benzodiazol-1-yl}-1-methylcyclobutan-1-ol

##STR02166##

[3206] To a solution of 1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-ol (Intermediate A-92, 250 mg, 0.873 mmol) and Cs.sub.2CO.sub.3 (0.854 g, 2.62 mmol) in DMF (1.747 mL) was added dibromo(.sup.2H.sub.4)ethane (0.838 g, 4.367 mmol). The reaction was heated at 90° C. for 3 h. The reaction was cooled to room temperature and diluted with EtOAc (30 mL), and then filtered through a pad of Celite. The filtrate was concentrated, and the filtrate was purified by flash silica gel chromatography (Biotage 25 g cartridge, 0-10% MeOH/DCM) to provide (cis)-3-{5-[2-bromo(1,1,2,2-.sup.2H.sub.4)ethoxy]-7-(trifluoromethyl)-1,3-benzodiazol-1-yl}-1-methylcyclobutan-1-ol. MS=397.2/399.2 [M+H].sup.+.

Step 2: 5-chloro-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl}oxy)(1,1,2,2-.SUP.2.H.SUB.4.)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR02167##

[3207] To a solution of (cis)-3-{5-[2-bromo(1,1,2,2-.sup.2H.sub.4)ethoxy]-7-(trifluoromethyl)-1,3-benzodiazol-1-yl}-1-methylcyclobutan-1-ol (257 mg, 0.647 mmol) and 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 0.177 g, 0.647 mmol, HCl salt) in DMF (1.29 mL) was added K.sub.2CO.sub.3 (0.268 g, 1.94 mmol) and KI (0.107 g, 0.647 mmol). The reaction was heated at 60° C. for 3 h, then was cooled to room temperature and the solution was diluted with H.sub.2O 1.5 mL) and MeCN 1.5 mL). The mixture was purified by reverse phase silica gel chromatography (Biotage 30 g C.sub.18 cartridge, 5-60% MeCN/H.sub.2O with 0.1% NH.sub.4OH) to give 5-chloro-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl}oxy)(1,1,2,2-.sup.2H.sub.4)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 463). .sup.1H NMR (500 MHz, DMSO-d.sub.6): δ 10.42 (s, 1H), 8.61 (s, 1H), 7.54 (d, J=2.4 Hz, 1H), 7.43 (d, J=2.1 Hz, 1H), 7.19 (d,

J=2.4 Hz, 1H), 7.16 (dd, J=8.2, 2.1 Hz, 1H), 6.78 (d, J=8.2 Hz, 1H), 5.25 (s, 1H), 4.51 (p, J=8.3 Hz, 1H), 2.92-2.83 (m, 2H), 2.70-2.60 (m, 2H), 2.59-2.48 (m, 4H), 1.78-1.69 (m, 2H), 1.68-1.59 (m, 2H), 1.27 (s, 3H). MS=553.2 [M+H].sup.+.

Alternative Procedure for Example 97

5-chloro-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl}oxy)(1,1,2,2-.SUP.2.H.SUB.4.)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 463)

##STR02168##

Step 1: 1-methyl-4-({2-[(4-methylbenzenesulfonyl)oxy](1,1,2,2-.SUP.2.H.SUB.4.)ethoxy}sulfonyl)benzene

##STR02169##

[3208] To a 0° C. mixture of (.sup.2H.sub.4)ethane-1,2-diol (5.00 g, 75.7 mmol) and TsCl (31.7 g, 166 mmol) in DCM (150 mL) was added dropwise TEA (26.3 mL, 189 mmol). The mixture was stirred at room temperature for 16 h. The mixture was poured into ice water (300 mL) and extracted with DCM (2×100 mL). The combined organic layers were washed with 10% w/w citric acid in H.sub.2O (200 mL) and brine (100 mL), dried over Na.sub.2SO.sub.4, and concentrated in vacuo. The residue was triturated in MTBE (150 mL) for 30 min. The mixture was filtered to collect the solid, which was dried in vacuo to provide 1-methyl-4-({2-[(4-methylbenzenesulfonyl)oxy](1,1,2,2-.sup.2H.sub.4)ethoxy}sulfonyl)benzene, which was used in the subsequent step without further purification. MS=374.9 [M+H].sup.+.

Step 2: 2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl}oxy)(1,1,2,2-.SUP.2.H.SUB.4.)ethyl 4-methylbenzene-1-sulfonate

##STR02170##

[3209] A mixture of 1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-ol (Intermediate A-92, 1.00 g, 3.49 mmol), 1-methyl-4-({2-[(4-methylbenzenesulfonyl)oxy](1,1,2,2-.sup.2H.sub.4)ethoxy}sulfonyl)benzene (3.92 g, 10.5 mmol) and K.sub.2CO.sub.3 (1.45 g, 10.5 mmol) in DMF (60 mL) was stirred at 50° C. for 16 h. After cooling to room temperature, the mixture was poured into H.sub.2O (100 mL) and extracted with EtOAc (3×50 mL). The combined organic layers were washed with brine (100 mL), dried over Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (Sepaflash 20 g cartridge, 0-80% EtOAc/Petroleum ether) to provide 2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl}oxy)(1,1,2,2-.sup.2H.sub.4)ethyl 4-methylbenzene-1-sulfonate. MS=489.1 [M+H].sup.+.

Step 3: 5-chloro-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl}oxy)(1,1,2,2-.SUP.2.H.SUB.4.)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 463)

##STR02171##

[3210] A mixture of 2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl}oxy)(1,1,2,2-.sup.2H.sub.4)ethyl 4-methylbenzene-1-sulfonate (700 mg, 1.43 mmol), 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 431 mg, 1.58 mmol, HCl salt) and NaHCO.sub.3 (602 mg, 7.16 mmol) in MeCN (30 mL) was stirred at 80° C. for 16 h. After cooling to room temperature, solids precipitated out. The solids were isolated by filtration, and the filter cake was further purified by reverse phase preparative HPLC (Phenomenex Luna C.sub.18 column, 5-50% MeCN:0.04% HCl in H.sub.2O) to give 5-chloro-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl}oxy)(1,1,2,2-.sup.2H.sub.4)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 463, HCl salt, 24/24 H). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 11.09 (br s, 0.5H), 10.81 (s, 0.5H), 10.73 (s, 0.5H), 10.49 (br s, 0.5H), 9.08-9.03 (m, 1H), 7.99 (s, 0.5H), 7.73-7.71 (m, 1H), 7.50-7.40 (m, 1H), 7.26-7.24 (m, 1H), 7.17 (s, 0.5H), 6.94-6.88 (m, 1H), 4.64-4.55 (m, 1H), 3.88-3.59 (m, 4H), 2.67-2.50 (m, 5H), 2.21-1.97 (m, 1H), 1.98-1.95 (m, 2H), 1.34 (s, 3H). MS=552.9

[M+H].sup.+.

Example 98

5-chloro-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-indazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 464)

##STR02172##

[3211] A mixture of 2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-indazol-5-yl}oxy)ethyl 4-methylbenzene-1-sulfonate (Intermediate A-123, 80.0 mg, 165 mol), 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 49.6 mg, 182 mol, HCl salt) and NaHCO₃ (69.4 mg, 826 mol) in MeCN (4 mL) was stirred at 80° C. for 16 h. The reaction mixture was cooled to room temperature and filtered. The filtrate was concentrated under reduced pressure. The filtrate was purified by reverse phase preparative HPLC (Phenomenex Luna C₁₈ column, 5-35% MeCN:0.04% HCl in H₂O) to give 5-chloro-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-indazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 464). ¹H NMR (400 MHz, DMSO-d₆, HCl salt): δ 10.80 (s, 0.5H), 10.73 (s, 0.5H), 10.60 (br s, 0.5H), 10.09 (s, 0.5H), 8.34 (d, J=2.8 Hz, 1H), 7.99 (s, 0.5H), 7.74 (s, 1H), 7.58 (dd, J=12.0 Hz, 2.0 Hz, 1H), 7.35-7.25 (m, 1H), 7.16 (d, J=1.6 Hz, 0.5H), 6.94-6.88 (m, 1H), 5.29 (br s, 1H), 4.80-4.73 (m, 1H), 4.54-4.53 (m, 2H), 3.90-3.75 (m, 5H), 3.45-3.35 (m, 1H), 2.76 (t, J=10.8 Hz, 2H), 2.50-2.45 (m, 3H), 2.24-2.12 (m, 1H), 2.06-1.86 (m, 2H), 1.16 (s, 3H). MS=549.2 [M+H].sup.+.

[3212] The following compounds in Table 38.20 were prepared according to procedures similar to steps described for Example 98 using the appropriate starting materials or common intermediates.

TABLE-US-00058 TABLE 38.20 Exact Mass Intermediates # Structure IUPAC Name [M + H].sup.+ Used 465 [02173]  2-oxo-1'-[2-({1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-indazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile Calc'd 540.2 Found 540.2 A-123 & B-9

Example 99

5-chloro-1'-[2-({2-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 466)

##STR02174##

Step 1: (cis)-N-[2-amino-5-bromo-3-(trifluoromethyl)phenyl]-3-hydroxy-3-methylcyclobutane-1-carboxamide

##STR02175##

[3213] To a solution of 3-hydroxy-3-methyl-cyclobutanecarboxylic acid (1.53 g, 11.8 mmol) in DCM (30 mL) was added DIEA (5.12 mL, 29.4 mmol) and HATU (5.37 g, 14.1 mmol). After stirring at room temperature for 20 min, 5-bromo-3-(trifluoromethyl)benzene-1,2-diamine (3.00 g, 11.76 mmol) was added. The mixture was stirred at room temperature for 2 h, then was poured into H₂O (50 mL) and extracted with EtOAc (3×30 mL). The combined organic layers were dried over anhydrous Na₂SO₄, filtered and concentrated in vacuo to give (cis)-N-[2-amino-5-bromo-3-(trifluoromethyl)phenyl]-3-hydroxy-3-methylcyclobutane-1-carboxamide.

MS=367.0/369.0 [M+H].sup.+.

Step 2: (cis)-3-[5-bromo-7-(trifluoromethyl)-1H-1,3-benzodiazol-2-yl]-1-methylcyclobutan-1-ol

##STR02176##

[3214] A solution of (cis)-N-[2-amino-5-bromo-3-(trifluoromethyl)phenyl]-3-hydroxy-3-methylcyclobutane-1-carboxamide (5.00 g, 13.6 mmol) in acetic acid (50 mL) was heated to 90° C. and stirred at 90° C. for 5 h. After cooling to 0° C., the reaction mixture was adjusted to pH=7 by addition of saturated aqueous Na₂CO₃ solution, then was extracted with EtOAc (3×30 mL). The combined organic layers were dried over anhydrous Na₂SO₄, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Sepaflash 40 g cartridge, 0-60% EtOAc/Petroleum ether) to provide (cis)-3-[5-bromo-7-(trifluoromethyl)-1H-1,3-benzodiazol-2-yl]-1-methylcyclobutan-1-ol. MS=349.0/350.9 [M+H].sup.+.

Step 3: 5-bromo-2-[(cis)-3-methyl-3-{[2-(trimethylsilyl)ethoxy]methoxy}cyclobutyl]-7-(trifluoromethyl)-1-{[2-(trimethylsilyl)ethoxy]methyl}-1H-1,3-benzodiazole

##STR02177##

[3215] To a 0° C. solution of (cis)-3-[5-bromo-7-(trifluoromethyl)-1H-1,3-benzodiazol-2-yl]-1-methylcyclobutan-1-ol (1.50 g, 4.30 mmol) in THE (20 mL) was added NaH (309 mg, 7.73 mmol, 60% in mineral oil) portion wise. The mixture was stirred at 0° C. for 30 min, and then a solution of SEMCl (1.52 mL, 8.59 mmol) in THE (8 mL) was added dropwise. The resulting mixture was warmed to room temperature and then stirred for 16 h. The reaction was cooled to 0-10° C., quenched with saturated aqueous NH₄Cl solution (20 mL), and then extracted with EtOAc (3×20 mL). The combined organic layers were dried over anhydrous Na₂SO₄, filtered, and concentrated in vacuo. The residue was purified by reverse phase preparative HPLC (Welch Xtimate C₁₈ column, 45-65% MeCN:10 mM NH₄HCO₃ in H₂O) to give 5-bromo-2-[(cis)-3-methyl-3-{[2-(trimethylsilyl)ethoxy]methoxy}cyclobutyl]-7-(trifluoromethyl)-1-{[2-(trimethylsilyl)ethoxy]methyl}-1H-1,3-benzodiazole. MS=609.0/611.0 [M+H]⁺.

Step 4: 5-(5,5-dimethyl-1,3,2-dioxaborinan-2-yl)-2-[(cis)-3-methyl-3-{[2-(trimethylsilyl)ethoxy]methoxy}cyclobutyl]-7-(trifluoromethyl)-1-{[2-(trimethylsilyl)ethoxy]methyl}-1H-1,3-benzodiazole

##STR02178##

[3216] A mixture of 5-bromo-2-[(cis)-3-methyl-3-{[2-(trimethylsilyl)ethoxy]methoxy}cyclobutyl]-7-(trifluoromethyl)-1-{[2-(trimethylsilyl)ethoxy]methyl}-1H-1,3-benzodiazole (300 mg, 492 mol), KOAc (72.4 mg, 738 mol), Pd(dppf)Cl₂ (18.0 mg, 24.6 mol), and bis(neopentyl glycolato)diboron (222 mg, 984 mol) in 1,4-dioxane (5 mL) was degassed and purged with N₂ (3×), and then the mixture was heated to 90° C. and stirred at 90° C. for 3 h under N₂ atmosphere. After cooling to room temperature, the reaction mixture was filtered, and the filtrate was concentrated in vacuo to give 5-(5,5-dimethyl-1,3,2-dioxaborinan-2-yl)-2-[(cis)-3-methyl-3-{[2-(trimethylsilyl)ethoxy]methoxy}cyclobutyl]-7-(trifluoromethyl)-1-{[2-(trimethylsilyl)ethoxy]methyl}-1H-1,3-benzodiazole, which was used in the subsequent step without further purification. MS=575.3 [M-C₅H₈+H]⁺.

Step 5: 2-[(cis)-3-methyl-3-{[2-(trimethylsilyl)ethoxy]methoxy}cyclobutyl]-7-(trifluoromethyl)-1-{[2-(trimethylsilyl)ethoxy]methyl}-1H-1,3-benzodiazol-5-ol

##STR02179##

[3217] To a solution of 5-(5,5-dimethyl-1,3,2-dioxaborinan-2-yl)-2-[(cis)-3-methyl-3-{[2-(trimethylsilyl)ethoxy]methoxy}cyclobutyl]-7-(trifluoromethyl)-1-{[2-(trimethylsilyl)ethoxy]methyl}-1H-1,3-benzodiazole (600 mg, 934 mol) in THE (6 mL) and H₂O (2 mL) was added Oxone (861 mg, 1.40 mmol). The mixture was stirred at room temperature for 2 h, then was quenched with saturated aqueous Na₂SO₃ solution (30 mL), and extracted with EtOAc (2×50 mL). The combined organic layers were dried over anhydrous Na₂SO₄, filtered, and concentrated in vacuo. The residue was purified by flash silica gel chromatography (Biotage 12 g cartridge, 0-19% EtOAc/Petroleum ether) to provide 2-[(cis)-3-methyl-3-{[2-(trimethylsilyl)ethoxy]methoxy}cyclobutyl]-7-(trifluoromethyl)-1-{[2-(trimethylsilyl)ethoxy]methyl}-1H-1,3-benzodiazol-5-ol. MS=547.2 [M+H]⁺.

Step 6: 5-(2-bromoethoxy)-2-[(cis)-3-methyl-3-{[2-(trimethylsilyl)ethoxy]methoxy}cyclobutyl]-7-(trifluoromethyl)-1-{[2-(trimethylsilyl)ethoxy]methyl}-1H-1,3-benzodiazole

##STR02180##

[3218] To a solution of 2-[(cis)-3-methyl-3-{[2-(trimethylsilyl)ethoxy]methoxy}cyclobutyl]-7-(trifluoromethyl)-1-{[2-(trimethylsilyl)ethoxy]methyl}-1H-1,3-benzodiazol-5-ol (130 mg, 238 mol) and 1,2-dibromoethane (206 µL, 2.38 mmol) in i-PrOH (5 mL) was added Cs₂CO₃ (232 mg, 713 mol). The mixture was heated to 60° C. and stirred at 60° C. for 12 h. After cooling to room temperature, the reaction mixture was filtered, and the filtrate was concentrated in vacuo.

The residue was purified by flash silica gel chromatography (Sepaflysh 4 g cartridge, 0-15% EtOAc/Petroleum ether) to provide 5-(2-bromoethoxy)-2-[(cis)-3-methyl-3-{[2-(trimethylsilyl)ethoxy]methoxy}cyclobutyl]-7-(trifluoromethyl)-1-{[2-(trimethylsilyl)ethoxy]methyl}-1H-1,3-benzodiazole. MS=653.1/655.1 [M+H].sup.+
Step 7: 5-chloro-1'-[2-({2-[(cis)-3-methyl-3-{[2-(trimethylsilyl)ethoxy]methoxy}cyclobutyl]-7-(trifluoromethyl)-1-{[2-(trimethylsilyl)ethoxy]methyl}-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR02181##

[3219] To a solution of 5-(2-bromoethoxy)-2-[(cis)-3-methyl-3-{[2-(trimethylsilyl)ethoxy]methoxy}cyclobutyl]-7-(trifluoromethyl)-1-{[2-(trimethylsilyl)ethoxy]methyl}-1H-1,3-benzodiazole (60.0 mg, 91.78 mol) and 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 30.1 mg, 110 mol, HCl salt) in MeCN (3 mL) was added NaHCO₃ (38.6 mg, 459 mol). The mixture was heated to 80° C. and stirred at 80° C. for 16 h. After cooling to room temperature, the reaction mixture was filtered and the filtrate was concentrated in vacuo to give 5-chloro-1'-[2-({2-[(cis)-3-methyl-3-{[2-(trimethylsilyl)ethoxy]methoxy}cyclobutyl]-7-(trifluoromethyl)-1-{[2-(trimethylsilyl)ethoxy]methyl}-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one, which was used in the subsequent step without further purification. MS=809.3 [M+H].sup.+.

Step 8: 5-chloro-1'-[2-({2-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 466)

##STR02182##

[3220] To a mixture of 5-chloro-1'-[2-({2-[(cis)-3-methyl-3-{[2-(trimethylsilyl)ethoxy]methoxy}cyclobutyl]-7-(trifluoromethyl)-1-{[2-(trimethylsilyl)ethoxy]methyl}-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (90.0 mg, 111 mol) in DCM (2 mL) was added TFA (165 µL, 2.22 mmol). The mixture was stirred at room temperature for 16 h, and then DCM was removed under reduced pressure. The residue was dissolved in THE (2 mL) and then 30% NH₃·H₂O in H₂O (285 µL, 2.22 mmol) was added. The mixture was stirred at room temperature for 1 h, then was concentrated under reduced pressure to remove THF. The residual mixture was purified by reverse phase preparative HPLC (Waters Xbridge BEH C₁₈ column, 30-60% MeCN:10 mM NH₄HCO₃ in H₂O) to give 5-chloro-1'-[2-({2-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 466). ¹H NMR (400 MHz, DMSO-d₆): δ 12.43 (br s, 1H), 10.49 (s, 1H), 7.50 (s, 1H), 7.29-7.20 (m, 2H), 7.06 (s, 1H), 6.84 (d, J=8.4 Hz, 1H), 5.15 (br s, 1H), 4.21 (t, J=5.6 Hz, 2H), 3.32-3.20 (m, 1H), 2.90-2.85 (m, 4H), 2.73-2.68 (m, 2H), 2.40-2.30 (m, 4H), 1.80-1.70 (m, 4H), 1.19 (s, 3H). MS=549.1 [M+H].sup.+.

Example 100

5-chloro-1'-(2-{[3-(hydroxymethyl)-2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-2,3-dihydro-1H-1,3-benzodiazol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 467) and 5-chloro-1'-[2-({2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-2,3-dihydro-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 468)

##STR02183##

Step 1: 5-bromo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-2,3-dihydro-1H-1,3-benzodiazol-2-one

##STR02184##

[3221] A solution of (cis)-3-{[2-amino-4-bromo-6-(trifluoromethyl)phenyl]amino}-1-methylcyclobutan-1-ol (General procedure for Intermediate A-92, Step 3, 2.00 g, 5.90 mmol) and carbonyl diimidazole (1.43 g, 8.85 mmol) in THE (20 mL) was warmed to 50° C. and stirred at 50°

C. for 16 h. After cooling to room temperature, the reaction mixture was poured into H.sub.2O (20 mL) and extracted with EtOAc (2×20 mL). The combined organic layers were concentrated in vacuo. The residue was dissolved into THF (30 mL) and H.sub.2O (10 mL) and LiOH.Math.H.sub.2O (1.01 g, 24.2 mmol) was added. The mixture was stirred at 50° C. for 2 h. After cooling to room temperature, the reaction mixture was poured into H.sub.2O (100 mL) and extracted with EtOAc (2×200 mL). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4, filtered, and concentrated in vacuo give 5-bromo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-2,3-dihydro-1H-1,3-benzodiazol-2-one, which was used in the subsequent step without further purification. MS=365.0/367.0 [M+H].sup.+.

Step 2: 6-bromo-3-[(cis)-3-hydroxy-3-methylcyclobutyl]-4-(trifluoromethyl)-1-[[2-(trimethylsilyl)ethoxy]methyl]-2,3-dihydro-1H-1,3-benzodiazol-2-one

##STR02185##

[3222] To a 0° C. mixture of 5-bromo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-2,3-dihydro-1H-1,3-benzodiazol-2-one (2.70 g, 7.39 mmol) in THE (30 mL) was added NaH (887 mg, 22.2 mmol, 60% in mineral oil) portion wise. (2-(chloromethoxy)ethyl)trimethylsilane (2.62 mL, 14.8 mmol) was added dropwise to the 0° C. reaction mixture. The mixture was stirred at 0° C. for 2 h, then was quenched with ice water (30 mL). After stirring at 0° C. for an additional 10 min, the mixture was extracted with EtOAc (3×30 mL). The combined organic layers were washed with brine (30 mL), dried over anhydrous Na.sub.2SO.sub.4, filtered, and concentrated under reduced pressure. The residue was purified by flash silica gel chromatography (Sepaflash 20 g cartridge, 0-60% EtOAc/Petroleum ether) to provide 6-bromo-3-[(cis)-3-hydroxy-3-methylcyclobutyl]-4-(trifluoromethyl)-1-[[2-(trimethylsilyl)ethoxy]methyl]-2,3-dihydro-1H-1,3-benzodiazol-2-one. MS=495.0/497.0 [M+H].sup.+.

Step 3: 6-(5,5-dimethyl-1,3,2-dioxaborinan-2-yl)-3-[(cis)-3-hydroxy-3-methylcyclobutyl]-4-(trifluoromethyl)-1-[[2-(trimethylsilyl)ethoxy]methyl]-2,3-dihydro-1H-1,3-benzodiazol-2-one

##STR02186##

[3223] A mixture of 6-bromo-3-[(cis)-3-hydroxy-3-methylcyclobutyl]-4-(trifluoromethyl)-1-[[2-(trimethylsilyl)ethoxy]methyl]-2,3-dihydro-1H-1,3-benzodiazol-2-one (1.20 g, 2.42 mmol), Pd(dppf)Cl.sub.2 (88.6 mg, 121 mol), KOAc (357 mg, 3.63 mmol) and bis(neopentyl glycolato)diboron (1.09 g, 4.84 mmol) in 1,4-dioxane (15 mL) was degassed and purged with N.sub.2 (3×), and then the mixture was stirred at 90° C. for 3 h under N.sub.2 atmosphere. After cooling to room temperature, the reaction mixture was filtered and the filtrate was concentrated under reduced pressure to give 6-(5,5-dimethyl-1,3,2-dioxaborinan-2-yl)-3-[(cis)-3-hydroxy-3-methylcyclobutyl]-4-(trifluoromethyl)-1-[[2-(trimethylsilyl)ethoxy]methyl]-2,3-dihydro-1H-1,3-benzodiazol-2-one, which was used in the subsequent step without further purification. MS=461.2 [M-C.sub.5H.sub.8+H].sup.+.

Step 4: 6-hydroxy-3-[(cis)-3-hydroxy-3-methylcyclobutyl]-4-(trifluoromethyl)-1-[[2-(trimethylsilyl)ethoxy]methyl]-2,3-dihydro-1H-1,3-benzodiazol-2-one

##STR02187##

[3224] To a solution of 6-(5,5-dimethyl-1,3,2-dioxaborinan-2-yl)-3-[(cis)-3-hydroxy-3-methylcyclobutyl]-4-(trifluoromethyl)-1-[[2-(trimethylsilyl)ethoxy]methyl]-2,3-dihydro-1H-1,3-benzodiazol-2-one (2.30 g, 4.35 mmol) in THE (16 mL) and H.sub.2O (8 mL) was added Oxone (4.01 g, 6.53 mmol). The mixture was stirred at room temperature for 1 h. The reaction mixture was quenched with saturated aqueous Na.sub.2SO.sub.3 solution (20 mL) and extracted with EtOAc (2×20 mL). The combined organic layers were dried over anhydrous Na.sub.2SO.sub.4, filtered and concentrated under reduced pressure to give 6-hydroxy-3-[(cis)-3-hydroxy-3-methylcyclobutyl]-4-(trifluoromethyl)-1-[[2-(trimethylsilyl)ethoxy]methyl]-2,3-dihydro-1H-1,3-benzodiazol-2-one, which was used in the subsequent step without further purification. MS=433.2 [M+H].sup.+.

Step 5: 6-(2-bromoethoxy)-3-[(cis)-3-hydroxy-3-methylcyclobutyl]-4-(trifluoromethyl)-1-[[2-(trimethylsilyl)ethoxy]methyl]-2,3-dihydro-1H-1,3-benzodiazol-2-one

##STR02188##

[3225] To a solution of 6-hydroxy-3-[(cis)-3-hydroxy-3-methylcyclobutyl]-4-(trifluoromethyl)-1-[[2-(trimethylsilyl)ethoxy]methyl]-2,3-dihydro-1H-1,3-benzodiazol-2-one (1.60 g, 3.70 mmol) and 1,2-dibromoethane (2.79 mL, 37.0 mmol) in i-PrOH (20 mL) was added Cs.sub.2CO.sub.3 (3.62 g, 11.1 mmol). The mixture was stirred at 60° C. for 12 h. After cooling to room temperature, the reaction mixture was filtered, and the filtrate was concentrated in vacuo. The residue was purified by flash silica gel chromatography (Sepaflash 20 g cartridge, 0-18% EtOAc/Petroleum ether) to provide 6-(2-bromoethoxy)-3-[(cis)-3-hydroxy-3-methylcyclobutyl]-4-(trifluoromethyl)-1-[[2-(trimethylsilyl)ethoxy]methyl]-2,3-dihydro-1H-1,3-benzodiazol-2-one. MS=539.1/541.0 [M+H].sup.+

Step 6: 5-chloro-1'-[2-({2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-3-[[2-(trimethylsilyl)ethoxy]methyl]-2,3-dihydro-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one

##STR02189##

[3226] To a solution of 6-(2-bromoethoxy)-3-[(cis)-3-hydroxy-3-methylcyclobutyl]-4-(trifluoromethyl)-1-[[2-(trimethylsilyl)ethoxy]methyl]-2,3-dihydro-1H-1,3-benzodiazol-2-one (100 mg, 185 mol) and 5-chloro-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Intermediate B-4, 50.6 mg, 185 mol, HCl salt) in MeCN (5 mL) was added NaHCO.sub.3 (77.9 mg, 927 mol). The mixture was warmed to 80° C. and stirred at 80° C. for 16 h. After cooling to room temperature, the reaction mixture was filtered, and the filtrate was concentrated in vacuo to give 5-chloro-1'-[2-({2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-3-[[2-(trimethylsilyl)ethoxy]methyl]-2,3-dihydro-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one, which was used in the subsequent step without further purification. MS=695.2 [M+H].sup.+.

Step 7: 5-chloro-1'-(2-{[3-(hydroxymethyl)-2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-2,3-dihydro-1H-1,3-benzodiazol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 467)

##STR02190##

[3227] To a mixture of 5-chloro-1'-[2-({2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-3-[[2-(trimethylsilyl)ethoxy]methyl]-2,3-dihydro-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (70.0 mg, 101 mol) in DCM (1 mL) was added TFA (74.8 µL, 1.01 mmol). The mixture was stirred at room temperature for 1 h, then concentrated in vacuo. The residue was purified by reverse phase preparative HPLC (Phenomenex Luna OBD C.sub.18 column, 5-35% MeCN:0.04% HCl in H.sub.2O) to give 5-chloro-1'-(2-{[3-(hydroxymethyl)-2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-2,3-dihydro-1H-1,3-benzodiazol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 467, HCl salt). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 10.82 (s, 0.5H), 10.74 (s, 0.5H), 10.41 (br s, 0.5H), 10.02 (br s, 0.5H), 8.00 (s, 0.5H), 7.40 (s, 1H), 7.38-7.30 (m, 1H), 7.16 (s, 0.5H), 7.06 (s, 1H), 6.94-6.90 (m, 1H), 6.62-6.50 (m, 1H), 5.28 (d, J=5.6 Hz, 2H), 5.27-5.15 (m, 1H), 4.52-4.50 (m, 2H), 4.30-4.25 (m, 1H), 3.95-3.50 (m, 6H), 3.21 (d, J=9.2 Hz, 2H), 2.50-2.40 (m, 1H), 2.20-2.17 (m, 3H), 2.18-1.90 (m, 2H), 1.23 (s, 3H). MS=595.1 [M+H].sup.+.

Step 8: 5-chloro-1'-[2-({2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-2,3-dihydro-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 468)

##STR02191##

[3228] To a solution of 5-chloro-1'-(2-{[3-(hydroxymethyl)-2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-2,3-dihydro-1H-1,3-benzodiazol-5-yl]oxy}ethyl)-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (170 mg, 286 mol) in THE (2 mL) was added 30%

NH.sub.3.Math.H.sub.2O in H.sub.2O (4.00 mL, 31.2 mmol). The mixture was stirred at room temperature for 2 h, then was concentrated in vacuo. The residue was purified by reverse phase preparative HPLC (Waters Xbridge OBD C.sub.18 column, 20-55% MeCN:10 mM NH.sub.4HCO.sub.3 in H.sub.2O) to give 5-chloro-1'-[2-({2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-2,3-dihydro-1H-1,3-benzodiazol-5-yl}oxy)ethyl]-1,2-dihydrospiro[indole-3,4'-piperidin]-2-one (Compound 468). .sup.1H NMR (400 MHz, DMSO-d.sub.6): δ 11.38 (s, 1H), 10.49 (s, 1H), 7.48 (d, J=2.0 Hz, 1H), 7.25-7.20 (m, 1H), 6.88 (s, 2H), 6.84 (d, J=8.0 Hz, 1H), 5.13 (s, 1H), 4.25-4.15 (m, 3H), 3.21 (t, J=9.2 Hz, 2H), 2.93-2.82 (m, 4H), 2.71-2.67 (m, 2H), 2.20-2.18 (m, 2H), 1.81-1.66 (m, 4H), 1.27 (s, 3H). MS=565.2 [M+H].sup.+.

[3229] The following compounds in Table 38.21 were prepared according to procedures similar to steps described for Example 100 using the appropriate starting materials or common intermediates.

TABLE-US-00059	TABLE 38.21	Exact Mass	#	Structure	IUPAC Name	[M + H].sup.+
Intermediates Used	469	[02192]		1'-(2-({3-(hydroxymethyl)-2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-2,3-dihydro-1H-1,3-benzodiazol-5-yl}oxy)ethyl)-2-oxo-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile	Calc'd 586.2 Found 586.1	
B-9	470	[02193]		2-oxo-1'-(2-({2-oxo-1-[(cis)-3-hydroxy-3-methylcyclobutyl]-7-(trifluoromethyl)-2,3-dihydro-1H-1,3-benzodiazol-5-yl}oxy)ethyl)-1,2-dihydrospiro[indole-3,4'-piperidine]-5-carbonitrile	Calc'd 556.2 Found 556.2	

BIOLOGICAL EXAMPLES

Example B1

[3230] This example shows that compounds of the present disclosure are able to inhibit calcium transport by APOL1.

[3231] A HEK293 clonal cell line was generated to stably express GCaMP6f, a genetically encoded calcium indicator, and inducibly express APOL1 G2 (HEK T-REx/GCaMP6f/APOL1 G2 K.sub.6.3). Cells were maintained in the following standard complete medium: DMEM with 4.5 g/L glucose and sodium pyruvate (BioWhittaker, Lonza, BE12-614F), supplemented with 10% FBS Performance Plus (Gibco, 16000044), 1% penicillin-streptomycin (BioWhittaker, DE17-602E), 2 mM ultraglutamine-1 (BioWhittaker cat. BE 17-605/U1), 50 μ g/mL Zeocin (InvivoGen, ant-zn), 2.5 μ g/mL Blastidin (InvivoGen, ant-bl-5), and 25 μ g/mL Hygromycin (InvivoGen, ant-hg). Standard propagation conditions consisted of plating 9×10^6 , 4×10^6 , 2×10^6 cells in a T225 flasks to be processed after 2, 3, or 4 days, respectively.

[3232] A source plate was generated containing 20 serially diluted compounds in DMSO (duplicate 8-point dose response). Next, 0.8 μ L of compounds were transferred from the source plate to a destination plate prefilled with 79.2 μ L of Ca.sup.2+ free Tyrode's buffer (130 mM NaCl, 5 mM KCl, 1 mM MgCl.sub.2, 5 mM NaHCO.sub.3, 20 mM HEPES at pH 7.4). The destination plate was placed on a plate shaker (5 seconds at 2000 rpm) to mix. This process resulted in a destination plate with 2 $\times$ concentrated compound solutions. All transfer and mixing steps were conducted with an CyBi®-Well dispenser.

[3233] Cells were split by gently washing with DPBS (Euroclone, ECB4004L), followed by a 5-minute incubation (humidified, 37° C. with 5% CO.sub.2) with trypsin-EDTA solution (Euroclone, ECB3052D). Detached cells were diluted with standard complete medium without selective agents, counted, and plated in a 384 MTP microplate (GR4332CPL, Twin Helix) (10,000 cells/well in 25 μ L/well) using a MATRIX WellMate dispenser. Plates were placed into a humidified incubator (37° C. with 5% CO.sub.2) overnight. The following day, 20 μ L of doxycycline (Sigma, D9891) at 20 ng/mL in standard complete medium was added to cells with a CyBi® Drop dispenser to induce APOL1 G2 expression. After a 6-hour incubation (humidified, 37° C. with 5% CO.sub.2), cells were washed 3 times with Ca.sup.2+ free Tyrode's Buffer (130 mM NaCl, 5 mM KCl, 1 mM MgCl.sub.2, 5 mM NaHCO.sub.3, 20 mM HEPES at pH 7.4) using a BIOTEK Microplate washer, such that 10 μ L of buffer remained in each well after the final wash. Assay plates were then stored at room temperature for 10 minutes. Next, 10 μ L of diluted compounds were transferred to the

assay plate from the 2× compound plate using a CyBi®-Well dispenser. Compound incubation was then carried out at room temperature for 10 minutes. The assay plate was transferred to the FLIPR.sup.TETRA and 20 µL of 10 mM Ca.sup.2+ (final concentration=5 mM) Tyrode's buffer was injected.

[3234] Table 39 below summarizes the data from this experiment. Unless otherwise specified, AC.sub.50 and values are reported as the geometric mean of at least 2 assay runs on separate days. Each run represents the average of a technical replicate, where each compound was assayed twice in the same plate. A superscript † symbol indicates a value from the average of a technical replicate from a single assay run, where each compound was assayed twice in the same plate.

[3235] The AC.sub.50 values in Table 39 below reflect the compound's ability to prevent calcium influx by inhibiting APOL1. As shown in the table, numerous compounds of the present disclosure are able to potentially inhibit APOL1-mediated calcium transport at sub micromolar concentrations. Compounds in Table 39 are referred to by the corresponding Compound Number in Table 1, which is also referred to in the synthetic examples. When one or more of the numbered compounds are identified by stereochemistry (for example, (R)- or (S)-), the specific stereoisomer for which data is provided in Table 39 may be identified by the elution order of such compound as described in the synthetic examples. For example, Compounds 39 and 40 are associated with Step 5 of Example 13, where Compound 39 is the first eluting enantiomer, and Compound 40 is the second eluting enantiomer in the chiral separation of Compound 38. For Table 39 below, n/a represents “not available”.

TABLE-US-00060				TABLE 39				APOL1				G2				FLIPR				AC.sub.50				Cmpd No.				(µM)				1				0.703				4				0.670				7																																																																															
0.601				8				0.833				9				1.31				10				1.27				11				1.05.sup.†				12				<0.457				13				<0.587				14				0.704				15				0.654				16				1.16				17				1.10																																															
18				0.240				19				0.481				20				0.348				21				<0.213				22				0.306				23				0.514				24				0.2				25				0.651.sup.†				26				0.943				27				0.51																																																			
28				<0.434				29				0.586.sup.†				30				0.409.sup.†				31				0.37.sup.†				32				0.262				33				0.218				34				0.168				35				0.142				36																																																															
0.223				37				0.327				38				0.307				39				0.745				40				0.657				41				0.709				42				0.316				43				0.507				44				0.352				45				0.746				46				0.434																																															
47				0.641				48				0.567				49				0.875.sup.†				50				0.622				51				0.953.sup.†				52				0.615.sup.†				53				0.838				54				0.804				55																																																															
<0.284				56				1.01				57				0.706				58				<0.320				59				0.879				60				0.652				61				0.686				62				0.958				63				0.461				64				0.455				65																																																			
0.639				66				0.461				67				0.351				68				<0.249				69				0.216				70				0.124				71				0.626				72				0.252				73				0.161				74				0.781				75																																																			
0.689				76				0.678				77				1.60.sup.†				78				0.522				79				0.687				80				0.375				81				0.549.sup.†				82				1.05.sup.†				83				0.718				84																																																											
1.45.sup.†				85				0.564				86				0.539				87				0.375				88				0.698				89				0.463				90				0.973				91				0.739				92				0.761				93				0.951				94																																																			
0.794				95				0.655.sup.†				96				0.552				97				0.823				98				0.6				99				0.643				100				0.503				101				0.634				102				0.671				103				0.591																																																							
104				0.408				105				0.392				106				0.756				107				0.712				108				0.86				109				1.19				110				0.772				111				0.919				112				0.746				113																																																							
0.524				114				0.529				115				0.472.sup.†				116				0.466.sup.†				117				0.396.sup.†				118				1.04.sup.†				119				1.01.sup.†				120				1.19.sup.†				121				0.508				122				0.616				123				0.796				124				0.727				125				0.729.sup.†				126				0.624.sup.†				127																			
0.542.sup.†				128				0.537.sup.†				129				0.352				130				0.525				131				0.464				132				0.492.sup.†				133				0.671.sup.†				134				0.879.sup.†				135				0.763.sup.†				136				0.80.sup.†				137				0.597.sup.†				138				1.22.sup.†				139				1.12.sup.†				140																											
0.807.sup.†				141				0.954.sup.†				142				0.845				143				0.701				144				0.708.sup.†				145				0.465.sup.†				146				0.279.sup.†				147				0.492				148				0.96				149				0.971				150				0.629				151				1.74				152				1.82				153				0.826.sup.†				154				0.869.sup.†				155											
0.452.sup.†				156				0.468.sup.†				157				0.933.sup.†				158				0.845.sup.†				159				1.01.sup.†				160				0.898				161				0.539.sup.†				162				0.603				163				0.528				164				0.555				165				0.558.sup.†				166				0.618.sup.†				167				0.467.sup.†				168																											
1.39.sup.†				169				1.21.sup.†				170				1.31.sup.†				171				0.598.sup.†				172				0.479.sup.†				173				0.908.sup.†				174				0.484.sup.†				175				0.381.sup.†				176				0.547.sup.†				177				0.67.sup.†				178				0.55.sup.†				179				0.655.sup.†				180																																			
0.756				181				0.656				182				0.574.sup.†				183				1.52.sup.†				184				1.28				185				0.993				186				0.524.sup.†				187				1.22.sup.†				188				0.513.sup.†				189				0.349.sup.†				190				n/a				191				1.17				192				0.291.sup.†				193				1.27				194																			
0.600.sup.†				195				0.604				196				0.696				197				0.889				198				0.631				199				1.13				200				1.2				201				0.909.sup.†				202				1.72.sup.†				203				1.19				204				0.869.sup.†				205				1.22.sup.†				206				n/a				207				n/a				208				1.55				209				1.24.sup.†				210			
1.54				211				1.29				212				1.36				213				1.12.sup.†				214				0.355				215				1.13				216				3.14.sup.†				217				0.866.sup.†				218				0.69.sup.†				219				0.835				220				0.679.sup.†				221				0.598				222				0.783				223				0.311.sup.†				224				0.83				225				0.847				226			
1.08.sup.†				227				1.11.sup.†				228				1.20.sup.†				229				0.792.sup.†				230				0.891.sup.†				231				1.02.sup.†				232				1.01				233				1.16				234				0.786.sup.†				235				1.00.sup.†				236				0.634				237				3.31				238				0.285.sup.†				239				1.54				240				n/a				241											

n/a 242 n/a 243 n/a 244 1.06 245 0.718 246 1.34 247 0.806.sup.† 248 0.82 249 0.433.sup.† 250
 0.559 251 0.546.sup.† 252 1.35.sup.† 253 1.22.sup.† 254 0.898.sup.† 255 2.64.sup.† 256
 0.261.sup.† 257 0.974.sup.† 258 1.05.sup.† 259 0.651 260 1.38 261 1.46 262 2.39.sup.† 263 2.54
 264 1.95.sup.† 265 0.697 266 0.862 272 0.921 273 2.10 274 0.722 275 0.640 276 1.75 277 n/a 278
 0.769 279 n/a 280 n/a 281 1.31 282 n/a 283 2.00 284 0.756 285 0.714 286 0.917 287 0.303 288
 0.369 289 0.552 290 0.253 291 1.19 292 1.30 293 n/a 294 n/a 295 n/a 296 n/a 297 0.937 298 n/a
 299 n/a 300 0.674 301 0.586 302 1.03 303 0.822 304 1.01 305 0.649 306 1.40 307 0.320 308 n/a
 309 n/a 310 n/a 311 n/a 312 n/a 313 n/a 314 0.968 315 0.330 316 1.15 317 0.792 318 n/a 319 0.378
 320 0.787 321 0.621 322 0.588 323 0.400 324 0.942 325 0.791 326 0.680 327 0.452 328 0.748 329
 n/a 330 n/a 331 n/a 332 n/a 333 0.486 334 2.00 335 n/a 336 n/a 338 n/a 339 n/a 340 n/a 341 1.06
 342 0.780 343 0.437 344 n/a 345 0.469 346 0.557 347 n/a 348 n/a 349 0.199 350 0.113 351 n/a 352
 0.597 353 n/a 354 n/a 355 n/a 356 2.49 357 0.412 358 n/a 359 n/a 360 n/a 361 n/a 362 n/a 363 n/a
 364 n/a 365 n/a 366 n/a 367 n/a 368 n/a 369 n/a 370 n/a 371 0.504 372 n/a 373 n/a 374 n/a 375 n/a
 376 n/a 377 n/a 378 n/a 379 n/a 380 n/a 381 n/a 382 n/a 383 n/a 384 n/a 385 n/a 386 n/a 387 n/a
 388 n/a 389 n/a 390 n/a 391 n/a 392 n/a 393 1.06 394 0.250 395 n/a 396 0.531 397 0.416 398 n/a
 399 n/a 400 n/a 401 0.603.sup.† 402 n/a 403 n/a 404 n/a 405 n/a 406 n/a 407 n/a 408 0.283.sup.†
 409 n/a 410 0.623.sup.† 411 0.436.sup.† 412 2.9.sup.† 413 0.251.sup.† 414 0.347.sup.† 415
 0.319.sup.† 416 0.449.sup.† 417 0.343.sup.† 418 0.309.sup.† 419 0.383.sup.† 420 0.892.sup.† 421
 0.564.sup.† 422 n/a 423 0.284.sup.† 424 0.895.sup.† 425 1.1.sup.† 426 0.597 427 0.708.sup.† 428
 1.2.sup.† 429 0.435.sup.† 430 0.518.sup.† 431 4.87.sup.† 432 2.56.sup.† 433 1.3.sup.† 434
 1.27.sup.† 435 1.17 436 n/a 437 0.612.sup.† 438 n/a 439 0.603 440 0.552 441 n/a 442 2.14.sup.†
 443 2.67.sup.† 444 n/a 445 1.11 446 1.18.sup.† 447 1.13 448 1.14.sup.† 449 0.609.sup.† 450
 2.27.sup.† 451 0.459.sup.† 452 n/a 453 0.935.sup.† 454 0.473.sup.† 455 1.25.sup.† 456 1.01.sup.†
 457 n/a 458 0.528.sup.† 459 0.37.sup.† 460 0.317.sup.† 461 0.676.sup.† 462 0.943.sup.† 463
 2.84.sup.† 464 1.19.sup.† 465 0.748.sup.† 466 n/a 467 n/a 468 0.867 469 n/a 470 1.11.sup.† 471
 n/a 472 n/a

Example B2

[3236] This example shows that the compounds of the present disclosure are able to reduce cell death caused by overexpression of APOL1.

[3237] A HEK293 clonal cell line overexpressing APOL1 G2 (HEK293/T-REx APOL1 G2/clone #2) was maintained in 1×DMEM-GlutaMax (Gibco, 10569-010) media with 1000 tetracycline-free FBS (Takara Bio USA, 631101), 5 µg/mL Blasticidin (Gibco, A1113903), and 100 µg/mL Zeocin (Invitrogen, R25001) in T75 flasks. In preparation for the assay, this media was aspirated and 2 mL of prewarmed TrypLE Express (Gibco, 12605-010) was added to a flask to detach cells. The flask was then incubated (humidified, 37° C. with 500 CO.sub.2) for 3-5 minutes. Afterwards, 8 mL of prewarmed cell assay media (1×DMEM-GlutaMax media with 10% tetracycline-free FBS) was added to the trypsinized cells. The suspension was gently mixed, and cells were counted using a Countess Cell Counting Chamber (Invitrogen). The suspension was diluted using cell assay media to generate a working stock solution (166,667 cells/mL). Using a MultiDrop Combi (Thermo Electron Corp), 30 µL (final cell density=5,000 cells/well) of the working stock solution was dispensed into each well of white 384-well assay ready plates (Nunc™ 164610) containing 6 ng/mL doxycycline, to induce APOL1 expression, and compound. All compounds were plated in a duplicate 8-point dilution series that consisted of 3-fold stepwise dilutions (0.5% DMSO final). Assay plates were incubated (humidified, 37° C. with 5% CO.sub.2) for 17 hours. After the incubation, the plates were equilibrated at room temperature for 1 hour. Next, 15 µl of CellTiter-Glo® reagent (Promega, G7570) was added to each well using a MultiDrop Combi. Plates were placed on an orbital shaker (500 rpm) for 5 minutes to induce cell lysis and then incubated at room temperature for 10 minutes. Luminescence was measured on an Envision plate reader.

Collaborative Drug Discovery software was utilized for graphing data. Plots were generated using a four parameter logistic curve fit.

[3238] Table 40 below provides the results from this experiment. Unless otherwise specified, EC.sub.50 values are reported as the geometric mean of at least 2 assay runs on separate days. Each run represents the average of a technical replicate, where each compound was assayed twice in the same plate. A superscript † symbol indicates a value from the average of a technical replicate from a single assay run, where each compound was assayed twice in the same plate. Compounds in Table 40 are referred to by the corresponding Compound Number in Table 1, which is also referred to in the synthetic examples. When one or more of the numbered compounds are identified by stereochemistry (for example, (R)- or (S)-), the specific stereoisomer for which data is provided in Table 40 may be identified by the elution order of such compound as described in the synthetic examples. Absolute stereochemistry of such compounds may be identified by methods known in the art. For Table 40 below, n/a represents “not available”.

[3239] Rescue EC.sub.50 values reported in Table 40 below represent the half-maximal effective concentration for reversal of cell death caused by overexpression of APOL1. This example demonstrates that compounds of the present disclosure are able to reduce cell death caused by overexpression of APOL1 at sub micromolar concentration.

TABLE-US-00061 TABLE 40 APOL1 G2 HEK293 Rescue Cmpd No. EC.sub.50 (μM) 1
0.120.sup.† 2 0.134.sup.† 3 0.142.sup.† 4 0.0719 5 0.213 6 0.121 7 0.279 8 0.598 9 1.76 10 0.0903
11 1.70 12 0.307 13 3.01 14 0.13 15 0.124 16 0.112 17 0.060 18 0.225 19 0.794 20 0.294 21 1.27
22 0.232 23 0.135 24 0.24 25 1.31 26 1.76 27 0.062 28 0.799 29 1.42 30 1.48 31 0.81 32 0.785 33
1.07 34 1.33 35 0.574 36 0.919 37 1.28 38 0.059 39 0.0567 40 0.072 41 0.097 42 0.343 43 0.0331
44 0.116 45 0.122 46 0.0728 47 0.148 48 0.199 49 0.0664 50 0.0249 51 0.214 52 0.214 53 0.0529
54 0.0903 55 0.983 56 0.105 57 0.19 58 1.31 59 0.213 60 0.0627 61 0.267 62 0.165 63 0.0511 64
0.163 65 0.0875 66 0.108 67 0.802 68 0.148 69 0.0826 70 1.05 71 0.345 72 0.0359.sup.† 73 0.527
74 0.167 75 0.114 76 0.0477 77 2.25 78 0.041 79 0.207 80 0.0657 81 0.28 82 0.0722 83 0.247 84
0.0649 85 0.0703 86 0.0465 87 0.103 88 0.121 89 0.0557 90 0.151 91 0.0245 92 0.0515 93 0.0909
94 0.14 95 0.246 96 0.0505 97 0.0446 98 0.0544.sup.† 99 0.0865 100 0.0465 101 0.0496 102
0.0314 103 0.0494 104 0.139 105 0.0407 106 0.0335 107 0.0308 108 0.0169 109 0.0969 110
0.0413 111 0.142 112 0.0472 113 0.0836 114 0.0465 115 0.0949 116 0.0818 117 0.0815 118 0.431
119 0.435 120 0.324 121 0.0812 122 0.0875 123 0.0795 124 0.297 125 0.355 126 0.273 127
0.0208 128 0.0427 129 0.0246 130 0.0923 131 0.068 132 0.153 133 0.203 134 0.199 135 0.211
136 0.0224 137 0.114 138 0.0782 139 0.133 140 0.172 141 0.048 142 0.0483 143 0.0369 144
0.081 145 0.0504 146 0.121 147 0.0631 148 0.107 149 0.101 150 0.228 151 0.397 152 0.405 153
0.0221 154 0.0671 155 0.0262 156 0.0559 157 0.0504 158 0.0657 159 0.168 160 0.0754 161
0.0464 162 0.0571 163 0.0246 164 0.103 165 0.0212.sup.† 166 0.0561.sup.† 167 0.0427.sup.† 168
0.343.sup.† 169 0.247 170 0.803.sup.† 171 0.100.sup.† 172 0.0991.sup.† 173 0.289.sup.† 174
0.104.sup.† 175 0.0631.sup.† 176 0.0469.sup.† 177 0.0782.sup.† 178 0.256.sup.† 179 0.164.sup.†
180 0.0221 181 0.0445 182 0.0996.sup.† 183 0.702.sup.† 184 0.498 185 0.113 186 0.0359.sup.†
187 0.268.sup.† 188 0.0392.sup.† 189 0.0591.sup.† 190 0.0547 191 0.13 192 0.0411 193 0.155 194
0.246 195 0.0394 196 0.0919 197 0.0323 198 0.0631 199 0.148 200 0.0872 201 0.609 202 0.315
203 0.187 204 1.55 205 0.472 206 >9.54 207 >10.0 208 0.238 209 3.07 210 0.0633 211 0.138 212
0.0814 213 0.189 214 0.079 215 0.0798 216 0.281 217 0.0326 218 0.0256 219 0.0263 220 0.053
221 0.0474 222 0.141 223 0.0978 224 0.0386 225 0.0326 226 0.043 227 0.047 228 0.101 229
0.128 230 0.0987 231 0.042 232 0.0392 233 0.0577 234 0.0803 235 0.0598 236 0.045 237 0.238
238 0.0852 239 0.0779 240 0.0526 241 0.0721 242 0.074 243 0.046 244 0.0704 245 0.139 246
0.562 247 0.092 248 0.0584 249 0.0268.sup.† 250 0.0492.sup.† 251 0.125.sup.† 252 0.471.sup.†
253 0.608.sup.† 254 0.175.sup.† 255 0.353.sup.† 256 0.0446.sup.† 257 0.256.sup.† 258
0.444.sup.† 259 0.0375 260 0.119 261 0.089 262 0.299 263 0.198 264 2.25 265 0.030 266 0.104
267 0.0378 268 0.0534 269 0.102 270 0.0256 271 0.152 272 0.0934 273 0.0885 274 0.034 275
0.027 276 0.101 277 0.034 278 0.023 279 0.060 280 0.080 281 0.056 282 0.130 283 0.099 284
0.199 285 0.200 286 0.181 287 0.0335 288 0.0341 289 0.0167 290 0.0182 291 0.0264 292 0.0232

293 0.294 294 0.0233 295 0.0436 296 0.0773 297 0.0318 298 0.0818 299 0.0213 300 0.10 301 0.086 302 0.0981 303 0.156 304 0.134 305 0.0854 306 0.110 307 0.0204 308 0.112 309 0.0559 310 0.103 311 0.0383 312 0.0243 313 0.0633 314 0.0243 315 0.0257 316 0.0327 317 0.0239 318 0.0232 319 0.0419 320 0.0221 321 0.0241 322 0.0185 323 0.0173 324 0.0256 325 0.0864 326 0.0312 327 0.0289 328 0.0341 329 0.0569 330 0.668 331 0.166 332 n/a 333 0.043 334 0.110 335 n/a 336 n/a 338 0.0994 339 0.0366 340 0.0302 341 0.0415 342 0.0902 343 0.162 344 0.194 345 0.0364 346 0.0343 347 n/a 348 n/a 349 0.0254 350 0.0188 351 0.166 352 0.0086 353 0.0123 354 0.0146 355 n/a 356 0.0506 357 0.0223 358 0.065 359 n/a 360 n/a 361 n/a 362 n/a 363 n/a 364 0.112 365 n/a 366 0.080 367 0.0877 368 n/a 369 n/a 370 n/a 371 0.0172 372 n/a 373 0.0534 374 0.101 375 n/a 376 0.0469 377 0.0859 378 0.388 379 0.842 380 n/a 381 n/a 382 n/a 383 n/a 384 n/a 385 n/a 386 n/a 387 n/a 388 n/a 389 n/a 390 n/a 391 n/a 392 0.0271 393 0.0805 394 0.024 395 0.0868 396 0.0405 397 0.0459 398 0.0201 399 0.0498 400 0.224 401 0.0367 402 0.0238 403 0.0957 404 0.129 405 0.00959 406 0.0583 407 0.0241 408 0.0231 409 0.0702 410 0.0641 411 0.0226 412 0.0453 413 0.0603 414 0.0367 415 0.0203 416 0.169 417 0.0492 418 0.0785 419 0.0309 420 0.0647 421 0.0345 422 0.0635 423 0.0593 424 0.0502 425 0.0906 426 0.0378 427 0.0185 428 0.0543 429 0.13 430 0.0729 431 0.0187 432 0.161 433 0.0148 434 0.0112 435 0.0129 436 0.0801 437 0.112 438 0.204 439 0.0149 440 0.0544 441 0.0574 442 <3.08E-03 443 0.0102 444 0.0224 445 0.00815 446 0.0332 447 0.0341.sup.† 448 0.112.sup.† 449 0.102 450 0.0055 451 0.0321 452 0.0282 453 0.0191 454 0.0241 455 0.0753 456 0.0282 457 0.278 458 0.0165 459 0.029 460 0.0541 461 0.0281 462 0.0178 463 0.00584 464 0.00502 465 0.00886 466 0.00426 467 0.00474 468 0.00645 469 0.0199 470 0.179 471 >3.30 472 n/a

Example B3

[3240] This example shows that the compounds of the present disclosure are able to reduce cell death caused by overexpression of APOL1.

[3241] Compounds were also assayed in a HEK293 clonal cell line overexpressing APOL1 G1 by a method similar to that shown in Example B2 above.

[3242] Table 41 below provides the results from this experiment. Unless otherwise specified, EC.sub.50 values are reported as the geometric mean of at least 2 assay runs on separate days. Each run represents the average of a technical replicate, where each compound was assayed twice in the same plate. A superscript † symbol indicates a value from the average of a technical replicate from a single assay run, where each compound was assayed twice in the same plate. A ¥ symbol represents compound concentration at 50% rescue estimated due to lack of sigmoidal curve preventing curve fitting. Compounds in Table 41 are referred to by the corresponding Compound Number in Table 1, which is also referred to in the synthetic examples. When one or more of the numbered compounds are identified by stereochemistry (for example, (R)- or (S)-), the specific stereoisomer for which data is provided in Table 41 may be identified by the elution order of such compound as described in the synthetic examples. Absolute stereochemistry of such compounds may be identified by methods known in the art. For Table 41 below, n/a represents “not available”. [3243] Rescue EC.sub.50 values reported in Table 41 below represent the half-maximal effective concentration for reversal of cell death caused by overexpression of APOL1. This example demonstrates that compounds of the present disclosure are able to reduce cell death caused by overexpression of APOL1 at sub micromolar concentration.

TABLE-US-00062 TABLE 41 APOL1 G1 HEK293 Rescue EC.sub.50 Cmpd No. (µM)

1	0.302	2	0.346	3	0.257	4	0.177	5	0.295	6	0.252	7	0.764	8	1.44	9	11.5.sup.¥	10	0.444	11	2.3	12	1.14	13	>30.0	14	0.476	15	0.279	16	0.295	17	0.16	18	0.663	19	4.81	20	0.815	21	>14.4	22	0.58	23	0.449	24	0.609	25	8.3.sup.¥	26	10.9.sup.¥	27	0.186	28	0.655	29	5.56	30	>22.5	31	1.57	32	1.86	33	>3.47	34	>1.96	35	0.488	36	>2.15	37	0.692	38	0.222	39	>0.247	40	0.294	41	0.307	42	1.08	43	0.131	44	0.478	45	0.387	46	0.253	47	0.399	48	0.445	49	0.26	50	0.0911	51	1.25	52	1.06	53	0.219	54	0.386	55	1.21	56	0.33	57	0.39	58	2.12	59	0.422	60	0.183	61	0.632	62	0.433	63	0.16	64	0.516	65	0.351	66	0.323	67	2.86	68	0.368	69	0.311	70	
---	-------	---	-------	---	-------	---	-------	---	-------	---	-------	---	-------	---	------	---	------------	----	-------	----	-----	----	------	----	-------	----	-------	----	-------	----	-------	----	------	----	-------	----	------	----	-------	----	-------	----	------	----	-------	----	-------	----	-----------	----	------------	----	-------	----	-------	----	------	----	-------	----	------	----	------	----	-------	----	-------	----	-------	----	-------	----	-------	----	-------	----	--------	----	-------	----	-------	----	------	----	-------	----	-------	----	-------	----	-------	----	-------	----	-------	----	------	----	--------	----	------	----	------	----	-------	----	-------	----	------	----	------	----	------	----	------	----	-------	----	-------	----	-------	----	-------	----	------	----	-------	----	-------	----	-------	----	------	----	-------	----	-------	----	--

>2.53 71 1.34 72 0.161 73 3.52 74 0.471 75 0.406 76 0.19 77 >4.76 78 0.128 79 0.533 80
 0.178 81 0.948 82 0.252 83 0.788 84 0.233 85 0.157 86 0.126 87 0.281 88 0.438 89 0.138 90 0.594
 91 0.099 92 0.101 93 0.23 94 0.21 95 0.457 96 0.148 97 0.143 98 0.183 99 0.205 100 0.134 101
 0.248 102 0.111 103 0.124 104 0.356 105 0.175 106 0.122 107 0.12 108 0.0497 109 0.321 110
 0.1 111 0.246 112 0.0775 113 0.176 114 0.128 115 0.199 116 0.234 117 0.174 118 2.1.sup.¥
 119 3.4.sup.¥ 120 >5.23 121 0.249 122 0.188 123 0.232 124 0.92 125 >4.86 126 1.01 127
 0.061 128 0.158 129 0.0728 130 0.253 131 0.186 132 0.685 133 0.885 134 0.991 135 1.14 136
 0.0494 137 0.249 138 0.253 139 0.383 140 0.6 141 0.204 142 0.192 143 0.125 144 0.246 145
 0.0985 146 0.221 147 0.183 148 0.282 149 0.246 150 1.54 151 2.5.sup.¥ 152 >4.17 153
 0.0962 154 0.252 155 0.067 156 0.153 157 0.152 158 0.252 159 0.588 160 0.244 161 0.135 162
 0.174 163 0.113 164 0.316 165 0.059 166 0.163 167 0.138 168 >7.84 169 1.03 170 2.3.sup.¥
 171 0.228 172 0.188 173 1.16.sup.†.sup. 174 0.253 175 0.153 176 0.157 177 0.132 178
 1.19.sup.†.sup. 179 0.484 180 0.0749 181 0.194 182 0.225 183 5.8.sup.¥ 184 2.5.sup.¥ 185
 0.231 186 0.0988 187 1.26 188 0.122 189 0.148 190 0.105 191 0.283 192 0.126 193 0.377 194
 0.414 195 0.072 196 0.207 197 0.0695 198 0.203 199 0.312 200 0.203 201 4.92 202 2.55 203
 0.457 204 1.27 205 1.76 206 4.31 207 >10.0 208 0.554 209 >2.84 210 0.153 211 0.38
 212 0.192 213 0.497 214 0.169 215 0.122 216 1.sup.¥ .sup. 217 0.108 218 0.125 219 0.116 220
 0.301 221 0.217 222 0.605 223 0.326 224 0.155 225 0.0517 226 0.138 227 0.21 228 0.426 229
 0.396 230 0.325 231 0.169 232 0.137 233 0.244 234 0.233 235 0.268 236 0.134 237 >3.25 238
 0.244 239 0.191 240 0.282 241 0.217 242 0.316 243 0.208 244 0.189 245 0.301 246 4.86 247
 0.27 248 0.219 249 0.0706 250 0.192 251 .sup. 0.323.sup.† 252 1.8.sup.¥ 253 4.4.sup.¥ 254
 2.04 255 2.9.sup.¥ 256 0.116 257 1.sup.¥ .sup. 258 1.3.sup.¥ 259 0.0925 260 0.248 261
 0.167 262 1.78 263 0.462 264 10.sup.¥ .sup. 265 0.132 266 0.353 267 0.137 268 0.211 269
 0.325 270 0.0607 271 0.34 272 0.148 273 0.206 274 0.0999 275 0.0627 276 0.228 277 0.117
 278 0.0926 279 0.234 280 0.235 281 0.164 282 0.383 283 0.211 284 0.775 285 0.292 286 0.272
 287 0.0858 288 0.0621 289 0.0506 290 0.0639 291 0.104 292 0.0515 293 .sup. 0.193.sup.†
 294 0.0656 295 0.112 296 0.159 297 0.119 298 0.218 299 0.0636 300 0.183 301 0.19 302 0.236
 303 0.351 304 0.286 305 0.172 306 0.245 307 0.07 308 0.196 309 0.134 310 0.267 311 0.1 312
 0.0665 313 0.135 314 0.0532 315 0.086 316 0.0994 317 0.0456 318 0.0659 319 0.118 320
 0.0539 321 0.0477 322 0.0386 323 0.0364 324 0.0408 325 0.197 326 0.104 327 0.0844 328
 0.0982 329 0.0961 330 1.5.sup.¥ 331 0.44 332 10.sup.¥ .sup. 333 0.0986 334 0.269 335
 n/a 336 n/a 338 0.213 339 n/a 340 n/a 341 n/a 342 n/a 343 0.431 344 0.254 345 0.079 346 0.0837
 347 n/a 348 n/a 349 0.0735 350 0.0439 351 0.299 352 0.00995 353 0.0345 354 0.0401 355
 n/a 356 .sup. 0.162.sup.† 357 0.0353 358 0.111 359 n/a 360 n/a 361 n/a 362 n/a 363 n/a 364
 0.239 365 n/a 366 0.136 367 0.145 368 n/a 369 n/a 370 n/a 371 0.0288 372 n/a 373 0.133 374
 0.216 375 n/a 376 0.129 377 0.172 378 1.sup.¥ .sup. 379 >10 380 n/a 381 n/a 382 n/a 383 n/a
 384 n/a 385 n/a 386 n/a 387 n/a 388 n/a 389 n/a 390 n/a 391 n/a 392 0.0899 393 0.159 394
 0.0777 395 0.157 396 0.085 397 0.105 398 0.374 399 0.117 400 0.678 401 0.0682 402 0.0357
 403 0.144 404 0.247 405 0.0192 406 0.109 407 0.0463 408 0.056 409 0.136 410 0.0933 411
 0.0556 412 0.0707 413 0.208 414 0.11 415 0.0692 416 0.382 417 0.104 418 0.215 419 0.108
 420 0.144 421 0.0833 422 0.165 423 0.116 424 0.112 425 0.271 426 0.0803 427 0.0358 428
 0.0896 429 0.2 430 0.123 431 0.037 432 0.288 433 0.0296 434 0.0148 435 0.0384 436 0.114
 437 0.186 438 0.242 439 0.0455 440 0.139 441 0.0961 442 0.0156 443 0.0308 444 0.0723
 445 0.0294 446 0.0825 447 0.0643 448 0.687 449 0.247 450 0.0149 451 0.115 452 0.0844
 453 n/a 454 n/a 455 n/a 456 0.0982 457 >3.30 458 0.0757 459 0.0955 460 0.0977 461
 0.0809 462 0.0534 463 0.0132 464 n/a 465 n/a 466 0.00871 467 0.0143 468 0.0145 469
 0.0487 470 0.0476 471 >3.30 472 n/a

Example B4

[3244] APOL1 G2 human immortalized podocyte viability assay for measurement of cytotoxicity reversal by compound (APOL1 G2 podocyte cell rescue assay). This example shows that the

compounds of the present disclosure are able to reverse cytotoxicity in human immortalized podocytes.

[3245] Cell Handling. The hTERT-immortalized kidney podocyte cell line was procured from the laboratory of Dr. Moin Saleem at the University of Bristol, UK (Nephrology 17 (2012) 525-531; doi:10.1111/j.1440-1797.2012.01619.x; herein incorporated by reference in its entirety). Parental and engineered cell lines were cultured in RPMI 1640 media (Gibco, 11875093) with 10% Tet System Approved FBS (Takara, 631101). The engineered cell line was maintained under selection (2.5 µg/mL puromycin). Cell lines were maintained and engineered at 33° C. Cell cultures were transferred to 37° C. for 10 to 14 days to initiate differentiation (Nephrology 17 (2012) 525-531; doi:10.1111/j.1440-1797.2012.01619.x; herein incorporated by reference in its entirety). During this time, media was refreshed every 3 days. After differentiation, cells engineered to inducibly express APOL1 G2 were used in the podocyte cell rescue assay.

[3246] Cell Line Engineering. The APOL1 G2 coding sequence was cloned into the pLVX-TetOne-Puro vector and verified by sequencing (Genscript Biotech). This construct was designed to have a C-terminal HiBiT tag. Lentiviral packaging of the vector was conducted using the Lenti-X Packaging Single Shots (VSVG) system according to manufacturer instructions (Takara Bio, 631275). A stable cell line was generated by transfecting the parental podocyte cell line with the concentrated virus in media with 5 µg/mL polybrene (Sigma, TR-1003-G). Media was changed the following day. 72 hours post transfection, 2.5 µg/mL puromycin (Gibco, A1113803) was added to the cells and cells were maintained in selection media thereafter. This APOL1 G2 stable cellular pool was subjected to a stringent limiting dilution to generate a pure stable clone (Podocyte/pLVX-TetOne APOL1 G2/clone D10) capable of inducible expression of APOL1 G2.

[3247] Assay Setup. T175 flasks containing differentiated podocytes engineered to inducibly express APOL1 G2 (Podocyte/pLVX-TetOne APOL1 G2/clone D10) were washed once with 15 mL of DPBS (Thermo Fisher, 14190-144). These flasks were each trypsinized with 3 mL of prewarmed TrypLE Express (Gibco, 12605-010) and incubated at 37° C. with 5% CO₂ until cells detached and neutralized with 7 mL of prewarmed assay media (RPMI 1640 media with 10% Tet System Approved FBS, no selection agent). Cells were pooled and the resulting suspension was gently mixed. The cell suspension was counted using a Countess Cell Counting Chamber (Invitrogen) and the cell concentration was adjusted to 240,000 cells/mL using assay media. Next, 25 µL of a 480 ng/mL working stock of doxycycline diluted in assay media was added to an assay ready plate. All experiments were performed in 384-well, white, solid bottom, tissue culture treated plates (Greiner, 781080). The assay ready compound plates were generated with duplicate 8-point compound dilution series that consisted of 3-fold stepwise dilutions (0.4% DMSO final). The plates were centrifuged at 1000 rpm for 1 minute. 25 µL of diluted cell suspension (final cell density=6,000 cells/well) was added to each well. The plates were centrifuged again at 1000 rpm for 1 minute and then incubated in a humidified incubator (37° C. with 5% CO₂). After 95 hours, the assay plates were removed from the incubator and allowed to equilibrate to room temperature for 1 hour. CellTiter-Glo® reagent (Promega, G7570) was prepared according to the manufacturer's instructions. 25 µL of CellTiter-Glo® reagent was added to each well. Assay plates were sealed with foil and mixed for 5 minutes on an orbital shaker (500 rpm) to induce cell lysis. Plates were centrifuged at 1000 rpm for 1 minute. 10 minutes after CellTiter-Glo® reagent addition, an Envision plate reader (Perkin Elmer) was used to measure the luminescent signal of each assay plate. Collaborative Drug Discovery software was utilized for graphing data.

[3248] Table 42 below provides the results from this experiment. Unless otherwise specified, EC₅₀ values are reported as the geometric mean of at least 2 assay runs on separate days. Each run represents the average of a technical replicate, where each compound was assayed twice in the same plate. A superscript † symbol indicates a value from the average of a technical replicate from a single assay run, where each compound was assayed twice in the same plate. Compounds in Table 42 are referred to by the corresponding Compound Number in Table 1, which is also referred

to in the synthetic examples. When one or more of the numbered compounds are identified by stereochemistry (for example, (R)- or (S)-), the specific stereoisomer for which data is provided in Table 42 may be identified by the elution order of such compound as described in the synthetic examples. Absolute stereochemistry of such compounds may be identified by methods known in the art. For Table 42 below, n/a represents “not available”.

[3249] Rescue EC.sub.50 values reported in Table 42 below represent the half-maximal effective concentration for reversal of cell death caused by overexpression of APOL1. This example demonstrates that compounds of the present disclosure are able to reduce cell death caused by overexpression of APOL1 at sub micromolar concentration.

TABLE-US-00063 TABLE 42 APOL1 G2 Podocyte Rescue EC.sub.50 Cmpd No. (μM)

1	0.0188	2	0.0107	3	0.0235	4	0.00916	5	>0.330	6	0.0477	7	n/a	8	n/a	9	n/a	10	n/a	11	n/a	12	n/a	13	n/a	14	n/a	15	n/a	16	n/a	17	n/a	18	n/a	19	n/a	20	n/a	21	n/a	22	n/a	23	n/a	24	0.0374	25	n/a	26	n/a	27	n/a	28	n/a	29	n/a	30	n/a	31	n/a	32	n/a	33	n/a	34	n/a	35	n/a	36	n/a	37	n/a	38	n/a	39	n/a	40	n/a	41	n/a	42	n/a	43	n/a	44	n/a	45	n/a	46	n/a	47	n/a	48	n/a	49	n/a	50	n/a	51	n/a	52	n/a	53	n/a	54	n/a	55	n/a	56	n/a	57	n/a	58	n/a	59	n/a	60	n/a	61	n/a	62	n/a	63	n/a	64	n/a	65	n/a	66	n/a	67	n/a	68	n/a	69	n/a	70	n/a	71	n/a	72	n/a	73	n/a	74	n/a	75	n/a	76	n/a	77	n/a	78	n/a	79	n/a	80	n/a	81	n/a	82	n/a	83	n/a	84	n/a	85	n/a	86	n/a	87	n/a	88	n/a	89	n/a	90	n/a	91	n/a	92	n/a	93	n/a	94	n/a	95	n/a	96	n/a	97	n/a	98	n/a	99	n/a	100	n/a	101	n/a	102	n/a	103	n/a	104	n/a	105	n/a	106	n/a	107	n/a	108	n/a	109	n/a	110	n/a	111	n/a	112	n/a	113	n/a	114	n/a	115	n/a	116	n/a	117	n/a	118	n/a	119	n/a	120	n/a	121	n/a	122	n/a	123	n/a	124	n/a	125	n/a	126	n/a	127	0.00539	128	n/a	129	n/a	130	n/a	131	n/a	132	n/a	133	n/a	134	n/a	135	n/a	136	n/a	137	n/a	138	n/a	139	n/a	140	n/a	141	n/a	142	n/a	143	n/a	144	n/a	145	n/a	146	n/a	147	n/a	148	n/a	149	n/a	150	n/a	151	n/a	152	n/a	153	n/a	154	n/a	155	n/a	156	n/a	157	n/a	158	n/a	159	n/a	160	n/a	161	n/a	162	n/a	163	n/a	164	n/a	165	n/a	166	n/a	167	n/a	168	n/a	169	n/a	170	n/a	171	n/a	172	n/a	173	n/a	174	n/a	175	n/a	176	n/a	177	n/a	178	n/a	179	n/a	180	n/a	181	n/a	182	n/a	183	n/a	184	n/a	185	n/a	186	n/a	187	n/a	188	n/a	189	n/a	190	n/a	191	n/a	192	n/a	193	n/a	194	n/a	195	n/a	196	n/a	197	n/a	198	n/a	199	n/a	200	n/a	201	n/a	202	n/a	203	n/a	204	n/a	205	n/a	206	n/a	207	n/a	208	n/a	209	n/a	210	n/a	211	n/a	212	n/a	213	n/a	214	n/a	215	n/a	216	n/a	217	n/a	218	n/a	219	n/a	220	n/a	221	n/a	222	n/a	223	n/a	224	n/a	225	n/a	226	n/a	227	n/a	228	n/a	229	n/a	230	n/a	231	n/a	232	0.00217	233	0.00425	234	n/a	235	n/a	236	n/a	237	n/a	238	n/a	239	0.0021	240	0.0139	241	n/a	242	0.0197	243	0.00411	244	n/a	245	n/a	246	n/a	247	n/a	248	n/a	249	n/a	250	n/a	251	n/a	252	n/a	253	n/a	254	n/a	255	n/a	256	n/a	257	n/a	258	n/a	259	0.00121	260	0.0103	261	0.00229.sup.†	262	n/a	263	n/a	264	n/a	265	n/a	266	n/a	267	0.00663	268	n/a	269	0.0304	270	0.00292	271	n/a	272	0.001.sup.†	273	0.0052	274	0.0188	275	0.00389	276	0.0104	277	n/a	278	n/a	279	n/a	280	n/a	281	n/a	282	n/a	283	0.00508	284	0.105	285	n/a	286	n/a	287	0.0177	288	0.0195.sup.†	289	0.00222	290	0.00276.sup.†	291	n/a	292	n/a	293	n/a	294	n/a	295	n/a	296	n/a	297	n/a	298	n/a	299	n/a	300	0.0333	301	0.0669	302	n/a	303	n/a	304	n/a	305	0.015	306	0.00194	307	0.00261.sup.†	308	n/a	309	n/a	310	n/a	311	n/a	312	n/a	313	n/a	314	0.00457.sup.†	315	n/a	316	n/a	317	0.00292	318	n/a	319	n/a	320	n/a	321	0.00765	322	n/a	323	0.00202	324	n/a	325	n/a	326	n/a	327	n/a	328	n/a	329	n/a	330	n/a	331	0.0857.sup.†	332	n/a	333	n/a	334	0.00958.sup.†	335	n/a	336	n/a	338	n/a	339	n/a	340	n/a	341	n/a	342	n/a	343	n/a	344	n/a	345	0.0158	346	0.0388	347	n/a	348	n/a	349	0.00681	350	0.0103	351	n/a	352	0.00112	353	n/a	354	n/a	355	n/a	356	0.0215	357	0.00404	358	0.024	359	n/a	360	n/a	361	n/a	362	n/a	363	n/a	364	n/a	365	n/a	366	n/a	367	n/a	368	n/a	369	n/a	370	n/a	371	0.0011	372	n/a	373	n/a	374	n/a	375	n/a	376	0.00621	377	0.0244	378	n/a	379	n/a	380	n/a	381	n/a	382	n/a	383	n/a	384	n/a	385	n/a	386	n/a	387	n/a	388	n/a	389	n/a	390	n/a	391	n/a	392	n/a	393	0.023	394	0.0169	395	0.137	396	0.0186	397	0.0143	398	n/a	399	n/a	400	n/a	401	0.00462	402	n/a	403	n/a	404	n/a	405	n/a	406	n/a	407	n/a	408	0.00705	409	n/a	410	n/a	411	0.00145	412	n/a	413	n/a	414	n/a	415	n/a	416	n/a	417	n/a	418	n/a	419	n/a	420	n/a	421	n/a	422	n/a	423	n/a	424	n/a	425	n/a	426	0.0041	427	0.00195	428	0.0122	429	n/a	430	n/a	431	n/a	432	n/a	433	0.00213	434	0.0015	435	0.00294	436	n/a	437	n/a	438	n/a	439	0.00392	440	0.015	441	n/a	442	0.00116	443	0.0025	444	n/a	445	n/a	446	n/a	447	n/a	448	n/a	449	n/a	450	0.000846	451	n/a	452	n/a	453	0.00181	454	0.00434	455	n/a	456	n/a	457	n/a	458	n/a
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n/a 459 n/a 460 n/a 461 n/a 462 n/a 463 n/a 464 0.000659 465 0.00122 466 n/a 467 n/a 468 n/a 469
n/a 470 n/a 471 n/a 472 n/a

Example B5

[3250] APOL1 G0/G1/G2 viability assay for measurement of cytotoxicity reversal by compound in trypanosomes (APOL1 G0/G1/G2 trypanosome cell rescue assay). This example shows that the compounds of the present disclosure are able to reverse cytotoxicity in trypanosomes.

[3251] APOL1 protein expression and purification. The mature form of APOL1 proteins, residues 28-398, were expressed from a pET28a vector with an N-terminal His-tag and TEV cleavage site. Proteins were expressed in *Escherichia coli* BL21-CodonPlus (DE3)-RIPL cells. Liter cultures of terrific broth were grown at 37° C. until an OD_{sub}600 of ~0.8 was reached and then induced with isopropyl β-D-1-thiogalactopyranoside (IPTG; final concentration of 500 μM). Afterwards, cultures were grown for 3 hours at 37° C. For protein purification, cell pellets were resuspended in lysis buffer (50 mM Tris, pH 8.5, 5 mM EDTA, 0.5 mM DTT, 0.5 mM PMSF) supplemented with a cocktail of protease inhibitors. Cells were lysed by sonication, centrifuged, and the resulting pellet was collected. The homogenized pellet was resuspended in wash buffer (50 mM Tris, pH 8.5, 0.5 M NaCl, 5 mM EDTA, 0.5 mM DTT, 0.5 mM PMSF, protease inhibitor cocktail) and collected by centrifugation. Inclusion body dissolution was then conducted as previously reported (PNAS 112(9) (2015) 2894-2899; www.pnas.org/cgi/doi/10.1073/pnas.1421953112; herein incorporated by reference in its entirety). The solubilized APOL1 protein was applied to a nickel column (HisTrap, GE Life Sciences) pre-equilibrated in buffer A (50 mM Tris, pH 8.5, 0.15 M NaCl, 1% zwittergent 3-14, protease inhibitor cocktail). The resin was washed with buffer B (50 mM Tris, pH 8.5, 0.15 M NaCl, 0.1% DDM) and TEV cleavage was conducted on column overnight. Afterwards, the column was washed with buffer B, followed by His-washing buffer (50 mM Tris, pH 8.5, 0.15 M NaCl, 0.1% DDM, 10 mM imidazole). The protein was then removed from the resin using His-elution buffer (50 mM Tris, pH 8.5, 0.15 M NaCl, 0.1% DDM, 250 mM imidazole). The sample was further purified by size exclusion chromatography using a Superdex 200 Increase column (GE Life Sciences) in 50 mM Tris, pH 8.5, 0.15 M NaCl, and 0.1% DDM buffer. All APOL1 proteins (G0, G1, and G2) were generated at Viva Biotech (Shanghai) Ltd.

[3252] Modified HMI-9 media preparation

(https://tryps.rockefeller.edu/trypsru2_culture_media_preparation.html; herein incorporated by reference in its entirety). Trypanosomes were cultured in modified HMI-9 media consisting of IMDM (ThermoFisher, 12440053), 10% heat-inactivated FBS (Gibco, 10082-147), 10% Serum Plus (Sigma-Aldrich, 14008C), 1×HMI-9 supplement stock, and 1% hypoxanthine stock. The 10×HMI-9 supplement stock was made by dissolving 280 mg bathocuproine disulfonic acid (Sigma-Aldrich, 146625), 1820 mg cysteine (add after bathocuproine) (Sigma-Aldrich, 30089), 1100 mg pyruvic acid (Sigma-Aldrich, 107360), 100 mg uracil (Sigma-Aldrich, U0750), 100 mg cytosine (Sigma-Aldrich, C3506) and 140 μL 2-mercaptoethanol (Sigma-Aldrich, M3148) in 1000 mL of water. The resulting solution was aliquoted and stored at -20° C. The hypoxanthine stock was made by dissolving 4 g of NaOH into 1000 mL of water. Afterwards, 13.6 g hypoxanthine (Sigma-Aldrich, H9377) was added to this mixture. The resulting solution was aliquoted and stored at -20° C.

[3253] Assay Setup. *Trypanosoma brucei* Lister 427 VSG221 (ATCC, PRA-382) cells were cultured in modified HMI-9 media. All experiments were performed in 384-well, white, solid bottom, tissue culture treated plates (Greiner, 781080). Assay ready plates were generated with duplicate 11-point compound dilution series that consisted of 2-fold stepwise dilutions (0.4% DMSO final). To each well, was added 20 μL of 2 g/mL of APOL1 G0, G1, or G2 recombinant protein in modified HMI-9 media using a MultiDrop Combi (final APOL1 protein concentration=1 g/mL). Trypanosomes were counted using a hemacytometer and diluted in modified HMI-9 media to a concentration of 1.25×10^{sup}.4 cell/mL. 20 μL of this trypanosome suspension was added to each well to give a total assay volume of 40 μL and a final cell count of 250 trypanosomes/well.

Plates were centrifuged at 1000 rpm for 1 minute and then incubated for 20 hours (humidified, 37° C. with 5% CO₂). After incubation, the plates were equilibrated at room temperature for 1 hour. Next, 20 µL of CellTiter-Glo® reagent (Promega, G7570) was added to each well. Plates were sealed and placed on an orbital shaker (500 rpm) for 5 minutes to induce cell lysis. The plates were centrifuged at 1000 rpm for 1 minute and then incubated at room temperature for an additional 10 minutes. Luminescence signal was measured on an Envision plate reader.

Collaborative Drug Discovery software was utilized for graphing data.

[3254] Table 43 below provides the results from this experiment. Unless otherwise specified, EC₅₀ values are reported as the geometric mean of at least 2 assay runs on separate days. Each run represents the average of a technical replicate, where each compound was assayed twice in the same plate. A superscript † symbol indicates a value from the average of a technical replicate from a single assay run, where each compound was assayed twice in the same plate. Compounds in Table 43 are referred to by the corresponding Compound Number in Table 1, which is also referred to in the synthetic examples. When one or more of the numbered compounds are identified by stereochemistry (for example, (R)- or (S)-), the specific stereoisomer for which data is provided in Table 43 may be identified by the elution order of such compound as described in the synthetic examples. Absolute stereochemistry of such compounds may be identified by methods known in the art. For Table 43 below, n/a represents “not available”.

[3255] Rescue EC₅₀ values reported in Table 43 below represent the half-maximal effective concentration for reversal of cell death caused by overexpression of APOL1. This example demonstrates that compounds of the present disclosure are able to reduce cell death caused by overexpression of APOL1 at sub micromolar concentration.

TABLE-US-00064 TABLE 43 APOL1 G1 APOL1 G2 APOL1 G0 Trypanosome Trypanosome																								
Trypanosome	Rescue	EC.sub.50	Rescue	EC.sub.50	Rescue	EC.sub.50	Cmpd No.	(μM)	(μM)	(μM)														
1	0.293	0.0896	0.225	2	0.191	0.0563	0.192	3	0.323	0.092	0.248	4	0.171	0.0519	0.116	5	0.171							
6	0.052	0.15	0.47	0.137	0.381	7	n/a	n/a	n/a	8	n/a	n/a	n/a	9	>4.00	1.32	>4.00	10	0.352.sup.†	0.0933				
11	0.351.sup.†	11	n/a	n/a	n/a	12	1.95	0.494	1.93	13	>4.00	>4.00	>4.00	14	0.488	0.127	0.378	15	0.342					
16	0.124.sup.†	0.336.sup.†	16	0.312	0.11	0.267	17	0.148	0.0367	0.108	18	1.54	0.234	0.69	19	n/a	n/a							
20	n/a	0.949	0.263	0.798	21	n/a	n/a	n/a	22	n/a	n/a	n/a	23	0.612.sup.†	0.113.sup.†	0.321	24							
25	0.759.sup.†	0.163.sup.†	0.513.sup.†	25	>4.00	1.23	>4.00	26	>4.00	>4.00	>4.00	27	0.269.sup.†											
28	0.0663.sup.†	0.201	28	n/a	n/a	n/a	29	n/a	n/a	n/a	30	n/a	n/a	n/a	31	n/a	n/a	n/a	32	n/a	n/a	n/a	33	n/a
34	n/a	n/a	34	n/a	n/a	n/a	35	>4.00	>4.00	0.406	1.26	43	0.0866	0.0194										
39	n/a	n/a	39	0.206	0.053	0.156	40	n/a	n/a	n/a	41	n/a	n/a	n/a	42	1.88	0.406	1.26	43	0.0866	0.0194			
44	0.0522	44	0.429	0.119	0.331	45	0.747	0.162	0.476	46	0.215	0.0505	0.156	47	n/a	n/a	n/a	48	n/a	n/a				
49	n/a	49	0.304	0.109	0.213	50	0.0449	0.0102	0.03	51	n/a	n/a	n/a	52	n/a	n/a	n/a	53	0.157	0.0478	0.156			
54	n/a	54	n/a	n/a	55	n/a	n/a	n/a	56	0.319	0.119	0.296	57	0.672	0.183	0.673	58	n/a	n/a	n/a	59	n/a	n/a	n/a
60	0.228	0.066	0.189	61	n/a	n/a	n/a	62	0.39	0.12	0.407	63	0.164.sup.†	0.0409.sup.†	0.118	64	0.712							
65	0.167	0.503	65	0.294	0.0968	0.318	66	0.322	0.0976	0.246	67	n/a	n/a	n/a	68	0.441	0.115	0.358	69					
70	0.184.sup.†	0.0465.sup.†	0.153.sup.†	70	>4.00	0.918.sup.†	>4.00	71	1.36.sup.†															
72	0.205.sup.†	0.814.sup.†	72	0.177	0.0457	0.128	73	n/a	n/a	n/a	74	n/a	n/a	n/a	75	0.578	0.146	0.383						
76	0.126	0.0355	0.106	77	>4.00	1.45	>4.00	78	0.0976	0.0326	0.0952	79	0.952	0.293	0.846	80	n/a							
81	n/a	n/a	81	n/a	n/a	n/a	82	0.344	0.113	0.254	83	n/a	n/a	n/a	84	0.144	0.05	0.101	85	0.19	0.0485	0.115		
86	0.137	0.0384	0.0964	87	0.383	0.0958	0.268	88	0.561	0.138	0.451	89	0.114	0.0327	0.101	90								
91	0.679	0.163	0.546	91	0.0827	0.0204	0.0546	92	0.0993	0.0285	0.0844	93	0.264	0.0569	0.156	94								
95	0.424	0.129	0.403	95	0.703	0.222	0.592	96	0.143	0.053	0.122	97	0.0984	0.0406	0.099	98	0.259							
99	0.0722	0.207	99	0.346	0.0901	0.216	100	0.113	0.0259	0.0635	101	0.2	0.0694	0.195	102	0.0505								
103	0.018	0.0473	103	0.17	0.0437	0.113	104	0.726	0.184	0.583	105	0.274	0.0752	0.217	106	0.0918								
107	0.0234	0.069	107	0.0935	0.023	0.0572	108	0.0307	0.00922	0.0233	109	0.385	0.098	0.32	110									
111	0.0877	0.0261	0.0714	111	0.362	0.105	0.282	112	0.0696	0.0183	0.0582	113	0.154	0.0467	0.124									
114	0.104	0.038	0.108	115	n/a	n/a	n/a	116	0.209	0.0692	0.149	117	0.423	0.117	0.309	118	n/a	n/a						

n/a 119 n/a n/a n/a 120 1.25 0.346 0.807 121 n/a n/a n/a 122 0.337 0.108 0.252 123 0.258 0.0769
0.178 124 n/a n/a n/a 125 n/a n/a n/a 126 n/a n/a n/a 127 0.0839 0.0294 0.0649 128 0.383 0.133
0.269 129 0.0695 0.0222 0.0652 130 n/a n/a n/a 131 0.176 0.0478 0.132 132 0.731 0.18 0.525 133
n/a n/a n/a 134 0.991 0.277 0.851 135 1.2 0.303 1.08 136 0.0371 0.0143 0.0315 137 0.454 0.119
0.322 138 0.585 0.165 0.326 139 0.537 0.147 0.335 140 0.653 0.164 0.408 141 n/a n/a n/a 142
0.0983 0.0353 0.0635 143 0.0641 0.0211 0.0411 144 0.358 0.101 0.224 145 0.118 0.0355 0.0865
146 0.427 0.119 0.346 147 0.231 0.0654 0.155 148 0.201 0.0618 0.14 149 0.174 0.0518 0.117 150
2.29 0.519 1.44 151 1.33 0.347 1.04 152 1.53 0.416 1.05 153 0.0667 0.0256 0.0522 154 0.569 0.16
0.328 155 0.0361 0.00965 0.0259 156 0.154 0.0448 0.0966 157 0.142 0.0451 0.109 158 0.142
0.0459 0.0942 159 n/a n/a n/a 160 0.156 0.0423 0.109 161 0.107 0.0333 0.0763 162 0.142 0.0509
0.143 163 0.09 0.0212 0.0559 164 0.349 0.0871 0.211 165 0.0338 0.0111 0.0287 166 0.112 0.0338
0.094 167 0.148 0.035 0.0984 168 0.638 0.191 0.496 169 0.918 0.295 0.619 170 n/a n/a n/a 171 n/a
n/a n/a 172 0.353 0.108 0.246 173 n/a n/a n/a 174 0.481 0.137 0.401 175 n/a n/a n/a 176 0.177
0.0446 0.12 177 0.314 0.0844 0.26 178 n/a n/a n/a 179 n/a n/a n/a 180 0.0477 0.0133 0.0352 181
0.232 0.0707 0.158 182 0.481 0.138 0.401 183 n/a n/a n/a 184 2.08 0.585 1.6 185 0.479 0.14 0.31
186 n/a n/a n/a 187 n/a n/a n/a 188 0.0965 0.0267 0.071 189 0.246 0.0614 0.167 190 0.104 0.0357
0.0921 191 0.304 0.104 0.251 192 0.142 0.0491 0.108 193 0.288 0.0702 0.2 194 n/a n/a n/a 195
0.0744 0.0198 0.0482 196 0.185 0.0603 0.16 197 0.193 0.0106 0.0303 198 0.142 0.0419 0.107 199
n/a n/a n/a 200 0.192 0.0724 0.185 201 n/a n/a n/a 202 n/a n/a n/a 203 0.583 0.227 0.565 204 n/a
n/a n/a 205 n/a n/a n/a 206 n/a n/a n/a 207 n/a n/a n/a 208 n/a n/a n/a 209 n/a n/a n/a 210 0.0711
0.0218 0.0638 211 0.527 0.159 0.48 212 0.103 0.0321 0.0696 213 n/a n/a n/a 214 0.43 0.0887 0.25
215 0.097 0.0224 0.0701 216 0.912 0.231 0.629 217 n/a n/a n/a 218 0.108 0.0316 0.0651 219 0.039
0.0105 0.0223 220 n/a n/a n/a 221 0.129 0.0295 0.075 222 n/a n/a n/a 223 n/a n/a n/a 224 0.0928
0.0327 0.066 225 0.0508 0.0129 0.0352 226 0.112 0.036 0.08 227 0.248 0.0616 0.142 228 0.363
0.0862 0.196 229 0.778 0.186 0.417 230 0.801 0.209 0.427 231 0.231 0.0667 0.135 232 0.157
0.0416 0.0976 233 0.134 0.0411 0.0817 234 0.475 0.146 0.272 235 0.452 0.132 0.27 236 0.866
0.397 0.766 237 0.196 0.0572 0.132 238 >4.00 0.763 1.59 239 0.0488 0.0148 0.04 240 n/a n/a n/a
241 0.258 0.0633 0.177 242 n/a n/a n/a 243 n/a n/a n/a 244 0.151 0.0493 0.141 245 0.379 0.102
0.265 246 n/a n/a n/a 247 0.246 0.081 0.186 248 0.126 0.0379 0.0862 249 0.0738 0.0221 0.0538
250 0.678 0.194 0.445 251 1.12 0.24 0.903 252 n/a n/a n/a 253 n/a n/a n/a 254 0.951 0.252 0.695
255 2.92 0.575 1.38 256 n/a n/a n/a 257 n/a n/a n/a 258 n/a n/a n/a 259 0.0922 0.0255 0.068 260
0.397 0.11 0.274 261 0.1 0.0296 0.084 262 n/a n/a n/a 263 0.32 0.0953 0.226 264 n/a n/a n/a 265
0.105 0.0231 0.0676 266 0.407 0.0771 0.236 267 0.115 0.0337 0.0954 268 0.121 0.0327 0.108 269
0.277 0.0797 0.207 270 0.0822 0.0253 0.0725 271 0.0507 0.0159 0.051 272 0.0732 0.0189 0.0589
273 0.0716 0.0258 0.0649 274 0.0584 0.0172 0.0593 275 0.0736 0.0214 0.0612 276 0.185 0.055
0.178 277 n/a n/a n/a 278 0.111 0.0255 0.114 279 0.283 0.0831 0.213 280 n/a n/a n/a 281 0.101
0.0281 0.0959 282 0.23 0.0598 0.181 283 0.235 0.0711 0.216 284 0.943 0.235 0.808 285 0.749
0.212 0.547 286 0.629 0.175 0.505 287 0.05.sup.† 0.0139.sup.† 0.0376.sup.† 288 0.0361.sup.†
0.00975.sup.† 0.0262.sup.† 289 0.0519 0.0157 0.0382 290 0.0692 0.024 0.0555 291 0.0797 0.0265
0.0631 292 0.0396 0.0108 0.0316 293 n/a n/a n/a 294 0.089 0.0214 0.0619 295 0.0402 0.0111
0.0326 296 n/a n/a n/a 297 0.0854 0.0315 0.0726 298 0.156 0.0566 0.119 299 n/a n/a n/a 300 0.543
0.16 0.375 301 0.447 0.134 0.327 302 0.498 0.142 0.356 303 n/a n/a n/a 304 n/a n/a n/a 305 0.265
0.0804 0.187 306 0.0508 0.015 0.037 307 0.0797 0.0272 0.0657 308 0.435 0.15 0.335 309 0.0706
0.0234 0.0582 310 n/a n/a n/a 311 0.0503 0.0138 0.0374 312 0.0344 0.0101 0.0282 313 0.0718
0.0202 0.0545 314 0.0349 0.0115 0.0254 315 n/a n/a n/a 316 n/a n/a n/a 317 0.0335 0.0115 0.033
318 n/a n/a n/a 319 0.0811 0.027 0.0697 320 n/a n/a n/a 321 0.0459 0.0143 0.0396 322 0.0349
0.0117 0.0293 323 0.0359 0.0118 0.032 324 0.035 0.00831 0.0289 325 0.269 0.0775 0.213 326
0.0944 0.0327 0.0886 327 0.0466 0.0134 0.0399 328 0.0921 0.0239 0.0799 329 0.24 0.0673 0.199
330 n/a n/a n/a 331 n/a n/a n/a 332 n/a n/a n/a 333 0.205.sup.† 0.074.sup.† 0.152.sup.† 334 n/a n/a
n/a 335 n/a n/a n/a 336 n/a n/a n/a 338 n/a n/a n/a 339 n/a n/a n/a 340 n/a n/a n/a 341 n/a n/a n/a

342 n/a n/a n/a 343 0.203.sup.† 0.0642.sup.† 0.174.sup.† 344 n/a n/a n/a 345 0.242.sup.†
0.0802.sup.† 0.161.sup.† 346 0.116.sup.† 0.0377.sup.† 0.0836.sup.† 347 n/a n/a n/a 348 n/a n/a n/a
349 0.0898.sup.† 0.0331.sup.† 0.0646.sup.† 350 n/a n/a n/a 351 n/a n/a n/a 352 0.0188 0.00822
0.0187 353 n/a n/a n/a 354 n/a n/a n/a 355 n/a n/a n/a 356 0.113.sup.† 0.0374.sup.† 0.101.sup.† 357
0.0916 0.0281 0.0612 358 0.0752 0.0313 0.0621 359 n/a n/a n/a 360 n/a n/a n/a 361 n/a n/a n/a 362
n/a n/a n/a 363 n/a n/a n/a 364 0.203 0.0923 0.155 365 n/a n/a n/a 366 0.0873 0.0324 0.0664 367
0.323 0.106 0.231 368 n/a n/a n/a 369 n/a n/a n/a 370 n/a n/a n/a 371 0.0219 0.00916 0.0188 372
n/a n/a n/a 373 n/a n/a n/a 374 n/a n/a n/a 375 n/a n/a n/a 376 0.115.sup.† 0.0399.sup.†
0.0956.sup.† 377 0.271.sup.† 0.0978.sup.† 0.205.sup.† 378 n/a n/a n/a 379 n/a n/a n/a 380 n/a n/a
n/a 381 n/a n/a n/a 382 n/a n/a n/a 383 n/a n/a n/a 384 n/a n/a n/a 385 n/a n/a n/a 386 n/a n/a n/a
387 n/a n/a n/a 388 n/a n/a n/a 389 n/a n/a n/a 390 n/a n/a n/a 391 n/a n/a n/a 392 0.0794.sup.†
0.0285.sup.† 0.07.sup.† 393 0.36.sup.† 0.12.sup.† 0.3.sup.† 394 n/a n/a n/a 395 0.312 0.106 0.24
396 n/a n/a n/a 397 0.269 0.0852 0.226 398 0.0522 0.018 0.0536 399 n/a n/a n/a 400 n/a n/a n/a
401 n/a n/a n/a 402 0.0427 0.0165 0.0336 403 n/a n/a n/a 404 0.554 0.21 0.425 405 0.0216 0.00806
0.02 406 0.128 0.0511 0.107 407 0.0997.sup.† 0.033.sup.† 0.0724 408 0.121 0.0436 0.0848.sup.†
409 n/a n/a n/a 410 0.269 0.0924 0.206 411 n/a n/a n/a 412 n/a n/a n/a 413 n/a n/a n/a 414 n/a n/a
n/a 415 n/a n/a n/a 416 n/a n/a n/a 417 n/a n/a n/a 418 n/a n/a n/a 419 n/a n/a n/a 420 n/a n/a n/a
421 n/a n/a n/a 422 n/a n/a n/a 423 n/a n/a n/a 424 n/a n/a n/a 425 n/a n/a n/a 426 0.0888 0.0367
0.071 427 0.0181 0.00662 0.0156 428 0.0719 0.0323 0.0633 429 n/a n/a n/a 430 n/a n/a n/a 431 n/a
n/a n/a 432 n/a n/a n/a 433 n/a n/a n/a 434 n/a n/a n/a 435 0.0272.sup.† 0.0145.sup.† 0.0271.sup.†
436 n/a n/a n/a 437 n/a n/a n/a 438 n/a n/a n/a 439 0.051.sup.† 0.0202.sup.† 0.0505.sup.† 440 0.205
0.0705 0.16 441 0.215 0.0783 0.144 442 n/a n/a n/a 443 n/a n/a n/a 444 n/a n/a n/a 445 n/a n/a n/a
446 n/a n/a n/a 447 n/a n/a n/a 448 n/a n/a n/a 449 n/a n/a n/a 450 n/a n/a n/a 451 n/a n/a n/a 452
n/a n/a n/a 453 n/a n/a n/a 454 n/a n/a n/a 455 n/a n/a n/a 456 n/a n/a n/a 457 n/a n/a n/a 458 n/a n/a
n/a 459 0.133 0.0498 0.101 460 0.118 0.0537 0.104 461 0.0755.sup.† 0.0288.sup.† 0.0692.sup.†
462 n/a n/a n/a 463 n/a n/a n/a 464 n/a n/a n/a 465 n/a n/a n/a 466 0.0306 0.0119 0.0194 467
0.00603 0.00224 0.00374 468 0.00559 0.0024 0.00372 469 0.0155 0.00533 0.00976 470 0.0124
0.00441 0.00882 471 n/a n/a n/a 472 n/a n/a n/a

Example B6

[3256] APOL1 G2 mouse renal model for measurement of albuminuria prevention by compound. This example shows that the compounds of the present disclosure are able to prevent albuminuria.

[3257] APOL1 transgenic (Tg) mice with a bacterial artificial chromosome (BAC) containing APOL1 G2 and its upstream and downstream genomic regions (Taconic, Model #13022) were bred to homozygosity. APOL1 G2 homozygous (G2 HOM) male mice, greater than 6 weeks of age were utilized in experiments. All studies were conducted under a protocol approved by the Institutional Animal Care and Use Committee (IACUC).

[3258] Baseline urine samples were collected over a 24-hour period 7 days prior to the start of the experiment. Urine volume was determined by weight and stored at -80° C. until use. Afterwards, baseline blood samples were collected via submandibular bleed into serum separate (SS) tubes (BD Microtainer). Serum samples were allowed to clot at room temperature for 30 minutes prior to centrifugation at 12000 rpm, 4° C. for 5 minutes. The concentration of APOL1 was determined in these samples using an APOL1 ELISA kit (Proteintech, KE00047) according to the manufacturer's instructions.

[3259] On the day of the experiment (0 hour), mice were individually identified, weighed, and assigned to one of the treatment groups. Treatment groups had similar average levels of serum APOL1. Mice were administered with two doses of compound test article and one dose of recombinant mouse interferon gamma (mIFN γ) (MilliporeSigma, Catalog #IF005) per day for 2 days. Compound test article was administered via oral gavage at 10 mL/kg body weight dose volume at 0, approximately 10, 24, and 34 hours. mIFN γ (1.5×10^{11} unit/kg body weight) was administered via intraperitoneal injection at 10 mL/kg body weight dose volume at 0 and 24 hours.

Urine collection was immediately initiated after the second dose of mIFN γ . Specifically, mice were individually placed in a metabolic cage (Lab Products) for 24 hours where urine and feces were collected into separate tubes. During this time, mice had access to an enrichment toy, gel diet (Bio-serve), and drinking water (Innovive). At the end of the experiment, mice were anesthetized under isoflurane and blood was collected via cardiac puncture followed with a cervical dislocation to ensure death.

[3260] Urine samples were analyzed for urine albumin and urine creatinine levels. The concentration of urine albumin was determined using a mouse albumin immunoperoxidase assay kit (Immunology Consultants Laboratory, E-90AL) according to the manufacturer's instructions. Urine creatinine was analyzed using LC-MS methodology at WuXi App Tec (DMPK, New Jersey, USA).

[3261] Selected compounds from Table 1 described by formula (I-E) were tested in this APOL1 G2 mouse renal model for measurement of albuminuria. The results are shown in Table 44.

TABLE-US-00065
TABLE 44 Compound Dose (route of Reduction in urine albumin/ Tested administration) creatinine ratio (uACR)
Compound A 0.3 mg/kg (PO, BID) 50%
Compound A 1 mg/kg (PO, BID) 58%
Compound A 3 mg/kg (PO, BID) 75%
Compound B 10 mg/kg (PO, BID) 67%
Compound B 50 mg/kg (PO, BID) 89%

[3262] All publications, patent applications, patents, and other references mentioned herein are expressly incorporated by reference in their entireties, to the same extent as if each were incorporated by reference individually.

[3263] It is to be understood that, while the disclosure has been described in conjunction with the above embodiments, the foregoing description and examples are intended to illustrate and not limit the scope of the disclosure. Other aspects, advantages and modifications within the scope of the disclosure will be apparent to those skilled in the art to which the disclosure pertains.

Claims

1. A compound of formula (II): ##STR02194## or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein: m is an integer from 0 to 4; n is an integer from 0 to 2; p is an integer from 0 to 10; R^{sup.1}, if present is, independently at each occurrence selected from the group consisting of halo, —CN, C_{sub.1-6}alkoxy, and C_{sub.1-6}alkyl, wherein the C_{sub.1-6}alkoxy of R^{sup.1} is optionally substituted with one or more halo, and the C_{sub.1-6}alkyl of R^{sup.1} is optionally substituted with one or more halo; R^{sup.2} is H, C_{sub.1-6}alkyl, C_{sub.3-10}cycloalkyl, or 3-15 membered heterocyclyl, wherein the C_{sub.1-6}alkyl of R^{sup.2} is optionally substituted with one or more deuterium, halo, —OH, —NH_{sub.2}, or C_{sub.1-6}alkoxy, and the C_{sub.3-10}cycloalkyl of R^{sup.2} is optionally substituted with one or more —OH; R^{sup.3}, if present, is C_{sub.1-6}alkyl; L^{sup.1} is C_{sub.1-6}alkylene, wherein the C_{sub.1-6}alkylene of L^{sup.1} is optionally substituted with one or more deuterium or C_{sub.1-6}alkyl, and wherein the C_{sub.1-6}alkyl is further optionally substituted with one or more —OH or C_{sub.1-6}alkoxy; L^{sup.2} is O or N(R'), wherein R^{sup.x} is H or C_{sub.1-6}alkyl; and either (1) L^{sup.3} is absent or is O, C_{sub.3-10}cycloalkyl, 3-10 membered heterocyclyl, or C_{sub.1-6}alkylene, wherein the C_{sub.3-10}cycloalkyl of L^{sup.3} is optionally substituted with one or more —OH or C_{sub.1-6}alkyl, the C_{sub.1-6}alkylene of L^{sup.3} is optionally substituted with one or more —OH or C_{sub.1-6}alkyl, wherein the C_{sub.1-6}alkyl is optionally substituted with one or more —OH, and the 3-10 membered heterocyclyl of L^{sup.3} is optionally substituted with one or more —OH or C_{sub.1-6}alkyl; X^{sup.1} and X^{sup.2} are each independently N or C(R^{sup.5}); and R^{sup.4} is: (i) —S(O)_{sub.2}—R^{sup.a}; (ii) 5-20 membered heteroaryl, wherein the 5-20 membered heteroaryl of R^{sup.4} is optionally substituted with one or more C_{sub.1-6}alkyl; (iii) —N(R^{sup.d})_{sub.2}, wherein R^{sup.d} is independently at each occurrence H, C_{sub.1-6}alkyl, or —S(O)_{sub.2}—R^{sup.a}, wherein the C_{sub.1-6}alkyl of R^{sup.d} is optionally substituted with one or more —OH, (iv) —

NS(O)—(C.sub.1-6alkyl).sub.2, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH, (v) —C(O)—N(R.sup.e).sub.2 wherein R.sup.e is independently at each occurrence H, C.sub.1-6 alkyl, or 3-10 membered heterocycle, wherein the 3-10 membered heterocycle of R.sup.e is optionally substituted with one or more oxo, or both R.sup.e together with the N to which they are attached are taken together to form a 3-10 membered heterocyclyl, wherein the 3-10 membered heterocyclyl is optionally substituted with one or more halo, oxo, —OH, NH.sub.2, —NH—S(O).sub.2—R.sup.a, or —S(O).sub.2—R.sup.a, (vi) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl, —OH, oxo or —S(O).sub.2R.sup.a, (vii) —S(O)—N(C.sub.1-6alkyl)-(C.sub.1-6alkyl), (viii) —CN, (ix) —(CH.sub.2).sub.qOH, wherein q is an integer from 0-6, (x) —C(O)—C.sub.1-6alkyl, or (xi) —P(O)(C.sub.1-6alkyl).sub.2; or (2) L.sup.3 is absent; and one of X.sup.1 and X.sup.2 is N or C(R.sup.5); and the other of X.sup.1 and X.sup.2 is N or C that is taken together with R.sup.4, and the atoms to which they are attached, to form a 5-10 membered heterocyclyl or a 5-20 membered heteroaryl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R.sup.b, wherein R.sup.b is, independently at each occurrence, selected from the group consisting of —OH, halo, oxo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10cycloalkyl, and 3-10 membered heterocyclyl, wherein the C.sub.1-6alkyl of R.sup.b is optionally substituted with one or more halo, OH, —S(O).sub.2—C.sub.1-6alkyl, or C.sub.3-10cycloalkyl, and wherein the C.sub.3-10cycloalkyl of the C.sub.1-6alkyl of R.sup.b is further optionally substituted with one or more C.sub.1-6alkyl or —OH and the C.sub.3-10cycloalkyl of R.sup.b is optionally substituted with one or more —OH, C.sub.3-10cycloalkyl, or C.sub.1-6alkyl, and wherein the C.sub.1-6alkyl of the C.sub.3-10cycloalkyl of R.sup.b is further optionally substituted with one or more —OH, deuterium, or halo, and the 5-20 membered heteroaryl is optionally substituted with one or more R.sup.e, wherein R.sup.e is, independently at each occurrence, selected from the group consisting of halo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10cycloalkyl, and 3-10 membered heterocyclyl, wherein the C.sub.1-6alkyl of R.sup.e is optionally substituted with one or more —S(O).sub.2—C.sub.1-6alkyl, the C.sub.3-10cycloalkyl of R.sup.e is optionally substituted with one or more —OH or C.sub.1-6alkyl, and the 3-10 membered heterocyclyl of R.sup.e is optionally substituted with one or more —OH or C.sub.1-6alkyl, and wherein the C.sub.1-6alkyl of the 3-10 membered heterocyclyl of R.sup.e is further optionally substituted with one or more —OH; R.sup.a is, independently at each occurrence: (i) C.sub.1-6alkyl optionally substituted with one or more halo, —OH, —S(O).sub.2—C.sub.1-6alkyl, or —N(C.sub.1-6alkyl)-C(O)—C.sub.1-6alkyl, (ii) C.sub.3-10cycloalkyl optionally substituted with one or more —OH, —C(O).sub.2—C.sub.1-6alkyl, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, or —C(O)—C.sub.3-10heterocyclyl, or C.sub.1-6alkyl, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH, (iii) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl, or (iv) NH(C.sub.1-6alkyl); R.sup.5 is, independently at each occurrence, H, halo, —CN, 3-10 membered heterocyclyl, C.sub.1-6alkyl, or C.sub.1-6alkoxy, wherein the C.sub.1-6alkyl of R.sup.5 is optionally substituted with one or more halo or —OH, and the C.sub.1-6alkoxy of R.sup.5 is optionally substituted with one or more halo; X.sup.3 is N or C(R.sup.6); X.sup.4 is N or C(R.sup.7); and R.sup.6 and R.sup.7 are each independently H or halo.

2. The compound of claim 1, wherein the compound is a compound of formula (I'): ##STR02195## or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein: m is an integer from 0 to 4; n is an integer from 0 to 2; p is an integer from 0 to 10; R.sup.1, if present is, independently at each occurrence selected from the group consisting of halo, —CN, C.sub.1-6alkoxy, or C.sub.1-6alkyl, wherein the C.sub.1-6alkoxy of R.sup.1 is optionally substituted with one or more halo, and the C.sub.1-6alkyl of R.sup.1 is optionally substituted with one or more halo; R.sup.2 is H, C.sub.1-6alkyl, C.sub.3-10cycloalkyl, or 3-15 membered

heterocyclyl, wherein the C.sub.1-6alkyl of R.sup.2 is optionally substituted with one or more halo, —OH, —NH.sub.2, or C.sub.1-6alkoxy, and the C.sub.3-10cycloalkyl of R.sup.2 is optionally substituted with one or more —OH; R.sup.3, if present, is C.sub.1-6alkyl; L.sup.1 is C.sub.1-6alkylene, wherein the C.sub.1-6alkylene of L.sup.1 is optionally substituted with one or more C.sub.1-6alkyl, and wherein the C.sub.1-6alkyl is further optionally substituted with one or more —OH or C.sub.1-6alkoxy; L.sup.2 is O or N(R'), wherein R.sup.x is H or C.sub.1-6alkyl; and either (1) L.sup.3 is absent or is O, C.sub.3-10cycloalkyl, 3-10 membered heterocyclyl, or C.sub.1-6alkylene, wherein the C.sub.3-10cycloalkyl is optionally substituted with one or more —OH, or C.sub.1-6alkyl, the C.sub.1-6alkylene of L.sup.3 is optionally substituted with one or more —OH, or C.sub.1-6alkyl, and the 3-10 membered heterocyclyl is optionally substituted with one or more —OH; X.sup.1 and X.sup.2 are each independently N or C(R.sup.5); and R.sup.4 is: (i) —S(O).sub.2—R.sup.a; (ii) 5-20 membered heteroaryl, wherein the 5-20 membered heteroaryl of R.sup.4 is optionally substituted with one or more C.sub.1-6alkyl; (iii) —N(R.sup.d).sub.2, wherein R.sup.d is independently at each occurrence H, C.sub.1-6 alkyl, or —S(O).sub.2—R.sup.a, wherein the C.sub.1-6alkyl of R.sup.d is optionally substituted with one or more —OH, (iv) —NS(O)—(C.sub.1-6alkyl).sub.2, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH, (v) —C(O)—N(R.sup.e).sub.2 wherein R.sup.e is independently at each occurrence H, C.sub.1-6 alkyl, or 3-10 membered heterocycle, wherein the 3-10 membered heterocycle is optionally substituted with one or more oxo, or both R.sup.e together with the N to which they are attached are taken together to form a 3-10 membered heterocyclyl, wherein the 3-10 membered heterocyclyl is optionally substituted with one or more halo, oxo, —OH, NH.sub.2, —NH—S(O).sub.2—R.sup.a, or —S(O).sub.2—R.sup.a, (vi) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl, —OH, oxo or —S(O).sub.2R.sup.a, (vii) —S(O)—N(C.sub.1-6alkyl)-(C.sub.1-6alkyl), (viii) —CN, (ix) —(CH.sub.2).sub.qOH, wherein q is an integer from 0-6, (x) —C(O)—C.sub.1-6alkyl, or (xi) —P(O)(C.sub.1-6alkyl).sub.2; or (2) L.sup.3 is absent; and one of X.sup.1 and X.sup.2 is N or C(R.sup.5); and the other of X.sup.1 and X.sup.2 is N or C that is taken together with R.sup.4, and the atoms to which they are attached, to form a 5-10 membered heterocyclyl or a 5-20 membered heteroaryl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R.sup.b, wherein R.sup.b is, independently at each occurrence, selected from the group consisting of halo, oxo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10cycloalkyl, and 3-10 membered heterocyclyl, wherein the C.sub.1-6alkyl of R.sup.b is optionally substituted with one or more halo, OH, —S(O).sub.2—C.sub.1-6alkyl, or C.sub.3-10cycloalkyl, and wherein the C.sub.3-10cycloalkyl of the C.sub.1-6alkyl of R.sup.b is further optionally substituted with one or more C.sub.1-6alkyl or —OH and the C.sub.3-10cycloalkyl of R.sup.b is optionally substituted with one or more —OH, C.sub.3-10 cycloalkyl, or C.sub.1-6alkyl, and wherein the C.sub.1-6alkyl of the C.sub.3-10cycloalkyl of R.sup.b is further optionally substituted with one or more —OH, and the 5-20 membered heteroaryl is optionally substituted with one or more R.sup.e, wherein R.sup.e is, independently at each occurrence, selected from the group consisting of halo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10cycloalkyl, and 3-10 membered heterocyclyl, wherein the C.sub.1-6alkyl of R.sup.c is optionally substituted with one or more —S(O).sub.2—C.sub.1-6alkyl, the C.sub.3-10cycloalkyl of R.sup.c is optionally substituted with one or more —OH, or C.sub.1-6alkyl, and the 3-10 membered heterocyclyl of R.sup.c is optionally substituted with one or more —OH, or C.sub.1-6alkyl, and wherein the C.sub.1-6alkyl of 3-10 membered heterocyclyl of R.sup.c is further optionally substituted with one or more —OH; R.sup.a is, independently at each occurrence: (i) C.sub.1-6alkyl optionally substituted with one or more halo, —OH, —S(O).sub.2—C.sub.1-6alkyl, or —N(C.sub.1-6alkyl)-C(O)—C.sub.1-6alkyl, (ii) C.sub.3-10cycloalkyl optionally substituted with one or more —OH, —C(O).sub.2—C.sub.1-6alkyl, —

C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, or —C(O)—C.sub.3-10heterocyclyl, or C.sub.1-6alkyl, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH, or (iii) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl; R.sup.5 is, independently at each occurrence, H, halo, —CN, 3-10 membered heterocyclyl, C.sub.1-6alkyl, or C.sub.1-6alkoxy, wherein the C.sub.1-6alkyl of R.sup.5 is optionally substituted with one or more halo, or —OH and wherein the C.sub.1-6alkoxy of R.sup.5 is optionally substituted with one or more halo; and R.sup.6 and R.sup.7 are each independently H or halo.

3. The compound of claim 1, or claim 2, wherein the compound is a compound of formula (I): ##STR02196## or a stereoisomer or a tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein: either (1) L.sup.3 is absent or is O, C.sub.3-10cycloalkyl, 3-10 membered heterocyclyl, or C.sub.1-6alkylene, wherein the C.sub.1-6alkylene of L.sup.3 is optionally substituted with one or more C.sub.1-6alkyl, and the 3-10 membered heterocyclyl is optionally substituted with one or more —OH; X.sup.1 and X.sup.2 are each independently N or C(R.sup.5); and R.sup.4 is: (i) —S(O).sub.2—R.sup.a; (ii) 5-20 membered heteroaryl, wherein the 5-20 membered heteroaryl of R.sup.4 is optionally substituted with one or more C.sub.1-6alkyl; (iii) —N(R.sup.d).sub.2, wherein R.sup.d is independently at each occurrence H, C.sub.1-6 alkyl, or —S(O).sub.2—R.sup.a, wherein the C.sub.1-6alkyl of R.sup.d is optionally substituted with one or more —OH, (iv) —NS(O)—(C.sub.1-6alkyl).sub.2, wherein the C.sub.1-6alkyl of R.sup.d is optionally substituted with one or more —OH, (v) —C(O)—N(R.sup.e).sub.2 wherein R.sup.e is independently at each occurrence H, C.sub.1-6 alkyl, or 3-10 membered heterocycle, wherein the 3-10 membered heterocycle is optionally substituted with one or more oxo, or both R.sup.e together with the N to which they are attached are taken together to form a 3-10 membered heterocyclyl, wherein the 3-10 membered heterocyclyl is optionally substituted with one or more halo, oxo, —OH, NH.sub.2, NH—S(O).sub.2—R.sup.a, or —S(O).sub.2—R.sup.a, (vi) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl or oxo, (vii) —S(O)—N(C.sub.1-6alkyl)-(C.sub.1-6alkyl), (viii) —CN; or (2) L.sup.3 is absent; and one of X.sup.1 and X.sup.2 is N or C(R.sup.5); and the other of X.sup.1 and X.sup.2 is N or C that is taken together with R.sup.4, and the atoms to which they are attached, to form a 5-10 membered heterocyclyl or a 5-20 membered heteroaryl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R.sup.b, wherein R.sup.b is, independently at each occurrence, selected from the group consisting of halo, oxo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10cycloalkyl, and 3-10 membered heterocyclyl, wherein the C.sub.1-6alkyl of R.sup.b is optionally substituted with one or more halo, OH, or —S(O).sub.2—C.sub.1-6alkyl, and the C.sub.3-10cycloalkyl of R.sup.b is optionally substituted with one or more —OH, and the 5-20 membered heteroaryl is optionally substituted with one or more R.sup.c, wherein R.sup.c is, independently at each occurrence, selected from the group consisting of halo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10 cycloalkyl, and 3-10 membered heterocyclyl, wherein the C.sub.1-6alkyl of R.sup.c is optionally substituted with one or more —S(O).sub.2—C.sub.1-6alkyl, and the C.sub.3-10cycloalkyl of R.sup.c is optionally substituted with one or more —OH; R.sup.a is, independently at each occurrence: (i) C.sub.1-6alkyl optionally substituted with one or more halo, —OH, —S(O).sub.2—C.sub.1-6alkyl, or —N(C.sub.1-6alkyl)-C(O)—C.sub.1-6alkyl, or (ii) C.sub.3-10cycloalkyl optionally substituted with one or more —OH, —C(O).sub.2—C.sub.1-6alkyl, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, or —C(O)—C.sub.3-10heterocyclyl, or C.sub.1-6alkyl, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH, or (iii) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl; R.sup.5 is, independently at each occurrence, H, halo, —CN, 3-10 membered heterocyclyl, C.sub.1-6alkyl, or C.sub.1-6alkoxy, wherein the C.sub.1-6alkyl of R.sup.5 is optionally substituted with one or more halo, or —OH and wherein the C.sub.1-6alkoxy is

optionally substituted with one or more halo; and R.sup.6 and R.sup.7 are each independently H or halo.

4. The compound of claim 1, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein L.sup.2 is —O—, such that the compound is a compound of formula (I-A): ##STR02197## or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing.

5. The compound of any one of claims 1-4, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein n is 1, or 2.

6. The compound of any one of claims 1-5, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R.sup.2 is H, C.sub.1-3alkyl, C.sub.3-6cycloalkyl, or 3-10 membered heterocyclyl, wherein the C.sub.1-3alkyl of R.sup.2 is optionally substituted with one or more deuterium, halo, —OH, —NH.sub.2, or C.sub.1-3alkoxy, and the C.sub.3-6cycloalkyl of R.sup.2 is optionally substituted with one or more —OH.

7. The compound of any one of claims 1-6, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein m is 0, 1, or 2.

8. The compound of any one of claims 1-7 or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R.sup.1 is halo, —CN, or —C.sub.1-3alkyl, wherein the C.sub.1-3alkyl of R.sup.1 is optionally substituted with one or more halo.

9. The compound of any one of claims 1-8, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein p is 0 or 1.

10. The compound of any one of claims 1-9, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R.sup.3 is C.sub.1-6alkyl.

11. The compound of any one of claims 1-10, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein L.sup.1 is C.sub.1-3alkylene, wherein the C.sub.1-3alkylene of L.sup.1 is optionally substituted with one or more deuterium or C.sub.1-3alkyl, wherein the C.sub.1-3alkyl is optionally substituted with —OH or C.sub.1-3alkoxy.

12. The compound of any one of claims 1-11, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein L.sup.1 is selected from the group consisting of ##STR02198## wherein, for each L.sup.1, * denotes the point of attachment to L.sup.2 and ** denotes the point of attachment to the remainder of the molecule.

13. The compound of any one of claims 1-12, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein L.sup.2 is O.

14. The compound of any one of claims 1-13, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein L.sup.3 is absent.

15. The compound of any one of claims 1-13, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein L.sup.3 is —O—, C.sub.3-10cycloalkyl, 3-10 membered heterocyclyl, or C.sub.1-6alkylene, wherein the C.sub.3-10cycloalkyl of L.sup.3 is optionally substituted with one or more —OH or C.sub.1-6alkyl, the C.sub.1-6alkylene of L.sup.3 is optionally substituted with one or more —OH or C.sub.1-6alkyl, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH, and the 3-10 membered heterocyclyl of L.sup.3 is optionally substituted with one or more —OH or C.sub.1-6alkyl.

16. The compound of any one of claims 1-13, and 15 or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein L.sup.3 is selected from the group consisting of —O—, ##STR02199##

17. The compound of any one of claims 1-16 or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R.sup.4 is S(O).sub.2-R.sup.a, 5-10 membered heteroaryl, —N(R.sup.d).sub.2, —NS(O)—(C.sub.1-3alkyl).sub.2, —C(O)—N(R.sup.e).sub.2, 3-6 membered heterocyclyl, —S(O)(N—C.sub.1-3alkyl)-(C.sub.1-3alkyl), —

CN, —OH, —C(O)—C.sub.1-3alkyl, or —P(O)(C.sub.1-3alkyl).sub.2, wherein the 5-20 membered heteroaryl of R.sup.4 is optionally substituted with one or more C.sub.1-3alkyl, and the 3-6 membered heterocyclyl optionally substituted with one or more —OH, oxo, C.sub.1-3alkyl, or —S(O).sub.2—R.sup.a.

18. The compound of claim 17 or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R.sup.a is C.sub.1-6alkyl, C.sub.3-10cycloalkyl, or 3-10 membered heterocyclyl, wherein the C.sub.1-6alkyl of R.sup.a is optionally substituted with one or more halo, —OH, —S(O).sub.2—C.sub.1-6alkyl, or —N(C.sub.1-6alkyl)—C(O)—C.sub.1-6alkyl, the C.sub.3-10cycloalkyl of R.sup.a is optionally substituted with one or more —OH, C(O).sub.2—C.sub.1-6alkyl, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —C(O)—C.sub.3-10heterocyclyl or C.sub.1-6alkyl, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH, and the 3-10 membered heterocyclyl of R.sup.a is optionally substituted with one or more C.sub.1-6alkyl.

19. The compound of claim 17, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R.sup.d is independently at each occurrence H, C.sub.1-6 alkyl, or —S(O).sub.2—R.sup.a, wherein the C.sub.1-6alkyl of R.sup.d is optionally substituted with one or more —OH, and wherein R.sup.a is C.sub.1-6alkyl.

20. The compound of claim 17, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R.sup.e is independently at each occurrence H, C.sub.1-6alkyl, or 3-10 membered heterocycle, wherein the 3-6 membered heterocycle is optionally substituted with one or more oxo.

21. The compound of claim 17 or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein both R.sup.e together with the N to which they are attached are taken together to form a 3-10 membered heterocyclyl, wherein the 3-10 membered heterocyclyl is optionally substituted with one or more halo, oxo, —OH, —NH.sub.2, —NH—S(O).sub.2—R.sup.a, or —S(O).sub.2—R.sup.a, wherein R.sup.a is C.sub.1-6alkyl.

22. The compound of any one of claims 1-21, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein each of X.sup.1 and X.sup.2 is C(R.sup.5).

23. The compound of any one of claims 1-21, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein each of X.sup.1 and X.sup.2 is N.

24. The compound of any one of claims 1-21, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein one of X.sup.1 and X.sup.2 is N and the other is C(R.sup.5).

25. The compound of any one of claims 1-21, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein one of X.sup.1 and X.sup.2 is N or CR.sup.5, and the other of X.sup.1 and X.sup.2 is N or C that is taken together with R.sup.4, and the atoms to which they are attached, to form a 5-10 membered heterocyclyl optionally substituted with one or more of R.sup.b.

26. The compound of claim 25, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R.sup.b is independently at each occurrence, selected from the group consisting of —OH, halo, oxo, C.sub.1-3alkyl, —C(O)—C.sub.1-3alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-3alkyl), —C(O)—N(C.sub.1-3alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-6cycloalkyl, and 3-6 membered heterocyclyl, wherein the C.sub.1-3alkyl of R.sup.b is optionally substituted with one or more halo, OH, —S(O).sub.2—C.sub.1-3alkyl, or C.sub.3-6cycloalkyl, and wherein the C.sub.3-6cycloalkyl of the C.sub.1-3alkyl of R.sup.b is further optionally substituted with one or more C.sub.1-3alkyl or —OH and the C.sub.3-6cycloalkyl of R.sup.b is optionally substituted with one or more —OH, C.sub.3-6cycloalkyl, or C.sub.1-3alkyl, and wherein the C.sub.1-3alkyl of the C.sub.3-6cycloalkyl of R.sup.b is further optionally substituted with one or more —OH, deuterium, or halo.

- 27.** The compound of any one of claims 1-21, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein one of X^{sup.1} and X^{sup.2} is N or CR^{sup.5}, and the other of X^{sup.1} and X^{sup.2} is N or C that is taken together with R^{sup.4}, and the atoms to which they are attached, to form a 5-10 membered heteroaryl optionally substituted with one or more R^{sup.c}.
- 28.** The compound of claim 27, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein, R^{sup.c} is independently at each occurrence, selected from the group consisting of halo, C_{sub.1-3}alkyl, —C(O)—C_{sub.1-3}alkyl, —C(O)—NH_{sub.2}, —C(O)—NH(C_{sub.1-3}alkyl), —C(O)—N(C_{sub.1-3}alkyl)_{sub.2}, —S(O)_{sub.2}—R^{sup.a}, C_{sub.3-6}cycloalkyl, and 3-6 membered heterocyclyl, wherein the C_{sub.1-6}alkyl of R^{sup.c} is optionally substituted with one or more —S(O)_{sub.2}—C_{sub.1-3}alkyl, the C_{sub.3-10}cycloalkyl of R^{sup.c} is optionally substituted with one or more —OH, or C_{sub.1-3}alkyl, and the 3-6 membered heterocyclyl of R^{sup.c} is optionally substituted with one or more —OH or C_{sub.1-3}alkyl, and wherein the C_{sub.1-3}alkyl of the 3-6 membered heterocyclyl of R^{sup.c} is further optionally substituted with one or more —OH.
- 29.** The compound of any one of claims 1-28, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein R^{sup.5} is independently at each occurrence, H, halo, —CN, 3-6 membered heterocyclyl, C_{sub.1-3}alkyl, or C_{sub.1-3}alkoxy, wherein the C_{sub.1-3}alkyl of R^{sup.5} is optionally substituted with one or more halo, or —OH and wherein the C_{sub.1-3}alkoxy is optionally substituted with one or more halo.
- 30.** The compound of claim 1, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, wherein the compound, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, is selected from Compounds 1-472 of Table 1.
- 31.** A method for preparing a compound of formula (II), as recited in claim 1, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, comprises a step of reacting a compound of formula (II'-A): ##STR02200## wherein: m is an integer from 0 to 4; n is an integer from 0 to 2; p is an integer from 0 to 10; R^{sup.1}, if present is, independently at each occurrence selected from the group consisting of halo, —CN, C_{sub.1-6}alkoxy, and C_{sub.1-6}alkyl, wherein the C_{sub.1-6}alkoxy of R^{sup.1} is optionally substituted with one or more halo, and the C_{sub.1-6}alkyl of R^{sup.1} is optionally substituted with one or more halo; R^{sup.2} is H, C_{sub.1-6}alkyl, C_{sub.3-10}cycloalkyl, or 3-15 membered heterocyclyl, wherein the C_{sub.1-6}alkyl of R^{sup.2} is optionally substituted with one or more deuterium, halo, —OH, —NH_{sub.2}, or C_{sub.1-6}alkoxy, and the C_{sub.3-10}cycloalkyl of R^{sup.2} is optionally substituted with one or more —OH; R^{sup.3}, if present, is C_{sub.1-6}alkyl; with: a compound of formula (II'-B): ##STR02201## wherein: the dashed line represents a single or double bond; Y^{sup.1} is halo, oxo, or a sulfonate ester L^{sup.1} is C_{sub.1-6}alkylene, wherein the C_{sub.1-6}alkylene of L^{sup.1} is optionally substituted with one or more deuterium or C_{sub.1-6}alkyl, and wherein the C_{sub.1-6}alkyl is further optionally substituted with one or more —OH or C_{sub.1-6}alkoxy; L^{sup.2} is O or N(R^{sup.x}), wherein R^{sup.x} is H or C_{sub.1-6}alkyl; and either (1) L^{sup.3} is absent or is O, C_{sub.3-10}cycloalkyl, 3-10 membered heterocyclyl, or C_{sub.1-6}alkylene, wherein the C_{sub.3-10}cycloalkyl of L^{sup.3} is optionally substituted with one or more —OH or C_{sub.1-6}alkyl, the C_{sub.1-6}alkylene of L^{sup.3} is optionally substituted with one or more —OH or C_{sub.1-6}alkyl, wherein the C_{sub.1-6}alkyl is optionally substituted with one or more —OH, and the 3-10 membered heterocyclyl of L^{sup.3} is optionally substituted with one or more —OH or C_{sub.1-6}alkyl; X^{sup.1} and X^{sup.2} are each independently N or C(R^{sup.5}); and R^{sup.4} is: (i) —S(O)_{sub.2}—R^{sup.a}; (ii) 5-20 membered heteroaryl, wherein the 5-20 membered heteroaryl of R^{sup.4} is optionally substituted with one or more C_{sub.1-6}alkyl; (iii) —N(R^{sup.d})_{sub.2}, wherein R^{sup.d} is independently at each occurrence H, C_{sub.1-6}alkyl, or —S(O)_{sub.2}—R^{sup.a}, wherein the C_{sub.1-6}alkyl of R^{sup.d} is optionally substituted with one or more —OH, (iv) —

NS(O)—(C.sub.1-6alkyl).sub.2, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH, (v) —C(O)—N(R.sup.e).sub.2 wherein R.sup.e is independently at each occurrence H, C.sub.1-6 alkyl, or 3-10 membered heterocycle, wherein the 3-10 membered heterocycle of R.sup.e is optionally substituted with one or more oxo, or both R.sup.e together with the N to which they are attached are taken together to form a 3-10 membered heterocyclyl, wherein the 3-10 membered heterocyclyl is optionally substituted with one or more halo, oxo, —OH, NH.sub.2, —NH—S(O).sub.2—R.sup.a, or —S(O).sub.2—R.sup.a, (vi) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl, —OH, oxo or —S(O).sub.2R.sup.a, (vii) —S(O)—N(C.sub.1-6alkyl)-(C.sub.1-6alkyl), (viii) —CN, (ix) —(CH.sub.2).sub.qOH, wherein q is an integer from 0-6, (x) —C(O)—C.sub.1-6alkyl, or (xi) —P(O)(C.sub.1-6alkyl).sub.2; or (2) L.sup.3 is absent; and one of X.sup.1 and X.sup.2 is N or C(R.sup.5); and the other of X.sup.1 and X.sup.2 is N or C that is taken together with R.sup.4, and the atoms to which they are attached, to form a 5-10 membered heterocyclyl or a 5-20 membered heteroaryl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R.sup.b, wherein R.sup.b is, independently at each occurrence, selected from the group consisting of —OH, halo, oxo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10cycloalkyl, and 3-10 membered heterocyclyl, wherein the C.sub.1-6alkyl of R.sup.b is optionally substituted with one or more halo, OH, —S(O).sub.2—C.sub.1-6alkyl, or C.sub.3-10cycloalkyl, and wherein the C.sub.3-10cycloalkyl of the C.sub.1-6alkyl of R.sup.b is further optionally substituted with one or more C.sub.1-6alkyl or —OH and the C.sub.3-10cycloalkyl of R.sup.b is optionally substituted with one or more —OH, C.sub.3-10cycloalkyl, or C.sub.1-6alkyl, and wherein the C.sub.1-6alkyl of the C.sub.3-10cycloalkyl of R.sup.b is further optionally substituted with one or more —OH, deuterium, or halo, and the 5-20 membered heteroaryl is optionally substituted with one or more R.sup.c, wherein R.sup.c is, independently at each occurrence, selected from the group consisting of halo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10cycloalkyl, and 3-10 membered heterocyclyl, wherein the C.sub.1-6alkyl of R.sup.c is optionally substituted with one or more —S(O).sub.2—C.sub.1-6alkyl, the C.sub.3-10cycloalkyl of R.sup.c is optionally substituted with one or more —OH or C.sub.1-6alkyl, and the 3-10 membered heterocyclyl of R.sup.c is optionally substituted with one or more —OH or C.sub.1-6alkyl, and wherein the C.sub.1-6alkyl of the 3-10 membered heterocyclyl of R.sup.c is further optionally substituted with one or more —OH; R.sup.a is, independently at each occurrence: (i) C.sub.1-6alkyl optionally substituted with one or more halo, —OH, —S(O).sub.2—C.sub.1-6alkyl, or —N(C.sub.1-6alkyl)-C(O)—C.sub.1-6alkyl, (ii) C.sub.3-10cycloalkyl optionally substituted with one or more —OH, —C(O).sub.2—C.sub.1-6alkyl, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, or —C(O)—C.sub.3-10heterocyclyl, or C.sub.1-6alkyl, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH, (iii) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl, or (iv) NH(C.sub.1-6alkyl); R.sup.5 is, independently at each occurrence, H, halo, —CN, 3-10 membered heterocyclyl, C.sub.1-6alkyl, or C.sub.1-6alkoxy, wherein the C.sub.1-6alkyl of R.sup.5 is optionally substituted with one or more halo or —OH, and the C.sub.1-6alkoxy of R.sup.5 is optionally substituted with one or more halo; X.sup.3 is N or C(R.sup.6); X.sup.4 is N or C(R.sup.7); and R.sup.6 and R.sup.7 are each independently H or halo; to give a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing.

32. The method of claim 31, wherein the compound of claim 1 is prepared by a step comprising: a) alkylation of an amine of formula (II'-A) with an alkyl halide, or sulfonate ester compound of formula (II'-B) in the presence of an inorganic base; or b) reductive amination of a ketone of formula (II'-B) with an amine of formula (II'-A).

33. The method of claim 32, wherein the inorganic base is selected from the group consisting of

potassium carbonate, and sodium bicarbonate.

34. The method of claim 32, wherein the reductive amination proceeds under the action of sodium triacetoxyborohydride, titanium tetraisopropoxide and acetic acid.

35. A method for preparing a compound of formula (II), as recited in claim 1, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, comprising a step of reacting a compound of formula (II'—C): ##STR02202## wherein: m is an integer from 0 to 4; n is an integer from 0 to 2; p is an integer from 0 to 10; R^{sup.1}, if present is, independently at each occurrence selected from the group consisting of halo, —CN, C_{sub.1-6}alkoxy, and C_{sub.1-6}alkyl, wherein the C_{sub.1-6}alkoxy of R^{sup.1} is optionally substituted with one or more halo, and the C_{sub.1-6}alkyl of R^{sup.1} is optionally substituted with one or more halo; R^{sup.2} is H, C_{sub.1-6}alkyl, C_{sub.3-10}cycloalkyl, or 3-15 membered heterocyclyl, wherein the C_{sub.1-6}alkyl of R^{sup.2} is optionally substituted with one or more deuterium, halo, —OH, —NH_{sub.2}, or C_{sub.1-6}alkoxy, and the C_{sub.3-10}cycloalkyl of R^{sup.2} is optionally substituted with one or more —OH; R^{sup.3}, if present, is C_{sub.1-6}alkyl; L^{sup.1} is C_{sub.1-6}alkylene, wherein the C_{sub.1-6}alkylene of L^{sup.1} is optionally substituted with one or more deuterium or C_{sub.1-6}alkyl, and wherein the C_{sub.1-6}alkyl is further optionally substituted with one or more —OH or C_{sub.1-6}alkoxy; and Y^{sup.2} is halo, —OH or —NH_{sub.2}; with: a compound of formula (II'—D): ##STR02203## wherein: Y^{sup.3} is —OH or —NH(R'), wherein each R^{sup.x} is independently H or C_{sub.1-6}alkyl; and either (1) L^{sup.3} is absent or is O, C_{sub.3-10}cycloalkyl, 3-10 membered heterocyclyl, or C_{sub.1-6}alkylene, wherein the C_{sub.3-10}cycloalkyl of L^{sup.3} is optionally substituted with one or more —OH or C_{sub.1-6}alkyl, the C_{sub.1-6}alkylene of L^{sup.3} is optionally substituted with one or more —OH or C_{sub.1-6}alkyl, wherein the C_{sub.1-6}alkyl is optionally substituted with one or more —OH, and the 3-10 membered heterocyclyl of L^{sup.3} is optionally substituted with one or more —OH or C_{sub.1-6}alkyl; X^{sup.1} and X^{sup.2} are each independently N or C(R^{sup.5}); and R^{sup.4} is: (i) —S(O)_{sub.2}—R^{sup.a}; (ii) 5-20 membered heteroaryl, wherein the 5-20 membered heteroaryl of R^{sup.4} is optionally substituted with one or more C_{sub.1-6}alkyl; (iii) —N(R^{sup.d})_{sub.2}, wherein R^{sup.d} is independently at each occurrence H, C_{sub.1-6}alkyl, or —S(O)_{sub.2}—R^{sup.a}, wherein the C_{sub.1-6}alkyl of R^{sup.d} is optionally substituted with one or more —OH, (iv) —NS(O)—(C_{sub.1-6}alkyl)_{sub.2}, wherein the C_{sub.1-6}alkyl is optionally substituted with one or more —OH, (v) —C(O)—N(R^{sup.e})_{sub.2} wherein R^{sup.e} is independently at each occurrence H, C_{sub.1-6}alkyl, or 3-10 membered heterocycle, wherein the 3-10 membered heterocycle of R^{sup.e} is optionally substituted with one or more oxo, or both R^{sup.e} together with the N to which they are attached are taken together to form a 3-10 membered heterocyclyl, wherein the 3-10 membered heterocyclyl is optionally substituted with one or more halo, oxo, —OH, NH_{sub.2}, —NH—S(O)_{sub.2}—R^{sup.a}, or —S(O)_{sub.2}—R^{sup.a}, (vi) 3-10 membered heterocyclyl optionally substituted with one or more C_{sub.1-6}alkyl, —OH, oxo or —S(O)_{sub.2}R^{sup.a}, (vii) —S(O)—N(C_{sub.1-6}alkyl)-(C_{sub.1-6}alkyl), (viii) —CN, (ix) —(CH_{sub.2})_{sub.q}OH, wherein q is an integer from 0-6, (x) —C(O)—C_{sub.1-6}alkyl, or (xi) —P(O)(C_{sub.1-6}alkyl)_{sub.2}; or (2) L^{sup.3} is absent; and one of X^{sup.1} and X^{sup.2} is N or C(R^{sup.5}); and the other of X^{sup.1} and X^{sup.2} is N or C that is taken together with R^{sup.4}, and the atoms to which they are attached, to form a 5-10 membered heterocyclyl or a 5-20 membered heteroaryl, wherein the 5-10 membered heterocyclyl is optionally substituted with one or more R^{sup.b}, wherein R^{sup.b} is, independently at each occurrence, selected from the group consisting of —OH, halo, oxo, C_{sub.1-6}alkyl, —C(O)—C_{sub.1-6}alkyl, —C(O)—NH_{sub.2}, —C(O)—NH(C_{sub.1-6}alkyl), —C(O)—N(C_{sub.1-6}alkyl)_{sub.2}, —S(O)_{sub.2}—R^{sup.a}, C_{sub.3-10}cycloalkyl, and 3-10 membered heterocyclyl, wherein the C_{sub.1-6}alkyl of R^{sup.b} is optionally substituted with one or more halo, OH, —S(O)_{sub.2}—C_{sub.1-6}alkyl, or C_{sub.3-10}cycloalkyl, and wherein the C_{sub.3-10}cycloalkyl of the C_{sub.1-6}alkyl of R^{sup.b} is further optionally substituted with one or more C_{sub.1-6}alkyl or —OH and the C_{sub.3-10}cycloalkyl of R^{sup.b} is optionally substituted with one or more —OH, C_{sub.3-10}cycloalkyl, or C_{sub.1-6}alkyl, and wherein the C_{sub.1-}

6alkyl of the C.sub.3-10cycloalkyl of R.sup.b is further optionally substituted with one or more —OH, deuterium, or halo, and the 5-20 membered heteroaryl is optionally substituted with one or more R.sup.e, wherein R.sup.e is, independently at each occurrence, selected from the group consisting of halo, C.sub.1-6alkyl, —C(O)—C.sub.1-6alkyl, —C(O)—NH.sub.2, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, —S(O).sub.2—R.sup.a, C.sub.3-10cycloalkyl, and 3-10 membered heterocyclyl, wherein the C.sub.1-6alkyl of R.sup.c is optionally substituted with one or more —S(O).sub.2—C.sub.1-6alkyl, the C.sub.3-10cycloalkyl of R.sup.e is optionally substituted with one or more —OH or C.sub.1-6alkyl, and the 3-10 membered heterocyclyl of R.sup.c is optionally substituted with one or more —OH or C.sub.1-6alkyl, and wherein the C.sub.1-6alkyl of the 3-10 membered heterocyclyl of R.sup.e is further optionally substituted with one or more —OH; R.sup.a is, independently at each occurrence: (i) C.sub.1-6alkyl optionally substituted with one or more halo, —OH, —S(O).sub.2—C.sub.1-6alkyl, or —N(C.sub.1-6alkyl)-C(O)—C.sub.1-6alkyl, (ii) C.sub.3-10cycloalkyl optionally substituted with one or more —OH, —C(O).sub.2—C.sub.1-6alkyl, —C(O)—NH(C.sub.1-6alkyl), —C(O)—N(C.sub.1-6alkyl).sub.2, or —C(O)—C.sub.3-10heterocyclyl, or C.sub.1-6alkyl, wherein the C.sub.1-6alkyl is optionally substituted with one or more —OH, (iii) 3-10 membered heterocyclyl optionally substituted with one or more C.sub.1-6alkyl, or (iv) NH(C.sub.1-6alkyl); R.sup.5 is, independently at each occurrence, H, halo, —CN, 3-10 membered heterocyclyl, C.sub.1-6alkyl, or C.sub.1-6alkoxy, wherein the C.sub.1-6alkyl of R.sup.5 is optionally substituted with one or more halo or —OH, and the C.sub.1-6alkoxy of R.sup.5 is optionally substituted with one or more halo; X.sup.3 is N or C(R.sup.6); X.sup.4 is N or C(R.sup.7); and R.sup.6 and R.sup.7 are each independently H or halo; to give a compound of formula (II), or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing.

36. The method of claim 35, wherein the compound of claim 1 is prepared by a step comprising: a) coupling of an alcohol compound of formula (II'—C) with a phenol compound of formula (II'-D), or a heterocyclic variant, under Mitsunobu-type reaction conditions; or b) reacting an alkyl halide compound of formula (II'—C) with a phenol or amine compound of formula (II'-D), in the presence of a catalyst.

37. The method of claim 36, wherein the Mitsunobi-type reaction conditions comprises coupling of an alcohol compound of formula (II'—C) with a phenol compound of formula (II'-D), or a heterocyclic variant, in the presence of triphenylphosphine and diisopropyl azodicarboxylate.

38. The method of claim 37, wherein the catalyst is silver oxide or potassium carbonate.

39. A pharmaceutical composition, comprising (i) a compound of any one of claims 1-30, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, and (ii) one or more pharmaceutically acceptable excipients.

40. A method of modulating APOL1 in a cell, comprising exposing the cell to a composition comprising an effective amount of a compound of any one or claims 1-30, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or a pharmaceutical composition of claim 39.

41. A method of inhibiting APOL1 in a cell, comprising exposing the cell to a composition comprising an effective amount of a compound of any one or claims 1-30 or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or a pharmaceutical composition of claim 39.

42. A method of treating an APOL1-mediated disease, disorder, or condition in an individual in need thereof, comprising administering to the individual a compound of any one of claims 1-30, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or a pharmaceutical composition of claim 39.

43. The method of claim 42, wherein the disease, disorder, or condition is selected from the group consisting of chronic kidney disease, focal segmental glomerulosclerosis (FSGS), hypertension-attributed kidney disease, human immunodeficiency virus-associated nephropathy (HIVAN),

sickle-cell nephropathy, lupus nephritis, diabetic kidney disease, APOL1-associated nephropathy, viral nephropathy, COVID-19 associated nephropathy, preeclampsia, and sepsis.

44. The method of claim 42 or claim 43, wherein the disease, disorder, or condition is a kidney disease.

45. The method of any one of claims 40-44, wherein the disease, disorder, or condition is a chronic kidney disease (CKD).

46. A method of delaying the development of an APOL1-mediated disease, disorder, or condition, comprising administering a compound of any one of claims 1-30, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or a pharmaceutical composition of claim 39, to an individual who is at risk of developing an APOL1-mediated disease, disorder, or condition.

47. The method of claim 46, wherein the APOL1-mediated disease, disorder, or condition is a kidney disease.

48. The method of claim 46 or claim 47, wherein the APOL1-mediated disease, disorder, or condition is a chronic kidney disease.

49. The method of claim 46, wherein the APOL1-mediated disease, disorder, or condition is selected from the group consisting of chronic kidney disease, focal segmental glomerulosclerosis (FSGS), hypertension-attributed kidney disease, human immunodeficiency virus-associated nephropathy (HIVAN), sickle-cell nephropathy, lupus nephritis, diabetic kidney disease, APOL1-associated nephropathy, viral nephropathy, COVID-19 associated nephropathy, preeclampsia, and sepsis.

50. The method of any one of claims 40-49, wherein the individual has an APOL1 mutation.

51. The method of claim 50, wherein the APOL1 mutation is a gain-of-function mutation.

52. The method of any one of claims 42-51, wherein a therapeutically effective amount of a compound of any one of claims 1-30, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or a pharmaceutical composition of claim 39, is administered.

53. A kit, comprising (i) a compound of any one of claims 1-30, or a stereoisomer or tautomer thereof, or a pharmaceutically acceptable salt of any of the foregoing, or a pharmaceutical composition of claim 39, and (ii) instructions for use in treating an APOL1-mediated disease, disorder, or condition in an individual in need thereof.

54. The kit of claim 53, wherein the disease, disorder, or condition is a kidney disease.

55. The kit of claim 53 or claim 54, wherein the disease, disorder, or condition is a chronic kidney disease (CKD).

56. The kit of any one of claims 53-55, wherein the disease, disorder, or condition is selected from the group consisting of chronic kidney disease, focal segmental glomerulosclerosis (FSGS), hypertension-attributed kidney disease, human immunodeficiency virus-associated nephropathy (HIVAN), sickle-cell nephropathy, lupus nephritis, diabetic kidney disease, APOL1-associated nephropathy, viral nephropathy, COVID-19 associated nephropathy, preeclampsia, and sepsis.

57. The kit of any one of claims 53-56, wherein the individual has an APOL1 mutation.

58. The kit of claim 57, wherein the APOL1 mutation is a gain-of-function mutation.
